

## PHARMACOLOGY 1 3

### CYP450 INDUCERS & INHIBITORS 4

|                                              |   |
|----------------------------------------------|---|
| Enzyme Inducers — "GPRS Cell Phone"          | 5 |
| Enzyme Inhibitors — "SICK FACES.COM"         | 6 |
| Interactions That Change How a Drug Is Taken | 8 |

### ADRENERGIC DRUGS 10

|                                                         |    |
|---------------------------------------------------------|----|
| Non-Selective Direct Agonists                           | 12 |
| α1-Selective Agonists                                   | 14 |
| α2-Selective Agonists — Centrally-Acting Sympatholytics | 15 |
| β1-Selective Agonists                                   | 16 |
| β2-Selective Agonists                                   | 17 |
| β3-Selective Agonists                                   | 19 |
| Dopamine-Receptor Agonists                              | 20 |
| Indirect & Mixed-Acting Sympathomimetics                | 21 |
| Selective α1-Blockers                                   | 23 |
| Non-Selective α-Blockers                                | 25 |
| Non-Selective β-Blockers                                | 26 |
| Cardioselective (β1) Blockers                           | 28 |
| β-Blockers with ISA                                     | 30 |
| α + β Blockers (Third-Generation)                       | 31 |
| Clinical Applications                                   | 32 |

### CHOLINERGIC AGENTS 34

|                                                           |    |
|-----------------------------------------------------------|----|
| Direct-Acting Cholinergic Agonists                        | 35 |
| Reversible Cholinesterase Inhibitors                      | 37 |
| Irreversible Cholinesterase Inhibitors (Organophosphates) | 39 |
| Cholinesterase-Inhibitor Poisoning — Treatment            | 40 |
| Ganglionic Stimulants — Nicotine                          | 41 |
| Ganglionic Blockers                                       | 42 |
| Muscarinic Antagonists (Antimuscarinics)                  | 43 |
| Neuromuscular Blockers — Non-Depolarizing                 | 45 |
| Neuromuscular Blockers — Depolarizing                     | 47 |

### DRUGS AFFECTING GI MOTILITY 48

|                                            |    |
|--------------------------------------------|----|
| Bulk-Forming Laxatives                     | 50 |
| Osmotic Laxatives                          | 51 |
| Stool-Wetting Laxatives & Emollients       | 53 |
| Chloride Secretion Activators              | 54 |
| Stimulant Laxatives                        | 55 |
| IBAT Inhibitors                            | 56 |
| Oral Rehydration Therapy                   | 57 |
| Absorbents, Bulk-Formers & Mucosal Coaters | 58 |
| Enkephalinase Inhibitors                   | 59 |
| Opioid Anti-Motility Agents                | 60 |
| Prokinetics — D2 Antagonists               | 61 |
| Prokinetics — 5-HT4 Agonists               | 62 |
| Prokinetics — Cholinergic Enhancers        | 63 |
| Anti-Spasmotics                            | 64 |

### DRUGS FOR ACID-PEPTIC DISEASES 66

|                                                       |    |
|-------------------------------------------------------|----|
| Antacids                                              | 68 |
| Anti-flatuses, Digestive Enzymes & Anti-reflux Agents | 69 |
| H <sub>2</sub> Antagonists                            | 70 |
| Proton Pump Inhibitors (PPIs)                         | 72 |
| Potassium-Competitive Acid Blockers (PCABs)           | 74 |
| Misoprostol                                           | 75 |
| Mucus-Enhancing Agents                                | 76 |
| Sucralfate                                            | 77 |
| Bismuth-Containing Drugs                              | 78 |

### DRUGS FOR INFLAMMATORY BOWEL DISEASE 79

|                                               |    |
|-----------------------------------------------|----|
| 5-Aminosalicylates (5-ASA)                    | 81 |
| Corticosteroids                               | 83 |
| Thiopurines                                   | 85 |
| Methotrexate                                  | 87 |
| Anti-TNF-α Biologics                          | 88 |
| Anti-Integrin (Vedolizumab)                   | 90 |
| Anti-IL-12/23 (Ustekinumab)                   | 91 |
| JAK Inhibitor (Tofacitinib)                   | 92 |
| Nutritional Therapy & Extraintestinal Disease | 93 |

### PROSTANOIDS, NSAIDS & LEUKOTRIENE ANTAGONISTS 94

|                                      |     |
|--------------------------------------|-----|
| NSAIDs — Shared Mechanism & Effects  | 96  |
| NSAID Adverse Effects & Interactions | 97  |
| Non-selective NSAIDs (tNSAIDs)       | 98  |
| Aspirin                              | 101 |
| COX-2 Selective Inhibitors (Coxibs)  | 102 |
| Paracetamol (Acetaminophen)          | 104 |
| Prostanoid Analogues                 | 105 |
| Leukotriene Modifiers                | 106 |
| PAF Antagonists                      | 107 |

### DRUGS FOR GOUT 108

|                             |     |
|-----------------------------|-----|
| Colchicine                  | 110 |
| NSAIDs                      | 112 |
| Corticosteroids             | 113 |
| IL-1 Inhibitors             | 114 |
| Xanthine Oxidase Inhibitors | 115 |
| Uricosurics                 | 117 |
| Uricase                     | 118 |
| Treatment Strategy          | 119 |

### DRUGS IN OA & RA 120

|                                                             |     |
|-------------------------------------------------------------|-----|
| Acetaminophen                                               | 121 |
| NSAIDs (oral)                                               | 122 |
| Topical Agents                                              | 124 |
| Tramadol                                                    | 125 |
| Duloxetine                                                  | 126 |
| Intra-articular Agents                                      | 127 |
| Glucosamine, Chondroitin & DMOADs                           | 128 |
| Glucocorticoids (RA)                                        | 129 |
| csDMARDs — Methotrexate                                     | 130 |
| csDMARDs — Sulfasalazine · Leflunomide · Hydroxychloroquine | 131 |
| bDMARDs — TNF Inhibitors                                    | 133 |
| bDMARDs — Interleukin Inhibitors                            | 134 |
| bDMARDs — B-cell & T-cell Targeted                          | 135 |
| tsDMARDs — JAK Inhibitors                                   | 136 |
| Comparison: RA vs OA                                        | 137 |

### HISTAMINE, ANTIHISTAMINES & SEROTONIN AGONISTS 138

|                                           |     |
|-------------------------------------------|-----|
| H1 Antihistamines — Mechanism & Effects   | 139 |
| 1st-Generation H1 Antihistamines          | 140 |
| 2nd-Generation H1 Antihistamines          | 142 |
| Clinical Uses of H1 Antihistamines        | 143 |
| Bethahistine (H3)                         | 144 |
| Triptans — 5-HT <sub>1B/1D</sub> Agonists | 145 |
| Ergots                                    | 147 |

### DIURETICS 149

|                                  |     |
|----------------------------------|-----|
| Carbonic Anhydrase Inhibitor     | 151 |
| Thiazide & Thiazide-like         | 152 |
| Loop Diuretic                    | 154 |
| K <sup>+</sup> -Sparing Diuretic | 155 |
| Osmotic Diuretic                 | 156 |
| Clinical Applications            | 157 |

### VASODILATORS 158

|                                           |     |
|-------------------------------------------|-----|
| Hydralazine                               | 160 |
| Minoxidil — K <sup>+</sup> Channel Opener | 161 |
| Sodium Nitroprusside                      | 162 |
| Organic Nitrates                          | 163 |
| PDE-5 Inhibitors                          | 165 |
| Endothelin Receptor Antagonists (ERAs)    | 168 |
| sGC Stimulators                           | 170 |
| Neprilysin Inhibitor / ARNI               | 171 |

### CALCIUM CHANNEL BLOCKERS 173

|                              |     |
|------------------------------|-----|
| Dihydropyridines (DHP)       | 175 |
| Phenylalkylamine — Verapamil | 177 |
| Benzothiazepine — Diltiazem  | 178 |
| Other Non-Dihydropyridines   | 179 |
| Comparison & Clinical Use    | 180 |

### THE RENIN-ANGIOTENSIN SYSTEM 182

|                                                  |     |
|--------------------------------------------------|-----|
| ACE Inhibitor                                    | 184 |
| Angiotensin Receptor Blocker (ARB)               | 186 |
| Direct Renin Inhibitor (DRI)                     | 188 |
| ARNI — Angiotensin Receptor–Neprilysin Inhibitor | 189 |

### ANTIHYPERTENSIVE AGENTS 190

|                                 |     |
|---------------------------------|-----|
| Diuretics in Hypertension       | 192 |
| Centrally Acting Sympatholytics | 193 |
| Adrenergic Neuron Blockers      | 194 |
| α- and Mixed αβ-Blockers        | 195 |
| β-Blockers in Hypertension      | 196 |
| Direct Vasodilators             | 197 |
| Angiotensin Blockade            | 198 |
| Why It Matters                  | 199 |

### ANTIANGINAL AGENTS 200

|                                                     |     |
|-----------------------------------------------------|-----|
| Organic Nitrates                                    | 201 |
| Calcium Channel Blockers (CCBs)                     | 203 |
| β-Blockers                                          | 205 |
| Ivabradine — Direct Bradycardic Agent               | 206 |
| Ranolazine — Late Na <sup>+</sup> Current Inhibitor | 207 |
| Trimetazidine — Metabolic Modulator                 | 208 |
| Combination Therapy                                 | 209 |

### ANTIARRHYTHMIC DRUGS 210

|                                                             |     |
|-------------------------------------------------------------|-----|
| Class IA — Na <sup>+</sup> Block, Intermediate Dissociation | 211 |
| Class IB — Na <sup>+</sup> Block, Fast Dissociation         | 212 |
| Class IC — Na <sup>+</sup> Block, Slow Dissociation         | 213 |
| Class II — β-Adrenergic Antagonists                         | 214 |
| Class III — K <sup>+</sup> Channel Blockers                 | 215 |
| Class IV — Ca <sup>2+</sup> Channel Blockers                | 216 |
| Unclassified Agents                                         | 217 |
| Drug-Induced QT Prolongation & Torsades                     | 218 |

### Clinical Use Principles 219

### PHARMACOLOGY OF HEART FAILURE 220

|                                                  |     |
|--------------------------------------------------|-----|
| Cardiac Glycosides                               | 221 |
| β <sub>1</sub> -Adrenergic Agonists              | 223 |
| Phosphodiesterase Inhibitors                     | 224 |
| Calcium Sensitizer                               | 225 |
| Loop Diuretics                                   | 226 |
| K <sup>+</sup> -Sparing Diuretics & MRAs         | 227 |
| Nitrovasodilators (NO Donors)                    | 228 |
| Hydralazine + Nitrate                            | 229 |
| ACE Inhibitors                                   | 230 |
| Angiotensin Receptor Blockers                    | 231 |
| ARNI (Angiotensin Receptor–Neprilysin Inhibitor) | 232 |
| Nesiritide (Recombinant BNP)                     | 233 |
| SGLT2 Inhibitors                                 | 234 |
| β-Adrenergic Blockers in HF                      | 235 |
| Ivabradine                                       | 236 |
| Other & Emerging Agents                          | 237 |
| What Actually Changes Mortality                  | 238 |

### ANTITHROMBOTICS 239

|                               |     |
|-------------------------------|-----|
| COX-1 Inhibitors (Aspirin)    | 240 |
| P2Y <sub>12</sub> Antagonists | 241 |
| GPIIb/IIIa Antagonists        | 244 |
| PDE Inhibitors                | 245 |
| PAR-1 Antagonists             | 246 |
| Heparins & Fondaparinux       | 247 |
| Warfarin                      | 249 |
| DOACs / NOACs                 | 250 |
| Thrombolytics / Fibrinolytics | 252 |
| Reversal & Clinical Use       | 253 |

### LIPID-LOWERING AGENTS 254

|                                              |     |
|----------------------------------------------|-----|
| Statins (HMG-CoA Reductase Inhibitors)       | 255 |
| Bempedoic Acid (ATP-Citrate Lyase Inhibitor) | 257 |
| PCSK9 Inhibitors                             | 258 |
| Ezetimibe (Cholesterol Absorption Inhibitor) | 260 |
| Bile Acid Sequestrants                       | 261 |
| Fibrates (Fibric Acid Derivatives)           | 262 |
| Niacin (Nicotinic Acid)                      | 264 |
| Omega-3 Fatty Acids                          | 265 |

## PHARMACOLOGY 2 266

### PRINCIPLES OF ANTIBIOTIC USE 268

|                                            |     |
|--------------------------------------------|-----|
| Host & Pathogen Factors                    | 269 |
| Bactericidal vs Bacteriostatic             | 271 |
| Time- vs Concentration-Dependent Killing   | 272 |
| Reaching the Site of Infection             | 273 |
| Empiric, Definitive & Prophylactic Therapy | 274 |
| Combination Therapy                        | 275 |
| Adverse Effects of Antimicrobials          | 276 |
| Antimicrobial Resistance                   | 277 |

### DRUGS AFFECTING CELL WALL SYNTHESIS 278

|                                       |     |
|---------------------------------------|-----|
| Penicillin-Binding Proteins (PBPs)    | 279 |
| Penicillins — Shared Properties       | 280 |
| Natural Penicillins                   | 281 |
| Penicillinase-Resistant Penicillins   | 283 |
| Aminopenicillins                      | 285 |
| Antipseudomonal Penicillins           | 287 |
| β-Lactamase Inhibitors & Combinations | 289 |
| Cephalosporins — Shared Properties    | 291 |
| Cephalosporins — 1st Generation       | 292 |
| Cephalosporins — 2nd Generation       | 293 |
| Cephalosporins — 3rd Generation       | 294 |
| Cephalosporins — 4th & 5th Generation | 296 |
| Carbapenems                           | 298 |
| β-Lactam Resistance                   | 300 |

### GLYCOPEPTIDES & LIPOGLYCOPEPTIDES 301

|                               |     |
|-------------------------------|-----|
| Glycopeptides                 | 302 |
| Lipoglycopeptides             | 304 |
| Glycopeptide Resistance (VRE) | 305 |
| Bacitracin                    | 306 |

### DRUGS AFFECTING THE CELL MEMBRANE 307

|            |     |
|------------|-----|
| Polymyxins | 308 |
| Daptomycin | 309 |

### INHIBITORS OF DNA, NUCLEIC ACID & PROTEIN SYNTHESIS 310

|                                              |     |
|----------------------------------------------|-----|
| Fluoroquinolones                             | 312 |
| Fluoroquinolones — Agent by Agent            | 315 |
| Metronidazole                                | 316 |
| Macrolides                                   | 317 |
| Macrolides — Agent by Agent                  | 320 |
| Ketolides (Telithromycin)                    | 321 |
| Lincosamides (Lincomycin & Clindamycin)      | 322 |
| Streptogramins (Quinupristin / Dalfopristin) | 324 |

|                                       |     |
|---------------------------------------|-----|
| Chloramphenicol                       | 325 |
| Aminoglycosides                       | 326 |
| Aminoglycosides — Dosing & Monitoring | 329 |

|                                       |     |
|---------------------------------------|-----|
| Tetracyclines                         | 330 |
| Glycylcyclines (Tigecycline)          | 332 |
| Oxazolidinones (Linezolid)            | 334 |
| Folate Synthesis Inhibitors (TMP-SMX) | 336 |

### ANTIMYCOBACTERIAL DRUGS 338

|                                          |     |
|------------------------------------------|-----|
| TB Regimens & When First-Line Fails      | 340 |
| Isoniazid (INH)                          | 341 |
| Rifamycins                               | 342 |
| Pyrazinamide                             | 344 |
| Ethambutol                               | 345 |
| Aminoglycosides (second-line)            | 346 |
| Fluoroquinolones (second-line)           | 347 |
| Oral Bacteriostatic Second-Line Agents   | 348 |
| Bedaquiline                              | 349 |
| Nitroimidazoles — Pretomanid & Delamanid | 350 |

|                                   |     |
|-----------------------------------|-----|
| Linezolid                         | 351 |
| Leprosy (Hansen's Disease)        | 352 |
| Dapsone                           | 353 |
| Clofazimine                       | 354 |
| Nontuberculous Mycobacteria (NTM) | 355 |

### ANTIFUNGAL DRUGS 356

|                     |     |
|---------------------|-----|
| Polyenes            | 358 |
| Allylamines         | 360 |
| Azoles — Imidazoles | 362 |
| Azoles — Triazoles  | 364 |
| Echinocandins       | 367 |
| Triterpenoids       | 369 |
| Flucytosine         | 370 |
| Griseofulvin        | 371 |
| Topical Antifungals | 372 |

ANTIPARASITIC DRUGS 373

Aminoquinolines — Chloroquine, Amodiaquine, Piperazine 375

Quinine & Quinidine 377

Mefloquine 378

Artemisinin & Derivatives (ACTs) 379

Other Heme-Polymerisation Inhibitors 381

8-Aminoquinolines — Primaquine & Tafenoquine 382

Atovaquone 384

Antimalarial Translation Inhibitors 385

Antimalarial Folate Inhibitors 386

Malaria in Practice 387

Nitroimidazoles — Metronidazole & Tinidazole 388

Pentamidine 390

Anti-Toxoplasma Agents 391

Benzimidazoles — Albendazole & Mebendazole 392

Ivermectin 394

Neuromuscular Paralysing Anthelmintics 395

Praziquantel 396

Niclosamide 397

Diethylcarbamazine (DEC) 398

Anthelmintics at a Glance 399

ANTI-HEPATITIS AGENTS 400

Interferons (IFN-α / PegIFN) 401

HBV Nucleoside / Nucleotide Analogues (NAs) 402

Ribavirin 404

HCV NS5B Polymerase Inhibitors 405

HCV NS5A Inhibitors 406

HCV NS3/4A Protease Inhibitors 407

Putting It Together 409

ANTIVIRALS & ANTI-HIV DRUGS 410

Antih herpes Agents 411

Anti-CMV Agents 413

Influenza — M2 Proton Channel Inhibitors 415

Influenza — Neuraminidase Inhibitors 416

Influenza — Polymerase Complex Inhibitors 417

Other Antivirals 418

HIV — Antiretroviral Therapy (ART) 419

NRTIs — Nucleoside/Nucleotide RT Inhibitors 420

NNRTIs — Non-Nucleoside RT Inhibitors 422

Protease Inhibitors (PIs, -navir) 423

Entry Inhibitors 425

INSTIs — Integrase Strand Transfer Inhibitors (-gravir) 426

HAAART & Pharmacokinetic Boosters 427

ANTIDIABETIC DRUGS 428

Insulin 429

GLP-1 Receptor Agonists 431

Dual GIP/GLP-1 Receptor Agonist 433

DPP-4 Inhibitors 434

Sulfonylureas 435

Meglitinides (Non-sulfonylureas) 437

Biguanide — Metformin 438

Thiazolidinediones (TZDs) 439

α-Glucosidase Inhibitors 441

SGLT2 Inhibitors 442

Choosing a Drug — ADA 2026 445

ESTROGENS, PROGESTINS & CONTRACEPTIVES 446

Estrogens 447

Selective Estrogen Receptor Modulators (SERMs) 450

Estrogen Receptor Antagonists 451

Aromatase Inhibitors 452

Progestins 453

Antiprogestins 455

Combined Oral Contraceptives (COC) 456

COC Use & Missed Pills 458

Progestin-Only Contraceptives (POC) 459

Emergency Contraception 460

THYROID HORMONES & ANTITHYROID DRUGS 461

Thyroid Hormone Replacement 463

Thionamides 465

Anion Inhibitors 467

Iodides 468

Radioactive Iodine (<sup>131</sup>I) 469

Adjunctive & Thyroid Storm Therapy 470

DRUGS REGULATING BONE-MINERAL HOMEOSTASIS 471

Calcium 472

Vitamin D Derivatives 474

Bisphosphonates 475

RANKL Inhibitors 477

Estrogens, STEARs & SERMs 478

Calcitonin 480

Parathyroid Hormone Analogues 481

Sclerostin Inhibitors 482

Vitamin K Analogues & Fluoride 483

Calcimimetics 484

Phosphate Binders 485

HYPOTHALAMIC-PITUITARY HORMONES 486

GnRH Agonists 487

GnRH Antagonists 488

Gonadotropins 489

Dopamine Agonists 490

Oxytocin & Oxytocin Antagonists 491

Growth Hormone 492

Somatostatin Analogs 493

Vasopressin & Analogs 494

Vasopressin Antagonists 495

ANABOLIC-ANDROGENIC STEROIDS & ANTI-ANDROGENS 496

Anabolic-Androgenic Steroids & Anti-Androgens 496

Testosterone & Testosterone Esters 498

17α-Alkylated Testosterone 500

Anabolic Steroids 502

AAS — Shared Effects, Toxicity & Doping 503

Steroid Synthesis Inhibitors 504

5α-Reductase Inhibitors 505

Steroidal Androgen Receptor Antagonists 506

Non-Steroidal Androgen Receptor Antagonists 507

ADRENAL STEROIDS & THEIR ANTAGONISTS 508

Mineralocorticoids 509

Mineralocorticoid Receptor Antagonists 510

Glucocorticoids — Actions & Adverse Effects 511

Glucocorticoids — Systemic 512

Glucocorticoids — Topical & Local 515

Glucocorticoid Antagonists 517

ANTICANCER AGENTS I 518

Alkylating Agents 520

Platinum Analogs 522

Antifolates 523

Ribonucleotide Reductase Inhibitor 525

Pyrimidine Analogs 526

Purine Analogs 528

Vinca Alkaloids 530

Taxanes 531

Topoisomerase I Inhibitors 533

Topoisomerase II Inhibitors 534

Anthracyclines 535

Bleomycin & Mitomycin C 537

L-Asparaginase 538

Toxicity, Resistance & the Big Picture 539

ANTICANCER AGENTS II — TARGETED THERAPY 540

EGFR Inhibitors — Small-Molecule TKIs 541

EGFR Inhibitors — Monoclonal Antibodies 542

How Therapeutic mAbs Actually Kill 543

HER2 Inhibitors 544

BCR-ABL Inhibitors 546

mTOR Inhibitors 548

BRAF Inhibitors 549

PARP Inhibitors 550

CDK4/6 Inhibitors 551

Angiogenesis (VEGF) Inhibitors 552

Proteasome Inhibitors 554

Cancer Vaccines 555

Immune Checkpoint Inhibitors 556

CAR T-Cell Therapy 557

Anti-Estrogens — SERMs & SERDs 558

Aromatase Inhibitors 559

Anti-Androgens 560

Glucocorticoids 561

Gene Expression Modulators 562

Target Map 563

IMMUNOMODULATORS 564

Induction Antibodies 565

B-Cell Targeted Therapy (Desensitization) 566

Corticosteroids 567

Calcineurin Inhibitors 568

Antimetabolites 569

mTOR Inhibitors 570

Costimulation Blocker 571

JAK Inhibitors 572

Maintenance Regimens 573

Immunostimulants 574

Vaccines 575

Intravenous Immunoglobulin (IVIG) 576

Specific Immunoglobulins 577

Myeloid Growth Factors & Erythropoietins 578

PHARMACOLOGY 3 584

DRUGS ACTING ON GABA-A RECEPTOR 586

Benzodiazepines 588

Benzodiazepine Tolerance & Withdrawal 591

Barbiturates 592

Z-Drugs 594

Etifoxine 595

Flumazenil 596

Clinical Applications 597

ANALGESICS I — OPIOIDS & PARACETAMOL 598

Weak Opioids 600

Strong Opioids 601

Partial Agonists & Mixed Agonist–Antagonists 604

Opioid Antagonists 605

Opioid Adverse Effects 606

Opioid Tolerance & Withdrawal 607

Opioid Pharmacokinetics & Equianalgesia 608

Paracetamol 610

Paracetamol Overdose & Antidote 611

CNS STIMULANTS AND DRUGS AFFECTING APPETITE 612

DRUG ABUSE AND ADDICTION 613

ANXIOLYTICS AND HYPNOTICS 614

Antidepressants for Anxiety 615

Benzodiazepines in Anxiety 617

Pregabalin & Gabapentin 618

Bupirone 619

Other Agents for Anxiety 620

Insomnia & Sleep Architecture 621

The Sleep–Wake Switch 622

Melatonin & Melatonin Receptor Agonists 623

Orexin Receptor Antagonists (DORAs) 624

Prescribing & Patient Information 625

ANTIDEPRESSANTS AND LITHIUM 626

Why Antidepressants Take Weeks 627

MAO Inhibitors (MAOIs and RIMAs) 628

Tricyclic Antidepressants (TCAs) 629

SSRIs 632

The 5-HT Effect Map 634

Serotonin Syndrome & Discontinuation 635

SARIs — Serotonin Antagonist and Reuptake Inhibitors 636

SNRIs 637

NDRIs 638

NaSSAs 639

Miscellaneous Antidepressants 640

Cross-Class Adverse Effects 641

Metabolism & Drug Interactions 642

Clinical Applications of Antidepressants 644

Lithium 645

Lithium — Interactions & Pregnancy 646

ANTIPSYCHOTICS 647

The D2 Map — One Receptor, Six Outcomes 648

Conventional (First-Generation) Antipsychotics 649

Extrapyramidal Side Effects 651

Hyperprolactinemia 652

Atypical (Second-Generation) Antipsychotics 653

The Three Atypical Families 655

D2 Receptor Partial Agonists 656

Comparative Adverse Effects 657

Serious & Drug-Specific Adverse Effects 659

Pharmacokinetics & Interactions 660

Clinical Applications 661

LOCAL AND GENERAL ANESTHETICS 662

NEURODEGENERATIVE DISEASE DRUGS 663

Alzheimer's Disease — The Cholinergic Story 664

Cholinesterase Inhibitors (ChEIs) 665

ChEI Adverse Effects & Interactions 666

NMDA Receptor Antagonist — Memantine 667

Anti-Amyloid Monoclonal Antibodies 668

Parkinson's Disease — The Dopaminergic Story 669

Levodopa + Dopa Decarboxylase Inhibitor 670

Levodopa Motor Fluctuation 672

MAO-B Inhibitors 673

COMT Inhibitors 674

Dopamine Receptor Agonists 675

Amantadine & Anticholinergics 676

ANTIEPILEPTICS 677

Na<sup>+</sup> Channel Blockers 679

Phenytoin 680

Carbamazepine & Oxcarbazepine 681

Lamotrigine 682

Ca<sup>2+</sup> Channel Blockers 683

GABA Enhancers 685

Broad Spectrum — Valproate 686

Broad Spectrum — Topiramate 687

AMPA Receptor Blocker — Perampanel 688

SV2A Modulators — Levetiracetam 689

Matching Drug to Seizure Type 690

Class-Wide Adverse Effects 691

Pharmacokinetics of the Conventional Four 692

Enzyme Induction & Inhibition 693

Beyond Epilepsy 694

MIGRAINE HEADACHE 695

Ergots 697

Triptans & Lasmiditan 698

Gepants — CGRP Receptor Antagonists 699

Abortive Therapy — Rules of Use 701

Preventive Therapy — Classical Oral Agents 703

Anti-CGRP Monoclonal Antibodies 705

ANALGESICS II — TWO SEPARATE TERRITORIES 708

Neuropathic Pain — the Disease 709

Sodium Channel Blockers 711

Gabapentinoids (α<sub>2</sub>-δ-1 Ligands) 712

Antidepressants — TCAs & SNRIs 713

Opioids — Tramadol & Methadone 714

Other Agents — Capsaicin, Ketamine, Cannabinoids 715

Centrally Acting Muscle Relaxants — Overview 716

Orphenadrine 718

Tolperisone & Eperisone 719

Benzodiazepines — Diazepam & Clonazepam 720

Baclofen 721

Tizanidine 722

Riluzole 723

Muscle Relaxants — Shared PK & Interactions 724

# Pharmacology 1

ภกปว 304 · ภาคต้น · autonomic → GI → inflammation → cardiovascular

# CYP450 Inducers & Inhibitors

Cytochrome P-450 (CYP) is the main enzyme system for drug metabolism — mostly **liver**, also gut and other organs. Six isozymes account for most known interactions: **CYP1A2 · 2C9 · 2C19 · 2D6 · 3A4 · 2E1**. A drug meets an isozyme in one of three roles, and two of those roles are the whole topic.

**Inducer** → ↑ **CYP activity** → ↓ **plasma level** → **treatment failure** · **Inhibitor** → ↓ **CYP activity** → ↑ **plasma level** → **toxicity**

## THREE ROLES AT THE ISOZYME

- **Substrate** — the drug *is* metabolised by that isozyme
- **Inducer** — "revs up" the isozyme system, giving it greater metabolic capacity
- **Inhibitor** — competes for the isozyme-binding site, rendering the isozyme inactive

## INDUCTION VS INHIBITION

|                  | INDUCTION                                       | INHIBITION                          |
|------------------|-------------------------------------------------|-------------------------------------|
| <b>Onset</b>     | 2 to 3 <b>weeks</b> to develop fully            | within 2 to 3 <b>days</b>           |
| <b>Off-set</b>   | slow to resolve after stopping                  | —                                   |
| <b>Result</b>    | ↓ effect of the affected drug                   | ↓ metabolism → <b>toxicity</b> ★    |
| <b>Frequency</b> | common                                          | ★ <b>more common than induction</b> |
| <b>Fix</b>       | raise the dose ( <i>needs good monitoring</i> ) | —                                   |

Both act on the same pathway: **phase I oxidation** by CYP isoenzymes.

## HOW YOU USE THE TWO LISTS

If a new drug is known to be an inducer, inhibitor, and/or substrate of a given isozyme, the interaction is **predictable** from the lists.

*Worked example from the lecture:* ciclosporin is a CYP3A4 **substrate** · rifampicin is a potent CYP3A4 **inducer** → it **lowers** ciclosporin levels · ketoconazole **inhibits** CYP3A4 → it **raises** them.

⚠ **CYP2D6, 2C9 and 2C19 are polymorphic** — some people are poor/slow metabolisers (≈5–10% white Caucasians, 0–2% Asian and black people). **CYP3A4 is not.**



# Enzyme Inducers — "GPRS Cell Phone"

## MECHANISM

The drug **increases the synthesis/activity of CYP isoenzymes**, so the affected drug is cleared faster. Takes **2–3 weeks** to develop completely and is **slow to resolve** after the inducer is stopped.

→ **Downstream:** ↑ metabolism of the affected drug → ↓ plasma concentration → **loss of effect**

## GPRS CELL PHONE

- **G** — Griseofulvin
- **P** — Phenytoin
- **R** — Rifampicin
- **S** — St. John's wort
- **C** — Carbamazepine
- **P** — Phenobarbitone
- **+** **Cigarette smoke**

## DRUGS

| DRUG                   | WHAT IT TREATS                  |
|------------------------|---------------------------------|
| <b>Griseofulvin</b>    | dermatophyte (fungal) infection |
| <b>Phenytoin</b>       | epilepsy                        |
| <b>Rifampicin</b>      | tuberculosis                    |
| <b>St. John's wort</b> | herbal — used for depression    |
| <b>Carbamazepine</b>   | epilepsy                        |
| <b>Phenobarbitone</b>  | epilepsy                        |
| <b>Cigarette smoke</b> | ★ not a drug — counts anyway    |

## WHAT FALLS WHEN YOU ADD AN INDUCER

★ **Decreased concentration of:** warfarin · quinidine · ciclosporin · losartan · oral contraceptives · methadone.

- Also inducing: **chlorinated hydrocarbon insecticides** (dicophane, lindane) and **smoking tobacco**
- **Rifampicin also induces P-glycoprotein** in the gut lining → ejects digoxin into the gut more vigorously → **digoxin plasma levels fall**

# Enzyme Inhibitors — "SICK FACES.COM"

## MECHANISM

The drug **competes for the isozyme-binding site**, inactivating it — so the affected drug is not cleared. Occurs within **2–3 days**, giving **rapid development of toxicity**. ★ **More common than induction**.

*If serum levels stay inside the therapeutic range the interaction may not be clinically important.*

→ **Downstream:** ↓ metabolism of the affected drug → ↑ plasma concentration → **toxicity**

## SICK FACES.COM

- **S** — Sulphonamides
- **I** — Isoniazid
- **C** — Cimetidine
- **K** — Ketoconazole
- **F** — Fluconazole
- **A** — Alcohol (*or Amiodarone*)
- **C** — Ciprofloxacin
- **E** — Erythromycin
- **S** — Sertraline
- **C** — Clarithromycin
- **O** — Omeprazole
- **M** — Metronidazole

*The slide prints the string and the list side by side, not the letter pairing — every drug above is taken from that list. The full list below has a few extras the letters don't reach.*

## DRUGS — THE FULL SLIDE LIST

| DRUG                                                                | WHAT IT TREATS                |
|---------------------------------------------------------------------|-------------------------------|
| <b>Sulphonamides</b>                                                | bacterial infection           |
| <b>Antifungals</b> — itraconazole · ketoconazole · fluconazole      | fungal infection              |
| <b>Macrolides</b> — clarithromycin · azithromycin · erythromycin    | bacterial infection           |
| <b>Ciprofloxacin</b>                                                | bacterial infection           |
| <b>Sertraline</b>                                                   | depression                    |
| <b>Cimetidine</b>                                                   | peptic ulcer / GERD           |
| <b>Omeprazole</b>                                                   | peptic ulcer / GERD           |
| <b>Metronidazole</b>                                                | anaerobic bacteria · protozoa |
| <b>Antivirals</b> — ritonavir · indinavir · nelfinavir · amprenavir | HIV                           |
| <b>Antiarrhythmics</b> — amiodarone · quinidine                     | arrhythmia                    |
| <b>Antidepressants</b> — fluoxetine · paroxetine                    | depression                    |
| <b>Isoniazid</b>                                                    | tuberculosis                  |
| <b>Alcohol</b>                                                      | —                             |

### THE LECTURE'S WORKED INTERACTIONS

- **Ketoconazole / erythromycin / grapefruit juice** → inhibit CYP3A4 → block terfenadine metabolism → **fatal cardiac arrhythmias (Torsades de pointes)**
- **Gemfibrozil (and other fibrates)** → CYP3A inhibition → statins not metabolised → **rhabdomyolysis**
- **Ritonavir · indinavir · nelfinavir · amprenavir · saquinavir** → inhibit CYP3A4 → ↑ **PDE-5 inhibitors** (sildenafil, tadalafil, vardenafil)
- **Cimetidine** — inhibits multiple CYPs → ↑ warfarin, theophylline, phenytoin → toxicity
- **Amiodarone** — inhibits many CYPs **and** P-glycoprotein → ↓ clearance of warfarin, digoxin, quinidine
- **Grapefruit juice** inhibits CYP3A4 **mainly in the gut** → ↓ first-pass metabolism of oral CCBs → ↑ levels

### CAUTION

- ⚠ **Ketoconazole also inhibits P-glycoprotein** — it prevents ritonavir efflux from the CNS, raising CSF levels. Same for **verapamil**, which blocks digoxin's ejection into the gut → **digoxin levels rise**.
- ⚠ An inhibitor's onset is days, an inducer's is weeks — **the timing of a new symptom tells you which one you're looking at.**

# Interactions That Change How a Drug Is Taken

## ★ ABSORPTION INTERACTIONS — THE ONES YOU FIX BY SEPARATING DOSES

| WHAT HAPPENS                                                                                                                   | THE CLASSIC PAIR                                                                                                                                                                | RESULT                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Chelation</b> — tetracyclines chelate <b>divalent and trivalent metal ions</b> ( <i>calcium, aluminium, bismuth, iron</i> ) | tetracycline + <b>antacids or dairy</b> ( <i>milk, yogurt, cheese</i> )                                                                                                         | complexes form → <b>poor absorption</b> → <b>reduced antibacterial effect</b>                                                |
| <b>Adsorption</b> — binds a large number of drugs indiscriminately                                                             | <b>activated charcoal or antacids</b> + almost anything                                                                                                                         | reduced absorption                                                                                                           |
| ↑ <b>gastric pH</b> — acidic drugs need acid to be absorbed                                                                    | <b>PPIs</b> ( <i>omeprazole, esomeprazole, pantoprazole, rabeprazole</i> ) or <b>H<sub>2</sub> blockers</b> ( <i>ranitidine</i> ) + acidic drugs ( <i>e.g. salicylic acid</i> ) | reduced absorption · ★ <b>ketoconazole</b> , a poorly soluble base, dissociates badly without acid → <b>absorption falls</b> |
| ↑ <b>gastric pH, the other direction</b> — basic drugs absorb better at high pH                                                | ranitidine + <b>triazolam</b>                                                                                                                                                   | <b>increased</b> triazolam absorption                                                                                        |
| <b>P-glycoprotein induction</b> — the transporter ejects drug back into the gut                                                | <b>rifampicin</b> + <b>digoxin</b>                                                                                                                                              | digoxin pumped back out → <b>plasma levels fall</b>                                                                          |

## ELIMINATION INTERACTIONS — URINARY PH DECIDES WHAT GETS REABSORBED

| DRUG                                                      | WITH                                    | WHAT HAPPENS                                                                                                                                                             |
|-----------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quinidine</b> ( <i>weak base, excreted unchanged</i> ) | antacids or <b>urinary alkalinisers</b> | urine pH ↑ → quinidine becomes <b>non-ionised and lipid-soluble</b> → diffuses into tubule cells → <b>reabsorbed</b> → <b>clearance falls</b> → <b>levels rise</b>       |
| <b>Aspirin</b> ( <i>weak acid</i> )                       | antacids                                | urine pH ↑ → aspirin becomes <b>ionised and lipid-insoluble</b> → can't diffuse back → <b>excretion increases</b> — ★ <b>this is used deliberately to treat overdose</b> |

- ★ **Same intervention, opposite outcomes, and the reason is just acid–base:** alkalinising the urine traps the *acid* (aspirin, excreted) and frees the *base* (quinidine, reabsorbed). If you remember which one is ionised at high pH, you never have to memorise the pairs.
- Food-drug interactions are their own category** alongside drug-drug, herb-drug, smoking, alcohol and disease. Beyond dairy: **vitamin K in broccoli and spinach reduces the effectiveness of warfarin.**
- 💡 Almost everything in the absorption table is fixed by **spacing the doses**, not by stopping a drug — which is the single most useful thing to be able to say at a counter.
- 💡 ⚠️ And the PPI trap is worth holding onto: a patient started on omeprazole can quietly lose their ketoconazole, with nothing to see except a fungal infection that won't clear.

Principle behind this: *[[pharm-pharmacokinetics]]* — that page is the memorised inventory; the principles page explains why induction and inhibition move a drug level.

Principle behind this: *[[pharm-drug-interactions]]* — The two lists in full — this page is deliberately the mechanism half and that one is the inventory. Read together or neither works.

*Principle behind this: [[pharm-pharmacogenomics]] — Genotype is the silent third variable in every interaction on that page: an inhibitor added to a poor metabolizer is not the same event as an inhibitor added to a normal one.*

*Source: [Drug interaction-for Pco-I.pdf](#) — ผศ.ดร.ภก. อานัฐชัย ม้ายอุเทศ, PYCH304 Pharmacology I, 90 slides (this page covers the metabolism block, slides 40–54, plus the absorption and elimination interaction blocks). Structures omitted.*

---

# Adrenergic Drugs

Drugs that **mimic** (sympathomimetic) or **block** (sympatholytic) the sympathetic nervous system — the NE/Epi system that controls the cardiovascular response to stress. Everything on the next pages is one question: **which adrenoceptor, and is the drug turning it on or off?**

**Tyrosine → DOPA → dopamine → NE → Epi · stored in vesicles (VMAT) → Ca<sup>2+</sup>-triggered exocytosis → adrenoceptor → terminated by uptake-1 (NET) · MAO / COMT**

## WORDS WORTH SEPARATING

- **Direct-acting** — binds the receptor itself
- **Indirect-acting** — displaces stored NE, or blocks its reuptake (NET); needs an intact nerve terminal
- **Mixed-acting** — does both (ephedrine)
- **ISA** — *intrinsic sympathomimetic activity*: a  $\beta$ -blocker that is really a **partial agonist**

## ADRENOCEPTORS — LOCATION VS EFFECT ★

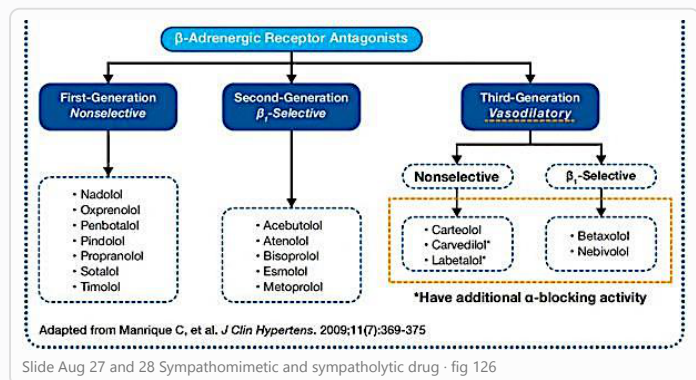
| RECEPTOR                     | LOCATION                                     | ACTION                                                      |
|------------------------------|----------------------------------------------|-------------------------------------------------------------|
| <b><math>\alpha_1</math></b> | most vascular smooth muscle                  | <b>contraction</b> → vasoconstriction                       |
|                              | pupillary dilator muscle                     | contraction → <b>mydriasis</b>                              |
|                              | prostate, prostatic/bladder trigone          | contraction → ↓ urine flow                                  |
|                              | pilomotor smooth muscle                      | erects hair                                                 |
| <b><math>\alpha_2</math></b> | adrenergic & cholinergic nerve terminals     | <b>inhibits transmitter release</b> ( <i>autoreceptor</i> ) |
|                              | platelets                                    | stimulates aggregation                                      |
|                              | adipocytes                                   | inhibits lipolysis                                          |
|                              | pancreatic $\beta$ cells                     | <b>inhibits insulin release</b>                             |
|                              | some vascular smooth muscle                  | contraction                                                 |
| <b><math>\beta_1</math></b>  | heart                                        | ↑ <b>force and rate</b> of contraction                      |
|                              | juxtaglomerular cells                        | ↑ <b>renin release</b>                                      |
| <b><math>\beta_2</math></b>  | respiratory, uterine, vascular smooth muscle | <b>relaxation</b>                                           |
|                              | skeletal muscle                              | promotes K <sup>+</sup> uptake                              |
|                              | liver                                        | activates glycogenolysis                                    |
|                              | pancreatic $\beta$ cells                     | <b>stimulates insulin release</b>                           |
| <b><math>\beta_3</math></b>  | bladder detrusor                             | relaxation                                                  |
|                              | fat cells                                    | activates lipolysis                                         |
| <b>D1</b>                    | renal, splanchnic, coronary vessels          | G <sub>s</sub> → ↑cAMP → <b>vasodilation</b>                |
| <b>D2</b>                    | nerve terminals, striatum                    | G <sub>i</sub> → inhibits adenylyl cyclase                  |



## RELATIVE RECEPTOR AFFINITIES ★

| DRUG                               | AFFINITY                                    |
|------------------------------------|---------------------------------------------|
| Phenylephrine, methoxamine         | $\alpha_1 > \alpha_2 > > > > \beta$         |
| Clonidine, methyldopa              | $\alpha_2 > \alpha_1 > > > > \beta$         |
| <b>Norepinephrine</b>              | $\alpha_1 = \alpha_2 ; \beta_1 > > \beta_2$ |
| <b>Epinephrine</b>                 | $\alpha_1 = \alpha_2 ; \beta_1 = \beta_2$   |
| Dobutamine                         | $\beta_1 > \beta_2 > > > \alpha$            |
| Isoproterenol                      | $\beta_1 = \beta_2 > > > \alpha$            |
| Salbutamol, terbutaline, ritodrine | $\beta_2 > > \beta_1 > > > \alpha$          |
| Dopamine                           | $D_1 = D_2 > > \beta > > \alpha$            |
| Fenoldopam                         | $D_1 > > D_2$                               |

## OVERVIEW MAPS



Slide Aug 27 and 28 Sympathomimetic and sympatholytic drug · fig 126

## Cardioselective Beta Blockers

**M**etoprolol  
**A**tenolol  
**N**ebivolol  
**B**isoprolol  
**A**cebutolol  
**B**etaxolol  
**E**smolol



Slide Aug 27 and 28 Sympathomimetic and sympatholytic drug · fig 127

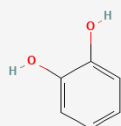
# Non-Selective Direct Agonists

## MECHANISM

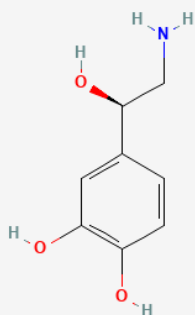
Bind  $\alpha$  and  $\beta$  receptors directly. The clinical picture is decided by **which receptor wins at that dose** — this is why the same drug can raise or lower BP.

→ **Epinephrine, dose-dependent:** low dose ( $\sim 0.01 \mu\text{g/kg/min}$ ) →  **$\beta_2$  predominates** → vasodilation → ↓ BP · high dose ( $> 0.2 \mu\text{g/kg/min}$ ) →  $\alpha + \beta$  together → ↑ BP ★ → **Downstream (all):**  $\alpha_1$  vasoconstriction ·  $\beta_1$  positive inotropic & chronotropic ·  $\beta_2$  bronchodilation → Catecholamines are destroyed fast by **MAO & COMT** ( $T_{1/2} \sim 2\text{--}3 \text{ min}$ ) and are **not orally active** — IV/SC only.

## THE SHARED CORE · WHAT CHANGES

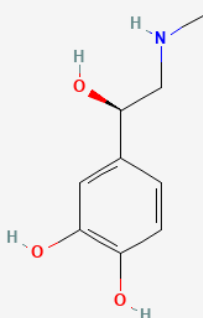


**Catechol — two adjacent OH.** ● this pair of hydroxyls is what COMT recognises. Any drug carrying it is destroyed in minutes and is not orally active — the entire PK of this class sits in these two groups.



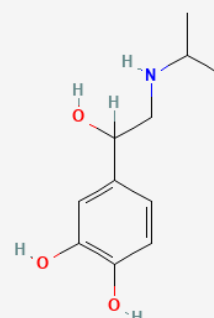
**Norepinephrine**

a bare  $\text{--NH}_2$  →  $\alpha_1$ ,  $\alpha_2$ ,  $\beta_1$ , almost no  $\beta_2$



**Epinephrine**

one N-methyl added →  $\beta_2$  appears; the substituent size on N is what tunes  $\beta$  selectivity



**Isoprenaline**

a big isopropyl on N → pure  $\beta$ , no  $\alpha$  at all

## MEDCHEM

**SAR** ★ The catechol is both the reason they work and the reason they can't be swallowed. The **3',4'-dihydroxy** pattern is required for full  $\alpha$  and  $\beta$  activity - and it is exactly what **COMT** methylates, so oral activity and duration are poor. It is also hydrophilic, so CNS penetration is poor. Add the requirements: at least **two carbons** between ring and nitrogen, an amine that is **protonated at physiological pH** ( $\text{pKa } 8.5\text{--}10$ ), and **R configuration at C1** for maximal direct activity.

**SAR** ● One number predicts the whole class: the size of the N-substituent. **H** →  $\alpha$  agonist (noradrenaline). **Methyl** → **non-selective** (adrenaline). **Isopropyl** →  **$\beta_1 + \beta_2$**  (isoprenaline). **tert-Butyl** →  **$\beta_2$** . **Arylalkyl (2–10 atoms)** →  **$\beta_2$ , longer duration, resists MAO** (salmeterol). Selectivity marches  $\alpha \rightarrow \beta \rightarrow \beta_2$  as the group grows.

## DRUGS

| DRUG                                                                                                  | WHAT IT TREATS                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Epinephrine</b> ( $\alpha 1 \alpha 2 \beta 1 \beta 2$ )                                            | <b>anaphylactic shock — drug of choice</b> ★ · cardiac arrest · prolongs local anaesthetics · topical haemostatic · open-angle glaucoma ( <i>dipivefrin</i> ) |
| <b>Norepinephrine</b> ( $\alpha 1 \alpha 2 \beta 1$ , little $\beta 2$ )                              | <b>vasopressor</b> — cardiogenic shock, hypotension                                                                                                           |
| <b>Isoproterenol</b> ( $\beta 1 = \beta 2$ )                                                          | inotropic agent · reverses $\beta$ -blocker overdose                                                                                                          |
| <b>Oxymetazoline · Xylometazoline · Tetrahydrozoline</b> ( <i>non-selective <math>\alpha</math></i> ) | topical <b>nasal decongestant</b> · red/itchy eye from allergy                                                                                                |

## USES

**Anaphylaxis** — epinephrine · **Shock / hypotension** — NE, epinephrine · **Cardiac arrest** — epinephrine · **Local vasoconstriction & haemostasis** — epinephrine · **Decongestion** — oxymetazoline, tetrahydrozoline

## WHY NE SLOWS THE HEART

NE raises systolic **and** diastolic BP through  $\alpha 1$ . The baroreceptor response fires a **vagal reflex** → **negative chronotropic effect** that cancels out its own  $\beta 1$  tachycardia.

## ADVERSE EFFECTS

- **Cerebral haemorrhage** · cardiac arrhythmias
- Throbbing headache, restlessness, tremor, palpitations
- Isoproterenol: marked vasodilation and reflex tachycardia
- Topical decongestants: **rebound congestion** with prolonged use

## ADMINISTRATION — EPINEPHRINE IS DOSE-DEPENDENT, IN BOTH DIRECTIONS

- **Cannot be given orally** — destroyed by COMT and MAO in the small intestine and liver, so it is **IV or SC only**. Subcutaneous absorption is slower than IV.
- ★ **The dose decides the receptor, and the receptor decides the blood pressure:**
  - **Low dose (0.01  $\mu\text{g/kg/min}$ )** →  $\beta 2$  predominates → vasodilation → **BP falls**
  - **High dose (> 0.2  $\mu\text{g/kg/min}$ )** → both  $\alpha$  and  $\beta$  → **BP rises**
- **Continuous IV infusion for hypotension: 0.05–1  $\mu\text{g/kg/min}$ .**
- **Anaphylactic shock: 0.3–0.5 mL of a 1:1000 solution. Local vasoconstriction: 1:200,000.**
- 💡 Half-life is **~2 minutes**, which is why it is run as an infusion and why stopping the pump *is* the antidote for overshoot — there is no oral route to undo.
- ⚠️ The same ampoule at the wrong dilution does the opposite thing. **1:1000 and 1:200,000 are not interchangeable strengths of one drug; they are two different jobs.**

# $\alpha$ 1-Selective Agonists

## MECHANISM

Selective  $\alpha$ 1 activation — no meaningful  $\beta$  effect, so you get pure vasoconstriction without cardiac stimulation.

→ **Downstream:** vascular smooth muscle **contraction** →  $\uparrow$  BP · **mydriasis** (*pupillary dilator*) · mucosal vasoconstriction → decongestion

## DRUGS

| DRUG                                                         | WHAT IT TREATS                                            |
|--------------------------------------------------------------|-----------------------------------------------------------|
| <b>Phenylephrine</b>                                         | nasal decongestant · mydriatic (ophthalmic) · vasopressor |
| <b>Midodrine</b> ( <i>prodrug</i> → <i>desglymidodrine</i> ) | <b>orthostatic hypotension</b> — the orally active option |

## ADVERSE EFFECTS

- Hypertension · myocardial infarction
- Rebound nasal congestion (phenylephrine)
- ⚠️ **Supine hypertension** with midodrine — do not dose near bedtime

## ADMINISTRATION — PHENYLEPHRINE & MIDODRINE

- **Phenylephrine as a vasopressor: 0.5–8  $\mu$ g/kg/min.** Also supplied as **eye drops** (mydriatic) and as **nasal drops/spray** (decongestant) — three routes, three completely different indications.
- **Midodrine** is the **orally effective**  $\alpha$ 1 agonist — a **prodrug** converted to **desglymidodrine** (*peak ~1 h, half-life 3 h, duration of action 4–6 h*), used for **orthostatic hypotension**.
- 🚫 **Midodrine's dosing instruction is a posture instruction:** its main problem is **supine hypertension**, so the patient stays **upright and avoids lying down for 4 hours after a dose** — and it is not taken at bedtime.
- 💡 That single rule is what makes midodrine workable at home. A patient who takes it and goes for a nap gets the adverse effect the drug was prescribed to prevent, in reverse.
- ⚠️ Topical  $\alpha$ 1 agonists cause **rebound nasal congestion** — the reason nasal decongestant sprays carry a few-days limit.

# $\alpha$ 2-Selective Agonists — Centrally-Acting Sympatholytics

## MECHANISM

An **agonist that acts as a sympatholytic** ★ — the paradox worth understanding. Stimulating  $\alpha$ 2 at the **nucleus tractus solitarius** in the lower brainstem reduces **sympathetic outflow**, and stimulating presynaptic  $\alpha$ 2 autoreceptors reduces **NE release**.

→ **Downstream**: ↓ sympathetic tone → ↓ peripheral resistance → ↓ **BP**, without postural hypotension → **In the eye**: ↓ aqueous humour production → ↓ IOP (*apraclonidine, brimonidine*) IV clonidine transiently raises BP first — direct  $\alpha$ 1 on vascular smooth muscle — before the central effect takes over.

## DRUGS

| DRUG                                               | WHAT IT TREATS                                            |
|----------------------------------------------------|-----------------------------------------------------------|
| <b>Clonidine</b>                                   | hypertension · <b>opioid / drug-withdrawal symptoms</b>   |
| <b>Methyldopa</b> (→ $\alpha$ -methyl-NE in brain) | hypertension — <b>the choice in pregnancy</b> ★           |
| <b>Guanfacine</b>                                  | hypertension · <b>ADHD</b> (sustained-release, ages 6–17) |
| <b>Guanabenz</b>                                   | hypertension                                              |
| <b>Moxonidine</b> ( $\alpha$ 2 + imidazoline I1)   | hypertension · adjunct analgesia                          |
| <b>Tizanidine</b>                                  | muscle relaxant (spasticity)                              |
| <b>Apraclonidine · Brimonidine</b>                 | <b>glaucoma</b> — topical, no BP effect                   |

## ADVERSE EFFECTS

- **Sedation, dizziness, dry mouth** ★ — the class signature
- Hypotension, syncope, bradycardia
- Sexual dysfunction
- Brimonidine crosses the BBB → hypotension, depression (*apraclonidine does not — but irritates the eye*)

## CAUTIONS

- ● **Never stop clonidine abruptly — rebound hypertension** ★

# β<sub>1</sub>-Selective Agonists

## MECHANISM

Selective **β<sub>1</sub>** stimulation at the heart, sparing β<sub>2</sub> — so you get contractility without much vasodilation or bronchial effect.

→ **Downstream: positive inotropic** → ↑ cardiac output, with less tachycardia than isoproterenol

## DRUGS

| DRUG              | WHAT IT TREATS                                                          |
|-------------------|-------------------------------------------------------------------------|
| <b>Dobutamine</b> | <b>acute heart failure</b> · post-cardiac-surgery low output · acute MI |

## ADVERSE EFFECTS

- Hypertension, myocardial infarction
- Hyperglycemia
- ⚠️ **Tachyphylaxis** on long-term use

## ADMINISTRATION — DOBUTAMINE

- **Not orally absorbed.** Given by **infusion pump**; **onset 1–2 min, half-life ~2 min, duration ~10 min, steady state within 10 minutes** of starting the infusion.
- **Initial dose 0.5–1.0 µg/kg/min · therapeutic range 2–20 µg/kg/min** (*usual working rate 2.5–10 µg/kg/min*).
- ⚠️ **Long-term use causes tachyphylaxis** — it is a short-term drug for cardiac decompensation after cardiac surgery, acute CHF or acute MI, not a maintenance one.
- 💡 Everything about this drug is minutes long, which is the point: the dose is titrated against the patient in front of you, and the effect is gone within ten minutes of stopping.

# β2-Selective Agonists

## MECHANISM

Selective **β2** activation → Gs → ↑cAMP → **smooth muscle relaxation**, mainly bronchial.

→ **Downstream:** bronchodilation · uterine relaxation · ↓ mediator release from mast cells (histamine, leukotriene) → **At high dose the selectivity fails** and β1 is stimulated → tachycardia. This is the source of every ADR below.

## THREE TIERS BY DURATION

| TIER         | DRUGS                                                                    | POINT                                                                                                                               |
|--------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>SABA</b>  | <b>Salbutamol (albuterol)</b> ,<br>terbutaline, procaterol,<br>fenoterol | prompt onset — <b>rescue</b> in acute attack                                                                                        |
| <b>LABA</b>  | <b>Salmeterol</b> , formoterol                                           | <b>never as monotherapy</b> ★ — always with an inhaled corticosteroid; salmeterol's slow onset makes it useless for an acute attack |
| <b>VLABA</b> | Indacaterol, vilanterol                                                  | once daily, COPD                                                                                                                    |

## DRUGS

| DRUG                            | WHAT IT TREATS                                                                                |
|---------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Salbutamol</b>               | asthma, obstructive airway disease                                                            |
| <b>Terbutaline</b>              | asthma, COPD, status asthmaticus · <b>premature labour</b> (practical choice)                 |
| <b>Salmeterol / Formoterol</b>  | asthma maintenance; formoterol also exercise-induced bronchospasm, COPD                       |
| <b>Indacaterol / Vilanterol</b> | COPD                                                                                          |
| <b>Ritodrine</b>                | uterine relaxant — largely abandoned, questionable perinatal benefit and ↑ maternal morbidity |

## ADVERSE EFFECTS

- **Tremor** ★ · **tachycardia**
- Restlessness, apprehension, anxiety

## CAUTIONS

- ⚠ Salbutamol / clenbuterol as illegal livestock growth agents (สารเร่งเนื้อแดง) — residues cause muscle cramps, tremor, agitation, headache, nausea. Banned in animal feed in Thailand and abroad.
- ⚠ Contraindicated where a stray β1 effect matters: cardiovascular disease, thyroid disease

## ADMINISTRATION — B2 AGONISTS BY FORMULATION

|       | DRUG                                                                                                       | FORMULATIONS                                                       | ONSET / DURATION                   |
|-------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------|
| SABA  | Salbutamol                                                                                                 | tab · aqueous solution · <b>DPI · MDI · solution for nebulizer</b> | prompt onset                       |
|       | Terbutaline                                                                                                | tab · syrup · sterile solution · solution for nebulizer            | rapid onset, persists <b>3–6 h</b> |
|       | Procaterol                                                                                                 | syrup · tab                                                        |                                    |
|       | Fenoterol                                                                                                  | inhaler ( <i>with ipratropium</i> )                                | prompt onset, lasts <b>4–6 h</b>   |
| LABA  | Salmeterol —  never alone | inhaler ( <i>with fluticasone</i> )                                | <b>slow onset</b>                  |
|       | Formoterol —  never alone | inhaler ( <i>with budesonide</i> )                                 | <b>&gt; 12 h</b>                   |
| VLABA | Indacaterol                                                                                                | oral inhalation ( <i>with tiotropium</i> )                         | <b>once daily</b> , fast onset     |
|       | Vilanterol                                                                                                 | oral inhalation ( <i>with fluticasone</i> )                        | once daily                         |

-  **Salmeterol's slow onset is why it must never be used for an acute attack** and why it is not monotherapy — a patient reaching for the LABA inhaler during bronchospasm is reaching for the wrong one. Salmeterol is the choice for **nocturnal asthma**; formoterol also covers **exercise-induced** bronchospasm.
-  **Terbutaline is the one with a route for every situation** — oral, subcutaneous or inhaled — which is why it turns up in status asthmaticus and, separately, in **relaxing the pregnant uterus**.
-  Check which inhaler is the rescue one at every counselling. It is the single most common patient-side error in this class, and the drugs are deliberately packaged to look alike.



# β3-Selective Agonists

## MECHANISM

β3 sits on the **detrusor muscle** of the bladder (also brown/white fat, gallbladder, ileum). Agonism relaxes the detrusor, raising storage capacity.

→ **Downstream:** ↓ involuntary detrusor contraction → ↓ **urgency & frequency**

## DRUGS

| DRUG              | WHAT IT TREATS                                                                          |
|-------------------|-----------------------------------------------------------------------------------------|
| <b>Mirabegron</b> | <b>overactive bladder (OAB)</b> · urinary incontinence — also combined with solifenacin |

## ADVERSE EFFECTS

- Hypertension
- ↑ urinary tract infection
- Headache

## INTERACTIONS

- Mirabegron is a **moderate CYP2D6 inhibitor**

# Dopamine-Receptor Agonists

## MECHANISM

Dopamine is the **precursor of NE and Epi** and hits dopaminergic *and* adrenergic receptors — so its effect is entirely **dose-dependent** ★:

| DOSE           | RECEPTOR | EFFECT                                                           |
|----------------|----------|------------------------------------------------------------------|
| 1–3 µg/kg/min  | D1       | renal/splanchnic <b>vasodilation</b> → ↑ renal blood flow, ↑ GFR |
| 3–10 µg/kg/min | β1       | <b>positive inotropic</b>                                        |
| >10 µg/kg/min  | α1 (+β1) | <b>vasoconstriction</b> , vasopressor                            |

→ **Downstream:** short T<sub>1/2</sub> (~2 min), onset 5 min, duration <10 min — an infusion drug only

## DRUGS

| DRUG                         | WHAT IT TREATS                                                            |
|------------------------------|---------------------------------------------------------------------------|
| <b>Dopamine</b>              | <b>cardiogenic shock</b> with low cardiac output + failing renal function |
| <b>Fenoldopam</b> (D1 >> D2) | <b>severe hypertension</b> in hospital — IV, for <b>no more than 48 h</b> |

## ADVERSE EFFECTS

- Fenoldopam: headache, flushing, dizziness, **tachycardia or bradycardia**

## ADMINISTRATION — DOPAMINE & FENOLDOPAM

- Dopamine: IV only** — by injection or continuous infusion, **2–5 µg/kg/min**, increased gradually to **20–50 µg/kg/min**. ★ It stimulates cardiac β1 **while increasing renal blood flow** — the opposite of norepinephrine, which reduces it and can cost the kidney.
- Fenoldopam:** given through a **calibrated infusion pump** at **0.01–1.6 µg/kg/min**, half-life ~10 minutes, **for no more than 48 hours**, for severe hypertension in hospitalised patients.
- 💡 The 48-hour ceiling and the calibrated-pump requirement are the two things that make fenoldopam an inpatient-only drug — there is no version of this that goes home with anyone.

# Indirect & Mixed-Acting Sympathomimetics

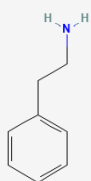
## MECHANISM

No receptor binding of their own — they raise **synaptic NE** instead. Two routes:

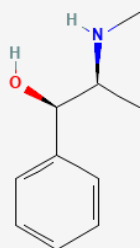
1. **Displacers** — enter the nerve ending and push stored catecholamine out (*amphetamine-like*)
2. **Reuptake inhibitors** — block **NET**, so released transmitter isn't cleared (*cocaine, atomoxetine*)

→ **Downstream:** ↑ BP, cardiac stimulation, arrhythmia · CNS: euphoria, alertness, ↓ fatigue, **anorexia**, insomnia, seizures at high dose → **Mixed-acting (ephedrine):** direct  $\alpha$  +  $\beta$  agonism **and** NE release — plus CNS penetration

## THE SHARED CORE · WHAT CHANGES

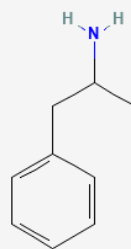


**Phenethylamine — the bare skeleton.** ● no catechol → COMT cannot touch it → orally active, long acting, and lipophilic enough to enter the CNS. Removing the two OH groups is the entire difference from the class above.



**Ephedrine**

orally active, hours not minutes, mixed direct + indirect



**Amphetamine**

an  $\alpha$ -methyl also blocks MAO → even longer, and straight into the CNS

## MEDCHEM

**SAR** ★ **Indirect action is what you get when you take the catechol away.** No phenolic OH means no hydrogen bonds to the receptor's serines → direct  $\alpha$  and  $\beta$  activity both fall, the molecule becomes lipophilic enough to enter the CNS, and it starts **blocking the presynaptic uptake-1 site** instead - displacing stored noradrenaline rather than mimicking it. Ephedrine, pseudoephedrine and amphetamine are all that trade.

**Metabolism** The  **$\alpha$ -methyl group** does two things at once: it **protects against MAO** (longer duration, oral activity) and it **lowers direct activity while raising affinity for the presynaptic uptake site** - i.e. it pushes the drug further toward indirect action. Small alkyls at that position are the standard way to make an oral sympathomimetic out of an injectable one.

**SAR** For the  $\beta_2$  branch the fix was different: keep one phenol and replace the other. **3',5'-diOH (resorcinol, terbutaline)**, **3'-CH<sub>2</sub>OH (salbutamol)** or **3'-formamide (formoterol)** with the 4'-OH intact →  $\beta_2$ -selective and **not a COMT substrate**. Note also **3'-OH matters more for  $\alpha_1$** , **4'-OH matters more for  $\beta$** .

## DRUGS

| DRUG                                                                           | WHAT IT TREATS                                                     |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Amphetamine · Methamphetamine · Dextroamphetamine</b> ( <i>displacers</i> ) | CNS stimulant, appetite suppression — abuse liability              |
| <b>Methylphenidate</b> ( <i>displacer</i> )                                    | <b>ADHD</b>                                                        |
| <b>Cocaine</b> ( <i>NET inhibitor</i> )                                        | local anaesthetic with vasoconstriction; otherwise a drug of abuse |
| <b>Atomoxetine</b> ( <i>NET inhibitor</i> )                                    | ADHD                                                               |
| <b>Ephedrine</b> ( <i>mixed</i> )                                              | decongestant · mild CNS stimulant · orthostatic hypotension        |
| <b>Pseudoephedrine</b> ( <i>mixed</i> )                                        | OTC oral decongestant                                              |

## ADVERSE EFFECTS

- Hypertension, palpitations, arrhythmia
- Insomnia, restlessness, dependence
- Urinary retention ( $\alpha$  at the bladder)
- Tolerance to nasal drops after ~10 days of use

## CAUTIONS & CONTRAINDICATIONS

-  **Phenylpropanolamine (PPA) was withdrawn** — rapid tolerance → dose escalation → **haemorrhagic stroke** ★
-  **Pseudoephedrine contraindicated in:** hypertension / coronary artery disease · hyperthyroidism · narrow-angle glaucoma · BPH · impaired eliminating organs · **MAO inhibitors now or in the last 2 weeks** · pregnancy, lactation, age <2 y
-  Pseudoephedrine and ephedra are **methamphetamine precursors** — a legal control issue, not just a clinical one

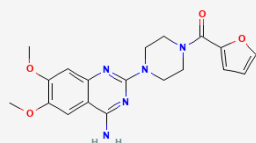
# Selective $\alpha_1$ -Blockers

## MECHANISM

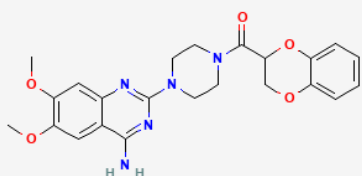
Competitive  $\alpha_1$  blockade. Because presynaptic  $\alpha_2$  is **left intact**, NE release is still auto-inhibited — so reflex tachycardia is far milder than with the non-selective blockers ★.

→ **Downstream:** arterial **and** venous dilation → ↓ peripheral resistance → ↓ **BP** · relaxation of **prostate and bladder-neck** smooth muscle → improved urine flow *Tamsulosin and silodosin prefer the  $\alpha_{1A}$  subtype that dominates the prostate → prostate-selective, less BP effect.*

## THE SHARED CORE · WHAT CHANGES

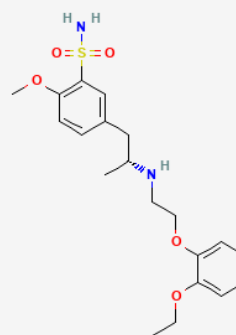


**Prazosin — the quinazoline core.** the quinazoline ring is the  $\alpha_1$  pharmacophore; the tail hung off it decides duration



**Doxazosin**

a benzodioxane tail →  $t_{1/2}$  ~22 h → once daily



**Tamsulosin**

● not a quinazoline at all — a sulfonamide that prefers  $\alpha_{1A}$  (prostate) over  $\alpha_{1B}$  (vessels), which is why it drops BP less

## MEDCHEM

**SAR** The shared core is the **4-amino-6,7-dimethoxyquinazoline** ring - that, not the tail, is the  $\alpha_1$  pharmacophore. The tails then set everything clinical: swapping prazosin's **furan** for a **tetrahydrofuran** gives terazosin's much longer duration, and opening the piperazine gives tamsulosin, which is **selective for  $\alpha_{1A}$**  and therefore has fewer cardiovascular effects.

## DRUGS

| DRUG                              | WHAT IT TREATS                                            |
|-----------------------------------|-----------------------------------------------------------|
| <b>Prazosin</b>                   | hypertension ( <i>short T<sub>1/2</sub> ~3 h</i> )        |
| <b>Terazosin</b>                  | hypertension · BPH                                        |
| <b>Doxazosin</b>                  | hypertension · BPH ( <i>longest-acting of the three</i> ) |
| <b>Alfuzosin</b>                  | BPH                                                       |
| <b>Tamsulosin</b> ( <i>α 1A</i> ) | <b>BPH</b> — least postural hypotension                   |
| <b>Silodosin</b> ( <i>α 1A</i> )  | BPH                                                       |
| <b>Indoramin · Urapidil</b>       | hypertension                                              |

## USES

**Hypertension** — usually combined with a diuretic or β-blocker to blunt the compensatory fluid retention and tachycardia · **BPH with urinary retention** — tamsulosin often with **finasteride** (5α-reductase inhibitor) to shrink the gland · **Raynaud's disease / vasospasm**

## ADVERSE EFFECTS

-  **First-dose phenomenon**  — abrupt syncope on the very first tablet
- **Postural (orthostatic) hypotension**, dizziness
- Headache, nasal congestion, oedema, flushing, nausea
- Alfuzosin: **QTc prolongation**

## CAUTIONS

-  Start at **1/3–1/2 of the usual dose and give it at bedtime** — this is how the first-dose phenomenon is prevented

## ADMINISTRATION — A1-BLOCKER DOSES & FORMS

| DRUG                                       | INDICATION        | DOSAGE FORM         | HALF-LIFE    |
|--------------------------------------------|-------------------|---------------------|--------------|
| <b>Alfuzosin</b>                           | BPH               | <b>ER tab 10 mg</b> |              |
| <b>Doxazosin</b>                           | hypertension, BPH | tab 2, 4 mg         | <b>20 h</b>  |
| <b>Prazosin</b>                            | hypertension      | tab 1, 2 mg         | <b>2–3 h</b> |
| <b>Tamsulosin</b>                          | BPH               | tab 0.4 mg          |              |
| <b>Terazosin</b>                           | hypertension, BPH | tab 1, 2, 5 mg      | 12 h         |
| <b>Silodosin</b> ( <i>selective α 1A</i> ) | BPH               | tab 4 mg            |              |

-  **The half-life column explains the dosing:** prazosin at 2–3 hours has to be given several times a day and is the worst offender for first-dose hypotension; doxazosin at 20 hours is a once-daily drug.
-  The BPH-only agents (tamsulosin, silodosin, alfuzosin) are the α1A-selective end — chosen precisely because they act on the prostate with less effect on blood pressure. That is also why they come in one strength.

# Non-Selective $\alpha$ -Blockers

## MECHANISM

Block  $\alpha_1$  **and**  $\alpha_2$ . Losing the presynaptic  $\alpha_2$  autoreceptor means **more NE is released**, which floods the still-open  $\beta_1$  receptors — hence the marked reflex tachycardia that makes this group unusable for routine hypertension ★.

→ **Downstream:** ↓ peripheral resistance → ↓ BP → baroreceptor reflex + extra NE → **tachycardia**, ↑ **cardiac output** → "**Epinephrine reversal**" ★ — with  $\alpha$  blocked, epinephrine's  $\beta_2$  vasodilation is unopposed, so epinephrine now **lowers** BP instead of raising it

## DRUGS

| DRUG                                                                                                 | WHAT IT TREATS                                                                           |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Phenoxybenzamine</b> ( $\alpha_1 > \alpha_2$ , irreversible / covalent, slow onset, long-lasting) | <b>pheochromocytoma</b> · Raynaud's disease                                              |
| <b>Phentolamine</b> ( $\alpha_1 = \alpha_2$ , competitive, rapid and transient)                      | hypertensive emergency in pheochromocytoma · male erectile dysfunction (with papaverine) |
| <b>Yohimbine · Tolazoline</b> ( $\alpha_2 \gg \alpha_1$ )                                            | little clinical use — orthostatic hypotension, erectile dysfunction                      |

## USES

**Pheochromocytoma** — the one real indication: an adrenal medullary tumour dumping Epi and NE, giving hypertension, tachycardia and arrhythmia · Toxicity from sympathomimetic drugs · **Reversible vasospasm** (Raynaud's)

## ADVERSE EFFECTS

- **Severe reflex tachycardia** → arrhythmia, myocardial infarction
- Orthostatic hypotension, nasal congestion, nausea, vomiting
- Phentolamine: diarrhoea, ↑ gastric acid secretion (*it is also an agonist at muscarinic and H1/H2 receptors*)
- Priapism when used for erectile dysfunction

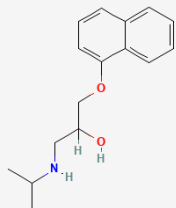
# Non-Selective $\beta$ -Blockers

## MECHANISM

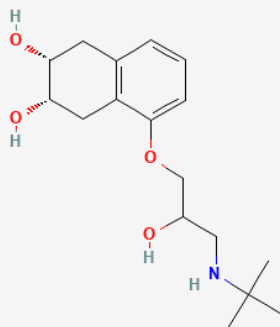
Competitive antagonism at  **$\beta_1$  and  $\beta_2$  equally**. **Propranolol is the prototype** and the comparator for every newer agent.

→ **Heart:** negative inotropic & chronotropic, slowed AV conduction, ↓ myocardial  $O_2$  demand → **Kidney:** ↓ **renin** release → ↓ angiotensin II — this is why peripheral resistance eventually *falls* even though acute  $\beta_2$  blockade raises it ★ → **Lung:**  $\beta_2$  blockade → **bronchoconstriction** · **Eye:** ↓ aqueous humour → ↓ IOP → **Also:** "membrane-stabilising" local anaesthetic action ( $Na^+$  channel block)

## THE SHARED CORE · WHAT CHANGES

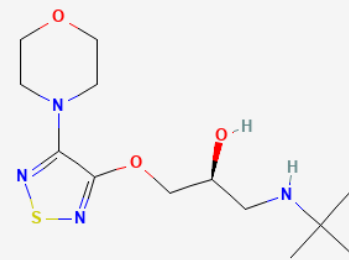


**Propranolol — the aryloxypropanolamine template.** an aromatic ring, an  $-O-CH_2-$  bridge, then the same amino-alcohol tail every  $\beta$ -blocker has. What changes between drugs is only the ring.



**Nadolol**

hydroxyls on the ring → water-soluble → renal clearance, no CNS effects, long  $t_{1/2}$



**Timolol**

a thiadiazole ring; small enough for eye drops — and systemic enough to cause bronchospasm from them

## MEDCHEM

**SAR ★ A  $\beta$ -blocker is a  $\beta$ -agonist with one extra bridge.** Inserting an **oxymethylene bridge** into an aryloxypropanolamine converts it into an **aryloxypropanolamine** - and that single insertion **inverts agonist into antagonist**. Propranolol is that edit applied to the catecholamine skeleton. The N-substituent still follows the agonist rule: an **isopropyl or tert-butyl** group is what makes it  $\beta$ -directed rather than  $\alpha$ .

## DRUGS

| DRUG                                     | WHAT IT TREATS                                                                                                                                                        |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Propranolol</b>                       | hypertension · angina · arrhythmia · <b>migraine prophylaxis</b> · <b>hyperthyroidism / thyroid storm</b> ★ · essential tremor · stage fright · infantile haemangioma |
| <b>Nadolol</b>                           | hypertension · <b>oesophageal varices</b> (with isosorbide mononitrate)                                                                                               |
| <b>Timolol</b>                           | <b>open-angle glaucoma</b> (eye drop) · post-MI survival                                                                                                              |
| <b>Pindolol · Carteolol · Penbutolol</b> | hypertension ( <i>these also carry ISA — see that page</i> )                                                                                                          |



## USES

**Hypertension · Ischaemic heart disease** — ↑ exercise tolerance, and timolol/propranolol/metoprolol improve post-MI survival · **Arrhythmia** — ↑ AV nodal refractory period, slows ventricular rate in AF/flutter · **Glaucoma · Hyperthyroidism** — propranolol also blocks T4 → T3 conversion · **Migraine prophylaxis** · **Portal hypertension / variceal bleeding · Alcohol and opioid withdrawal**

## ADVERSE EFFECTS

- **Bronchospasm** ★ — the β<sub>2</sub> consequence; life-threatening in asthma
- **Bradycardia**, AV block, worsening heart failure
- **Masks hypoglycaemia** ★ (no tachycardia, no tremor) and impairs recovery from it — dangerous in type 1 DM
- **Dyslipidemia** — ↑ TG and VLDL, ↓ HDL
- CNS: fatigue, nightmares, insomnia, depression (*worse with lipophilic agents; less with nadolol, atenolol*)
- Sexual dysfunction · cold extremities · worsens Raynaud's
- Overdose: hypotension, bradycardia, prolonged AV conduction, widened QRS, seizures, hypoglycaemia, bronchospasm — **treat with isoproterenol, glucagon, atropine ± pacing**

## INTERACTIONS

- **Verapamil** (or other cardiodepressants) → severe hypotension, bradycardia, heart failure
- **Aluminium salts, cholestyramine, colestipol** ↓ absorption
- **Phenytoin, rifampin, phenobarbital, smoking** (enzyme induction) ↓ propranolol levels
- **Cimetidine, furosemide, chlorpromazine** ↑ propranolol levels
- Additive with any other antihypertensive

## CAUTIONS

- **Never stop abruptly after long-term use** ★ — β-receptor up-regulation → rebound angina, MI, sudden death. Taper, especially short-T<sub>1/2</sub> drugs (propranolol, metoprolol).
- **Contraindicated in asthma / COPD** — use a cardioselective agent if a β-blocker is unavoidable
- ⚠️ Decompensated heart failure still dependent on sympathetic drive

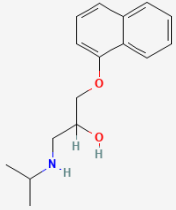
# Cardioselective ( $\beta_1$ ) Blockers

## MECHANISM

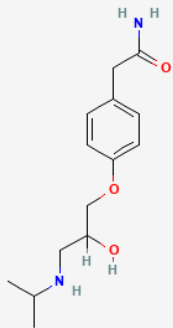
Same competitive blockade, but **50–100× more affinity for  $\beta_1$  than  $\beta_2$**  — sparing the bronchi.

→ **Downstream:** cardiac effects of a  $\beta$ -blocker with much less bronchoconstriction and less interference with glucose and lipid metabolism → ⚠️ **Selectivity is relative, not absolute — it is lost at high dose** ★

## THE SHARED CORE · WHAT CHANGES

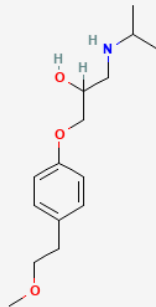


**Propranolol — the non-selective parent.** selectivity comes from putting a PARA substituent on the ring; without it the drug hits  $\beta_2$  as well



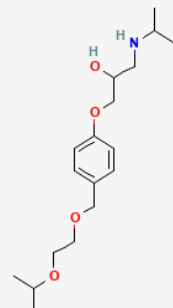
**Atenolol**

a polar amide para-substituent →  $\beta_1$ -selective AND hydrophilic → no CNS entry



**Metoprolol**

a lipophilic para chain →  $\beta_1$ -selective but does enter the CNS; CYP2D6-metabolised



**Bisoprolol**

the most  $\beta_1$ -selective of the three; balanced clearance

## MEDCHEM

**SAR** Same aryloxypropanolamine backbone;  **$\beta_1$  selectivity comes from a para substituent on the aryl ring**, and the N keeps its isopropyl group. **Nebivolol is the outlier** - alongside  $\beta_1$  blockade it **induces endothelial nitric oxide synthase**, so it vasodilates through NO rather than through  $\alpha$ -blockade.

**SAR** The mixed blockers show what the aryl half controls. **Labetalol** has an arylalkyl group that lets it also occupy  $\alpha_1$ : its **R,R isomer is the  $\beta$  antagonist**, its **R,S isomer the selective  $\alpha_1$  antagonist**, and its **m-carboxamide adds a little  $\beta_2$  agonism** - the built-in reason it is less bronchoconstricting than propranolol. **Carvedilol's S isomer is the  $\alpha+\beta$  antagonist**, its R isomer the  $\alpha_1$  antagonist, and its **carbazole ring is a genuine free-radical scavenger**.

## DRUGS

| DRUG                          | WHAT IT TREATS                                                                                  |
|-------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Atenolol</b>               | hypertension · angina                                                                           |
| <b>Metoprolol</b>             | hypertension · angina · post-MI · <b>heart failure</b>                                          |
| <b>Bisoprolol</b>             | hypertension · <b>heart failure</b>                                                             |
| <b>Acebutolol</b>             | hypertension · arrhythmia ( <i>also has ISA</i> )                                               |
| <b>Esmolol</b>                | IV only — perioperative tachycardia, SVT ( $T_{1/2} \sim 10 \text{ min}$ , hydrolysed in blood) |
| <b>Betaxolol</b>              | hypertension · glaucoma                                                                         |
| <b>Celiprolol · Nebivolol</b> | hypertension — also vasodilating (third-generation)                                             |

## USES

The  $\beta$ -blocker of choice when the patient **also has asthma/COPD, diabetes, or dyslipidemia** · Heart failure with reduced ejection fraction — metoprolol, bisoprolol · Acute rate control — esmolol

## ADVERSE EFFECTS

- As for non-selective  $\beta$ -blockers, **minus most of the respiratory effect**
- Bradycardia, fatigue, cold extremities
- Metoprolol is a **CYP2D6** substrate — response varies with genotype

# β-Blockers with ISA

## MECHANISM

These are **partial agonists**: they occupy the β receptor and give a weak sub-maximal stimulation while still blocking endogenous NE/Epi from binding. That resting stimulation is what softens the class ADRs.

→ **Downstream**: less fall in heart rate and cardiac output · **less disturbance of lipid and glucose metabolism** ★ · theoretically some β2 agonism → mild bronchodilation

## DRUGS

| DRUG                          | WHAT IT TREATS                                                        |
|-------------------------------|-----------------------------------------------------------------------|
| <b>Pindolol</b>               | hypertension — the classic ISA agent                                  |
| <b>Acebutolol</b>             | hypertension · arrhythmia ( <i>β1-selective and ISA</i> )             |
| <b>Carteolol · Celiprolol</b> | hypertension ( <i>celiprolol = β1 blockade + β2 partial agonism</i> ) |

## USES

**Hypertension in a patient with dyslipidemia or diabetes** ★ · Hypertension where **bradycardia** is already a problem (*e.g. a resting HR in the 50s*) · Cautiously, hypertension with respiratory disease

## CAUTIONS

- ⚠ A pure antagonist — **not** an ISA agent — is what works in **hyperthyroidism**; β-blockers are not interchangeable

# $\alpha + \beta$ Blockers (Third-Generation)

## MECHANISM

Block  $\beta_1$ ,  $\beta_2$  **and**  $\alpha_1$  ( $\beta_1 = \beta_2 \geq \alpha_1 > \alpha_2$ ). The  $\alpha_1$  component removes the peripheral vasoconstriction that plain  $\beta$ -blockade would leave behind, so BP falls harder and faster.

→ **Downstream:** ↓ peripheral resistance **and** ↓ cardiac work · **no adverse effect on lipids or glucose** ★

## DRUGS

| DRUG              | WHAT IT TREATS                                                                            |
|-------------------|-------------------------------------------------------------------------------------------|
| <b>Labetalol</b>  | <b>hypertensive emergency</b> · hypertension <b>in pregnancy</b> (replaces hydralazine) ★ |
| <b>Carvedilol</b> | <b>heart failure with reduced ejection fraction</b> ★ · hypertension                      |

## HOW THE VASODILATING B-BLOCKERS ACTUALLY DILATE

| MECHANISM                                         | DRUGS                                                             |
|---------------------------------------------------|-------------------------------------------------------------------|
| <b>Nitric oxide production</b>                    | Celiprolol, <b>nebivolol</b> , carteolol, bopindolol, nipradilol  |
| <b><math>\beta_2</math> agonism</b>               | Celiprolol, carteolol, bopindolol                                 |
| <b><math>\alpha_1</math> antagonism</b>           | <b>Carvedilol, labetalol</b> , bucindolol, bevantolol, nipradilol |
| <b><math>\text{Ca}^{2+}</math> entry blockade</b> | Carvedilol, betaxolol, bevantolol                                 |
| <b><math>\text{K}^+</math> channel opening</b>    | Tilisolol                                                         |
| <b>Antioxidant activity</b>                       | Carvedilol                                                        |

## USES

**Hypertensive emergency** — IV labetalol · **Pregnancy hypertension** — labetalol · **HFrEF** — carvedilol · Hypertension complicated by dyslipidemia or diabetes

# Clinical Applications

## SYMPATHOMIMETICS — WHERE THEY GET USED

| TARGET                                      | DRUGS                                                                |
|---------------------------------------------|----------------------------------------------------------------------|
| <b>Shock / hypotension / cardiac arrest</b> | epinephrine · norepinephrine · dopamine · dobutamine · phenylephrine |
| <b>Anaphylaxis</b>                          | <b>epinephrine</b>                                                   |
| <b>Nasal &amp; ocular decongestion</b>      | phenylephrine · oxymetazoline · tetrahydrozoline · pseudoephedrine   |
| <b>Asthma / COPD</b>                        | salbutamol · terbutaline · procaterol · salmeterol · formoterol      |
| <b>Glaucoma &amp; mydriasis</b>             | brimonidine · apraclonidine · phenylephrine                          |
| <b>Uterine relaxation</b>                   | terbutaline · ritodrine                                              |
| <b>Hypertension</b>                         | clonidine · methyldopa · fenoldopam                                  |
| <b>Withdrawal craving (opioid, alcohol)</b> | clonidine                                                            |
| <b>Overactive bladder</b>                   | mirabegron                                                           |
| <b>ADHD</b>                                 | methylphenidate · atomoxetine · guanfacine                           |

## SYMPATHOLYTICS — WHERE THEY GET USED

| TARGET                                                        | DRUGS                                                                  |
|---------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Hypertension</b>                                           | prazosin · doxazosin · most $\beta$ -blockers · labetalol · carvedilol |
| <b>Pheochromocytoma</b>                                       | phenoxybenzamine · phentolamine                                        |
| <b>BPH / urinary retention</b>                                | tamsulosin · silodosin · alfuzosin · terazosin · doxazosin             |
| <b>Angina &amp; post-MI</b>                                   | propranolol · metoprolol · timolol · atenolol                          |
| <b>Arrhythmia</b>                                             | propranolol · esmolol · sotalol                                        |
| <b>Heart failure (HFrEF)</b>                                  | <b>carvedilol</b> · metoprolol · bisoprolol                            |
| <b>Glaucoma</b>                                               | timolol · betaxolol                                                    |
| <b>Hyperthyroidism</b>                                        | propranolol                                                            |
| <b>Migraine prophylaxis · essential tremor · stage fright</b> | propranolol                                                            |
| <b>Oesophageal varices</b>                                    | nadolol · propranolol                                                  |

Principle behind this: *[[pharm-drug-action]]* — The sympathomimetics are the defined GPCR example — receptor → G protein → adenylyl cyclase. Also where  $\alpha/\beta$  subtype selectivity is decided.

Principle behind this: *[[pharm-cns-basics]]* —  $\alpha 2$  as the presynaptic autoreceptor, NET as the reuptake target, and the locus coeruleus as the single-source divergent nucleus.



# Cholinergic Agents

Every drug here works on **one transmitter — acetylcholine (ACh)** — by hitting one of three points: the **receptor** (agonist or antagonist), the **enzyme that destroys ACh** (cholinesterase inhibitors), or **release itself**. Which *receptor subtype* an organ carries is what decides the effect, so the receptor table below is the whole lecture in one grid.

**ACh released → binds mAChR (G-protein) or nAChR (ion channel) → hydrolysed by AChE (synapse) / BuChE (plasma) → signal off**

## RECEPTORS — THE MASTER TABLE ★

| RECEPTOR                 | TYPE                              | MAIN LOCATION                                                                                       | EFFECT OF STIMULATION                                                                                       |
|--------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>M1</b><br>(neural)    | Gq →<br>↑IP <sub>3</sub> /DAG     | CNS cortex, hippocampus ·<br>autonomic ganglia · gastric parietal<br>& salivary glands              | <b>excitation</b> — learning/memory,<br>gastric acid, ganglionic depolarization                             |
| <b>M2</b><br>(cardiac)   | Gi → ↓cAMP,<br>↑K <sup>+</sup>    | heart SA & AV node · smooth<br>muscle · presynaptic nerve endings                                   | <b>inhibition</b> — ↓ heart rate, ↓ AV<br>conduction, ↓ atrial contractility;<br>autoreceptor ↓ ACh release |
| <b>M3</b><br>(glandular) | Gq →<br>↑IP <sub>3</sub> /DAG     | exocrine glands · GI, airway &<br>bladder smooth muscle · vascular<br>endothelium                   | <b>excitation</b> — ↑ secretion, smooth-<br>muscle contraction, miosis                                      |
| <b>M4</b>                | Gi → ↓cAMP                        | brain — forebrain, striatum, cortex                                                                 | inhibits neurotransmitter release                                                                           |
| <b>M5</b>                | Gq →<br>↑IP <sub>3</sub> /DAG     | substantia nigra, VTA                                                                               | modulates dopamine release ·<br>cerebral vasodilation                                                       |
| <b>Nn</b>                | ligand-gated<br>cation<br>channel | <b>all autonomic ganglia</b><br>(sympathetic <i>and</i> parasympathetic)<br>· adrenal medulla · CNS | depolarizes the postganglionic<br>neuron · catecholamine release                                            |
| <b>Nm</b>                | ligand-gated<br>cation<br>channel | <b>skeletal neuromuscular junction,</b><br>postjunctional                                           | end-plate depolarization → <b>skeletal<br/>muscle contraction</b>                                           |

Pattern worth holding: M1 · M3 · M5 = Gq (excitatory), M2 · M4 = Gi (inhibitory). Tissues carry several subtypes; one dominates.

## TWO ENZYMES, NOT ONE

- **AChE (acetylcholinesterase)** — sits in the synapse. The real target.
- **BuChE / pseudocholinesterase** — plasma, liver, many tissues. Low specificity for ACh, not tied to cholinergic synapses — but it is what destroys **succinylcholine** before it reaches the NMJ.

## DUMBBELLS — EXCESS CHOLINERGIC EFFECT ★

|                               |                   |                               |                     |
|-------------------------------|-------------------|-------------------------------|---------------------|
| <b>D</b> iarrhea              | <b>U</b> rination | <b>M</b> iosis / muscle spasm | <b>B</b> radycardia |
| <b>B</b> ronchospasm          | <b>E</b> mesis    | <b>L</b> acrimation           | <b>L</b> ethargy    |
| <b>S</b> alivation & sweating |                   |                               |                     |

*This is the toxidrome for every cholinomimetic — direct agonist overdose, ChE inhibitor, organophosphate.*



# Direct-Acting Cholinergic Agonists

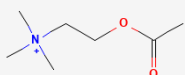
## MECHANISM

Bind **muscarinic (and some nicotinic) receptors directly** — no enzyme step, no dependence on endogenous ACh. Two chemical families: **choline esters** (ACh, methacholine, carbachol, bethanechol) and **alkaloids** (pilocarpine, muscarine, arecoline; synthetic: cevimeline, oxotremorine).

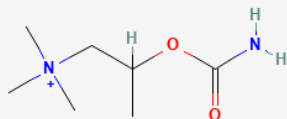
→ **Downstream:** the choline esters differ mainly in **how fast AChE destroys them** — ACh is destroyed instantly (+++), bethanechol and carbachol resist it entirely, which is why only those are useful systemically.

→ **Selectivity:** bethanechol/pilocarpine = muscarinic only · carbachol keeps some nicotinic activity.

## THE SHARED CORE · WHAT CHANGES

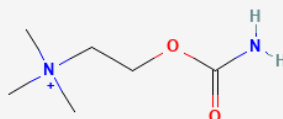


**Acetylcholine.** a quaternary N plus an ester. ● the ester is the problem — cholinesterase destroys it in seconds. Every drug below fixes that.



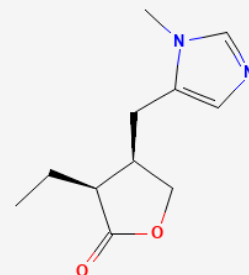
### Bethanechol

ester → **CARBAMATE** (resists AChE) plus a **β-methyl** (kills nicotinic activity) → long-acting and muscarinic-only



### Carbachol

carbamate only → resists AChE but keeps nicotinic action too



### Pilocarpine

a plant alkaloid, not an ester at all — ● uncharged, so unlike the others it crosses membranes and enters the eye and CNS

## MEDCHEM

**SAR** The pharmacophore is exact and unforgiving: a **quaternary ammonium** ( $4^\circ > 3^\circ > 2^\circ > 1^\circ$ ; swapping N for As, P, S or Se lowers activity), **methyl groups on that nitrogen and nothing bigger**, an **ethylene bridge of exactly two carbons** between N and O, and an **ester-like oxygen** for the hydrogen bond. Lengthen or shorten the bridge and activity falls.

**SAR** ★ **Carbachol and bethanechol exist because a carbamate is not an ester.** Replacing acetylcholine's acyloxy group with a **carbamate** keeps the electronics ( $\text{NH}_2$  and  $\text{CH}_3$  are isosteric here, so the pharmacology is unchanged) but makes the molecule **resistant to cholinesterase** - the drug survives long enough to be useful. Adding a **β-methyl** on top (bethanechol, methacholine) buys **muscarinic selectivity**; an **α-methyl** instead flips it the other way, to nicotinic > muscarinic.

**Stability** ⚠ Pilocarpine has **two independent ways to die** - **epimerisation** and **hydrolysis**, both giving inactive product - which is why the eye drops are stored cold.

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------------|
| <b>Acetylcholine</b> | topical only — induction of <b>miosis during eye surgery</b>                                           |
| <b>Bethanechol</b>   | ★ <b>urinary retention</b> (post-op, post-partum) · GI atony — the gut & bladder drug                  |
| <b>Carbachol</b>     | glaucoma · miosis in cataract surgery                                                                  |
| <b>Pilocarpine</b>   | ★ <b>glaucoma</b> (↓ intraocular pressure) · <b>xerostomia</b> — Sjögren's, post head-&-neck radiation |
| <b>Cevimeline</b>    | dry mouth / dry eyes in Sjögren's — selective <b>M1 &amp; M3</b> on lacrimal & salivary epithelium     |
| <b>Methacholine</b>  | diagnostic only (bronchial challenge)                                                                  |

## EFFECTS BY SYSTEM

**Heart** ↓ rate (negative chronotropic) · ↓ AV conduction (dromotropic) · ↓ force (inotropic), plus **vasodilation** · **Lung** bronchoconstriction + ↑ secretions (M3) · **Bladder** detrusor contraction, ↑ voiding pressure (M2) · **GI** ↑ tone, ↑ peristalsis, ↑ secretion (M2, M3) · **Eye** miosis + ciliary contraction (near accommodation) · **Glands** sweat, saliva, tears

## ADVERSE EFFECTS

- Diaphoresis (ภาวะหลังเหงื่อมาก), salivation
- Diarrhea, abdominal cramps, nausea/vomiting
- Sensation of tightness in the bladder
- Difficulty with visual accommodation
- Hypotension

## CONTRAINDICATIONS

- ● **Asthma / COPD** — bronchoconstriction
- ● **Urinary or GI tract obstruction** — you are contracting against a blocked outlet
- ⚠ Acid-peptic disease · bradycardia or hypotension · **hyperthyroidism** (*may precipitate atrial fibrillation*)
- **Overdose antidote: parenteral atropine** + respiratory/cardiovascular support

## ADMINISTRATION — PILOCARPINE

- Used as **pilocarpine hydrochloride eye drops**, and as a ★ **sustained-release ocular insert releasing 20 or 40 µg/hour**, for **open-angle glaucoma**.
- Also used **orally** for dry mouth and dry eyes, including in **Sjögren's syndrome**.
- 💡 The insert exists because drops have to be remembered several times a day and a steady release controls intraocular pressure better than a spike-and-trough. Same drug, better curve.

# Reversible Cholinesterase Inhibitors

## MECHANISM

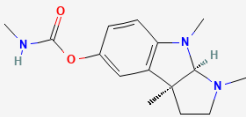
Inhibit **AChE**, so ACh accumulates and acts longer at **every** cholinergic synapse — muscarinic, ganglionic and NMJ at once. **Indirect-acting**: they need endogenous ACh to be there.

→ **Short-acting (minutes)**: edrophonium binds the AChE active site reversibly; cleared renally fast. →

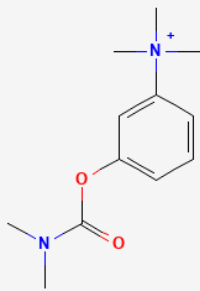
**Medium (30 min–6 h)**: carbamates **carbamylate** the enzyme — a much slower-to-hydrolyse complex. →

**Getting into the brain matters**: physostigmine & rivastigmine cross the BBB (*tertiary amines*); neostigmine & pyridostigmine largely don't (*quaternary*), so they act peripherally at the NMJ.

## THE SHARED CORE · WHAT CHANGES

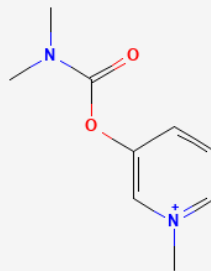


**Physostigmine**. a carbamate that the enzyme attacks and is then stuck with for minutes instead of microseconds. ● uncharged → it crosses into the CNS



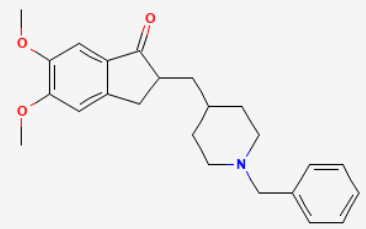
### Neostigmine

a quaternary N — permanently charged → ● CANNOT enter the CNS. Same mechanism, and the charge alone decides the clinical use



### Pyridostigmine

quaternary as well, longer acting → myasthenia maintenance



### Donepezil

not a carbamate — a reversible binder, uncharged and CNS-selective, built for Alzheimer's

## MEDCHEM

**SAR ● One principle organises the entire field, from neurotransmitter to drug to nerve agent: how fast the modified enzyme regenerates. Acetyl > carbamyl > phosphoryl.** Acetylcholine acetylates the serine and the enzyme frees itself in microseconds. A **carbamate** (physostigmine, neostigmine, pyridostigmine, rivastigmine) carbamylates it and regeneration takes minutes to hours - that gap *is* the therapeutic effect. An **organophosphate** phosphorylates it and it effectively never comes back ('aging'). Same chemistry, three different intents.

**SAR ★ Tertiary versus quaternary decides where the drug acts.** Physostigmine is a **tertiary amine** → oral absorption and **crosses the BBB** (glaucoma, anticholinergic overdose, Alzheimer's). Neostigmine and pyridostigmine are **quaternary** → poor absorption, **no central activity**, and activity independent of pH - which is exactly what myasthenia gravis wants. **Edrophonium** does not carbamylate at all: it just binds the anionic site with its quaternary N plus one phenol hydrogen bond, giving the **rapid onset and short duration** that make it a diagnostic agent rather than a treatment.

## DRUGS

| DRUG                                          | WHAT IT TREATS                                                                                               |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Edrophonium</b>                            | ★ <b>diagnosis</b> of myasthenia gravis — 5–15 min action                                                    |
| <b>Neostigmine</b>                            | myasthenia gravis (oral) · <b>reversal of non-depolarizing NMB</b> · antimuscarinic overdose · post-op ileus |
| <b>Pyridostigmine</b>                         | myasthenia gravis — longer-acting than neostigmine · NMB reversal (IV)                                       |
| <b>Physostigmine</b>                          | glaucoma / ↓ intraocular pressure — acts at postganglionic parasympathetic junctions, enters CNS             |
| <b>Donepezil · Rivastigmine · Galantamine</b> | mild–moderate <b>Alzheimer's disease</b>                                                                     |

## USES

**Eye** — glaucoma, miosis (physostigmine) · **Myasthenia gravis** — edrophonium to diagnose, neostigmine/pyridostigmine to treat · **Alzheimer's** — donepezil, rivastigmine, galantamine · **Antimuscarinic (atropine) intoxication** — neostigmine · **Reversal of neuromuscular block**

## TOXICITY — SPLIT IT BY RECEPTOR

- **Muscarinic:** nausea, vomiting, diarrhea, salivation, sweating, vasodilation, **bradycardia**, bronchoconstriction
- **Nicotinic:** twitching, generalised spasm → ● **depolarization blockade** → **respiratory paralysis**

## THE ALZHEIMER'S THREE — WHY THEY WERE DESIGNED DIFFERENTLY

- **Donepezil** — selective for **AChE over BuChE**; low hepatotoxicity and low interaction potential
- **Rivastigmine** — slow hydrolysis of the carbamylated enzyme; inhibits both AChE and BuChE
- **Galantamine** — also **allosterically binds nAChR** — dual cholinergic action

# Irreversible Cholinesterase Inhibitors (Organophosphates)

## MECHANISM

**Phosphorylate** AChE. The phosphorylated enzyme is extremely stable — activity stays depressed for **weeks to months**, until new enzyme is synthesised. Functionally irreversible.

→ ● **"Aging"**: one phosphorus–oxygen bond of the inhibitor breaks, which *strengthens* the phosphorus–enzyme bond permanently. **After aging, no reactivator can rescue the enzyme** — this is why the oxime antidote must be given early.

## AGENTS

| AGENT                                      | WHERE IT COMES FROM                            |
|--------------------------------------------|------------------------------------------------|
| <b>Parathion · Malathion · Ecothiopate</b> | insecticides — the real-world poisoning source |
| <b>Dyflos (isofluorophate)</b>             | experimental / ophthalmic                      |
| <b>Sarin</b>                               | nerve gas                                      |

## TOXIC REACTIONS

- **Muscarinic**: miosis, blurred vision, headache · lacrimal/salivary/bronchial hypersecretion · bronchial & visceral smooth-muscle contraction · **bradycardia, hypotension**
- **Nicotinic**: skeletal muscle **fasciculation** → **weakness** · tachycardia, unstable blood pressure
- **CNS**: mixed excitation and depression → ● **respiratory and circulatory collapse** (*the cause of death*)

# Cholinesterase-Inhibitor Poisoning — Treatment

## THE TWO-DRUG LOGIC ★

The problem is **too much ACh at too many receptors**, so you attack it from both ends:

1. **Atropine** — blocks the *muscarinic* effects (secretions, bradycardia, bronchoconstriction). It does **nothing** for the nicotinic fasciculation or muscle weakness.
2. **Pralidoxime (2-PAM)** — a **cholinesterase reactivator**: pulls the phosphate off AChE and regenerates free enzyme. Also **prevents aging**.

→ **Timing is everything**: 2-PAM only works **before the aging process completes**.

## ANTIDOTES & SUPPORTIVE CARE

| STEP                   | WHAT TO DO                                                                |
|------------------------|---------------------------------------------------------------------------|
| <b>Decontamination</b> | activated charcoal                                                        |
| <b>Supportive</b>      | airway, ventilation — respiratory failure is what kills                   |
| <b>Antimuscarinic</b>  | ★ <b>Atropine</b> IV ( <i>paediatric case in lecture: 0.05 mg/kg IV</i> ) |
| <b>Reactivators</b>    | <b>Pralidoxime (2-PAM)</b> · Obidoxime · Diacetylmonoxime (DAM)           |

## CAUTIONS

- ● Atropine alone will not reverse **respiratory muscle paralysis** — ventilate.
- ⚠ Give the oxime **early**; after aging it is useless.

# Ganglionic Stimulants — Nicotine

## MECHANISM

Agonist at **Nn receptors** in autonomic ganglia and the adrenal medulla — so it stimulates **both** sympathetic and parasympathetic ganglia at once, giving a mixed picture.

→ **Downstream:** tachycardia + ↑ blood pressure · ↑ GI motility & secretion · ↑ bronchial, salivary and sweat secretion → ★ **Sustained stimulation** → **depolarization block** — the receptor stops responding. Same mechanism as succinylcholine. → **CNS:** ↑ alertness, memory facilitation, appetite suppression, and the reward pathway → **addiction**.

## USES

**No therapeutic use except smoking cessation** — nicotine replacement therapy (gum, patch, nasal spray, inhaler, lozenge, sublingual).

## ADMINISTRATION — NICOTINE REPLACEMENT FORMS

- **Gum · transdermal patch · nasal spray · inhaler · lozenges · sublingual tablets.**
- 💡 The forms split by what they replace: **the patch covers the steady background craving**, while the gum, spray, inhaler and lozenges are the fast ones for a sudden urge. Most people who fail on one form alone are missing the half they didn't get.

# Ganglionic Blockers

## MECHANISM

Block **Nn receptors** at **all** autonomic ganglia — sympathetic, parasympathetic *and* enteric.

→ **Downstream**: each organ loses whichever autonomic tone was dominant, which is exactly why these drugs are clinically unusable today — the effect is unpredictable and everywhere at once.

## DRUGS & RELATED TRANSMISSION BLOCKERS

| DRUG                                                                   | NOTE                                                                                                                           |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Hexamethonium</b> ·<br><b>Trimethaphan</b> ·<br><b>Mecamylamine</b> | classic Nn ganglion blockers                                                                                                   |
| <b>Botulinum toxin</b>                                                 | blocks nerve-stimulated <b>ACh release</b> — FDA-approved for <b>strabismus</b> (ตาเหล่) and <b>blepharospasm</b> (ตาปิดเกร็ง) |
| <b>Hemicholinium</b>                                                   | blocks <b>choline transport</b> into the neuron ( <i>experimental</i> )                                                        |



# Muscarinic Antagonists (Antimuscarinics)

## MECHANISM

**Competitive** inhibition at muscarinic receptors — little effect on nicotinic receptors except at very high dose. Prototype: **atropine**, a belladonna alkaloid (*Atropa belladonna*, *Datura stramonium*).

→ **Where the drug goes decides what it treats:** lipid-soluble agents (*tertiary amines*) reach the eye, gut and **brain** — hence the anti-parkinsonian ones; poorly absorbed agents (*quaternary*) stay peripheral, mostly GI and airway. → **Selectivity:** atropine, scopolamine, oxybutynin, tolterodine, trospium = **non-selective** · **solifenacin & darifenacin** = **M3-selective** (fewer CNS/cardiac effects — the bladder answer).

## DRUGS

| DRUG                                           | WHAT IT TREATS                                                                                                     |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Atropine</b>                                | ★ <b>sinus bradycardia</b> · organophosphate poisoning · pupil dilation · drying secretions pre-anaesthesia        |
| <b>Hyoscine (scopolamine)</b>                  | <b>motion sickness</b> · drying secretions · endoscopy/GI radiology (smooth-muscle relaxation)                     |
| <b>Tropicamide · Cyclopentolate</b>            | mydriasis for eye exam                                                                                             |
| <b>Ipratropium · Tiotropium</b>                | ★ <b>asthma &amp; COPD</b> (inhaled) — dominant effect is ↓ airway secretion; tiotropium longer & better tolerated |
| <b>Benzhexol (trihexyphenidyl) · Bztropine</b> | <b>parkinsonism</b> — the CNS-penetrating ones                                                                     |
| <b>Dicycloverine (dicyclomine)</b>             | antispasmodic in <b>irritable bowel syndrome</b>                                                                   |
| <b>Oxybutynin · Tolterodine · Trospium</b>     | overactive bladder — non-selective                                                                                 |
| <b>Solifenacin · Darifenacin</b>               | overactive bladder / urinary incontinence — <b>M3-selective</b> , fewer side effects                               |

## EFFECTS BY SYSTEM — THE MIRROR IMAGE OF THE AGONISTS

**Eye** mydriasis + cycloplegia → photophobia, blurred near vision, dry eye · **Heart** ↑ rate (blocks M2 at SA node), ↑ AV conduction; **BP barely changes**, but high dose gives "**atropine flush**" vasodilation · **Lung** bronchodilation + dry mucosa · **GI** ↓ motility, ↓ gastric secretion · **Bladder/biliary** ↓ tone, antispasmodic · **Glands** ↓ secretion, dry mouth, **sweating blocked** → **body temperature** ↑ · **CNS** minimal at normal dose; toxic dose = **central excitation then respiratory failure**

## ADVERSE EFFECTS

- **Xerostomia** (ปากแห้ง) ★ — the signature complaint
- Constipation · dyspepsia
- **Blurred vision**
- ● **Cognitive impairment** — **delirium and confusion in the elderly**
- Anhidrosis → hyperthermia; dry, flushed skin

## CONTRAINDICATIONS

- **Uncontrolled angle-closure glaucoma** — mydriasis shuts the drainage angle
- **Urinary tract obstruction / benign prostatic hyperplasia** — precipitates retention
- **GI obstruction**
- **Overdose antidote: neostigmine** (a cholinesterase inhibitor — raise ACh to out-compete the blocker)

## ADMINISTRATION — ANTIMUSCARINICS BY ROUTE

- **Scopolamine for motion sickness:** effective orally or by injection, ★ **but its action is short, so it is given as a transdermal patch lasting about 3 days.** Best used *preventively*, before travel.
- **Ipratropium** is FDA-approved as a **nasal spray/inhaler** for rhinitis — ★ **it can be given at a higher dose than atropine with fewer systemic adverse effects**, which is the entire reason it replaced it here.
- **Atropine eye drops** (*and cyclopentolate*) for mydriasis and cycloplegia.

## Antimuscarinics for overactive bladder

| DRUG                     | BIOAVAILABILITY                                                             | T <sub>1/2</sub> (H) | CYP450                     | FORM                         | ADULT DOSE                                  |
|--------------------------|-----------------------------------------------------------------------------|----------------------|----------------------------|------------------------------|---------------------------------------------|
| <b>Oxybutynin</b>        | <b>6%</b> — food has no effect                                              | 2–5                  | 3A4                        | IR · ER · <b>patch · gel</b> | 10–20 mg · 5–30 mg · <b>3.9 mg</b> · 100 mg |
| <b>Tolterodine</b>       | 77% — food has no effect                                                    | 2–9.6                | 2D6                        | IR · ER                      | 2–4 mg · 4 mg                               |
| <b>Trospium chloride</b> | ● <b>&lt; 10%</b> , and a <b>high-fat meal cuts absorption by about 70%</b> | 20                   | <b>not CYP-metabolised</b> | IR · ER                      | 20–40 mg · 60 mg                            |
| <b>Solifenacin</b>       | <b>90%</b> — food has no effect                                             | <b>55</b>            | 3A4                        | IR                           | 5–10 mg                                     |
| <b>Darifenacin</b>       | 15–20% — food has no effect                                                 | 13–19                | 3A4, 2D6                   | ER                           | 7.5–15 mg                                   |
| <b>Fesoterodine</b>      | 52% — food has no effect                                                    | 7                    | 3A4, 2D6                   | ER                           | 4–8 mg                                      |

IR = immediate-release tablet · ER = extended-release tablet or capsule. ⚠ The source table lists a second half-life figure for tolterodine (6.9–18 h) and trospium (35 h) whose column isn't resolvable from the extracted text — most likely the extended-release form. Treat the values above as the immediate-release figures.

- ● **Trospium is the only one with a food rule, and it's a big one** — under 10% gets in to begin with, and a fatty meal removes most of that. Everything else in the table is food-independent.
- ★ **Solifenacin's 55-hour half-life is why it is comfortably once daily**, and why a missed dose matters far less than with immediate-release oxybutynin.
- 💡 Oxybutynin's patch and gel exist to dodge first-pass metabolism — 6% oral bioavailability is also where the dry mouth comes from, so the non-oral routes are a tolerability move, not a convenience one.
- ⚠ **Contraindicated in** urinary tract or GI obstruction, uncontrolled angle-closure glaucoma, and BPH.

# Neuromuscular Blockers — Non-Depolarizing

## MECHANISM

**Competitively** block ACh binding at the **Nm receptor** of the motor end plate. Prototype: **curare** (D-tubocurarine). Muscle relaxes **without preceding fasciculation**.

→ **Order of paralysis**: jaw and eye muscles first, **diaphragm last** — and the diaphragm recovers first. → **Poorly absorbed orally, no CNS entry** → IV only; mostly renally eliminated. Duration ~15 min–2 h. → ★ **Reversible by anticholinesterases** — this is the single sharpest difference from the depolarizing group.

## DRUGS

| DRUG                | WHAT IT TREATS                                                                                                                                                                                                         |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tubocurarine</b> | surgical muscle relaxation — slow · 1–2 h · hypotension + <b>histamine release</b> → bronchoconstriction. ADR-heavy, barely used now                                                                                   |
| <b>Pancuronium</b>  | surgical muscle relaxation, long cases — medium (1–3 min) · 1–2 h · tachycardia, hypertension                                                                                                                          |
| <b>Vecuronium</b>   | surgical muscle relaxation · ICU ventilation — medium · 30–40 min · <b>low ADR</b> , widely used; can cause prolonged paralysis                                                                                        |
| <b>Atracurium</b>   | ★ <b>surgical muscle relaxation in hepatic or renal failure</b> — medium · 30–40 min · <b>spontaneous degradation, the only one needing no dose adjustment in renal failure</b> ; transient hypotension from histamine |
| <b>Mivacurium</b>   | muscle relaxation for short procedures — fast (<2 min) · <b>short ~15 min</b> · atracurium-like but quicker                                                                                                            |
| <b>Rocuronium</b>   | ★ <b>rapid-sequence intubation — the alternative to SCh</b> · surgical muscle relaxation — fast (1.5–2 min) · 20–35 min · high dose acts in ~1 min                                                                     |

## USES

**Adjunct to anaesthesia** — skeletal muscle relaxation · **Facilitating intubation**

## ADVERSE EFFECTS

- **Prolonged apnoea / respiratory failure** — the dangerous one
- Cardiovascular collapse with agents that also block cardiac muscarinic receptors or ganglia (*tubocurarine, gallamine*)
- Histamine release → bronchial secretions, transient hypotension (*atracurium and vecuronium are the NMJ-specific ones — least of this*)

## INTERACTIONS

- **Synergism (deeper block)**: general anaesthetics (*isoflurane, enflurane, halothane*) · **aminoglycosides** (*gentamicin, kanamycin, amikacin* — they block ACh release) · local anaesthetics (*lidocaine*) · **calcium channel blockers**
- **Antagonism**: anticholinesterases (*neostigmine*) — that's the reversal, not an accident
- ⚠ Do **not** combine with a depolarizing blocker — they antagonize each other
- ● With **halothane** → risk of **malignant hyperthermia** → treat with **dantrolene**

## REVERSAL ★

- **Neostigmine** — raises ACh in the synaptic cleft to out-compete the blocker. Give **atropine alongside** to block the unwanted muscarinic effects. Cannot reverse a *deep* block.
- **Sugammadex** — a cyclodextrin; the first **selective relaxant binding agent**. **Encapsulates rocuronium/vecuronium 1:1 in the plasma** (not just the cleft) → inactive complex → renally excreted. Faster, fewer unwanted effects, and works **at any depth of block**. Limit: rare anaphylaxis.

# Neuromuscular Blockers — Depolarizing

## MECHANISM

**Succinylcholine (suxamethonium, SCh)** — effectively two ACh molecules joined at their acetyl groups. It **agonises** the Nm receptor and holds the end plate depolarized, so the muscle cannot repolarize and respond again: **depolarization block**, preceded by visible **fasciculation**.

→ **Ultra-short**: hydrolysed fast by plasma **pseudocholinesterase (BuChE)** — and since the NMJ has almost none of that enzyme, plasma BuChE level is what determines how much drug ever reaches the junction. →

● **Anticholinesterases make it WORSE, not better** — the mirror image of the non-depolarizing group.

## USES

**Induction of anaesthesia and airway intubation** — first choice for rapid intubation, more effective than rocuronium for tube insertion.

## ADVERSE EFFECTS

- ● **Hyperkalemia** ★ —  $K^+$  leaves the cell during depolarization
- ↑ **Intraocular pressure** — caution in eye injury or cataract
- **Bradycardia** — SCh is ACh-like enough to stimulate parasympathetic ganglia and SA-node muscarinic receptors
- ● **Prolonged paralysis** in patients with genetically **low pseudocholinesterase**
- ● **Malignant hyperthermia** (*esp. with halothane*) → cool the patient + **dantrolene**
- **Myalgia** post-op (neck, back, abdomen) from the fasciculation — ~1–20%, dose-related
- Mild histamine release; anaphylaxis rare but commoner than with other relaxants. Safe in asthma.

## CAUTIONS

- ● **No specific antidote** — management is supportive (ventilate) until it is metabolised
- ⚠ Low-dose tubocurarine before SCh is used in practice to prevent phase-1 fasciculation

→ **Non-depolarizing vs depolarizing, the exam contrast:**

|                              | NON-DEPOLARIZING         | DEPOLARIZING (SCH)               |
|------------------------------|--------------------------|----------------------------------|
| <b>Initial fasciculation</b> | none                     | <b>yes, transient</b>            |
| <b>Effect on end plate</b>   | prevents depolarization  | <b>persistent depolarization</b> |
| <b>Anticholinesterase</b>    | <b>reverses it</b>       | no effect / worsens              |
| <b>Antidote</b>              | neostigmine · sugammadex | <b>none</b>                      |

*Principle behind this: [[pharm-drug-action]] — ACh at the nicotinic receptor is the ionotropic archetype; galantamine inhibits acetylcholinesterase; pralidoxime is the chemical antagonist that regenerates it after organophosphate poisoning.*

*Principle behind this: [[pharm-cns-basics]] — Donepezil's peripheral cholinergic effects — cramping, nausea, diarrhoea, muscle cramps — as the price of a central action. M1/M2/nAChR distinctions.*

*Source: 2568\_Sept 3\_Slide\_Drugs used in Parasympathetic systems.pdf — อ.วิลาสินี, PSYCH304 Pharmacology I, 60 slides. Structures omitted.*

# Drugs Affecting GI Motility

The lower GI tract absorbs water and electrolytes and then evacuates the rest. When **motility** or **luminal water** goes wrong you get **diarrhoea · constipation · IBS** — and four drug families exist to push each variable back: **laxatives · anti-diarrhoeals · prokinetics · anti-spasmodics**.

↓ **motility** ± ↓ **colonic water** → **constipation** ← *laxatives, prokinetics* · ↑ **motility** ± ↑ **colonic water** → **diarrhoea** ← *anti-diarrhoeals* · **pain** ± **either** → **IBS** ← *anti-spasmodics*

## ENTERIC RECEPTORS — AND WHICH CLASS HITS EACH ★

| RECEPTOR             | WHERE                                        | EFFECT WHEN ACTIVATED                                           | DRUGS THAT TARGET IT                                                      |
|----------------------|----------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>M3</b>            | excitatory interneuron · smooth muscle       | ↑ peristalsis, smooth-muscle contraction                        | <b>cholinergic enhancers</b> (raise ACh) · <b>anti-spasmodics</b> (block) |
| <b>5-HT1A</b>        | IPAN                                         | ↓ peristalsis, ↓ excitatory mediator release                    | —                                                                         |
| <b>5-HT3</b>         | EPAN                                         | ↑ <b>nausea, vomiting</b> , abdominal pain, intestinal movement | blocked by <b>metoclopramide, cisapride, mosapride</b>                    |
| <b>5-HT4</b>         | presynaptic IPAN                             | ↑ ACh release from the excitatory neuron                        | <b>5-HT4 agonists</b> (prokinetics)                                       |
| <b>D2</b>            | presynaptic myenteric motor neuron, upper GI | ↓ ACh release ( <i>a brake</i> )                                | <b>D2 antagonists</b> — release the brake                                 |
| <b>D2 at the CTZ</b> | chemoreceptor trigger zone                   | drives <b>nausea &amp; vomiting</b>                             | <b>domperidone, metoclopramide</b> — this is their antiemetic action      |
| <b>μ opioid</b>      | presynaptic cholinergic neuron               | ↓ ACh release → ↓ motility                                      | <b>loperamide, diphenoxylate</b>                                          |
| <b>δ opioid</b>      | enterocyte                                   | ↓ adenylyl cyclase → ↓ water secretion                          | <b>racecadotril</b> ( <i>indirect — raises enkephalins</i> )              |

## LAXATIVES BY MECHANISM

| WHAT IT MOVES               | CLASS                        | DRUGS                                             |
|-----------------------------|------------------------------|---------------------------------------------------|
| ↑ <b>water/electrolytes</b> | Osmotic                      | magnesium, lactulose, PEG, sodium phosphate, NaCl |
| ↑ water/electrolytes        | Stool-wetting / emollient    | glycerin, mineral oil, docusate                   |
| ↑ water/electrolytes        | Chloride channel-2 activator | lubiprostone                                      |
| ↑ <b>motility</b>           | Bulk-forming                 | psyllium                                          |
| ↑ motility                  | 5-HT4 agonist                | prucalopride                                      |
| ↑ <b>both</b>               | Stimulant                    | bisacodyl, senna, castor oil                      |
| ↑ both                      | IBAT inhibitor               | elobixibat                                        |

## TWO WORDS WORTH SEPARATING

- **Laxative** — enhances evacuation of **formed** faecal material
- **Cathartic / purgative** — enhances evacuation of **unformed** faecal material (*stronger, faster*)

# Bulk-Forming Laxatives

## MECHANISM

An **indigestible, hydrophilic colloid** that absorbs water and swells into a bulky gel.

→ **Downstream:** distends the colon → stretch triggers **peristalsis** · softer, larger stool

## DRUGS

| DRUG                                      | WHAT IT TREATS       |
|-------------------------------------------|----------------------|
| <b>Psyllium</b> ( <i>ispaghula husk</i> ) | constipation · IBS-C |

## USES

Constipation · haemorrhoids · constipation-predominant IBS · ★ **The safest laxative class in almost every patient** — it never gets absorbed

## ADVERSE EFFECTS

- Bloating and flatulence

## CAUTIONS

- ⚠ Take **with food** — best immediately after meals — and **drink a lot of water**
- ● **Not safe in obstructive colon, megacolon, megarectum** — you are adding bulk to a bowel that cannot move it



# Osmotic Laxatives

## MECHANISM

Poorly absorbed solutes stay in the lumen and **hold water osmotically** ("saline laxatives").

→ **Downstream:** ↑ luminal water → softer, bulkier stool → ↑ transit → **Lactulose specifically:** a non-absorbable disaccharide the **colonic bacteria** hydrolyse into short-chain fatty acids, which add their own osmotic load

## DRUGS

| DRUG                                        | WHAT IT TREATS                                                                                                        |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Magnesium hydroxide</b>                  | constipation                                                                                                          |
| <b>Lactulose</b>                            | constipation                                                                                                          |
| <b>Polyethylene glycol (PEG / macrogol)</b> | ★ <b>low dose</b> = constipation · <b>high dose + electrolytes</b> = bowel prep for colonoscopy / large-bowel surgery |
| <b>Sodium phosphate</b>                     | occasional constipation · bowel cleansing                                                                             |

## USES

Constipation · **Bowel cleansing** before colonoscopy, surgery, radiation — PEG, sodium phosphate · ★ **PEG is the one drug where the dose changes the indication**, not just the intensity

## ADVERSE EFFECTS

- **Electrolyte imbalance** — sodium phosphate especially
- **Hypermagnesemia** — magnesium hydroxide
- Bloating and flatulence — lactulose

## CAUTIONS

- ⚠ **Magnesium:** renal insufficiency, cardiac disease, existing electrolyte imbalance → hypermagnesemia risk
- ● **Lactulose is contraindicated in galactosemia** — it *is* a galactoside

## ADMINISTRATION — OSMOTIC LAXATIVE PREPARATIONS

| PREPARATION                                            | COMPOSITION & HOW IT'S GIVEN                                                                                                                                                                                            |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Magnesium hydroxide</b> ( <i>milk of magnesia</i> ) | oral suspension · ⚠️ <b>risk of hypermagnesaemia</b> in renal insufficiency, cardiac disease or electrolyte imbalance                                                                                                   |
| <b>Lactulose</b>                                       | oral solution · 🚫 <b>not in galactosaemia</b> · also combined with dextrin                                                                                                                                              |
| <b>Macrogol 4000 for constipation</b>                  | <b>10 g sachet — dissolve one sachet in a glass of water</b>                                                                                                                                                            |
| <b>Balanced PEG for bowel prep</b>                     | <b>macrogol 4000 118 g + KCl 1.485 g + NaCl 2.93 g + NaHCO<sub>3</sub> 3.37 g + sodium sulfate 11.37 g — ⭐ dissolve the contents of one full bag in water to make 2 L</b> , for colonoscopy prep or large-bowel surgery |
| <b>Sodium phosphate</b>                                | per 5 mL: <b>monobasic 2.4 g + dibasic 0.9 g</b> · occasional constipation and bowel cleansing · ⚠️ <b>electrolyte imbalance</b>                                                                                        |

- ⭐ **The same polymer does two completely different jobs at two completely different scales** — 10 g in a glass for constipation, **118 g in two litres** for a colonoscopy. It is the volume, not the drug, that empties the bowel.
- 💡 The added salts in the bowel-prep bag aren't filler; they are there so that pulling two litres of water through the gut doesn't strip the patient's electrolytes with it.

# Stool-Wetting Laxatives & Emollients

## MECHANISM

Don't change motility — change the **stool itself**, letting water and fat penetrate it.

→ **Glycerin**: hygroscopic + lubricant (*suppository — acts locally in the rectum*) → **Docusate**: a **surfactant** — lubricates the stool *and* stimulates fluid/electrolyte secretion → **Mineral oil**: penetrates the stool and **blocks water reabsorption** from it

## DRUGS

| DRUG                                                     | WHAT IT TREATS                               |
|----------------------------------------------------------|----------------------------------------------|
| <b>Glycerin</b>                                          | constipation — local, rectal                 |
| <b>Docusate</b> ( <i>dioctyl sodium sulfosuccinate</i> ) | constipation — often combined with bisacodyl |
| <b>Mineral oil</b> ( <i>liquid paraffin</i> )            | constipation                                 |

## ADVERSE EFFECTS

- **Glycerin**: local discomfort, burning, hyperaemia, bleeding
-  **Mineral oil**: **pneumonitis** (*aspiration*) and ↓ **absorption of fat-soluble vitamins**

# Chloride Secretion Activators

## MECHANISM

A **PGE1 derivative**. Activates the **EP4 receptor** → ↑ adenylyl cyclase → opens **chloride channel-2 (ClC-2)** on the apical membrane.

→ **Downstream**: ↑ Cl<sup>-</sup> secretion → ↑ fluid secretion into the lumen → ↑ colonic transit → **Bonus**: restores tight junctions → ↓ intestinal permeability → mucosal barrier recovery

## DRUGS

| DRUG                | WHAT IT TREATS                                                    |
|---------------------|-------------------------------------------------------------------|
| <b>Lubiprostone</b> | chronic constipation · IBS-C · <b>opioid-induced constipation</b> |

## USES

Chronic idiopathic constipation · IBS-C · ★ **Opioid-induced constipation** — it works *downstream* of the μ receptor, so an opioid can't block it

# Stimulant Laxatives

## MECHANISM

⚠️ **The definite mechanism is unknown.** Believed to **directly irritate the colonic mucosa** (mild inflammation) and thereby switch on both motility and secretion.

→ **Downstream:** ↑ motility **and** ↑ water/electrolyte secretion — the strongest of the two-hit classes →

**Senna:** a prodrug — colonic bacteria generate the active stimulator **anthranol**

## DRUGS

| DRUG              | WHAT IT TREATS                |
|-------------------|-------------------------------|
| <b>Bisacodyl</b>  | constipation · bowel emptying |
| <b>Senna</b>      | constipation                  |
| <b>Castor oil</b> | constipation                  |

## ADVERSE EFFECTS

- 🚫 **Dependence with long-term use** — and **destruction of the enteric nervous system**
- **Melanosus coli** and **cathartic colon** after chronic senna use
- Abdominal cramping

## CAUTIONS

- ⚠️ **Bisacodyl is enteric-coated for a reason** — never chew or crush it, and don't take it with antacids. Antacids raise gastric pH, the coating dissolves early → gastric irritation and cramping
- ⚠️ Short-term, as-needed use only

# IBAT Inhibitors

## MECHANISM

Blocks the **ileal bile acid transporter (IBAT)** — so bile acids are not reclaimed in the terminal ileum and spill into the colon, where they are themselves laxative.

→ **Downstream:** ↑ colonic bile acids → ↑ water/electrolyte secretion **and** ↑ motility → **Beyond the bowel:** ↓ LDL · ↑ GLP-1 (*bile acids stop being recycled*)

## DRUGS

| DRUG              | WHAT IT TREATS       |
|-------------------|----------------------|
| <b>Elobixibat</b> | chronic constipation |

## USES

Chronic constipation · Take **15–30 min before a meal** — you have to block the transporter *before* bile acids are secreted

## ADVERSE EFFECTS

- Abdominal pain
- Diarrhoea

## CAUTIONS

-  **Contraindicated in severe liver impairment** — and mechanistically it would work poorly anyway: no bile acid production, nothing to redirect 

# Oral Rehydration Therapy

## MECHANISM

Not an anti-motility drug at all — it treats the thing that actually kills in diarrhoea. **Na<sup>+</sup> and glucose are co-absorbed by the SGLT1 symporter**; water then follows by osmotic equilibrium.

→ **Downstream**: the glucose is not nutrition — it is the **carrier that drags Na<sup>+</sup>, and therefore water, across the enterocyte** ★

## DRUGS

| DRUG                                | WHAT IT TREATS                                                         |
|-------------------------------------|------------------------------------------------------------------------|
| <b>Oral rehydration salts (ORS)</b> | prevention & treatment of dehydration from diarrhoea and mild vomiting |

## USES

First-line in **all** acute diarrhoea · Composition is fixed by **WHO/UNICEF** recommendation · The **reduced-osmolality** formula is the current standard

# Absorbents, Bulk-Formers & Mucosal Coaters

## MECHANISM

Act **inside the lumen only** — never absorbed. Three variations on the same idea:

1. **Absorbents** — huge surface area binds toxins, bile salts, water
  2. **Bulk-forming** — polysaccharide swells in water → ↓ stool fluidity
  3. **Mucosal coating** — binds the glycoprotein of colonic mucus to form a **protective barrier**
- **Downstream:** firmer stool, less irritant contact — motility itself is untouched

## DRUGS

| DRUG                                                     | WHAT IT TREATS                                                                    |
|----------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Activated charcoal</b>                                | diarrhoea · toxin binding                                                         |
| <b>Kaolin</b> ( <i>kaolinite clay</i> )                  | diarrhoea — usually with pectin                                                   |
| <b>Pectin</b>                                            | diarrhoea — reduces stool fluidity                                                |
| <b>Cholestyramine</b> ( <i>bile-salt binding resin</i> ) | <b>bile-salt diarrhoea</b> · hypercholesterolaemia · pruritus from biliary stasis |
| <b>Diocetahedral smectite</b>                            | diarrhoea — absorbent <b>plus</b> mucosal barrier                                 |

## USES

Acute non-specific diarrhoea · ★ **Cholestyramine is the one with three different indications running off one mechanism** — it binds bile salts, which fixes bile-salt diarrhoea, lowers cholesterol, and relieves the itch of biliary stasis

## ADVERSE EFFECTS

- **Activated charcoal:** black stool (*harmless, but tell the patient*)
- **Cholestyramine:** constipation, bloating

## INTERACTIONS

- ● **They absorb other drugs too.** Separate from every other oral drug by **at least 2 hours** ★

## ADMINISTRATION — ABSORBENTS AND COATERS

- **Activated charcoal** — powder · causes **black stool**.
- **Kaolin** — suspension, with **pectin**. **Pectin** — suspension.
- **Cholestyramine** — powder · ⚠ causes **constipation and bloating**.
- **Diocetahedral smectite** — powder.
- ● **Take all of these at least 2 hours apart from any other drug.** They bind indiscriminately; whatever else the patient swallows at the same time is bound too.
- 💡 This is the counselling line that decides whether the rest of someone's regimen works. A patient given charcoal or cholestyramine alongside their usual tablets is quietly having those tablets removed.



# Enkephalinase Inhibitors

## MECHANISM

A **prodrug**, activated by non-specific hydrolases, that inhibits **peripheral enkephalinase** — so endogenous enkephalins survive longer.

→ **Downstream:** ↑ enkephalins → ↑ **δ (delta) opioid receptor** activation → ↓ adenylyl cyclase → ↓ **water secretion** → ★ **Purely anti-secretory: no direct effect on bowel movement** — which is why it doesn't cause the constipation that opioid anti-motility drugs do

## DRUGS

| DRUG                | WHAT IT TREATS                        |
|---------------------|---------------------------------------|
| <b>Racecadotril</b> | acute diarrhoea — secretory component |

# Opioid Anti-Motility Agents

## MECHANISM

**Peripheral  $\mu$  (mu) receptor agonists.**  $\mu$  is Gi-coupled and sits on the **presynaptic cholinergic neuron** — activating it cuts ACh release onto the M3 receptor of the muscle.

→ **Downstream:** ↓ colonic movement · ↓ water content of stool (*more contact time to reabsorb*) · ↑ **anal sphincter tone**

## DRUGS

| DRUG                 | WHAT IT TREATS                                         |
|----------------------|--------------------------------------------------------|
| <b>Loperamide</b>    | acute & chronic non-infectious diarrhoea               |
| <b>Diphenoxylate</b> | diarrhoea — <b>not currently available in Thailand</b> |

## USES

Symptomatic control of watery, non-bloody diarrhoea · **Loperamide is the workhorse** — it stays peripheral at normal doses

## ADVERSE EFFECTS

- **Constipation** — the mechanism, overshoot
- Anticholinergic effects
- **Diphenoxylate at high dose crosses into the CNS** → central opioid adverse effects

## CAUTIONS

-  **Never in bloody diarrhoea**  — slowing the gut retains the invasive pathogen and its toxin
-  **Diphenoxylate is a narcotic** — formulated as a fixed-dose combination with **atropine** specifically to make abuse unpleasant. The atropine is a deterrent, not a therapeutic ingredient

# Prokinetics — D2 Antagonists

## MECHANISM

D2 on the presynaptic myenteric motor neuron is a **brake** — it *reduces* ACh release in the upper GI tract. Blocking D2 **releases the brake**.

→ **Downstream:** ↑ ACh release → ↑ coordinated motility & transit in the upper GI tract → **At the chemoreceptor trigger zone:** blocking D2 = **antiemetic** ★ → **Metoclopramide is also a 5-HT<sub>4</sub> agonist + 5-HT<sub>3</sub> antagonist; itopride also inhibits acetylcholinesterase** — same output, extra routes

## DRUGS

| DRUG                  | WHAT IT TREATS                                                                                               |
|-----------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Domperidone</b>    | dyspepsia · GERD · gastroparesis · <b>nausea &amp; vomiting</b> · off-label postpartum lactation stimulation |
| <b>Metoclopramide</b> | dyspepsia · GERD · gastroparesis · <b>nausea &amp; vomiting</b>                                              |
| <b>Itopride</b>       | dyspepsia · GERD · gastroparesis                                                                             |

## USES

**Functional dyspepsia · GERD · gastroparesis · Nausea and vomiting** — domperidone, metoclopramide ·  
⚠️ Weaker than **PPIs** for GERD — prokinetics only speed clearance, they don't touch acid production

## ADVERSE EFFECTS

- **Extrapyramidal effects — metoclopramide only** ★. It **crosses the BBB**, so it blocks D2 in the **nigrostriatal** pathway. Domperidone does not cross → this is the single biggest difference between them
- **Hyperprolactinaemia** — metoclopramide, domperidone **and** itopride. The **tuberoinfundibular** pathway sits *outside* the BBB, so even a non-penetrating drug reaches it
- **QT prolongation — domperidone**, via inhibition of the cardiac **HERG** potassium channel

## INTERACTIONS

- **Domperidone** — CYP3A4 substrate (*inhibitors raise its level → QT risk*)
- **Metoclopramide** — CYP2D6 substrate

# Prokinetics — 5-HT<sub>4</sub> Agonists

## MECHANISM

5-HT<sub>4</sub> sits **presynaptically on the IPAN** and its job is to **increase ACh release** from the excitatory neuron. Agonise it and you amplify normal peristalsis rather than forcing it.

→ **Downstream:** ↑ ACh → ↑ coordinated peristalsis along the whole gut → **Cisapride and mosapride also block 5-HT<sub>3</sub>** — which is why they trim nausea as well

## DRUGS

| DRUG                | WHAT IT TREATS                                         |
|---------------------|--------------------------------------------------------|
| <b>Mosapride</b>    | dyspepsia · GERD · gastroparesis                       |
| <b>Prucalopride</b> | <b>constipation</b> — a pure 5-HT <sub>4</sub> agonist |
| <b>Cisapride</b>    | withdrawn in many countries — cardiac risk             |

## USES

**Dyspepsia, GERD, gastroparesis** — mosapride · **Chronic constipation** — prucalopride · ★ Same receptor, different indication: **where in the gut you need the push decides which drug**, which is why prucalopride is a laxative and domperidone is not

## ADVERSE EFFECTS

- Abdominal pain / cramping, **diarrhoea** — the mechanism, overshoot
- ↓ **transit time** = ↓ **absorption window for other oral drugs**
- ● **Cisapride: QT prolongation** via **HERG** channel inhibition — the reason it was withdrawn

## INTERACTIONS

- **CYP3A4 substrates** — cisapride, mosapride

# Prokinetics — Cholinergic Enhancers

## MECHANISM

Two cholinergic moves at once: **acetylcholinesterase inhibition** (*ACh is not broken down*) plus **M2 antagonism**.

→ **Downstream:** more ACh available at **M3** → ↑ smooth-muscle contraction and peristalsis → ★ **No D2 blockade at all** — so it is **not** an antiemetic and cannot substitute for domperidone

## DRUGS

| DRUG              | WHAT IT TREATS                   |
|-------------------|----------------------------------|
| <b>Acotiamide</b> | dyspepsia · GERD · gastroparesis |

## USES

Functional dyspepsia · GERD · gastroparesis · Give **15–30 min before meals**

## ADVERSE EFFECTS

- Cholinergic: **diarrhoea**
- ↓ absorption time for other oral drugs
- ● **Cholinergic crisis in overdose**

## INTERACTIONS

- ⚠ **Other cholinergic drugs** — effects stack toward crisis

# Anti-Spasmodics

## MECHANISM

Two routes, same result — **less spastic contraction of GI smooth muscle**:

1. **Anti-muscarinics** — block **M1/M3** on enteric neurons and on the smooth muscle itself, so ACh can't drive contraction
2. **Smooth-muscle relaxants** — **calcium antagonism and/or phosphodiesterase inhibition** acting directly on the intestinal smooth-muscle cell, bypassing the receptor entirely

→ **Downstream**: ↓ spasm → ↓ colicky abdominal pain. *Prokinetics raise ACh-driven motility; anti-spasmodics lower it — giving them together is pharmacological self-cancellation* ★

## DRUGS

| DRUG                                                                    | WHAT IT TREATS                              |
|-------------------------------------------------------------------------|---------------------------------------------|
| <b>Hyoscine-N-butylbromide</b> ( <i>anti-muscarinic</i> )               | abdominal colic · IBS pain · visceral spasm |
| <b>Clidinium bromide</b> ( <i>anti-muscarinic, + chlordiazepoxide</i> ) | IBS pain                                    |
| <b>Dicyclomine</b> ( <i>anti-muscarinic, + simethicone</i> )            | IBS pain, with bloating                     |
| <b>Mebeverine</b> ( <i>smooth-muscle relaxant</i> )                     | IBS pain                                    |
| <b>Alverine</b> ( <i>+ simethicone</i> )                                | IBS pain, with bloating                     |
| <b>Drotaverine</b>                                                      | visceral spasm                              |
| <b>Pinaverium bromide</b>                                               | IBS pain                                    |

## USES

**Abdominal pain from diarrhoea or IBS** — the headline use · Spasm of the **gall bladder, urinary bladder, myometrium**

## ADVERSE EFFECTS

- **Anticholinergic**: dry mouth · visual disturbance · **urinary retention** · **constipation** · tachycardia
- **Reduced muscle tone** (*all classes*): urinary retention, constipation
- **Central**: dizziness, drowsiness, hallucination — only from agents that reach the brain

## CAUTIONS

- ★ **Quaternary amines** — hyoscine-N-butylbromide, clidinium, pinaverium, glycopyrronium — carry a permanent positive charge, so they barely cross the BBB. **Fewer central ADRs, but also no central action: this is exactly why hyoscine-N-butylbromide cannot treat motion sickness**
- ⚠ **Care in a patient who is already constipated** — you are adding constipation on purpose
- ⚠ **Atropine is not used as a GI anti-spasmodic** — a tertiary amine with no tissue selectivity means systemic and central anticholinergic effects for a local problem

Principle behind this: *[[pharm-cns-basics]]* — The CTZ is outside the BBB and carries D2, 5-HT3 and M1 — the entire antiemetic target set, reachable without brain penetration.



# Drugs for Acid-Peptic Diseases

Acid-peptic (acid-related) disease is never acid alone — it's **acid acting on a mucosa whose defence has been weakened**. Every drug here shifts one side of that balance: **offenders** (acid · *H. pylori* · ethanol · NSAIDs · oxidative stress) versus **protectors** (alkaline mucus · prostaglandins · blood flow).

**Vagus (ACh→M) · gastrin · histamine from ECL cell (→H<sub>2</sub>) → parietal cell activated → tubular vesicles fuse with the secretory canaliculus → H<sup>+</sup>/K<sup>+</sup>-ATPase (proton pump) → HCl into the lumen → mucosal injury if defence is down**

## THE DISEASES THESE DRUGS TREAT

- **Dyspepsia / functional dyspepsia** — indigestion with no structural cause
- **Gastritis** — inflamed gastric mucosa
- **Peptic ulcers** — *gastric* or *duodenal*; complication = **upper GI bleeding (UGIB)**
- **GERD** — non-erosive reflux disease (NERD) · erosive esophagitis (EE) · **Barrett's esophagus**

## THE 4 DRUG GROUPS AT A GLANCE

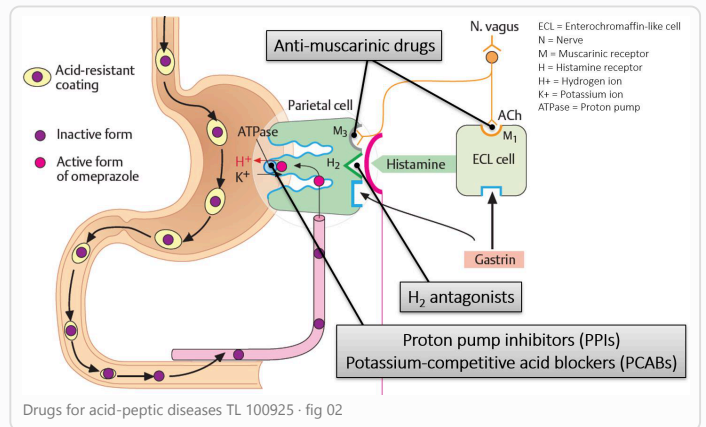
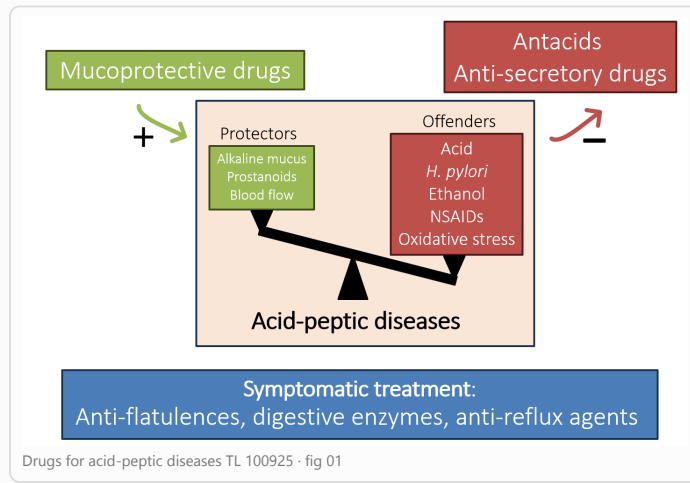
| GROUP                 | WHAT IT DOES                                | WHERE IT SITS IN THE CHAIN                                          |
|-----------------------|---------------------------------------------|---------------------------------------------------------------------|
| <b>Antacids</b>       | chemically neutralise acid already secreted | <i>after</i> the pump — in the lumen                                |
| <b>Anti-secretory</b> | stop the parietal cell secreting            | H <sub>2</sub> receptor · proton pump · K <sup>+</sup> site         |
| <b>Mucoprotective</b> | rebuild the defensive side                  | mucus, HCO <sub>3</sub> <sup>-</sup> , blood flow, physical coating |
| <b>Adjuncts</b>       | symptom relief alongside the above          | gas, digestion, reflux barrier                                      |

## ANTI-SECRETORY DRUGS, WEAKEST → STRONGEST

| CLASS                                             | TARGET                                                      | NOTE                                           |
|---------------------------------------------------|-------------------------------------------------------------|------------------------------------------------|
| <b>Anti-muscarinics</b><br>( <i>pirenzepine</i> ) | M receptor on the parietal cell                             | historic — no longer used for acid disease     |
| <b>H<sub>2</sub> antagonists</b>                  | H <sub>2</sub> receptor — blocks ECL-cell histamine only    | other stimuli (gastrin, ACh) still get through |
| <b>PPIs</b>                                       | H <sup>+</sup> /K <sup>+</sup> -ATPase, <b>irreversible</b> | final common step — no stimulus can bypass it  |
| <b>PCABs</b>                                      | K <sup>+</sup> binding site of the pump                     | faster and deeper than PPIs                    |



## OVERVIEW MAPS



# Antacids

## MECHANISM

**Weak bases** that react chemically with gastric HCl — **acid neutralisation**, raising gastric pH. No absorption needed, so the effect is immediate and purely local.

**Rapid reacting:**  $\text{NaHCO}_3 \cdot \text{Mg}(\text{OH})_2$  **Slow reacting:**  $\text{CaCO}_3 \cdot \text{Al}(\text{OH})_3$

→ **Downstream:** ↑ gastric pH → symptom relief · but **short-lived** — acid keeps being secreted, so the acid-neutralising capacity (ANC) of a dose has to match the HCl the meal will trigger.

## DRUGS

| DRUG                                       | WHAT IT TREATS                       |
|--------------------------------------------|--------------------------------------|
| <b><math>\text{NaHCO}_3</math></b>         | rapid symptomatic relief             |
| <b><math>\text{Mg}(\text{OH})_2</math></b> | rapid relief — laxative pull         |
| <b><math>\text{CaCO}_3</math></b>          | slower, longer relief                |
| <b><math>\text{Al}(\text{OH})_3</math></b> | slower, longer relief — constipating |

## USES

**Symptomatic relief only** — pain · heartburn · difficulty swallowing — in **GERD** · **dyspepsia** · **peptic ulcers**.

★ Marketed products are almost always **≥2 weak bases combined**: better pH control, and the ADRs cancel ( **$\text{Mg}^{2+}$  diarrhoea vs  $\text{Al}^{3+}$  constipation**).

## ADVERSE EFFECTS — READ OFF THE ION

- $\text{OH}^-$  → metabolic alkalosis
- $\text{HCO}_3^- / \text{CO}_3^{2-}$  → metabolic alkalosis + **gastric distention and belching** ( $\text{CO}_2$  released)
- $\text{Na}^+$  → fluid retention, hypertension
- $\text{Ca}^{2+}$  → hypercalcaemic problems
- $\text{Mg}^{2+}$  → **diarrhoea**
- $\text{Al}^{3+}$  → **constipation**
- ⚠ All of these get pronounced in **chronic kidney disease** — the ions can't be cleared

## INTERACTIONS

- ↑ gastric & urinary pH → ↓ **itraconazole absorption** (needs acid to dissolve)
- **Chelation** with **tetracyclines** · **fluoroquinolones** · **iron** ★
- **Fix:** separate the antacid from the other drug by > 2 hours

## ADMINISTRATION

- Take **1 hour after a meal** — peak acid secretion — or as needed
- Severe symptoms: every **30–60 minutes**
- **Chew tablets** before swallowing — faster disintegration, faster onset
- 💡 It relieves symptoms, it doesn't heal anything — daily use past ~2 weeks means they should see a doctor.

# Anti-flatulences, Digestive Enzymes & Anti-reflux Agents

## MECHANISM — THREE DIFFERENT JOBS, ALL ADJUNCTIVE

1. **Simethicone** (*dimethicone stabilised with silicon dioxide*) — a non-systemic **anti-foaming surfactant**: drains liquid out of foam films and ruptures thick films, so small gas bubbles coalesce and can be expelled. Inert, insoluble, non-toxic.
  2. **Pancreatin** — exogenous **lipase + amylase + protease**, replacing/supplementing what the pancreas isn't making.
  3. **Alginate** — a **raft-forming agent**: in gastric acid it precipitates into a gel that floats on the stomach contents and **physically displaces the postprandial acid pocket** in the proximal stomach.
- **Downstream**: gas expelled · food actually digested · a mechanical lid on reflux — **none of them touch acid secretion**

## DRUGS

| DRUG                                                         | WHAT IT TREATS                                         |
|--------------------------------------------------------------|--------------------------------------------------------|
| <b>Simethicone</b>                                           | bloating · flatulence · gas pain                       |
| <b>Pancreatin</b> ( $\pm$ <i>Aspergillus oryzae</i> enzymes) | pancreatic insufficiency · pancreatitis · maldigestion |
| <b>Alginate / sodium alginate</b>                            | GERD — used <b>combined with an antacid</b>            |
| <b>Activated charcoal</b>                                    | claimed for gas — ⚠ <b>efficacy not evidenced</b>      |

## ADMINISTRATION

- **Simethicone** — chew the tablet, faster onset. **Dosage forms**: plain tablet, and **fixed-dose combinations** with other drugs
- **Digestive enzymes** — take **with food** (*no food, nothing to digest*)
- **Alginate** — take **with food**, so the raft forms on top of the meal
- 💡 Alginate — take it after the meal and don't lie flat straight away, so the raft sits on top of the food instead of refluxing with it.

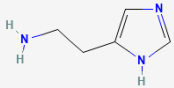
# H<sub>2</sub> Antagonists

## MECHANISM

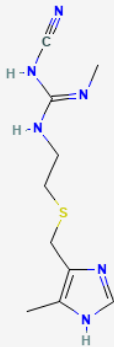
**Selective competitive antagonism at the H<sub>2</sub> receptor** of the parietal cell — blocking the histamine released by **ECL cells**, so the cell is stimulated less and secretes less acid.

→ **Downstream:** suppresses **basal and meal-stimulated** secretion → **The ceiling:** gastrin and vagal ACh still reach the parietal cell, so acid suppression is **partial** — this is exactly the gap PPIs were built to close ★

## THE SHARED CORE · WHAT CHANGES

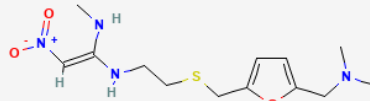


**Histamine — the natural ligand.** an imidazole plus a short amine chain. The first H<sub>2</sub> blockers were built by keeping the ring and lengthening the chain.



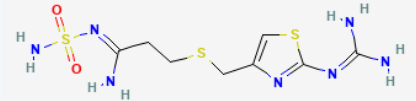
**Cimetidine**

kept the imidazole — ● and that imidazole binds CYP450 heme, which is why cimetidine is the interaction problem and the others aren't



**Ranitidine**

imidazole swapped for a furan → same H<sub>2</sub> block, no CYP inhibition



**Famotidine**

a thiazole → most potent, no CYP effect

## MEDCHEM

**SAR** The class is deliberately **polar** - the opposite design choice from the H<sub>1</sub> antihistamines, and the reason it doesn't sedate. Measured logP runs from **-0.64 (famotidine) to 0.40 (cimetidine)**. The architecture is a heteroaromatic ring, a **four-atom spacer**, and a basic terminal group whose basicity is deliberately damped by an electron-withdrawing substituent.

**Interaction** ● **Cimetidine's whole drug-interaction reputation lives in one ring.** Its **imidazole competes with the CYP450 histidine for the haem Fe<sup>2+</sup>**, so it inhibits many isoforms including 3A4. Ranitidine (furan), famotidine (thiazole) and nizatidine were built by **replacing that imidazole** - same pharmacology, no CYP problem. This is a clean case of an ADR being designed out.

## DRUGS

| DRUG              | WHAT IT TREATS                                     |
|-------------------|----------------------------------------------------|
| <b>Cimetidine</b> | peptic ulcers · GERD — the one with the extra ADRs |
| <b>Famotidine</b> | peptic ulcers · GERD · Zollinger-Ellison syndrome  |

## USES

Peptic ulcers · GERD · Zollinger-Ellison syndrome · Versus PPIs: **less potent**, but **faster onset** and a **better long-term safety profile**

## ADVERSE EFFECTS

- ★ **Tachyphylaxis** — tolerance develops with continued use. The signature H<sub>2</sub>-blocker problem; PPIs don't do this.
- Long-term consequences of the **absence of acid** (*shared with every acid suppressant*)
- **Cimetidine only: anti-androgenic + estrogenic** effects — gynecomastia, impotence

## INTERACTIONS

- ● **Cimetidine inhibits multiple CYPs** — a broad interaction risk. Famotidine does not.

## ADMINISTRATION

- **Dosage forms: PO and IV**
- **May be taken with or without food** — no timing rule at all, unlike a PPI ★
- 💡 Nothing to explain about meals — but it fades with continuous use (tachyphylaxis), so it isn't a drug to stay on indefinitely.

# Proton Pump Inhibitors (PPIs)

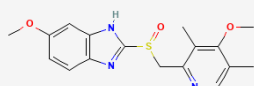
## MECHANISM

Prodrugs that accumulate in the **acidic secretory canaliculus**, convert, and bind the **H<sup>+</sup>/K<sup>+</sup>-ATPase covalently via an S-S bond — irreversible**. A blocked pump is dead no matter which stimulus arrives, which is why PPIs beat H<sub>2</sub> antagonists.

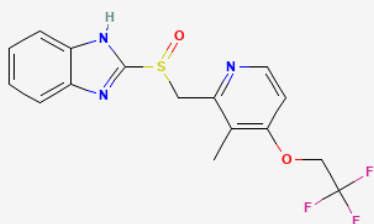
→ **Downstream:** acid secretion stops until **new pump protein** is synthesised → **Two consequences that drive every dosing rule:** the drug is **acid-labile (pKa 3–5)** and it can only bind an **already-activated** pump



## THE SHARED CORE · WHAT CHANGES

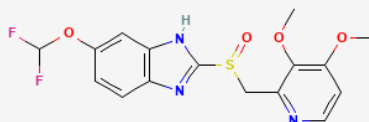


**Omeprazole — benzimidazole + pyridine.** ● inactive as drawn. Only in the parietal cell's acid canaliculus does it rearrange into a sulfenamide that bonds covalently to the pump — an acid-activated prodrug, which is why it must be enteric-coated and why it works long after it has left the blood.



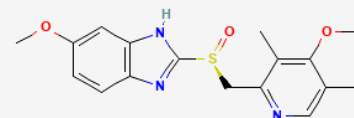
**Lansoprazole**

a CF<sub>3</sub> side chain → faster activation



**Pantoprazole**

difluoromethoxy → activates more slowly and selectively; fewest interactions



**Esomeprazole**

the single S-isomer of omeprazole — same molecule, less CYP2C19 variability

## MEDCHEM

**Prodrug** ★ Every PPI is a prodrug that can only assemble itself inside an acid space. In the parietal-cell canaliculus the pyridine protonates, the molecule undergoes **intramolecular nucleophilic attack** to form a reactive sulfenamide, and that species forms a **covalent disulfide with Cys813 and/or Cys822** of H<sup>+</sup>/K<sup>+</sup>-ATPase. Selectivity for the stomach is a consequence of the chemistry needing pH ~1, not of any tissue targeting.

**SAR** ● Why duration differs across the class - and it is counter-intuitive. Cys813 is near the surface; **Cys822 is buried deep**. A PPI that activates *more slowly* has time to diffuse in and reach **Cys822** - and only **Cys813 is reachable by glutathione**, so a Cys822 adduct essentially cannot be reversed. Recovery then needs new enzyme synthesis. Slower activation, longer effect.

**Physicochem** The pyridine pK<sub>a1</sub> sets how fast it triggers: **rabeprazole 4.53 > omeprazole 4.06 > lansoprazole = pantoprazole 3.83**. An electron-*donating* substituent raises the nitrogen's nucleophilicity → faster onset; electron-withdrawing groups slow it. ⚠ The class is **acid-labile** in the stomach lumen, hence enteric coating and MUPS pellets, and is cleared by **CYP2C19** - the origin of the clopidogrel interaction. Single-isomer versions (esomeprazole, dexlansoprazole) are **equipotent**; only the metabolism differs.

## DRUGS

| DRUG                   | WHAT IT TREATS                                                                   |
|------------------------|----------------------------------------------------------------------------------|
| <b>Omeprazole</b>      | ulcers · GERD — inhibits CYP2C19                                                 |
| <b>Esomeprazole</b>    | ulcers · GERD — inhibits CYP2C19                                                 |
| <b>Lansoprazole</b>    | ulcers · GERD · <i>H. pylori</i> eradication                                     |
| <b>Dexlansoprazole</b> | GERD                                                                             |
| <b>Pantoprazole</b>    | ulcers · GERD                                                                    |
| <b>Rabeprazole</b>     | ulcers · GERD — <b>non-enzymatic</b> metabolism, no CYP2C19 polymorphism issue ★ |

## USES

Acid-related **dyspepsia** · **gastric & duodenal ulcers** · **NSAID-associated ulcers** — treatment *and* prophylaxis · symptomatic **GERD** · reflux / **erosive esophagitis** · maintenance of a healed ulcer or esophagitis · **Zollinger-Ellison syndrome** · ***H. pylori* eradication** — a PPI plus antibacterials

## ADVERSE EFFECTS — THE LONG-TERM LIST

- ↓ **mineral & vitamin absorption** (*no acid, no solubilisation*):
  - **Ca<sup>2+</sup>** → osteoporosis · **Mg<sup>2+</sup>** → hypomagnesaemia and neuronal dysfunction · **Fe<sup>3+</sup>** → iron-deficiency anaemia
- **Opportunistic infection** (*acid is a barrier*): ● ***C. difficile*** colitis & diarrhoea · infection-related ventilator-associated complications → pneumonia
- ★ **Acid rebound after discontinuation**
- Debated associations: hypergastrinaemia · chronic kidney disease · cardiovascular disease · dementia

## INTERACTIONS

- **Omeprazole & esomeprazole inhibit CYP2C19** — matters most for **clopidogrel**, a prodrug that *needs* CYP2C19 to become active ★
- **Drugs that need acid to be absorbed** — itraconazole · atazanavir · rilpivirine · nelfinavir · gefitinib · nilotinib · erlotinib
- **CYP2C19 genetic polymorphism** changes PPI exposure between patients (*rabeprazole is the exception*)

## ADMINISTRATION — ALL OF IT FOLLOWS FROM THE MECHANISM

- ★ Take **~30 minutes before a meal**: the meal activates the pumps, and the drug needs to be there when they're active
- Plasma **half-life is short** ( $\leq 2$  h) — but the effect isn't, because the binding is irreversible
- ★ **Maximal effect only after 3–5 doses** — each dose catches only the pumps active at that moment. Don't judge it on day one.
- ⚠ Acid-labile, so it always ships in an acid-protective form — never crush/expose the protected form
- ⚠ Anything that stops pumps activating (*e.g. co-given H<sub>2</sub> antagonist*) **reduces PPI efficacy** — stacking anti-secretory classes is not additive
- **Dosage forms** — **PO and IV, all acid-protective**: enteric-coated pellets in a capsule · enteric-coated tablet · **MUPS** (multi-unit pellet system) tablet · **FDT** (fast-disintegrating) tablet · **DR** (dual-release) capsule
- **Lansoprazole — approved doses**: acid-related dyspepsia **15–30 mg OD × 2–4 wk** · gastric ulcer **30 mg OD × 8 wk** · duodenal ulcer **30 mg OD × 4 wk** · NSAID-associated ulcer — treatment **15–30 mg OD × 4–8 wk**, prophylaxis **15–30 mg OD**
- 💡 Two things to stress when dispensing: **30 minutes before breakfast, every day**, and **don't stop when they feel better** — it only reaches full effect after 3–5 doses.

# Potassium-Competitive Acid Blockers (PCABs)

## MECHANISM

Blocks the **K<sup>+</sup> site of the H<sup>+</sup>/K<sup>+</sup>-ATPase** — the pump needs K<sup>+</sup> coming in to push H<sup>+</sup> out, so denying it K<sup>+</sup> stops secretion. **Not a prodrug**: with a **pKa of 9.3** it is stable in acid, needs no enteric protection, and does not depend on the pump being pre-activated.

→ **Downstream**: acid suppression that is **faster in onset and greater in extent** than a PPI ★ → **Why it exists**: built for the situations where PPIs are too slow or not deep enough

## DRUGS

| DRUG              | WHAT IT TREATS                                                                              |
|-------------------|---------------------------------------------------------------------------------------------|
| <b>Vonoprazan</b> | acid-peptic disease where speed/depth of suppression matters · <i>H. pylori</i> eradication |

## THE PHARMACOKINETICS THAT EXPLAIN THE BEHAVIOUR

- **T<sub>max</sub> 1.5–2 h**, rapid-release — **food doesn't clinically matter** (*unlike PPIs*)
- **Gastric wall concentration** >> **plasma concentration** — it parks where it works
- **Plasma t<sub>1/2</sub> ≈ 7.7 h** (*vs ≤ 2 h for PPIs*), **Vd ≈ 1050 L**
- Metabolised **mainly by CYP3A4** — not the CYP2C19 pathway PPIs use

## ADVERSE EFFECTS

- Shares the whole **long-term acid-suppression** list with PPIs (*malabsorption, opportunistic infection, rebound*) — and being the **most potent** suppressor, the risk sits at the top of that range ⚠

## ADMINISTRATION

- **Food does not clinically matter** — acid-stable (pKa 9.3), rapid-release, **T<sub>max</sub> 1.5–2 h**. No enteric coating, no pre-activated pump to catch, so **no meal timing rule** ★
- 💡 Before or after food, either is fine. If they've been on a PPI, say it out loud — the 30-minutes-before rule no longer applies here.



# Misoprostol

## MECHANISM

A synthetic **methyl analogue of PGE<sub>1</sub>** — the missing prostaglandin when NSAIDs block COX. Activates the **EP3 receptor (G<sub>i</sub>-coupled)** on two cell types:

- **Supporting cells:** ↑ mucus · ↑ HCO<sub>3</sub><sup>-</sup> · ↑ mucosal blood flow
- **Parietal cells:** ↓ HCl

→ **Downstream:** it is the only agent here that **restores defence and cuts acid at once** — and the only one that directly replaces what an NSAID took away ★

## DRUGS

| DRUG               | WHAT IT TREATS                                   |
|--------------------|--------------------------------------------------|
| <b>Misoprostol</b> | <b>prevention of NSAID-induced peptic ulcers</b> |

## ADVERSE EFFECTS — COMMON AND SEVERE

- **Diarrhoea in ~30%**, with or without abdominal pain/cramps — the reason it's rarely tolerated
- **Exacerbation of inflammatory bowel disease**

## CAUTIONS & CONTRAINDICATIONS

- **CONTRAINDICATED IN PREGNANCY** ★ — a prostaglandin analogue stimulates uterine contraction; it is used clinically *to induce abortion*

## ADMINISTRATION

- Take **with meals, QID** — the dosing frequency is part of why compliance is poor
- 💡 Warn about the diarrhoea up front (~30%) or they'll quietly stop taking it · 🚫 ask about pregnancy every single time before dispensing.

# Mucus-Enhancing Agents

## MECHANISM

Three drugs, three routes, one endpoint: **strengthen the mucosal defensive barrier** rather than fight acid.

- **Rebamipide** — ↑ endogenous prostaglandins, ↓ oxidative stress
- **Teprenone** — ↑ **alkaline mucus** and general defensive mechanisms
- **Irsogladine** — inhibits **phosphodiesterase** → ↑ cAMP and ↑ nitric oxide

→ **Downstream**: thicker alkaline mucus + better blood flow → mucosa survives the acid it's already making. Because they don't suppress acid, they **add on top of** an anti-secretory drug rather than compete with it.

## DRUGS

| DRUG                       | WHAT IT TREATS                                                                                                                                    |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rebamipide</b><br>(TID) | gastric ulcers · gastric mucosal lesions in acute gastritis / acute exacerbation of chronic gastritis · <b>prevention of NSAID-induced ulcers</b> |
| <b>Teprenone</b><br>(TID)  | gastric ulcers · gastric mucosal lesions in acute gastritis / acute exacerbation of chronic gastritis                                             |
| <b>Irsogladine</b><br>(OD) | gastric ulcers · gastric mucosal lesions in acute gastritis / acute exacerbation of chronic gastritis                                             |

## USES

**Gastric ulcers · erosion, bleeding, redness and oedema** of the gastric mucosa in acute gastritis and in the acute exacerbation of chronic gastritis · **rebamipide only** — NSAID-ulcer prevention

# Sucralfate

## MECHANISM

A **sulfate–aluminium derivative of sucrose**. It is **activated by HCl**: in acid it polymerises into a sticky, viscous paste that binds the **ulcer base** and coats it. Acts **purely locally** — barely absorbed.

→ **Downstream**: a physical barrier over damaged mucosa → **The catch that sets the dosing**: it needs acid to work, so it goes in on an **empty stomach** — never alongside food or an antacid ★

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                                            |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Sucralfate</b> | peptic ulcers · <b>oral mucositis</b> ( <i>suspension</i> ) · <b>radiation proctitis / rectal cancer</b> ( <i>enema</i> ) |

## ADVERSE EFFECTS — COMMON BUT NON-SEVERE

- **Constipation (~2%)** — the aluminium
- **Bezoar formation** — the paste can congeal into a mass

## INTERACTIONS

- ★ ↓ **absorption of other drugs** — it coats them too. **Avoid co-administration**; space them out.

## ADMINISTRATION

- Take **1 hour before meals, BID–QID** — acid present, food absent
- **Dosage forms**: tablet · **suspension** (*oral mucositis*) · **enema** (*radiation proctitis*)
- 💡 Space it ~2 hours from their other medicines — it coats those too. If they're on several, map the timing out for them rather than saying "separate them".

# Bismuth-Containing Drugs

## MECHANISM

Three actions at once, which is why one drug covers ulcers, infection and diarrhoea:

1. **Binds the ulcer base and mucus** — a protective coat — plus a **weak antacid** effect
2. **Antibacterial** — against *H. pylori* and *E. coli*
3. **Anti-secretory + anti-inflammatory** — from the **salicylate** half

→ **Chemistry: bismuth subsalicylate** is decomposed by gastric HCl into **bismuth oxychloride + salicylic acid** — the two halves then do two different jobs ★ (Other salts: *subnitrate · subcarbonate · subcitrate*.)

## DRUGS

| DRUG                                                  | WHAT IT TREATS                                                                                                            |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Bismuth subsalicylate</b>                          | peptic ulcers — especially in <i>H. pylori</i> eradication · acute & traveller's diarrhoea · nausea, dyspepsia, heartburn |
| <b>Bismuth subcitrate / subnitrate / subcarbonate</b> | peptic ulcers · <i>H. pylori</i> eradication                                                                              |

## ADVERSE EFFECTS — COMMON BUT NON-SEVERE

- **Darkening of stool and tongue** ★ — harmless **bismuth sulfide**. Warn the patient or they'll panic.
- **Hypersensitivity reactions** — the salicylate
- ● **Reye syndrome** — the salicylate. Same reason you don't give children aspirin.

## INTERACTIONS

- ↓ **absorption of tetracycline** (*bismuth suspension*) — awkward, since tetracyclines appear in *H. pylori* regimens alongside it

## ADMINISTRATION

- Take **with or without meals, QID**
- 💡 Warn that stool and tongue will turn black — harmless, but alarming if nobody said so · ● avoid in children with flu or chickenpox (salicylate → Reye syndrome).

Principle behind this: *[[pharm-drug-action]]* — Antacids ( $\text{NaHCO}_3$ ) are pure chemistry, no receptor — non-specific action in its simplest form.

Principle behind this: *[[pharm-drug-interactions]]* — PPIs and H<sub>2</sub>-blockers raising gastric pH and wrecking ketoconazole absorption; cimetidine as a multi-CYP inhibitor. Acid suppression is an interaction machine.

Principle behind this: *[[pharm-pharmacogenomics]]* — CYP2C19 is the PPI enzyme, and the one with an explicit percentage dose rule attached.

Source: *Drugs for acid-peptic diseases TL 100925.pdf* — อ.ธีรรัตน์ (รศ.ดร.ภก.ธีรรัตน์ เหลืองมั่งคอง), PYCH304 Pharmacology I, 33 slides + teaching notes. Structures omitted.

# Drugs for Inflammatory Bowel Disease

IBD = a **chronic** condition from **inappropriate mucosal immune activation** — *not* an autoimmune disease. The current model: an inappropriate immune response to the gut's **own endogenous microbiota**, on a background of defective immune regulation + genetic mutation + exogenous triggers. Two subtypes, **Crohn's disease (CD)** and **ulcerative colitis (UC)** — and which one you're treating changes the drug.

**bacterial products cross a leaky mucosal barrier → innate epithelial receptors fire → cytokines/chemokines recruit immune cells → NF-κB → Th1/Th17 (CD) or atypical Th2/Th17 (UC) → TNF-α → chronic inflammation**

## CROHN'S DISEASE VS ULCERATIVE COLITIS

|                        | ULCERATIVE COLITIS                                                                            | CROHN'S DISEASE                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Depth</b>           | mucosal inflammation only                                                                     | transmural                                                                                      |
| <b>Where</b>           | rectum outward — proctitis → proctosigmoiditis → distal (left-sided) → extensive → pancolitis | anywhere — ileocolitis, ileitis, gastroduodenal, jejunoileitis, granulomatous colitis, perianal |
| <b>Age of onset</b>    | 15–35 yrs <b>and</b> 55–70 yrs (bimodal)                                                      | variable                                                                                        |
| <b>Bloody stool</b>    | <b>common</b> ★                                                                               | less bloody than UC                                                                             |
| <b>Malnutrition</b>    | less common                                                                                   | <b>common</b> ★                                                                                 |
| <b>Other symptoms</b>  | stool urgency, ↑ bowel movements, mucus, nocturnal BMs, tenesmus, constipation (proctitis)    | abdominal pain, weight loss, fatigue/malaise, abdominal mass, perianal fissure/fistula          |
| <b>T-cell profile</b>  | atypical Th2 + Th17 · IL-1, IL-6, <b>IL-13</b> , TNF-α                                        | Th1 + Th17 · IFN-γ, IL-1, IL-6, IL-8, <b>IL-12</b> , TNF-α                                      |
| <b>Antibody marker</b> | <b>pANCA</b> (70%)                                                                            | <b>ASCA</b> — anti- <i>Saccharomyces cerevisiae</i> (50%)                                       |
| <b>Smoking</b>         | <b>protective</b> ★                                                                           | <b>worsens it</b> ★                                                                             |
| <b>5-ASA works?</b>    | <b>yes — first line</b>                                                                       | <b>no</b> ★                                                                                     |
| <b>Relapse</b>         | 90% relapse in the first 2 years                                                              | —                                                                                               |

*Both share: insidious/subacute onset, extraintestinal manifestations, fever + tachycardia when severe, leukocytosis, ↓ Hct/Hb, ↑ ESR/CRP, (+) fecal occult blood. ~10% is "indeterminate colitis" — can't be told apart.*

## CLINICAL SEVERITY — THE SCORING THAT PICKS THE DRUG

|                     | MILD     | MODERATE     | SEVERE   |
|---------------------|----------|--------------|----------|
| Bloody stools / day | <4       | 4–6          | >6       |
| Temperature         | afebrile | intermediate | >37.8 °C |
| Pulse               | normal   | intermediate | >90      |
| Hemoglobin (g/L)    | >110     | 105–110      | <105     |
| ESR (mm/h)          | <20      | 20–30        | >30      |

## THE TREATMENT PYRAMID — CLIMB ONLY WHEN THE STEP BELOW FAILS

| STEP | CLASS                                                                             |
|------|-----------------------------------------------------------------------------------|
| 1    | <b>5-ASA</b> ( <i>UC only</i> )                                                   |
| 2    | <b>Corticosteroid</b>                                                             |
| 3    | <b>Immune-modifying therapy</b> — thiopurines, methotrexate                       |
| 4    | <b>Biological therapy</b> — anti-TNF, anti-integrin, anti-IL-12/23, JAK inhibitor |
| 5    | <b>Surgery</b>                                                                    |

## TWO GOALS, TWO DIFFERENT DRUG SETS

- **Induction of remission** (acute episode) — kill signs/symptoms clinically *and* endoscopically, suppress inflammation
  - **Maintenance of remission** — prevent relapse, prevent surgery/hospitalisation, prevent malnutrition & treatment ADRs
- ★ **A drug good at one is not automatically good at the other** — corticosteroids induce but never maintain; thiopurines maintain but are too slow to induce.

## ETIOLOGY — WHAT MAKES IT WORSE

- **Genetics** — family history is a strong predictor, more so in **CD**. **NOD2** (CD, Caucasian, <10%) binds intracellular bacterial peptidoglycan → activates NF-κB. Also **ATG16L1**, **IRGM**; shared with UC: **JAK2**, **IL-23R** (*protective* — *Th17 maturation*)
- **Microbes** — may trigger IBD; the body reacts against normal flora, which is why **antibiotics can improve symptoms**
- **Drugs** — **NSAIDs** disrupt GI mucosal integrity and may promote IBD · contraceptives (cohort data)
- **Psychosocial** — major lifestyle change, daily stress worsen symptoms
- **Smoking** — protective in UC, worsening in CD

# 5-Aminosalicylates (5-ASA)

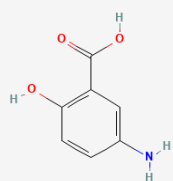
## MECHANISM

**Not COX inhibition** ★ — that's the trap. Several anti-inflammatory actions at once: inhibits **IL-1** and **TNF- $\alpha$**  production · inhibits the **lipxygenase** pathway · **scavenges free radicals** · inhibits **NF- $\kappa$ B** via **PPAR- $\gamma$** .

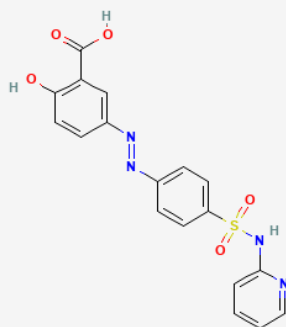
**Delivery is the whole design:** the **azo bond** prevents absorption in stomach and small bowel; colonic bacterial **azoreductase** cleaves it, releasing active 5-ASA (mesalamine) *where the disease is*.

→ **Downstream:** local mucosal anti-inflammation in the colon, minimal systemic exposure → **Sulfasalazine splits into:** 5-ASA (*the active half*) + **sulfapyridine** (*the half that causes the ADRs*) ★

## THE SHARED CORE · WHAT CHANGES

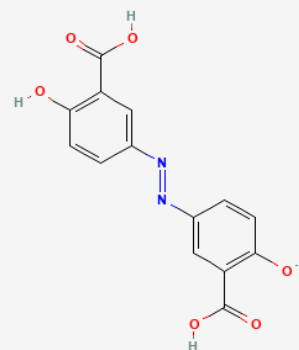


**Mesalamine (5-ASA).** the active molecule — and ● it is absorbed almost completely in the upper gut, so given plainly it never reaches the colon. Every drug below is a delivery trick for this one molecule.



**Sulfasalazine**

5-ASA azo-bonded to sulfapyridine; colonic bacteria break the azo bond → released exactly where it's needed. The sulfa half causes most of the ADRs



**Olsalazine**

two 5-ASA molecules joined to each other — same bacterial trigger, no sulfa half

## DRUGS

| DRUG                      | WHAT IT TREATS                                                                    |
|---------------------------|-----------------------------------------------------------------------------------|
| <b>Sulfasalazine</b>      | mild-moderate UC — 5-ASA + sulfapyridine carrier                                  |
| <b>Mesalamine (5-ASA)</b> | mild-moderate UC — the active molecule, delivered by coating instead of a carrier |
| <b>Olsalazine</b>         | UC — <b>2 mesalamine molecules</b> azo-linked, no sulfa                           |
| <b>Balsalazide</b>        | UC — mesalamine + <b>4-ABA</b> , an inert carrier                                 |

## USES

**First-line for mild–moderate UC** ★ · effective for **induction** of UC remission · effective for **maintenance** of UC remission · **NOT effective in Crohn's disease** — neither induction nor maintenance ★

**Active UC:** 2–3 g/day, once daily is as effective as divided dosing; higher doses for more severe disease (4.8 > 2.4 g/day in *moderate* UC — ASCEND II/III; no dose-response in *mild* disease). ★ **Every UC patient, at every severity, should be offered oral + enema 5-ASA combined** — superior to oral alone even in pancolitis (RR of no remission 0.65). **Maintenance:** ≥2 g/day orally; rectal 5-ASA for distal UC. High dose if they needed ≥2 steroid courses in the past year, are steroid-dependent/refractory, or are escalating to a thiopurine.

## ADVERSE EFFECTS

- **Dose-related (sulfapyridine)** — headache, fatigue, nausea
- **Allergic** — rash, **Stevens-Johnson syndrome**, hepatitis, **bone marrow suppression**
- ↓ **sperm number and motility** — ★ **reversible**
- ● **Nephrotoxicity** (1:4000 patient-years) — median 3 years before renal impairment. **Monitor renal function at baseline, repeat q2–3 months.** Treat by withdrawing the drug + high-dose oral corticosteroid
- **Olsalazine specifically** — high incidence of **secretory diarrhea**

## CAUTIONS

- ⚠ **Sulfonamide allergy** — sulfasalazine is out; the newer azo drugs (**olsalazine, balsalazide**) and plain coated mesalamine are safe
- ⚠ **Coated/sustained-release forms: do not crush, chew, or break** — the coating *is* the targeting (pH >7 release at terminal ileum & colon)
- ⚠ **Adherence is the main real-world failure** — support and education are part of the prescription

## ADMINISTRATION — THE 5-ASA FORMULATION IS THE DRUG

| FORMULATION                                                                | HOW IT RELEASES                                                                                                                               | SITE                                  |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>pH-sensitive release tablet</b><br>( <i>enteric-coated mesalamine</i> ) | liberated at <b>pH &gt; 7</b>                                                                                                                 | <b>terminal ileum and colon</b>       |
| <b>MMX multi-matrix tablet</b>                                             | 5-ASA in a lipophilic matrix inside a hydrophilic matrix, encapsulated by a <b>pH-sensitive polymer releasing at pH &gt; 7</b>                | colon                                 |
| <b>Sustained-release capsule</b>                                           | ★ <b>the capsule disintegrates in the stomach;</b> ethylcellulose-coated beads let water in, dissolve the 5-ASA and diffuse it into the lumen | along the whole small bowel and colon |

● **Do not crush, chew or break — any of them.** The coating is the targeting mechanism; destroy it and the drug is absorbed before it reaches the inflamed bowel.

**Doses:** - **Active mild-to-moderate UC: 2–3 g/day** — ★ **once daily is as effective as divided doses.** - **pH-sensitive / MMX tablets: 2.4–4.8 g/day acute, once daily · 1.6–4.8 g/day maintenance.** - Higher doses for more severe disease — **4.8 g/day beat 2.4 g/day in moderate UC (72% vs 59%),** but made no difference in mild disease. - ⚠ **Monitor renal function** at baseline and **every 2–3 months** during treatment (*nephrotoxicity, sulfapyridine*).

- 💡 The once-daily finding is the useful one to hand a patient — a 4.8 g/day regimen sounds like a lot of tablets until they hear it is one sitting, not four.



# Corticosteroids

## MECHANISM

Broad suppression of the mucosal inflammatory cascade. Used **systemically** or **delivered to the site of action** — the newer agents are engineered for high first-pass loss so the drug acts in the gut and barely reaches the rest of the body.

→ **Downstream:** rapid induction of remission → ★ **NOT effective for maintenance of remission** — induce, then taper over many weeks. Never a long-term drug.

## DRUGS

| DRUG                               | WHAT IT TREATS                                                   |
|------------------------------------|------------------------------------------------------------------|
| <b>Prednisolone (oral)</b>         | active UC/CD not responding to 5-ASA · colonic CD · 20–60 mg/day |
| <b>Hydrocortisone (IV)</b>         | severe disease · 300 mg/day                                      |
| <b>Methylprednisolone (IV)</b>     | severe disease · 40–60 mg/day                                    |
| <b>Budesonide MMX</b>              | mild–moderate <b>left-sided/extensive UC</b> · 9 mg daily        |
| <b>Ileal-release budesonide</b>    | mild–moderate <b>ileocecal CD</b> · 9 mg daily × 8 wks           |
| <b>Beclomethasone dipropionate</b> | active left-sided/extensive UC · 5 mg daily                      |

## USES

**UC mild–moderate** — budesonide MMX 9 mg (6 mg is no better than placebo; **ileal-release budesonide does not work in UC** — inferior to placebo and 5-ASA) · beclomethasone 5 mg ≈ 2.4 g 5-ASA, and better when *added* to 5-ASA · **UC moderate–severe** — prednisolone 40 mg daily, single dose = split dose with less adrenal suppression; **no response at 2 weeks** → **escalate to biologics or admit**; taper over 6–8 weeks · **Ileocecal CD mild–moderate** — ileal-release budesonide 9 mg × 8 wks, as effective as prednisolone; taper over 1–2 wks · **Colonic CD (any severity) + severe ileocolonic CD** — 8-week systemic course, start 40 mg, taper 5 mg weekly

★ **Prednisolone beats 5-ASA at inducing UC remission but is reserved** for 5-ASA failure or intolerance — the side effects aren't worth it otherwise.

## ADVERSE EFFECTS

- Weight gain · **fat redistribution** · hyperglycemia / glucose intolerance
- **Osteoporosis** · cataract
- Emotional changes — sleep and mood disturbance
- Acne, edema, dyspepsia
- **50% of patients get a short-term steroid-related adverse event** ★
- Going above 40–60 mg/day adds **no benefit, only ADRs**
- ★ **Budesonide is the exception** — systemic ADRs 33% vs 55% for conventional steroids, and **no adrenal suppression, no significant BMD loss** (high first-pass metabolism)

## CAUTIONS

- ⚠️ **Short-term only (weeks)**, then taper gradually — abrupt stop and long courses are both wrong
- ⚠️ **Response splits patients into three groups:** steroid-**responsive** 40% · steroid-**dependent** 30–40% · steroid-**unresponsive** 15–20%. The last two are the ones who need escalation ⭐
- ⚠️ In CD, **ileal-release budesonide only helps proximal colonic disease** — no evidence in more distal colonic inflammation, and no MMX trials in CD at all

## ADMINISTRATION — STEROID DOSES IN IBD

| DRUG                                                                       | DOSE                                                                    |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Oral prednisolone</b> ( <i>active disease not responding to 5-ASA</i> ) | <b>20–60 mg/day</b>                                                     |
| <b>IV hydrocortisone</b> ( <i>severe disease</i> )                         | <b>300 mg/day</b>                                                       |
| <b>IV methylprednisolone</b> ( <i>severe disease</i> )                     | <b>40–60 mg/day</b>                                                     |
| <b>Budesonide MMX</b> ( <i>mild-to-moderate UC</i> )                       | ⭐ <b>9 mg daily — 6 mg was no better than placebo</b>                   |
| <b>Beclomethasone dipropionate</b>                                         | <b>5 mg daily × 4 weeks, then alternate weeks for a further 4 weeks</b> |

- 🟡 **Short courses only — weeks — and never for maintenance.** Once remission is induced the dose is **tapered gradually over many weeks**, not stopped.
- ⭐ **Budesonide MMX is dose-specific, not dose-ranged:** 9 mg works, 6 mg doesn't, and the **ileal-release** budesonide doesn't induce remission in UC at all — it is inferior to both placebo and 5-ASA. Same molecule, wrong delivery site.
- 💡 Budesonide MMX has a **33% vs 55%** systemic adverse-effect rate against conventional steroids and no adrenal suppression or bone-density hit — that trade is the whole reason to reach for it.

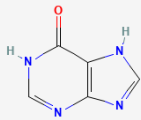
# Thiopurines

## MECHANISM

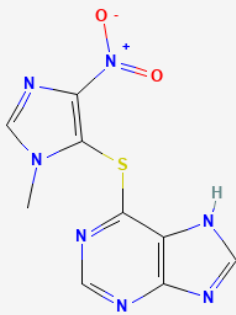
**Purine analogues — both are prodrugs.** Impair purine biosynthesis, which inhibits **proliferation of immune cells** and reduces IBD-associated GI inflammation.

→ **Downstream:** steroid-sparing — cuts long-term corticosteroid use → **Slow onset: 3–12 months** ★  
— which is exactly why they're useless in *active* IBD and used for **maintenance only**

## THE SHARED CORE · WHAT CHANGES

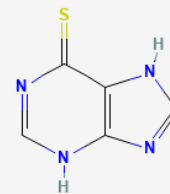


**Hypoxanthine — the purine being faked.** the drugs below are this with sulfur in place of oxygen — see [[pharm-pharmacogenomics]] for why TPMT decides the dose



### Azathioprine

a prodrug: an imidazole group is cleaved off non-enzymatically to release mercaptopurine



### 6-Mercaptopurine

the active thiopurine; ● in blood-forming cells xanthine oxidase is absent, so TPMT is the only way out — which is why TPMT variants are dangerous here specifically

## DRUGS

| DRUG                         | WHAT IT TREATS                               |
|------------------------------|----------------------------------------------|
| <b>Azathioprine</b>          | maintenance of remission in <b>UC and CD</b> |
| <b>Mercaptopurine (6-MP)</b> | maintenance of remission in <b>UC and CD</b> |

## USES

**Maintenance of remission in UC · Maintenance of remission in CD · Reducing long-term corticosteroid exposure** in steroid-dependent patients · **Combination with infliximab** — reduces immunogenicity and gives superior outcomes (SONIC, PANTS)

*Evidence: NNT of 5 to prevent one recurrence (absolute risk reduction 23%); azathioprine vs placebo failure 44% vs 65%.*

## ADVERSE EFFECTS

★ The ADR you get depends on the patient's TPMT (thiopurine methyltransferase) activity — this is the exam fact:

- **Low/absent TPMT** → metabolism shunts toward **6-thioguanine nucleotides** → **bone marrow suppression**. *Monitor CBC weekly.*
- **Rapid metabolizer** → metabolism shunts toward **6-MMP** → **hepatotoxicity + reduced therapeutic response**. *Monitor LFTs.*

## INTERACTIONS

- **Allopurinol** ★ — inhibits xanthine oxidase, diverting metabolism toward 6-thioguanine → **augmented immunosuppression**. If the patient is already on allopurinol, **cut the thiopurine to 25% of standard dose**
- ⚠ Warn every patient on mercaptopurine about **gout / hyperuricemia medications**

# Methotrexate

## MECHANISM

Inhibits **dihydrofolate reductase** → impaired DNA synthesis. Also **decreases IL-1 production**.

→ **Downstream:** immunosuppression + anti-inflammatory effect → ★ **The whole point of this class in IBD is the CD/UC split:** it works in Crohn's, and it does nothing in UC.

## DRUGS

| DRUG                | WHAT IT TREATS                                                                |
|---------------------|-------------------------------------------------------------------------------|
| <b>Methotrexate</b> | <b>maintenance of remission in CD</b> — ≥15 mg weekly. <b>No role in UC</b> ★ |

## USES

**Crohn's disease** — may be used for **maintenance of remission**, dose **at least 15 mg weekly** (OR 2.4 for benefit in maintenance) · **Ulcerative colitis** — ★ **no role in induction or maintenance**. Three separate lines of evidence agree: Cochrane 2015 doesn't support maintenance use, the European double-blind RCT found it no better than placebo, and MERIT found no difference vs placebo at week 48.

**Route matters:** oral methotrexate has **reduced and extremely variable bioavailability**; **subcutaneous and intramuscular are equivalent to parenteral** — so use SC at higher doses.

# Anti-TNF- $\alpha$ Biologics

## MECHANISM

Monoclonal antibodies that bind **TNF- $\alpha$**  — **both the free/soluble form and the membrane-bound form** — and block its interaction with the cell-surface receptor. **TNF- $\alpha$  is the major contributor to the inflammatory process** in both UC and CD, which is why this is the workhorse biologic class.

→ **Downstream:** neutralised TNF signalling → mucosal healing, steroid-free remission → ⚠️

**Immunogenicity:** these are proteins, so the patient makes antibodies against them — which is why they're **combined with a thiopurine** ⭐

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                                        |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Infliximab</b> | moderate–severe <b>UC and CD</b> · <b>chimeric</b> IgG1 (25% mouse / 75% human) · IV 5 mg/kg                          |
| <b>Adalimumab</b> | moderate–severe <b>UC and CD</b> · fully <b>recombinant human</b> IgG1 · SC · <b>less immunogenic than infliximab</b> |
| <b>Golimumab</b>  | moderate–severe <b>UC</b> · recombinant human mAb · SC                                                                |

## USES

**Moderate–severe UC** — all three · **Moderate–severe CD** — infliximab, adalimumab · **Steroid-dependent UC** — infliximab gives 47% steroid-free remission at 1 year, 77% colectomy-free at ~41 months · **Post-infliximab failure** — adalimumab still works (GAIN trial) · ⭐ **In CD, infliximab is recommended combined with a thiopurine** for both induction and maintenance — SONIC showed combination beats either alone, PANTS showed the immunomodulator cuts immunogenicity.

*Dosing shape for all three: loading doses (wk 0, 2, 6 for infliximab; 160→80 mg for adalimumab), then maintenance.*

## CAUTIONS

- ⚠️ **Infliximab is chimeric** → **most immunogenic** of the three; adalimumab is fully human and less so
- ⚠️ Network meta-analyses give **conflicting data** on which of the three is actually most effective — don't claim a winner
- ⚠️ Choosing any biologic weighs: route · speed of induction response (need for bridging therapy?) · immunogenicity & need for combination · **side effects including cancer risk** · persistence · infusion facilities & TDM availability · **overall cost**

## ADMINISTRATION — ANTI-TNF REGIMENS

| DRUG       | ROUTE        | REGIMEN                                                                                                                  |
|------------|--------------|--------------------------------------------------------------------------------------------------------------------------|
| Infliximab | IV infusion  | 5 mg/kg at weeks 0, 2 and 6 — ★ 10 mg/kg gave a similar response, so 5 mg/kg is the dose                                 |
| Adalimumab | subcutaneous | 160 mg at week 0 → 80 mg at week 2 → 40 mg fortnightly                                                                   |
| Golimumab  | subcutaneous | 200 mg then 100 mg — the 400/200 mg arm gained almost nothing ( <i>response 54.9% vs 51%; remission 17.9% vs 17.8%</i> ) |

- ★ **The whole class front-loads:** a big induction dose, a second smaller one at week 2, then maintenance. Missing the induction schedule is not the same as missing a maintenance dose.
- ★ **Doubling the dose buys very little in any of the three** — that is the practical read of the trial numbers, and it is why 5 mg/kg and 200/100 mg are the standards.
- 💡 Infliximab is an infusion (a hospital visit); adalimumab and golimumab are self-injected at home. For a patient weighing up options, that logistical difference usually matters more than the efficacy gap, which is small.

# Anti-Integrin (Vedolizumab)

## MECHANISM

Recombinant monoclonal antibody that binds  **$\alpha 4\beta 7$  integrin**, blocking the **entry of inflammation-stimulating lymphocytes** into gut tissue.

→ **Downstream:** ★ **gut-selective** — it stops trafficking *into the intestine* specifically, which is why safety looked no different from placebo in GEMINI I

## DRUGS

| DRUG               | WHAT IT TREATS                                                                |
|--------------------|-------------------------------------------------------------------------------|
| <b>Vedolizumab</b> | moderate–severe <b>UC</b> ( <i>and used in CD</i> ) · IV 300 mg, day 1 and 15 |

## USES

**Moderate–severe UC** · Formally **recommended after anti-TNF failure** (induction and maintenance) · ★ but the head-to-head **VARSITY** trial (vedolizumab vs adalimumab, 769 patients) put it ahead at week 52 — clinical remission **31.3% vs 22.5%** ( $p=0.006$ ), mucosal healing **39.7% vs 27.7%** ( $p=0.0005$ ), steroid-free remission no different — **supporting vedolizumab as a first-line biologic for UC failing conventional therapy**.

*GEMINI I: wk6 response 53.1% vs 26.3% placebo; wk52 maintenance 60.7% vs 26.6%.*

## CAUTIONS

- ⚠ **Prior anti-TNF exposure reduces vedolizumab efficacy** (VICTORY consortium) — anti-TNF-naïve patients do much better (wk14 response 79.1%)



# Anti-IL-12/23 (Ustekinumab)

## MECHANISM

Monoclonal antibody against the shared **p40 subunit of IL-12 and IL-23** → inhibits both cytokines → **inhibits T-cell activation**.

→ **Downstream:** blocks the Th1 (IL-12) and Th17 (IL-23) arms — the two lineages driving IBD inflammation

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                         |
|--------------------|--------------------------------------------------------------------------------------------------------|
| <b>Ustekinumab</b> | moderate–severe <b>CD</b> — induction and maintenance · IV induction (130 mg or 6 mg/kg) then 90 mg SC |

## USES

★ **Crohn's disease is where it's established** — recommended for **induction and maintenance in CD**, in both anti-TNF-naïve and anti-TNF-failed patients (UNITI-1/2, IM-UNITI: wk44 remission 48.8–53.1%; 41.1% even in anti-TNF-refractory).

**Ulcerative colitis** — UNIFI showed real benefit (wk8 remission ~15.5%, endoscopic healing ~26%; wk44 maintenance 38.4% 8-weekly / 43.8% 12-weekly vs 24% placebo, working in both anti-TNF-naïve and refractory), **but at the time of this lecture it had no licence/approval in UC and needs more study**.

*Safety in UNIFI: no difference vs placebo — no malignancies, opportunistic infections or TB reported.*

## ADMINISTRATION — USTEKINUMAB

- **Induction: 130 mg or 6 mg/kg IV.** Week-8 clinical remission was near-identical (15.6% vs 15.5%), though clinical response favoured the weight-based dose (61.8% vs 51.3%).
- **Maintenance: 8-weekly or 12-weekly**, with week-44 remission of **38.4%** and **43.8%** respectively versus 24% on placebo.
- ⚠️ **Not licensed or approved for UC** — the data are there, the indication is not.
- 💡 The 12-weekly arm did at least as well as the 8-weekly one, which is unusual and patient-friendly: fewer doses without paying for it in remission rates.

# JAK Inhibitor (Tofacitinib)

## MECHANISM

★ **A small molecule, not an antibody** — this is what separates it from everything above. Blocks Janus kinase → prevents **phosphorylation and activation of STAT** → inhibits immune cell function.  
→ **Downstream:** broad cytokine-signalling blockade, **orally active** → ★ **No immunogenicity** — a small molecule, so no anti-drug antibodies, no need for a combination immunomodulator

## DRUGS

| DRUG               | WHAT IT TREATS                                                      |
|--------------------|---------------------------------------------------------------------|
| <b>Tofacitinib</b> | moderate–severe <b>UC</b> , <b>oral</b> — induction and maintenance |

## USES

Approved for **induction and maintenance of moderate-to-severely active UC** where conventional treatment or anti-TNF **failed or wasn't tolerated**. 10 mg twice daily confirmed in OCTAVE-1/2 (wk8 remission 18.5% / 16.6%; **23.7% in anti-TNF-naïve**). OCTAVE-SUSTAIN at 1 year: remission 34.4% on 5 mg, 40.6% on 10 mg, 11.1% placebo. **Retreatment works after a break in therapy.**

## ADVERSE EFFECTS

- ↑ **risk of infection**, notably **herpes zoster** ★
- Overall ADR rate did not differ from placebo in trials (*the lecture flags this as needing further data*)

## CAUTIONS

- ⚠ **Zoster vaccination ≥4 weeks before starting** — required for age >70, and >50 if high risk ★
- ● **Do not use the high dose in patients at high risk of pulmonary embolus**

# Nutritional Therapy & Extraintestinal Disease

## EXCLUSIVE ENTERAL NUTRITION (EEN) — A CROHN'S-ONLY TOOL

★ **The primary treatment option in pediatric CD**, because it avoids corticosteroids in children. Suggested for inducing remission in **mild–moderate CD**. Needs **strict adherence**; symptom relief within **10 days**, continue up to **8 weeks** for mucosal healing. Meta-analyses: **as effective as corticosteroids in pediatric patients** (73% ITT) — **but not in adults**.

● **Dietary intervention has no role in UC.**

## EXTRAINTESTINAL MANIFESTATIONS — THE ONES THAT SPLIT UC FROM CD

| SYSTEM        | MANIFESTATION                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------|
| <b>Joint</b>  | sacroiliitis · ankylosing spondylitis                                                           |
| <b>Ocular</b> | episcleritis · uveitis · iritis (blurred vision) · eye pain · photophobia                       |
| <b>Skin</b>   | pyoderma gangrenosum · erythema nodosum                                                         |
| <b>Kidney</b> | nephrolithiasis — <b>oxalate stones more common in CD, uric-acid stones more common in UC</b> ★ |
| <b>Liver</b>  | <b>CD:</b> ↑ gallstone formation · <b>UC:</b> sclerosing cholangitis, cholangiocarcinoma ★      |

Source: *IBD.pdf* — ผศ.ดร.ภก. อาณัฐชัย ม้ายอุเทศ, PYCH304 Pharmacology I, 41 slides. Structures omitted.

# Prostanoids, NSAIDs & Leukotriene Antagonists

**Autacoids** — made fast on demand, act locally, die fast. The lipid ones are **eicosanoids** (*eikosi* = twenty carbons), almost all cut from **arachidonic acid**. Everything in this lecture either blocks a step in that cascade or *is* a synthetic copy of one of its products.

**Tissue injury** → **phospholipase A<sub>2</sub>** (← **corticosteroids/annexin**) → **arachidonic acid** → **COX-1/COX-2** (← **NSAIDs**) → **PGG<sub>2</sub>** → **PGH<sub>2</sub>** → **isomerases** → **PGE<sub>2</sub> · PGD<sub>2</sub> · PGF<sub>2</sub>α · PGI<sub>2</sub> · TxA<sub>2</sub>** <sup>L</sup> **5-LOX** (← **zileuton**) → **5-HPETE** → **LTA<sub>4</sub>** → **LTB<sub>4</sub> · CysLTs (LTC<sub>4</sub>/D<sub>4</sub>/E<sub>4</sub>)** (← **CysLTR1 antagonists**)

## COX ISOZYMES — THE WHOLE THERAPEUTIC INDEX IN ONE TABLE ★

| ISOZYME      | WHERE                                                                                 | REGULATION                                         | WHY IT MATTERS                                                                                             |
|--------------|---------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>COX-1</b> | most cell types, <b>constitutive</b>                                                  | always on — "housekeeping"                         | gastric mucus/bicarbonate · platelet TxA <sub>2</sub> · renal flow. <b>Blocking it = the ADRs</b>          |
| <b>COX-2</b> | endothelium, macrophages, synovial fibroblasts, mast cells, chondrocytes, osteoblasts | <b>induced</b> by cytokines & inflammatory stimuli | the inflammation itself + endothelial PGI <sub>2</sub> . <b>Blocking it = the benefit, and the CV risk</b> |
| <b>COX-3</b> | brain & spinal cord ( <i>less: heart, endothelium, monocytes</i> )                    | splice variant of COX-1 (keeps intron-1)           | pain perception · <b>sensitive to acetaminophen</b>                                                        |

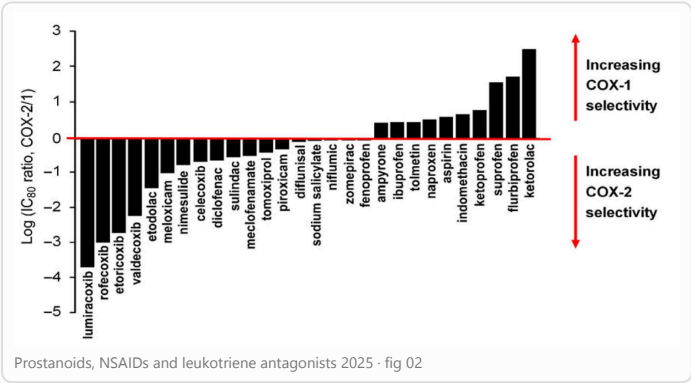
## WHAT THE PROSTANOIDS ACTUALLY DO

| SYSTEM                | PROSTANOID                                               | EFFECT                                                                              |
|-----------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Reproductive</b>   | PGE <sub>2</sub> , PGF <sub>2</sub> α                    | uterine contraction ( <b>oxytocic</b> ) · fertility, penile erection                |
| <b>Cardiovascular</b> | <b>TxA<sub>2</sub> / PGI<sub>2</sub></b>                 | platelet aggregation + vasoconstriction / <b>anti</b> -aggregation + vasodilation ★ |
|                       | PGE <sub>2</sub> , PGI <sub>2</sub>                      | arterial vasodilation · <b>keeps ductus arteriosus patent</b>                       |
| <b>Respiratory</b>    | PGE <sub>2</sub> / PGF <sub>2</sub> α, TxA <sub>2</sub>  | bronchodilation / bronchoconstriction                                               |
| <b>Renal</b>          | PGE <sub>2</sub> , PGI <sub>2</sub>                      | ↑ renal blood flow & GFR · renin release · opposes ADH                              |
| <b>GI</b>             | PGE <sub>2</sub> , PGI <sub>2</sub>                      | <b>cytoprotection</b> of the mucosa                                                 |
| <b>CNS</b>            | PGE <sub>2</sub> / PGD <sub>2</sub>                      | <b>fever</b> & pain / sleep                                                         |
| <b>Eye</b>            | PGD <sub>2</sub> , PGE <sub>2</sub> , PGF <sub>2</sub> α | drainage of aqueous humour                                                          |

## TWO LABELS WORTH SEPARATING

- **tNSAIDs** — traditional / non-selective. Compete **reversibly** with arachidonic acid at COX.
- **Coxibs** — the selective COX-2 inhibitors (diarylheterocycles).
- ★ **Aspirin is the exception in both directions:** it inhibits COX **irreversibly**, by acetylation.

OVERVIEW MAPS



# NSAIDs — Shared Mechanism & Effects

## MECHANISM

Inflamed tissue induces **COX-2** → floods the area with **PGE<sub>2</sub> and PGI<sub>2</sub>**, which raise local blood flow, vascular permeability and leukocyte infiltration. NSAIDs **competitively and reversibly inhibit COX** (*aspirin: irreversible acetylation*) → ↓ prostanoid synthesis.

→ **Downstream:** the six pharmacological effects below are all just "less prostaglandin, in a different tissue."

## SIX EFFECTS, ONE MECHANISM

| EFFECT                          | THE PROSTANOID BEING REMOVED                                                                                                                             |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Anti-inflammatory</b>        | PGE <sub>2</sub> /PGI <sub>2</sub> driving vasodilation, permeability, leukocyte influx                                                                  |
| <b>Analgesic</b>                | PG sensitising peripheral nociceptors <b>and</b> dorsal-horn neurons — hence a <b>central</b> component too                                              |
| <b>Antipyretic</b>              | pyrogens (IL-1, TNF-α, IL-6) → hypothalamic endothelial <b>PGE<sub>2</sub></b> → <b>EP3</b> → ↑ cAMP → ↑ temperature set-point                           |
| <b>Antiplatelet</b>             | platelet COX-1 → <b>TxA<sub>2</sub></b> ( <i>aspirin, low dose</i> )                                                                                     |
| <b>Tocolytic</b>                | PGE <sub>2</sub> /PGF <sub>2α</sub> myometrial contraction — <b>indomethacin</b> , off-label                                                             |
| <b>Closes ductus arteriosus</b> | PGE <sub>2</sub> → <b>EP4</b> → adenylate cyclase → vasodilation keeping it open. Remove it and it shuts — <b>indomethacin, ibuprofen lysine</b> for PDA |

## PHARMACOKINETICS — WHY THEY CONCENTRATE EXACTLY WHERE YOU WANT THEM ★

Hydrophobic **weak acids, pKa ~3–5**. Ionised at plasma pH 7.4, but at the **inflamed site (pH ~5.5 or lower)** more drug is un-ionised → crosses into cells and **accumulates there**.

→ **Downstream:** duration of action outlasts plasma  $t_{1/2}$  (*diclofenac:  $t_{1/2}$  1–2 h in normal tissue, 5–7 h in inflamed tissue*) · 95–99% albumin-bound · CNS levels sufficient for central analgesia · synovial fluid ≥ half plasma · hepatic metabolism (*celecoxib CYP2C9 · etoricoxib CYP3A4 · parecoxib → hydrolysis → valdecoxib → CYP3A4/2C9*), then glucuronidation and renal excretion.

*More soluble salts absorb faster (diclofenac  $K^+ > Na^+$ ) — but faster absorption only sometimes means faster analgesia (naproxen  $Na^+$  and ibuprofen soft-gel: no clear onset advantage).*

## PLASMA HALF-LIFE, THREE BUCKETS

| $T_{1/2}$       | DRUGS                                                                                  |
|-----------------|----------------------------------------------------------------------------------------|
| <b>1–6 h</b>    | aspirin (0.25) · diclofenac · ibuprofen · mefenamic acid · flurbiprofen · indomethacin |
| <b>6–12 h</b>   | ketorolac · etodolac · sulindac · diflunisal · <b>celecoxib</b>                        |
| <b>&gt;12 h</b> | naproxen · meloxicam · nabumetone · <b>etoricoxib</b> · <b>piroxicam (30–86 h)</b>     |

## CLINICAL APPLICATIONS

**Fever** (*pick a rapid-onset agent*) · **Inflammation & pain of low-to-moderate intensity, mainly somatic** — headache incl. **migraine (1st-line for the attack)**, OA/RA/acute gout, post-operative & dental pain · some **visceral pain** (dysmenorrhoea) · **not effective in neuropathic pain** · **Thromboembolism prevention** (low-dose aspirin) · **Kawasaki disease** · **Preterm labour / PDA closure** (indomethacin) · **Pre-eclampsia prevention** in high-risk pregnancy (low-dose aspirin, started before 16 weeks)

# NSAID Adverse Effects & Interactions

## ADVERSE EFFECTS

- **GI** ★ — the commonest, worst with tNSAIDs. Losing mucosal **PGE<sub>2</sub>** removes mucus, bicarbonate and mucosal blood flow; the drugs are weak acids so they also irritate directly. Nausea, vomiting, abdominal pain, indigestion → **peptic ulcer, perforation, GI bleeding**. Coxibs: less, not none.
- **Renal** — both COX-1 and COX-2 make PGE<sub>2</sub>/PGI<sub>2</sub> that **dilate the afferent arteriole** (maintaining GFR) and promote Na<sup>+</sup>/water excretion. Block them → **Na<sup>+</sup>/water retention, hyperkalemia, renal vasoconstriction** → edema, **hypertension (SBP ↑ ~5 mmHg within 1–2 weeks)**, worsened heart failure, **hemodynamic AKI**.
- **Cardiovascular** ★ — platelet **COX-1** → **TxA<sub>2</sub>** (pro-thrombotic) is balanced against endothelial **COX-2** → **PGI<sub>2</sub>** (anti-thrombotic). COX-2-selective drugs remove PGI<sub>2</sub> only → ↑ **MI and ischemic stroke**. *This is why rofecoxib and valdecoxib were withdrawn.*
- **Urate** — **low-dose** aspirin upregulates **URAT1** → ↑ urate reabsorption → recurrent gout attacks. (*High dose >5 g/day is the opposite: uricosuric.*)
- **Reye's syndrome** ★ — aspirin/salicylate for **viral fever in under-20s**: acute encephalopathy, liver dysfunction, fatty infiltration of liver and viscera. Mechanism unclear.
- **Hypersensitivity** — see the caution block.

## HYPERSENSITIVITY: CROSS-REACTIVE VS TRULY ALLERGIC ★

|                     | NON-ALLERGIC (CROSS-REACTIVITY)                                                                                            | ALLERGIC (SELECTIVE)                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>trigger</b>      | ≥ 2 NSAIDs of <b>different</b> chemical groups                                                                             | NSAIDs of the <b>same</b> chemical group, often one drug                           |
| <b>presentation</b> | NSAID- <b>exacerbated respiratory</b> disease (asthma) · NSAID-exacerbated <b>cutaneous</b> disease · urticaria–angioedema | single-NSAID urticaria/angioedema or <b>anaphylaxis</b> · delayed hypersensitivity |
| <b>why</b>          | COX blocked → arachidonic acid <b>shunted to leukotrienes</b> (~10% of tNSAID users)                                       | true immune sensitisation                                                          |

## INTERACTIONS

- **ACEIs / ARBs** — ↓ hypotensive effect · ● ↑ hyperkalemia
- **Warfarin** — ● ↑ bleeding risk
- **Corticosteroids & SSRIs** — ↑ GI complications
- **Lithium** — most NSAIDs ↓ its renal excretion → **lithium toxicity**. ⚠ **Sulindac is the exception — it lowers plasma lithium.**

## CAUTIONS

- ⚠ **Severe hepatic or renal disease** — generally not recommended
- ⚠ **CVD patients needing analgesia**: try **paracetamol or a weak opioid (tramadol) first**. If an NSAID is unavoidable, use a **COX-1-preferring** one (naproxen, ibuprofen) at the lowest effective dose with GI vigilance, and **avoid diclofenac and the coxibs**. ★

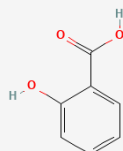
# Non-selective NSAIDs (tNSAIDs)

## MECHANISM

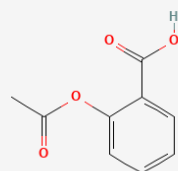
Reversible, competitive COX inhibition with **no clean isozyme preference** — "non-selective" is a spectrum, not a category. Where a drug sits on that spectrum predicts its ADR profile.

→ **Downstream:** COX-1-leaning → more GI injury, more antiplatelet effect · COX-2-leaning → more CV risk

## THE SHARED CORE · WHAT CHANGES

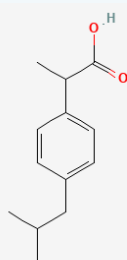


**Salicylic acid — the ancestor.** ● every classical NSAID is a WEAK ACID. That single property explains the shared problems: they are un-ionised in stomach acid, cross into gastric cells, become trapped there, and they concentrate in inflamed (acidic) tissue.



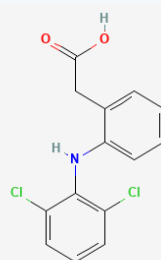
### Aspirin

an acetyl group added →  
● it hands that acetyl to COX's serine — irreversible, unlike every other NSAID here



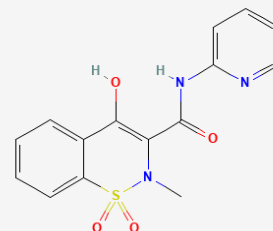
### Ibuprofen

propionic acid family — reversible, short, the safest GI profile



### Diclofenac

acetic acid family, pKa 3.9 → concentrates hard in inflamed joints; tissue t½ far exceeds plasma t½



### Piroxicam

an oxicam (enolic acid) → t½ ~50 h → once daily and the worst GI record

## MEDCHEM

**SAR** ● The GI damage has TWO independent causes and only one is COX-1. The second is **ion trapping**: these are weak acids, so at gastric pH 2 they are **un-ionised and cross into the mucosal cell**, where pH 7.4 ionises them and they can no longer leave. The drug concentrates inside the cell and damages it directly. That is why an enteric coat helps and a COX-sparing dose does not.

**SAR** Salicylates: the **COOH and the phenolic OH are both required** (remove or substitute either and activity drops), **substitution at C4 kills it outright**, and halogens raise potency and toxicity together. Aspirin's pKa is **3.5** against salicylic acid's **2.9** - acetylating the OH removes the intramolecular hydrogen bond that made the parent more acidic. ★ And aspirin is itself a prodrug in part: plasma esterase hydrolyses it to salicylic acid, a **reversible** COX inhibitor, while the intact molecule is what **irreversibly acetylates Ser530**.

**SAR** Arylpropionic acids: the **(S) enantiomer is ~100× more active**, and the α-methyl group both raises anti-inflammatory activity and lowers side effects. ★ **About 60% of (R)-ibuprofen is enzymatically inverted to (S) in the body** - the 'wrong' half of the racemate is largely recycled into the right one, which is why selling the single isomer never mattered much here.

**Prodrug** Sulindac is a prodrug - **reduced** (not oxidised) to sulindac sulfide by methionine sulfoxide reductase, and oxidised back to the inactive sulfone by FMO. ⚠ **Diclofenac's rare idiosyncratic hepatotoxicity** is traced to **quinone imine** metabolites forming covalent protein adducts, and it carries the highest cardiovascular risk of the non-selective NSAIDs. **The oxicams were designed specifically to have no carboxylic acid at all** - an enol instead - to escape the rapid glucuronidation and short half-lives of the acids.



## SEVEN CHEMICAL FAMILIES

| FAMILY                                       | DRUGS                                                                            |
|----------------------------------------------|----------------------------------------------------------------------------------|
| <b>Salicylic acid derivatives</b>            | <b>aspirin</b> · diflunisal · sulfasalazine                                      |
| <b>Acetic acid — indole/indene</b>           | <b>indomethacin</b> · sulindac · etodolac                                        |
| <b>Acetic acid — heteroaryl</b>              | <b>diclofenac</b> · ketorolac · tolmetin                                         |
| <b>Arylpropionic acid</b>                    | <b>ibuprofen</b> · naproxen · ketoprofen · flurbiprofen · fenoprofen · oxaprozin |
| <b>Fenamates (anthranilic acid)</b>          | <b>mefenamic acid</b> · eclofenamic acid                                         |
| <b>Enolic acid — oxicams / pyrazolidones</b> | <b>piroxicam</b> · tenoxicam / phenylbutazone · oxyphenbutazone                  |
| <b>Alkanones</b>                             | <b>nabumetone</b>                                                                |

## WHERE THEY SIT ON THE COX-1 ↔ COX-2 SPECTRUM ★

| LEANING                 | DRUGS                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------|
| <b>COX-1-preferring</b> | aspirin · indomethacin · naproxen · ibuprofen                                            |
| <b>Roughly equal</b>    | diflunisal · sodium salicylate · fenoprofen                                              |
| <b>COX-2-preferring</b> | <b>diclofenac</b> ( <i>≈ celecoxib's selectivity</i> ) · sulindac · meloxicam · etodolac |

## DRUGS

| DRUG                         | WHAT IT TREATS                                                                      |
|------------------------------|-------------------------------------------------------------------------------------|
| <b>Ibuprofen</b>             | mild–moderate pain · fever · inflammation · (as lysine salt, IV) <b>PDA closure</b> |
| <b>Naproxen</b>              | inflammatory pain, OA/RA — <b>the safer NSAID in CVD patients</b>                   |
| <b>Diclofenac</b>            | inflammatory pain, migraine ⚠ COX-2-leaning → avoid in CVD                          |
| <b>Indomethacin</b>          | acute gout · <b>tocolysis</b> · <b>PDA closure</b>                                  |
| <b>Mefenamic acid</b>        | dysmenorrhoea, mild–moderate pain                                                   |
| <b>Piroxicam / meloxicam</b> | chronic arthritis — long t <sub>1/2</sub> , once daily                              |
| <b>Ketorolac</b>             | short-term moderate–severe post-operative pain                                      |

## ADMINISTRATION — NSAID HALF-LIVES AND SALT FORMS

| T½ BAND     | DRUGS (HOURS)                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------|
| 1–6 h       | aspirin <b>0.25</b> · diclofenac 1–2 · ibuprofen 2–2.5 · mefenamic acid 3–4 · flurbiprofen 3–6 · indomethacin 4–5 |
| > 6 to 12 h | ketorolac 2.4–8.6 · diflunisal 5–20 · etodolac 7 · sulindac 7 · celecoxib 11–16                                   |
| > 12 h      | naproxen 12–15 · meloxicam 13–20 · etoricoxib 20–30 · nabumetone 22.5–30 · <b>piroxicam 30–86</b>                 |

**Salt and formulation change the speed, not the ceiling:** - **Diclofenac potassium** dissolves better than the sodium salt → faster tablet disintegration, faster absorption, **faster onset of analgesia**. - **Naproxen sodium** is absorbed faster than plain naproxen and reaches a higher peak sooner — ⚠️ **but there is no clear evidence either form relieves pain faster** (*e.g. moderate-to-severe dysmenorrhoea*). - **Ibuprofen soft gelatin capsules** likewise reach levels faster.

- ★ **The half-life band is the dosing frequency.** Piroxicam at 30–86 hours is once daily and takes days to clear after a GI bleed; ibuprofen at 2–2.5 hours is out of the system by the evening.
- 💡 The salt-form difference is real pharmacokinetically and thin clinically — worth knowing so you don't oversell the "fast-acting" version at the counter.

# Aspirin

## MECHANISM

The only **irreversible** COX inhibitor — it **acetylates** the enzyme. Platelets have no nucleus, so they cannot resynthesise COX-1: one dose disables them **for their lifetime**. Low-dose aspirin acetylates platelets **presystemically, within minutes**, before hepatic first-pass removes it — which is why a small dose hits platelets without touching endothelial COX-2.

→ **Downstream:** ↓ platelet **TxA<sub>2</sub>** → ↓ aggregation & vasoconstriction · endothelial **PGI<sub>2</sub>** spared

## USES

**Antiplatelet** — **80–300 mg/day**: prevention & treatment of thromboembolic events (IHD, stroke) ·

**Kawasaki disease** — high dose first for anti-inflammatory + antipyretic effect, then low dose (from ~2–3 days after afebrile) for 6–8 weeks to protect the coronary arteries · **Pre-eclampsia prevention** in high-risk pregnancy, started **before 16 weeks** · **Colorectal cancer primary prevention** — ages 50–59 with ≥10% 10-year CVD risk, no bleeding risk, ≥10-year horizon

## ADVERSE EFFECTS (BEYOND THE SHARED NSAID LIST)

- **Reye's syndrome** ★ — under-20s given aspirin for **viral** fever
- **Gout attacks** — low dose upregulates URAT1 → urate retention
- Bleeding — the irreversible platelet effect cuts both ways

## CAUTIONS

- **Never for fever in children/adolescents with a viral illness**
- ⚠ Monitor serum urate if given alongside urate-lowering therapy

## ADMINISTRATION — LOW-DOSE ASPIRIN AND URATE

- **Low-dose aspirin upregulates URAT1** (*the urate/anion exchanger*), **raising serum urate** and **increasing the risk of recurrent gout attacks**.
- **With concomitant use:** monitor **serum urate**, and ★ **dose adjustment of urate-lowering therapy may be especially important**.
- 💡 This is an easy one to miss because both drugs are "correct" — the aspirin is cardioprotective and nobody wants to stop it, so the answer is to move the allopurinol, not the aspirin.

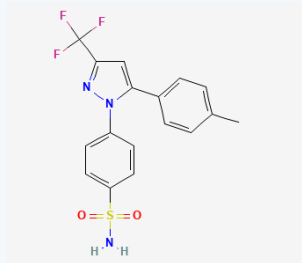
# COX-2 Selective Inhibitors (Coxibs)

## MECHANISM

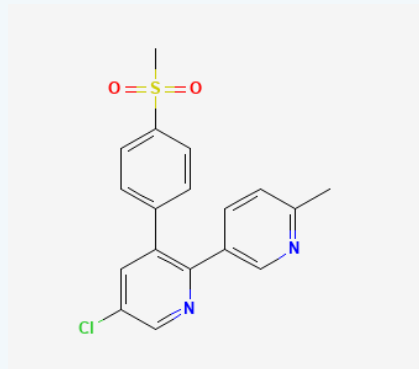
Diarylheterocycles that spare **COX-1**. Inflammation is COX-2-driven, so efficacy is retained while gastric PGE<sub>2</sub> and platelet TxA<sub>2</sub> are largely left alone.

→ **Downstream:** ↓ GI injury (*less, not zero*) · **no antiplatelet effect** · ● endothelial **PGI<sub>2</sub> suppressed while platelet TxA<sub>2</sub> continues** → **prothrombotic imbalance** ★

## THE SHARED CORE · WHAT CHANGES



**Celecoxib.** ● selectivity is pure geometry: COX-2's channel has a side pocket COX-1 lacks, and the bulky sulfonamide arm here only fits that pocket. Too big for COX-1 = no gastric COX-1 block.



**Etoricoxib**

a bigger sulfone → even more COX-2-selective → the strongest cardiovascular warning of the group

## MEDCHEM

**SAR** ★ The entire selectivity of this class rests on one methyl group's worth of difference between two enzymes. Residue 523 is **isoleucine in COX-1** and **valine in COX-2**; valine is smaller, which opens a **side pocket** (Val434, Arg513, Val523) that COX-1 does not have. Every coxib is a **diaryl heterocycle** carrying a **sulfonamide (SO<sub>2</sub>NH<sub>2</sub>)** or **methanesulfone (SO<sub>2</sub>CH<sub>3</sub>)** positioned to insert into that pocket. No pocket, no selectivity.

**SAR** ● Which coxib cross-reacts with a sulfa allergy is decided by that same group. **Celecoxib and parecoxib carry a sulfonamide** → **cross-reactivity with antibiotic sulfonamides**. Etoricoxib and rofecoxib use a **methanesulfone** instead and do not. This is answerable from the structure alone.

**SAR** ⚠ The cardiotoxicity that withdrew **rofecoxib and valdecoxib** is attributed to **hERG potassium-channel inhibition** (hERG runs cardiac repolarisation → arrhythmia). For rofecoxib specifically the deck names a **reactive species that binds cellular proteins including hERG** - and explicitly notes that reaction **does not apply to the other coxibs**.

## DRUGS

| DRUG                      | WHAT IT TREATS                                                |
|---------------------------|---------------------------------------------------------------|
| <b>Celecoxib</b>          | OA/RA and inflammatory pain where GI risk is the concern      |
| <b>Etoricoxib</b>         | OA/RA, acute gout, inflammatory pain                          |
| <b>Parecoxib</b>          | injectable <b>prodrug of valdecoxib</b> — post-operative pain |
| ~~Rofecoxib, valdecoxib~~ | <b>withdrawn — MI/stroke</b>                                  |
| ~~Lumiracoxib~~           | <b>withdrawn — hepatotoxicity</b>                             |

## CAUTIONS & THE SULFONAMIDE QUESTION ★

- **Cardiovascular disease** — the class effect that killed two of them
- ⚠ **Sulfonamide allergy:** the evidence says coxibs **lack** the two structural features that drive sulfonamide antibiotic allergy (the N1 ring-nitrogen substituent and the N4 aromatic amine), so clinical cross-reaction is **not expected**. But the prescribing information disagrees:
  - **Celecoxib, parecoxib** (*true sulfonamides,  $SO_2NH_2$* ) — **contraindicated**
  - **Etoricoxib** (*a sulfone,  $SO_2$* ) — **not contraindicated**

# Paracetamol (Acetaminophen)

## MECHANISM

A **COX inhibitor** like the NSAIDs, but the isozyme that matters is **COX-3** — the COX-1 splice variant found in brain and spinal cord, which is specifically sensitive to acetaminophen. That central-only site is why it shares the antipyretic and analgesic effects but sits outside the NSAID ADR profile.

→ **Downstream:** ↓ hypothalamic PGE<sub>2</sub> → **antipyretic** · central **analgesia**

## USES

**Fever · Mild-to-moderate pain · ★ First-choice analgesic ahead of NSAIDs in cardiovascular disease patients** (*with weak opioids such as tramadol as the other option*)

## NOTE

- ⚠ This lecture only touches paracetamol in passing (COX-3 sensitivity, antipyresis, the CVD-patient preference). Hepatotoxicity/NAPQI is **not** covered here — get it from the liver/toxicology lecture rather than assuming it belongs to this one.

# Prostanoid Analogues

## MECHANISM

Not inhibitors — **synthetic copies of the prostanoids themselves**, given where you *want* the prostaglandin effect. Read the target from the receptor table: whatever PGE<sub>1</sub>/PGE<sub>2</sub>/PGF<sub>2</sub>α/PGI<sub>2</sub> does physiologically is the indication.

→ **Downstream:** smooth-muscle relaxation (vascular, corpus cavernosum, pulmonary artery) · myometrial contraction (oxytocic) · GI mucus & Cl<sup>-</sup> secretion · ↑ aqueous humour outflow

## DRUGS

| DRUG                                                             | WHAT IT TREATS                                                                                                     |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Alprostadil</b> (PGE <sub>1</sub> )                           | <b>erectile dysfunction</b> — relaxes corpus cavernosal smooth muscle                                              |
| <b>Lubiprostone</b> (PGE <sub>1</sub> )                          | <b>constipation</b> — activates a Cl <sup>-</sup> channel                                                          |
| <b>Misoprostol</b> (PGE <sub>1</sub> )                           | <b>mucoprotection</b> (↑ GI mucus — the antidote to NSAID ulceration) · <b>oxytocic</b> (myometrial contraction) ★ |
| <b>Dinoprostone</b> (PGE <sub>2</sub> )                          | <b>oxytocic</b> — cervical ripening / labour induction                                                             |
| <b>Latanoprost, bimatoprost, travoprost</b> (PGF <sub>2</sub> α) | <b>glaucoma</b> — ↑ outflow of aqueous humour                                                                      |
| <b>Carboprost</b> (PGF <sub>2</sub> α)                           | <b>oxytocic</b>                                                                                                    |
| <b>Epoprostenol, iloprost, treprostinil</b> (PGI <sub>2</sub> )  | <b>pulmonary arterial hypertension</b> — relaxes the pulmonary artery                                              |

## CAUTIONS

- **Misoprostol is an abortifacient** — its oxytocic action is the flip side of its gastroprotection ★

# Leukotriene Modifiers

## MECHANISM

The **other** branch of the cascade. **5-LOX** turns arachidonic acid into 5-HPETE → LTA<sub>4</sub> → **LTB<sub>4</sub>** and the **cysteinyl leukotrienes (LTC<sub>4</sub>, LTD<sub>4</sub>, LTE<sub>4</sub>)**. Via CysLTR1/2 and LTB<sub>4</sub> these drive chemokine production, immune cell activation and recruitment, **mucus secretion** and **bronchoconstriction** — and 5-LOX plus CysLTR1 sit mainly on airway leukocytes and smooth muscle.

Two ways in: 1. **CysLTR1 antagonists** — block the receptor 2. **5-LOX inhibitor** — block synthesis upstream → **Downstream:** ↓ bronchoconstriction, mucus and airway inflammation. *This also explains why COX blockade can trigger asthma: arachidonic acid gets shunted down this branch.* ★

## DRUGS

| DRUG                                                              | WHAT IT TREATS                    |
|-------------------------------------------------------------------|-----------------------------------|
| <b>Montelukast</b> (CysLTR1 antagonist, t <sub>1/2</sub> 2.7–7 h) | asthma · <b>allergic rhinitis</b> |
| <b>Pranlukast</b> (CysLTR1 antagonist, t <sub>1/2</sub> 3–9 h)    | asthma · <b>allergic rhinitis</b> |
| <b>Zafirlukast</b> (CysLTR1 antagonist, t <sub>1/2</sub> 10 h)    | asthma                            |
| <b>Zileuton</b> (5-LOX inhibitor, t <sub>1/2</sub> 2.5 h)         | asthma                            |

## USES

**Asthma** — allergen-challenged · exercise-exacerbated · ★ **aspirin-exacerbated** (the shunt mechanism makes this the logical class) · **Allergic rhinitis** — montelukast and pranlukast only

## ADVERSE EFFECTS

- Headache
- Abdominal pain
- **Liver enzyme elevations**
- ⚠ Possible association with **Churg–Strauss syndrome** in some asthma patients — vasculitis with eosinophilia reducing organ perfusion. Mechanism unclear.



# PAF Antagonists

## MECHANISM

**Platelet activating factor** is a *different* lipid-derived autacoid — not an eicosanoid, not from the COX/LOX split — that drives allergic reactions and **bronchial hyperresponsiveness**.

→ **Downstream:** blocking the PAF receptor removes a mediator that antihistamines alone leave untouched

## DRUGS

| DRUG              | WHAT IT TREATS                                           |
|-------------------|----------------------------------------------------------|
| <b>Rupatadine</b> | <b>allergic rhinitis · chronic spontaneous urticaria</b> |

## WHY RUPATADINE IS DUAL-ACTING

An **inverse agonist at the histamine H<sub>1</sub> receptor** *and* a specific, dose-dependent inhibitor of **PAF receptors**. Metabolised by **CYP3A4** into active metabolites that retain antihistamine activity — which is why its duration outlasts its **4–5 h** plasma half-life.

*Principle behind this: [[pharm-pharmacokinetics]] — diclofenac (pKa 3.9) concentrates itself in inflamed tissue by ionization, and its tissue half-life (5–7 h) is far longer than its plasma half-life (1–2 h) — which is what sets the dosing interval.*

*Principle behind this: [[pharm-drug-action]] — Aspirin inhibits COX irreversibly — the practical face of non-competitive/insurmountable blockade: platelet function returns only as platelets are replaced. Paracetamol is the high-TI counterexample.*

*Principle behind this: [[pharm-drug-toxicity]] — Aspirin overdose — pyloric spasm and bezoars wrecking normal absorption kinetics, and urinary alkalinisation as the way out (→ [[pharm-drug-interactions]]).*

*Source: Prostanoids, NSAIDs and Leukotriene antagonists\_2025.pdf — อ.สุรศักดิ์, PYCH304 Pharmacology I, 42 slides. Structures omitted.*

# Drugs for Gout

**Hyperuricemia** = serum urate > **6.8 mg/dL** (408  $\mu\text{mol/L}$ ) — the solubility limit. (*Thai guideline 2012: male > 7, female > 6.*) It comes from **overproduction** or **underexcretion**. Hyperuricemia alone can be silent, but > **90% of gout patients are hyperuricemic** — and it tracks with cardiovascular disease, metabolic syndrome, CKD and type-2 DM.

**Purine** (*diet · tissue nucleic acid · de novo synthesis*) → hypoxanthine —PNP→ xanthine —  
XO→ uric acid → renal excretion

Urate above solubility → MSU crystals in joints (*knee, toe, elbow*) → NLRP3 inflammasome → IL-1 $\beta$  → acute flare · long-term deposition → tophi → chronic gout

## THREE WORDS WORTH SEPARATING

- **Uricosstatic** — blocks uric acid **synthesis** (XOIs)
- **Uricosuric** — raises renal uric acid **excretion**
- **Uricase** — **degrades** existing urate to allantoin
- **Tophus** (ปุ่มโทฟัส) — subcutaneous MSU deposit; the marker of chronic gout

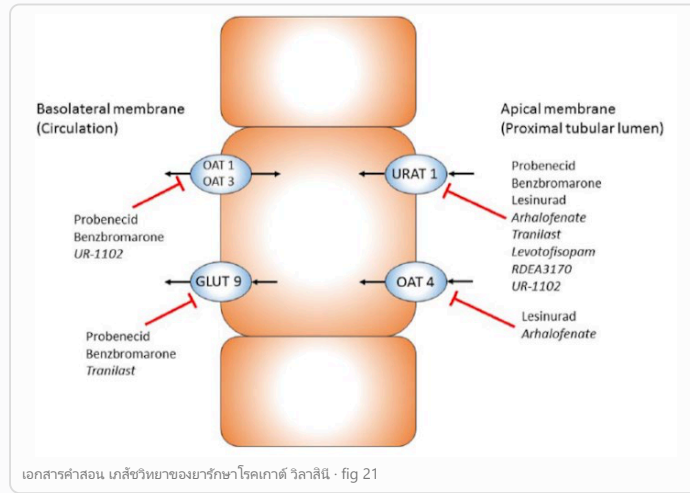
## THE SPLIT THAT RUNS THE WHOLE LECTURE

| GOAL                                                                                              | WHEN                                      | CLASSES                                                 |
|---------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------|
| <b>Relieve the acute flare</b> — anti-inflammatory only, does nothing to urate                    | within <b>12 h</b> of onset               | Colchicine · NSAIDs · Corticosteroids · IL-1 inhibitors |
| <b>Lower urate in chronic gout (ULT)</b> — dissolves the crystal load, useless for the pain today | long-term, target <b>sUA &lt; 6 mg/dL</b> | XO inhibitors · Uricosurics · Uricase                   |

## WHERE EACH ULT CLASS HITS THE CHAIN

| CLASS                              | POINT OF ATTACK                                          | LINE            |
|------------------------------------|----------------------------------------------------------|-----------------|
| <b>XO inhibitor (uricosstatic)</b> | blocks hypoxanthine → xanthine → uric acid               | <b>first</b>    |
| <b>Uricosuric</b>                  | blocks URAT1 reabsorption in proximal tubule             | second / add-on |
| <b>Uricase</b>                     | converts existing uric acid → <b>allantoin</b> (soluble) | refractory only |

## OVERVIEW MAPS



# Colchicine

## MECHANISM

Binds intracellular  **$\beta$ -tubulin** → colchicine–tubulin complex → **prevents polymerization into microtubules** → the neutrophil loses its cytoskeleton, so it cannot migrate, chemotax or phagocytose. Also inhibits formation of **leukotriene B<sub>4</sub> and IL-1 $\beta$** .

→ **Downstream:** ↓ neutrophil recruitment into the joint → pain relief within 24 h. → **Timing matters:** efficacy drops sharply if given more than 24 h after the flare starts ★

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                                                                |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Colchicine</b> | acute gout flare · flare <b>prophylaxis</b> while starting ULT · familial Mediterranean fever · (FDA 2023) reducing atherosclerotic CV events |

## USES

**Acute flare** — 1.2 mg at the first sign, then 0.6 mg one hour later. ★ **Low dose beats high dose:** the AGREE trial showed the same efficacy as the old 4.8 mg regimen with significantly fewer ADRs · **Flare prophylaxis** — 0.5–1 mg/day alongside ULT for the first 6 months

## ADVERSE EFFECTS

- **GI** ★ — diarrhea (23%), nausea/vomiting (4–17%), stomachache. *The dose-limiting one.*
- Fatigue, headache
- Neuritis, **myopathy**
- Alopecia
- ● **Bone marrow suppression**

## INTERACTIONS

- ● **Colchicine is a CYP3A4 and P-gp substrate** ★ — CYP3A4/P-gp inhibitors (*clarithromycin, ketoconazole, cyclosporine, diltiazem, erythromycin, fluconazole*) push it into toxicity. **Narrow therapeutic index**, so this is not theoretical.
- **Statins** → rhabdomyolysis risk

## CAUTIONS

- ⚠ **Renal impairment** — eliminated renally; avoid at **CrCl < 30 mL/min**, dose-adjust otherwise
- ⚠ Severe hepatic impairment — no safety data
- ⚠ Pregnancy category C

## ADMINISTRATION — THE DOSES, AND THE DOSE ADJUSTMENTS THAT MATTER MORE

- **Acute flare:** 1.2 mg at the first sign, then 0.6 mg one hour later (US/FDA). ⚠️ **The Thai gout guideline words it as 1 mg immediately, then 0.5 mg one hour later**, and separately recommends **0.6 mg 2–4 times/day** — explicitly warning **against** high or frequent dosing, because diarrhoea is what you get ★
- **Prophylaxis:** 0.6 mg, 1–3 times daily · **Familial Mediterranean fever:** 1.2–2.4 mg/day (adults and children > 12), given in one or two doses
- **PK:** rapidly absorbed, absolute bioavailability ~45%, **food has no effect** · renal elimination · ● **narrow therapeutic index** · **P-gp and CYP3A4 substrate**
- ● **Dose adjustment — this is the exam-and-life-critical block:**
  - **CrCl < 30 mL/min** — do not repeat more than **once in 2 weeks**
  - **Dialysis** — 0.6 mg, not repeated more than once in 2 weeks (*prophylaxis: 0.3 mg twice weekly, monitor closely*)
  - **Hepatic impairment** — do not repeat more than once in 2 weeks
  - **With a STRONG CYP3A4 inhibitor** (*atazanavir, clarithromycin, darunavir/ritonavir, indinavir, itraconazole, ketoconazole*) — 0.6 mg then **0.3 mg** an hour later, **no repeat within 3 days**
  - **With a MODERATE CYP3A4 inhibitor** (*amprenavir, aprepitant, diltiazem, erythromycin, fluconazole*) — 1.2 mg, **no repeat within 3 days**
  - **Prophylaxis alongside a P-gp or strong CYP3A4 inhibitor** — 0.3 mg every other day, or 0.3 mg once daily
- 💡 Ask what else they take **before** you hand this over. Clarithromycin or itraconazole plus a normal colchicine dose is one of the classic serious interactions, and the patient will never volunteer it.

# NSAIDs

## MECHANISM

COX inhibition → ↓ prostaglandin → ↓ pain and inflammation in the crystal-inflamed joint. **No effect on urate.** All NSAIDs perform equally in gout — pick on tolerability, not potency.

→ ● **Except aspirin:** it acts on urate, and in the wrong direction at normal doses. **Low-dose aspirin binds URAT1** → ↓ urate excretion → ↑ **sUA**. Only high dose (> 3 g/d) is uricosuric. ★

## DRUGS

| DRUG                | WHAT IT TREATS                                          |
|---------------------|---------------------------------------------------------|
| <b>Naproxen</b>     | acute flare · low dose for flare prophylaxis during ULT |
| <b>Indomethacin</b> | acute flare                                             |
| <b>Diclofenac</b>   | acute flare                                             |
| ● <b>Aspirin</b>    | <b>not used</b> — low dose raises serum urate           |

## USES

**Acute flare** — first-line alongside colchicine · **Flare prevention** during the first months of urate-lowering therapy

## ADVERSE EFFECTS

- **GI** — irritation, ulcer, bleeding (*co-prescribe a PPI*)
- Long-term high dose → **chronic renal insufficiency**

## CAUTIONS

- ⚠ **Avoid in severe renal impairment**

# Corticosteroids

## MECHANISM

Broad anti-inflammatory and immunosuppressive action on the crystal-driven inflammatory cascade.

→ **Downstream:** short-course oral steroid is **as effective as oral NSAIDs** in an acute flare, and no more dangerous over that timeframe. The value is having an option when the other two are blocked.

## DRUGS

| DRUG                           | WHAT IT TREATS                                      |
|--------------------------------|-----------------------------------------------------|
| <b>Prednisolone</b>            | acute flare — 30–35 mg/day for 3–5 days, then taper |
| <b>Betamethasone</b>           | acute flare — single IM dose                        |
| <b>Methylprednisolone</b>      | acute flare — taper over ~6 days                    |
| <b>Triamcinolone acetonide</b> | acute flare — single IM dose                        |

## USES

★ **The escape route** — for patients who cannot tolerate or are contraindicated to NSAIDs *and* colchicine (*renal failure, GI bleeding*) · **Intra-articular injection** for acute **monoarthritis** — fast, safe, high efficacy · **Combined** with colchicine or NSAIDs in severe polyarticular flares · Low-dose prednisolone ( $< 10$  mg/day) as flare prophylaxis during ULT

## ADVERSE EFFECTS

- **Short course:** dysphoria & mood disorder ·  $\uparrow$  blood glucose · immune suppression · fluid retention
- **Long term:** hypokalemia · hyperglycemia +  $\downarrow$  insulin secretion · hypertension · infection susceptibility · peptic ulcer & perforation · osteoporosis · myopathy · cataracts · growth retardation
- **Cushing's syndrome** — moon face, buffalo hump
- ⚠️ **Abrupt withdrawal after long use** → fever, myalgia, arthralgia

## INTERACTIONS

- Antagonize **oral hypoglycemics and insulin**
- **CYP3A4 inducers** (*rifampicin, phenytoin, barbiturates*) → faster steroid breakdown, lost effect
- **NSAIDs** → GI bleeding and ulcer risk stacks

## CAUTIONS

- 🚫 **Do not use during active infection**
- ⚠️ Diabetes · peptic ulcer · osteoporosis · hypothyroidism · hypertension
- ⚠️ Pregnancy category C — **category D in the first trimester**

## ADMINISTRATION

- **Oral prednisolone 30–35 mg/day for 5 days** is the dose the Thai guideline gives for an acute flare — or an **intra-articular injection** where a single joint is involved
- 💡 Five days and stop; no taper needed at this length. Say that explicitly, because patients who have had steroids before often assume they should taper and stretch the course out.

# IL-1 Inhibitors

## MECHANISM

MSU crystals activate the **NLRP3 inflammasome**, which releases **IL-1 $\beta$**  — the key mediator of gouty inflammation. ★ These agents block IL-1 $\beta$ , cutting the pathway at its source rather than downstream.

→ **Downstream**: specific for MSU crystal-induced inflammation. Not a urate-lowering drug — pair with ULT.

## DRUGS

| DRUG               | WHAT IT TREATS                                                |
|--------------------|---------------------------------------------------------------|
| <b>Anakinra</b>    | frequent flares when the first-line three are contraindicated |
| <b>Rilonacept</b>  | same                                                          |
| <b>Canakinumab</b> | same                                                          |

## USES

**Last-line for the acute flare** (EULAR 2016) — only after colchicine, NSAIDs and corticosteroids have failed or are all contraindicated (*e.g. renal failure*), in patients with frequent flares



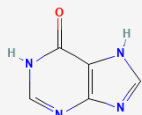
# Xanthine Oxidase Inhibitors

## MECHANISM

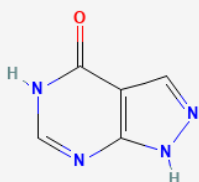
Inhibit **xanthine oxidase**, the enzyme running both final steps of the chain — so hypoxanthine and xanthine never become uric acid. "**Uricostatic**": **cut production, not excretion**. First-line ULT.

→ **Allopurinol**: purine analogue, **competitive** inhibitor; converted to the active metabolite **oxypurinol**; **renal** elimination. → **Febuxostat**: **non-purine**, highly specific, non-competitive; **hepatic** elimination — which is what makes it the option when the kidney is the problem.

## THE SHARED CORE · WHAT CHANGES

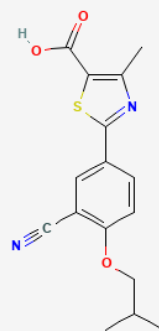


**Hypoxanthine — the real substrate.** xanthine oxidase turns this into xanthine, then uric acid



### Allopurinol

● hypoxanthine with two ring atoms swapped — an isomer. The enzyme oxidises it into oxypurinol, which then never leaves. The drug's metabolite is the real drug.



### Febuxostat

not a purine at all → no purine-pathway side effects, and ⚠ safe in the allopurinol-hypersensitive

## DRUGS

| DRUG               | WHAT IT TREATS                                             |
|--------------------|------------------------------------------------------------|
| <b>Allopurinol</b> | <b>first-line</b> ULT in chronic gout — cheap, effective   |
| <b>Febuxostat</b>  | second-line — allopurinol failure, intolerance, or allergy |

## ALLOPURINOL VS FEBUXOSTAT ★

|                         | ALLOPURINOL                                                                     | FEBUXOSTAT                                                    |
|-------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Type</b>             | purine analogue, competitive                                                    | non-purine, highly specific                                   |
| <b>Active form</b>      | oxypurinol                                                                      | parent drug                                                   |
| <b>Elimination</b>      | <b>renal</b>                                                                    | <b>hepatic</b>                                                |
| <b>Hypersensitivity</b> | ● <b>HLA-B*58:01</b> — SCAR/DRESS in Thai, Han Chinese, Korean, African descent | rare, but severe reactions have been reported                 |
| <b>Caution in</b>       | severe renal impairment, eGFR < 30                                              | <b>heart failure NYHA III–IV</b> , hepatic failure, eGFR < 30 |
| <b>Downside</b>         | 1/3 of patients never reach sUA < 6                                             | <b>expensive</b> · cardiovascular concern                     |

### USES

**Chronic gout / hyperuricemia** — first choice in anyone with normal renal function · ★ **"Start slow, go slow"** — allopurinol from ≤ 100 mg/day, titrate every 2–4 weeks up to 800–900 mg until **sUA < 6 mg/dL** (< 5 in severe or tophaceous gout)

### ADVERSE EFFECTS

- **Allopurinol hypersensitivity syndrome (AHS)** ★ — 2–8%: rash, eosinophilia, leukocytosis, fever, hepatitis, **progressive kidney failure**. Can escalate to Stevens–Johnson syndrome.
- GI disturbance
- **Febuxostat**: muscle pain, GI discomfort, skin rash, diarrhea, slight ↑ liver enzymes

### INTERACTIONS

- **Azathioprine, mercaptopurine** — cytotoxics normally cleared *by* XO, so blocking XO makes them accumulate to toxic levels. **Both drugs.**
- **Warfarin** — with allopurinol

### CAUTIONS

- ★ **Test for HLA-B\*58:01 before starting allopurinol** — the allele is common in Thai patients and it is the single biggest predictor of severe cutaneous reaction
- ⚠ Allopurinol: avoid at **eGFR < 30**; below that, dose by creatinine clearance
- ⚠ Febuxostat: heart failure (NYHA III–IV), hepatic failure, eGFR < 30

### ADMINISTRATION — "START SLOW, GO SLOW"

|                                                                                                                                                                                                                                                             | ALLOPURINOL                                                                       | FEBUXOSTAT                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>start</b>                                                                                                                                                                                                                                                | <b>100 mg/day</b> , titrate every <b>2–4 weeks</b> to target                      | <b>20–80 mg PO OD</b>                                  |
| <b>escalate to</b>                                                                                                                                                                                                                                          | up to <b>800–900 mg/day</b> ( <i>usual 300 mg/day</i> )                           | <b>120 mg/day</b> after 2 weeks if target missed       |
| <b>target</b>                                                                                                                                                                                                                                               | <b>sUA &lt; 6 mg/dL</b> — 65% reach it on febuxostat                              | same                                                   |
| <b>elimination</b>                                                                                                                                                                                                                                          | <b>renal</b> → adjust by CrCl, <b>avoid in renal failure</b>                      | <b>hepatic</b>                                         |
| <b>watch</b>                                                                                                                                                                                                                                                | ● <b>HLA-B*5801</b> — Korean, Chinese and <b>Thai</b> populations, SJS/DRESS risk | HF (NYHA III–IV), hepatic failure, <b>eGFR &lt; 30</b> |
| <ul style="list-style-type: none"> <li>● 💡 Two things patients get wrong about starting urate-lowering therapy: it can <b>trigger a flare</b> as urate mobilises, and it is <b>for life, not for the attack</b>. Both are why people abandon it.</li> </ul> |                                                                                   |                                                        |

# Uricosurics

## MECHANISM

Inhibit the **urate-reabsorbing transporters** — chiefly **URAT1**, plus GLUT9 and the organic anion transporters (OAT1/3/4) — in the **proximal tubule**. Urate that would have been reabsorbed stays in the filtrate.

→ **Downstream:** ↑ renal uric acid excretion → ↓ sUA · **but** ↑ **urate concentration in the collecting tubule** → **stone risk**. That single consequence drives every caution below.

## DRUGS

| DRUG                  | WHAT IT TREATS                                                                                              |
|-----------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Probenecid</b>     | chronic gout — inhibits URAT1, GLUT9, <b>OAT1 &amp; OAT3</b> ( <i>hence the interactions</i> )              |
| <b>Benzbromarone</b>  | chronic gout — URAT1 + GLUT9, <b>more potent than probenecid</b> ; usable in mild–moderate renal impairment |
| <b>Sulfinpyrazone</b> | withdrawn in many countries — severe ADRs                                                                   |
| <b>Lesinurad</b>      | URAT1 + OAT4; approved 2015, <b>withdrawn</b>                                                               |
| <b>Dotinurad</b>      | newest agent (2023)                                                                                         |

## USES

**Second-line / alternative ULT** — for patients who cannot tolerate XOIs · **Combination with an XO** when monotherapy fails to reach target (*all recent guidelines back this*) · Best candidates: **age < 60, normal renal function, no stone history**

## ADVERSE EFFECTS

- ★ **Uric acid stone formation** — the class ADR, straight from the mechanism
- **Benzbromarone:** ↑ liver enzymes, **deaths from hepatic failure** → withdrawn in some countries ●
- **Probenecid:** efficacy falls as GFR falls — near-useless in CKD

## INTERACTIONS

- **Probenecid inhibits OAT1/OAT3** → blocks tubular secretion of **NSAIDs, β-lactams, heparin**

## CAUTIONS

- **Do not use** with known renal calculi or **CKD stage > 3**
- ⚠️ Counsel **hydration > 1 L/day** — the entire defence against stones
- ⚠️ Monitor **urinary** uric acid (not just serum)
- ⚠️ Urine alkalinization is **conditionally recommended against** in patients on uricosurics

## ADMINISTRATION

- **Benzbromarone 50–200 mg daily** · **Probenecid 250–500 mg twice daily** · **Sulfinpyrazone 200–800 mg daily** (*withdrawn in many countries — severe ADRs*) · **Lesinurad** approved 2015, **withdrawn** · **Dotinurad** approved 2023
- ⚠️ Best suited to patients **< 60 years, normal renal function, no history of urinary stones**
- ● **Every drug in this class raises urate concentration in the collecting tubules** → **stone risk**. **Patients must be advised to drink plenty of fluids** ★
- 💡 "Drink plenty of water" is the whole counselling point here, and it needs a number attached or it gets ignored — tell them what a day's intake actually looks like.

# Uricase

## MECHANISM

★ **Humans lack uricase** — animals have it, we don't, which is exactly why we accumulate urate and they don't. Pegloticase is a **recombinant PEGylated uricase** that supplies the missing enzyme, converting uric acid → **allantoin**, which is far more water-soluble and simply washes out.

→ **Downstream:** dissolves deposited urate, including tophi — the only class that clears an existing crystal burden fast.

## DRUGS

| DRUG               | WHAT IT TREATS                           |
|--------------------|------------------------------------------|
| <b>Pegloticase</b> | refractory / severe tophaceous gout — IV |
| <b>Rasburicase</b> | withdrawn for gout                       |

## USES

**Refractory gout only** — severe tophaceous disease or damaging arthropathy where XOIs, uricosurics and everything else have failed to hit the sUA target

## ADVERSE EFFECTS

- **Infusion reactions** ★ — chest tightness, flushing, dyspnea
- **Immunogenicity** — anti-pegloticase antibodies develop, and as titres rise both **efficacy falls and infusion reactions get worse**. This is the drug's ceiling.
- Very high cost

# Treatment Strategy

## WHEN TO START URATE-LOWERING THERAPY

(Thai Rheumatism Association 2555 / EULAR)

| TRIGGER                           | DETAIL                                                                    |
|-----------------------------------|---------------------------------------------------------------------------|
| <b>Recurrent flares</b>           | more than 2–3 gout flares per year                                        |
| <b>Chronic arthritis or tophi</b> | subcutaneous tophi present                                                |
| <b>Radiographic change</b>        | bone/joint changes consistent with gout                                   |
| <b>Renal impairment</b>           | especially if continuous diuretic therapy is needed                       |
| <b>Stones</b>                     | urinary calculi or uric acid nephropathy                                  |
| <b>High risk</b>                  | first attack before age 40 · sUA > 8 mg/dL · cardiac or renal comorbidity |

## ★ WHY COLCHICINE OR AN NSAID GOES WITH THE ALLOPURINOL

Starting ULT **provokes flares** — dropping serum urate destabilizes existing deposits. So co-prescribe low-dose **colchicine 0.5–1 mg/day** (or a low-dose NSAID, or prednisolone < 10 mg/day) for the **first 3–6 months** of ULT. Watch the colchicine–statin interaction while doing it.

## COMORBID DRUGS THAT MAKE GOUT WORSE

- ⚠ **Loop diuretics & thiazides** ↓ urate excretion — in hypertensive gout patients use **losartan** or a **calcium channel blocker** instead
- ⚠ **Low-dose aspirin** ↓ urate excretion via URAT1
- For coexisting dyslipidemia, **statins or fenofibrate** are the preferred choices
- Other agents with incidental urate-lowering effect: losartan, oral vitamin C, fenofibrate

Principle behind this: [[pharm-drug-toxicity]] — Allopurinol is on the SJS list, and HLA-B\*58:01 appears as a genotype of interest in the pharmacogenomics lecture — see [[pharm-pharmacogenomics]] for the HLA half.

Principle behind this: [[pharm-pharmacogenomics]] — Allopurinol inhibits xanthine oxidase — which removes the thiopurine backup route, and is a named HLA-B\*58:01 association. Two separate genetic collisions on one drug.

Source: 2568\_For Lecture\_pharmacology1\_ Gout (1 h).pdf — อ.วิลาสินี, PYCH304 Pharmacology I, 33 slides. Structures omitted.

# Drugs in OA & RA

Two diseases that both end in a wrecked joint by completely different routes. **OA** is *degenerative* — cartilage wears out, so drugs only buy **symptom control**. **RA** is *autoimmune* — an immune drive you can actually switch off, so drugs aim at **remission and stopping erosion**. Everything below follows from that split.

**OA:** mechanical load + aging → chondrocyte stress → IL-1, TNF-α, IL-6 → catabolic > anabolic → cartilage loss + subchondral bone remodeling → pain

**RA:** autoimmunity → T-cell & B-cell activation → TNF-α, IL-6, IL-1 → synovitis / pannus → cartilage & bone erosion → deformity

## OA VS RA — PATHOLOGY & GOAL

|                       | OA (ข้อเสื่อม)                                                                       | RA (ข้ออักเสบรูมาตอยด์)                                                                                |
|-----------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Nature</b>         | chronic degenerative joint disorder                                                  | autoimmune disease, unknown etiology                                                                   |
| <b>Core lesion</b>    | cartilage destruction · subchondral bone remodeling · synovial membrane inflammation | immune attack on joint lining → pannus → cartilage & bone damage                                       |
| <b>Inflammation</b>   | none or mild, local                                                                  | <b>local and systemic</b>                                                                              |
| <b>Treatment goal</b> | ↓ pain & swelling · improve joint function, minimize disability · ↑ quality of life  | ↓ pain & inflammation · <b>prevent joint damage &amp; deformity</b> · remission / low disease activity |
| <b>Drug logic</b>     | analgesic ladder — ★ <b>symptomatic only</b>                                         | <b>DMARDs</b> modify disease; NSAIDs/steroids only cover symptoms                                      |

## OA PHARMACOLOGICAL LADDER (ACR / EULAR / OARSI)

| STEP                                                                                  | DRUGS                                                                      |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>First-line</b>                                                                     | <b>Acetaminophen</b> — mild to moderate pain                               |
| <b>Alternative first-line</b> ( <i>acetaminophen contraindicated or ineffective</i> ) | topical or oral <b>NSAIDs</b> · <b>IA corticosteroid</b> · <b>tramadol</b> |
| <b>Second-line</b>                                                                    | <b>tramadol</b> · <b>duloxetine</b> · <b>IA hyaluronic acid</b>            |
| <b>Adjunct / unproven</b>                                                             | glucosamine · chondroitin · DMOADs                                         |

# Acetaminophen

## MECHANISM

Central **analgesic and antipyretic** — ★ **no anti-inflammatory activity**. That's why it works on OA pain but does nothing for the synovitis of RA.

→ **Metabolism**: mainly liver — **CYP2E1** (main), CYP1A2, CYP3A4 → the toxic intermediate **NAPQI**, normally detoxified by glutathione (GSH) → **Overdose or ↓ GSH**: NAPQI accumulates → **hepatotoxicity**. Half-life 1.5–2.5 h stretches to 4–8 h.

## DRUGS

| DRUG                               | WHAT IT TREATS                                               |
|------------------------------------|--------------------------------------------------------------|
| <b>Acetaminophen (paracetamol)</b> | first-line mild–moderate OA pain · good risk–benefit profile |

## USES

**Mild to moderate OA pain** — first-line · 500–1000 mg, 3–4 times/day (*single dose ceiling 1000 mg*)

## ADVERSE EFFECTS

- **Hepatotoxicity** — dose-dependent, the whole safety story of this drug

## INTERACTIONS

- **Warfarin** — prolonged 4000 mg/day → **elevated INR**
- ● **Chronic alcohol misuse** — induces CYP2E1 *and* depletes GSH *and* impairs NAPQI detox. Triple hit.

## CAUTIONS & DOSING LIMITS

- ★ **Max 4000 mg/day** in adults (*some guidelines say 3000 mg/day to cut liver risk*) · children 50–75 mg/kg/day
- ⚠ **Elderly or liver disease: 2000–3000 mg/day**
- ● **Contraindicated** in active liver disease / severe hepatic impairment; mild impairment — use with caution
- ⚠ Severe renal impairment (CrCl ≤30 mL/min) — extend the dosing interval, reduce total daily dose

## ADMINISTRATION

- **Dose**: 500–1000 mg, **3–4 times a day** — **maximum single dose 1000 mg**
- **Maximum daily**: adult **4000 mg** (*some guidelines say 3000 mg to cut hepatotoxicity risk*) · children **50–75 mg/kg/day** · ● **elderly or liver disease — 2000–3000 mg/day**
- **PK**: oral bioavailability **88%** · peak **1.5 h** ·  $t_{1/2}$  **1.5–2.5 h**, stretching to **4–8 h** in overdose or impaired metabolism · hepatic **CYP2E1** (main), 1A2, 3A4 · ~90% eliminated within 24 h
- 💡 The number that matters at a counter is the **combined** daily total. Patients stack plain paracetamol on top of cold-and-flu sachets without counting it — ask what else they're taking.

# NSAIDs (oral)

## MECHANISM

Inhibit **cyclo-oxygenase** → ↓ prostaglandins → analgesia **plus** anti-inflammatory effect (the thing acetaminophen can't do).

→ **COX-1 inhibition** is the gastric mucosa problem · **COX-2 selectivity** spares the stomach but shifts risk to the **cardiovascular** side → **In RA**: ★ relieves pain, swelling and stiffness but **does NOT prevent joint damage** — never a substitute for a DMARD

## DRUGS

| DRUG                                         | WHAT IT TREATS                                                     |
|----------------------------------------------|--------------------------------------------------------------------|
| <b>Ibuprofen</b> ( <i>non-selective</i> )    | moderate–severe OA pain · RA symptoms                              |
| <b>Naproxen</b> ( <i>non-selective</i> )     | same — ★ <b>lower CV risk</b> , the safer pick in cardiac patients |
| <b>Diclofenac</b> ( <i>non-selective</i> )   | moderate–severe OA pain                                            |
| <b>Meloxicam</b> ( <i>non-selective</i> )    | RA symptoms                                                        |
| <b>Celecoxib</b> ( <i>COX-2 selective</i> )  | OA / RA when GI risk is the concern                                |
| <b>Etoricoxib</b> ( <i>COX-2 selective</i> ) | OA / RA when GI risk is the concern                                |

## NON-SELECTIVE VS COX-2 SELECTIVE

|                           | NON-SELECTIVE                                                  | COX-2 SELECTIVE                                     |
|---------------------------|----------------------------------------------------------------|-----------------------------------------------------|
| <b>GI</b>                 | ● dyspepsia, ulcer, bleeding                                   | <b>less GI toxicity</b>                             |
| <b>CV</b>                 | lower ( <i>naproxen lowest</i> )                               | ● <b>increased CV events</b>                        |
| <b>Contraindicated in</b> | active peptic ulcer · bleeding disorder · <b>heart failure</b> | active peptic ulcer · <b>cardiovascular disease</b> |

## USES

**Moderate to severe OA pain** — first-line at that severity · **RA** — symptomatic anti-inflammatory cover only

## ADVERSE EFFECTS

- **GI: dyspepsia** → **gastric ulceration & bleeding** (*COX-1 driven*)
- Rashes — urticaria, up to **toxic epidermal necrolysis**
- **Acute renal failure** · ↑ blood pressure · worsening heart failure in chronic use
- ↑ risk of cardiovascular events
- Hepatotoxicity — rare, but check LFTs on prolonged therapy

## INTERACTIONS

- ● **Anticoagulants (warfarin)** → ↑ bleeding risk — **monitor INR**
- **Antihypertensives** — ACEIs, ARBs, diuretics: NSAIDs **reduce their efficacy**

## CAUTIONS

- ★ **High GI risk** (elderly, ulcer history) on a non-selective NSAID → **add a PPI**
- ⚠ Monitor renal function in long-term users, especially with pre-existing renal disease
- ⚠ Monitor blood pressure on COX-2 selective agents



## ADMINISTRATION — DOSE AND CEILING

| DRUG                      | USUAL DOSE                               | MAXIMUM/DAY    |
|---------------------------|------------------------------------------|----------------|
| <b>Ibuprofen</b>          | 200–400 mg q 4–6 h                       | <b>3200 mg</b> |
| <b>Naproxen</b>           | 250–500 mg q 12 h                        | <b>1500 mg</b> |
| <b>Diclofenac</b>         | 50 mg 2–3 times daily                    | <b>150 mg</b>  |
| <b>Celecoxib</b> (COX-2)  | 200 mg OD ( <i>may increase to BID</i> ) | <b>400 mg</b>  |
| <b>Etoricoxib</b> (COX-2) | 30 or 60 mg OD                           | <b>120 mg</b>  |

- 💡 Give it **with or just after food**, at the **lowest dose for the shortest time** — and screen for the two things that turn a routine NSAID into a problem: ulcer history and cardiovascular risk.

# Topical Agents

## MECHANISM

- **Topical NSAID** — local COX inhibition at the joint, minimal systemic exposure
- **Capsaicin** — with **repeated use** it **depletes substance P**, so the nociceptive signal fades  
→ **Downstream**: effective only for **localized, superficial** pain · capsaicin needs **days to weeks** before relief is noticeable

## DRUGS

| DRUG                         | WHAT IT TREATS                                     |
|------------------------------|----------------------------------------------------|
| <b>Diclofenac gel (1–3%)</b> | ★ first-line mild–moderate <b>knee and hand OA</b> |
| <b>Ketoprofen gel (2.5%)</b> | same                                               |
| <b>Topical capsaicin</b>     | localized pain — elbow, knee, ankle, wrist, feet   |

## USES

**Mild to moderate knee & hand OA** — first-line, ★ especially in **older adults and those with comorbidities** who can't take the systemic hit · apply to the affected area **3–4 times daily**

## ADVERSE EFFECTS

- Topical NSAID: dry skin, local irritation, rash, dermatitis at the application site
- Capsaicin: **initial burning sensation** (*mild, fades with continued use*) · redness, itching, swelling

## CAUTIONS

- ⚠ **Never apply capsaicin to broken or irritated skin**

# Tramadol

## MECHANISM

**Dual mechanism** — a **weak opioid agonist** *plus* a **serotonin & norepinephrine reuptake inhibitor**.

→ **CYP2D6** converts it to **O-desmethyl-tramadol (M1)**, the *active* metabolite: ★ **poor metabolizer** → less M1 → **inadequate analgesia** · **ultrarapid metabolizer** → more M1 → **toxicity**

## DRUGS

| DRUG            | WHAT IT TREATS                                                                              |
|-----------------|---------------------------------------------------------------------------------------------|
| <b>Tramadol</b> | moderate–severe OA pain — second-line / adjunct when acetaminophen and NSAIDs aren't enough |

## USES

**Moderate to severe OA pain** — 50 mg every 4–6 h, **max 400 mg/day** · alternative first-line when acetaminophen is contraindicated

## ADVERSE EFFECTS

- GI: nausea, vomiting, constipation · CNS: dizziness, drowsiness
- ● **Serotonin syndrome** with other serotonergic drugs — agitation, confusion, fever, shivering, sweating, muscle rigidity
- **Respiratory depression** — high dose or with other CNS depressants
- ★ **Dependence & abuse** — abrupt stop → withdrawal (anxiety, insomnia, sweating, tremor). **Taper.**

## INTERACTIONS

- **CYP2D6 inhibitors** (*amiodarone, fluoxetine, paroxetine, duloxetine, bupropion*) → less M1 → ★ **less pain relief**, not more
- **CYP3A4 inhibitors** (*ketoconazole, erythromycin, clarithromycin, ritonavir*) → ↑ tramadol → ↑ side effects
- **CYP3A4 inducers** (*rifampin, phenytoin, carbamazepine*) → ↓ tramadol → ↓ efficacy

## CAUTIONS

- ⚠ **Elderly**: start low, **max 300 mg/day**
- ⚠ **Renal impairment** (CrCl <30 mL/min) — reduce dose · **hepatic impairment** — adjust

## ADMINISTRATION

- **Dose 50 mg every 4–6 hours** · **maximum 400 mg/day**
- **PK**: oral bioavailability **75%** · protein binding **20%** · metabolised by **CYP2D6, CYP3A4, CYP2B6** to **M1 (O-desmethyl-tramadol)**, the active metabolite · excreted renally (*30% unchanged, 60% metabolites*)
- ★ **CYP2D6 phenotype decides the outcome**: poor metabolisers make less M1 → **inadequate pain relief**; ultrarapid metabolisers make more → **higher toxicity risk**
- 💡 Two questions before handing it over: anything else serotonergic (*SSRI/SNRI, triptan*), and any other tramadol-containing product — the combination products are how people double-dose it.

# Duloxetine

## MECHANISM

**SNRI** — serotonin-norepinephrine reuptake inhibitor acting on **central pain modulation**, not on the joint itself.

→ **Onset:** relief begins in **1–2 weeks**, full effect takes **4–6 weeks** — set expectations or the patient quits early

## DRUGS

| DRUG              | WHAT IT TREATS                                                        |
|-------------------|-----------------------------------------------------------------------|
| <b>Duloxetine</b> | chronic musculoskeletal pain — <b>knee OA</b> · chronic low back pain |

## USES

**Chronic OA pain not controlled** by acetaminophen or NSAIDs · patients who **can't tolerate NSAIDs or tramadol** (GI, cardiovascular, or addiction concerns) · 30 mg OD → 60 mg/day, **max 60 mg/day**

## ADVERSE EFFECTS

- Common: headache, **nausea** (*settles over days to weeks*), drowsiness, fatigue, dry mouth, constipation, diarrhea
-  Serious: **suicidality · serotonin syndrome · hepatotoxicity**

## INTERACTIONS

- **CYP2D6 inhibitors** → ↑ duloxetine levels (*it's a CYP2D6 substrate*)
-  **MAO inhibitors** → hypertensive crisis or serotonin syndrome
- **Alcohol** → ↑ liver damage risk — advise limiting or avoiding

## ADMINISTRATION

- **Initial 30 mg once daily**; after **1–2 weeks** may increase to **60 mg/day. Maximum 60 mg/day**
-  It isn't a rescue analgesic — the benefit takes weeks, so tell them not to judge it on day three, and not to stop it abruptly.

# Intra-articular Agents

## MECHANISM

- **IA corticosteroid** — local anti-inflammatory + analgesic, delivered straight into the inflamed joint
  - **IA hyaluronic acid** — a **lubricant and shock absorber**; improves the **quality & quantity of synovial fluid**
- **Corticosteroid onset:** relief in **24–48 h**, maximal at **1–2 weeks** → **Hyaluronic acid:** a **single injection** can give **6 months to 1 year** of relief

## DRUGS

| DRUG                                                            | WHAT IT TREATS                                                                                                                        |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>IA corticosteroid</b><br>(triamcinolone, methylprednisolone) | significant local inflammation & pain in one joint — knee, hip, shoulder · a <b>local RA flare</b> without changing the whole regimen |
| <b>IA hyaluronic acid</b>                                       | <b>knee OA</b> — second-line, when pain persists                                                                                      |

## USES

**Local inflammation in a specific joint** · **alternative to oral medication** · **short-term relief** that lets the patient actually do physical therapy

## ADVERSE EFFECTS

- Local: injection-site pain or swelling · **joint infection**
-  **Cartilage damage with repetitive injection**  → **maximum 3–4 injections per year in a single joint**
- Systemic corticosteroid long-term → ↓ bone density, **osteoporosis**

## POST-INJECTION CARE

-  Rest the joint for the **first 24 hours** · avoid strenuous activity
-  **Monitor for signs of infection**

# Glucosamine, Chondroitin & DMOADs

## MECHANISM

- **Glucosamine** — precursor for **glycosaminoglycan** synthesis, a major cartilage component → promotes cartilage repair
  - **Chondroitin** — itself a major cartilage component; ↓ **activity of the enzymes that destroy cartilage and collagen**, may stimulate repair
  - **DMOADs** (*Disease-Modifying Osteoarthritis Drugs*) — the aspiration: **block pro-inflammatory cytokines and matrix-degrading enzymes** to change structure, not just symptoms
- **Reality check:** ★ **effectiveness is mixed** for both supplements, and glucosamine has **no FDA-approved indication** in OA

## DRUGS

| DRUG               | WHAT IT TREATS                                                            |
|--------------------|---------------------------------------------------------------------------|
| <b>Glucosamine</b> | mild–moderate OA — 500 mg TID (1,500 mg/day), typically <b>3–6 months</b> |
| <b>Chondroitin</b> | mild–moderate OA — 200–400 mg BID–TID (800–1200 mg/day)                   |
| <b>DMOADs</b>      | structural modification in OA — the class OA still doesn't really have    |

## ADVERSE EFFECTS

- **Glucosamine:** epigastric pain, heartburn, diarrhea, nausea
- **Chondroitin:** stomach upset, headache, increased intestinal gas, diarrhea

## ADMINISTRATION

- **Glucosamine** — **500 mg TID or 1,500 mg/day**, typically used for **3–6 months** in mild-to-moderate OA
- 💡 Set the expectation before they commit: **no FDA-approved indication and mixed evidence**. Six months of a supplement is a real cost, and they should know what they're buying.

# Glucocorticoids (RA)

## MECHANISM

Broad suppression of inflammation — inhibits inflammation **and** slows disease progression, with a **short onset of action**.

→ **The role:** ★ **bridge therapy** — quick relief of acute inflammation **while waiting for DMARDs to kick in** (csDMARDs take weeks to months). **Not a long-term drug**.

## DRUGS

| DRUG                                                                 | WHAT IT TREATS                                                                       |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Prednisolone</b> ( <i>oral</i> )                                  | acute RA inflammation — 5–10 mg/day, up to 60 mg/day if needed                       |
| <b>Triamcinolone / methylprednisolone</b> ( <i>intra-articular</i> ) | a <b>local flare in one joint</b> — controls it without touching the overall regimen |

## USES

**Quick relief of acute RA inflammation · bridging** until DMARDs take effect · use at the **lowest dose and as infrequently as possible**

## ADVERSE EFFECTS (LONG-TERM USE)

- **Osteoporosis**
- Weight gain · hypertension · **hyperglycemia**
- **Cushing's syndrome**

## CAUTIONS

- ★ **Not suitable for long-term use** — the whole point is to get off it
- ⚠ Patients on low-dose prednisone, **with or without** osteoporosis risk factors, need **bone densitometry (DEXA)**
- ⚠ **Bisphosphonates** (alendronate, risedronate, ibandronate) + adequate **calcium & vitamin D** to prevent/treat steroid osteoporosis

# csDMARDs — Methotrexate

## MECHANISM

Two pathways, one drug.

1. **Folate pathway** — inhibits **DHFR · MTHFR · TYMS · ATIC** → ↓ nucleotide and DNA synthesis → blocks **rapidly dividing cells**, including immune cells
2. **Adenosine signaling** — **adenosine accumulates** → binds adenosine receptors on immune cells  
→ **Net effect**: suppresses **T-cell & B-cell activation** · inhibits **neutrophil adhesion and recruitment** · ↓ cytokine production (**TNF- $\alpha$ , IL-6, IL-8**)

## DRUGS

| DRUG                      | WHAT IT TREATS                                                                         |
|---------------------------|----------------------------------------------------------------------------------------|
| <b>Methotrexate (MTX)</b> | ★ <b>first-line DMARD</b> for RA — rapid onset, high efficacy, cheap. The anchor drug. |

## USES

RA, first-line · 7.5–25 mg **ONCE A WEEK** ● · onset of action **4–6 weeks**

## ADVERSE EFFECTS

- **Mucosal ulcers**, nausea, vomiting, loss of appetite
- Hair thinning / hair loss
- **Bone marrow suppression** (*the reason for folate supplementation*)
- **Hepatotoxicity** — ↑ LFTs, risk of **fibrosis / cirrhosis** with long-term use

## INTERACTIONS

- **NSAIDs, salicylates, trimethoprim, PPIs, warfarin** → ↑ **MTX levels** ★ (*and RA patients are on NSAIDs constantly*)
- **Chloramphenicol, tetracycline** → ↓ intestinal absorption → ↓ MTX levels
- ● **Alcohol, leflunomide** → ↑ liver toxicity

## CAUTIONS & MONITORING

- ● **WEEKLY dosing, not daily** — daily dosing is the classic lethal error
- ● **Teratogenic — contraindicated in pregnancy**
- ★ **Folate 5–7 mg weekly** for all patients, to prevent bone marrow suppression
- ⚠ **Baseline**: CBC, LFTs, renal function, **chest X-ray** (*lung disease risk*) · **then CBC/LFT/renal every 1–3 months**

## ADMINISTRATION

- ● **7.5–25 mg ONCE A WEEK** — weekly, never daily ★ · **onset 4–6 weeks**
- **PK**: oral bioavailability **60%** · protein binding **50%** · mainly hepatic metabolism · mainly renal excretion
- **Folate 5–7 mg weekly** should be considered in all patients, to prevent bone-marrow suppression
- ⚠ **Hepatotoxicity** — ↑ LFTs, fibrosis/cirrhosis with long-term use · ● **teratogenic, contraindicated in pregnancy**
- 💡 This is the most dangerous dispensing error on the page: **weekly mistaken for daily is fatal**. Say "once a week", name the day, and have the patient repeat it back to you.



# csDMARDs — Sulfasalazine · Leflunomide · Hydroxychloroquine

## MECHANISM

- **Sulfasalazine** — exact mechanism unclear. Proposed: ↓ T-cell proliferation & cytokine release · inhibition of **NF-κB** and prostaglandin synthesis · antioxidant (scavenges ROS)
- **Leflunomide** — its active metabolite **teriflunomide** inhibits **DHODH**, an enzyme of **pyrimidine synthesis** → ↓ proliferation of **activated T-cells**
- **Hydroxychloroquine** — inhibits **toll-like receptor (TLR) signaling** and reduces **antigen presentation**

## DRUGS

| DRUG                      | WHAT IT TREATS                                                                                                                            |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sulfasalazine</b>      | RA — <b>second choice</b> ; ★ preferred over MTX in <b>liver disease, hepatitis B or C</b> , and <b>safe in pregnancy &amp; lactation</b> |
| <b>Leflunomide</b>        | RA — efficacy <b>similar to MTX</b> , the alternative when MTX fails or isn't tolerated                                                   |
| <b>Hydroxychloroquine</b> | RA — cheap, <b>moderate efficacy</b> ; acceptable in pregnancy ( <i>category C</i> )                                                      |

## ONSET & EFFICACY

| DRUG                      | ONSET      | EFFICACY VS MTX    |
|---------------------------|------------|--------------------|
| <b>Methotrexate</b>       | 4–6 weeks  | reference standard |
| <b>Leflunomide</b>        | 4–6 weeks  | <b>similar</b>     |
| <b>Sulfasalazine</b>      | 1–3 months | <b>lower</b>       |
| <b>Hydroxychloroquine</b> | 2–6 months | moderate           |

## ADVERSE EFFECTS

- **Sulfasalazine**: nausea, vomiting, loss of appetite, dyspepsia, rash, headache, ★ **orange-coloured urine/skin** (*harmless, but warn them*). Low myelosuppression & hepatotoxicity.
- **Leflunomide**: nausea, diarrhea, **hypertension**, alopecia, ↑ liver enzymes, abdominal & back pain, low bone marrow suppression
- **Hydroxychloroquine**: nausea, diarrhea, alopecia, ● **visual disturbance** → **retinopathy** if severe

## CAUTIONS & MONITORING

- ● **Sulfasalazine contraindicated: G6PD deficiency** (*hemolysis*) and **sulfa allergy** — screen G6PD first
- ● **Leflunomide is teratogenic** — **contraindicated in pregnancy**
- ⚠ **Sulfasalazine monitoring**: CBC, liver, renal at baseline then every 1–3 months
- ⚠ **Leflunomide monitoring**: LFTs, CBC, **blood pressure**

## ADMINISTRATION — DOSE AND HOW LONG BEFORE IT WORKS

| DRUG                      | DOSE                                                                             | ONSET             |
|---------------------------|----------------------------------------------------------------------------------|-------------------|
| <b>Sulfasalazine</b>      | 500 mg OD, or 1–3 g/day                                                          | <b>1–3 months</b> |
| <b>Leflunomide</b>        | <b>loading 100 mg OD × 3 days</b> → maintenance <b>10–20 mg/day</b> by tolerance | 4–6 weeks         |
| <b>Hydroxychloroquine</b> | 200–400 mg OD or BID                                                             | <b>2–6 months</b> |

- **Hydroxychloroquine is pregnancy category C** — use in pregnancy is considered acceptable
- 💡 All three are slow, one of them by half a year. Say the onset out loud when dispensing, or the patient quits at week three convinced it does nothing.

# bDMARDs — TNF Inhibitors

## MECHANISM

**TNF- $\alpha$**  is a pro-inflammatory cytokine produced by **macrophages and lymphocytes** and a key driver of systemic inflammation in RA. These agents block its activity.

→ **Position:** the **first biologics approved for RA**; biologics are used **when csDMARDs fail** → **Onset:** rapid — improvement within **2–4 weeks** → ★ **Naming rule:** **-ximab** = chimeric mAb · **-zumab** = humanized mAb · **-mumab** = fully human mAb

## DRUGS

| DRUG                | WHAT IT TREATS           |
|---------------------|--------------------------|
| <b>Etanercept</b>   | RA after csDMARD failure |
| <b>Adalimumab</b>   | RA after csDMARD failure |
| <b>Infliximab</b>   | RA after csDMARD failure |
| <b>Golimumab</b>    | RA after csDMARD failure |
| <b>Certolizumab</b> | RA after csDMARD failure |

## ADVERSE EFFECTS

- **Increased risk of infection — tuberculosis and fungal infections** ★
- **Infusion reactions**
- **Heart failure exacerbation**
- Rare: **demyelinating disease**

## CAUTIONS

- **TB screening is MANDATORY before starting TNF therapy**
- Screen for **hepatitis B** before starting

# bDMARDs — Interleukin Inhibitors

## MECHANISM

- **Tocilizumab (anti-IL-6)** — binds **both soluble and membrane-bound IL-6 receptors**, shutting down IL-6-mediated signaling
- **Anakinra (anti-IL-1)** — a **recombinant human IL-1 receptor antagonist (hu rIL-1Ra)**; blocks **IL-1 binding to its receptor**, preventing its pro-inflammatory actions

→ **Onset:** tocilizumab **4–8 weeks** · anakinra **2–4 weeks**

## DRUGS

| DRUG                                                | WHAT IT TREATS                                  |
|-----------------------------------------------------|-------------------------------------------------|
| <b>Tocilizumab</b> ( <i>IL-6 receptor</i> )         | RA after csDMARD failure                        |
| <b>Anakinra</b> ( <i>IL-1 receptor antagonist</i> ) | RA after csDMARD failure — 100 mg SC once daily |

## ADVERSE EFFECTS

- **Both:**  increased risk of **infection** · **neutropenia** (*check blood counts regularly*)
- **Tocilizumab also:** thrombocytopenia · ↑ liver transaminases ·  **hyperlipidemia / lipid abnormalities**
- **Anakinra also: injection-site reactions** — erythema, itching, discomfort (*resolve over 1–2 months*)

## ADMINISTRATION

- **Tocilizumab (IL-6)** — **IV:** 4 mg/kg q 4 weeks, then **8 mg/kg** q 4 weeks by response · **SC: 162 mg** every other week (*< 100 kg*) or **weekly** (*≥ 100 kg*). **Onset 4–8 weeks**
- **Anakinra (IL-1)** — **100 mg SC once daily**. Onset 2–4 weeks
-  Injection-site reactions with anakinra are near-universal early on and settle over 1–2 months — warn first, or the patient stops a working drug in week two.

# bDMARDs — B-cell & T-cell Targeted

## MECHANISM

- **Rituximab** — targets **CD20 on B-cells** → **B-cell depletion** → reduced autoimmune activity
  - **Abatacept** — binds **CD80 & CD86** on antigen-presenting cells, blocking their interaction with **CD28** on T-cells → ★ **removes the co-stimulatory signal**, so T-cells never fully activate
- **Onset:** abatacept responses typically within **3 months**

## DRUGS

| DRUG                                                      | WHAT IT TREATS                                                  |
|-----------------------------------------------------------|-----------------------------------------------------------------|
| <b>Rituximab</b> ( <i>anti-CD20</i> )                     | RA — 1000 mg IV ×2, two weeks apart, repeated every 6–12 months |
| <b>Abatacept</b> ( <i>T-cell co-stimulation blocker</i> ) | RA — SC weekly or IV weight-based loading                       |

## ADVERSE EFFECTS

- **Rituximab:** ● infections, especially **viral reactivation (hepatitis B)** · **infusion reaction** — fever, chills, hypotension
- **Abatacept:** increased infection risk · infusion reaction (*more with IV*)

## CAUTIONS

- ★ **Rituximab:** complete **all immunizations before** starting — ● **avoid live virus vaccines**
- ⚠ **Abatacept:** respiratory infection (**pneumonia**) more common in **COPD** patients · **TB screening recommended**

## ADMINISTRATION

- **Rituximab (anti-CD20)** — **1000 mg IV, two infusions two weeks apart**
- **Abatacept (T-cell co-stimulation)** — dosed by **weight band**: < 60 kg → 500 mg · 60–100 kg → 750 mg · > 100 kg → 1000 mg

# tsDMARDs — JAK Inhibitors

## MECHANISM

Block **Janus kinase (JAK)** enzymes, which transduce the signals of the cytokine receptors driving **immune cell activation and inflammation** — one drug interrupting many cytokines at once.

→ ★ **Oral dosage** — the only *targeted* option that isn't an injection, which is most of its appeal

## DRUGS

| DRUG                | WHAT IT TREATS             |
|---------------------|----------------------------|
| <b>Tofacitinib</b>  | RA — oral targeted therapy |
| <b>Baricitinib</b>  | RA — oral targeted therapy |
| <b>Upadacitinib</b> | RA — oral targeted therapy |

## ADVERSE EFFECTS

- **Serious infections** — especially **herpes zoster** ★ (*the signature JAK-inhibitor infection*)
- **Thrombosis** — deep vein thrombosis, pulmonary embolism
- Elevated liver enzymes
- Lipid abnormalities

# Comparison: RA vs OA

## CLINICAL FEATURES

| CHARACTERISTIC                | RA                                | OA                                 |
|-------------------------------|-----------------------------------|------------------------------------|
| <b>Speed of onset</b>         | rapid — weeks to months           | slow — years                       |
| <b>Women : men</b>            | 3 : 1                             | 1–1.5 : 1                          |
| <b>Usual age of onset</b>     | 30–50 years                       | over 50                            |
| <b>Joints affected</b>        | <b>small joints</b>               | <b>large weight-bearing joints</b> |
| <b>Joint symptoms</b>         | pain, swelling, warmth, stiffness | pain, <b>bony enlargement</b>      |
| <b>Inflammation</b>           | local <b>and systemic</b>         | none or mild                       |
| <b>Morning stiffness</b>      | ★ <b>60 min or longer</b>         | <b>less than 30 min</b>            |
| <b>Joint pattern</b>          | <b>symmetric</b>                  | symmetric or asymmetric            |
| <b>ESR</b>                    | elevated                          | normal                             |
| <b>Synovial fluid</b>         | leukocytosis, cloudy              | mild leukocyte                     |
| <b>Systemic manifestation</b> | yes                               | no                                 |

## THE TWO DRUG MENUS

| OA — PHARMACOLOGICAL                           | RA — PHARMACOLOGICAL                                                                             |
|------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Acetaminophen · NSAIDs · Tramadol · Duloxetine | NSAIDs · Corticosteroids                                                                         |
| IA corticosteroid · IA hyaluronic acid         | <b>csDMARDs</b> — MTX, sulfasalazine, leflunomide, hydroxychloroquine                            |
| Glucosamine · Chondroitin · DMOADs             | <b>bDMARDs</b> — TNF, IL, B-cell, T-cell agents · <b>tsDMARDs</b> — JAK inhibitors               |
| ★ all <b>symptomatic</b>                       | ★ DMARDs are <b>disease-modifying</b> , with a <b>slow onset</b> — NSAIDs/steroids cover the gap |

Source: 20250925 PCo1\_Drug in OA\_RA\_HandoutV2.pdf — อ.ภชรัตน์, PYCH304 Pharmacology I, 73 slides. Structures omitted.

# Histamine, Antihistamines & Serotonin Agonists

**Histamine** ( $\beta$ -aminoethylimidazole) is a biologically active amine — an **autacoid**. It drives **allergic inflammation**, immunomodulation, **gastric acid secretion**, and modulates neurotransmitter release. Antihistamine activity was first shown in **1937**; **>40 H1 antihistamines** are marketed worldwide.

**Histamine → H1 receptor → eNOS/NO vasodilation · ↑ capillary permeability · sensory-nerve itch → allergic symptoms**

## HISTAMINE RECEPTOR SUBTYPES

| SUBTYPE   | WHERE IT LIVES                                               | SIGNALLING                   | BLOCKERS (ALL INVERSE AGONISTS)             |
|-----------|--------------------------------------------------------------|------------------------------|---------------------------------------------|
| <b>H1</b> | smooth muscle, endothelium, brain                            | Gq → ↑ IP <sub>3</sub> , DAG | mepyramine, triprolidine, cetirizine        |
| <b>H2</b> | gastric mucosa, cardiac muscle, mast cells, brain            | Gs → ↑ cAMP                  | cimetidine, ranitidine                      |
| <b>H3</b> | presynaptic auto- & heteroreceptors: brain, myenteric plexus | Gi → ↓ cAMP                  | tiprolisant, pitolisant, <b>betahistine</b> |
| <b>H4</b> | eosinophils, neutrophils, CD4 T cells                        | Gi → ↓ cAMP                  | thioperamide                                |

## SEROTONIN RECEPTORS THAT MATTER IN MIGRAINE

| RECEPTOR                                         | WHERE                                                           | WHAT AN AGONIST DOES THERE                                            |
|--------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>5-HT<sub>1B</sub></b>                         | dural blood vessels ( <i>dilated in migraine</i> )              | <b>constricts</b> them — also the source of the cardiovascular ADRs ★ |
| <b>5-HT<sub>1D</sub> / 5-HT<sub>1F</sub></b>     | trigeminal endings & trigeminocervical complex (TCC), brainstem | <b>blocks pain transmission</b> , ↓ CGRP release                      |
| <b>5-HT<sub>1F</sub></b><br>( <i>selective</i> ) | same central sites, no vascular effect                          | the <b>ditans</b> (lasmiditan) — analgesia without vasoconstriction   |
| <b>α<sub>1</sub>-adrenoceptor</b>                | systemic vessels                                                | ergots hit this too — extra, unwanted vasoconstriction                |



# H1 Antihistamines — Mechanism & Effects

## MECHANISM

**Reversible competitive inhibitors at the H1 receptor** — but reclassified as **inverse agonists** ★: the H1 receptor has **constitutive activity** and triggers downstream events even with no ligand bound. An inverse agonist has preferential affinity for the **inactive** conformation and stabilises it there.

→ **Downstream**: blocks eNOS-driven vasodilation · blocks the plasma leak at post-capillary venules · blocks sensory-nerve itch — *it does not remove histamine, it silences the receptor*

## WHAT ACTUALLY HAPPENS, TISSUE BY TISSUE

| SITE                      | HISTAMINE VIA H1                                                                                            | + ANTIHISTAMINE                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Respiratory smooth muscle | bronchoconstriction                                                                                         | ⚠ <b>barely reduced</b> — leukotrienes & PAF also drive it                            |
| Vascular endothelium      | stimulates eNOS → ↑ NO → vasodilation ( <i>nasal congestion, red eye, hypotension, flushing, headache</i> ) | ↓ nasal congestion, ↓ red eye, ↓ BP swings                                            |
| Post-capillary venules    | endothelial cells contract & separate → protein + fluid extravasation ( <i>edema, weal, rhinorrhea</i> )    | ★ <b>strongly blocked</b> → ↓ edema, weal, nasal discharge                            |
| Exocrine glands           | ↑ parasympathetic activity → rhinorrhea                                                                     | 1st gen dries it further ( <i>anticholinergic</i> ); 2nd gen has no muscarinic effect |
| Sensory nerve endings     | itch + axon reflex → indirect vasodilation → flare                                                          | ↓ itch ( <i>skin, eyes, ears, nose, palate, throat</i> ), ↓ flare                     |

*Immediate hypersensitivity: edema and itch are well suppressed; hypotension and bronchoconstriction are not. Local anaesthetic effect exists (promethazine most active) but needs higher concentrations than H1 blockade.*

## 1ST VS 2ND GENERATION — THE WHOLE SPLIT

|                  | 1ST GEN (SEDATING)                 | 2ND GEN (NON-SEDATING)                                                                      |
|------------------|------------------------------------|---------------------------------------------------------------------------------------------|
| BBB              | <b>lipophilic</b> → crosses freely | <b>lipophobic</b> , ionised at physiological pH, <b>P-gp substrate</b> → poor penetration ★ |
| Sedation         | all of them                        | minimal — <i>cetirizine is the exception</i>                                                |
| Muscarinic block | many, strongly                     | none ( <i>cetirizine minimal</i> )                                                          |
| Duration         | <b>4–6 h</b>                       | <b>12–24 h</b>                                                                              |
| Metabolism       | CYPs                               | loratadine → desloratadine (CYP3A4/2D6); cetirizine, fexofenadine, bilastine ≈ none         |
| Excretion        | urine, as metabolites              | mostly <b>unchanged</b> — cetirizine & loratadine in urine, <b>fexofenadine in feces</b>    |

*Shared PK: well absorbed from the GI tract · onset 15–30 min · Tmax 1–3 h · ★ skin concentration peaks fast and persists ≥36 h even when plasma is nearly zero.*

# 1st-Generation H1 Antihistamines

## MECHANISM

H1 inverse agonist **plus off-target promiscuity** — this is the class's whole personality. H1 receptors share ~**45% homology with muscarinic receptors**, so many 1st-gen agents block M receptors too; the phenothiazines add **α-adrenergic blockade**.

→ **Downstream:** BBB crossing → **sedation** · M block → dry mouth, blurred vision, retention · α block → **postural hypotension**

## DRUGS

| DRUG                          | WHAT IT TREATS                                                                                                                    |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Chlorpheniramine (CPM)</b> | allergic rhinitis · urticaria — an alkylamine, <b>least sedating</b> of the class                                                 |
| <b>Diphenhydramine</b>        | allergy · motion sickness · <b>antipsychotic-induced EPS</b> — strongly sedating, potent anticholinergic                          |
| <b>Dimenhydrinate</b>         | motion sickness, vertigo ( <i>diphenhydramine + 8-chlorotheophylline</i> )                                                        |
| <b>Promethazine</b>           | allergy · motion sickness — <b>strongest anticholinergic</b> , most active local anaesthetic, worst postural hypotension          |
| <b>Hydroxyzine</b>            | urticaria & itchy skin disorders — enters skin readily, sustained high skin concentration                                         |
| <b>Cyproheptadine</b>         | <b>appetite stimulation</b> — via its antiserotonin activity                                                                      |
| <b>Doxepin</b>                | chronic urticaria — ★ ~ <b>800× more potent antihistamine than diphenhydramine</b> ; also H2 blocker + a tricyclic antidepressant |

## BRAIN H1 OCCUPANCY ON PET — THE SEDATION LADDER

| DRUG (DOSE)                           | MEAN CNS H1 OCCUPANCY |
|---------------------------------------|-----------------------|
| Bilastine 20 mg · Fexofenadine 120 mg | ≈ <b>0%</b>           |
| Ebastine 10 mg · Terfenadine 60 mg    | 10–17%                |
| Azelastine 1 mg                       | 20%                   |
| <b>Cetirizine 20 mg</b>               | 26%                   |
| Hydroxyzine 25 mg                     | 54%                   |
| <b>d-Chlorpheniramine 2 → 5 mg</b>    | ● <b>60–77% → 86%</b> |

## ADVERSE EFFECTS

- ★ **CNS depression** — sedation, somnolence, morning hangover, ↓ alertness, dizziness. *Impaired cognitive and psychomotor performance can happen without perceptible sedation.*
- **CNS stimulation** — paradoxical excitation in infants & young children (*alkylamines e.g. CPM most often*): restlessness, irritability, hyperactivity, insomnia. ● **Overdose** → **convulsions, particularly in infants.**
- **Anticholinergic (common)**: dry mouth · dry eye, mydriasis, blurred vision · urinary retention · constipation
- **Anticholinergic (less common)**: memory deficit
- **Postural hypotension** —  $\alpha$ -blockade, especially the phenothiazines (*promethazine*)
- Rare: weight gain · drug allergy (*commonly from topical use*) · allergic dermatitis

## CAUTIONS & CONTRAINDICATIONS

- ⚠ Caution — all anticholinergic-driven: **narrow-angle glaucoma** · **symptomatic prostatic hypertrophy** · pregnancy
- ⚠ **Avoid tasks needing alertness until the effect is known — drivers, pilots**
- ● **Lactation**: excreted in small amounts into breast milk; 1st gen can cause **irritability, drowsiness and respiratory depression in the nursing infant**
- **Pregnancy category B**: chlorpheniramine, diphenhydramine (*and 2nd gen cetirizine, loratadine*)
- **Contraindicated**: hypersensitivity to antihistamines

# 2nd-Generation H1 Antihistamines

## MECHANISM

Same H1 inverse agonism — engineered to **stay out of the brain**: lipophobic, **ionised at physiological pH**, and actively pumped out by **P-glycoprotein**. No muscarinic affinity.

→ **Downstream**: the H1 effects without the sedation or the dry mouth · **12–24 h duration**

## DRUGS

| DRUG                                                                      | WHAT IT TREATS                                                                                                                                                                 |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cetirizine</b>                                                         | allergic rhinitis · <b>urticaria &amp; skin disease</b> — active metabolite of hydroxyzine, enters skin readily. ⚠️ The one that still sedates ( <i>30% CNS H1 occupancy</i> ) |
| <b>Levocetirizine</b>                                                     | as cetirizine — the active enantiomer: <b>2× the H1 affinity</b> of cetirizine, ~30× dextrocetirizine, longer dissociation half-life                                           |
| <b>Loratadine</b>                                                         | allergic rhinitis · urticaria — CYP3A4/2D6 → <b>desloratadine</b> , which is more potent                                                                                       |
| <b>Desloratadine</b>                                                      | as loratadine — the active metabolite, given directly                                                                                                                          |
| <b>Fexofenadine</b>                                                       | allergic rhinitis · urticaria — active metabolite of terfenadine; <b>excreted in the feces</b>                                                                                 |
| <b>Bilastine</b>                                                          | allergic rhinitis — <b>no hepatic metabolism</b> , ≈ zero CNS H1 occupancy                                                                                                     |
| <b>Rupatadine · Ebastine · Mizolastine · Acrivastine</b>                  | allergic rhinitis · urticaria                                                                                                                                                  |
| <b>Azelastine · Olopatadine · Levocabastine · Emedastine · Epinastine</b> | <b>topical</b> — conjunctival / nasal allergy                                                                                                                                  |

*Levocetirizine, desloratadine and fexofenadine — the active metabolites/enantiomers — are sometimes called 3rd generation.*

## ADVERSE EFFECTS

- Sedation **minimal** — ⚠️ except **cetirizine**, which has a higher incidence of drowsiness than the rest of the class
- **No anticholinergic effects** (*cetirizine: minimal*) — so no dry mouth, retention or mydriasis
- Rare: weight gain · drug allergy · allergic dermatitis

## CAUTIONS

- 🚫 **Terfenadine and astemizole had their approval withdrawn** — the parent drugs, not their metabolites (*fexofenadine remains in use*)
- ⚠️ Even "non-sedating" agents: check alertness before driving/flying

# Clinical Uses of H1 Antihistamines

## USES

**Allergic rhinitis** — rhinorrhea, nasal itching, sneezing, nasal congestion · **Allergic conjunctivitis** — red eye, eye itching · **Urticaria** · Itching of palate, throat, ears · Insect bites & stings · **Drug allergy** · **Common cold** · **Motion sickness & vertigo** · **Appetite stimulation** — cyproheptadine · **Antipsychotic-induced EPS** — diphenhydramine, on its anticholinergic activity

## CHOOSING THE DRUG

| SITUATION                               | WHAT TO REACH FOR                                                                                                                                                |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chronic urticaria</b>                | high-dose 2nd gen · combined <b>H1 + H2</b> blockade sometimes works · <b>doxepin</b> is effective                                                               |
| <b>Skin disease / itch</b>              | <b>hydroxyzine or cetirizine</b> — they enter skin and hold high concentration ★                                                                                 |
| <b>Motion sickness, vertigo</b>         | <b>promethazine</b> (D2, H1, M1) · <b>dimenhydrinate</b> & <b>diphenhydramine</b> (H1, M1) — ★ the effect is <b>central anticholinergic</b> , not antihistaminic |
| <b>Pilot / driver / heavy machinery</b> | <b>bilastine</b> — no CNS occupancy                                                                                                                              |
| <b>Hepatic or renal impairment</b>      | bilastine · desloratadine · fexofenadine                                                                                                                         |

## THE LINE THAT GETS TESTED

- **Not for anaphylaxis or angioedema as primary treatment** ★ — H1 antihistamines are **adjuvant therapy only** there. Edema and itch are suppressed; **hypotension and bronchoconstriction are not**.

# Betahistine (H3)

## MECHANISM

Three parallel actions, one endpoint — **faster functional recovery from vestibular disorder**:

1. **CNS** — H3 **inverse agonist** → rebalances vestibular-nucleus activity + brain arousal
2. **Vascular tree** — H1 **agonist** → ↑ **cochlear blood flow**
3. **Peripheral labyrinth** — H3 inverse agonist + **H4 antagonist** → ↓ vestibular input

→ **Downstream:** ★ note the inversion — this is a histamine *agonist* used therapeutically, the mirror image of the rest of this lecture

## DRUGS

| DRUG               | WHAT IT TREATS                 |
|--------------------|--------------------------------|
| <b>Betahistine</b> | vertigo / vestibular disorders |

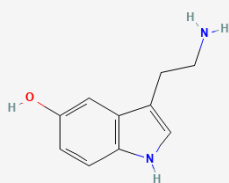
# Triptans — 5-HT<sub>1B/1D</sub> Agonists

## MECHANISM

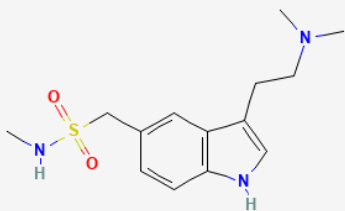
Two hits on the same migraine:

1. **5-HT<sub>1B</sub>** on the **dilated dural blood vessels** → **vasoconstriction**
  2. **5-HT<sub>1B/1D</sub>** on trigeminal endings and the **trigeminocervical complex (TCC)** in the brainstem → **blocks pain transmission** and ↓ **CGRP** release, so the thalamus/cortex never get the pain & nausea signal
- **Downstream:** ★ the same 5-HT<sub>1B</sub> constriction that aborts the headache also constricts **coronary** arteries — that is the entire safety problem

## THE SHARED CORE · WHAT CHANGES

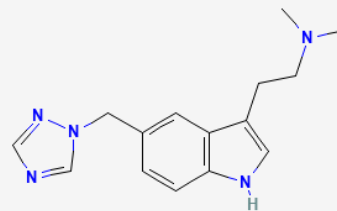


**5-HT (serotonin) — the natural ligand.** an indole with a short amine tail. Every triptan is this indole with a bulky group added to force 1B/1D selectivity.



**Sumatriptan**

a sulfonamide-methyl group → very polar → ● poor BBB entry and only ~15% oral bioavailability



**Rizatriptan**

a triazole instead → more lipophilic → better absorbed, faster onset

## DRUGS

| DRUG                         | WHAT IT TREATS                                                                                                                        |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sumatriptan</b>           | acute migraine — $t_{1/2}$ 2 h, MAO-metabolised, oral bioavailability only <b>14%</b>                                                 |
| <b>Rizatriptan</b>           | acute migraine — $t_{1/2}$ 2 h, MAO                                                                                                   |
| <b>Zolmitriptan</b>          | acute migraine — $t_{1/2}$ 3 h, CYP450 + MAO                                                                                          |
| <b>Almotriptan</b>           | acute migraine — $t_{1/2}$ 3.5 h, CYP450 + MAO, <b>bioavailability 69%</b> (best)                                                     |
| <b>Eletriptan</b>            | acute migraine — $t_{1/2}$ 5 h, <b>CYP3A4</b>                                                                                         |
| <b>Naratriptan</b>           | acute migraine — $t_{1/2}$ 5–6.3 h, <b>renal 70%</b>                                                                                  |
| <b>Frovatriptan</b>          | acute migraine — ★ $t_{1/2}$ <b>25 h</b> , the long-acting one; renal 50%                                                             |
| <b>Lasmiditan</b><br>(ditan) | acute migraine — selective <b>5-HT<sub>1F</sub></b> agonist: central analgesia <b>without</b> the 5-HT <sub>1B</sub> vasoconstriction |

### ADVERSE EFFECTS

- **Hypertension** — 5-HT<sub>1B</sub> receptor
- ● **Coronary vasoconstriction** ★ — 5-HT<sub>1B</sub> receptor

### INTERACTIONS

- **Triptans + MAO-A inhibitors** — sumatriptan and rizatriptan are MAO-cleared, so the block stacks

### CAUTIONS

- ● **Contraindicated in patients with cardiovascular risk factors** — driven by the coronary vasoconstriction



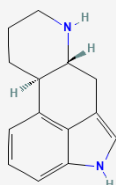
# Ergots

## MECHANISM

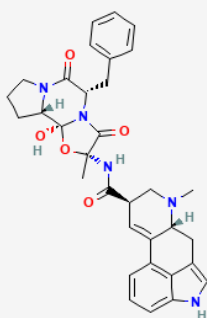
**Ergotamine** is an  $\alpha_1$ -adrenoceptor agonist **and** a 5-HT<sub>1</sub> receptor agonist — it constricts the dilated dural vessels like a triptan, but with none of the selectivity.

→ **Downstream:** ● the  $\alpha_1$  component adds systemic vasoconstriction on top of the 5-HT<sub>1B</sub> effect — worse hypertension, and vessels far from the head get constricted too

## THE SHARED CORE · WHAT CHANGES

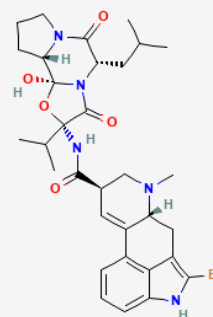


**Ergoline.** ● the reason ergots are so dirty: this ring system contains the shapes of serotonin, dopamine AND noradrenaline at once, so one molecule hits 5-HT, D2 and  $\alpha_1$  receptors



**Ergotamine**

the  $\alpha_1$  activity is what constricts vessels far from the head → limb ischaemia



**Bromocriptine**

one Br added and the same skeleton becomes a D2 agonist for prolactinoma

## DRUGS

| DRUG              | WHAT IT TREATS |
|-------------------|----------------|
| <b>Ergotamine</b> | acute migraine |

## ADVERSE EFFECTS

- **Hypertension** — 5-HT<sub>1B</sub> **plus**  $\alpha_1$ -receptor agonism
- ● **Coronary vasoconstriction** — 5-HT<sub>1B</sub>
- ● **Teratogenic** ★ — this is the ergot-specific one; triptans do not carry it

## INTERACTIONS

- ● **Ergotamine + CYP3A4 inhibitors** ★ — e.g. **protease inhibitors (anti-HIV drugs)**.  
**Contraindicated for concomitant use:** ergotamine accumulates and causes **acute ischemia of the limb**, leading to gangrene

## CAUTIONS & CONTRAINDICATIONS

- ● **Contraindicated:** cardiovascular risk factors · **pregnancy** (*teratogenic*) · concurrent CYP3A4 inhibitors

*Principle behind this: [[pharm-drug-action]] — Histamine's bronchoconstriction antagonised by epinephrine is the textbook physiologic antagonist — different receptor, opposite effect.*

Principle behind this: [\[\[pharm-drug-interactions\]\]](#) — Terfenadine + a CYP3A4 inhibitor → fatal torsades — the interaction that removed a drug from the market.

Principle behind this: [\[\[pharm-drug-toxicity\]\]](#) — The Type I mediators — histamine, leukotrienes, prostaglandins — are the drug targets of that page. Anaphylaxis is where this page and that one meet.

Principle behind this: [\[\[pharm-cns-basics\]\]](#) — Diphenhydramine's four-quadrant table is the whole Concept 3, and 5-HT<sub>3</sub> at the CTZ is the antiemetic target.

Source: [Histamine anti-histamines and serotonin agonists\\_2025.pdf](#) — อ.สุรศักดิ์ (Surasak Wichaiyo), PYCH304 Pharmacology I, 36 slides. Structures omitted.

---

# Diuretics

Agents that promote **diuresis** — increased urine production by excreting excess water and electrolytes (mainly  $\text{Na}^+$ ). Used in **heart failure · kidney disease · liver cirrhosis · hypertension**.

↑ renal  $\text{Na}^+$  &  $\text{H}_2\text{O}$  excretion → ↓ intravascular volume → ↓ preload & stroke volume → ↓ cardiac output → ↓ BP

## TWO WORDS WORTH SEPARATING

- **Natriuretic** — renal sodium excretion
- **Aquaretic** — excretion of solute-free water

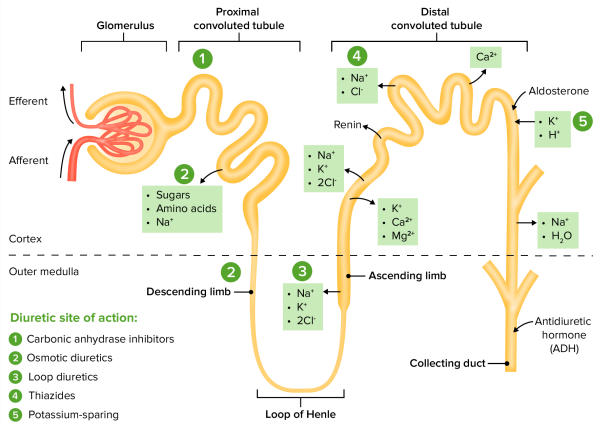
## THE 5 CLASSES AT A GLANCE

| CLASS                                  | SITE OF ACTION                           | MECHANISM                                                        |
|----------------------------------------|------------------------------------------|------------------------------------------------------------------|
| <b>Carbonic anhydrase inhibitor</b>    | proximal convoluted tubule               | inhibit carbonic anhydrase                                       |
| <b>Thiazide / thiazide-like</b>        | distal convoluted tubule                 | inhibit $\text{Na}^+-\text{Cl}^-$ symporter                      |
| <b>Loop</b>                            | thick ascending limb, loop of Henle      | inhibit $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ cotransporter      |
| <b><math>\text{K}^+</math>-sparing</b> | late distal tubule, collecting duct      | $\text{Na}^+$ channel inhibitor <b>or</b> aldosterone antagonist |
| <b>Osmotic</b>                         | proximal tubule, loop, collecting tubule | osmotic effect                                                   |

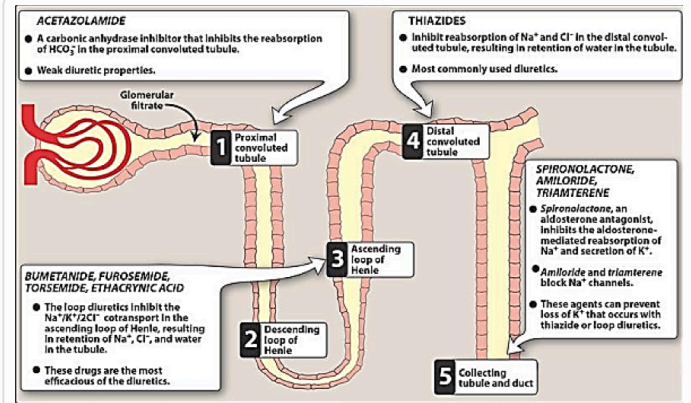
## URINARY ELECTROLYTES & BLOOD PH

| CLASS                        | $\text{NaCl}$ | $\text{NaHCO}_3$ | $\text{K}^+$ | BLOOD PH |
|------------------------------|---------------|------------------|--------------|----------|
| Carbonic anhydrase inhibitor | +             | +++              | +            | ↓        |
| Thiazide / thiazide-like     | ++            | +                | +            | ↑        |
| Loop                         | ++++          | 0                | +            | ↑        |
| $\text{K}^+$ -sparing        | +             | +                | –            | ↓        |
| Osmotic                      | +             | +                | +            | N/A      |

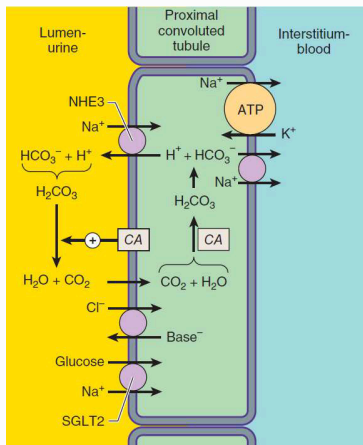
## OVERVIEW MAPS



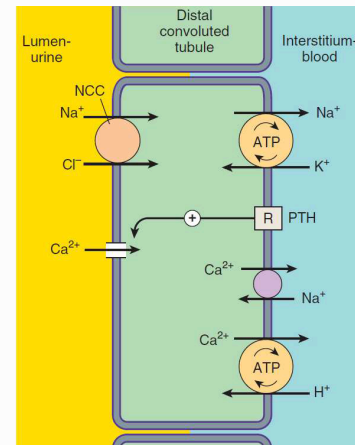
KUB 5-6 Water and electrolytes excretion KT2025 · fig 68



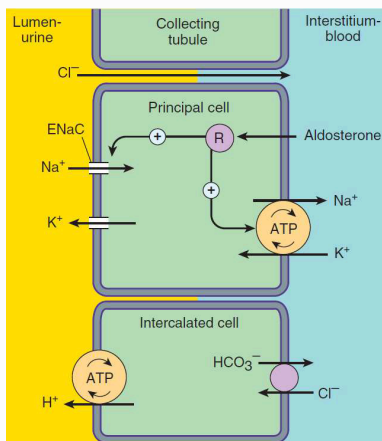
Antihypertensive Agents · fig 26



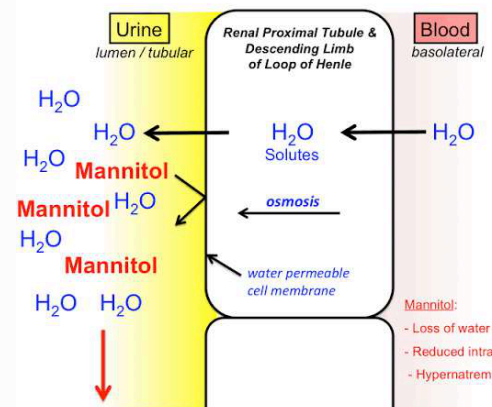
Diuretics Warisara 2025 · fig 05



Diuretics Warisara 2025 · fig 11



Diuretics Warisara 2025 · fig 19



Diuretics Warisara 2025 · fig 24

# Carbonic Anhydrase Inhibitor

## MECHANISM

Inhibits **carbonic anhydrase** in the **proximal convoluted tubule** — so  $\text{NaHCO}_3$  is not reabsorbed. (CA type IV = membrane · CA type II = cytosol)

→ **Downstream:**  $\uparrow \text{Na}^+$ ,  $\text{HCO}_3^-$ ,  $\text{H}_2\text{O}$  excretion ·  $\uparrow \text{K}^+$  excretion at collecting duct · **metabolic acidosis** · urine pH  $\uparrow$  → **In the eye:**  $\downarrow$  aqueous humour →  $\downarrow$  **intraocular pressure** — this is why it treats glaucoma

## DRUGS

| DRUG                 | WHAT IT TREATS                                                |
|----------------------|---------------------------------------------------------------|
| <b>Acetazolamide</b> | glaucoma · altitude sickness prevention · metabolic alkalosis |
| <b>Brinzolamide</b>  | glaucoma                                                      |
| <b>Dorzolamide</b>   | glaucoma                                                      |

## USES

**Glaucoma** — acetazolamide, brinzolamide, dorzolamide · **Metabolic alkalosis** — acetazolamide · **Acute mountain sickness** — acetazolamide

## ADVERSE EFFECTS

- Metabolic acidosis
- Renal  $\text{K}^+$  wasting (*prevent: KCl or a  $\text{K}^+$ -sparing diuretic*)
- Renal calculi — alkaline urine precipitates calcium phosphate
- Hyperammonemia in cirrhotics
- Paresthesia, somnolence, depression, fatigue

## CAUTIONS

- ⚠ **Do not use in sulfonamide hypersensitivity**
- ⚠ Chronic use depletes body  $\text{HCO}_3^-$  stores

# Thiazide & Thiazide-like

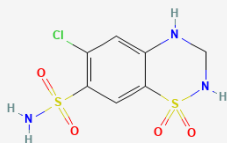
## MECHANISM

Inhibits the **Na<sup>+</sup>-Cl<sup>-</sup> symporter** in the luminal membrane of the **distal convoluted tubule**.

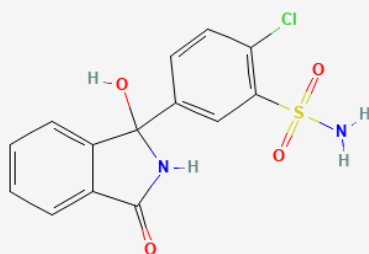
→ **Downstream:** ↑ Na<sup>+</sup> & Cl<sup>-</sup> excretion · ↑ K<sup>+</sup> excretion at collecting duct · ↓ **uric acid excretion**

*Thiazide-like has the same action and ADR profile — developed for longer duration.*

## THE SHARED CORE · WHAT CHANGES

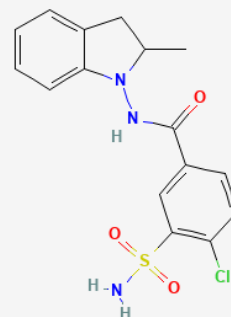


**Hydrochlorothiazide — the benzothiadiazine core.** ● note the sulfonamide group — it is a carbonic-anhydrase inhibitor's group, which is why thiazides came out of that drug family and why sulfa allergy is asked about



**Chlorthalidone**

'thiazide-like' — the ring is different, the sulfonamide isn't;  $t_{1/2}$  ~2× longer



**Indapamide**

an indoline instead of the thiazide ring → less metabolic effect, some vasodilation

## MEDCHEM

**SAR** ● The unsubstituted sulfonamide is the whole drug. Remove it and there is **no activity at all** - it works as an isostere of carbonic acid, which is also why thiazides retain weak carbonic-anhydrase activity and why this group sits at the centre of the **sulfa cross-allergy** question. R1 must be an **electron-withdrawing group** (-Cl, -CF<sub>3</sub>); with -H there is almost nothing, and electron-*donating* groups (-CH<sub>3</sub>, -OCH<sub>3</sub>) actively reduce activity. -CF<sub>3</sub> adds lipid solubility and duration.

**SAR** ★ Why chlorothiazide is dosed at 500 mg and hydrochlorothiazide at 25 mg. Saturating the 3,4-double bond - thiazide → **hydrothiazide** - is worth **about tenfold in activity**. Same molecule otherwise. N-alkylation and an R2 haloalkyl/aralkyl/thioether push potency and duration further (methyclothiazide: 2.5 mg, >24 h).

**Physicochem** The sulfonamide protons are acidic (**pKa 6.7–9.5**), which is what allows a water-soluble sodium salt. The thiazide-*like* agents keep the same unsubstituted sulfonamide but drop the thiadiazine ring entirely - chlorthalidone is a **1-oxo-isoindoline**, indapamide a benzamide - and that redesign is where their very long half-lives come from.

## DRUGS

| DRUG                              | WHAT IT TREATS       |
|-----------------------------------|----------------------|
| <b>Hydrochlorothiazide (HCTZ)</b> | hypertension · edema |
| <b>Indapamide</b>                 | hypertension · edema |
| <b>Chlorthalidone</b>             | hypertension · edema |

## USES

**Mild hypertension** — monotherapy or combined with ACEIs / ARBs /  $\beta$ -blockers · **Edema** — heart failure, cirrhosis, renal insufficiency

## ADVERSE EFFECTS

- **Hypokalemia** ★, hyponatremia, hypochloremia, hypomagnesemia
- Ventricular rhythm disturbance (*from the hypokalemia*)
- **Hyperuricemia** — caution in gout
- Impaired glucose tolerance — caution in DM
- $\uparrow$  cholesterol & TG — caution in hyperlipidemia
- Rarely: vertigo, headache, nausea, vomiting

## INTERACTIONS

- $\downarrow$  effect of **uricosurics, sulfonyleureas, insulin**
- **NSAIDs**  $\downarrow$  renal blood flow  $\rightarrow$   $\downarrow$  diuretic effect
- **Digoxin**  $\rightarrow$  toxicity risk, driven by the hypokalemia ★

## CAUTIONS

- ⚠ **Do not use in sulfonamide hypersensitivity**

## ADMINISTRATION — THIAZIDE PREPARATIONS

| DRUG                             | DOSAGE FORMS                   |
|----------------------------------|--------------------------------|
| <b>Hydrochlorothiazide</b>       | tab 25, 50 mg · syrup 10 mg/mL |
| <b>Chlorthalidone + atenolol</b> | tab 12.5/50 mg                 |
| <b>Indapamide</b>                | tab 2.5 mg                     |

- 💡 The **syrup** is the one worth remembering — it is what makes HCTZ usable in a child or anyone who can't swallow tablets, and it isn't obvious that a drug this old has one.

# Loop Diuretic

## MECHANISM

Blocks the  $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$  cotransporter in the **ascending limb of the loop of Henle** — where the bulk of  $\text{Na}^+$  is reabsorbed. **The most potent class ("high ceiling")**.

→ **Downstream:**  $\uparrow \text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Cl}^-$  excretion ·  $\downarrow$  uric acid excretion · **venodilation** via nitric oxide release

## DRUGS

| DRUG                   | WHAT IT TREATS                            |
|------------------------|-------------------------------------------|
| <b>Furosemide</b>      | edema · acute pulmonary edema             |
| <b>Torsemide</b>       | edema                                     |
| <b>Bumetanide</b>      | edema                                     |
| <b>Ethacrynic acid</b> | edema — the <b>non-sulfonamide</b> option |

## USES

**Edema** — heart failure, cirrhosis, nephritic syndrome · **Acute pulmonary edema** · Chronic renal failure · **Hyperkalemia** · Hypertension (*not first-choice*)

## ADVERSE EFFECTS

- **OTOTOXICITY** ★ — tinnitus, hearing impairment, vertigo, deafness. *Risk: rapid IV of a large dose. This is what separates loop from thiazide.*
- **Hypokalemia** ★, hyponatremia, hypochloremia, hypomagnesemia
- Hypokalemic metabolic alkalosis
- **Hyperuricemia** — caution in gout
- Impaired glucose tolerance ·  $\uparrow$  cholesterol & TG
- Dehydration, hypotension

## INTERACTIONS

- $\downarrow$  effect of **uricosurics, sulfonylureas, insulin**
- **NSAIDs**  $\downarrow$  diuretic effect
- ● **Aminoglycosides or cisplatin** →  $\uparrow$  ototoxicity

## CAUTIONS

- ⚠ Sulfonamide hypersensitivity (*furosemide, bumetanide, torsemide* — use *ethacrynic acid* instead)

## ADMINISTRATION — LOOP DIURETIC PREPARATIONS

| DRUG              | DOSAGE FORMS                                                                                                  |
|-------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Furosemide</b> | tab <b>40 mg</b> and <b>500 mg</b> · injection <b>20 mg/2 mL</b> , <b>250 mg/25 mL</b> · syrup <b>2 mg/mL</b> |
| <b>Torsemide</b>  | tab <b>10 mg</b>                                                                                              |

- ● **Furosemide comes as both a 40 mg and a 500 mg tablet.** That is a twelve-fold difference in the same box shape — the 500 mg is for renal failure, and picking it up by mistake is not a small error.
- 💡 The IV forms exist because in acute pulmonary oedema, gut absorption is exactly what you can't rely on.



# K<sup>+</sup>-Sparing Diuretic

## MECHANISM

Two routes, same result — at the **late distal tubule & collecting duct**:

1. **Na<sup>+</sup> channel inhibitor (NCC)** — blocks the Na<sup>+</sup> channel
2. **Aldosterone receptor antagonist (ARA / MRA)** — antagonizes aldosterone (*which normally raises Na<sup>+</sup> channel and Na<sup>+</sup>-K<sup>+</sup> ATPase activity*)

→ **Downstream**: ↑ Na<sup>+</sup> & Cl<sup>-</sup> excretion · ↓ K<sup>+</sup> excretion → **hyperkalemia**

## DRUGS

| DRUG                        | WHAT IT TREATS                                                |
|-----------------------------|---------------------------------------------------------------|
| <b>Spironolactone</b> (ARA) | hypertension · edema · <b>Conn's syndrome</b> · heart failure |
| <b>Eplerenone</b> (ARA)     | heart failure                                                 |
| <b>Amiloride</b> (NCC)      | hypertension · edema                                          |
| <b>Triamterene</b> (NCC)    | edema                                                         |

## USES

**Hypertension** — amiloride, spironolactone · **Edema** · **Primary hyperaldosteronism (Conn's)** — spironolactone · **Heart failure** — spironolactone, eplerenone

## ADVERSE EFFECTS

-  **HYPERKALEMIA**  — the defining ADR, especially in renal impairment. Mirror-image of every other class.
- Dehydration, hyponatremia (*occasional*) · Metabolic acidosis
- GI upset — nausea, vomiting, diarrhea
-  **Spironolactone is a steroid** → also binds androgen & progestogen receptors:
  - **Male**: gynecomastia, impotence — ***does not happen with eplerenone***
  - **Female**: menstrual irregularity, hair growth

## INTERACTIONS

-  **ACEIs or K<sup>+</sup> supplements** → serious hyperkalemia risk

## CAUTIONS

-  Renal impairment — hyperkalemia risk climbs sharply

## ADMINISTRATION — K<sup>+</sup>-SPARING PREPARATIONS

| DRUG                    | DOSAGE FORMS                                 |
|-------------------------|----------------------------------------------|
| <b>Spironolactone</b>   | tab <b>25, 100 mg</b> · syrup <b>2 mg/mL</b> |
| <b>Amiloride + HCTZ</b> | tab <b>5/50 mg</b>                           |

-  Amiloride is dispensed here only as a fixed combination with HCTZ — which tells you how it is used: as the potassium-sparing half of a thiazide, not as a diuretic in its own right.

# Osmotic Diuretic

## MECHANISM

Pharmacologically **inert, freely filtered** at the glomerulus, **not reabsorbed** — so it stays in the tubule and holds water with it. Raises urine volume without dragging out much electrolyte: "**aquaretic**."

→ **Downstream:** ↑↑ urine volume & flow rate · minor  $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Cl}^-$ ,  $\text{HCO}_3^-$ , phosphate excretion → ↑ **serum osmolality:** pulls fluid **out of the eye** (↓ IOP) and **out of the brain** (↓ cerebral edema)

## DRUGS

| DRUG            | WHAT IT TREATS                                                                                          |
|-----------------|---------------------------------------------------------------------------------------------------------|
| <b>Mannitol</b> | prevent oliguria / acute renal failure · ↓ <b>intraocular pressure</b> · ↓ <b>intracranial pressure</b> |

## USES

Maintain renal blood flow (*prevent oliguria / ARF*) · Reduce **intraocular pressure** · Reduce **intracranial pressure** & cerebral edema

## ADVERSE EFFECTS

- Dehydration — headache, nausea, vomiting
- Pulls water **out of cells** → ↑ **ECF volume** — dangerous if the circulation is already overloaded

## CAUTIONS & CONTRAINDICATIONS

- ⚠ **Precaution:** heart failure, pulmonary congestion
- 🚫 **Contraindicated:** anuria · severe renal disease · **active intracranial bleeding** · severe pulmonary congestion/edema · severe heart failure

# Clinical Applications

## WHERE DIURETICS GET USED

| STATE                | CONDITIONS                                                         |
|----------------------|--------------------------------------------------------------------|
| <b>Edematous</b>     | heart failure · kidney disease / renal failure · hepatic cirrhosis |
| <b>Non-edematous</b> | hypertension · nephrolithiasis · renal & cardiac protection        |

*Principle behind this: [[pharm-pharmacokinetics]] — several of these have to reach the tubular lumen by active tubular secretion before they can work.*

*Principle behind this: [[pharm-drug-action]] — Mannitol is the non-specific (osmotic) target, with no binding site to block; thiazides are the flat dose–response curve.*

*Principle behind this: [[pharm-drug-interactions]] — Probenecid inhibiting OATs (the deliberate version — it's how penicillin levels were once prolonged); furosemide's hypokalaemia feeding digoxin toxicity; amiloride + ACEI → hyperkalaemia.*

*Source: Diuretics\_Warisara\_2025.pdf — อ.วริศรา, PYCH304 Pharmacology I, 39 slides. Structures omitted.*

# Vasodilators

Drugs that **dilate blood vessels** by relaxing vascular smooth muscle. Dilating **arterial (resistance)** vessels drops afterload → ↓ systemic vascular resistance → ↓ BP. Dilating **venous (capacitance)** vessels drops preload → ↓ venous pressure.

↑ **cGMP** · ↑ **cAMP** · ↓ **intracellular Ca<sup>2+</sup>** → **vascular smooth muscle relaxes** → ↓ **afterload and/or**  
↓ **preload** → ↓ **BP & ↓ myocardial O<sub>2</sub> demand**

## THE ONLY THREE WAYS IN

Every vasodilator relaxes the smooth muscle cell by one of these — know the route and the drug's whole profile follows: - ↑ **cGMP** — nitrates, nitroprusside, hydralazine, PDE-5 inhibitors, sGC stimulators, ARNI - ↑ **cAMP** - ↓ **intracellular Ca<sup>2+</sup>** — CCBs, and K<sup>+</sup> channel openers indirectly (hyperpolarisation closes Ca<sup>2+</sup> channels)

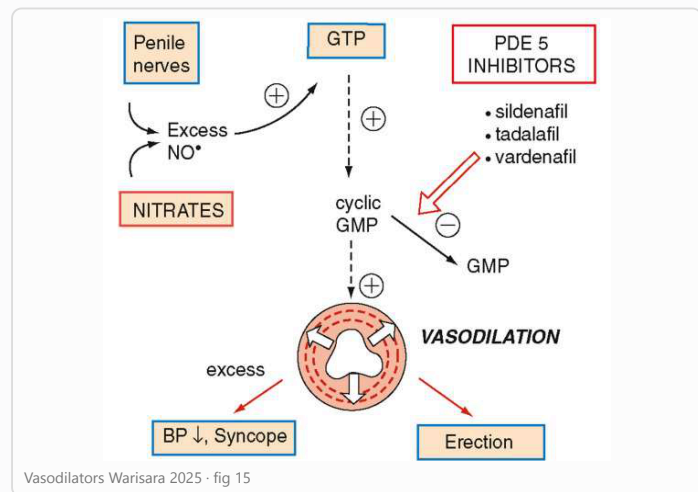
## FULL CLASSIFICATION (LECTURE'S LIST)

| CLASS                                   | DRUGS                                          | COVERED HERE              |
|-----------------------------------------|------------------------------------------------|---------------------------|
| <b>α<sub>1</sub>-AR antagonists</b>     | prazosin, doxazosin, terazosin                 | <i>(separate lecture)</i> |
| <b>Ca<sup>2+</sup> channel blockers</b> | nifedipine, diltiazem, verapamil               | <i>(separate lecture)</i> |
| <b>RAAS blockers</b>                    | ACEIs (-pril), ARBs (-sartan), DRI (aliskiren) | <i>(separate lecture)</i> |
| <b>Hydralazine</b>                      | hydralazine                                    | ✓                         |
| <b>K<sup>+</sup> channel openers</b>    | minoxidil, pinacidil, nicorandil               | ✓                         |
| <b>Sodium nitroprusside</b>             | nitroprusside                                  | ✓                         |
| <b>Organic nitrates</b>                 | NTG, ISDN, ISMN                                | ✓                         |
| <b>PDE-5 inhibitors</b>                 | sildenafil, vardenafil, tadalafil, avanafil    | ✓                         |
| <b>Endothelin receptor antagonists</b>  | bosentan, ambrisentan, macitentan              | ✓                         |
| <b>sGC stimulators</b>                  | riociguat, vericiguat                          | ✓                         |
| <b>Neprilysin inhibitors</b>            | sacubitril                                     | ✓                         |

## WHICH VESSEL EACH ONE HITS ★

| VESSEL               | DRUGS                   | CONSEQUENCE                                                                    |
|----------------------|-------------------------|--------------------------------------------------------------------------------|
| <b>Arterial only</b> | hydralazine · minoxidil | ↓ TPR, ↓ afterload — <b>orthostatic hypotension is less</b>                    |
| <b>Venous mainly</b> | organic nitrates        | ↑ venous capacitance, ↓ preload — <b>orthostatic hypotension &amp; syncope</b> |
| <b>Mixed (both)</b>  | sodium nitroprusside    | venous pooling + ↓ arterial resistance                                         |

## OVERVIEW MAPS



# Hydralazine

## MECHANISM

**Direct relaxation of arteriolar smooth muscle**, plus **NO release from arteriolar endothelial cells** → ↑ cGMP.

→ **Downstream:** vasodilation → ↓ TPR → ↓ BP → **Arterioles only:** veins are untouched, so **orthostatic hypotension is less** than with nitrates

## DRUGS

| DRUG               | WHAT IT TREATS                                                         |
|--------------------|------------------------------------------------------------------------|
| <b>Hydralazine</b> | hypertensive emergency <b>in pregnancy</b> · heart failure (with ISDN) |

## USES

**Hypertensive emergency in pregnancy** ★ — *not* a primary antihypertensive otherwise · **Heart failure** as **hydralazine + isosorbide dinitrate (H + ISDN)** — ESC guideline: consider in Black/African-American patients with LVEF < 35%, and in HFrEF patients intolerant of ACEIs/ARBs because of renal dysfunction or hyperkalemia

## ADVERSE EFFECTS

- **Reflex tachycardia** — baroreceptor reflex answering the BP drop. *Prevent: co-administer a  $\beta$ -blocker.*
- ★ **SLE-like syndrome** — sore throat, fever, rash, muscle & joint aches. Mechanism unknown; appears after **long-term use (> 6 months)**.

## ACETYLATOR STATUS (WHY THE DOSE VARIES BY PERSON)

Hydralazine is **N-acetylated by NAT2**, so bioavailability tracks acetylator phenotype: **fast acetylators (Asians) ≈ 16%** · **slow acetylators (Black & Caucasian) ≈ 35%**. Well absorbed, but low F overall from hepatic first-pass metabolism.

# Minoxidil — K<sup>+</sup> Channel Opener

## MECHANISM

Minoxidil is a **prodrug**: sulfotransferase converts it to **minoxidil N-O sulfate** (the active form), which opens the **ATP-modulated K<sup>+</sup> channel**.

→ **Downstream**: K<sup>+</sup> efflux → membrane **hyperpolarises** → no depolarisation → **voltage-gated Ca<sup>2+</sup> channels stay closed** → contraction inhibited → **Result**: arteriolar vasodilation → ↓ TPR → ↓ BP · ↑ blood flow to **skin**, skeletal muscle, GI tract, heart

## DRUGS

| DRUG              | WHAT IT TREATS                                                                          |
|-------------------|-----------------------------------------------------------------------------------------|
| <b>Minoxidil</b>  | severe hypertension refractory to other agents · <b>androgenetic alopecia</b> (topical) |
| <b>Pinacidil</b>  | (class member, K <sup>+</sup> channel opener)                                           |
| <b>Nicorandil</b> | (class member, K <sup>+</sup> channel opener)                                           |

## USES

**Severe hypertension** that responds poorly to other antihypertensives — *avoided in women because of the hair growth* · **Androgenetic alopecia** — topical, applied to the affected area BID, needs **at least 4 months** of therapy

## ADVERSE EFFECTS

- **Salt & water retention.** *Prevent: co-administer a diuretic.*
- **Reflex tachycardia & ↑ contractility** — sympathetic/baroreceptor reflex. *Prevent: co-administer a β-blocker.*
- ★ **Hypertrichosis (hirsutism)** on long-term use — from ↑ cutaneous blood flow. Hair on face, back, arms, legs. **This ADR is also the indication** for the topical form.

# Sodium Nitroprusside

## MECHANISM

A nitrate compound built from a **nitrosyl (NO) group**, **5 cyanide (CN) groups** and **iron**. It **releases NO spontaneously — no enzyme required** → NO activates guanylyl cyclase → ↑ cGMP → relaxation. A very potent vasodilator.

→ **Downstream:** dilates **both arterioles and venules** → venous pooling + ↓ arterial resistance → **The 5 cyanides are the catch:** metabolism yields NO and cyanide ●

## DRUGS

| DRUG                        | WHAT IT TREATS                                                                              |
|-----------------------------|---------------------------------------------------------------------------------------------|
| <b>Sodium nitroprusside</b> | hypertensive emergencies ( <b>not</b> in pregnancy) · controlled hypotension during surgery |

## USES

**Hypertensive emergencies — except pregnancy** (*that's hydralazine's slot*) · Controlled hypotension during surgery · **Short-term only, max ~72 hr**

## ADVERSE EFFECTS

- Excessive vasodilation and hypotension
- **CYANIDE TOXICITY** ★ — metabolic acidosis, arrhythmia, severe hypotension. Caused by prolonged use, fast infusion rate, or renal dysfunction. **Antidote: hydroxocobalamin (vitamin B<sub>12</sub>)**.
- Methemoglobinemia** — haemoglobin Fe<sup>2+</sup> (ferrous) → methaemoglobin Fe<sup>3+</sup> (ferric)

## CAUTIONS

- ⚠ **Unstable molecule** — decomposes in strongly alkaline conditions and on light exposure. **Prepare fresh and check the appearance.**
- Continuous IV infusion only. Onset **30 s**, peak **2 min**, effect gone **within 3 min** of stopping.
- Cyanide-toxicity thresholds: > 500 mcg/kg total, > 2–5 mcg/kg/min, or the max rate 10 mcg/kg/min for > 10 min
- ⚠ Renal or liver disease → toxicity risk climbs



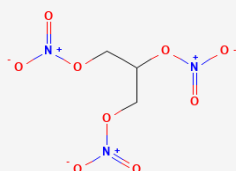
# Organic Nitrates

## MECHANISM

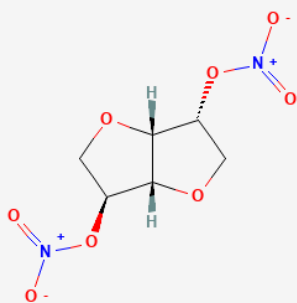
The **nitro group is reduced to nitric oxide (NO)** by specific enzymes and reductants (e.g. thiols) — *unlike nitroprusside, this one needs enzymatic conversion*. NO activates guanylyl cyclase → ↑ cGMP → vascular smooth muscle relaxation. NO is also an **antiplatelet** agent.

→ **Major effect: venodilation** ★ → ↑ venous capacitance → ↓ **preload** → **Minor (high dose only): arterial dilation** → ↓ TPR → ↓ afterload → **Net:** ↓ myocardial oxygen demand — which is how it relieves angina

## THE SHARED CORE · WHAT CHANGES

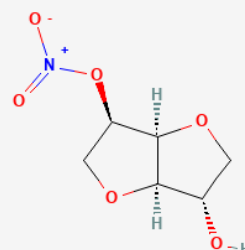


**Glycerol trinitrate.** three nitrate esters on glycerol — an NO donor once enzymes release it. ● hugely lipophilic → destroyed by first pass → hence sublingual



**Isosorbide dinitrate**

two nitrate groups on a rigid sugar → survives orally, but is itself a prodrug of the mononitrate



**Isosorbide mononitrate**

one nitrate — ● no first-pass metabolism at all, ~100% bioavailable

## MEDCHEM

**Physicochem** They are **esters of nitric and nitrous acid** - which means they are **hydrolysed by moisture**, the reason nitroglycerin tablets are a storage problem and not just a stability footnote. They are also lipophilic and low-boiling, volatile enough to have been given by inhalation.

**Metabolism** ★ **Mitochondrial aldehyde dehydrogenase (ALDH2) is the activating enzyme**, stripping nitrate groups off. Each removal costs activity: isosorbide dinitrate → mononitrates with reduced activity; nitroglycerin → 1,2- or 1,3-dinitrate with **less than 10%** of the parent's activity.

**SAR** ● **Nitrate tolerance is a chemistry problem, not a receptor problem.** Activation **consumes thiols**, and continuous exposure depletes them - which is why tolerance appears within **1–2 days** of continuous use and why a nitrate-free interval works.

## DRUGS

| DRUG                                       | WHAT IT TREATS                                     |
|--------------------------------------------|----------------------------------------------------|
| <b>Nitroglycerin (NTG)</b>                 | angina pectoris · heart failure · acute MI         |
| <b>Isosorbide dinitrate (ISDN)</b>         | angina pectoris · heart failure (with hydralazine) |
| <b>Isosorbide 5-mononitrate (ISMN)</b>     | angina pectoris                                    |
| <b>Pentaerythritol tetranitrate (PETN)</b> | angina pectoris                                    |

## USES

Angina pectoris · Heart failure · Acute MI

### WHY THE ROUTE DIFFERS BY DRUG ★

- **NTG & ISDN** — heavy hepatic **first-pass** metabolism → very low oral bioavailability. Bypass it with **sublingual** or **transdermal**.
- **ISMN** — **no first-pass metabolism** → **F = 100% orally**. This is why the mononitrate is the oral maintenance nitrate.
- NTG and ISDN are metabolised into **active** molecules with **longer half-lives than the parent** (NTG  $t_{1/2}$  3 min · ISDN 10 min · ISMN 280 min).

## ADVERSE EFFECTS

- **Orthostatic hypotension, syncope** — from the ↓ venous capacitance
- **Reflex tachycardia** — baroreceptor reflex
- **Most common: headache, dizziness**
- Occasional: flushing, nausea, vomiting
- **Contact dermatitis** — 40% of transdermal patch users
- **Methemoglobinemia** — rare but serious. *Antidotes: methylene blue, hydroxocobalamin.*
- ★ **NITRATE TOLERANCE** — from continuous or frequent exposure. **Fix: a nitrate-free period of at least 8 hr** — tablets TID with the night free, and remove the 24-hr patch overnight.

## INTERACTIONS

- ● **PDE-5 inhibitors (sildenafil, tadalafil, vardenafil)** ★ — both raise cGMP, so the effect is additive → **life-threatening hypotension. Nitrates in any form are contraindicated with them.**

## ADMINISTRATION — NITRATE FORMS, AND THE NITRATE-FREE PERIOD

| DRUG                                       | DOSAGE FORMS                                                   |
|--------------------------------------------|----------------------------------------------------------------|
| <b>Glyceryl trinitrate (NTG/GTN)</b>       | injection 10, 50 mg/10 mL · <b>transdermal patch</b> · aerosol |
| <b>Isosorbide dinitrate (ISDN)</b>         | tablet 10, 20, 30 mg · ★ <b>sublingual tablet 5 mg</b>         |
| <b>Isosorbide 5-mononitrate (ISMN)</b>     | tablet 20, 60 mg · <b>SR capsule 20, 60 mg</b>                 |
| <b>Pentaerythritol tetranitrate (PETN)</b> | tablet 10 mg                                                   |

- ● **The dosing schedule IS the tolerance strategy.** Continuous exposure produces tolerance, so every regimen is built around a **nitrate-free period of at least 8 hours**:
  - **Tablets** → **three times daily, with the gap at night**
  - **24-hour transdermal patch** → **take the patch off overnight**
- 💡 That night-free gap is the single instruction patients get wrong — a patch left on around the clock quietly stops working, and it looks like the angina is worsening rather than the drug failing.
- ⚠️ **Contact dermatitis affects 40% of patch users** — rotating the site is part of the counselling, not a nicety.

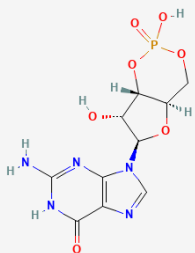
# PDE-5 Inhibitors

## MECHANISM

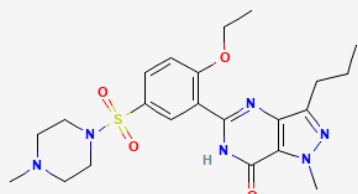
Selectively inhibit **phosphodiesterase 5**, the enzyme that degrades **cGMP** in the penis. Sexual stimulation releases NO from corpus cavernosum tissue → guanylate cyclase → cGMP; blocking PDE5 stops cGMP being broken back down to GMP.

→ **Downstream:** ↑ cGMP → smooth muscle relaxation → arterial dilation + expansion of the sinusoidal spaces of the corpus cavernosum → compressed veins → **erection** → **Same cGMP logic in the lung:** pulmonary vasodilation — hence the PAH indication

## THE SHARED CORE · WHAT CHANGES

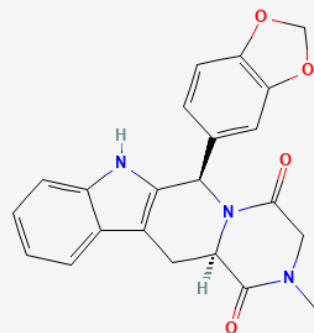


**cGMP — the substrate being protected.** PDE-5 destroys this. Every drug below is a flat molecule shaped enough like it to occupy the enzyme's pocket instead.



**Sildenafil**

a purine-like core; ⚠️ slight PDE-6 (retina) cross-reaction → the blue-tinge visual effect



**Tadalafil**

a completely different rigid ring system →  $t_{1/2}$  17.5 h and no PDE-6 problem

## DRUGS

| DRUG              | WHAT IT TREATS                                                |
|-------------------|---------------------------------------------------------------|
| <b>Sildenafil</b> | erectile dysfunction · <b>pulmonary arterial hypertension</b> |
| <b>Tadalafil</b>  | ED · <b>BPH — tadalafil only</b> ★ · PAH                      |
| <b>Vardenafil</b> | ED                                                            |
| <b>Avanafil</b>   | ED                                                            |

## USES

**Erectile dysfunction** — all four · **Benign prostatic hyperplasia** — tadalafil only · **Pulmonary arterial hypertension** — sildenafil, tadalafil (WHO Group 1, NYHA class II–III, to improve exercise ability and symptoms)

## WHAT ACTUALLY SEPARATES THEM

| DRUG              | ONSET     | DURATION      | FOOD                |
|-------------------|-----------|---------------|---------------------|
| <b>Avanafil</b>   | 15–30 min | 4–8 hr        | no interaction      |
| <b>Sildenafil</b> | 30–60 min | 4–8 hr        | <b>take fasting</b> |
| <b>Vardenafil</b> | 30 min    | 4–8 hr        | <b>take fasting</b> |
| <b>Tadalafil</b>  | 45 min    | ★ up to 36 hr | no interaction      |

*Efficacy is > 65% for all four — the choice is onset, duration and food, not potency.*

## PDE SUBTYPE SELECTIVITY — WHERE THE ODD ADRS COME FROM

- **PDE6** — in the **retina**, controls visual transduction. Inhibiting it causes **blurred vision, flashing lights, altered colour perception (blue/green)**. Incidence: sildenafil  $\approx$  vardenafil > avanafil > tadalafil.
- **PDE11** — in testes and prostate, but PDE-5 inhibitors have **no effect on spermatogenesis or testicular function**.

## ADVERSE EFFECTS

- **Hypotension**
- Vision loss (*rare*) · Hearing loss (*rare*)
- Heart attack, stroke, arrhythmia in patients with heart disease (*rare*)
- **Priapism** — persistent, usually painful erection lasting > 4 hr (*rare*)

## INTERACTIONS

- ● **Organic nitrates** — **absolute contraindication** ★, additive cGMP → life-threatening hypotension
- **CYP3A4** — the main interaction pathway (sildenafil; tadalafil also CYP2C9)
- Frequency restricted: **not more than once per day**

## ADMINISTRATION — PDE-5 INHIBITORS

### For erectile dysfunction

|                 | SILDENAFIL                 | TADALAFIL      | VARDENAFIL                 | AVANAFIL       |
|-----------------|----------------------------|----------------|----------------------------|----------------|
| <b>Start</b>    | 50 mg                      | 10 mg          | 10 mg                      | 100 mg         |
| <b>Max</b>      | 100 mg/day                 | 20 mg/day      | 20 mg/day                  | 200 mg/day     |
| <b>Onset</b>    | 30–60 min                  | 45 min         | 30 min                     | ★ 15–30 min    |
| <b>Duration</b> | 4–8 h                      | ★ up to 36 h   | 4–8 h                      | 4–8 h          |
| <b>Food</b>     | ● interacts — take fasting | no effect      | ● interacts — take fasting | no effect      |
| <b>Alcohol</b>  | no interaction             | no interaction | no interaction             | no interaction |

**Dosage forms:** sildenafil tab 25, 50, 100 mg · avanafil tab 50, 100, 200 mg · tadalafil tab 2.5, 5, 10, 20 mg · vardenafil tab 2.5, 5, 10, 20 mg

● **Not more than once per day**, any of them — the frequency restriction is part of the dosing, not a caution.

### For pulmonary arterial hypertension — different drug, different dose

|                           | SILDENAFIL                                                              | TADALAFIL                                         |
|---------------------------|-------------------------------------------------------------------------|---------------------------------------------------|
| <b>Dose</b>               | oral <b>20 mg three times daily</b> · IV <b>10 mg three times daily</b> | oral <b>40 mg once daily</b>                      |
| <b>Hepatic impairment</b> | no adjustment                                                           | <b>mild/moderate: start 20 mg · severe: avoid</b> |
| <b>Renal impairment</b>   | no adjustment                                                           | <b>severe: avoid</b>                              |
| <b>Dosage form</b>        | tab 20 mg · IV 10 mg/12.5 mL                                            | tab 20 mg                                         |

- 💡 The PAH dose and the ED dose are not the same drug in the patient's mind but they are on the shelf — 20 mg three times a day for the lungs versus 50 mg as needed. Check the indication before you check the strength.
- 💡 For sildenafil and vardenafil, "take it on an empty stomach" is the difference between working and not; a heavy meal is the most common reason a patient reports failure. Tadalafil and avanafil don't care.

# Endothelin Receptor Antagonists (ERAs)

## MECHANISM

**Endothelin-1 (ET-1)**, made mainly by endothelial cells, acts at **ETA and ETB** receptors on vascular smooth muscle. In **PAH**, endothelial dysfunction raises ET-1 → vasoconstriction, inflammation, fibrosis, hypertrophy & cell proliferation → the vessel narrows and occludes. ERAs block those receptors.

→ **Downstream:** ↓ vasoconstriction (*direct, and via renin / Ang II / sympathetic facilitation*) · ↓ inflammation · ↓ fibrosis · ↓ proliferation & remodelling → **Why it matters:** ET-1 drives the *structural* disease, not just the tone — so ERAs treat the remodelling

## DRUGS

| DRUG                                              | WHAT IT TREATS |
|---------------------------------------------------|----------------|
| <b>Ambrisentan</b> — <b>selective ETA</b> , 200:1 | PAH            |
| <b>Bosentan</b> — dual ETA/ETB, 20:1              | PAH            |
| <b>Macitentan</b> — dual ETA/ETB, 50:1            | PAH            |

## USES

**Pulmonary arterial hypertension (WHO Group 1)** with **NYHA functional class II–IV** symptoms — to improve exercise ability and clinical symptoms

## ADVERSE EFFECTS

(*class-wide, all three*) - ● **Hepatotoxicity** ★ - **Anemia** - **Peripheral edema**

## INTERACTIONS

- **Ambrisentan** — CYP3A4 inhibitor
- **Bosentan** — CYP2C9 & 3A4 substrate / inducer / inhibitor
- **Macitentan** — CYP3A4 substrate / inducer / inhibitor

## CAUTIONS

- ⚠ **Hepatic impairment** — **not recommended for any of the three**
- Renal impairment — **no dose adjustment needed** (all three clear hepatically; only macitentan excretes renally)

## PAH — THE DISEASE THESE TREAT

**Mean PAP > 25 mmHg at rest** (> 30 mmHg with exercise) with **PCWP < 15 mmHg**, measured by right heart catheterization. Symptoms: shortness of breath, dizziness, chest pain, edema (leg/ankle/abdomen), cyanosis, fast heartbeat. Long-term it drives vascular proliferation and remodelling → ↑ pulmonary vascular resistance → **right ventricular failure**.

## ADMINISTRATION — ENDOTHELIN RECEPTOR ANTAGONISTS

|                           | AMBRISENTAN        | BOSENTAN                                                 | MACITENTAN        |
|---------------------------|--------------------|----------------------------------------------------------|-------------------|
| <b>Dose</b>               | 5–10 mg once daily | ★ 62.5 mg twice daily × 4 weeks, then 125 mg twice daily | 10 mg once daily  |
| <b>Dosage form</b>        | tab 5, 10 mg       | tab 62.5, 125 mg                                         | tab 10 mg         |
| <b>Bioavailability</b>    | 90%                | 50%                                                      | 74%               |
| <b>Food effect</b>        | none               | none                                                     | none              |
| <b>Hepatic impairment</b> | ● not recommended  | ● not recommended                                        | ● not recommended |
| <b>Renal impairment</b>   | no adjustment      | no adjustment                                            | no adjustment     |

- ★ **Bosentan is the only one that is titrated** — half dose for the first four weeks, then doubled.
- ● **All three are off the table in hepatic impairment and all three are hepatotoxic**, yet none of them needs a renal adjustment. That asymmetry is the whole monitoring plan: watch the liver, not the kidney.
- 💡 Food is irrelevant for the entire class, so timing can be built around whatever the patient will actually remember.

# sGC Stimulators

## MECHANISM

**Direct stimulation of soluble guanylate cyclase** → cGMP formation. Note the contrast: nitrates supply NO to sGC; these stimulate the enzyme itself, **NO-independently**.

→ **Downstream:** ↑ cGMP → vasodilation → **Consequence:** stacking them on nitrates or PDE-5 inhibitors piles onto the same pathway ●

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                 |
|-------------------|------------------------------------------------------------------------------------------------|
| <b>Riociguat</b>  | pulmonary arterial hypertension                                                                |
| <b>Vericiguat</b> | chronic symptomatic <b>heart failure with EF &lt; 45%</b> — reduces CV death & hospitalisation |

## USES

**PAH** — riociguat · **Chronic symptomatic HFrEF (EF < 45%)** — vericiguat, to reduce the risk of cardiovascular death and hospitalisation

## ADVERSE EFFECTS

- **Hypotension** (*both*)
- **Riociguat:** ↑ liver enzymes, nausea/vomiting, flushing
- **Vericiguat:** anemia

## INTERACTIONS

- ● **Nitrates** and **PDE-5 inhibitors** — same cGMP pathway, additive hypotension
- Riociguat is metabolised by **CYP1A1, 3A4, 2C8**; vericiguat by **glucuronidation**

## ADMINISTRATION — SGC STIMULATORS ARE TITRATED, NEVER STARTED AT TARGET

|                        | RIOCIGUAT                                                                                           | VERICIGUAT                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Dose</b>            | <b>1 mg three times daily</b> , titrated up <b>every 2 weeks</b> to <b>2.5 mg three times daily</b> | <b>2.5 mg once daily</b> , doubled <b>every 2 weeks</b> to <b>10 mg once daily</b> |
| <b>Bioavailability</b> | 94%                                                                                                 | <b>93% — taken with food</b>                                                       |
| <b>Half-life</b>       | 5–10 h                                                                                              | 30 h                                                                               |
| <b>Metabolism</b>      | CYP1A1, 3A4, 2C8                                                                                    | glucuronidation                                                                    |

- ● **Both are contraindicated with nitrates and with PDE-5 inhibitors** — same final pathway, additive hypotension.
- 💡 The two-week step is there because hypotension is dose-related; a patient who feels nothing at the starting dose still shouldn't jump ahead, and one who feels dizzy usually needs the step held, not the drug stopped.



# Neprilysin Inhibitor / ARNI

## MECHANISM

**Natriuretic peptides (ANP, BNP, CNP)** are released by the heart in response to pressure and volume overload and are **degraded by neprilysin (neutral endopeptidase)**. Inhibiting neprilysin lets them accumulate.

→ **NP effects:** natriuresis & diuresis · vasodilation · **RAAS & sympathetic suppression** · ↑ RBF & GFR · antihypertrophic · antifibrotic · anti-inflammatory → **Why it can't be given alone:** blocking neprilysin raises ANP **but BP doesn't fall and Ang II rises** (shown with candoxatril) — so it is combined with a RAAS blocker  
→ **ARNI = neprilysin inhibitor + ARB:** sacubitril / valsartan

## DRUGS

| DRUG                                 | WHAT IT TREATS                                                                                                               |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Sacubitril / valsartan (ARNI)</b> | heart failure, NYHA class II–IV with <b>reduced ejection fraction</b> · pediatric HF with LV systolic dysfunction (> 1 year) |
| <b>Sacubitril</b> ( <i>alone</i> )   | not effective as monotherapy                                                                                                 |
| <b>Candoxatril</b>                   | ( <i>the proof-of-concept failure — raised ANP, no BP effect</i> )                                                           |

## USES

**Heart failure with reduced ejection fraction**, NYHA class II–IV · **Pediatric heart failure** with systemic left ventricular systolic dysfunction (age > 1 year)

## THE THREE NATRIURETIC PEPTIDES

|                 | ANP               | BNP                | CNP                        |
|-----------------|-------------------|--------------------|----------------------------|
| <b>Location</b> | atria             | ventricle          | vascular endothelial cells |
| <b>Trigger</b>  | atrial distension | ventricular volume | shear stress               |
| <b>Receptor</b> | NPR-A             | NPR-A              | NPR-B                      |

*All three are cleared via NPR-C (receptor-mediated internalisation) and degraded enzymatically by neprilysin.*

## ADVERSE EFFECTS

- **Hyperkalemia** ★ (*the ARB half*)
- **Kidney dysfunction**
- **Hypotension**

## CAUTIONS & CONTRAINDICATIONS

- ● **Pregnancy Category X**
- ⚠ **Severe renal impairment** — 50% dose reduction
- ⚠ **Severe hepatic impairment** — not recommended

#### ADMINISTRATION — SACUBITRIL/VALSARTAN

- **Start 49/51 mg twice daily, then double to the target maintenance dose of 97/103 mg twice daily.**
- **Dosage forms:** tab 24/26, 49/51, 97/103 mg.
- ⚠ **Severe renal impairment — 50% dose reduction. Severe hepatic impairment — not recommended.**
- 💡 The 24/26 mg strength exists for the patients who can't start at 49/51 — low blood pressure, renal impairment, or coming off a high ACE-inhibitor dose. It's a starting rung, not a maintenance dose.
- 🚫 The valsartan half is why this is **Pregnancy Category X** and why potassium and renal function get watched — it behaves like an ARB, because it contains one.

---

Source: [Vasodilators\\_warisara\\_2025.pdf](#) — อ.วริศรา, PYCH304 Pharmacology I, 52 slides. Structures omitted.

---

# Calcium Channel Blockers

Drugs that bind the  **$\alpha_1$  subunit of the L-type voltage-gated  $\text{Ca}^{2+}$  channel** and reduce how often it opens on depolarization  $\rightarrow$   **$\downarrow$  transmembrane  $\text{Ca}^{2+}$  current**. Also called calcium antagonists / calcium entry blockers. Used in **hypertension · angina · SVT · Raynaud's**.

**$\downarrow \text{Ca}^{2+}$  influx  $\rightarrow$   $\downarrow$  intracellular  $\text{Ca}^{2+}$   $\rightarrow$   $\downarrow \text{Ca}^{2+}$ -calmodulin  $\rightarrow$   $\downarrow$  MLCK  $\rightarrow$  vascular smooth muscle relaxation  $\rightarrow$   $\downarrow$  TPR  $\rightarrow$   $\downarrow$  BP**

## ★ VASCULAR VS CARDIAC — THE WHOLE EXAM IN ONE TABLE

|                                            | DHP (NIFEDIPINE-TYPE)           | NON-DHP (VERAPAMIL, DILTIAZEM) |
|--------------------------------------------|---------------------------------|--------------------------------|
| <b>Main target</b>                         | vascular smooth muscle          | heart <b>and</b> vessels       |
| <b>Vasodilation</b>                        | ++++ (arterioles >> veins)      | ++                             |
| <b>Negative inotropy</b> (contractility)   | ++                              | +++                            |
| <b>Negative chronotropy</b> (SA rate)      | +                               | +++                            |
| <b>Negative dromotropy</b> (AV conduction) | +                               | +++ (verapamil > diltiazem)    |
| <b>Reflex sympathetic response</b>         | +++ $\rightarrow$ tachycardia ★ | +                              |
| <b>Signature use</b>                       | hypertension, Raynaud's         | <b>SVT</b> , rate control      |

## CARDIOVASCULAR EFFECT, SCORED 0 (NONE) $\rightarrow$ 5 (MOST PROMINENT)

| DRUG                                 | VASODILATION | $\downarrow$ CONTRACTILITY | $\downarrow$ SA AUTOMATICITY | $\downarrow$ AV CONDUCTION |
|--------------------------------------|--------------|----------------------------|------------------------------|----------------------------|
| Nifedipine · Felodipine · Amlodipine | 5            | 1                          | 1                            | <b>0</b>                   |
| Nicardipine                          | 5            | 0                          | 1                            | <b>0</b>                   |
| <b>Diltiazem</b>                     | 3            | 2                          | 5                            | 4                          |
| <b>Verapamil</b>                     | 4            | 4                          | 5                            | <b>5</b>                   |

The DHP column of zeros is the point: they don't touch the AV node.

## THREE WORDS THE LECTURE LEANS ON

- **Inotrope** — contractility
- **Chronotrope** — heart rate (SA node)
- **Dromotrope** — AV-node conduction

## WHICH $\text{Ca}^{2+}$ CHANNEL, AND WHERE

| TYPE                             | ACTIVATION              | LOCATION                                         | FUNCTION                                                |
|----------------------------------|-------------------------|--------------------------------------------------|---------------------------------------------------------|
| <b>L</b> ( <i>long-lasting</i> ) | HVA, slow inactivation  | cardiac myocytes, smooth muscle, endocrine cells | <b>excitation–contraction coupling</b> — the CCB target |
| <b>T</b> ( <i>transient</i> )    | LVA, rapid inactivation | neurons, cardiac & smooth muscle                 | pacemaking, repetitive firing                           |
| <b>N</b> ( <i>neuronal</i> )     | HVA, moderate           | nerve terminals                                  | neurotransmitter release                                |
| <b>P/Q · R</b>                   | HVA                     | nerve terminals, neuronal bodies                 | transmitter release, repetitive firing                  |

*Ligand-gated  $\text{Ca}^{2+}$  channels ( $\text{IP}_3$  receptor, ryanodine receptor) sit on the SR and are not CCB targets.*

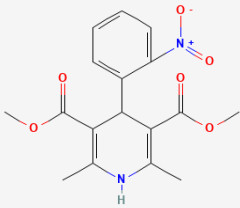
# Dihydropyridines (DHP)

## MECHANISM

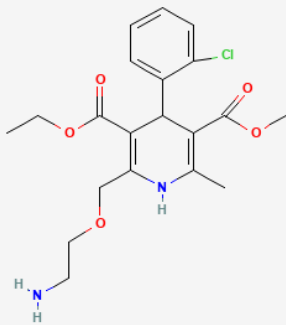
Binds the **transmembrane segments of domains III and IV** of the  $\alpha_1$  subunit. **Highly selective for vascular smooth muscle** — at usual doses the heart is barely touched.

→ **Downstream:** arteriolar dilation (*arterioles far more sensitive than veins*) → ↓ TPR → ↓ BP → **The catch:** the BP drop triggers a **baroreflex** → **reflex tachycardia & positive inotropy** ★ — which is why DHPs are not the drug for an arrhythmia.

## THE SHARED CORE • WHAT CHANGES

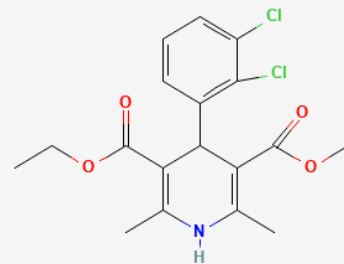


**Nifedipine — the DHP prototype.** a dihydropyridine ring with an aryl group hanging perpendicular to it. ● that geometry is what makes DHPs vascular-selective while verapamil and diltiazem hit the heart



**Amlodipine**

a basic charged side chain → sticks in the membrane  
→ t<sub>1/2</sub> 30–50 h, no reflex tachycardia



**Felodipine**

two Cl on the aryl ring → even more vascular-selective

## MEDCHEM

**SAR** ★ **The 1,4-dihydropyridine ring must stay exactly as it is** - oxidise it to a pyridine or reduce it to a piperidine and the activity is **gone**. That is not trivia: CYP oxidation of the DHP ring to the pyridine *is* the inactivating metabolic step for the whole class.

**SAR** The geometry is specific. **N1 and the phenyl para position must be unsubstituted**; the C4 aryl ring has to sit **perpendicular** to the DHP ring (a bulky ortho group locks it there, which is why **ortho beats meta** for EWGs like Cl, F, NO<sub>2</sub>); esters at C3 and C5 optimise activity, and making those two esters **different** creates a chiral centre at C4 that improves selectivity. ⚠ Swap the C3 ester for a nitro group and the drug becomes a channel **activator** instead of a blocker.

**Physicochem** ● **Amlodipine's long, gentle action is a charge story.** Most DHPs are **neutral** - the 1,4-DHP ring delocalises the nitrogen's electrons, so it isn't basic. Amlodipine carries a separate ionisable amine (**pKa 8.9**), so it is **positively charged at physiological pH**, which raises binding affinity and slows everything down: slow onset, long duration, no baroreflex tachycardia. ⚠ Practical consequence: mixing a basic CCB with sodium bicarbonate raises the pH and **precipitates the drug**.

## DRUGS

| DRUG                                   | WHAT IT TREATS                                                                                        |
|----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Nifedipine</b>                      | hypertension · angina ( <i>stable, Prinzmetal's, unstable</i> ) · <b>Raynaud's — the usual choice</b> |
| <b>Amlodipine</b>                      | hypertension · angina — <i>once-daily, half-life 35–45 h</i>                                          |
| <b>Felodipine</b>                      | hypertension · angina                                                                                 |
| <b>Nicardipine</b>                     | hypertension · angina                                                                                 |
| <b>Lercanidipine</b>                   | hypertension — <b>nephroprotective</b> , less peripheral edema                                        |
| <b>Manidipine</b> ( <i>L/T-type</i> )  | hypertension — less vasodilatory edema                                                                |
| <b>Cilnidipine</b> ( <i>L/N-type</i> ) | hypertension — also blocks sympathetic N-type channels                                                |
| <b>Lacidipine</b>                      | hypertension                                                                                          |
| <b>Nimodipine</b>                      | <b>ischemic neurological deficit after subarachnoid hemorrhage</b>                                    |

## USES

**Hypertension** — arterial vasodilation ↓ TPR · **Angina** — coronary vasodilation ↑ O<sub>2</sub> supply · **Raynaud's phenomenon** — ↑ blood flow to extremities (nifedipine) · **Post-SAH ischemic deficit** — nimodipine · Migraine prevention

## ADVERSE EFFECTS

- **Vascular:** flushing, **headache, peripheral/ankle edema** ★, rash (*cutaneous vasculitis*)
- **Reflex tachycardia** — the mirror image of the non-DHPs
- Hypotension
- **Non-vascular smooth muscle:** gingival hyperplasia · aggravated GI reflux · urinary retention

## INTERACTIONS

- All are **hepatic CYP3A4 substrates** — expect CYP3A4 inhibitors/inducers to move the levels
- No active metabolites (*unlike the non-DHPs*)

## WORTH KNOWING ABOUT INDIVIDUAL AGENTS

- **Amlodipine** — charged at physiologic pH → strong electrostatic binding to membrane phospholipid, forming a membrane reservoir > **10,000× the surrounding aqueous concentration**. That's the long duration. Pleiotropic: ↑ NO production, antioxidant, regulates membrane fluidity & cholesterol deposition.
- **Lercanidipine** — highly lipophilic, stored in the membrane → **slow onset, persistent effect**; short plasma half-life but long duration. Dilates **both afferent and efferent** arterioles (nephroprotection); antioxidant; LVH regression reported.
- **Manidipine (L/T)** — T-type channels dominate microvessels; blocking both **equalizes hydrostatic pressure across the capillary bed** via equal arteriolar *and* venular dilation → **less vasodilatory edema**.
- ⚠ **Enantiomers matter:** (S)-amlodipine >1000× and (S)-felodipine >13× stronger than their (R) forms.

# Phenylalkylamine — Verapamil

## MECHANISM

Binds **transmembrane segment 6 of domain IV**. The **most cardioselective** CCB — strongest effect on SA and AV nodal tissue, where recovery of the slow  $\text{Ca}^{2+}$  channel is what sets the rate.

→ **Downstream:** ↓ contractility (*negative inotropy*) → ↓ CO → ↓ BP · ↓ SA rate → **bradycardia** · ● ↓ **AV conduction** → **terminates supraventricular arrhythmias** ★ → **Vasodilation is real but secondary**, so reflex tachycardia barely appears.

## DRUGS

| DRUG             | WHAT IT TREATS                                                                                                                                                   |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Verapamil</b> | <b>paroxysmal SVT</b> · atrial fibrillation / flutter ( <i>rate control</i> ) · angina · hypertension · <b>hypertrophic cardiomyopathy</b> · migraine prevention |

## USES

**Paroxysmal SVT** and **AF / atrial flutter** — the defining indication · **Angina** — ↓ HR and contractility cut  $\text{O}_2$  demand, plus coronary dilation · **Hypertension** — ↓ TPR *and* ↓ CO · **Hypertrophic cardiomyopathy** · Migraine prophylaxis

## ADVERSE EFFECTS

- **Bradycardia** · **heart block** · **arrhythmia** — the conduction effects ★
- **Heart failure** and hypotension — from the negative inotropy
- ★ **Constipation** — the classic verapamil ADR (*non-vascular smooth muscle*)
- Aggravated GI reflux · urinary retention · gingival hyperplasia
- Flushing, headache, edema (*milder than DHP*)

## INTERACTIONS

- ● **Digoxin** — verapamil **blocks P-glycoprotein** → ↓ renal clearance of digoxin → toxicity risk ★
- P-gp blockade also **partially reverses multidrug resistance** in cancer cells
- CYP3A4 substrate; has an **active metabolite (norverapamil)**
- ⚠ (S)-verapamil is ~20× more potent than (R) for negative dromotropy on the AV node

## CAUTIONS

- ⚠ **Heart failure** — the negative inotropy can decompensate it
- ⚠ Existing bradycardia or AV block

# Benzothiazepine — Diltiazem

## MECHANISM

Binds the **cytoplasmic bridge between domains III and IV** — a third, distinct site. Sits **between** DHP and verapamil: real nodal suppression, less negative inotropy.

→ **Downstream:** ↓ SA automaticity (*as strong as verapamil*) · ↓ AV conduction (*less than verapamil*) · moderate vasodilation · mild negative inotropy

## DRUGS

| DRUG             | WHAT IT TREATS                                                                                                                                                              |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diltiazem</b> | angina ( <i>stable, Prinzmetal's, unstable</i> ) · <b>paroxysmal SVT</b> · atrial fibrillation / flutter · hypertension · hypertrophic cardiomyopathy · migraine prevention |

## USES

**Angina** — the balanced profile suits it: coronary dilation ↑ supply while ↓ HR and contractility ↓ demand · **SVT · AF / flutter · Hypertension** · Hypertrophic cardiomyopathy · Migraine prophylaxis

## ADVERSE EFFECTS

- **Bradycardia, heart block, arrhythmia** — less than verapamil but real
- Hypotension, heart failure
- Flushing, headache, edema, rash
- Gingival hyperplasia · aggravated GI reflux · urinary retention

## INTERACTIONS

- **CYP3A4** substrate, with an **active metabolite**

## CAUTIONS

- ⚠ Heart failure, bradycardia, AV block — same logic as verapamil, lower magnitude



# Other Non-Dihydropyridines

## MECHANISM

Same L-type blockade, different chemical families — the lecture lists them for completeness rather than working through them.

→ **Downstream:** as for the non-DHP group above

## DRUGS

| DRUG                                              | WHAT IT TREATS                                                |
|---------------------------------------------------|---------------------------------------------------------------|
| <b>Bepridil</b> ( <i>diarylaminopropylamine</i> ) | <b>refractory angina</b>                                      |
| <b>Cinnarizine</b> ( <i>diphenylpiperazine</i> )  | <i>class listed only — no indication given in the lecture</i> |
| <b>Flunarizine</b> ( <i>diphenylpiperazine</i> )  | <i>class listed only — no indication given in the lecture</i> |

# Comparison & Clinical Use

## THE FOUR PROTOTYPES, PHARMACOKINETICALLY

|                          | NIFEDIPINE     | AMLODIPINE     | VERAPAMIL                          | DILTIAZEM      |
|--------------------------|----------------|----------------|------------------------------------|----------------|
| <b>Bioavailability</b>   | 45–75%         | 60–65%         | 25–35%                             | 40%            |
| <b>Protein binding</b>   | >90%           | >95%           | 90%                                | 80–85%         |
| <b>Elimination</b>       | hepatic CYP3A4 | hepatic CYP3A4 | hepatic CYP3A4                     | hepatic CYP3A4 |
| <b>Active metabolite</b> | no             | no             | <b>yes</b> ( <i>norverapamil</i> ) | <b>yes</b>     |
| <b>Half-life</b>         | 2–5 h          | <b>35–45 h</b> | 3–7 h ( <i>SR ~11</i> )            | 4–6 h          |
| <b>Onset</b>             | 20 min         | —              | 30 min                             | 30–60 min      |
| <b>Dosing</b>            | TID–QID        | <b>daily</b>   | TID–QID                            | TID–QID        |

## WHICH CCB FOR WHICH CONDITION

| INDICATION                           | DHP ( <i>NIFEDIPINE, FELODIPINE, AMLODIPINE</i> ) | VERAPAMIL · DILTIAZEM |
|--------------------------------------|---------------------------------------------------|-----------------------|
| Hypertension                         | ✓                                                 | ✓                     |
| Raynaud's phenomenon                 | ✓                                                 | —                     |
| Angina — stable                      | ✓                                                 | ✓                     |
| Angina — Prinzmetal's                | ✓                                                 | ✓                     |
| Angina — unstable                    | ✓                                                 | ✓                     |
| <b>Paroxysmal SVT</b>                | —                                                 | ✓ ★                   |
| <b>Atrial fibrillation / flutter</b> | —                                                 | ✓ ★                   |
| Hypertrophic cardiomyopathy          | —                                                 | ✓                     |
| Migraine ( <i>prevention</i> )       | ✓                                                 | ✓                     |
| Post-SAH ischemic deficit            | ✓ ( <i>nimodipine</i> )                           | —                     |
| Cluster headache prophylaxis         | ✓                                                 | ✓                     |

## WHY IT WORKS, IN ONE LINE EACH

**Angina** — coronary vasodilation ↑ O<sub>2</sub> **supply**; ↓ HR and contractility ↓ O<sub>2</sub> **demand** · **Hypertension** — BP = CO × TPR; arterial dilation ↓ TPR, fewer impulses + less contractility ↓ CO · **Raynaud's** — arterial dilation ↑ flow to fingers and toes, which spasm on cold or stress · **SVT** — AV-node conduction block terminates the re-entry

## THE ADR MAP — FOUR BUCKETS

- **Vascular smooth muscle:** flushing · edema · headache · rash (*cutaneous vasculitis*)
- **Myocardial:** hypotension · heart failure · bradycardia
- **Conduction:** arrhythmia · heart block
- **Non-vascular smooth muscle:** constipation (*verapamil*) · aggravated GI reflux · urinary retention · gingival hyperplasia

*Principle behind this: [[pharm-drug-interactions]] — Grapefruit juice inhibits gut-wall CYP3A4 → oral CCB levels rise. Also verapamil, which inhibits P-gp and raises hepatic blood flow.*

*Principle behind this: [[pharm-drug-toxicity]] — Nifedipine's hypotension tracks concentration — Type A toxicity as pure extension of the intended effect.*

*Source: CCBs.pdf — อ.กฤษณ์, PYCH304 Pharmacology I, 18 slides. Structures omitted.*

---

# The Renin–Angiotensin System

Three drug classes all lower BP by attacking **the same chain at three different points**. They are not interchangeable: *where* you cut decides which peptides pile up behind the cut, and that decides the side-effect profile.

**Angiotensinogen → *renin* → Ang I → *ACE* → Ang II → AT1 → vasoconstriction · aldosterone · remodelling → ↑ BP**

## THE THREE PLAYERS

- **Angiotensinogen** —  $\alpha$ 2-globulin precursor, mostly from **liver**. Stimulated by inflammation, insulin, estrogens, glucocorticoids, thyroid hormone.
- **Renin** — aspartyl protease from the **juxtaglomerular apparatus**. ★ **Its blood level is the rate-limiting step of the whole system.**
- **ACE (= kininase II)** — dicarboxypeptidase; highest concentration in **lung vascular endothelium**. Two jobs: makes Ang II **and** destroys **bradykinin**. That second job is the whole ACEI side-effect story.

## WHAT CONTROLS RENIN RELEASE

| PATHWAY                    | TRIGGER                                               | EFFECT  |
|----------------------------|-------------------------------------------------------|---------|
| <b>Renal baroreceptor</b>  | ↓ perfusion pressure in preglomerular vessels         | ↑ renin |
| <b>Macula densa</b>        | ↓ Cl <sup>-</sup> sensed at the macula densa          | ↑ renin |
| <b>β1 adrenergic</b>       | β1 activation on JG cells → ↑ cAMP                    | ↑ renin |
| <b>Feedback (negative)</b> | Ang II on <b>AT1</b> at JG cells · Ang II-driven ↑ BP | ↓ renin |

## AT1 VS AT2 — OPPOSITE SIGNS

|            | AT1R                                                                                     | AT2R                                                                                     |
|------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| net effect | <b>hypertensive</b>                                                                      | <b>antihypertensive</b>                                                                  |
| actions    | vasoconstriction · fibrosis · VSMC inflammation · oxidative stress · cardiac hypertrophy | vasodilation · antifibrotic · anti-inflammatory · ↓ oxidative stress · antiproliferative |

*Also in the expanded RAS: Mas (Ang 1–7) and MrgD (alamandine) — both vasodilator/antifibrotic; AT4 — neuroprotection, cognition; PRR (pro-renin receptor) — hypertrophy, fibrosis, apoptosis.*

## ★ THE COMPARISON TABLE — WHAT EACH CLASS DOES TO THE CHAIN

|                                    | DRI            | ACE INHIBITOR  | ARB               |
|------------------------------------|----------------|----------------|-------------------|
| Plasma renin (PRC)                 | ↑              | ↑              | ↑                 |
| <b>Plasma renin activity (PRA)</b> | ↓              | ↑              | ↑                 |
| Angiotensin I                      | ↓              | ↑              | ↑                 |
| <b>Angiotensin II</b>              | ↓              | ↓              | ↑                 |
| Angiotensin (1–7)                  | ↓              | ↑              | ↑                 |
| <b>Bradykinin</b>                  | ↔              | ↑              | ↔                 |
| AT1R                               | not stimulated | not stimulated | <b>blocked</b>    |
| AT2R                               | not stimulated | not stimulated | <b>stimulated</b> |

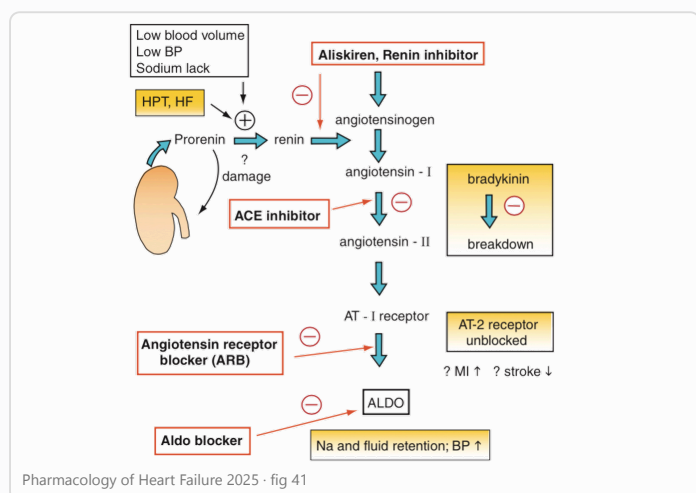
★ **PRC up but PRA down is the DRI fingerprint** — the enzyme protein accumulates, its activity doesn't.

★ **ARBs are the only class where Ang II rises** — that surplus is what drives AT2.

## THE CLASS-WIDE RULES

- **Stop all three in pregnancy** — fetal & neonatal injury, including fetal death (2nd & 3rd trimester).
- **Bilateral renal artery stenosis** (or stenosis in a single remaining kidney) → acute renal failure. Ang II constricts the **efferent** arteriole more than the afferent to hold GFR up; remove Ang II and GFR collapses.
- **Hyperkalemia** — less aldosterone means less K<sup>+</sup> excretion. Every class does it.
- **Never stack them: ACEI + ARB + DRI combinations add toxicity, not benefit.**

## OVERVIEW MAPS



# ACE Inhibitor

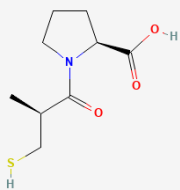
## MECHANISM

Blocks **ACE**, so Ang I is not converted to **Ang II**. Because ACE *is* kinase II, the same block also **saves bradykinin** — a vasodilator that would normally be destroyed.

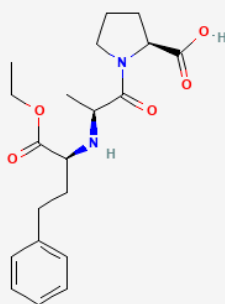
→ **Downstream:** ↓ Ang II → ↓ vasoconstriction, ↓ aldosterone → ↓ **BP** · ↑ natriuresis → **Bradykinin** ↑: extra antihypertensive + protective effect — *and* the cough and the angioedema → **No negative feedback:** ↑ renin release and ↑ Ang I → more **Ang(1–9) and Ang(1–7)** vasodilator peptides

*Most are ester prodrugs — 100–1000× less potent than the active acid, but far better oral bioavailability. Captopril and lisinopril are not prodrugs.\*\**

## THE SHARED CORE · WHAT CHANGES

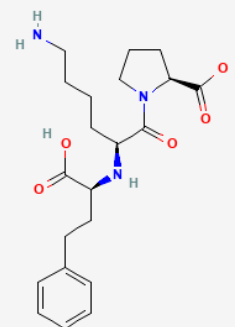


**Captopril — the first one.** designed from a snake-venom peptide: a zinc-binding group + a proline. ● the –SH is what grabs ACE's zinc, and it is also blamed for the rash and taste disturbance



**Enalapril**

–SH replaced by a carboxyl (no rash), and it is an ● **ESTER prodrug** — the liver has to hydrolyse it to enalaprilat



**Lisinopril**

a lysine side chain → water-soluble, NOT a prodrug, renal clearance

## MEDCHEM

**SAR** ACE inhibitors were **designed, not discovered**, to a six-point specification: a **zinc-binding group**, a **negative charge** for the enzyme's cationic site, a hydrogen-bond donor and acceptor, a hydrophobic group, **L-amino-acid (S) stereochemistry**, and the right spacing between them. The zinc-binder splits the class into three: **sulfhydryl** (captopril), **dicarboxylate** (most of them), **phosphinate** (fosinopril). The COOH is non-negotiable - it is what binds the cationic site, and it mimics the transition state of peptide hydrolysis.

**SAR** ● **Captopril's problems are all traceable to its –SH.** Sulfhydryl binds zinc better than anything else - and it also causes the **rash and metallic taste/loss of taste**, and it **forms disulfides and dimers**, which shortens the duration of action. Every later ACE inhibitor traded a little zinc affinity for not being a thiol.

**Prodrug** ★ **Why almost all of them are prodrugs, and why lisinopril isn't.** Esterifying the COOH (or the phosphinate) is what makes the molecule orally absorbable - enalapril → enalaprilat, and so on.

**Lisinopril is a di-zwitterion with a net neutral charge**, so it is absorbed as-is; no prodrug needed. Lisinopril and enalaprilat are then excreted unchanged.

**Physicochem** All are amphoteric except captopril and fosinopril, which are simply acidic (pKa 2.5–3.5).

## DRUGS

| DRUG                                          | WHAT IT TREATS                                                        |
|-----------------------------------------------|-----------------------------------------------------------------------|
| <b>Captopril</b>                              | hypertension · heart failure — active drug, $t_{1/2}$ ~2 h (shortest) |
| <b>Enalapril</b>                              | hypertension · heart failure — prodrug of <b>enalaprilat</b>          |
| <b>Lisinopril</b>                             | hypertension · heart failure — active drug, $t_{1/2}$ ~12 h           |
| <b>Ramipril · Benazepril · Quinapril</b>      | hypertension · heart failure · post-MI · diabetic nephropathy         |
| <b>Perindopril · Trandolapril · Moexipril</b> | hypertension — long-acting prodrugs                                   |
| <b>Fosinopril</b>                             | hypertension — the <b>phosphorus-containing</b> one                   |

## USES

**Hypertension** — ↓ systemic vascular resistance and mean/systolic/diastolic BP, *except* high BP from **primary aldosteronism** · **Left ventricular systolic dysfunction / heart failure** — delays progression, ↓ sudden death and MI, ↓ hospitalisation · **Acute myocardial infarction** · **Diabetes mellitus with renal failure** — prevents/delays progression of renal disease · **Scleroderma renal crisis** — improves survival · Patients **at risk of cardiovascular events**

## ADVERSE EFFECTS

- ★ **Dry cough** — 5–20%. From accumulated **bradykinin, substance P, prostaglandins** in the lung. The signature ACEI ADR and the usual reason to switch to an ARB.
- **Angioedema** — 0.1–0.5%. Swelling of nose, throat, mouth, glottis, larynx, lips, tongue. Airway-threatening.
- **Initial-dose hypotension** — a steep BP fall after the very first dose
- **Hyperkalemia** — rare if renal function is normal
- **Acute renal failure** — bilateral renal artery stenosis · single-kidney stenosis · heart failure · volume depletion
- ● **Fetopathic potential** — fetal/neonatal injury and death, 2nd & 3rd trimester
- Maculopapular **skin rash** · neutropenia · anemia

## INTERACTIONS

| AGENT                                  | WHAT HAPPENS                       |
|----------------------------------------|------------------------------------|
| <b>NSAIDs</b>                          | ↓ the antihypertensive response    |
| <b>K<sup>+</sup>-sparing diuretics</b> | ● worsen ACEI-induced hyperkalemia |
| <b>K<sup>+</sup> supplements</b>       | ● worsen ACEI-induced hyperkalemia |
| <b>Digoxin</b>                         | ACEIs ↑ plasma digoxin             |
| <b>Lithium</b>                         | ACEIs ↑ plasma lithium             |
| <b>Capsaicin</b>                       | worsens the ACEI cough             |

## CAUTIONS & CONTRAINDICATIONS

- ● **Contraindicated:** renal artery stenosis · pregnancy
- ⚠ **Caution:** impaired renal function · hypovolemia

# Angiotensin Receptor Blocker (ARB)

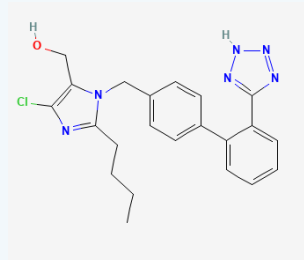
## MECHANISM

**Competitive antagonist at the AT1 receptor** — high affinity, **>1,000-fold selective for AT1 over AT2**.

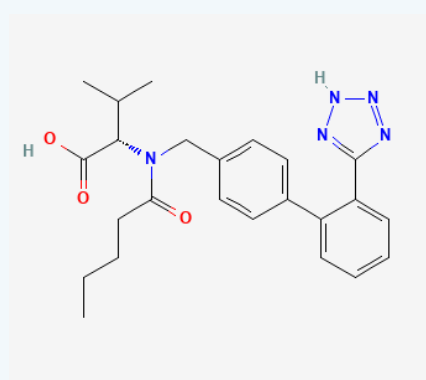
Blocks the *receptor*, not the enzyme, so it doesn't care how the Ang II was made.

→ **Downstream:** ↓ vascular smooth muscle contraction · ↓ rapid & slow pressor responses · ↓ aldosterone secretion · ↓ noradrenergic neurotransmission · ↓ cardiac hypertrophy & hyperplasia → **Ang II rises** and, with AT1 blocked, spills onto **AT2** → extra vasodilation and antifibrosis

## THE SHARED CORE · WHAT CHANGES

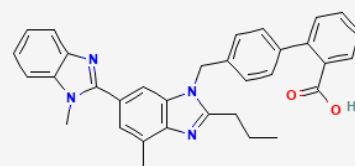


**Losartan — the first sartan.** biphenyl + tetrazole. The tetrazole is a carboxylic-acid mimic that survives metabolism, which is why sartans are oral and long-acting



**Valsartan**

the imidazole opened into an acyl amino acid — no prodrug step needed



**Telmisartan**

a big lipophilic double benzimidazole →  $t_{1/2}$  ~24 h, the longest of the class

## MEDCHEM

**SAR** A sartan is a **drawn copy of angiotensin II's business end**, residue by residue: the **n-butyl chain mimics Ile**, the **imidazole (or bioisostere) mimics His**, and the acidic group mimics **Asp or the Tyr phenol**. The **tetrazole and the COOH have to sit ortho to each other** on the biphenyl - move them and the affinity goes. The  $IC_{50}$  jump the deck prints is from 15  $\mu\text{mol/L}$  to 0.0013  $\mu\text{mol/L}$  across that optimisation: roughly ten-thousand-fold.

**Physicochem** Why **tetrazole rather than a second carboxylic acid**. Tetrazole has  $pK_a \approx 6$  (>90% ionised at 7.4) versus COOH's 3–4 (fully ionised), and its **charge is spread over the ring** rather than concentrated - better binding affinity, plus better stability, lipophilicity and bioavailability than the acid it replaced.

**Prodrug** Candesartan cilexetil and olmesartan medoxomil are esters hydrolysed during absorption - the active drug never exists in the tablet. Azilsartan's 5-oxo-oxadiazole is a further tweak for affinity, lipophilicity and bioavailability.

**Impurity** ● The valsartan/NDMA recalls came from the tetrazole ring's synthesis, and this is the deck's own worked example. Building the tetrazole uses **sodium azide**, which is quenched with **sodium nitrite + acid = nitrous acid**. Nitrous acid plus any dialkylamine (from the drug or from a formamide solvent) gives a **nitrosamine - NDMA, a carcinogen**. Not a manufacturing accident so much as a predictable consequence of that route.



## WHY ARBS WERE DEVELOPED — THE FOUR HOLES IN ACE INHIBITION

1. Competitive ACE inhibition causes a **reactive rise in renin and Ang I** that can overcome the block.
2. ★ **Ang II is also made by non-ACE routes** — **chymase** (heart, kidney), **tonin**, **cathepsin G** — “ACE escape.” An ACEI cannot touch these; an ARB can.
3. ACE is **nonspecific** — it also handles **bradykinin**, **enkephalin**, **neurotensin**, **substance P**, so blocking it makes all of them accumulate.
4. Those accumulated substrates are exactly what causes the ACEI-specific ADRs (cough, angioedema).

## DRUGS

| DRUG                                                            | WHAT IT TREATS                                                                                                                                         |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Losartan</b>                                                 | hypertension · <b>diabetic nephropathy</b> · <b>stroke prophylaxis in HTN + LVH</b> — also blocks the thromboxane A2 receptor → ↓ platelet aggregation |
| <b>Valsartan</b>                                                | hypertension · <b>heart failure / post-MI LV dysfunction</b>                                                                                           |
| <b>Candesartan</b>                                              | hypertension · <b>heart failure</b>                                                                                                                    |
| <b>Irbesartan</b>                                               | hypertension · <b>diabetic nephropathy</b>                                                                                                             |
| <b>Telmisartan</b>                                              | hypertension — longest $t_{1/2}$ (~24 h)                                                                                                               |
| <b>Azilsartan</b> ·<br><b>Olmesartan</b> ·<br><b>Eprosartan</b> | hypertension                                                                                                                                           |

## USES

**Hypertension** — all ARBs are approved; BP-lowering **comparable to ACEIs with a better ADR profile** ·  
**Heart failure** — valsartan, candesartan; also ↓ cardiovascular mortality in stable patients after MI ·  
**Renoprotection in diabetes** — irbesartan, losartan approved for diabetic nephropathy · **Stroke prophylaxis** — losartan, in hypertension with left ventricular hypertrophy

## ADVERSE EFFECTS

- Generally **well tolerated** — ★ **cough and angioedema are both less frequent than with ACEIs** (*no bradykinin accumulation*)
- **Hyperkalemia**
- **Initial-dose hypotension**
- **Acute renal failure** — where renal function depends on the RAS (e.g. renal artery stenosis): oliguria, progressive azotemia
- **Angioedema** — still possible, just rarer
- ● **Teratogenic** — discontinue in pregnancy

## CAUTIONS & CONTRAINDICATIONS

- ● **Contraindicated:** renal insufficiency · pregnancy
- ⚠ ACEI vs ARB in one line: **ARBs suppress AT1 activation more completely** (they cover ACE-independent Ang II) **and leave AT2 free to be stimulated**; ACEIs uniquely raise **bradykinin** — which is both their bonus effect and their ADR profile.

# Direct Renin Inhibitor (DRI)

## MECHANISM

A **potent competitive inhibitor of renin** — blocks the cleavage of **Ang I from angiotensinogen**, i.e. the **first and rate-limiting** step of the entire cascade.

→ **Downstream:** ↓ Ang I and ↓ Ang II → ↓ BP → **Fingerprint:** ★ ↑ **plasma renin concentration** but ↓ **plasma renin activity** — the only class that does this → **Bradykinin unchanged** — ACE is untouched, which is why cough is much rarer than with ACEIs

## DRUGS

| DRUG             | WHAT IT TREATS                                                                                  |
|------------------|-------------------------------------------------------------------------------------------------|
| <b>Aliskiren</b> | hypertension — a low-molecular-weight nonpeptide; the <b>only DRI approved for clinical use</b> |

*First-generation renin inhibitors (enalkiren, zankiren, remikiren, CGP38560A) lowered Ang II but died in trials — low potency, poor bioavailability, short  $t_{1/2}$ .*

## USES

**Hypertension** — the only indication. ⚠ **Therapeutic value unclear: no evidence of superiority over ACEIs or ARBs.**

## ADVERSE EFFECTS

*Well tolerated — adverse events mild or comparable to placebo, no gender difference.*

- Mild GI symptoms — abdominal pain, dyspepsia
- Headache · dizziness · fatigue
- Nasopharyngitis · upper respiratory tract infection
- Back pain
- Angioedema
- Cough — **much less common than with ACE inhibitors**

## CONTRAINDICATIONS

- ● **Diabetic nephropathy · pregnancy · renal insufficiency**

# ARNI — Angiotensin Receptor–Neprilysin Inhibitor

## MECHANISM

**Valsartan + sacubitril** — an ARB bolted to a **neprilysin** inhibitor. Neprilysin is a zinc-dependent metallopeptidase that degrades **ANP, BNP, bradykinin** and helps break down Ang II; inhibiting it lets the natriuretic peptides survive.

→ **Downstream:** AT1 blockade (valsartan) + natriuretic, vasodilatory, antiproliferative effects (sacubitril)

## DRUGS

| DRUG                                 | WHAT IT TREATS                                                       |
|--------------------------------------|----------------------------------------------------------------------|
| <b>Valsartan/sacubitril</b> (LCZ696) | <b>heart failure with reduced ejection fraction</b> — first in class |

## CONTRAINDICATIONS

- **Never with an ACE inhibitor or an ARB** — do not co-prescribe
- **History of angioedema** on an ACEI or ARB

*Principle behind this: [[pharm-drug-interactions]] — Trimethoprim behaves like a K-sparing diuretic → hyperkalaemia with an ACEI or ARB. A "harmless antibiotic" that isn't.*

*Source: ACE inhibitors, ARBs & DRIs.pdf — อ.กฤษณ์, PYCH304 Pharmacology I, 20 slides. Structures omitted.*

# Antihypertensive Agents

The **most common cardiovascular disease**, rising with age (*50% of people aged 60–69 have it*). Defined as a **sustained BP  $\geq 140/90$  mmHg** on repeated, reproducible measurements. This page is the **map** — physiology, classification, and the classes not covered elsewhere.

**BP = CO  $\times$  TPR** — every antihypertensive drug attacks one of those two terms, and nothing else.

## ESSENTIAL VS SECONDARY

- **Essential (>90%)** — no specific cause found. **Cannot be cured, only controlled.**
- **Secondary (<10%)** — an underlying disease or drug is the cause. **Can be mitigated or cured** by correcting that condition.

## BP CATEGORIES (THAI GUIDELINE 2019 · ESH/ESC 2018)

| CATEGORY                    | SBP        | DBP        |
|-----------------------------|------------|------------|
| Optimal                     | <120       | <80        |
| Normal                      | 120–129    | 80–84      |
| High normal                 | 130–139    | 85–89      |
| <b>Grade 1 HT</b>           | 140–159    | 90–99      |
| <b>Grade 2 HT</b>           | 160–179    | 100–109    |
| <b>Grade 3 HT</b>           | $\geq 180$ | $\geq 110$ |
| <b>Isolated systolic HT</b> | $\geq 140$ | <90        |

## WHAT ACTUALLY SETS BP — 4 ANATOMICAL SITES

| SITE               | ORGAN      | CONTRIBUTES                           |
|--------------------|------------|---------------------------------------|
| <b>Resistance</b>  | arterioles | arteriolar tone → <b>TPR</b>          |
| <b>Capacitance</b> | venules    | venous tone → filling pressure        |
| <b>Pump output</b> | heart      | heart rate, contractility → <b>CO</b> |
| <b>Volume</b>      | kidneys    | blood volume → CO                     |

Controlled by **CNS/sympathetic nerves** and **renin–angiotensin–aldosterone**.

## TWO CONTROL LOOPS — DIFFERENT TIMESCALES

| LOOP                       | SPEED            | HOW IT WORKS                                                                                                                                                                                                                                                                    |
|----------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Baroreceptor reflex</b> | moment-to-moment | $\uparrow$ BP → baroreceptors in <b>carotid sinus &amp; aortic arch</b> fire → inhibit vasomotor centre → $\downarrow$ sympathetic to heart ( $\downarrow$ HR) + vasodilation → $\downarrow$ CO, $\downarrow$ R. $\downarrow$ BP does the mirror image.                         |
| <b>Kidney / RAAS</b><br>★  | long-term        | $\downarrow$ BP → renin → angiotensin I → (ACE) → <b>angiotensin II</b> → vasoconstriction ( $\uparrow$ TPR) <b>and</b> aldosterone → salt retention ( $\uparrow$ blood volume, $\uparrow$ CO) → BP rises. <b>The kidney is primarily responsible for long-term BP control.</b> |

## THE 4 DRUG FAMILIES — AND WHERE EACH LIVES

| FAMILY                         | HOW IT LOWERS BP                                       | FULL DETAIL                                          |
|--------------------------------|--------------------------------------------------------|------------------------------------------------------|
| <b>1. Diuretics</b>            | ↓Na <sup>+</sup> , ↓blood volume                       | [[drug-class-diuretics]] · HTN-specific notes below  |
| <b>2. Sympatholytics</b>       | ↓TPR, ↓cardiac function, ↑capacitance (venous pooling) | <b>this page</b> — central · neuron-blocking · α · β |
| <b>3. Direct vasodilators</b>  | relax vascular smooth muscle → ↓TPR, ↑capacitance      | <b>this page</b> · CCBs → [[drug-class-ccb]]         |
| <b>4. Angiotensin blockers</b> | ↓TPR, ↓blood volume (potentially)                      | [[drug-class-ace-arb-dri]] · summary below           |

# Diuretics in Hypertension

## WHY THE MECHANISM CHANGES OVER TIME ★

**Initial:** ↓Na<sup>+</sup> & water retention → ↓blood volume → ↓CO → ↓BP (*TPR may actually increase at first*).

**After 6–8 weeks: CO returns toward normal**, and the sustained BP drop is now carried by ↓**peripheral resistance** — Na<sup>+</sup> is believed to raise vascular resistance by increasing **vessel stiffness** and **neural reactivity**.

→ **Downstream:** same BP, two different haemodynamic reasons depending on when you look.

## WHICH DIURETIC, WHICH PATIENT

| DRUG                              | WHAT IT TREATS                                          |
|-----------------------------------|---------------------------------------------------------|
| <b>Hydrochlorothiazide (HCTZ)</b> | mild–moderate HT — <b>10–15 mmHg</b> drop               |
| <b>Furosemide</b>                 | severe HT                                               |
| <b>Spironolactone</b>             | added to avoid excessive <b>K<sup>+</sup> depletion</b> |

## ADVERSE EFFECTS (*HTN-RELEVANT*)

- **Hypokalemia** — every class *except* K<sup>+</sup>-sparing
- ● **Hyperkalemia** with K<sup>+</sup>-sparing agents, **especially alongside ACEIs or ARBs**
- Magnesium depletion
- Impaired glucose tolerance · ↑ serum lipids
- ↑ uric acid → **may precipitate gout**

Full class detail: [[drug-class-diuretics]]

# Centrally Acting Sympatholytics

## MECHANISM

Act **in the brain** by binding **presynaptic  $\alpha_2$ -adrenergic receptors** — reducing sympathetic outflow from the **vasomotor centre in the brain stem**.

→ **Downstream:**  $\downarrow$ CO and  $\downarrow$ TPR — the drug works centrally, the effect is peripheral.

## METHYLDOPA — THE PRO-DRUG CHAIN ★

An analog of **L-dopa**, so it hijacks the NE synthesis pathway:

**Methyldopa** → (*L-aromatic amino acid decarboxylase*) → (*dopamine  $\beta$ -oxidase*) →  **$\alpha$ -methylnorepinephrine** → substitutes for NE in the adrenergic nerve vesicle → acts as an **agonist at presynaptic  $\alpha_2$**  →  $\downarrow$ NE release →  $\downarrow$ vasoconstrictor signals.

→ **Downstream:**  $\downarrow$ TPR, with a **variable** reduction in HR and CO.

## DRUGS

| DRUG              | WHAT IT TREATS                                                                         |
|-------------------|----------------------------------------------------------------------------------------|
| <b>Methyldopa</b> | ★ <b>hypertension in pregnancy</b> — the modern main use, chosen for efficacy + safety |
| <b>Clonidine</b>  | resistant HT not responding to other combinations                                      |
| <b>Guanabenz</b>  | hypertension                                                                           |
| <b>Guanfacine</b> | hypertension                                                                           |

## USES

**Methyldopa** — widely used in the past, now primarily **pregnancy-associated hypertension**; generally given **with a diuretic** because it expands plasma volume · Efficacy ~**20/13 mmHg** in mild HT, ~**38/22 mmHg** in severe HT · **Clonidine / guanabenz / guanfacine** stimulate the  **$\alpha_2A$  subtype** in the brain stem →  $\downarrow$ CO ( $\downarrow$ HR + *relaxation of capacitance vessels*) and  $\downarrow$ TPR · Clonidine ~**25/15 mmHg** at maximum tolerated dose · **Not a leading monotherapy** — limited evidence that it reduces cardiovascular outcomes.

## ADVERSE EFFECTS

**Methyldopa - Sedation** — the most common, worst at onset, transient - ★ **(+) Coombs test** in  $\geq 20\%$  of patients treated for a year — reversible - Dry mouth · parkinsonian signs · hyperprolactinemia - May precipitate **bradycardia and sinus arrest**

**Clonidine group - Dry mouth and sedation** — the common two - GI disturbance, weight gain, nightmares, insomnia, anxiety, depression - Urinary retention, impotence

## CAUTIONS

- **Never stop clonidine suddenly** ★ — abrupt discontinuation causes a **life-threatening hypertensive crisis**, driven by rebound sympathetic activity

# Adrenergic Neuron Blockers

## MECHANISM

Act at the **adrenergic nerve ending**, preventing normal physiologic **release of NE** from postganglionic sympathetic neurons. Two different sabotage routes:

1. **Guanethidine** — replaces NE inside transmitter vesicles as a "**false neurotransmitter**" → gradual **depletion of NE stores**
  2. **Reserpine** — blocks uptake & storage of biogenic amines into synaptic vesicles by interfering with **VMAT**
- **Downstream:** guanethidine → ↓ arteriolar **and** venous tone (↓TPR) · reserpine → depletion of **NE, DA and 5-HT** centrally *and* peripherally → ↓CO and ↓TPR

## DRUGS

| DRUG                | WHAT IT TREATS                                                           |
|---------------------|--------------------------------------------------------------------------|
| <b>Guanethidine</b> | hypertension — peripheral only                                           |
| <b>Reserpine</b>    | hypertension — an alkaloid from the roots of <i>Rauwolfia serpentina</i> |

## ADVERSE EFFECTS

**Guanethidine** — postural hypotension · diarrhea · sexual dysfunction (*impaired ejaculation*)

**Reserpine** — mostly from CNS + GI actions: - **Brain:** sedation, lassitude, **depression** - **GI:** mild diarrhea, cramps, ↑ gastric acid secretion

## INTERACTIONS

- ★ Guanethidine **needs the NE transporter (NET)** to get into the neuron — so anything blocking catecholamine uptake **abolishes its antihypertensive effect:** cocaine · amphetamine · TCAs



# $\alpha$ - and Mixed $\alpha\beta$ -Blockers

## MECHANISM

Block  $\alpha_1$  receptors on vessels → arteriolar and venous dilation → ↓TPR.

→ **Downstream:** ↓BP with **less reflex tachycardia** than non-selective  $\alpha$ -blockade, because presynaptic  $\alpha_2$  is left intact ( $\alpha_1 \gg \alpha_2$ ) and NE feedback inhibition still works. → **Mixed  $\alpha\beta$  agents:** the  $\beta$ -blockade caps the reflex, so HR barely rises.

## DRUGS

| DRUG              | WHAT IT TREATS                                                     |
|-------------------|--------------------------------------------------------------------|
| <b>Prazosin</b>   | hypertension — $\alpha_1$ selective                                |
| <b>Doxazosin</b>  | hypertension                                                       |
| <b>Terazosin</b>  | hypertension                                                       |
| <b>Labetalol</b>  | hypertension — mixed, $\beta_1 = \beta_2 \geq \alpha_1 > \alpha_2$ |
| <b>Carvedilol</b> | hypertension — mixed                                               |

# β-Blockers in Hypertension

## MECHANISM

Two hits from one drug: block **β<sub>1</sub> on the heart** → ↓CO, and block **β<sub>1</sub> on juxtaglomerular cells** → ↓renin → ↓angiotensin II → ↓aldosterone → ↓Na<sup>+</sup> & water retention → ↓blood volume.

→ **Downstream:** ↓CO · ↓blood volume · ↓peripheral resistance → ↓BP · the signature triad is ↓HR, ↓BP, ↓renin.

## DRUGS

| DRUG                                      | WHAT IT TREATS                                                   |
|-------------------------------------------|------------------------------------------------------------------|
| <b>Propranolol · nadolol · timolol</b>    | HT — <b>non-selective</b> (β <sub>1</sub> and β <sub>2</sub> )   |
| <b>Metoprolol · atenolol · alprenolol</b> | HT — <b>β<sub>1</sub>-selective</b> , ★ possibly safer in asthma |
| <b>Pindolol · acebutolol · penbutolol</b> | HT — <b>with ISA</b> ; ↓BP but only moderately ↓HR               |

## ADVERSE EFFECTS

- ↑ **Triglycerides**, ↓ **HDL** — *the ISA group has little or no effect*
- Masks / worsens hypoglycemia — risk ↑ in **diabetics on insulin**

## INTERACTIONS

- **NSAIDs (e.g. indomethacin)** blunt the antihypertensive effect of propranolol, and probably of the other β-blockers
- Drugs that inhibit **AV conduction** — additive block

## CAUTIONS

- ● **Avoid in asthma · SA or AV nodal dysfunction** · combined with other AV-conduction inhibitors
- ● **Never stop abruptly** ★ — sudden discontinuation gives a "**withdrawal syndrome**"; taper over **10–14 days**

# Direct Vasodilators

## MECHANISM

Relax **vascular smooth muscle** directly — no receptor upstream, no neuron involved.

→ **Downstream:** ↓TPR and ↑capacitance → ↓BP · but the intact **baroreflex fires back** with reflex tachycardia and Na<sup>+</sup>/water retention, which is why these are almost always **combined** with a β-blocker and a diuretic.

## FIVE WAYS TO RELAX A VESSEL

| DRUG                                        | MECHANISM                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hydralazine</b>                          | directly relaxes <b>arteriolar</b> smooth muscle — precise mechanism <b>not fully understood</b> ; may inhibit <b>IP<sub>3</sub>-induced Ca<sup>2+</sup> release</b> from intracellular stores |
| <b>Nitroglycerin · isosorbide dinitrate</b> | release of <b>nitric oxide (NO)</b>                                                                                                                                                            |
| <b>Amlodipine · verapamil · diltiazem</b>   | reduction of <b>Ca<sup>2+</sup> influx</b> → [[drug-class-ccb]]                                                                                                                                |
| <b>Minoxidil · nicorandil</b>               | <b>K<sup>+</sup> channel openers</b> — hyperpolarize the smooth muscle membrane                                                                                                                |
| <b>Fenoldopam</b>                           | <b>dopamine D<sub>1</sub> receptor</b> activation — a peripheral <b>arteriolar</b> dilator                                                                                                     |

## VASCULAR SMOOTH MUSCLE IS THE COMMON TARGET

| DRUG                 | WHAT IT TREATS                                                                                      |
|----------------------|-----------------------------------------------------------------------------------------------------|
| <b>Hydralazine</b>   | hypertension                                                                                        |
| <b>Minoxidil</b>     | hypertension                                                                                        |
| <b>Nitroprusside</b> | hypertension — acts on vascular smooth muscle                                                       |
| <b>Diazoxide</b>     | hypertension — acts on vascular smooth muscle                                                       |
| <b>Fenoldopam</b>    | hypertension — peripheral arteriolar dilation                                                       |
| <b>Trimethaphan</b>  | hypertension — the odd one out: a <b>ganglion blocker</b> at sympathetic ganglia, not a vasodilator |

# Angiotensin Blockade

## MECHANISM

Three points of attack on one cascade: **angiotensinogen** → (**renin**) → **angiotensin I** → (**ACE**) → **angiotensin II** → **vasoconstriction + aldosterone**.

★ **ACE is also kininase II** — so inhibiting it *also* stops **bradykinin** breakdown. Accumulated bradykinin → ↑prostaglandin synthesis → vasodilation → ↓PVR. **That extra vasodilator arm is the ACEI's bonus — and its cough.**

→ **Downstream:** ↓vasoconstriction · ↓aldosterone → ↓Na<sup>+</sup> & water retention · ↓BP

## THE THREE SUBCLASSES

| SUBCLASS                      | DRUGS                                                    | MECHANISM                                                                                                |
|-------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Direct renin inhibitor</b> | <b>Aliskiren</b>                                         | inhibits renin enzyme activity → ↓Ang I <b>and</b> Ang II, ↓aldosterone                                  |
| <b>ACE inhibitors</b>         | <b>Captopril · enalapril · lisinopril · ramipril</b>     | ↓angiotensin II → ↓vasoconstriction, ↓aldosterone, ↑ <b>bradykinin</b>                                   |
| <b>ARBs</b>                   | <b>Losartan · candesartan · irbesartan · telmisartan</b> | block the <b>AT<sub>1</sub> receptor</b> → ↓vasoconstriction, ↓aldosterone — <b>no bradykinin effect</b> |

Full class detail: [[drug-class-ace-arb-dri]]

## ADVERSE EFFECTS (ACEIS AND ARBS TOGETHER)

- **Severe hypotension after the first doses** in **hypovolemic** patients
- Acute renal failure
- **Hyperkalemia**
- ★ **Dry cough** — **ACE inhibitors only** (*bradykinin; ARBs don't do this*)

## INTERACTIONS

- ● **K<sup>+</sup> supplements or K<sup>+</sup>-sparing diuretics** → hyperkalemia

## CAUTIONS

- ● **Contraindicated in pregnancy** ★

# Why It Matters

## CONSEQUENCES OF UNCONTROLLED HYPERTENSION

| ORGAN          | WHAT HAPPENS                                                                                        |
|----------------|-----------------------------------------------------------------------------------------------------|
| <b>Heart</b>   | pumping against high BP thickens the muscle → restricted flow → <b>heart failure / heart attack</b> |
| <b>Brain</b>   | reduced supply → rapid loss of brain function / <b>stroke</b>                                       |
| <b>Kidney</b>  | damaged vessels can't filter → dangerous accumulation of fluid and waste → <b>renal failure</b>     |
| <b>Eye</b>     | damaged retinal vessels → <b>retinopathy</b> , loss of vision                                       |
| <b>Vessels</b> | leading cause of <b>atherosclerosis</b> → peripheral vascular disease                               |

## TROUGH : PEAK RATIO

The goal of modern therapy is BP reduction sustained over **>24 hours with once-daily dosing**. ★ The **FDA proposed a trough:peak ratio >50%** before an antihypertensive can be approved for once-a-day dosing — i.e. the effect just before the next dose must still be **more than half** the peak effect.

*Principle behind this: [[pharm-drug-action]] —  $\beta$ -blockers hitting  $\beta_1$  (wanted, SA node) and  $\beta_2$  (unwanted, bronchoconstriction) — one drug, two receptors, and the side effect falls straight out of it. Hydralazine's steep dose-response vs a flat one.*

*Source: Antihypertensive Agents.pdf — อ.กฤษฏา, PYCH304 Pharmacology I, 25 slides. Structures omitted.*

# Antianginal Agents

**Angina pectoris** is the principal symptom of **ischemic heart disease** — sudden severe substernal pain or pressure radiating to arm and neck. The single cause is an **imbalance between myocardial O<sub>2</sub> demand and O<sub>2</sub> supply**. Every antianginal drug fixes one side of that equation: ↓ **demand** (preload, afterload, heart rate, contractility) or ↑ **supply** (dilate coronary arteries/arterioles). Risk factors are the standard CV set — age · smoking · DM · dyslipidemia · hypertension · obesity · inactivity · family history.

**O<sub>2</sub> supply (coronary flow) < O<sub>2</sub> demand (wall stress × HR × contractility) → myocardial ischemia → substernal chest pain → angina**

## WHAT SETS EACH SIDE OF THE BALANCE

| SIDE                        | DETERMINANTS                                                                                                                                                                  |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>O<sub>2</sub> demand</b> | <b>wall stress</b> (intraventricular pressure, ventricular volume, wall thickness) · <b>heart rate</b> · <b>contractility</b> · myocardial metabolism (glucose vs fatty acid) |
| <b>O<sub>2</sub> supply</b> | <b>coronary blood flow</b> — perfusion pressure (aortic diastolic pressure) & <b>duration of diastole</b> · coronary vascular resistance                                      |
| <b>vascular tone</b>        | arteriolar tone → <i>systolic</i> wall stress · venous tone → <i>diastolic</i> wall stress                                                                                    |

## THE 3 TYPES OF ANGINA

| TYPE                                                     | CAUSE                                                      | SYMPTOMS                                                                                                                     |
|----------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Stable</b> ( <i>classic, chronic</i> ) — most common  | atherosclerotic plaque                                     | <b>angina of effort</b> ★ — none at rest, pain on exertion/stress. Predictable, moderate, <5 min, relieved by rest or SL NTG |
| <b>Unstable</b>                                          | plaque + <b>rupture</b> → platelet activation → thrombus   | <b>angina at rest</b> ★ — unpredictable, more severe, prolonged                                                              |
| <b>Variant</b> ( <i>vasospastic, Prinzmetal's</i> ) — 2% | <b>vasospasm</b> — hypercontractile vascular smooth muscle | very painful, typically midnight → morning                                                                                   |

## BEYOND THE ANTIANGINALS THEMSELVES

- **Co-therapy for long-term vascular protection** — antithrombotic (aspirin ± clopidogrel) · statin · ACEI/ARB (*protects from MI*)
- **Surgery** — CABG (coronary artery bypass graft) · PTCA/PCA (percutaneous transluminal coronary angioplasty)

# Organic Nitrates

## MECHANISM

Metabolised to **NO**, which activates **guanylate cyclase** → ↑ **cGMP** → **myosin light-chain phosphatase (MLCP)** → smooth muscle relaxation.

→ **Major effect: venodilation** ★ → ↓ **preload** & ventricular end-diastolic pressure → ↓ wall stress → ↓ **O<sub>2</sub> demand** → **Minor effect (large dose only):** arterial dilation → ↓ PVR → ↓ afterload → **Supply side:** ↑ coronary blood flow — redistributes blood to the ischemic area, ↑ collateral flow

*NTG and ISDN are themselves active, and are metabolised to active molecules with longer half-lives than the parent.*

## DRUGS

| DRUG                                       | WHAT IT TREATS                                                                                          |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Nitroglycerin (NTG)</b>                 | acute attack (SL, spray) · IV for <b>severe recurrent unstable angina</b> · transdermal for prophylaxis |
| <b>Isosorbide dinitrate (ISDN)</b>         | acute attack (SL) · oral for prophylaxis                                                                |
| <b>Isosorbide 5-mononitrate (5-ISMN)</b>   | prophylaxis — the <b>oral</b> nitrate                                                                   |
| <b>Pentaerythritol tetranitrate (PETN)</b> | prophylaxis, oral                                                                                       |

## WHY THE ROUTE DECIDES THE DRUG

|                      | NTG                                 | ISDN        | 5-ISMN      |
|----------------------|-------------------------------------|-------------|-------------|
| half-life            | 3 min                               | 10 min      | 280 min     |
| oral bioavailability | <1%                                 | 20%         | <b>100%</b> |
| first pass           | <b>high</b> — must bypass the liver | <b>high</b> | low         |

**Acute therapy:** SL NTG or ISDN — onset 1–3 min, lasts 20–30 min · **Maintenance:** slowly absorbed forms (oral, buccal, transdermal), 6–10 h — *but these are what cause tolerance*

## USES

**All types of angina** · acute MI · CHF · Acute attack — injection, sublingual, spray · Prophylaxis — sustained/slow release, transdermal

## ADVERSE EFFECTS

- **Orthostatic hypotension, syncope**
- **Reflex tachycardia** ★ — ↑ HR ↑ contractility, which partly undoes the benefit (*the reason nitrates are combined with a β-blocker*)
- Headache, dizziness, flushing
- ● **Methemoglobinemia** — rare but serious

## NITRATE TOLERANCE & CAUTIONS

- ★ **Nitrate tolerance** — from continuous or frequent exposure. **Fix: an 8-hour nitrate-free period.** Tablets TID with the night free · 24-h transdermal patch removed overnight
- ● **PDE-5 inhibitors** (sildenafil, tadalafil, vardenafil) — both raise cGMP → catastrophic hypotension

## ADMINISTRATION — NITRATE DOSES BY FORM, AND HOW THE PATIENT ACTUALLY USES THEM

| DRUG & ROUTE                         | DOSE                                           | DURATION  |
|--------------------------------------|------------------------------------------------|-----------|
| Nitroglycerin, SL                    | 0.15–1.2 mg                                    | 10–30 min |
| Isosorbide dinitrate, SL             | 2.5–5 mg                                       | 10–60 min |
| Nitroglycerin, oral sustained-action | 6.5–13 mg every 6–8 h                          | 6–8 h     |
| Nitroglycerin, slow-release buccal   | 1–2 mg every 4 h                               | 3–6 h     |
| Nitroglycerin, transdermal           | 10–25 mg per 24 h ( <i>one patch per day</i> ) | 8–10 h    |
| Isosorbide dinitrate, oral           | 10–60 mg every 4–6 h                           | 4–6 h     |
| Isosorbide mononitrate, oral         | 20 mg every 12 h                               | 6–10 h    |
| PETN, oral                           | 50 mg every 12 h                               | 10–12 h   |

★ **The route splits by job:** injection, sublingual and spray are for the **acute attack**; sustained-release, slow-release and transdermal are for **prophylaxis**.

### ● What the patient is told

- **Sublingual ISDN:** use it at the **first sign of chest pain, sit down and stay still**, put the tablet **under the tongue and let it dissolve slowly — do not swallow it**. It should work **within 3 minutes**; if there's no improvement **within 5 minutes, take one more**. **If the pain persists after a total of 3 tablets → go to hospital.**
- **Transdermal patch:** apply **one patch each morning** and leave it on for **12 to 14 hours** — the exact dose, number of doses and interval are individual, but the **nitrate-free period every day is not**.
- **Tablets:** three times daily, **with the free gap at night**.
- 💡 "Sit down first" is in there because the drug drops the blood pressure of someone already in pain — standing through it is how a nitrate turns into a syncope. ⚠️ And the **3-tablet stop rule** is the part that matters most: it is the line between angina and an MI being treated at home.



# Calcium Channel Blockers (CCBs)

## MECHANISM

Inhibit **L-type** ( $\pm$  **T-type**) **Ca<sup>2+</sup> channels** → less Ca<sup>2+</sup> entry into vascular smooth muscle and cardiac cells.

→ **Vascular**: relax **arterial** smooth muscle → ↓ afterload. **Dilates coronary arteries and peripheral arterioles — but NOT veins** ★ *(the mirror image of nitrates, which are venous)* → **Cardiac**: negative **inotropic** (↓ inward Ca<sup>2+</sup> during the plateau phase) and negative **chronotropic** (depress SA & AV conduction) — **non-DHP only** → **DHPs** have little cardiac effect but cause **reflex tachycardia** → ↑ O<sub>2</sub> demand

*Some DHPs are regionally selective — nimodipine for cerebral vessels, nicardipine for coronary vessels.*

## DHP VS NON-DHP — VASCULAR VS CARDIAC

| COMPOUND                            | CORONARY VASODILATION | ↓ CONTRACTILITY | ↓ SA NODE | ↓ AV NODE |
|-------------------------------------|-----------------------|-----------------|-----------|-----------|
| <b>Verapamil</b> ( <i>non-DHP</i> ) | ++++                  | ++++            | +++++     | +++++     |
| <b>Diltiazem</b> ( <i>non-DHP</i> ) | +++                   | ++              | +++++     | ++++      |
| <b>Nifedipine</b> ( <i>DHP</i> )    | +++++                 | +               | +         | 0         |
| <b>Nicardipine</b> ( <i>DHP</i> )   | +++++                 | 0               | +         | 0         |

★ **Vascular effects: DHP > non-DHP · Cardiac effects: non-DHP >>> DHP**

## DRUGS

| DRUG                                | WHAT IT TREATS                                        |
|-------------------------------------|-------------------------------------------------------|
| <b>Amlodipine</b> ( <i>DHP</i> )    | angina · hypertension                                 |
| <b>Nifedipine</b> ( <i>DHP</i> )    | angina · hypertension · Raynaud's                     |
| <b>Nicardipine</b> ( <i>DHP</i> )   | angina · hypertension                                 |
| <b>Felodipine</b> ( <i>DHP</i> )    | hypertension · Raynaud's                              |
| <b>Diltiazem</b> ( <i>non-DHP</i> ) | angina · hypertension · Raynaud's                     |
| <b>Verapamil</b> ( <i>non-DHP</i> ) | angina · hypertension · <b>arrhythmias</b> · migraine |

## USES

**Stable · unstable · variant angina** — the one class that covers variant angina safely · ⚠ **Not first-line** — used in patients unresponsive to  $\beta$ -blockers

## ADVERSE EFFECTS

- **Major**: ↓ contractility · **AV block, bradycardia, cardiac arrest** (*non-DHP*)
- Minor: hypotension, dizziness, **peripheral edema**, flushing, constipation

## INTERACTIONS

- **$\beta$ -blocker + non-DHP** ★ — additive negative inotropic + chronotropic → **bradycardia, AV block**
- **Verapamil / diltiazem** ↑ **plasma digoxin** → AV block

## CONTRAINDICATIONS

- ⚠️ **Non-DHP:** digitalis toxicity · concurrent  $\beta$ -blocker · **SA or AV block**
- ⚠️ **DHP:** aortic stenosis

## ADMINISTRATION — CCB DOSES IN ANGINA

| DRUG                | ORAL F (%) | HALF-LIFE (H) | DOSAGE                                              |
|---------------------|------------|---------------|-----------------------------------------------------|
| <b>Amlodipine</b>   | 65–90      | <b>30–50</b>  | 5–10 mg orally <b>once daily</b>                    |
| <b>Felodipine</b>   | 15–20      | 11–16         | 5–10 mg orally once daily                           |
| <b>Nicardipine</b>  | 35         | 2–4           | 20–40 mg orally <b>every 8 h</b>                    |
| <b>Nifedipine</b>   | 45–70      | 4             | 3–10 mcg/kg IV · 20–40 mg orally every 8 h          |
| <b>Nitrendipine</b> | 10–30      | 5–12          | 20 mg orally once or twice daily                    |
| <b>Diltiazem</b>    | 40–65      | 3–4           | 75–150 mcg/kg IV · 30–80 mg orally <b>every 6 h</b> |
| <b>Verapamil</b>    | 20–35      | 6             | 75–150 mcg/kg IV · 80–160 mg orally every 8 h       |

- ★ **Amlodipine's 30–50 hour half-life is why it is the once-daily one** and why a missed dose matters less than with nicardipine at every 8 hours. The dosing frequency in this table is just the half-life column restated.
- 💡 A patient who can't manage three doses a day is the patient who ends up on amlodipine, regardless of what the guideline prefers.

# β-Blockers

## MECHANISM

Block cardiac  $\beta_1$  receptors — purely a **demand-side** drug (no coronary vasodilation).

→ **Downstream:** ↓ **double product** ( $HR \times BP$ ) ★ · limits **exercise-induced** rises in contractility · slower HR = **longer diastole** → more time for coronary perfusion

*Subclass modifiers: ISA (intrinsic sympathomimetic activity) relaxes arterioles · some agents (carvedilol, nebivolol) add NO-mediated direct vasodilation.*

## DRUGS

| DRUG                                                                        | WHAT IT TREATS                                    |
|-----------------------------------------------------------------------------|---------------------------------------------------|
| <b>Propranolol, timolol, nadolol, alprenolol</b><br>(non-selective)         | angina · post-MI — ⚠ <b>never variant angina</b>  |
| <b>Pindolol, carteolol, penbutolol, acebutolol</b> (ISA)                    | ⚠ <b>not for severe angina</b> — partial agonists |
| <b>Labetalol, carvedilol</b> (non-selective + $\alpha_1$ )                  | added vasodilation                                |
| <b>Atenolol, metoprolol, bisoprolol, esmolol</b><br>( $\beta_1$ -selective) | stable & unstable angina · post-MI                |
| <b>Nebivolol</b> ( $\beta_1$ -selective + NO)                               | added vasodilation                                |

## USES

**Post-MI · stable angina · unstable angina** — first-line demand reduction

## ADVERSE EFFECTS

- **Bradycardia** (caution: heart block)
- ↓ cardiac contractility (caution: heart failure)
- **Bronchospasm** ★ — especially non-selective (caution: asthma, COPD)
- Impaired glucose tolerance (caution: DM) and ● **masked hypoglycemic symptoms**
- Dyslipidemia — ↑ TG, ↓ HDL
- CNS penetration, especially **propranolol** — insomnia, depression, hallucination

## CONTRAINDICATIONS

- ● **Variant angina** ★ — non-selective  $\beta$ -blockers leave coronary  $\beta_2$  unopposed → **induce vasospasm**
- ⚠ Sinus bradycardia, heart block
- ⚠ Asthma, severe COPD
- ⚠ **Severe angina + an ISA agent** — partial agonism drops HR and contractility less than a full antagonist

# Ivabradine — Direct Bradycardic Agent

## MECHANISM

**Selective  $I_f$  ("funny current")  $Na^+$  channel blocker in the SA node** — slows the pacemaker and nothing else.

→ **Downstream:** ↓ heart rate → ↓  $O_2$  demand + longer diastole. **No effect on GI or bronchial smooth muscle** — which is exactly its advantage over  $\beta$ -blockers.

## DRUGS

| DRUG              | WHAT IT TREATS                                                  |
|-------------------|-----------------------------------------------------------------|
| <b>Ivabradine</b> | anginal attack — efficacy similar to CCBs and $\beta$ -blockers |

## USES

Angina **when  $\beta$ -blockers are not tolerated or are contraindicated** — e.g. asthma, COPD

# Ranolazine — Late Na<sup>+</sup> Current Inhibitor

## MECHANISM

Inhibits the **slow inward (late) Na<sup>+</sup> current, I<sub>Na</sub>** in ischemic myocardium.

→ **Downstream:** ↓ cardiac contractility → ↓ O<sub>2</sub> demand, without touching heart rate or blood pressure

## DRUGS

| DRUG              | WHAT IT TREATS                                      |
|-------------------|-----------------------------------------------------|
| <b>Ranolazine</b> | angina — US FDA approved, <b>reserve agent only</b> |

## USES

Only when there is **no response to other antianginals** — the QT risk keeps it last in line

## CAUTIONS

-  **QT prolongation**  — the reason it's a reserve drug
-  **CYP3A4** inducers / inhibitors
-  Chronic liver disease

# Trimetazidine — Metabolic Modulator

## MECHANISM

**pFOX inhibitor** — partially inhibits **fatty acid oxidation** in the myocardium.

→ **Why that helps:** ischemic myocardium shifts its metabolism toward fatty acid oxidation, which costs **more O<sub>2</sub> per ATP produced**. Partially blocking that enzyme pushes the tissue back toward glucose oxidation and improves its metabolic status. **The only class that changes the metabolism instead of the haemodynamics.**

## DRUGS

| DRUG                 | WHAT IT TREATS                               |
|----------------------|----------------------------------------------|
| <b>Trimetazidine</b> | <b>stable angina</b> — demonstrated efficacy |

# Combination Therapy

## WHY COMBINING FIXES EACH DRUG'S OWN FLAW

|                                                    | HEART RATE             | AFTERLOAD       | PRELOAD          | CONTRACTILITY          | EJECTION TIME   |
|----------------------------------------------------|------------------------|-----------------|------------------|------------------------|-----------------|
| <b>Nitrates alone</b>                              | <i>reflex increase</i> | decrease        | decrease         | <i>reflex increase</i> | decrease        |
| <b><math>\beta</math>-blocker or CCB alone</b>     | decrease*              | decrease        | <i>increase</i>  | decrease*              | <i>increase</i> |
| <b>Nitrate + <math>\beta</math>-blocker or CCB</b> | <b>decrease</b>        | <b>decrease</b> | none or decrease | <b>none</b>            | <b>none</b>     |

*Italics = the undesirable effect. \* DHPs may do the opposite, via reflex sympathetic tone. ★ The combination cancels the nitrate's reflex tachycardia and the  $\beta$ -blocker's rise in preload.*

## RECOMMENDED COMBINATIONS

**Nitrate +  $\beta$ -blocker** (*the standard pairing*) · **Nitrate + non-DHP CCB** (*when  $\beta$ -blockers can't be used*) · **Nitrate +  $\beta$ -blocker + CCB**

## COMBINATIONS TO AVOID

- ⚠ **Nitrate + DHP CCB** — both drop BP and both trigger reflex tachycardia
- **$\beta$ -blocker + non-DHP CCB** — additive negative inotropy and chronotropy → bradycardia, AV block

Source: [Antianginal Agents\\_Warisara\\_2025.pdf](#) — อ.วรวิศร, PYCH304 Pharmacology I, 38 slides. Structures omitted.

# Antiarrhythmic Drugs

The heart is a **mechanical** pump run by an **electrical** controller. **Pacemaker cells** (SA node, AV node, His–Purkinje) have **automaticity** — they fire on their own; **non-pacemaker cells** (atrial & ventricular myocytes) don't, they just contract. Antiarrhythmics work by editing the cardiac action potential of one or both.

**Block a channel → alter phase 4 slope / threshold / max diastolic potential / APD → ↓ abnormal automaticity & re-entry → rhythm restored**

## THE 4 CLASSES AT A GLANCE

| CLASS                                         | TARGET                                                                                 | EFFECT ON THE ACTION POTENTIAL                                       |
|-----------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>I</b> — Na <sup>+</sup> channel blockers   | Na <sup>+</sup> channel — phase 0 (ventricle), phase 4 <i>f</i> <sub>H</sub> (SA node) | ↓ phase 0 upstroke & conduction; APD depends on dissociation rate    |
| <b>II</b> — β-blockers                        | β-adrenergic receptor → phase 4                                                        | ↓ slope of phase 4 · ↑ <b>ERP of the AV node</b>                     |
| <b>III</b> — K <sup>+</sup> channel blockers  | K <sup>+</sup> channel — phase 3 repolarization                                        | <b>prolongs repolarization</b> → ↑ <b>APD &amp; ERP</b> (and the QT) |
| <b>IV</b> — Ca <sup>2+</sup> channel blockers | L-type Ca <sup>2+</sup> channel — phase 0 of <i>pacemaker</i> cells                    | ↓ AP upstroke at SA/AV node → <b>prolonged AV nodal conduction</b>   |

## CLASS I SUBCLASSES — IT'S ALL ABOUT HOW FAST THE DRUG LETS GO ★

| SUBCLASS  | NA <sup>+</sup> CHANNEL DISSOCIATION | EFFECT ON APD               | DRUGS                                       |
|-----------|--------------------------------------|-----------------------------|---------------------------------------------|
| <b>IA</b> | intermediate                         | <b>prolong APD</b>          | quinidine, procainamide, disopyramide       |
| <b>IB</b> | fast                                 | <b>shorten APD</b>          | lidocaine, mexiletine, tocainide, phenytoin |
| <b>IC</b> | slow                                 | <b>little effect on APD</b> | flecainide, propafenone, encainide          |

## HOW ANY ANTIARRHYTHMIC CAN ACT

A drug changes rhythm by altering one of four things: - **Rate of phase 4 depolarization** — decrease the slope - **Threshold potential** — raise it - **Maximum diastolic potential** — make it more negative - **Action potential duration** — increase it

All four converge on: ↓ **rate of spontaneous AP firing**.

## THE ARRHYTHMIAS BEING TREATED

| ORIGIN             | TYPES                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>SA node</b>     | sinus arrhythmia · sinus bradycardia (<60/min) · sinus tachycardia (>100/min) · sinus arrest                                   |
| <b>Atrial</b>      | PACs · atrial tachycardia · atrial <b>flutter</b> (250–400/min, regular) · atrial <b>fibrillation</b> (400–700/min, irregular) |
| <b>Ventricular</b> | PVCs · ventricular tachycardia · ventricular flutter · ventricular fibrillation                                                |



# Class IA — Na<sup>+</sup> Block, Intermediate Dissociation

## MECHANISM

**Direct:** inhibit Na<sup>+</sup> channels with **intermediate dissociation** → **prolongs APD**. **Indirect:** anticholinergic effect → ↑ AV conduction, ↑ contractility.

→ **Downstream:** depress pacemaker rate · ↓ conduction & excitability in atria, His–Purkinje, ventricle → ⚠️

**Prolonged APD = prolonged QT** — this is why class IA is a torsades class

## DRUGS

| DRUG                | WHAT IT TREATS                                                                                                             |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Quinidine</b>    | conversion of atrial flutter/fibrillation + maintenance of sinus rhythm · PSVT · premature atrial/ventricular contractions |
| <b>Procainamide</b> | PVCs · <b>life-threatening ventricular tachycardia</b> · atrial flutter/fibrillation conversion                            |
| <b>Disopyramide</b> | PVCs · ventricular tachycardia · atrial flutter/fibrillation conversion                                                    |

## USES

**Atrial flutter / fibrillation** — conversion and maintenance of normal sinus rhythm · **PSVT** · **Premature contractions (PAC / PVC)** · **Life-threatening VT** — procainamide

## ADVERSE EFFECTS

- **Quinidine** — **GI is the common one:** diarrhea, nausea, vomiting
- **Cinchonism** ★ — dizziness, tinnitus, headache (*quinidine's signature*)
- ● **Cardiovascular toxicity:** prolonged QT → **torsade de pointes** · ↑ AV conductance · hypotension · **quinidine syncope**
- Hypokalemia · thrombocytopenia · muscle weakness
- **Procainamide:** same as quinidine but **fewer anticholinergic effects**; ● **lupus-like syndrome** on prolonged use ★
- **Disopyramide:** same as quinidine but **more anticholinergic effects** — GI and CVS

## INTERACTIONS

- ● **Digoxin** — quinidine-induced **digoxin toxicity** ★
- **Warfarin** · **verapamil** · **cimetidine** · **tricyclic antidepressants** · **neuromuscular blockers**

## CONTRAINDICATIONS

- ⚠️ Partial or complete **AV block**
- ⚠️ Severe renal disease · digitalis-induced arrhythmia · **myasthenia gravis** (*anticholinergic*)
- ⚠️ History of drug-induced **torsades de pointes**
- ⚠️ **Procainamide additionally: SLE**

## WORTH KNOWING

**Procainamide** → **NAPA** (N-acetylprocainamide) by hepatic acetylation. NAPA is an **active metabolite** with a longer half-life and carries **K<sup>+</sup> block (class III)** as well as **Na<sup>+</sup> block** — so procainamide effectively runs two mechanisms at once.

# Class IB — Na<sup>+</sup> Block, Fast Dissociation

## MECHANISM

Na<sup>+</sup> channel block with **fast dissociation** — on and off the channel quickly, so it preferentially hits rapidly firing / depolarized (i.e. ischaemic ventricular) tissue.

→ **Downstream:** ↓ depolarization & automaticity of **ventricular** cells · **shortens APD** · ventricle-selective — little atrial effect

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                                                    |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Lidocaine</b>  | <b>drug of choice for acute ventricular tachycardia</b> · ventricular arrhythmia after cardiac surgery, MI, or cardiac glycosides |
| <b>Mexiletine</b> | ventricular arrhythmia                                                                                                            |
| <b>Tocainide</b>  | ventricular arrhythmia                                                                                                            |
| <b>Phenytoin</b>  | ventricular arrhythmia                                                                                                            |

## USES

**Acute ventricular tachycardia** — lidocaine is first choice, **rapid and safe by IV** · Ventricular arrhythmia from **cardiac surgery, MI, cardiac glycoside toxicity** · ⚠ **Not for routine/prophylactic use — increases mortality**

## ADVERSE EFFECTS

- ★ **CNS toxicity** — confusion, twitching, **seizure** (*the limiting toxicity of lidocaine*)
- Less cardiotoxic** than the other class I drugs
- Hypotension at large doses

## WHY LIDOCAINE IS IV ONLY

**Extensive first-pass metabolism** → oral is useless; give IV. **Short t<sub>1/2</sub> (1–2 h)**. Reduce the dose in liver disease.

# Class IC — Na<sup>+</sup> Block, Slow Dissociation

## MECHANISM

Na<sup>+</sup> channel block with **slow dissociation** — the drug stays on the channel, so conduction is depressed hard at all heart rates. **Little effect on APD.**

→ **Downstream:** ↓ depolarization & automaticity of ventricular cells · marked ↓ conduction velocity → ●

**The same slow dissociation that makes it potent makes it proarrhythmic**

## DRUGS

| DRUG               | WHAT IT TREATS                                                                          |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Flecainide</b>  | sustained ventricular tachycardia · PSVT & paroxysmal AF unresponsive to other measures |
| <b>Propafenone</b> | sustained ventricular tachycardia · PSVT & paroxysmal AF unresponsive to other measures |
| <b>Encainide</b>   | ventricular arrhythmia                                                                  |

## USES

**Sustained ventricular tachycardia · PSVT** and **paroxysmal atrial fibrillation** — only when unresponsive to other measures (*reserve agents, because of the proarrhythmia*)

## ADVERSE EFFECTS

- **Proarrhythmic** — **increases mortality ~2.5×** ★ (*the reason IC is a last-resort class*)
- **Negative inotropy** — flecainide
- **Propafenone:** GI (common) — **metallic taste**, constipation · CVS — proarrhythmia, negative inotropy · CNS — dizziness, headache, visual changes

## CONTRAINDICATIONS

- **2nd- or 3rd-degree heart block without a pacemaker**
- **Cardiogenic shock · recent MI · hypersensitivity**

## PROPAFENONE IS NOT A PURE IC

Its structure resembles **propranolol**, so it carries **weak β-blockade (II)**, **weak prolongation of repolarization (III)**, and **weak Ca<sup>2+</sup> channel blockade (IV)** on top of the Na<sup>+</sup> block. Extensive first-pass hepatic metabolism.

# Class II — $\beta$ -Adrenergic Antagonists

## MECHANISM

Block  **$\beta$ -adrenergic receptors** on pacemaker cells → removes the sympathetic drive on the inward  **$I_f$  current** of phase 4.

→ **Downstream:** ↓ slope of **phase 4** depolarization → ↓ automaticity · ↑ **effective refractory period (ERP) of the AV node** → slows ventricular rate in supraventricular arrhythmias ★

## DRUGS

| DRUG               | WHAT IT TREATS                                                                        |
|--------------------|---------------------------------------------------------------------------------------|
| <b>Propranolol</b> | supraventricular tachyarrhythmias — <b>reduces sudden arrhythmic death after MI</b> ★ |
| <b>Acebutolol</b>  | premature ventricular beats                                                           |
| <b>Esmolol</b>     | acute arrhythmias in <b>surgery / emergency</b> — very short-acting, IV               |

## USES

**Supraventricular tachyarrhythmias · Ventricular arrhythmia induced by catecholamines or digitalis**  
(the arrhythmias that are adrenergically driven are exactly the ones this class owns)

# Class III — K<sup>+</sup> Channel Blockers

## MECHANISM

Block the **K<sup>+</sup> channels** carrying the outward current of **phase 3** (rapid repolarization) — i.e. inhibitors of repolarization.

→ **Downstream:** ↑ **APD** and ↑ **ERP** → re-entry circuits can't sustain themselves → ● **Prolonged repolarization = prolonged QT = torsades risk, built into the mechanism**

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                                      |
|--------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Amiodarone</b>  | recurrent <b>ventricular fibrillation</b> · unstable <b>VT</b> · atrial fibrillation · supraventricular arrhythmias |
| <b>Dronedarone</b> | atrial fibrillation                                                                                                 |
| <b>Sotalol</b>     | ventricular & supraventricular arrhythmias                                                                          |
| <b>Ibutilide</b>   | atrial fibrillation / flutter conversion                                                                            |
| <b>Dofetilide</b>  | atrial fibrillation / flutter                                                                                       |

## USES

**Recurrent ventricular fibrillation** · **Unstable ventricular tachycardia** · **Atrial fibrillation** · Supraventricular arrhythmias

## ADVERSE EFFECTS — AMIODARONE HITS EVERY ORGAN ★

- **Cardiac:** ↓ SA/AV node function · ↓ contractility · hypotension
- ● **Pulmonary:** pneumonitis → **pulmonary fibrosis** (*the one that kills*)
- **Thyroid:** hyper- or hypothyroidism (*it's iodine-rich and blocks T4→T3 conversion*)
- **Hepatic:** elevated liver enzymes
- **CNS:** peripheral neuropathy, headache, ataxia, tremor
- **Other:** **corneal microdeposits** · skin discoloration · testicular dysfunction

## INTERACTIONS

- **Amiodarone inhibits CYP2D6, 2C9 and 3A4** — it raises the level of a long list of co-prescribed drugs ★
- Additive QT prolongation with any other QT-prolonging drug

## WHY AMIODARONE IS A CLASS OF ITS OWN

It is labelled **class III (K<sup>+</sup> block is the main action)** but it also blocks **Na<sup>+</sup> channels (I)**, **β-receptors (II)** and **Ca<sup>2+</sup> channels (IV)** — all four Vaughan Williams classes in one molecule. Extracardiac: **peripheral vasodilation + blocks peripheral T4 → T3 conversion**. Pharmacokinetics: slow, variable absorption · **active desmethyl metabolite** · very long t<sub>1/2</sub>, slow onset.

# Class IV — $\text{Ca}^{2+}$ Channel Blockers

## MECHANISM

Block L-type  $\text{Ca}^{2+}$  channels. In SA/AV nodal cells the **phase 0 upstroke is carried by  $\text{Ca}^{2+}$** , not  $\text{Na}^+$  — so this class hits the nodes specifically. **Non-DHP only** (verapamil, diltiazem); dihydropyridines are vascular and don't do this.

→ **Downstream:** ↑ AV nodal conduction time · slowed AP upstroke → ↓ conduction velocity · suppresses **early and delayed afterdepolarizations** · extracardiac **vasodilation**

## DRUGS

| DRUG             | WHAT IT TREATS                                                               |
|------------------|------------------------------------------------------------------------------|
| <b>Verapamil</b> | supraventricular tachycardia · rate control in atrial fibrillation / flutter |
| <b>Diltiazem</b> | supraventricular tachycardia · rate control in atrial fibrillation / flutter |

## USES

**Supraventricular tachycardia** · **Reduce ventricular rate in atrial fibrillation & flutter** · Rarely used in ventricular tachycardia

## PRECAUTIONS

- ⚠ **AV block** — the drug already slows nodal conduction; excessive reduction stalls it

# Unclassified Agents

## OUTSIDE VAUGHAN WILLIAMS

| DRUG                            | USE                                                |
|---------------------------------|----------------------------------------------------|
| <b>Adenosine</b>                | supraventricular tachycardia                       |
| <b>Magnesium sulfate</b>        | ventricular fibrillation                           |
| <b>Adrenaline (epinephrine)</b> | cardiac arrest                                     |
| <b>Isoprenaline</b>             | heart block                                        |
| <b>Digoxin</b>                  | rapid atrial fibrillation                          |
| <b>Calcium chloride</b>         | ventricular tachycardia due to <b>hyperkalemia</b> |

### ADENOSINE — MECHANISM

Activates the **A1 receptor** on cardiac tissue → ↑ **K<sup>+</sup> conductance** and ↓ **Ca<sup>2+</sup>-mediated conduction**.

→ **Downstream:** depresses the **SA node** and **AV conduction** → **terminates re-entry** ★

### ADENOSINE — USE & CAUTIONS

**PSVT** — **rapid IV bolus** (*very short-lived, so it must go in fast*) · **Side effects:** dyspnea, flushing, heart block

· ⚠ **Less effective with adenosine-receptor blockers** — **theophylline or caffeine**

### DIGOXIN — MECHANISM

**Direct:** inhibits **Na<sup>+</sup>-K<sup>+</sup> ATPase** → ↑ contractility (*it's an inotrope first*). **Indirect** — **vagomimetic:** ↑ vagal tone · ↓ central sympathetic outflow · ↓ myocardial response to β-AR stimulation · ↓ RAAS activation.

→ **Downstream:** slows cardiac conduction & heart rate → **rate control in atrial flutter / fibrillation**

# Drug-Induced QT Prolongation & Torsades

## THE NUMBERS ★

- **Long QT syndrome:**  $QT_c > 440 \text{ ms}$  (men),  $> 460 \text{ ms}$  (women)
- $QT_c > 500 \text{ ms}$  → **real risk of torsade de pointes**
- **Torsade de pointes (TdP)** — literally "*twisting of the points*": polymorphic VT that can degenerate into VF

## DRUGS THAT CAN CAUSE TDP

- **Antiarrhythmics:** disopyramide, procainamide, quinidine, **amiodarone**
- **Prokinetics:** cisapride, domperidone
- **Antibiotics:** macrolides (clarithromycin, erythromycin) · co-trimoxazole · fluoroquinolones (moxifloxacin, gatifloxacin) · voriconazole
- **Antipsychotics/antidepressants:** phenothiazines (chlorpromazine, thioridazine) · TCAs · haloperidol · atypicals (quetiapine, ziprasidone)
- **Miscellaneous:** digoxin · corticosteroids · **hydrochlorothiazide** · quinine, chloroquine · tacrolimus · methadone

## RISK FACTORS FOR DRUG-INDUCED TDP

- ● **Hypokalemia** ( $K^+ < 3.5 \text{ mEq/L}$ ) — *e.g. from a thiazide* ★
- ● **Hypomagnesemia** ( $Mg^{2+} < 1.5 \text{ mEq/L}$ )
- **Bradycardia** ( $HR < 60 \text{ bpm}$ )
- **Congestive heart failure**
- High drug concentration · **rapid IV infusion** of a QT-prolonging drug
- Baseline QT prolongation



# Clinical Use Principles

## BEFORE AND AFTER STARTING AN ANTIARRHYTHMIC

| STAGE         | WHAT TO DO                                                                                                                                                                                                                                                                                            |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Before</b> | <b>Eliminate the cause</b> — correct electrolytes ( $K^+$ , $Mg^{2+}$ ); treat hyperthyroidism / underlying cardiac disease; review current drug therapy<br><br>Make a <b>firm diagnosis</b> · establish the <b>baseline QTc</b> · <b>question whether therapy is needed at all</b> (benefit vs risk) |
| <b>After</b>  | Drug selection — lowest side effects, proper dose & route, individual risk factors<br><br>Watch <b>concurrent drugs</b> · keep <b><math>K^+</math> and <math>Mg^{2+}</math></b> in range · <b>monitor the QTc</b>                                                                                     |

Principle behind this: *[[pharm-drug-interactions]]* — Quinidine retained by alkaline urine; amiodarone as a broad CYP + P-gp inhibitor; dofetilide and the torsades risk.

Principle behind this: *[[pharm-drug-toxicity]]* — Quinidine-induced thrombocytopenia — same Type II mechanism, different blood line.

Source: *Antiarrhythmic drugs - Warisara\_2025.pdf* — อ.วรวิศรา, PYCH304 Pharmacology I, 32 slides. Structures omitted.

# Pharmacology of Heart Failure

HF is a **clinical syndrome**: symptoms/signs from a structural or functional cardiac change, plus either an **elevated natriuretic peptide** or objective congestion. The heart can't supply enough blood → tissue hypoxia → **fatigue · dyspnea on exertion, orthopnea · fluid retention & pitting edema**. Treatment goals: relieve symptoms · slow/reverse myocardial deterioration · **reduce mortality**.

↓ **cardiac output** → **neurohormonal activation (sympathetic + RAAS)** → ↑ **HR & contractility · vasoconstriction · Na<sup>+</sup>/H<sub>2</sub>O retention** → ↑ **preload & afterload** → **remodeling & myocyte loss** → ↓ **cardiac output** ★

## THE THREE WORDS THAT DECIDE THE DRUG

- **HFrEF** — systolic dysfunction; EF typically <40%. Myocyte loss & remodeling · neurohormonal activation · calcium dysregulation. *The pump is weak.*
- **HFpEF** — diastolic dysfunction; EF preserved but the LV can't relax and fill. Myocardial stiffness (collagen/fibrosis) · endothelial dysfunction · systemic inflammation from comorbidities (DM, HT). *The pump is stiff.*
- **BNP & NT-proBNP** — the **biomarkers** for HF. Natriuretic peptides themselves cause vasodilation, natriuresis, diuresis.

## THE FOUR PILLARS OF GDMT (HFREF)

| PILLAR                   | REPRESENTATIVE DRUGS                                    | WHAT IT DOES                                                                                      |
|--------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>ARNI / ACEI / ARB</b> | sacubitril–valsartan · "-pril"s · losartan, valsartan   | block RAAS → ↓ afterload & preload · <b>ACEI improves survival</b>                                |
| <b>β-blocker</b>         | metoprolol succinate, bisoprolol, carvedilol, nebivolol | ↓ inotropy & chronotropy, ↓ renin, reverses remodeling · <b>survival benefit additive to ACEI</b> |
| <b>MRA</b>               | spironolactone                                          | aldosterone antagonism · <b>the only diuretic that improves survival (NYHA III/IV)</b>            |
| <b>SGLT2 inhibitor</b>   | the "-gliflozin"s                                       | ↓ volume & preload · ↓ fibrosis, inflammation, oxidative stress · ketone energy shift             |

## ACUTE HF — THE 2×2 THAT PICKS THE DRUG

|                              | NO CONGESTION (DRY)                         | CONGESTION (WET)                                       |
|------------------------------|---------------------------------------------|--------------------------------------------------------|
| <b>good perfusion (warm)</b> | Warm & Dry — <i>stable</i>                  | Warm & Wet — <b>diuretics, vasodilators</b>            |
| <b>low perfusion (cold)</b>  | Cold & Dry — <b>inotropes, vasodilators</b> | Cold & Wet — <b>diuretics, inotropes, vasodilators</b> |

*Congestion at rest = orthopnea, ↑ JVP, rales, S3 gallop, edema. Low perfusion = narrow pulse pressure, cool extremities, hypotension.*

# Cardiac Glycosides

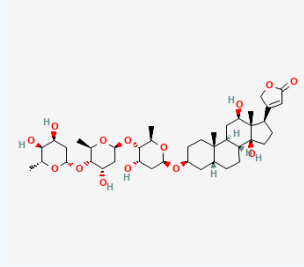
## MECHANISM

Inhibits the **Na<sup>+</sup>/K<sup>+</sup> ATPase** pump on the cardiac cell membrane → ↑ intracellular Na<sup>+</sup> → the **Na<sup>+</sup>/Ca<sup>2+</sup> exchanger** can no longer extrude Ca<sup>2+</sup> → intracellular Ca<sup>2+</sup> accumulates.

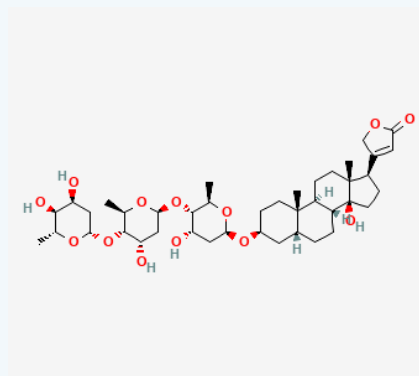
→ **Downstream:** ↑ contractility (inotropy) → ↑ stroke volume & cardiac output — but a **weak inotrope** →

**Also:** mild systemic/coronary/venous vasoconstriction (*especially IV*) · mild diuresis from ↓ renal Na<sup>+</sup> reabsorption

## THE SHARED CORE · WHAT CHANGES



**Digoxin.** three parts: a steroid, an unsaturated lactone (does the Na<sup>+</sup>/K<sup>+</sup>-ATPase inhibiting) and a sugar chain (pharmacokinetics only)



### Digitoxin

- one -OH fewer → far more lipophilic → hepatic clearance and a t<sub>1/2</sub> of days instead of ~36 h. One oxygen atom decides the entire dosing problem.

## MEDCHEM

**SAR** Three hard requirements on the aglycone, and they are geometric. Rings **A–B** and **C–D** must be **cis-fused**, which bends digoxin into a **U shape** - that shape is what fits Na<sup>+</sup>/K<sup>+</sup>-ATPase. The **C14 hydroxyl must be free**, and the **α,β-unsaturated 17-lactone is required for binding**. Saturate that lactone and the drug stops working.

**Physicochem** ★ **The sugars don't bind anything - they set the pharmacokinetics.** The number of hydroxyls on the sugar chain decides lipophilicity, and the more lipophilic glycoside is **absorbed faster and lasts longer** because it is excreted more slowly in urine. That, not any difference in potency at the pump, is the digoxin-versus-digitoxin story.

## DRUGS

| DRUG             | WHAT IT TREATS                                                           |
|------------------|--------------------------------------------------------------------------|
| <b>Digoxin</b>   | symptomatic HFrEF · rate control in <b>atrial fibrillation / flutter</b> |
| <b>Digitoxin</b> | as above                                                                 |

## USES

**Congestive heart failure** — symptomatic HFrEF only · **Antidysrhythmic** — slows ventricular rate (<100 bpm) in atrial flutter/fibrillation · ★ **Little to no effect on long-term survival** — not recommended in NYHA class I/II unless there is AF.

## ADVERSE EFFECTS

- **GI — most common:** nausea, vomiting, diarrhea, anorexia
- **Cardiac:** any brady- or tachydysrhythmia (*exception: AF with rapid ventricular response*)
- **CNS:** confusion, **visual disturbance — yellow halos around lights** ★

## INTERACTIONS

- ↑ **digoxin level** ● — quinidine · amiodarone · flecainide · verapamil · **spironolactone** · triamterene · indomethacin · cyclosporine
- ↓ **digoxin level** — antacids · kapectate · bran · phenytoin · propafenone · neomycin · sulfamethoxazole · phenobarbital

## CAUTIONS

- ● **Narrow therapeutic index** ★ —  $t_{1/2} = 1.5\text{--}2$  days, so levels build up easily
- ⚠ Excretion is proportional to **GFR** — renal impairment → toxicity

## ADMINISTRATION — DIGOXIN

- **Available as IV, oral tablets, soft gel capsules and elixir.**
- $t_{1/2}$  1.5–2 days — ● **easy to accumulate**, and **excretion is proportional to GFR.**
- ● **Narrow therapeutic index.** The long half-life plus renal clearance means a falling GFR raises levels without anyone changing the dose.
- 💡 The elixir matters more than it looks: dosing in this range is fine-grained, and a tablet strength can be too blunt an instrument for a small or elderly patient.

# $\beta_1$ -Adrenergic Agonists

## MECHANISM

Selective  **$\beta_1$  agonist** →  $\uparrow$  cAMP → activates protein kinases →  $\uparrow$   $\text{Ca}^{2+}$  in the cell and sarcoplasmic reticulum.  
→ **Downstream:**  $\uparrow$  contractility →  $\uparrow$  stroke volume · much **less heart-rate response** than other  $\beta_1$  agonists →  
**Mild  $\beta_2$  stimulation:** vasodilation →  $\downarrow$  **afterload**

## DRUGS

| DRUG                                | WHAT IT TREATS                                                             |
|-------------------------------------|----------------------------------------------------------------------------|
| <b>Dobutamine</b>                   | acute decompensated HF — continuous IV infusion, $t_{1/2} \approx 2.5$ min |
| <b>Dopamine</b>                     | inotropic support                                                          |
| <b>Epinephrine / Norepinephrine</b> | inotropic + pressor support                                                |

## USES

**Acute decompensated heart failure** — short-term inotropic support, continuous IV infusion only.

## ADVERSE EFFECTS

(*unusual at therapeutic doses*) - Atrial / ventricular dysrhythmias - May precipitate **myocardial ischemia**

# Phosphodiesterase Inhibitors

## MECHANISM

Inhibits enzymatic **hydrolysis of cAMP** → cAMP accumulates → activates protein kinases → ↑  $\text{Ca}^{2+}$  in the sarcoplasmic reticulum. Acts on **both heart and vasculature** — an "inodilator."

→ **Downstream:** ↑ cardiac contractility → ↑ stroke volume · arterial **and** venous vasodilation → ↓ **afterload & preload**

## DRUGS

| DRUG             | WHAT IT TREATS                                |
|------------------|-----------------------------------------------|
| <b>Milrinone</b> | acute decompensated HF — constant IV infusion |
| <b>Amrinone</b>  | acute decompensated HF                        |

## ADVERSE EFFECTS

- Hypotension
- **Thrombocytopenia**
- Dysrhythmias
- ● **May be associated with increased mortality** — this is what limits its use ★

# Calcium Sensitizer

## MECHANISM

**Levosimendan** works by three separate routes:

1. ↑ sensitivity of cardiomyocyte **troponin C** to ionic calcium → **inotropy**
2. Opens **ATP-dependent K<sup>+</sup> channels** on vascular smooth muscle → **vasodilation**
3. Opens ATP-dependent K<sup>+</sup> channels in **mitochondria** → **cardioprotection**

→ **Downstream:** inotropy + vasodilation = an **inodilator** ★ Because it *sensitizes* rather than *mobilizes* calcium, it does **not** raise myocardial O<sub>2</sub> consumption, and it prevents apoptosis & remodeling.

## DRUGS

| DRUG                | WHAT IT TREATS              |
|---------------------|-----------------------------|
| <b>Levosimendan</b> | decompensated heart failure |

# Loop Diuretics

## MECHANISM

Inhibit the  **$\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$  cotransporter** of the ascending limb — where the bulk of  $\text{Na}^+$  is reabsorbed.

→ **Downstream:** potent  $\text{Na}^+$  & water excretion → ↑ urine volume → ↓ **preload** (*Normally <1% of filtered sodium is excreted.*)

## DRUGS

| DRUG                   | WHAT IT TREATS                   |
|------------------------|----------------------------------|
| <b>Furosemide</b>      | HF congestion · edematous states |
| <b>Bumetanide</b>      | edema                            |
| <b>Torsemide</b>       | edema                            |
| <b>Ethacrynic acid</b> | edema                            |

## USES

★ **First-line therapy for heart failure** and other edematous states — symptom relief (congestion), not a mortality drug.

## ADVERSE EFFECTS

- **Hypovolemia**, hyponatremia, **hypokalemia**, hypomagnesemia
- **Metabolic alkalosis** (hypokalemic)
- Hyperuricemia · hyperglycemia · hyperlipidemia
- Hypercalcemia
- ● **Ototoxicity** — at very high doses



# K<sup>+</sup>-Sparing Diuretics & MRA

## MECHANISM

Weak diuretics acting at the **late distal tubule & collecting duct** — two routes:

1. **Spironolactone** — competitive **aldosterone antagonist**; blocks aldosterone-driven Na<sup>+</sup> reabsorption and K<sup>+</sup>/H<sup>+</sup> excretion
2. **Amiloride & triamterene** — block the **Na<sup>+</sup> channel**; since K<sup>+</sup>/H<sup>+</sup> secretion is exchanged for Na<sup>+</sup>, that secretion stops too

→ **Downstream:** ↑ Na<sup>+</sup> excretion · ↓ K<sup>+</sup> excretion → **hyperkalemia**

## DRUGS

| DRUG                  | WHAT IT TREATS                                                  |
|-----------------------|-----------------------------------------------------------------|
| <b>Spironolactone</b> | ★ the only diuretic that improves survival in NYHA class III/IV |
| <b>Amiloride</b>      | edema — <b>no</b> survival benefit                              |
| <b>Triamterene</b>    | edema — <b>no</b> survival benefit                              |

## ADVERSE EFFECTS

- ● **Hyperkalemia** ★ — the class-defining ADR
- **Metabolic acidosis**

## INTERACTIONS

- **Spironolactone / triamterene** ↑ **digoxin levels** ●
- With **ACEI** or K<sup>+</sup> supplements → compounded hyperkalemia

# Nitrovasodilators (NO Donors)

## MECHANISM

Metabolized to **nitric oxide** → vasodilation.

→ **Nitroglycerin**: primarily a **venodilator** at therapeutic doses → ↓ **preload**; high doses add arterial dilation  
→ ↓ preload **and** afterload → **Sodium nitroprusside**: a **balanced** arterial–venous dilator from the start → ↓ preload **and** afterload

## DRUGS

| DRUG                        | WHAT IT TREATS                                                          |
|-----------------------------|-------------------------------------------------------------------------|
| <b>Nitroglycerin</b>        | acute HF / congestion — $t_{1/2} \approx 5$ min                         |
| <b>Isosorbide dinitrate</b> | oral nitrate — $t_{1/2}$ 1–4 h                                          |
| <b>Sodium nitroprusside</b> | acute HF — continuous IV infusion only; onset 2 min, $t_{1/2}$ 5–10 min |

## ADVERSE EFFECTS

- **Nitroglycerin**: headache — most common · dizziness, lightheadedness · hypotension
- **Nitroprusside**: hypotension — most common
- ● Nitroprusside is metabolized to **cyanide** → **thiocyanate**; rarely **cyanide poisoning**

# Hydralazine + Nitrate

## MECHANISM

**Arteriolar vasodilator** — mechanism unknown.

→ **Downstream:** ↓ afterload. Paired with a nitrate (isosorbide dinitrate) the combination covers **both** arterial and venous sides.

## DRUGS

| DRUG                                                | WHAT IT TREATS                                    |
|-----------------------------------------------------|---------------------------------------------------|
| <b>Hydralazine</b> (+ <i>isosorbide dinitrate</i> ) | HF in patients who <b>cannot tolerate an ACEI</b> |

## USES

★ **The ACEI-intolerant route.** With nitrates it **increases survival** — comparable to ACEIs.

## ADVERSE EFFECTS

- Hypotension — most common
- Rare: **drug-induced lupus erythematosus**

# ACE Inhibitors

## MECHANISM

Inhibit generation of **angiotensin II**, and inhibit **bradykinin breakdown**.

→ **Downstream:** ↓ arteriolar vasoconstriction · ↓ renal NaCl reabsorption · ↓ ADH release → **Net:** arterial vasodilation → ↓ **afterload and** ↓ **preload** · ★ **improves survival in heart failure**

## DRUGS

| DRUG                                                                  | WHAT IT TREATS               |
|-----------------------------------------------------------------------|------------------------------|
| <b>Enalapril · Lisinopril · Ramipril</b>                              | heart failure · hypertension |
| <b>Captopril · Benazepril · Quinapril · Fosinopril · Trandolapril</b> | heart failure · hypertension |

*Rule of thumb from the lecture: anything ending in "-pril."*\*\*\*

## ADVERSE EFFECTS

- **Dry cough** — **20% of patients** ★ (the bradykinin effect)
- ● **Angioedema** — ACEIs are the **most common cause of angioedema in adults**
- **Hyperkalemia**
- CNS: dizziness, fatigue, headache, insomnia, weakness
- CV: hypotension, angina, tachycardia
- GI: taste disturbance, anorexia, nausea, diarrhea
- GU: proteinuria, impotence, **renal failure**
- Rash · **agranulocytosis / neutropenia (captopril only)**

## INTERACTIONS

- **K<sup>+</sup> supplements** → hyperkalemia
- **K<sup>+</sup>-sparing diuretics** → hyperkalemia ●
- **NSAIDs** → acute renal failure + hyperkalemia

# Angiotensin Receptor Blockers

## MECHANISM

Block **angiotensin II at its receptor** — downstream of where ACEIs act, and with **no effect on bradykinin metabolism**.

→ **Downstream:** same haemodynamics as ACEI (↓ afterload & preload) → **The whole point:** no bradykinin build-up → ★ **unlikely to cause cough or angioedema**

## DRUGS

| DRUG               | WHAT IT TREATS               |
|--------------------|------------------------------|
| <b>Losartan</b>    | heart failure · hypertension |
| <b>Valsartan</b>   | heart failure · hypertension |
| <b>Candesartan</b> | heart failure · hypertension |

## USES

**Heart failure** — may increase survival, but **usually not first-line**; the ACEI-intolerant alternative · **Hypertension**.

## ADVERSE EFFECTS

- Broadly the **same profile as ACEI**, minus the cough and angioedema
- Hypotension · hyperkalemia · renal impairment

# ARNI (Angiotensin Receptor–Neprilysin Inhibitor)

## MECHANISM

One molecule, two jobs — **sacubitril/valsartan (LCZ696)**:

1. **Valsartan** — ARB → blocks the **maladaptive** arm (angiotensin II: vasoconstriction,  $\text{Na}^+$  retention, remodeling)
  2. **Sacubitril** — **neprilysin inhibitor** → neprilysin normally degrades natriuretic peptides, so blocking it **preserves the adaptive arm** (ANP/BNP/CNP)
- **Downstream**: ↓ activity of maladaptive mechanisms **and** ↑ activity of adaptive ones — natriuresis/diuresis + venous & arterial vasodilation → ↓ **preload & afterload** ★

## DRUGS

| DRUG                                 | WHAT IT TREATS                  |
|--------------------------------------|---------------------------------|
| <b>Sacubitril/valsartan (LCZ696)</b> | HFrEF — the ARNI pillar of GDMT |

## CAUTIONS

- ⚠ Contains an ARB — the ACEI/ARB cautions carry over (hypotension, hyperkalemia, renal impairment)
- ⚠ **Never combine with an ACEI** — overlapping RAAS blockade plus bradykinin accumulation

# Nesiritide (Recombinant BNP)

## MECHANISM

Recombinant **BNP**. Natriuretic peptides — **ANP** (atria), **BNP** (ventricles), **CNP** (vascular endothelium) — are released in response to ↑ intravascular volume / chamber dilation and act on natriuretic peptide receptors in vasculature, kidney and CNS.

→ **Downstream:** ↑ natriuresis & diuresis · venous **and** arterial vasodilation → ↓ **preload & afterload**

## DRUGS

| DRUG              | WHAT IT TREATS                                       |
|-------------------|------------------------------------------------------|
| <b>Nesiritide</b> | acute decompensated HF — continuous IV infusion only |

## ADVERSE EFFECTS

- Hypotension — most common
- May be associated with **renal failure**
- ● **May worsen mortality in heart failure** ★

# SGLT2 Inhibitors

## MECHANISM

Block sodium–glucose cotransporter 2 in the proximal tubule → ↑ sodium and water excretion. The cardiac benefit is **far more than the glucose-lowering** it was developed for.

→ **In HFrEF**: ↓ blood volume & pressure → ↓ cardiac workload · ↓ **myocardial fibrosis** → better elasticity · ↓ inflammation · ↓ oxidative stress · **shifts myocardial fuel to ketones** · balances  $\text{Na}^+/\text{K}^+$  → fewer arrhythmias  
→ **In HFpEF**: ↓ volume & pressure → **improved diastolic filling** · ↓ inflammation & fibrosis → **less stiffness** · ↓ oxidative stress · ketone energy efficiency ★ *(one of the few classes that helps HFpEF)*

## DRUGS

| DRUG                 | WHAT IT TREATS             |
|----------------------|----------------------------|
| <b>Dapagliflozin</b> | HFrEF · HFpEF · T2DM · CKD |
| <b>Empagliflozin</b> | HFrEF · HFpEF · T2DM · CKD |

*The lecture names the class and shows the trials, not the individual agents — these are the HF-approved "-gliflozin"s.*



# β-Adrenergic Blockers in HF

## MECHANISM

Counterintuitive — you block the heart in a failing heart — but the target is **chronic catecholamine toxicity**, not acute haemodynamics.

→ **Downstream:** ↓ inotropy & chronotropy (↓ afterload) · **inhibits renin release** (↓ preload) → **The real mechanism:** attenuates the **cytotoxic & signaling effects of circulating catecholamines** → **up-regulates β<sub>1</sub> receptors** → restores receptor function → ★ **improves myocardial remodeling**

## DRUGS

| DRUG                        | WHAT IT TREATS                                                                            |
|-----------------------------|-------------------------------------------------------------------------------------------|
| <b>Metoprolol succinate</b> | HFrEF — β <sub>1</sub> -selective                                                         |
| <b>Bisoprolol</b>           | HFrEF — β <sub>1</sub> -selective                                                         |
| <b>Nebivolol</b>            | HFrEF — β <sub>1</sub> -selective <b>with NO-potentiating effect</b>                      |
| <b>Carvedilol</b>           | HFrEF — β <sub>1</sub> , β <sub>2</sub> <b>and α<sub>1</sub></b> blocker, NO-potentiating |

## USES

- ★ **Only these four are approved in HF** — all four improve survival, and the benefit is **additive to ACEI**.
- **Not for acute therapy** — the benefit only develops with long-term use.

## INTERACTIONS

- All are **CYP2D6 substrates** — *except* **bisoprolol (CYP3A4)** and **carvedilol (CYP2C9)** ★

# Ivabradine

## MECHANISM

Selectively and dose-dependently inhibits the **I<sub>f</sub> current** ("f" for *funny*) carried by **HCN channels**, which are highly expressed in the **sinoatrial node**. I<sub>f</sub> is a mixed Na<sup>+</sup>–K<sup>+</sup> inward current activated by hyperpolarization and modulated by the autonomic nervous system — it sets pacemaker rate.

→ **Downstream:** ↓ pacemaker activity → **pure heart-rate reduction, no effect on contractility** ★ → longer diastole → more time for myocardial blood flow

## DRUGS

| DRUG              | WHAT IT TREATS                                                                        |
|-------------------|---------------------------------------------------------------------------------------|
| <b>Ivabradine</b> | stable chronic HFrEF, <b>LVEF ≤35%</b> , resting <b>HR ≥70</b> in <b>sinus rhythm</b> |

## CAUTIONS

- ⚠ **Sinus rhythm only** — it works at the SA node, so it does nothing useful in AF
- ⚠ Reserved for patients still tachycardic despite maximally tolerated β-blockade

# Other & Emerging Agents

## DRUGS

| DRUG                                                                  | WHAT IT TREATS                                                                                                                         |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vericiguat</b> — soluble <b>guanylate cyclase stimulator</b>       | LVEF <45%, NYHA II–IV; <b>second-line</b> in patients still symptomatic after first-line therapy, mainly after a recent decompensation |
| <b>Omecamtiv mecarbil</b> — selective <b>cardiac myosin activator</b> | ● development <b>halted by the manufacturer</b>                                                                                        |

## ALSO WORTH TREATING

**Iron deficiency** and **skeletal muscle wasting** have significant impact on exercise capacity, quality of life and survival — not footnotes.

# What Actually Changes Mortality

## SURVIVAL BENEFIT VS SYMPTOM RELIEF

| IMPROVES SURVIVAL                                 | SYMPTOM / HAEMODYNAMICS ONLY                     | MAY WORSEN OUTCOMES                      |
|---------------------------------------------------|--------------------------------------------------|------------------------------------------|
| ACEI · ARNI · ARB ( <i>probable</i> )             | Loop diuretics — first-line for congestion       | <b>Milrinone</b> — ↑ mortality signal    |
| β-blockers ( <i>the four</i> ) — additive to ACEI | Digoxin — no long-term survival effect           | <b>Nesiritide</b> — may worsen mortality |
| <b>Spironolactone</b> — NYHA III/IV               | Dobutamine, nitrates, nitroprusside — acute only |                                          |
| SGLT2 inhibitors                                  | Amiloride, triamterene                           |                                          |
| Hydralazine + nitrate (ACEI-intolerant)           |                                                  |                                          |

Principle behind this: *[[pharm-drug-action]]* — Digoxin — low TI and a non-receptor target ( $\text{Na}^+/\text{K}^+$ -ATPase). Both halves of this page in one drug.

Principle behind this: *[[pharm-drug-interactions]]* — Digoxin is the most interacted-with drug in the lecture — rifampicin induces P-gp (level ↓), verapamil inhibits it (level ↑), amiodarone cuts clearance, furosemide's hypokalaemia raises digoxin binding, cholestyramine blocks its absorption. Five separate mechanisms, one drug.

Source: [Pharmacology\\_of\\_Heart\\_Failure\\_2025.pdf](#) — อ.อนันต์ชัย, PYCH304 Pharmacology I, 53 slides. Structures omitted.

# Antithrombotics

Drugs that stop a **thrombus** (a clot that forms in a vessel and stays there) or an **embolus** (one that travels). Three families: **antiplatelets** · **anticoagulants** · **thrombolytics**. ★ The split that decides which family you reach for: **arterial thrombosis is platelet-rich ("white clot")** → **antiplatelet** · **venous thrombosis is fibrin-rich ("red clot")** → **anticoagulant**.

vessel injury → platelet adhesion → activation (TxA<sub>2</sub> · ADP-P2Y<sub>12</sub> · thrombin-PAR-1) → GPIIb/IIIa activation → aggregation || coagulation cascade → FXa → thrombin (FIIa) → fibrinogen → fibrin → stable clot → (*plasminogen* → *plasmin* → *lysis*)

## WHERE EACH GROUP HITS THE CHAIN — AND WHAT YOU MONITOR

| GROUP                        | TARGET IN THE CHAIN                                       | MONITOR                                    |
|------------------------------|-----------------------------------------------------------|--------------------------------------------|
| <b>COX-1 inhibitor</b>       | TxA <sub>2</sub> synthesis in the platelet                | none                                       |
| <b>P2Y12 antagonist</b>      | ADP receptor → platelet activation                        | none                                       |
| <b>GPIIb/IIIa antagonist</b> | the <b>final common step</b> — fibrinogen cross-linking   | none                                       |
| <b>PDE inhibitor</b>         | ↑ cAMP/cGMP → suppresses activation + vasodilates         | none                                       |
| <b>PAR-1 antagonist</b>      | thrombin's receptor on the platelet                       | none                                       |
| <b>UFH</b>                   | AT-mediated: FXIIa, FXIa, <b>FXa</b> , FIXa, <b>FIIa</b>  | <b>aPTT</b> (intrinsic), 1.5–2.5 × control |
| <b>LMWH / fondaparinux</b>   | AT-mediated: <b>FXa</b> >> <b>FIIa</b> (fonda = FXa only) | none routinely                             |
| <b>Warfarin</b>              | hepatic synthesis of vit-K-dependent factors              | <b>INR</b> (extrinsic), target <b>2–3</b>  |
| <b>DOACs</b>                 | thrombin <i>or</i> FXa, directly                          | none                                       |
| <b>Thrombolytics</b>         | plasminogen → plasmin → <b>dissolves formed fibrin</b>    | none                                       |

## TWO BALANCES WORTH HOLDING IN YOUR HEAD

- **Procoagulant vs anticoagulant** — tip one way → **thrombosis**, the other → **bleeding**. Every drug here is a deliberate push toward the bleeding side.
- **Fibrin isn't only pathology** — it is required for wound healing, which is why anticoagulation costs something real.

# COX-1 Inhibitors (Aspirin)

## MECHANISM

**Irreversible, selective** acetylation of platelet **COX-1** — and the platelet has **no nucleus**, so it cannot resynthesise the enzyme. Blocked at the **pre-systemic circulation**, before first-pass.

→ **Downstream:** ↓ **TxA<sub>2</sub>** (which normally drives vasoconstriction + aggregation) · endothelial **PGI<sub>2</sub> is spared** (COX-2-derived) → the balance tips to vasodilation and no aggregation → **The PK paradox** ★: plasma t<sub>1/2</sub> only **15–30 min**, but the effect lasts the **platelet's 7–10-day lifespan**

## DRUGS

| DRUG                            | WHAT IT TREATS                                                             |
|---------------------------------|----------------------------------------------------------------------------|
| <b>Aspirin (ASA)</b> , low-dose | arterial thrombosis — primary & secondary prevention                       |
| <b>Triflusal</b>                | same space; COX-1 inhibition is only <i>one</i> of its possible mechanisms |

## USES & DOSING LOGIC

Inhibits arachidonic-acid-induced aggregation by ~**70–90%** · **10–15% of platelets are replaced daily**, so by 48 h off the drug **20–30%** already respond to TxA<sub>2</sub> normally — which is why it is **dosed daily** and why **most elective surgery needs no discontinuation**.

## ADVERSE EFFECTS

- **Peptic ulcer · GI bleeding** ★ — gastric COX-1 → ↓ PGE<sub>2</sub> → loses gastroprotection (↓ mucus, ↓ bicarbonate, ↓ mucosal blood flow), **plus** direct local irritation
- ↑ **bleeding risk** — the on-target consequence, ↓TxA<sub>2</sub> with PGI<sub>2</sub> unchanged
- Allergy

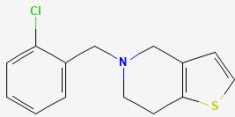
# P2Y12 Antagonists

## MECHANISM

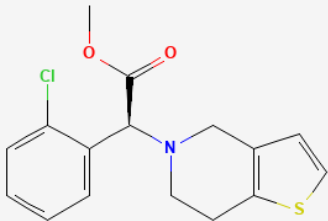
Block the platelet **ADP receptor P2Y12**, cutting off the amplification loop that follows initial activation. Two structural families: 1. **Thienopyridines** — **irreversible** (ticlopidine, clopidogrel, prasugrel); all are **prodrugs**  
2. **Non-thienopyridines** — **reversible** (ticagrelor, cangrelor)

→ **Downstream:** ↓ ADP-induced aggregation, graded by drug — 40–50% → ~100%

## THE SHARED CORE · WHAT CHANGES

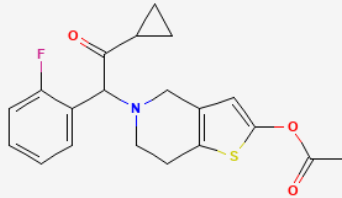


**Ticlopidine — the first thienopyridine.** ● a prodrug. CYP oxidises it to a thiol that forms a permanent disulfide with the P2Y12 receptor — so the platelet is finished for its whole life



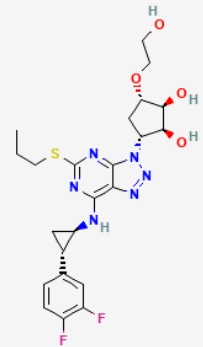
### Clopidogrel

same scaffold, safer marrow profile; ⚠ needs CYP2C19 to activate — poor metabolisers get less effect



### Prasugrel

activated by esterases instead → ● no CYP2C19 dependence, faster and more consistent



### Ticagrelor

not a thienopyridine at all — active as given and **REVERSIBLE**, so effect fades with the drug

## MEDCHEM

**Prodrug** ● The split in this class is prodrug versus not, and it explains everything clinical. The **thienopyridines** (ticlopidine, clopidogrel, prasugrel) are **prodrugs**: they must be oxidised to a **thiol metabolite** that forms a **disulfide bond with P2Y12** - covalent, so inhibition lasts the platelet's life. **Ticagrelor and cangrelor are non-thienopyridine ATP analogues** that bind directly and **reversibly**, with no activation step at all.

**Metabolism** ★ **Clopidogrel's variability is a two-step activation problem.** It needs **CYP2C19** (with 1A2 and 2B6), which is why the 2C19 poor-metaboliser genotype - and CYP2C19-inhibiting PPIs - blunt it, and why a large share of each dose is diverted to the inactive carboxy metabolite instead. **Prasugrel is activated by esterase plus a single CYP step**, which is why it is faster and far less variable. ⚠ Ticlopidine's **M5 and M6 metabolites are linked to its life-threatening neutropenia** - a metabolite problem, not a receptor one.

## DRUGS

| DRUG                                | WHAT IT TREATS                                                                       |
|-------------------------------------|--------------------------------------------------------------------------------------|
| <b>Ticlopidine</b>                  | the first thienopyridine — displaced by its blood dyscrasias                         |
| <b>Clopidogrel</b>                  | PCI · ACS · ACS-PCI · SCAD                                                           |
| <b>Prasugrel</b>                    | ACS-PCI                                                                              |
| <b>Ticagrelor</b>                   | ACS · ACS-PCI                                                                        |
| <b>Cangrelor</b> ( <i>IV only</i> ) | PCI · <b>bridging</b> in high-ischaemic-risk patients going to an invasive procedure |

## THE COMPARISON TABLE ★

|                         | CLOPIDOGREL  | PRASUGREL    | TICAGRELOR                 | CANGRELOR         |
|-------------------------|--------------|--------------|----------------------------|-------------------|
| Max inhibition (%)      | 40–50        | 70–80        | 80–90                      | 100               |
| Bioavailability (%)     | 50           | 80           | 36                         | 100               |
| Half-life (active met.) | 30–60 min    | 2–15 h       | 7–9 h ( <i>met. 12 h</i> ) | 3–6 min           |
| Reversibility           | irreversible | irreversible | <b>reversible</b>          | <b>reversible</b> |
| Onset (loading)         | 2–6 h        | 30 min       | 30 min                     | 2 min             |
| Frequency               | OD           | OD           | <b>BID</b>                 | IV infusion       |
| Duration                | 3–10 d       | 7–10 d       | 3–5 d                      | 1–2 h             |
| Stop before surgery     | ≥ 5 d        | ≥ 7 d        | ≥ 3 d                      | 1 h               |

## ADVERSE EFFECTS

- **Ticlopidine** — nausea/vomiting/diarrhoea, and the serious set (all resolve on stopping): ● **neutropenia** · **aplastic anaemia** · **TTP** · ↑ liver enzymes
- **Clopidogrel** — lower bleeding risk than aspirin, far lower dyscrasia than ticlopidine
- **Prasugrel** — ● **highest bleeding risk** of the group; less TTP than ticlopidine
- **Ticagrelor** — ★ **dyspnoea**, dizziness, headache, from **adenosine reuptake inhibition**, not from P2Y<sub>12</sub>. Bleeding comparable to clopidogrel
- **Cangrelor** — bleeding comparable to clopidogrel

## INTERACTIONS & PHARMACOGENETICS

- ★ **Clopidogrel is bioactivated by CYP2C19** — loss-of-function alleles **CYP2C19\*2 (681G→A)** and **\*3 (636G→A)** blunt the response
- **CYP2C19 inhibitors** — **omeprazole**, **esomeprazole** — reduce clopidogrel's antiplatelet activity
- **Ticagrelor** — metabolised by **CYP3A4**, excreted in faeces

## CAUTIONS

- ● **Prasugrel contraindicated** in prior stroke/TIA, or age ≥ 75
- ⚠ Ticlopidine has **non-linear PK** —  $t_{1/2}$  12.6 h initially, stretching to **4–5 days** on long-term use



## ADMINISTRATION — P2Y12 ANTAGONISTS COMPARED

|                                            | CLOPIDOGREL         | PRASUGREL                                            | TICAGRELOR                                         | CANGRELOR          |
|--------------------------------------------|---------------------|------------------------------------------------------|----------------------------------------------------|--------------------|
| <b>Max platelet inhibition</b>             | 40–50%              | 70–80%                                               | 80–90%                                             | <b>100%</b>        |
| <b>Bioavailability</b>                     | 50%                 | 80%                                                  | <b>36%</b>                                         | 100%               |
| <b>t<sub>1/2</sub> (active metabolite)</b> | 30–60 min           | distribution 30–60 min;<br>elimination <b>2–15 h</b> | <b>7–9 h</b> ( <i>active met.</i><br><i>12 h</i> ) | <b>3–6 min</b>     |
| <b>Binding</b>                             | <b>irreversible</b> | <b>irreversible</b>                                  | reversible                                         | reversible         |
| <b>Onset (loading dose)</b>                | <b>2–6 h</b>        | 30 min                                               | 30 min                                             | ★ <b>2 min</b>     |
| <b>Frequency</b>                           | once daily          | once daily                                           | <b>twice daily</b>                                 | <b>IV infusion</b> |
| <b>Duration of effect</b>                  | 3–10 days           | 7–10 days                                            | 3–5 days                                           | <b>1–2 h</b>       |
| ● <b>Stop before non-acute surgery</b>     | <b>≥ 5 days</b>     | <b>≥ 7 days</b>                                      | <b>≥ 3 days</b>                                    | <b>1 hour</b>      |

- **The stop-before-surgery row is the one that gets people hurt.** It tracks binding reversibility, not half-life: the irreversible drugs need the platelet pool to turn over, which is why prasugrel needs a week and ticagrelor only three days.
- ★ **Cangrelor's whole reason to exist is the bottom two rows** — full inhibition in 2 minutes, gone in 1–2 hours. It is the bridging drug for someone at high ischaemic risk heading into a procedure.
- 💡 Ticagrelor is twice daily and the others are once — an easy thing to get wrong on a discharge script, and a missed second dose leaves a real gap given its 7–9 hour half-life.

# GP1Ib/IIa Antagonists

## MECHANISM

Block **GP1Ib/IIa (integrin  $\alpha$ IIb $\beta$ 3)**, the receptor that binds fibrinogen and physically bridges platelets. This is the **final common pathway** — so it works regardless of which agonist started the activation.

→ **Downstream:** aggregation blocked, but **platelet signalling/activation is untouched** — that is the definitional difference from every other antiplatelet.

## DRUGS

| DRUG                | WHAT IT TREATS                                                                                 |
|---------------------|------------------------------------------------------------------------------------------------|
| <b>Abciximab</b>    | Fab fragment of a humanised chimeric mAb — blocks by <b>steric hindrance</b> , not competition |
| <b>Eptifibatide</b> | cyclic heptapeptide ( <b>KGD</b> sequence) — <b>competitive</b> ; 98% renal, unchanged         |
| <b>Tirofiban</b>    | tyrosine derivative — <b>competitive</b> ; 60–70% renal, unchanged                             |

## ADVERSE EFFECTS

- Bleeding — the strongest antiplatelet blockade available
- **Abciximab:** hypersensitivity reaction (*it's an antibody fragment*)

## CAUTIONS

- ⚠ **Eptifibatide and tirofiban are renally cleared as active drug** — renal impairment accumulates them

# PDE Inhibitors

## MECHANISM

Phosphodiesterases degrade cyclic nucleotides — **PDE-3** degrades **cAMP** (platelets), **PDE-5** degrades **cGMP** (vascular smooth muscle). Inhibit them and the cyclic nucleotide accumulates.

→ **Downstream**: ↑ cAMP in platelets → **antiplatelet** · ↑ cGMP in VSMC → **vasodilation** → **Both also inhibit adenosine reuptake**, which adds to both effects

## DRUGS

| DRUG                                           | WHAT IT TREATS                                                                  |
|------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Dipyridamole</b> ( <i>PDE-3 and PDE-5</i> ) | secondary prevention of <b>stroke</b>                                           |
| <b>Cilostazol</b> ( <i>potent PDE-3</i> )      | secondary prevention of <b>stroke</b> · <b>intermittent claudication</b> in PAD |

## USES

**Secondary stroke prevention** — the shared role of both · **Intermittent claudication (PAD)** — cilostazol, where the vasodilator half of the mechanism is the point

## ADVERSE EFFECTS

- **Dipyridamole** — headache, typically at the start of therapy
- **Cilostazol** — headache, loose stool, **palpitation** (*PDE-3 is also cardiac*)

## INTERACTIONS

- **Cilostazol** is metabolised by **CYP3A5**; **dipyridamole** by **glucuronidation**. Half-life **10 h** for both.

# PAR-1 Antagonists

## MECHANISM

**PAR-1** is the platelet's thrombin receptor and mediates activation at **low** thrombin concentrations; **PAR-4** lacks the thrombin-binding site and only responds at **high** concentrations. Blocking PAR-1 therefore removes the physiologically dominant thrombin signal.

→ **Downstream:** ↓ thrombin-induced aggregation by **80% at 2 h** · **thrombin's enzymatic activity is untouched** — fibrin still forms, only the platelet stops listening

## DRUGS

| DRUG             | WHAT IT TREATS                                                                                    |
|------------------|---------------------------------------------------------------------------------------------------|
| <b>Vorapaxar</b> | secondary prevention after <b>prior MI</b> or in <b>PAD</b> — FDA-approved, <b>not first line</b> |

## CAUTIONS

-  **Contraindicated in prior stroke**
-  Reversible but **dissociates very slowly** — half-life **150–300 h**, so "reversible" is not "quickly undone"
- Metabolised by **CYP3A4**

# Heparins & Fondaparinux

## MECHANISM

**Indirect** — they bind **antithrombin (AT)** and accelerate it enormously. Chain length decides which factors AT can then reach: only a **long** chain can bridge AT to thrombin, whereas the **pentasaccharide** alone is enough to inactivate FXa.

→ **UFH**: inhibits FXIIa, FXIa, **FXa**, FIXa, **FIIa** — anti-FXa:FIIa = **1.0** → **LMWH**: **FXa** > **FIIa** (ratio 2.2–3.9) · also releases **TFPI** from endothelium → **Fondaparinux**: synthetic pentasaccharide, **AT-binding site only** → **FXa exclusively**

## DRUGS

| DRUG                          | WHAT IT TREATS                                                 |
|-------------------------------|----------------------------------------------------------------|
| <b>UFH</b>                    | acute anticoagulation where titration and fast reversal matter |
| <b>Enoxaparin</b>             | VTE · ACS — <b>IV loading in STEMI</b>                         |
| <b>Nadroparin, Dalteparin</b> | VTE prophylaxis & treatment                                    |
| <b>Fondaparinux</b>           | VTE — the option when HIT risk must be near zero               |

## UFH VS LMWH VS FONDAPARINUX ★

|                         | UFH                          | ENOXAPARIN      | NADROPARIN      | DALTEPARIN      | FONDAPARINUX    |
|-------------------------|------------------------------|-----------------|-----------------|-----------------|-----------------|
| Target                  | FXIIa, FXIa, FXa, FIXa, FIIa | FXa > FIIa      | FXa > FIIa      | FXa > FIIa      | <b>FXa only</b> |
| anti-FXa : FIIa         | 1.0                          | 3.9             | 3.5             | 2.2             | —               |
| Bioavailability (%)     | 30                           | 91              | 67              | 83              | <b>100</b>      |
| Half-life (h)           | 1                            | 4.5             | 3               | 2.3             | <b>17</b>       |
| Renal excretion, active | no                           | yes             | yes             | yes             | yes             |
| Antidote                | <b>protamine</b>             | protamine (60%) | protamine (60%) | protamine (60%) | ● <b>none</b>   |
| HIT (%)                 | < 5                          | < 1             | < 1             | < 1             | case reports    |

## WHY YOU'D PICK LMWH

**More predictable effect · no lab monitoring · 70–90% s.c. bioavailability · BID or OD dosing · lower HIT rate.** UFH keeps its place where a **1-hour half-life and full protamine reversal** are worth the aPTT titration.

## ADVERSE EFFECTS

- **Bleeding**
- ● **HIT** — heparin-induced thrombocytopenia. LMWH < 1% but **cross-reacts with UFH**, so it is not a rescue
- **Osteoporosis** — long-term use > **1 month**

## CAUTIONS

- ⚠️ **UFH has dose-dependent kinetics:** low dose = saturable **zero-order** clearance by plasma enzymes, endothelium and macrophages; high dose = **non-saturable renal** clearance.  $t_{1/2}$  stretches 30 min → ~2.5 h
- ⚠️ **Fondaparinux offset takes 2–4 days** after stopping, and there is **no antidote**
- ⚠️ Everything except UFH is renally excreted as active drug

## ADMINISTRATION — LMWH AND DABIGATRAN

- **LMWH: 70–90% bioavailability after subcutaneous injection**, plasma  $t_{1/2}$  **3–4 h** (*mean residence time for anti-FXa 5–7 h*), **dosed once or twice daily**. ★ **No laboratory monitoring needed** — the effect is predictable, unlike unfractionated heparin. **Enoxaparin IV loading is used in STEMI**.
- **Dabigatran: usual dosing 110–150 mg twice daily.**
- 💡 "No monitoring" is the entire clinical argument for LMWH over UFH, and it comes straight from that 70–90% predictable subcutaneous absorption. Nothing about UFH's dose–response is that reliable.

# Warfarin

## MECHANISM

Blocks **VKORC1**, the enzyme that regenerates reduced vitamin K — so the  $\gamma$ -carboxylation of factors **II, VII, IX, X** (and proteins C & S) cannot happen. It disables **synthesis**, not circulating factor.

→ **Downstream**: effect appears only as existing factors decay — **onset 3–7 days**, so a **rapid-acting anticoagulant must cover the start** ★ → **Monitor INR (extrinsic pathway), target 2–3**

## DRUGS

| DRUG            | WHAT IT TREATS                                                |
|-----------------|---------------------------------------------------------------|
| <b>Warfarin</b> | VTE · AF · mechanical valves — long-term oral anticoagulation |

## ADVERSE EFFECTS

- **Bleeding** — the dose-limiting effect, and why INR exists
- **Warfarin-induced skin necrosis** ★ — a **paradoxical** reaction on **day 3–8**, because protein C (short half-life) falls before the procoagulant factors do
- **Purple toe syndrome**
- (*serious ADRs run 1–10%*)

## INTERACTIONS & PHARMACOGENETICS

- ★ **CYP2C9 poor metabolisers** — \*2 (144R→C), \*3 (359I→L) — clear warfarin slowly
- ★ **VKORC1 (1639G→A)** — a more sensitive target, so a lower dose suffices
- ● The interaction list is enormous (CYP inducers/inhibitors, antibiotics, **certain NSAIDs, dose-related**) — assume any new drug moves the INR until proven otherwise

## REVERSAL

- **Antidote: vitamin K<sub>1</sub> (phytonadione)** — slow, because it works by restoring synthesis
- **For severe bleeding, replace the factors: 3-PCC · 4-PCC · FFP · rFVIIa**

# DOACs / NOACs

## MECHANISM

**Direct** — they bind the enzyme itself, no antithrombin needed, and reach **both free and clot-bound** target. All are **competitive, reversible, dose-dependent**, and all are **P-gp substrates**.

→ **Direct thrombin inhibitor**: dabigatran, > **100-fold selective** for thrombin over other serine proteases → **Direct FXa inhibitors**: ≈ **10,000-fold** (rivaroxaban) to > **86,000-fold** (betrixaban) FXa-selective → **No routine monitoring** — that's the whole clinical argument against warfarin

## DRUGS

| DRUG                                                    | WHAT IT TREATS                                                                              |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Dabigatran etexilate</b>                             | the <b>only oral</b> direct thrombin inhibitor — a prodrug, because dabigatran is too polar |
| <b>Desirudin · Lepirudin · Bivalirudin · Argatroban</b> | injectable direct thrombin inhibitors                                                       |
| <b>Rivaroxaban · Apixaban · Edoxaban · Betrixaban</b>   | direct FXa inhibitors — VTE, AF                                                             |
| (Ximelagatran)                                          | <b>withdrawn — hepatotoxicity</b>                                                           |

## PK THAT ACTUALLY CHANGES DECISIONS

|                      | DABIGATRAN | RIVAROXABAN            | APIXABAN      | EDOXABAN | BETRIXABAN |
|----------------------|------------|------------------------|---------------|----------|------------|
| Bioavailability (%)  | <b>6–7</b> | 80–100                 | 50            | 62       | 34         |
| Tmax (h)             | 1.5–3      | 2–4                    | 1–3           | 1–2      | 3–4        |
| Half-life (h)        | 12–17      | 7–11                   | 8–14          | 10–14    | 19–27      |
| Protein binding (%)  | 30–35      | 90–95                  | 87            | 40–59    | 60         |
| CYP metabolism       | < 2%       | <b>CYP3A4</b> , CYP2J2 | <b>CYP3A4</b> | < 4%     | < 1%       |
| Renal, unchanged (%) | <b>80</b>  | 33                     | 27            | 35–50    | 11         |
| P-gp substrate       | yes        | yes                    | yes           | yes      | yes        |

⚠ **All of them need dose reduction in renal impairment** — dabigatran most of all, at 80% renal.

## ADVERSE EFFECTS

- **Bleeding**
- ★ **Dabigatran — dyspepsia/oesophagitis**: heartburn, chest pain, upper abdominal pain soon after a dose. Believed to be the **tartaric acid** in the capsule sticking to the oesophagus. *Fix: ≥ 150 mL water, stay upright ≥ 30 min, add a PPI or H2RA; if it persists, switch to apixaban, rivaroxaban or warfarin.*
- Dabigatran, RE-LY trial — slightly **higher MI rate** than warfarin (0.72–0.74%/yr vs 0.53%/yr)



### INTERACTIONS — ALL OF IT RUNS THROUGH P-GP ± CYP3A4

-  **↑ exposure (bleeding):** P-gp / CYP3A4 **inhibitors** — verapamil, diltiazem, dronedarone, amiodarone, quinidine, clarithromycin, erythromycin, ketoconazole, itraconazole, posaconazole, ritonavir & other PIs, ciclosporin, tacrolimus, **grapefruit**
-  **↓ exposure (clot):** P-gp / CYP3A4 **inducers** — rifampicin, carbamazepine, phenytoin, barbiturates, dexamethasone, **St John's wort**
-  **Rivaroxaban and apixaban are the CYP3A4-dependent pair;** edoxaban is essentially non-CYP — so CYP interactions matter far less for it

### REVERSAL

- A **specific antidote is currently approved for rivaroxaban and apixaban**
-  Otherwise reversal rests on stopping the drug and supportive/factor replacement

# Thrombolytics / Fibrinolytics

## MECHANISM

The only group that **destroys a clot already formed** — everything else merely prevents one. They convert **plasminogen** → **plasmin**, which digests fibrin.

→ **Direct (tPA family)**: alteplase, tenecteplase, reteplase — act on plasminogen directly, **fibrin-selective** → **Indirect**: streptokinase — a **bacterial ( $\beta$ -haemolytic streptococcal) enzyme** that must first form a complex with plasminogen; **non-fibrin-specific**, so it lyses systemically

## DRUGS

| DRUG                                              | WHAT IT TREATS                                                                            |
|---------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Streptokinase</b>                              | ACS · PE                                                                                  |
| <b>Alteplase</b> ( <i>recombinant human tPA</i> ) | <b>acute ischaemic stroke</b> · ACS · acute PE · central venous catheter occlusion        |
| <b>Tenecteplase</b>                               | acute ischaemic stroke · ACS                                                              |
| <b>Reteplase</b>                                  | ACS — <span style="color: red;">●</span> <b>contraindicated in acute ischaemic stroke</b> |

## COMPARISON ★

|                        | STREPTOKINASE                                 | ALTEPLASE | TENECTEPLASE | RETEPLASE      |
|------------------------|-----------------------------------------------|-----------|--------------|----------------|
| Action                 | <b>indirect</b>                               | direct    | direct       | direct         |
| Fibrin specificity     | <b>non-specific</b>                           | ++        | +++          | +              |
| Clot penetration       | —                                             | +         | +            | +++            |
| PAI-1 resistance       | —                                             | ++        | ++++         | +              |
| t <sub>1/2</sub> (min) | 20                                            | < 5       | 20           | 13–16          |
| Excretion              | liver                                         | liver     | liver        | liver & kidney |
| Hypersensitivity       | <span style="color: red;">●</span> <b>yes</b> | no        | no           | no             |

## ADVERSE EFFECTS

- **Intracranial haemorrhage (ICH)** and **death** — the reason these are used under strict criteria
- Systemic bleeding — worst with **non-fibrin-specific** streptokinase
- Streptokinase: hypersensitivity** — anti-streptokinase antibody stays elevated for **up to 4 years**, so it cannot simply be repeated ★

# Reversal & Clinical Use

## ANTIDOTE MAP

| DRUG                         | ANTIDOTE                                                                                         |
|------------------------------|--------------------------------------------------------------------------------------------------|
| <b>UFH</b>                   | <b>protamine</b> — full effect                                                                   |
| <b>LMWH</b>                  | protamine — <b>only ~60%</b>                                                                     |
| <b>Fondaparinux</b>          |  none           |
| <b>Warfarin</b>              | <b>vitamin K<sub>1</sub> (phytonadione)</b> ; severe bleed → <b>3-PCC / 4-PCC / FFP / rFVIIa</b> |
| <b>Rivaroxaban, apixaban</b> | specific reversal agent, approved                                                                |
| <b>Antiplatelets</b>         | none — you wait out the platelet turnover ( <i>see the discontinuation table</i> )               |

## THE THREE CLINICAL ROLES

**Prophylaxis / primary prevention** · **Treatment** of an established thrombus · **Secondary prevention** — stopping recurrence, which is where most of these drugs actually live.

*Principle behind this: [[pharm-pharmacokinetics]] — clopidogrel is a prodrug requiring CYP2C19 — prodrug activation and CYP polymorphism in a single drug.*

*Principle behind this: [[pharm-drug-action]] — Warfarin is the low-TI drug, which is why it's monitored rather than simply dosed.*

*Principle behind this: [[pharm-drug-interactions]] — Warfarin displaced by chloral hydrate, induced away by rifampicin, inhibited up by cimetidine and amiodarone, opposed by dietary vitamin K — and heparin reversed by protamine (chemical antagonism). Plus the smoking/clopidogrel case.*

*Principle behind this: [[pharm-drug-toxicity]] — Warfarin variability = CYP2C9 (metabolism) + VKORC1 (target) — genetics on both sides of the drug–receptor pair.*

*Principle behind this: [[pharm-pharmacogenomics]] — Warfarin is the flagship — CYP2C9 for clearance and VKORC1 for target sensitivity, both polymorphic, both with Thai frequencies, against a narrow therapeutic window. Clopidogrel is the other side: a prodrug needing CYP2C19.*

*Source: Pharmacology of Antithrombotics\_2025\_SW.pdf — อ.สุรศักดิ์ (Surasak Wichaiyo), PYCH304 Pharmacology I, 60 slides. Structures omitted.*

# Lipid-lowering Agents

Lipoproteins carry cholesterol (free + esterified), **triglycerides** and phospholipids, wrapped in **apolipoproteins** that give structure and act as receptor ligands / enzyme cofactors. Lowering **LDL-C** is the goal because LDL-C drives **atherosclerosis** — and CVD/CHD mortality tracks it.

↓ **hepatic cholesterol** → ↑ **LDL receptor expression** → ↑ **LDL clearance from plasma** → ↓ **LDL-C** → ↓ **atherosclerotic risk**

## THE THREE LIPID PATHWAYS (ALL COORDINATED BY THE LIVER)

- **Exogenous** — dietary lipid → chylomicrons → remnants → liver
- **Endogenous** — hepatic *de novo* synthesis → VLDL → IDL → LDL. ★ **The liver removes ~75% of all plasma LDL** via its LDL receptors — which is why almost every drug here works by raising LDL-receptor number.
- **Reverse** — HDL collects cholesterol from tissue and returns it to the liver

## THE 8 CLASSES AT A GLANCE

| CLASS                                  | LDL-C                      | HDL-C                               | TG            |
|----------------------------------------|----------------------------|-------------------------------------|---------------|
| <b>Statin</b> (HMG-CoA reductase inh.) | ↓↓↓ 24–63%                 | ↑ 6–12%                             | ↓ 10–29%      |
| <b>Bempedoic acid</b> (ACLY inh.)      | ↓↓ 25–30%                  | —                                   | —             |
| <b>PCSK9 inhibitor</b>                 | ↓↓↓ 58–64%                 | —                                   | —             |
| <b>Ezetimibe</b> (absorption inh.)     | ↓ 15–20%                   | —                                   | —             |
| <b>Bile acid sequestrant</b>           | ↓ 10–20%                   | —                                   | ● ↑           |
| <b>Fibrate</b>                         | —                          | ↑ ~15%                              | ↓↓↓ up to 50% |
| <b>Niacin</b>                          | ↓ 25% ( <i>high dose</i> ) | ★ ↑↑ <b>30–40%</b> — best available | ↓↓ 35–50%     |
| <b>Omega-3 (EPA/DHA)</b>               | ⚠ ↑                        | —                                   | ↓ ~20%        |

## WHICH PROBLEM, WHICH DRUG

| PRIMARY TARGET                                    | CLASSES                                                                       |
|---------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Primarily ↓ LDL-C</b>                          | statin · bempedoic acid · PCSK9 inhibitor · ezetimibe · bile acid sequestrant |
| <b>Primarily ↓ TG</b>                             | fibrate · niacin · omega-3                                                    |
| <b>Familial hypercholesterolemia (self-study)</b> | MTP inhibitor · apo B-100 synthesis inhibitor · ANGPTL3 inhibitor             |

# Statins (HMG-CoA Reductase Inhibitors)

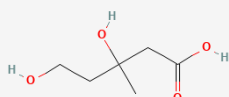
## MECHANISM

A **mevalonic-acid-like moiety** that **competitively inhibits HMG-CoA reductase** — the early, **rate-limiting** step of cholesterol biosynthesis.

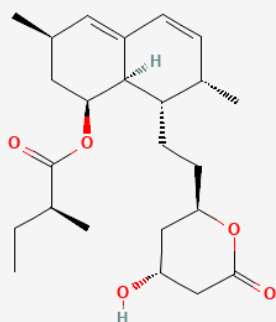
→ **Downstream:** ↓ hepatic cholesterol → ↑ LDL receptors → ↓ **LDL-C (the major effect)** · ↓ VLDL production (*cholesterol is a required VLDL component*) · **pleiotropic effects** beyond lipids

*Simvastatin and lovastatin are inactive lactones — the liver must convert them to their  $\beta$ -hydroxy acids.*

## THE SHARED CORE · WHAT CHANGES

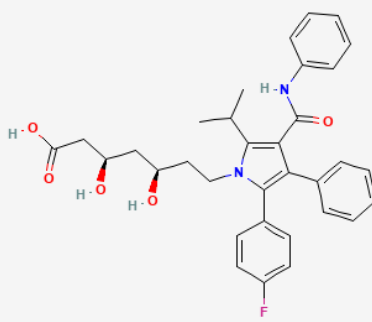


**Mevalonate — the product being mimicked.** ● every statin carries an arm shaped like this. That is the whole mechanism: the enzyme grabs the fake and stops working.



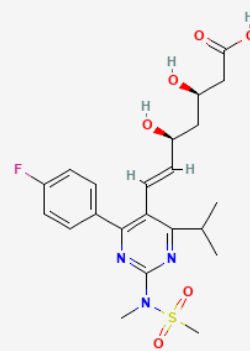
**Lovastatin**

fungal, and a ● **LACTONE** prodrug — the ring must open in the body to expose the acid arm



**Atorvastatin**

synthetic, already an open acid, F-phenyl added → more potent, CYP3A4-metabolised



**Rosuvastatin**

a polar sulfonamide → hydrophilic → liver-selective via OATP uptake, barely touched by CYP3A4

## MEDCHEM

**SAR** The pharmacophore is the **3,5-dihydroxyheptanoic acid** arm and its absolute stereochemistry (**3R,5R**); the two-carbon spacer between C5 and the ring system cannot be shortened or lengthened. Everything else is the anchor: **type 1** statins use the natural **decalin ring** (lovastatin, simvastatin, pravastatin), **type 2** use a synthetic heteroaromatic ring with a **p-fluorophenyl** and an isopropyl group (atorvastatin, rosuvastatin, fluvastatin, pitavastatin).

**Prodrug** Lovastatin and simvastatin are **lactone prodrugs** - the ring has to be opened to the hydroxy acid before anything happens.

**Physicochem** ★ **Hepatoselectivity is just logP.** Pravastatin cLogP **1.44** and rosuvastatin **0.42** are hydrophilic, so they need active hepatic uptake and stay out of muscle; simvastatin **4.42**, fluvastatin 4.13 and lovastatin 4.07 diffuse everywhere. That single number is most of the myopathy-risk conversation.

**Metabolism** ● **This is why some statins are bedtime drugs and some aren't.** Lovastatin, simvastatin (CYP3A4) and fluvastatin (CYP2C9) produce **inactive** metabolites and have short half-lives → **take at bedtime**, when cholesterol synthesis peaks. **Atorvastatin's metabolites are active** and rosuvastatin largely resists CYP inactivation → **take any time**. Pravastatin bypasses CYP but degrades in stomach acid.

## DRUGS

| DRUG                | WHAT IT TREATS                                                                                               |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Rosuvastatin</b> | hypercholesterolemia — <b>high-intensity</b> , ↓LDL 63% · long $t_{1/2}$                                     |
| <b>Atorvastatin</b> | hypercholesterolemia — <b>high-intensity</b> , ↓LDL 50% · long $t_{1/2}$                                     |
| <b>Pitavastatin</b> | hypercholesterolemia — ↓LDL 48% · ★ <b>statin of choice in severe renal dysfunction</b> (no dose adjustment) |
| <b>Simvastatin</b>  | hypercholesterolemia — ↓LDL 41%                                                                              |
| <b>Pravastatin</b>  | hypercholesterolemia — ↓LDL 34% · better muscle tolerability                                                 |
| <b>Lovastatin</b>   | hypercholesterolemia — ↓LDL 34% · absorption ↑ with food                                                     |
| <b>Fluvastatin</b>  | hypercholesterolemia — ↓LDL 24%                                                                              |

## USES

**Hypercholesterolemia / ↑ LDL-C · ASCVD risk reduction** — dosed by *intensity*: high-intensity ↓LDL ≥50% · moderate 30–49% · low <30%

## PK WORTH KEEPING ★

- **Extensive first-pass hepatic uptake via OATP1B1** — the transporter that explains the interactions below
- Metabolites retain HMG-CoA reductase activity — **except fluvastatin and pravastatin**
- 70% of metabolites excreted by liver → **feces**
- **Dosing time:** statins with  $t_{1/2} \leq 4 \text{ h}$  → **take in the evening** (hepatic cholesterol synthesis peaks at night). **Atorvastatin and rosuvastatin** have long  $t_{1/2}$  → any time of day, whichever helps adherence — and that longer  $t_{1/2}$  contributes to their greater efficacy.

## ADVERSE EFFECTS

- ● **Myopathy** ★ — the class-defining ADR. *Rosuvastatin and pravastatin may be better tolerated — consider them in a patient with myalgias on another statin.*
- Hepatotoxicity — **very rare**

## INTERACTIONS

- ● **CYP3A4 inhibitors** → ↑ statin levels → **may precipitate statin-induced myopathy**
- ● **Gemfibrozil** → OATP1B1-mediated ↑ statin exposure **plus** additive muscle toxicity
- ★ **Pravastatin, pitavastatin, fluvastatin, rosuvastatin** are *not* extensively CYP3A4-metabolised → **less likely to cause myopathy** with a predisposing drug

# Bempedoic Acid (ATP-Citrate Lyase Inhibitor)

## MECHANISM

Inhibits **ATP-citrate lyase (ACLY)**, a step **upstream of HMG-CoA reductase** in hepatic cholesterol synthesis. It is a **prodrug requiring activation by hepatic ACSVL1** — an enzyme muscle lacks.

→ **Downstream:** ↓ hepatic cholesterol synthesis → ↑ LDL receptor activity → ↓ **LDL-C 25–30%** · ↓ hs-CRP ~20% → **Why it matters:** liver-only activation → **low/no muscle toxicity relative to statins**

## DRUGS

| DRUG                  | WHAT IT TREATS                                                                                          |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| <b>Bempedoic acid</b> | hypercholesterolemia — ★ especially the <b>statin-intolerant</b> patient (ASCVD risk ↓, CLEAR-OUTCOMES) |

## USES

↑ **LDL-C** · **ASCVD risk reduction in statin-intolerant patients** · orally active,  $t_{1/2}$  15–24 h

## ADVERSE EFFECTS

- **Headache** — most common, 10–20%
- Constipation (7%)
- Myalgia (1–7%) — notably *low* for a lipid drug

# PCSK9 Inhibitors

## MECHANISM

**PCSK9** is a protease that binds the hepatocyte **LDL receptor** and sends it for **lysosomal degradation**. Block PCSK9 and the receptor is recycled instead of destroyed.

→ **Downstream:** ↑ LDL receptors on hepatocytes → ↓ **LDL-C 58–64%** · pleiotropic effects → ★ **The statin link:** statins activate **SREBP-2**, which raises LDL receptors *but also raises PCSK9* — a built-in brake. Adding a PCSK9 inhibitor removes it, so the two are strongly complementary.

## DRUGS

| DRUG                       | WHAT IT TREATS                                                                                                               |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Evolocumab</b><br>(mAb) | ↑ LDL-C — ↓ LDL 64% · SC Q2W or monthly · max PCSK9 suppression 85–95%                                                       |
| <b>Alirocumab</b><br>(mAb) | ↑ LDL-C — ↓ LDL 58% · SC Q2W                                                                                                 |
| <b>Inclisiran</b> (siRNA)  | ↑ LDL-C — ★ dosed <b>day 1, month 3, then every 6 months</b> (plasma $t_{1/2}$ only ~9.6 h but liver tissue $t_{1/2}$ 270 h) |

## USES

↑ **LDL-C** — LDL reduction **equivalent to statins**, added on when statin alone is not enough or not tolerated  
· onset: nadir LDL-C at ~**14–21 days**

## ADVERSE EFFECTS

- **Injection-site reactions** — the most frequent, <10%
- Slight ↑ infection risk (*as with other monoclonal antibodies*) — nasopharyngitis, UTI, URI
- Neurocognitive effects — small, <1%

## INTERACTIONS

- ★ **None expected** — being antibodies/siRNA, they sidestep CYP and transporter interactions entirely



## ADMINISTRATION — PCSK9 INHIBITORS

|                              | ALIROCUMAB                                           | EVOLOCUMAB                                          |
|------------------------------|------------------------------------------------------|-----------------------------------------------------|
| <b>Dose &amp; route</b>      | <b>75–150 mg SC every 2 weeks</b>                    | <b>140 mg SC every 2 weeks or 420 mg SC monthly</b> |
| <b>Bioavailability</b>       | 85%                                                  | 72%                                                 |
| <b>Tmax</b>                  | 3–7 days (median)                                    | 3–4 days (median)                                   |
| <b>Time to steady state</b>  | 4–6 weeks                                            | <b>12 weeks</b>                                     |
| <b>Half-life</b>             | 17–20 days — <b>12 days when given with a statin</b> | 11–17 days                                          |
| <b>Max PCSK9 suppression</b> | 4–8 h                                                | 4 h (85–95% suppression)                            |
| <b>Time to LDL-C nadir</b>   | 15 days                                              | 14 days (140 mg) · 21 days (420 mg)                 |
| <b>LDL-C reduction</b>       | <b>58%</b> (150 mg)                                  | <b>64%</b> (420 mg)                                 |

- ★ **Evolocumab's monthly 420 mg is simply three fortnightly doses given at once** — same drug, and the choice is adherence, not efficacy.
- ⚠ **Alirocumab's half-life drops from 17–20 days to 12 when co-administered with a statin** — the combination changes the antibody's own kinetics, which is not obvious.
- 💡 An LDL nadir at two to three weeks means **rechecking lipids at one week tells you nothing**. Wait for the nadir before judging whether it worked.

# Ezetimibe (Cholesterol Absorption Inhibitor)

## MECHANISM

Inhibits **NPC1L1**, the intestinal sterol transporter — so dietary/biliary cholesterol is not taken into the enterocyte. Active form is **ezetimibe glucuronide**.

→ **Downstream:** ↓ cholesterol delivered to liver by chylomicron remnants → ↑ hepatic LDL receptor expression → ↓ **LDL-C 15–20%** → **Compensation:** hepatic cholesterol **biosynthesis rises** in response — which is exactly what a statin blocks

## DRUGS

| DRUG             | WHAT IT TREATS                             |
|------------------|--------------------------------------------|
| <b>Ezetimibe</b> | ↑ LDL-C — usually <b>added to a statin</b> |

## USES

↑ **LDL-C** · ★ **Dual therapy with a statin is more than additive in logic:** ezetimibe blocks the absorption that statins increase, the statin blocks the synthesis that ezetimibe increases — each cancels the other's escape route

## PK WORTH KEEPING

- Highly **water-insoluble**; glucuronidated in the intestinal epithelium, then **enterohepatic recirculation**
- ~70% excreted in feces, ~10% in urine
- **No dose adjustment** in mild hepatic dysfunction or *any* grade of renal insufficiency

## INTERACTIONS

- ⚠ **Bile acid sequestrants inhibit ezetimibe absorption** — separate the doses

# Bile Acid Sequestrants

## MECHANISM

Highly **positively charged, non-absorbable** resins that bind **negatively charged bile acids** in the gut and carry them out in the stool — interrupting the enterohepatic circulation that normally recovers **>95%** of bile acids.

→ **Downstream:** ↓ bile acid pool → ↑ hepatic bile acid synthesis → ↓ hepatic cholesterol → ↑ **LDL receptors (same endpoint as statins)** → ↓ **LDL-C 10–20%** → ● **The backfire:** the liver compensates by upregulating **HMG-CoA reductase** (↑ cholesterol synthesis) and ↑ **triglyceride synthesis**

## DRUGS

| DRUG                  | WHAT IT TREATS |
|-----------------------|----------------|
| <b>Cholestyramine</b> | ↑ LDL-C        |
| <b>Colestipol</b>     | ↑ LDL-C        |
| <b>Colesevelam</b>    | ↑ LDL-C        |

## USES

↑ **LDL-C** — modest effect, no systemic absorption, so useful where systemic drug exposure is unwanted

## ADVERSE EFFECTS

- **Gritty slurry** — cholestyramine/colestipol powder is genuinely unpleasant to drink → adherence problem
- GI: **bloating, dyspepsia, constipation**
- ● **Hypertriglyceridemia** ★ — the one class here that makes TG *worse*

## INTERACTIONS

- ★ **Reduces absorption of other drugs** — non-specific binding. Give other drugs **1 h before or 3–4 h after** the resin.

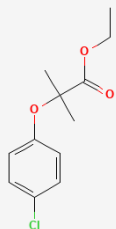
# Fibrates (Fibric Acid Derivatives)

## MECHANISM

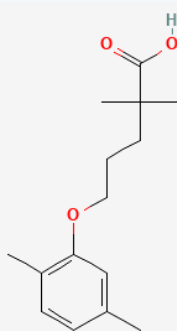
**PPAR $\alpha$  agonists** — the nuclear receptor that governs fatty-acid handling.

→ **Downstream:** ↑ **TG clearance** (lipoprotein lipase activity) · ↓ hepatic TG synthesis → ↓ **TG up to 50%** · ↑ **HDL-C ~15%** → **Extra:** most fibrates are **antithrombotic** — inhibit coagulation, enhance fibrinolysis

## THE SHARED CORE · WHAT CHANGES

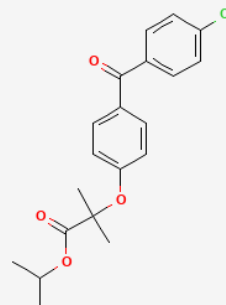


**Clofibrate — the parent.** an isobutyric acid on a chlorophenyl ether. The acid group is what binds PPAR- $\alpha$



**Gemfibrozil**

● inhibits the glucuronidation of statins → the classic rhabdomyolysis pairing



**Fenofibrate**

an ester prodrug → safer with statins than gemfibrozil

## MEDCHEM

**SAR** The scaffold is **phenoxyisobutyric acid** and both halves are required - remove the isobutyric acid or the phenoxy and the activity goes. The only real variable is the R group and the linker length ( $n = 0-3$ ), which set lipophilicity, absorption, hepatic extraction and how fast the drug is metabolised.

**Prodrug Fenofibrate is an ester prodrug** hydrolysed to fenofibric acid. Its cLogP is **5.24** against gemfibrozil's 3.9 - and that extra lipophilicity is exactly why fenofibrate is once daily and gemfibrozil is twice.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                          |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Fenofibrate</b> | hypertriglyceridemia — ★ <b>better than gemfibrozil at raising HDL</b> · $t_{1/2}$ 20 h |
| <b>Gemfibrozil</b> | hypertriglyceridemia — $t_{1/2}$ 1–2 h · ● <b>the one with the statin interaction</b>   |
| <b>Bezafibrate</b> | hypertriglyceridemia — $t_{1/2}$ 1–2 h                                                  |
| <b>Clofibrate</b>  | hypertriglyceridemia                                                                    |

## USES

★ **Drugs of choice for severe hypertriglyceridemia (TG > 1000 mg/dL) and chylomicronemia syndrome** — lowering TG **prevents pancreatitis** · **Type III hyperlipoproteinemia (dysbetalipoproteinemia)** — the most sensitive responders, both TG and cholesterol fall dramatically · Mild hypertriglyceridemia (TG < 400 mg/dL)

## PK WORTH KEEPING

- Absorbed rapidly and >90% **when given with a meal**; T<sub>max</sub> 1–4 h
- **>95% protein-bound**, almost entirely to albumin — this is the warfarin problem
- Excreted mainly as **glucuronides in urine** → ⚠ **excretion impaired in renal failure**

## ADVERSE EFFECTS

- GI — most common, <5%
- **Myalgias**
- Minor ↑ liver transaminases and alkaline phosphatase
- **Gallstones** ★ — bile becomes cholesterol-supersaturated

## INTERACTIONS

- ● **Warfarin** → ↑ effect (*albumin displacement*) — bleeding risk
- ● **Gemfibrozil + statin** ★ — **double hit**: ↑ statin levels (PK, via OATP1B1) **and** additive myopathy (PD)

# Niacin (Nicotinic Acid)

## MECHANISM

A water-soluble B vitamin that acts as a vitamin only once converted to **NAD/NADP** — but ★ **only niacin itself lowers lipids**, and only at **doses far above the vitamin dose**. Acts on adipocyte **HCA2 (hydroxy-carboxylic acid receptor 2)** and hepatic **DGAT2**.

→ **Downstream:** ↓ adipose lipolysis (HSL/ATGL) → ↓ free fatty acids to liver → ↓ VLDL/TG synthesis → **Net:** ↓ **TG 35–50%** · ↑ **HDL-C 30–40%** — **the most effective agent available for raising HDL** ★ · ↓ LDL-C up to 25% at high dose

## DRUGS

| DRUG                        | WHAT IT TREATS                                                                                |
|-----------------------------|-----------------------------------------------------------------------------------------------|
| <b>Niacin (crystalline)</b> | hypertriglyceridemia (2–6 g/d, as effective as fibrates/statins) · elevated LDL-C · low HDL-C |

## USES

**Hypertriglyceridemia · Elevated LDL-C** · the go-to when the problem is **low HDL-C** · t½ 60 min → **BID–TID dosing**

## ADVERSE EFFECTS

- ★ **Flushing and skin reactions** — most common, and the main reason patients quit
- Dyspepsia, nausea, vomiting, diarrhea
- **Hepatotoxicity** — ↑ AST/ALT, ↓ serum albumin
- **Myopathy when combined with a statin** ●
- **Hyperglycemia** — caution in DM
- **Hyperuricemia** — caution in gout

# Omega-3 Fatty Acids

## MECHANISM

**EPA and DHA ethyl esters** reduce hepatic **VLDL-triglyceride** production and secretion.

→ **Downstream:** ↓ **TG** · ↓ hs-CRP (~40% with icosapent ethyl) · ⚠️ **LDL may rise** → **Outcome data:** REDUCE-IT — icosapent ethyl 2 g BID in patients with fasting TG 135–499 mg/dL *despite statin*: TG ↓20%, ischemic events including CV death ↓25–30%

## DRUGS

| DRUG                                   | WHAT IT TREATS                                                                |
|----------------------------------------|-------------------------------------------------------------------------------|
| <b>Icosapent ethyl</b>                 | hypertriglyceridemia + residual CV risk on a statin — <b>EPA only, no DHA</b> |
| <b>EPA/DHA ethyl esters (fish oil)</b> | hypertriglyceridemia — 3–4 g/d <b>with food</b>                               |

## USES

**Hypertriglyceridemia** — FDA indication at **TG ≥ 500 mg/dL** · **residual CV risk** on optimal statin therapy

## PK WORTH KEEPING

- Absorbed by the small intestine; **oxidised in the liver like any dietary fatty acid**
- Elimination  $t_{1/2}$  ~**50–80 h**

## ADVERSE EFFECTS

- Nausea, **fishy burps**, dyspepsia
- ⚠️ **Increased LDL** — the class's own contradiction
- Arthralgia

## INTERACTIONS

- 🚫 **Anticoagulants** — omega-3 **prolongs bleeding time**; monitor

*Principle behind this: [[pharm-pharmacokinetics]] — statins need OATP uptake transporters to enter hepatocytes at all — an uptake step, not a barrier, as the rate-limiting factor.*

*Principle behind this: [[pharm-drug-interactions]] — Gemfibrozil + statin → rhabdomyolysis via CYP3A inhibition, and cholestyramine as the dispositional antagonist.*

*Principle behind this: [[pharm-pharmacogenomics]] — Statin + gemfibrozil + SLCO1B1 → myopathy. The transporter genotype turns a known interaction into a personal risk.*

*Source: Pharmacology of Lipid-Lowering Agents\_SW\_2025.pdf — อ.สุรศักดิ์, PYCH304 Pharmacology I, 50 slides. Structures omitted.*

# Pharmacology 2

ภกปว 305 · ภาคปลาย · anti-infectives → endocrine → oncology → immune





# Principles of Antibiotic Use

The framework every antibacterial class hangs off. Not a drug class — the set of questions you answer **before** picking a drug: which bug, where is it, what does the host tolerate, and how do you dose it so the killing actually happens.

**Which organism? → Where is the infection? → Which drug reaches it & kills it? → What dosing pattern drives that killing? → What will resist or harm?**

## TWO WORDS WORTH SEPARATING

- ยาปฏิชีวนะ (**antibiotic**) — a substance **produced by and isolated from one microorganism** that inhibits the growth of, or destroys, another
- ยาต้านจุลชีพ (**antimicrobial agent**) — the wider set: isolated from microbes, **semi-synthetic** from those, or **purely chemically synthesised** — anything that inhibits or destroys microbes

*Every antibiotic is an antimicrobial; not every antimicrobial is an antibiotic.*

## THE GOAL: SELECTIVE TOXICITY

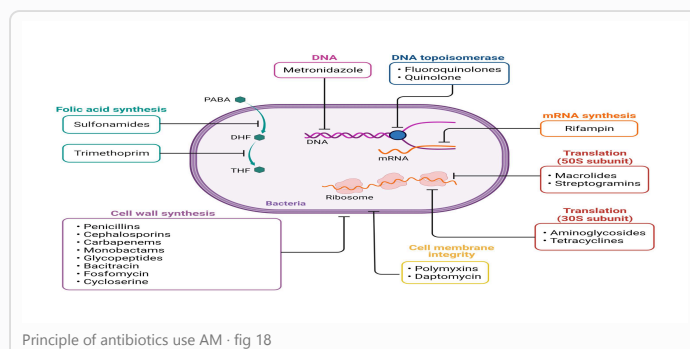
Kill or inhibit the microbial cell **without simultaneously damaging host tissue**. Two ways to get there:

- Target a genetic or biochemical pathway **unique to the pathogen** (e.g. *bacterial cell-wall synthesis inhibitors* — *host cells have no cell wall*)
  - Target a **protein isoform unique to the pathogen** (e.g. *dihydrofolate reductase inhibitors*)
- **Downstream**: the wider the gap between microbial and human target, the safer the drug — ADRs come from where that gap is narrow.

## THE THREE FACTOR GROUPS ★

| GROUP           | WHAT YOU ASSESS                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| <b>Host</b>     | physiological conditions · underlying disease · concomitant agents · epidemiologic exposure |
| <b>Pathogen</b> | species · virulence factors · susceptibility · resistance mechanism                         |
| <b>Site</b>     | organ/system · barrier integrity · defence mechanism · tissue perfusion                     |
| <b>Drug</b>     | spectrum · target / mode of action · PK & tissue penetration · adverse-event profile        |

## OVERVIEW MAPS



# Host & Pathogen Factors

## HOST FACTORS THAT CHANGE THE CHOICE

| FACTOR                        | WHY IT MATTERS                                                                                                                                                |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physiological</b>          | newborn, paediatric, elderly, pregnancy, lactation                                                                                                            |
| <b>Hypersensitivity</b>       | ● can be severe or life-threatening                                                                                                                           |
| <b>Genetic</b>                | G6PD deficiency                                                                                                                                               |
| <b>Underlying disease</b>     | AIDS, cancer, transplant, diabetes, malnutrition → <b>immunocompromised</b> ;<br>kidney/liver disease → <b>impaired elimination or detoxification</b>         |
| <b>Concomitant agents</b>     | immunosuppressants · CYP450 inhibitors/inducers, antacids, probenecid · synergistic adverse effects                                                           |
| <b>Epidemiologic exposure</b> | sick family member · new sexual partner · recent hospitalisation · recent travel · <b>previous antibiotic exposure</b> → <b>expect a resistant organism</b> ★ |

## SITE OF INFECTION NAMES THE LIKELY ORGANISM

| ORGANISM                      | TYPE          | KEY POINT                                                                                                                        |
|-------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>S. aureus</b>              | Gram +        | skin (pus), eye, lung, food poisoning, TSS · <b>MRSA = worldwide problem</b>                                                     |
| <b>S. pyogenes</b><br>(GAS)   | Gram +        | skin, upper respiratory tract · TSLS · exotoxin (β-haemolytic), biofilm                                                          |
| <b>S. pneumoniae</b>          | Gram +        | sinusitis, pneumonia · facultative anaerobe                                                                                      |
| <b>S. agalactiae</b><br>(GBS) | Gram +        | GI microbiota, polysaccharide capsule → <b>neonatal infection</b>                                                                |
| <b>E. coli</b>                | Gram –        | large-intestine microbiota (vitamin K) → <b>urinary tract infection</b>                                                          |
| <b>P. aeruginosa</b>          | Gram –        | ● <b>multidrug-resistant</b> , biofilm, encapsulated · hospital-acquired / opportunistic (burns, cystic fibrosis)                |
| <b>H. influenzae</b>          | Gram –        | opportunistic — pneumonia, meningitis, URT infection                                                                             |
| <b>N. gonorrhoeae</b>         | Gram –        | STD, eye infection · <b>survives inside neutrophils</b>                                                                          |
| <b>C. jejuni</b>              | Gram –        | animal/bird faeces · diarrhoea → bloody stools · entero- + cytotoxin                                                             |
| <b>C. trachomatis</b>         | intracellular | STD, eye infection → conjunctivitis, blindness · elementary bodies infect, reticulate bodies replicate only inside the host cell |
| <b>M. tuberculosis</b>        | special       | mycolic acid (wax) coat, high O <sub>2</sub> requirement, <b>slow growth</b> · intracellular in macrophages                      |
| <b>M. pneumoniae</b>          | atypical      | very small, <b>no peptidoglycan cell wall</b> · mimics host surface molecules · "walking pneumonia"                              |

*Salmonella · Shigella · Clostridium spp. are named in the lecture without further detail.*

### WHY THE SITE MATTERS PRACTICALLY

Certain bacteria infect particular sites or fluids · site + likely organism is what makes **empiric** selection possible before cultures return · site also determines **dose and route** — serum concentration vs tissue concentration, both judged against the **MIC** (minimum inhibitory concentration).

# Bactericidal vs Bacteriostatic

## WHICH DRUGS SIT WHERE

| BACTERICIDAL — KILLS                            | BACTERIOSTATIC — INHIBITS GROWTH |
|-------------------------------------------------|----------------------------------|
| $\beta$ -lactams · glycopeptides · lipopeptides | macrolides                       |
| aminoglycosides · quinolones                    | tetracyclines · tigecycline*     |
| metronidazole · rifampicin                      | lincosamides · fusidic acid      |
| polymyxins · trimethoprim                       | chloramphenicol                  |
| linezolid*                                      | oxazolidinones / linezolid*      |

*\*Linezolid and tigecycline appear on both sides of the lecture's own lists.*

## WHEN THE DIFFERENCE ACTUALLY MATTERS ★

- **Immunocompetent host** — bactericidal and bacteriostatic are **equivalent** for most infections; the immune system finishes the job
- **Immunocompromised · serious infection · poor blood flow** — choose **bactericidal**, because you need a rapid cure and there's no host help coming

# Time- vs Concentration-Dependent Killing

## THE TWO KILLING PATTERNS

- **Time-dependent** — killing continues **as long as serum concentration stays above the MIC**. Raising the peak buys nothing; keeping it up buys everything.
- **Concentration-dependent** — killing **increases with drug concentration**. This is what makes **once-daily dosing regimens** work: one big peak beats several small ones.

→ **Downstream:** the pattern dictates the regimen — time-dependent drugs want frequent doses or extended infusions, concentration-dependent drugs want a large single dose.

## PK/PD TARGET BY PATTERN ★

| PATTERN                                    | MAXIMISE                  | DRUGS                                                                       |
|--------------------------------------------|---------------------------|-----------------------------------------------------------------------------|
| <b>Concentration-dependent</b>             | $C_{max}/MIC$ · $AUC/MIC$ | aminoglycosides · fluoroquinolones · metronidazole · polymyxin · daptomycin |
| <b>Time-dependent</b>                      | $T > MIC$                 | penicillins · cephalosporins · monobactams                                  |
| <b>Concentration- &amp; time-dependent</b> | $AUC_{0-24}/MIC$          | macrolides · clindamycin · azithromycin · vancomycin                        |

## POST-ANTIBIOTIC EFFECT (PAE)

**Persistent suppression of bacterial growth after drug levels have fallen below the MIC** — it reflects the time bacteria need to get back to logarithmic growth.

**PAE = T – C** · T = time to increase 1 log<sub>10</sub> **after drug removal** · C = time to increase 1 log<sub>10</sub> **without any drug**

Proposed mechanisms: slow recovery from reversible non-lethal damage to cell structures · drug persisting at its binding site or in the periplasmic space · new enzymes must be synthesised before growth resumes.

# Reaching the Site of Infection

## GETTING THE DRUG TO WHERE THE BUG IS

| CONSIDERATION              | THE POINT                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Route — oral vs IV</b>  | decided by bioavailability, severity, site of infection, GI function                                                                                                              |
| <b>Tissue penetration</b>  | fluoroquinolones reach <b>high prostate concentrations</b> — <i>except moxifloxacin</i>                                                                                           |
| <b>Blood–brain barrier</b> | crossing needs <b>lipophilicity · low MW · low protein binding · low ionisation</b> . ⚠<br>1st & 2nd-gen cephalosporins, macrolides, aminoglycosides = <b>limited</b> penetration |
| <b>Special barriers</b>    | CSF · ocular fluid · <b>abscess cavity</b> — ● aminoglycosides are <b>inactivated by the low pH and low oxygen of an abscess</b>                                                  |

# Empiric, Definitive & Prophylactic Therapy

## THE THREE STRATEGIES ★

| STRATEGY            | PATHOGEN STATUS    | HOW YOU CHOOSE                                                                                                                        |
|---------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Empiric</b>      | not yet identified | <b>early</b> intervention to improve outcome · <b>broad spectrum</b> , based on clinical-trial experience and likely organisms        |
| <b>Definitive</b>   | already identified | <b>specific</b> drug, <b>narrow spectrum</b> , appropriate duration                                                                   |
| <b>Prophylactic</b> | no infection yet   | prevent an initial infection or its recurrence after cure · triggered by predisposing factors — surgery/procedures, patient condition |

## WORKED EXAMPLE — INFECTION ACQUIRED AT SEA

Exposure history narrows the organism list: *Vibrio vulnificus* (Gram –, capsule, MARTX exotoxin) causes **rapidly progressing cellulitis and sepsis** · also *Candidatus Pelagibacter ubique* and *Synechococcus* spp.

## V. VULNIFICUS — TREATMENT BY PRESENTATION

| PRESENTATION           | DRUGS                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------|
| <b>Gastroenteritis</b> | ciprofloxacin <b>or</b> norfloxacin <b>or</b> tetracycline                                       |
| <b>Wound infection</b> | (ceftazidime <b>or</b> cefotaxime) + (doxycycline <b>or</b> ciprofloxacin)                       |
| <b>Septicaemia</b>     | (ceftazidime <b>or</b> cefotaxime) + (doxycycline <b>or</b> ciprofloxacin) + ICU supportive care |



# Combination Therapy

## MONO VS COMBINATION

| APPROACH           | REASON                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Monotherapy</b> | fewer drug interactions, medication errors, missed doses — the default                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Combination</b> | <p><b>polymicrobial infection</b> — cover the common known/suspected pathogens (not every possible one), e.g. intra-abdominal abscess = Gram(–) + anaerobes + enterococci</p> <p>↓ <b>dose-related toxicity</b> — flucytosine + amphotericin B lowers amphotericin B nephrotoxicity</p> <p>↓ <b>emergence of resistance</b> — e.g. tuberculosis</p> <p><b>enhance activity (synergism)</b></p> <p><b>broader empiric cover</b> in life-threatening infection</p> |

## SYNERGISM — >4-FOLD REDUCTION IN MIC VS EITHER DRUG ALONE ★

| COMBINATION                                            | WHY IT WORKS                                            |
|--------------------------------------------------------|---------------------------------------------------------|
| <b>Trimethoprim + sulfamethoxazole</b>                 | blocks <b>two steps</b> in the folic acid pathway       |
| <b>β-lactam + β-lactamase inhibitor</b>                | inhibits <b>enzymatic inactivation</b> of the β-lactam  |
| <b>Penicillin/ampicillin + gentamicin/streptomycin</b> | cell-wall damage <b>increases aminoglycoside uptake</b> |

## ANTAGONISM — THE PAIRING TO AVOID

- **Cell-wall inhibitor + bacteriostatic agent** → combined killing is **less** than either drug alone. *Why: cell-wall-active drugs need bacteria to be actively growing — the bacteriostatic drug stops exactly that.*

# Adverse Effects of Antimicrobials

## อาการไม่พึงประสงค์

- **Hypersensitivity** — urticaria · anaphylaxis · **TEN** · **DRESS** · maculopapular exanthema · angioedema
- **Direct toxicity** — nephrotoxicity · hepatotoxicity
- **Superinfection** — ● **pseudomembranous colitis** following broad-spectrum antibiotic use

## CAUTIONS

- ⚠ A documented hypersensitivity is a **host factor**, not a footnote — it can be severe or life-threatening
- ⚠ Broad spectrum is not "safer coverage" — it is what wipes out the microbiota and lets superinfection in

# Antimicrobial Resistance

## NATURAL VS ACQUIRED ★

| TYPE                      | HOW                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Natural (inherent)</b> | Gram(−) <b>LPS</b> outer membrane blocks entry · the organism simply <b>lacks the target or the transporter</b> |
| <b>Acquired</b>           | <b>mutation</b> , or <b>gene transfer</b>                                                                       |

## HOW RESISTANCE GENES MOVE

- **Vertical** — passed to daughter cells on division
- **Horizontal** — passed between organisms, by three routes:
  - **Transformation** — uptake of free DNA
  - **Transduction** — carried by a bacteriophage
  - **Conjugation** — direct cell-to-cell transfer

## เชื่อด้อยๆ — THE NAMED PROBLEM ORGANISMS

| ABBREVIATION | WHAT IT IS                                                                               |
|--------------|------------------------------------------------------------------------------------------|
| <b>ESBL</b>  | Extended-Spectrum Beta-Lactamases — ● <b>carbapenems are reserved for ESBL infection</b> |
| <b>MRSA</b>  | Methicillin-Resistant <i>Staphylococcus aureus</i>                                       |
| <b>VRE</b>   | Vancomycin-Resistant <i>Enterococci</i>                                                  |
| <b>FQRP</b>  | Fluoroquinolone-Resistant <i>Pseudomonas aeruginosa</i>                                  |

**ESBL producers, mostly hospital-acquired:** *E. coli* · *K. pneumoniae* · *P. aeruginosa* · *A. baumannii* · *S. typhimurium* · Enterobacteriaceae

Source: 1.Principle of antibiotics use AM.pdf — อ.อาณัฐชัย (Arnatchai Maiuthed), Pharmacology II, 48 slides. Structures omitted.

# Drugs Affecting Cell Wall Synthesis

**β-lactam antibiotics.** Four structural families, **one target**: the **penicillin-binding proteins (PBPs)** that cross-link peptidoglycan. Everything that differs between them — spectrum, β-lactamase survival, where they reach — comes from the **side chain**, not from a different target.

**β-lactam binds PBP → transpeptidase blocked → peptidoglycan not cross-linked → cell wall fails**

## THE B-LACTAM FAMILY

| FAMILY                | CORE   | MEMBERS                                                                                                                                                    |
|-----------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Penicillins</b>    | penam  | Pen G · Pen V · ampicillin · amoxicillin · methicillin · nafcillin · oxacillin · cloxacillin · dicloxacillin · flucloxacillin · piperacillin · ticarcillin |
| <b>Cephalosporins</b> | cephem | 1st → 5th generation (see the generation pages)                                                                                                            |
| <b>Carbapenems</b>    | —      | imipenem · meropenem · ertapenem · doripenem · biapenem · panipenem                                                                                        |
| <b>Monobactams</b>    | —      | aztreonam · tigemonam · carumonam · nocardicin A                                                                                                           |

## TWO THINGS THE SLIDES ONLY SHOW AS PICTURES

- **Monobactams** — named in the family chart, but their spectrum slide is figure-only. Nothing textual to summarise.
- ⚠️ **กลไกการแพ้ยากลุ่ม penicillin** (penicillin hypersensitivity) — 8 slides, all diagrams. Go back to the original deck for this one; it is not recoverable from the text.

# Penicillin-Binding Proteins (PBPs)

## THE TARGET ITSELF

**High-molecular-weight PBPs** — anchored to the inner membrane by a transmembrane helix, enzymatic domain pointing into the **periplasm**: - **Class A** (PBP1a, 1b, 1c) — *two* domains: **glycosyltransferase** + **transpeptidase** - **Class B** (PBP2, PBP3) — a single **transpeptidase** domain

**Low-molecular-weight PBPs** — anchored by an amphipathic helix, one enzymatic domain: **endopeptidase** (PBP4, 7) or **carboxypeptidase** (PBP5, 6, 6b) → cell separation, peptidoglycan maturation and recycling.

→ **Downstream**:  $\beta$ -lactams occupy the **transpeptidase** site → cross-linking stops. ★ Change the PBP's *affinity* and the drug stops working — that is exactly MRSA (PBP2a) and penicillin-resistant pneumococci/enterococci.

## WHY THE SIZE SPLIT MATTERS

- **HMW PBPs** — build the wall. **This is what  $\beta$ -lactams kill.**
- **LMW PBPs** — trim and recycle the wall. Not the killing target.

# Penicillins — Shared Properties

## WHY EACH PENICILLIN HITS DIFFERENT BUGS

Same PBP target every time. The **R side chain** decides two things — can it get *in*, and does  $\beta$ -lactamase *see* it:

- **Pen G / Pen V** — hydrophobic **benzene** side chain → **cannot pass gram-negative porins**
  - **Antistaphylococcal** — **bulky** residues physically block staphylococcal  $\beta$ -lactamase from binding
  - **Aminopenicillins** — an extra **amino group** → more hydrophilic → **passes the porins** of enteric gram-negative rods
  - **Antipseudomonal** — a **polar** side chain → even better porin passage
- **Downstream:** spectrum = porin penetration +  $\beta$ -lactamase survival. Nothing about the target changes.

## THE FOUR SUBGROUPS

| SUBGROUP                       | SPECTRUM | COVERS                                                                                  |
|--------------------------------|----------|-----------------------------------------------------------------------------------------|
| <b>Natural</b>                 | narrow   | Streptococcus spp. · MSSA · anaerobes                                                   |
| <b>Penicillinase-resistant</b> | narrow   | S. aureus · S. epidermidis · penicillinase producers                                    |
| <b>Aminopenicillins</b>        | broad    | Listeria · H. influenzae · E. coli · N. gonorrhoeae                                     |
| <b>Antipseudomonal</b>         | broad    | aminopenicillin coverage + <b>P. aeruginosa</b> · Enterobacter · Klebsiella · anaerobes |

## PHARMACOKINETICS THAT CHANGE DECISIONS

| POINT                                | DETAIL                                                                                                                                               |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acid-stable → oral</b>            | aminopenicillins (ampicillin, amoxicillin) · penicillinase-resistant (methicillin, nafcillin, oxacillin, cloxacillin, dicloxacillin, flucloxacillin) |
| <b>Highly protein-bound (90–97%)</b> | nafcillin, oxacillin, cloxacillin, dicloxacillin, flucloxacillin — Pen G and ampicillin much less                                                    |
| <b>Food</b>                          | take 1–2 h before or after a meal — ★ <b>amoxicillin is the exception</b>                                                                            |
| <b>Depot IM</b>                      | benzathine & procaine penicillin — delayed absorption → prolonged blood levels                                                                       |
| <b>Distribution</b>                  | tissue $\approx$ serum, but <b>poor into eye, prostate and across the BBB</b>                                                                        |
| <b>Excretion</b>                     | mainly renal, largely unchanged → <b>renal failure prolongs <math>t_{1/2}</math></b>                                                                 |

## INTERACTIONS

- ★ **Probenecid** blocks renal tubular secretion of penicillins → **prolonged duration of action**

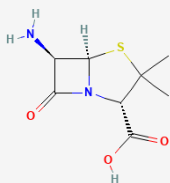
# Natural Penicillins

## MECHANISM

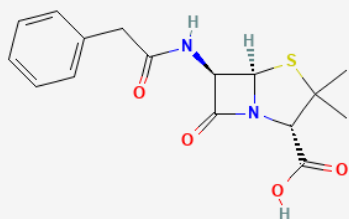
Binds PBPs like every  $\beta$ -lactam — but the **hydrophobic benzene R side chain** cannot cross the porins of the gram-negative outer membrane, and nothing shields the  $\beta$ -lactam ring from staphylococcal  $\beta$ -lactamase.

→ **Downstream:** narrow spectrum · most gram-negatives **naturally resistant** · killed off by penicillinase

## THE SHARED CORE · WHAT CHANGES

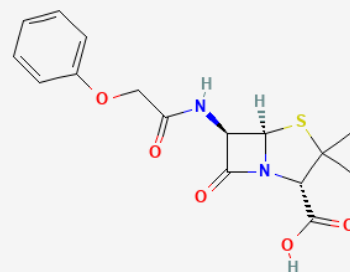


**6-APA — the penam core.**  $\beta$ -lactam fused to thiazolidine. Every penicillin is this with a different group on the left-hand N;  $\beta$ -lactamase attacks the four-membered ring.



**Penicillin G**

flat benzene side chain — hydrophobic, cannot pass gram-negative porins



**Penicillin V**

an added O makes it acid-stable → the oral one

## MEDCHEM

**SAR** The R group hangs off the acylamino side chain at C6 - that is the **only** place the class is edited. C2 carries a carboxylic acid, which is what lets the drug be made into a salt or an ester; the active stereochemistry is fixed at **2S,5R,6R** and the bicycle is **not planar**, which is what strains the ring and makes it reactive.

**Stability** ● **This is why Pen G is an injection and Pen V is a tablet** - the page gives them the same spectrum but not the reason. Below pH 3 the stomach opens the ring (penicillenic → penilloic acid); above pH 8 base hydrolysis does the same. Pen V's side-chain oxygen is an **electron-withdrawing group**, which pulls electron density away from the ring and makes it survive gastric acid.

**Physicochem** The C2 acid is why the depot forms exist: **Na<sup>+</sup>/K<sup>+</sup> salts** dissolve for IV, while the **benzathine** and **procaine** salts are deliberately insoluble, so an IM dose dissolves slowly and lasts.

## DRUGS

| DRUG                | WHAT IT TREATS                        |
|---------------------|---------------------------------------|
| <b>Penicillin G</b> | Streptococcus spp. · MSSA · anaerobes |
| <b>Penicillin V</b> | Streptococcus spp. · MSSA · anaerobes |

### CAUTIONS

- ⚠️ **Penicillinase producers defeat this class outright** — that is what the antistaphylococcal group exists for
- ⚠️ Many gram-negatives are resistant *by default*: LPS blocks the porins, the drug never reaches the PBPs



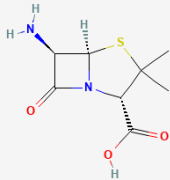
# Penicillinase-Resistant Penicillins

## MECHANISM

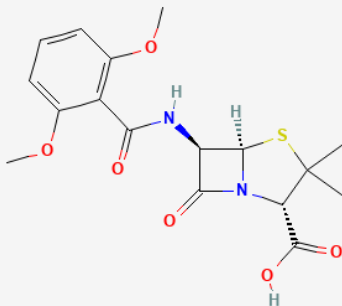
**Bulky residues on the R side chain** sterically prevent **staphylococcal  $\beta$ -lactamase** from binding — the ring survives long enough to reach the PBPs.

→ **Downstream:** narrow, **staph-directed** spectrum · the classic "antistaphylococcal" penicillins

## THE SHARED CORE · WHAT CHANGES

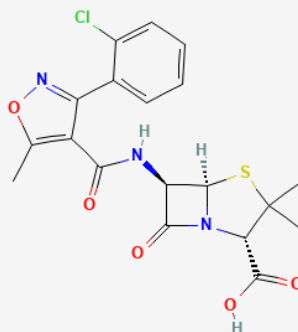


**6-APA — same core.** the ring is unchanged; only the shielding around it changes



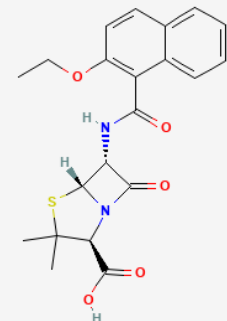
**Methicillin**

two bulky methoxy groups  
physically block  $\beta$ -lactamase from  
reaching the ring



**Cloxacillin**

bulky isoxazoyl + Cl — same trick,  
and acid-stable enough to give  
orally



**Nafcillin**

bulky naphthalene; hepatic  
clearance, unlike the rest

## MEDCHEM

**SAR** The bulk is not decoration - it is an **isoxazoyl ring** (plus halogens on the cloxacillin family) sitting right beside the  $\beta$ -lactam, close enough to block the enzyme's approach but not the PBP's.

**Resistance** ⚠️ The deck's blunt version: **methicillin induced the resistance that killed it** - pressure from this drug is what selected MRSA in *S. aureus*.

## DRUGS

| DRUG                  | WHAT IT TREATS                                                                           |
|-----------------------|------------------------------------------------------------------------------------------|
| <b>Methicillin</b>    | <i>Staphylococcus aureus</i> · <i>S. epidermidis</i> — including penicillinase producers |
| <b>Nafcillin</b>      | same                                                                                     |
| <b>Oxacillin</b>      | same                                                                                     |
| <b>Cloxacillin</b>    | same                                                                                     |
| <b>Dicloxacillin</b>  | same                                                                                     |
| <b>Flucloxacillin</b> | same                                                                                     |

### CAUTIONS

-  **MRSA defeats the whole class**  — resistance is not enzymatic, it is an **altered target**: `mecA` expresses **PBP2a**, which has low affinity for  $\beta$ -lactams. No inhibitor can fix that.
-  All highly protein-bound (90–97%) — acid-stable, so oral works

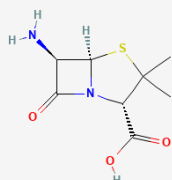
# Aminopenicillins

## MECHANISM

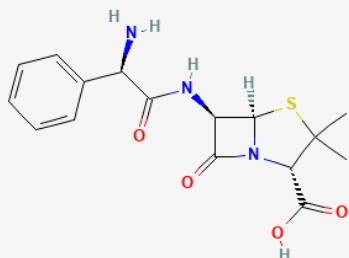
An **additional amino group** on the side chain raises hydrophilicity → the drug **passes through the porins** in the outer membrane of some enteric gram-negative rods.

→ **Downstream:** broad spectrum — gram-positive cover *plus* selected gram-negatives · **no  $\beta$ -lactamase protection**

## THE SHARED CORE · WHAT CHANGES

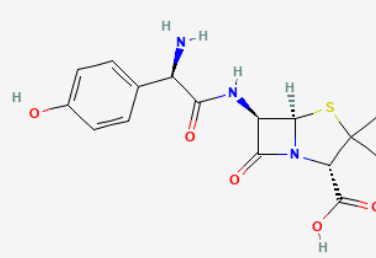


**6-APA — same core.** the extra  $\text{-NH}_2$  on the side chain is the whole difference



**Ampicillin**

the  $\text{-NH}_2$  makes it hydrophilic → passes enteric gram-negative porins



**Amoxicillin**

ampicillin + one  $\text{-OH}$  → far better oral absorption, and food doesn't matter

## MEDCHEM

**SAR** The  $\alpha$ -amino group does **two** jobs, and the page only names one. Besides opening the porins, it also stabilises the ring in acid - which is what makes this the first orally reliable broad-spectrum penicillin.

**Physicochem** ★ The reason food doesn't interfere with amoxicillin but does with ampicillin is one atom: the **p-OH** on amoxicillin's phenyl ring. The page states the fact; this is the cause.

**Prodrug** **Bacampicillin and sultamicillin are not drugs** - they are esters that gut and plasma **esterases** cleave back to ampicillin. The ester exists purely to get absorbed; sultamicillin delivers ampicillin *and* sulbactam from one molecule.

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                    |
|----------------------|-----------------------------------------------------------------------------------|
| <b>Ampicillin</b>    | Listeria monocytogenes · Haemophilus influenzae · E. coli · Neisseria gonorrhoeae |
| <b>Amoxicillin</b>   | same — ★ the one penicillin whose absorption <b>food does not interfere with</b>  |
| <b>Bacampicillin</b> | same                                                                              |
| <b>Ciclacillin</b>   | same                                                                              |

### CAUTIONS

- ⚠  **$\beta$ -lactamase producers inactivate them** — this is why they are sold paired with clavulanate or sulbactam
- ⚠ Ampicillin is ~90% renally excreted → adjust in renal failure

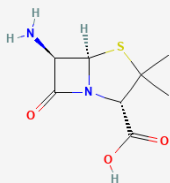
# Antipseudomonal Penicillins

## MECHANISM

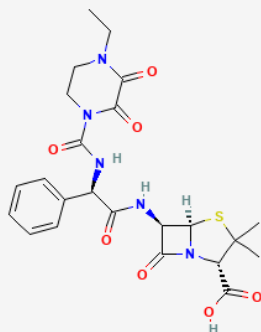
The side chain is **polar**, which increases its ability to cross the outer-membrane porins of some gram-negative bacteria — further than the aminopenicillins reach.

→ **Downstream:** broad spectrum, **including** *Pseudomonas aeruginosa*

## THE SHARED CORE · WHAT CHANGES

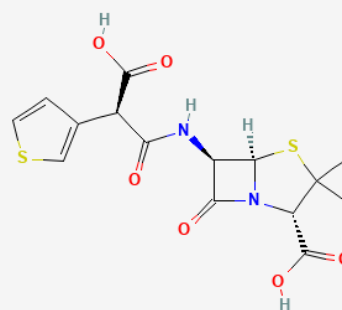


**6-APA — same core.** a large polar tail buys entry through *Pseudomonas* porins



**Piperacillin**

polar piperazinedione tail — best porin passage of the family



**Ticarcillin**

carboxyl on a thiophene — the earlier, weaker version of the same idea

## MEDCHEM

**SAR** 'Polar side chain' is two different chemistries doing the same job:  **$\alpha$ -carboxy** (carbenicillin, ticarcillin) and **acylureido** (piperacillin, mezlocillin). The acylureido group reaches further - it adds *Enterobacteriaceae* and *B. fragilis* on top of *Pseudomonas*.

**Stability** Both subfamilies are **acid-labile**, which is why the class is IV. The one oral exception is **indanyl carbenicillin**, an ester prodrug - and it only ever reaches useful concentration in urine.

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                                                  |
|----------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Piperacillin</b>  | aminopenicillin spectrum + <b>P. aeruginosa</b> · <i>Enterobacter</i> spp. · <i>Klebsiella</i> spp. · anaerobes |
| <b>Ticarcillin</b>   | same                                                                                                            |
| <b>Carbenicillin</b> | same                                                                                                            |
| <b>Azlocillin</b>    | same                                                                                                            |
| <b>Mezlocillin</b>   | same                                                                                                            |
| <b>Temocillin</b>    | same                                                                                                            |

### CAUTIONS

- ⚠ Still  **$\beta$ -lactamase-labile** — the useful clinical form is piperacillin/**tazobactam**
- ⚠ Renally excreted (carbenicillin 85–95%, ticarcillin 75–85%, piperacillin 60–80%)

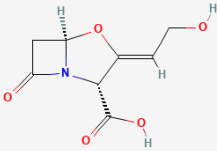
# $\beta$ -Lactamase Inhibitors & Combinations

## MECHANISM

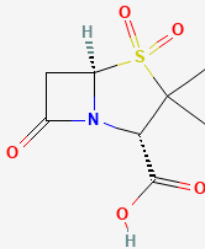
An **irreversible suicide inhibitor**: the inhibitor is itself cleaved by  $\beta$ -lactamase and stays stuck on the enzyme, destroying it in the process.

→ **Downstream**: the partner  $\beta$ -lactam survives long enough to reach the PBPs — spectrum restored against  **$\beta$ -lactamase producers only**. ● Useless against **altered-PBP** resistance (MRSA) and against efflux.

## THE SHARED CORE · WHAT CHANGES

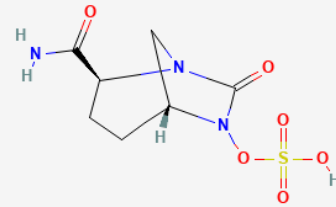


**Clavulanate**. a  $\beta$ -lactam with almost no antibacterial power — its job is to be destroyed instead of the partner drug (suicide inhibition)



**Sulbactam**

penam sulfone; same sacrificial ring



**Avibactam**

● not a  $\beta$ -lactam at all — a diazabicyclooctane, which is why it survives enzymes that chew the others

## MEDCHEM

**SAR** Two generations, split by whether the inhibitor is itself a  $\beta$ -lactam. **1st gen has a  $\beta$ -lactam core** - clavulanic acid is an **oxapenam** (the ring sulfur swapped for oxygen, plus a hydroxyethylidene), sulbactam and tazobactam are **penam sulfones**. **2nd gen is not a  $\beta$ -lactam at all**: avibactam is a **diazabicyclooctane**, which is why it reaches  $\beta$ -lactamases (some CRE) that the older three cannot.

**SAR** Clavulanic acid has **no activity against PBPs or transpeptidase** - it is a decoy and nothing else. That is the design: all of its reactivity is spent on the enzyme.

**Prodrug** The oral combinations are formulated as **ester prodrugs**; the injectable ones are simply the two drugs mixed.

## COMBINATIONS

| DRUG                           | WHAT IT TREATS                                                                                       |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Amoxicillin/clavulanate</b> | streptococci · MSSA · enterococci · Enterobacteriaceae · anaerobes                                   |
| <b>Ampicillin/sulbactam</b>    | same, + <b>A. baumannii</b>                                                                          |
| <b>Piperacillin/tazobactam</b> | same, + <b>P. aeruginosa</b> + <b>A. baumannii</b>                                                   |
| <b>Cefoperazone/sulbactam</b>  | Enterobacteriaceae · P. aeruginosa · A. baumannii                                                    |
| <b>Ceftolozane/tazobactam</b>  | Enterobacteriaceae incl. some <b>ESBL</b> · P. aeruginosa · A. baumannii incl. some MDR              |
| <b>Ceftazidime/avibactam</b>   | Enterobacteriaceae incl. some <b>ESBL and some CRE</b> · P. aeruginosa · A. baumannii incl. some MDR |

## ADVERSE EFFECTS

- **Amoxicillin/clavulanate** — as for ampicillin/amoxicillin, **plus hepatotoxicity** ★
- **Ampicillin/sulbactam** — as for ampicillin/amoxicillin
- **Piperacillin/tazobactam** — **thrombocytopenia**
- **Cefoperazone/sulbactam** — **coagulopathy**
- **Ceftolozane/tazobactam** — renal impairment
- **Ceftazidime/avibactam** — seizure (*from the ceftazidime*) · hypersensitivity reaction

## ADMINISTRATION — B-LACTAMASE INHIBITOR COMBINATION DOSES

| DRUG                                | RECOMMENDED DOSE                                                                      |
|-------------------------------------|---------------------------------------------------------------------------------------|
| <b>Amoxicillin/clavulanate</b>      | <b>IV: 1000/200 q 8h · PO: 500/125 mg three times daily or 875/125 mg twice daily</b> |
| <b>Ampicillin/sulbactam (IV)</b>    | <b>2/1 g q 6h</b>                                                                     |
| <b>Piperacillin/tazobactam (IV)</b> | <b>4.5 g q 6–8h — q 6h if <i>P. aeruginosa</i> is suspected</b>                       |

- ★ **The two oral amoxicillin/clavulanate options carry the same daily amoxicillin but different clavulanate loads** — 875/125 twice daily gives less clavulanate per day than 500/125 three times daily, which is the practical reason the twice-daily form is easier on the gut.
- 💡 ⚠️ **The slide writes the IV amoxicillin/clavulanate dose as "1000/200 mcg", which doesn't sit with the milligram–gram scale of every other dose in the table. Read it as the milligram strength and **check it against the paper before quoting the unit.****
- 💡 Piperacillin/tazobactam shortening to q 6h for pseudomonas is a **time-above-MIC** move, not a potency one — same dose, more often, because the drug is time-dependent.



# Cephalosporins — Shared Properties

## THE GENERATIONS

| GEN            | MEMBERS                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------|
| <b>1st</b>     | cefazolin · cephalexin · cefadroxil · cephalothin · cephapirin · cephradine                          |
| <b>2nd</b>     | cefuroxime · cefaclor · cefamandole · cefoxitin · cefmetazole · cefotetan                            |
| <b>3rd</b>     | ceftriaxone · cefotaxime · ceftazidime · cefixime · cefdinir · ceftibuten · ceftizoxime · moxalactam |
| <b>4th–5th</b> | cefepime · ceftiprome · ceftobiprole · ceftaroline                                                   |

## WHAT EACH GENERATION ADDS (READ OFF THE LECTURE'S OWN CLINICAL-USE TABLES)

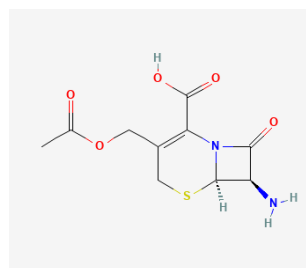
| GEN            | WHAT IT BUYS YOU                                                        |
|----------------|-------------------------------------------------------------------------|
| <b>1st</b>     | streptococci, MSSA — nothing more                                       |
| <b>2nd</b>     | + <b>H. influenzae</b> , community-acquired Enterobacteriaceae          |
| <b>3rd</b>     | + <b>PRSP</b> , CNS penetration, and (ceftazidime) <b>P. aeruginosa</b> |
| <b>4th–5th</b> | + <b>MRSA</b> (ceftaroline)                                             |

## PHARMACOKINETICS

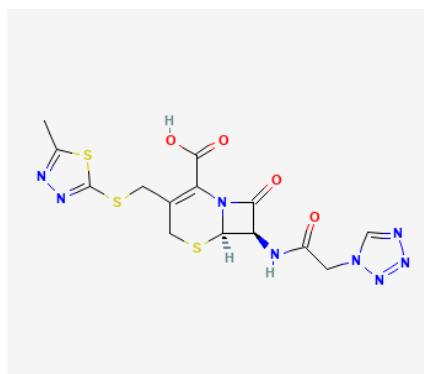
| POINT               | DETAIL                                                                                                                                                |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Distribution</b> | well distributed in body fluids · <b>all cross the placenta and enter breast milk</b>                                                                 |
| <b>CSF</b>          | ★ therapeutic levels only with 3rd gen — <b>ceftriaxone, cefotaxime</b>                                                                               |
| <b>Excretion</b>    | 80–90% unchanged in urine — ★ <b>except cefoperazone &amp; ceftriaxone</b> , mainly <b>biliary</b> → <b>no dose adjustment in renal insufficiency</b> |

# Cephalosporins — 1st Generation

## THE SHARED CORE • WHAT CHANGES

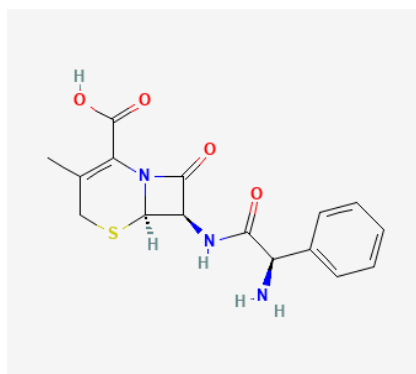


**7-ACA — the cephem core.**  $\beta$ -lactam fused to a SIX-membered dihydrothiazine, not five. That one extra atom is why cephalosporins resist penicillinase.



**Cefazolin**

parenteral workhorse — surgical prophylaxis



**Cephalexin**

the oral 1st-gen

### MEDCHEM

**SAR** The cephem nucleus is  $\beta$ -lactam + **dihydrothiazine** (a 6-ring, where the penicillins have a 5-ring thiazolidine). Three positions are edited and they do different things: **C7 acylamino** = spectrum, **C7 methoxy** = enzyme resistance, **C3** = pharmacokinetics.

**SAR** ★ **This is the oral-vs-IV split in the class.** An  **$\alpha$ -amino** group on the side chain (cephalexin, cefadroxil, cefaclor) confers acid stability → oral. An **acetoxymethyl** at C3 (cephalothin, cephalixin) makes the ring more reactive toward PBPs → more potent, but parenteral. Cephadrine is cephalexin with the benzene swapped for **1,4-cyclohexadiene**.

**Metabolism** The acetoxymethyl group is the weakness: **esterases hydrolyse it to a hydroxyl, which then spontaneously lactonizes with the C2 acid into an inactive lactone.** The drug destroys itself in the body without any bacterial help.

### DRUGS

| DRUG                                                      | WHAT IT TREATS                                                      |
|-----------------------------------------------------------|---------------------------------------------------------------------|
| <b>Cefazolin</b>                                          | streptococci · MSSA — the IV workhorse                              |
| <b>Cephalexin</b>                                         | streptococci · MSSA — the oral one                                  |
| <b>Cefadroxil · Cephalothin · Cephapirin · Cephadrine</b> | same generation; the lecture lists them without a separate spectrum |

### USES

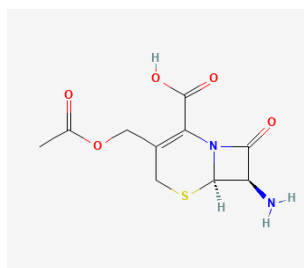
Gram-positive cover — **streptococci · MSSA**. No H. influenzae, no Enterobacteriaceae, no Pseudomonas.

### CAUTIONS

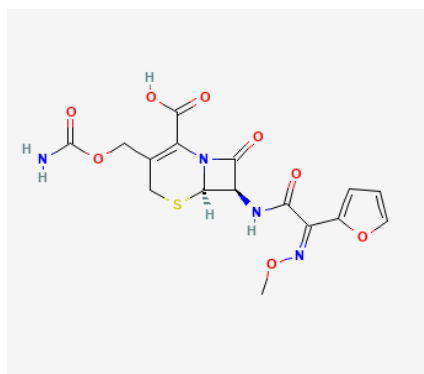
- ⚠ Enterococci are **naturally resistant** to 1st–5th generation cephalosporins — their PBPs differ (higher cell-wall lipid content)

# Cephalosporins — 2nd Generation

## THE SHARED CORE · WHAT CHANGES

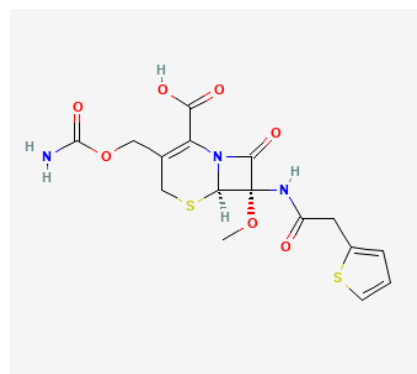


**7-ACA — same core.** side-chain bulk starts buying  $\beta$ -lactamase stability



**Cefuroxime**

methoxyimino group — the first real  $\beta$ -lactamase resistance step



**Cefoxitin**

7 $\alpha$ -methoxy (a cephamycin) → anaerobe cover

### MEDCHEM

**SAR** The generation is defined by an **oximether** at the  $\alpha$ -carbon (cefuroxime). ● The **syn** geometry of that oxime is the whole point - it physically shields the  $\beta$ -lactam from **cephalosporinase**. The anti isomer does not protect it.

**SAR** The cephamycin branch (cefoxitin, cefotetan) works differently: a **7 $\alpha$ -methoxy** group blocks the enzyme sterically, and its **carbamate** at C3 resists the esterases that dismantle the 1st-gen acetoxymethyl. That is why cefoxitin holds up against anaerobes.

**Metabolism** ● The **NMTT side chain is the source of two ADRs, and it is a structural feature, not a drug-specific quirk**. Cephalosporins carrying **N-methyltetrazole-thiomethyl** at C3 (cefamandole, cefotetan, cefoperazone) release it on metabolism: it causes **hypoprothrombinaemia** → bleeding risk with anticoagulants, and an **Antabuse-like alcohol intolerance** (flushing, vomiting, tachycardia, hypotension).

**Prodrug** **Cefuroxime axetil** is an ester prodrug - esterification is what buys the GI absorption.

### DRUGS

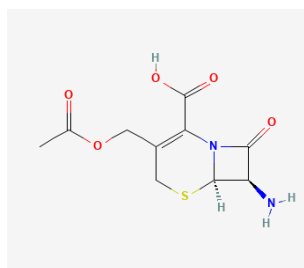
| DRUG                                                                | WHAT IT TREATS                                                                     |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Cefuroxime</b>                                                   | streptococci · MSSA · <b>H. influenzae</b> · community-acquired Enterobacteriaceae |
| <b>Cefaclor · Cefamandole · Cefoxitin · Cefmetazole · Cefotetan</b> | same generation; the lecture lists them without a separate spectrum                |

### USES

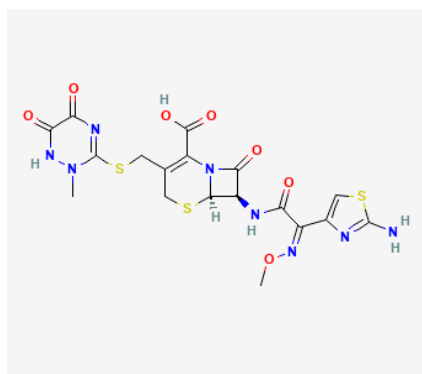
The step up from 1st gen: keeps the gram-positive cover and **adds H. influenzae + community-acquired gram-negative rods**. Available IV and PO.

# Cephalosporins — 3rd Generation

## THE SHARED CORE · WHAT CHANGES

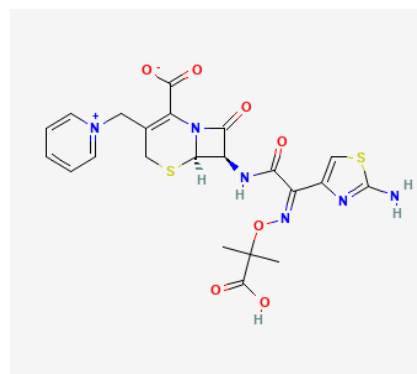


**7-ACA — same core.** the aminothiazolyl-oxime side chain is the 3rd-gen signature



**Ceftriaxone**

aminothiazole + oxime → gram-negative reach and a long  $t_{1/2}$



**Ceftazidime**

a charged carboxypropyl on the oxime → *Pseudomonas*

## MEDCHEM

**SAR** Two groups together define the generation: the **syn-oximether** (cephalosporinase protection, as in 2nd gen) **plus an aminothiazolyl** ring on the C7 side chain, which raises affinity for gram-negative PBPs. That combination is what buys the Enterobacteriaceae spectrum.

**Metabolism** **Cefotaxime is inactivated by the body, not by bacteria** - esterases strip its acetoxy group and the product lactonizes into an inactive form. That is the pharmacokinetic reason it is dosed more often than ceftriaxone, which has no such group.

**Prodrug** **Cefixime and cefdinir** are the members built for oral use; the rest of the generation needs a line.

## DRUGS

| DRUG                                    | WHAT IT TREATS                                                                                                   |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Ceftriaxone</b>                      | streptococci · MSSA · <b>PRSP</b> · CA-Enterobacteriaceae — ★<br>reaches the <b>CSF</b>                          |
| <b>Cefotaxime</b>                       | same — also reaches the CSF                                                                                      |
| <b>Cefixime · Cefdinir · Cefditoren</b> | the oral 3rd-generation options                                                                                  |
| <b>Ceftazidime</b>                      | CA-Enterobacteriaceae · <b>P. aeruginosa</b> · <b>B. pseudomallei</b> —<br>⚠️ <b>poor against gram-positives</b> |
| <b>Cefoperazone/sulbactam</b>           | Enterobacteriaceae · P. aeruginosa · A. baumannii                                                                |

## USES

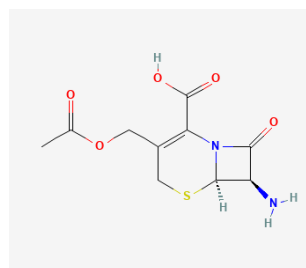
**CNS infection** — ceftriaxone, cefotaxime (dosed more frequently for CNS) · **Pseudomonas / melioidosis** — ceftazidime · **Renal impairment** — ceftriaxone and cefoperazone are biliary-cleared, so no adjustment

## ADVERSE EFFECTS

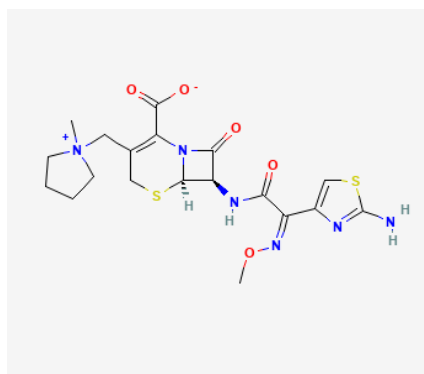
- **Ceftazidime** — ● **seizure**
- **Cefoperazone/sulbactam** — coagulopathy
- **Pseudocholelithiasis** (biliary sludging) and **kernicterus in newborns** — *the slide prints these in the cephalosporin ADR column; they belong to ceftriaxone, whose biliary clearance and bilirubin displacement cause them*

# Cephalosporins — 4th & 5th Generation

## THE SHARED CORE · WHAT CHANGES

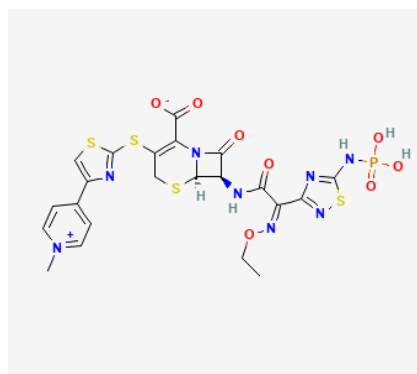


**7-ACA** — **same core**. a quaternary N makes the molecule zwitterionic → faster porin entry



**Cefepime**

zwitterion — through porins fast, poor  $\beta$ -lactamase substrate



**Ceftaroline**

the side chain that finally binds PBP2a → MRSA

### MEDCHEM

**SAR** ★ **4th gen = 3rd gen plus a permanent positive charge at C3** (N-methylpyrrolidinium in cefepime, pyridinium in ceftiofame). Gram-negative porins are lined with **phosphate groups**, so they actively pull cations through - the drug becomes a zwitterion and crosses the outer membrane far faster than a 3rd-gen does. Charge, not spectrum chemistry, is the upgrade.

**SAR** Ceftaroline's 5th-gen features are three separate parts: a **1,2,4-thiadiazole** for diffusion through the gram-negative outer membrane, a **1,3-thiazole** which is what delivers the MRSA (PBP2a) activity, and an **N-methylpyridinium** for porin passage.

**Prodrug** Ceftaroline is given as **ceftaroline fosamil** - a **phosphate ester** added purely for solubility, cleaved in plasma.

### DRUGS

| DRUG                                       | WHAT IT TREATS                                                                                                 |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Ceftaroline</b>                         | ★ gram-positive cocci <b>including MRSA</b> · CA-Enterobacteriaceae — ⚠ <b>no activity against enterococci</b> |
| <b>Cefepime · Cefpirome · Ceftobiprole</b> | listed in the class chart; the lecture's spectrum slide for them is figure-only                                |

### USES

**Ceftaroline** is the reason this generation exists — ★ the only  $\beta$ -lactam here that covers **MRSA**, because it binds the altered **PBP2a** the rest of the class cannot.

### ADVERSE EFFECTS

- **Ceftaroline** — positive **Coombs test** · **eosinophilic pneumonia**

## CAUTIONS

- ⚠ The newer  $\beta$ -lactamase-inhibitor pairs (**ceftolozane/tazobactam**, **ceftazidime/avibactam**) sit on the  $\beta$ -lactamase-inhibitor page, not here

## ADMINISTRATION — CEPHALOSPORIN DOSES BY GENERATION

| GENERATION | DRUG                               | RECOMMENDED DOSE                                  |
|------------|------------------------------------|---------------------------------------------------|
| 1st        | <b>Cefazolin</b> (IV)              | <b>1–2 g q 8h</b>                                 |
|            | <b>Cephalexin</b> (PO)             | <b>0.25–1 g three to four times daily</b>         |
| 2nd        | <b>Cefuroxime</b> (IV, PO)         | <b>IV: 0.75–1.5 g q 8h · PO: 0.25–0.5 g q 12h</b> |
| 3rd        | <b>Ceftriaxone</b> (IV)            | ★ <b>2 g q 24h — but q 12h for CNS infection</b>  |
|            | <b>Cefotaxime</b> (IV)             | <b>1–2 g q 8h</b>                                 |
|            | <b>Cefixime</b> (PO)               | 0.2–0.4 g q 12h                                   |
|            | <b>Cefdinir</b> (PO)               | 0.3 g q 12h                                       |
|            | <b>Cefditoren</b> (PO)             | 0.2–0.4 g q 12h                                   |
| 5th        | <b>Ceftaroline</b> (IV)            | <b>600 mg q 8–12h</b>                             |
|            | <b>Ceftolozane/tazobactam</b> (IV) | <b>1.5 g q 8h</b>                                 |
|            | <b>Ceftazidime/avibactam</b> (IV)  | <b>2.5 g q 8h</b>                                 |

- ★ **Ceftriaxone once daily is its whole selling point, and CNS infection is the one place that breaks** — twice daily, because the barrier the drug has to cross is the reason the interval halves, not the bug.
- ⚠ Adverse effects that come with the dose: **cefazolin/cephalexin — pseudocholelithiasis and kernicterus in newborns · ceftaroline — positive Coombs test, eosinophilic pneumonia · ceftolozane/tazobactam — renal impairment · ceftazidime/avibactam — seizure (from the ceftazidime) and hypersensitivity.**
- 💡 Everything IV in this table sits at **q 8h**, which is what a short half-life plus time-dependent killing looks like. Ceftriaxone is the exception that gets remembered *because* it's the exception.

# Carbapenems

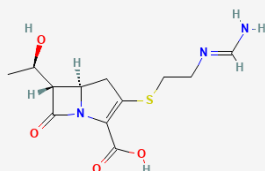
## MECHANISM

Three properties stack into the broadest  $\beta$ -lactam spectrum: 1. **Small and charged** → they use special **porins** of the gram-negative outer membrane to reach the PBPs 2. Their structure **resists cleavage by most  $\beta$ -lactamases** 3. They bind a **broad range of PBPs** across many different bacteria

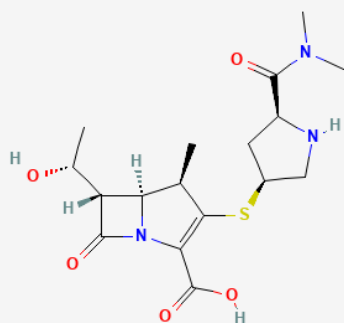
→ **Downstream:** gram-positive + gram-negative aerobes + Enterobacteriaceae + *P. aeruginosa* + anaerobes.

● **Reserve drugs — severe infection only** (ยาที่สงวนไว้).

## THE SHARED CORE • WHAT CHANGES

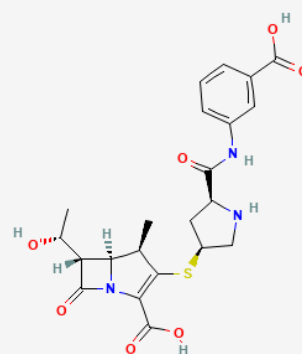


**Imipenem — the carbapenem core.** C replaces S in the five-membered ring, and the side chain is **trans** — that geometry is why nearly every  $\beta$ -lactamase fails on it



**Meropenem**

● the 1 $\beta$ -methyl group resists renal DHP-1 → no cilastatin needed



**Ertapenem**

a benzoic acid tail → high protein binding → once daily, no *Pseudomonas*

## MEDCHEM

**SAR** Three edits create the class's  $\beta$ -lactamase resistance, and they are all on the ring itself rather than the side chain. The **6-hydroxyethyl** group sits in *trans* configuration (penicillins are *cis*) and physically shields the ring; the **C2 double bond** raises ring strain, so the  $\beta$ -lactam is *more* reactive toward PBPs while being *less* accessible to enzymes.

**SAR** Thienamycin, the natural parent, was unusable - its free **amino group attacks its own ring** (aminolysis) in solution. Replacing it with an **N-formimidoyl** group gave imipenem. The later members add a **thio-pyrrolidine**, which is what confers the DHP-1 resistance that lets meropenem be given without cilastatin.

**Physicochem** ★ **Why ertapenem is once daily and the others are not:** it carries a **benzoic acid** group, which drives plasma protein binding right up and stretches the half-life. Same class, one substituent, a completely different dosing schedule.



## DRUGS

| DRUG                                    | WHAT IT TREATS                                                                                                                                      |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Imipenem/cilastatin</b>              | Enterobacteriaceae incl. <b>ESBL</b> · <i>P. aeruginosa</i> · <i>A. baumannii</i> incl. MDR · enterococci · anaerobes — binds <b>PBP2 and PBP1a</b> |
| <b>Meropenem</b>                        | same range, + <b>imipenem-resistant <i>P. aeruginosa</i></b> — bacteremia, meningitis, ruptured appendix                                            |
| <b>Ertapenem</b>                        | Enterobacteriaceae including <b>most ESBL</b>                                                                                                       |
| <b>Doripenem · Biapenem · Panipenem</b> | other members of the class                                                                                                                          |

## IMIPENEM VS MEROPENEM ★

|                             | IMIPENEM                                               | MEROPENEM                                    |
|-----------------------------|--------------------------------------------------------|----------------------------------------------|
| <b>DHP-I in body tissue</b> | destroyed by it → <b>must be given with cilastatin</b> | resistant → no cilastatin needed             |
| <b><i>P. aeruginosa</i></b> | standard                                               | ★ works on <b>imipenem-resistant</b> strains |
| <b>Gram-positive cocci</b>  | stronger                                               | weaker                                       |
| <b>Meningitis</b>           | ● <b>not recommended</b>                               | used for it                                  |
| <b>PK</b>                   | adjust in renal disease                                | unchanged by IV vs IM, by dose, or by age    |

## USES

Severe / multidrug-resistant infection · **ESBL producers** · Distribution is excellent to most organs — ⚠ **except the gallbladder / biliary tract** (imipenem)

## ADVERSE EFFECTS

- **Imipenem** — **seizure** ★. Neurotoxic, worst in patients with existing brain lesions.
- **Ertapenem** — dizziness, confusion

## INTERACTIONS

- **Meropenem × valproic acid** — carbapenems drop serum valproate, so seizure control can break

## NOT COVERED — KNOW THE HOLES

- ⚠ **Not active:** *Stenotrophomonas* · *Flavobacterium* · **MRSA**
- ⚠ **Not active:** *Chlamydia* · *Mycoplasma* · *Legionella* (*no peptidoglycan target worth attacking*)

# β-Lactam Resistance

## WHY THEY STOP WORKING — FOUR ROUTES

1. **Enzymatic inactivation — the most common.** β-lactamase from *S. aureus*, *H. influenzae*, *E. coli* hits **penicillins** >> **cephalosporins**. **ESBLs** (*P. aeruginosa*, *Enterobacter* spp.) hit **both**.
  2. **Impaired penetration (natural).** Gram-negative outer membrane / LPS blocks the porins — the drug never reaches the PBPs.
  3. **Altered target PBPs — low β-lactam affinity.** MRSA (**PBP2a**, *mecA*) · penicillin-resistant pneumococci and enterococci.
  4. **Efflux.** Gram-negatives pump the β-lactam back out across the outer membrane.
- **Downstream:** ★ β-lactamase inhibitors fix **route 1 only**. Routes 2–4 need a different drug.

## WHY GRAM-POSITIVES STAY EASIER

Their thick peptidoglycan layers **let β-lactams diffuse straight through** to PBPs sitting on the outer surface of the cytoplasmic membrane. There is no outer membrane and no porin bottleneck.

## NATURALLY RESISTANT BEFORE YOU EVEN START

- ⚠ Many **gram-negatives** — resistant to penicillin G and oxacillin because LPS blocks the porins
- ⚠ **Enterococcus** — its PBPs differ from other gram-positives (higher cell-wall lipid content) → resistant to penicillins **and 1st–5th generation cephalosporins**

## THE TWO MDR NAMES THE LECTURE CALLS OUT

| PATHOGEN                       | WHY IT MATTERS                                                                                                                                                                                                                              |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acinetobacter baumannii</b> | aerobic gram-negative · environmental and hospital · opportunistic — immunocompromised and ICU patients · <b>ESKAPE</b> · a majority of nosocomial infections                                                                               |
| <b>MRSA</b>                    | aerobic gram-positive · <b>changed PBP</b> → <b>low β-lactam affinity</b> · bloodstream infection, pneumonia, surgical site, abscess, sinusitis, food poisoning · <b>not more virulent than MSSA — just harder to treat</b> · <b>ESKAPE</b> |

Principle behind this: *[[pharm-drug-action]]* — Penicillins as the specific microbial-target case; the selective toxicity idea rests on this page's specific/non-specific split.

Principle behind this: *[[pharm-drug-toxicity]]* — Penicillin-induced haemolytic anaemia — the Type II cytolytic example, and the drug most often blamed for allergy it didn't cause.

Source: 2.Drugs affecting cell wall synthesis AM.pdf — ผศ.ดร.ภก. อาณัฐชัย ม้ายอุเทศ, PYPH305 Pharmacology 2, 54 slides. Structures omitted.

# Glycopeptides & Lipoglycopeptides

Big (MW ≈ 1500 Da) **bactericidal** cell-wall inhibitors that work on **Gram-positives only**. Clinical use keeps rising because of resistant organisms — above all **MRSA**.

**Bind the D-Ala–D-Ala terminus of the growing peptidoglycan → block transglycosylation & cross-linking → no further peptidoglycan elongation → cell lysis**

## THE TWO GROUPS

| GROUP                   | DRUGS                                  | WHAT MAKES IT DIFFERENT                                                                                                    |
|-------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Glycopeptide</b>     | vancomycin · teicoplanin               | two sugar moieties + an <b>aglycone</b> . Aglycone = the pharmacologic activity; sugars = hydrophilicity + dimer formation |
| <b>Lipoglycopeptide</b> | telavancin · dalbavancin · oritavancin | <b>semisynthetic</b> — a hydrophobic moiety bolted on, which buys a <b>very long half-life</b>                             |

## SPECTRUM OF ACTION

|                           | ORGANISMS                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Covered — Gram (+)</b> | <b>MSSA &amp; MRSA</b> ★ · coagulase-negative staphylococci · <i>S. pneumoniae</i> (incl. PRSP) · viridans & group streptococci · <i>Enterococcus</i> sp. · <i>Corynebacterium</i> , <i>Bacillus</i> , <i>Listeria</i> , <i>Actinomyces</i> · <i>Clostridium</i> sp. (incl. <b><i>C. difficile</i></b> ) · <i>Peptococcus</i> , <i>Peptostreptococcus</i> |
| <b>Not covered</b>        | ● <b>No activity against Gram (–) aerobes or anaerobes</b> (the molecule is too big to get through the Gram-negative outer membrane)                                                                                                                                                                                                                      |

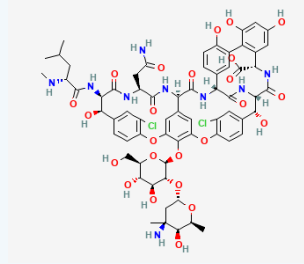
# Glycopeptides

## MECHANISM

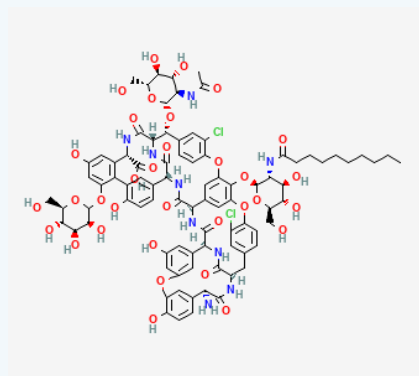
Binds the **D-Ala-D-Ala terminus** of the growing peptidoglycan chain, physically blocking **transglycosylation** and **cross-linking**. Note it binds the *substrate*, not an enzyme — which is why  $\beta$ -lactam resistance mechanisms don't touch it.

→ **Downstream:** peptidoglycan can't elongate or cross-link → **cell lysis** (bactericidal) · acts **only on actively dividing bacteria**, and more slowly than penicillin

## THE SHARED CORE · WHAT CHANGES



**Vancomycin.** a rigid cup of cross-linked amino acids that grips the D-Ala-D-Ala terminus itself — ● it binds the SUBSTRATE, not the enzyme. Resistance = the bug changes to D-Ala-D-Lac and the cup no longer fits.



### Teicoplanin

same cup + a lipid tail → far longer  $t_{1/2}$

## MEDCHEM

**SAR** The binding is countable: vancomycin's heptapeptide backbone makes **five hydrogen bonds** to the D-Ala4-D-Ala5 terminus. It is a peptide gripping a peptide - which is why nothing about  $\beta$ -lactamase touches it.

**Resistance** ● **VRE, at the molecular level: five hydrogen bonds become four.** The organism mutates the terminal **D-Ala5 to D-Lac**, deleting one N-H donor. Losing a single hydrogen bond out of five collapses the binding affinity - the drug still fits, it just no longer holds.

## DRUGS

| DRUG                          | WHAT IT TREATS                                                                                                                                                                  |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vancomycin</b> (IV)        | MSSA · <b>MRSA</b> · enterococci ( <i>except VRE</i> ) · <i>L. monocytogenes</i> · anaerobic cocci · staphylococci that make $\beta$ -lactamase or resist nafcillin/methicillin |
| <b>Vancomycin</b> (oral)      | <b><i>C. difficile</i> only</b> ★ — it is barely absorbed from the GI tract, so oral use is purely local                                                                        |
| <b>Teicoplanin</b> (IV or IM) | same spectrum as vancomycin · long $t_{1/2}$ <b>45–70 h</b> → <b>once-daily</b> dosing                                                                                          |

## VANCOMYCIN — THE NUMBERS THAT MATTER

|                           |                                                                     |
|---------------------------|---------------------------------------------------------------------|
| <b>Dose</b>               | 25–30 mg/kg loading, then <b>15–20 mg/kg q12h</b>                   |
| <b>Distribution</b>       | wide; <b>CSF 7–30%</b> of serum — only with meningeal inflammation  |
| <b>Elimination</b>        | <b>90% by glomerular filtration</b> ; $t_{1/2} \approx 6 \text{ h}$ |
| ● <b>Renal failure</b>    | $t_{1/2}$ stretches to <b>6–10 days</b> → dose must be adjusted     |
| <b>Hemodialysis</b>       | removes ~50% of the drug                                            |
| ★ <b>Toxicity ceiling</b> | keep <b>peak serum</b> < <b>60 µg/mL</b> to minimise ototoxicity    |

### ADVERSE EFFECTS

- **Red-man syndrome** ★ — flushing, pruritus, erythematous rash, hypotension, myalgia. **Driven by the rate of IV infusion, not by allergy** → infuse over **≥ 60 min**. Resolves spontaneously on stopping; give an antihistamine, or pretreat in patients who've had it.
- **Nephrotoxicity** and **ototoxicity** — rare on monotherapy; much more likely alongside other nephro-/oto-toxic drugs, or in renal impairment
- **Thrombophlebitis** at the infusion site

### INTERACTIONS

- ✓ **Synergy with gentamicin or streptomycin** against *Enterococcus faecium* and *E. faecalis*
- ● Other **nephrotoxic / ototoxic** drugs → stacked toxicity

### CAUTIONS

- ⚠ **Renal impairment** — the half-life goes from hours to days; dose-adjust or you accumulate straight into toxicity
- ⚠ Kills only **dividing** organisms, and slowly — don't expect penicillin-speed response

### ADMINISTRATION — VANCOMYCIN DOSING

- **Loading dose 25–30 mg/kg, then 15–20 mg/kg every 12 hours (IV).**
- ⚠ **Red man syndrome** (*infusion-rate related*) and **nephrotoxicity**.
- 💡 The loading dose exists because at 15–20 mg/kg alone it takes days to reach a therapeutic level — and in MRSA bacteraemia those are the days that decide the outcome.

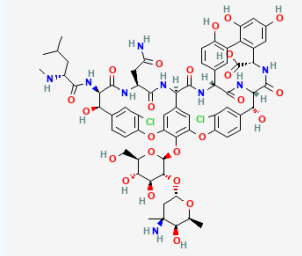
# Lipoglycopeptides

## MECHANISM

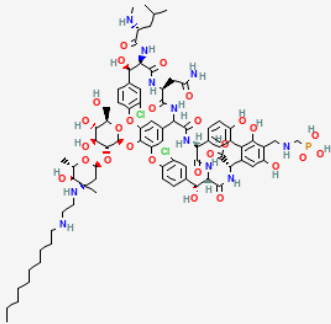
Same **D-Ala-D-Ala** binding as the glycopeptides — these are **semisynthetic derivatives** with a hydrophobic moiety added.

→ **Downstream:** the added lipophilic tail doesn't change the target, it changes the **pharmacokinetics** — **dramatically longer half-life**, hence very infrequent dosing

## THE SHARED CORE · WHAT CHANGES

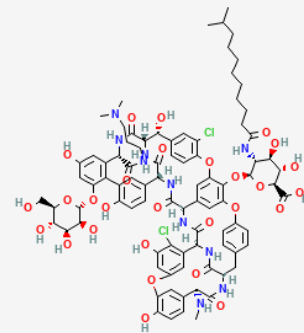


**Vancomycin — the parent.** the lipid tail below is the only real addition



**Telavancin**

the tail anchors it in the membrane → a second mechanism (depolarisation) on top of the first



**Dalbavancin**

tail + high protein binding →  $t_{1/2}$  in days, single-dose therapy

## DRUGS

| DRUG                                          | WHAT IT TREATS                                                                                   |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Telavancin</b> (from <i>vancomycin</i> )   | complicated <b>skin &amp; soft-tissue infection</b> · $t_{1/2}$ 8 h → once daily                 |
| <b>Dalbavancin</b> (from <i>teicoplanin</i> ) | complicated <b>skin &amp; soft-tissue infection</b> · $t_{1/2}$ 6–11 days ★ → <b>once weekly</b> |
| <b>Oritavancin</b>                            | named by the lecture as a lipoglycopeptide; no further detail given                              |

# Glycopeptide Resistance (VRE)

## MECHANISM

The target is the terminal **D-alanyl-D-alanine**. Resistant organisms simply **change the last residue**, which **loses a hydrogen bond** to the drug.

→ **Downstream:** sharply **reduced affinity** → **much higher MIC** → vancomycin-resistant enterococcus (**VRE**)

## THE SIX PHENOTYPES — VANA–VANE AND VANG

| ALTERED TERMINUS          | PHENOTYPES         |
|---------------------------|--------------------|
| <b>D-alanyl-D-lactate</b> | VanA · VanB · VanD |
| <b>D-alanyl-D-serine</b>  | VanC · VanE · VanG |

*Multiple genes underlie the different phenotypes.*

# Bacitracin

## MECHANISM

A **polypeptide** — different target from the glycopeptides. It interferes with the **dephosphorylation of the membrane carrier molecule** (C55-isoprenyl pyrophosphate / bactoprenol pyrophosphate) that ferries peptidoglycan subunits across the membrane.

→ **Downstream:** the carrier can't recycle → **cell-wall synthesis in Gram-positives is disrupted**

## DRUGS

| DRUG                           | WHAT IT TREATS                                                                                                                        |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bacitracin</b><br>(topical) | Gram-positive skin infection — <i>S. aureus</i> , <i>S. epidermidis</i> , <i>S. pyogenes</i> ; combined with polymyxin B and neomycin |

## ADMINISTRATION — BACITRACIN

- Supplied as an **ointment, in combination with polymyxin B and neomycin**.
- 💡 The triple combination is the whole point — bacitracin covers Gram-positives, polymyxin B covers Gram-negatives, and neomycin broadens it further. No single one of them would do as a topical.

Source: 3.Glycopeptide and Lipoglycopeptide AM.pdf — ผศ.ดร.ภก. อาณัฐชัย ม้ายอุเทศ, PYPH303 Pharmacology 2, 19 slides. Structures omitted.



# Drugs Affecting the Cell Membrane

Two classes that kill by **wrecking the bacterial membrane** instead of a synthesis pathway. They split cleanly by spectrum: **polymyxins hit Gram-negative**, **daptomycin hits Gram-positive** — and each is held back for resistant organisms.

## THE TWO AT A GLANCE

|                      | POLYMYXINS                                                            | DAPTOMYCIN                                  |
|----------------------|-----------------------------------------------------------------------|---------------------------------------------|
| <b>Spectrum</b>      | Gram <b>negative</b>                                                  | Gram <b>positive</b>                        |
| <b>Membrane hit</b>  | outer membrane → inner membrane                                       | inner membrane, Ca <sup>2+</sup> -dependent |
| <b>Result</b>        | membrane disorganised → lysis                                         | rapid depolarisation, K <sup>+</sup> efflux |
| <b>Reserved for</b>  | MDR <i>P. aeruginosa</i> · carbapenemase-producing Enterobacteriaceae | MRSA · VRSA · VRE                           |
| <b>Signature ADR</b> | ● nephrotoxicity + neurotoxicity                                      | myopathy / ↑ CPK                            |

# Polymyxins

## MECHANISM

Amphipathic peptides — a **hydrophobic fatty-acyl tail** plus a **hydrophilic peptide ring** — so they act like a **detergent** on the Gram-negative envelope. They first **disorganise the outer membrane**, then disrupt the physical integrity of the **inner membrane**.

→ **Downstream:** membrane permeability lost → contents leak → **cell death**

*Irony worth remembering ★ — they're produced by a Gram-positive organism (Paenibacillus polymyxa, via non-ribosomal peptide synthetase) but are used only against Gram-negatives.*

## DRUGS

| DRUG                          | WHAT IT TREATS                                                                                                  |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Polymyxin B</b>            | Gram-negative infection — multidrug-resistant <i>P. aeruginosa</i> · carbapenemase-producing Enterobacteriaceae |
| <b>Colistin (polymyxin E)</b> | same — MDR Gram-negative infection                                                                              |

## ADVERSE EFFECTS

- **Nephrotoxicity** ★ — ↑ permeability of the **renal tubular epithelial** cell membrane (same trick, wrong cell) → acute tubular necrosis, proteinuria, hematuria, azotemia, oliguria
- **Neurotoxicity** ★ — blocks **ACh release from the presynaptic** terminal → facial & peripheral paresthesia, weakness, visual disturbance, ataxia, **neuromuscular blockade**
- **Respiratory arrest**
- *Both are dose-dependent and reversible.*

## INTERACTIONS

- ⚠ **Avoid co-administration with other nephrotoxic or neurotoxic drugs** — the two toxicities stack

# Daptomycin

## MECHANISM

Binds **calcium ions** in the cell membrane and inserts — causing **rapid depolarisation** with **K<sup>+</sup> efflux**.  
**Concentration-dependent bactericidal** activity.

→ **Downstream:** loss of membrane potential → intracellular **DNA, RNA and protein synthesis** all shut down → cell death

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Daptomycin</b> | serious Gram-positive infection — MRSA/VRSA · VRE · <i>S. aureus</i> bacteremia · complicated skin & soft-tissue infection |

## USES

**Spectrum:** MSSA & MRSA, VRSA · coagulase-negative staphylococci · *S. pneumoniae* (incl. PRSP), viridans & group streptococci · *E. faecium* & *E. faecalis*, VRE · **relatively inactive against Gram-negative aerobes**

**Reserved for serious / complicated resistant infection:** complicated skin & soft-tissue infection (MSSA, MRSA, *S. pyogenes*) · ***S. aureus* bacteremia** · accumulating data in MRSA/VRE endocarditis, meningitis, osteomyelitis · catheter-related bacteremia

## ADVERSE EFFECTS

- **Myopathy with ↑ CPK** ★ — the one to monitor
- Nausea, vomiting, diarrhea
- Headache
- Injection-site reactions
- Rash

## INTERACTIONS

- ⚠ **HMG-CoA reductase inhibitors (statins)** → ↑ incidence of **myopathy**

## CAUTIONS

- ● **Never use for pneumonia** ★ — daptomycin is **inactivated by pulmonary surfactant**, so it has no activity in lung parenchyma
- Parenteral only · poor penetration into **CNS and bone**
- Renally excreted — **dose adjustment required in renal impairment**; *not* removed by hemodialysis

Source: 4.Polymyxin AM.pdf — ผศ.ดร.ภก. อาณัฐชัย ม้ายอุเทศ, PYPH303 Pharmacology 2, 11 slides. Structures omitted.

# Inhibitors of DNA, Nucleic Acid & Protein Synthesis

Everything here attacks the bacterial **information machinery** rather than the cell wall — either the enzymes that unwind and copy DNA, the folate pathway that feeds DNA precursors, or the **ribosome** itself. Selectivity comes from the bacterial ribosome being **70S (30S + 50S)** where ours is 80S.

## WHERE EACH CLASS BINDS

| TARGET                                      | CLASSES                                                                                   |
|---------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>DNA gyrase / topoisomerase IV</b>        | Fluoroquinolones                                                                          |
| <b>DNA directly</b> (reduced nitro radical) | Metronidazole                                                                             |
| <b>Folate pathway</b> (DNA precursors)      | Sulfamethoxazole + Trimethoprim                                                           |
| <b>30S ribosome</b>                         | Aminoglycosides · Tetracyclines / Glycylcyclines · Spectinomycin                          |
| <b>50S ribosome</b>                         | Macrolides · Ketolides · Lincosamides · Streptogramins · Chloramphenicol · Oxazolidinones |

## CIDAL VS STATIC — AND WHAT THE KILLING DEPENDS ON

| CLASS                         | EFFECT                                          | KINETICS                                  |
|-------------------------------|-------------------------------------------------|-------------------------------------------|
| <b>Fluoroquinolones</b>       | bactericidal                                    | concentration-dependent (AUC/MIC)         |
| <b>Metronidazole</b>          | bactericidal                                    | concentration-dependent                   |
| <b>Aminoglycosides</b>        | bactericidal                                    | concentration-dependent + long <b>PAE</b> |
| <b>Macrolides / Ketolides</b> | bacteriostatic ( <i>cidal at high conc.</i> )   | time-dependent with PAE                   |
| <b>Clindamycin</b>            | bacteriostatic ( <i>cidal at high conc.</i> )   | —                                         |
| <b>Streptogramins</b>         | bacteriostatic ( <i>cidal vs some</i> )         | A + B are <b>synergistic</b>              |
| <b>Tetracyclines</b>          | bacteriostatic                                  | —                                         |
| <b>Linezolid</b>              | bacteriostatic ( <i>cidal vs some</i> )         | time-dependent                            |
| <b>TMP-SMX</b>                | each alone static → <b>combination is cidal</b> | sequential blockade                       |

#### TERMS THIS LECTURE LEANS ON

- **PAE (post-antibiotic effect)** — suppression of bacterial growth persisting *after* drug levels fall below MIC. Justifies once-daily aminoglycoside dosing.
- **MLS<sub>B</sub>** — macrolide–lincosamide–streptogramin B; a single ribosomal methylation (*erm*) knocks out all three at once.
- **Atypicals** — *Legionella* · *Chlamydia* · *Mycoplasma* · *Ureaplasma*. FQs, macrolides and tetracyclines all cover them; cell-wall drugs cannot.

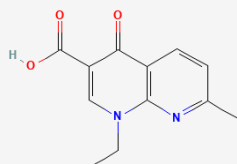
# Fluoroquinolones

## MECHANISM

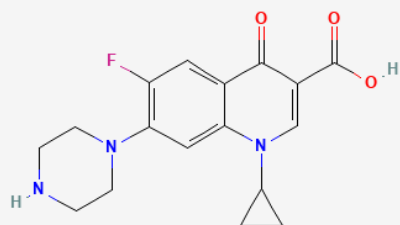
Inhibit the bacterial **topoisomerases** that manage DNA supercoiling:

- **DNA gyrase** — removes excess positive supercoiling → primary target in **gram-negatives**
  - **Topoisomerase IV** — separates interlinked daughter DNA → primary target in **gram-positives**
- **Downstream: DNA repair, replication and transcription** all blocked → **bactericidal**, concentration-dependent (AUC/MIC drives efficacy)

## THE SHARED CORE · WHAT CHANGES

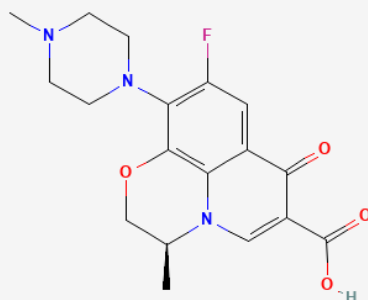


**Nalidixic acid — the parent quinolone.** the 4-oxo-3-carboxylic acid pair is the pharmacophore that grips DNA gyrase. No fluorine yet — urinary use only.



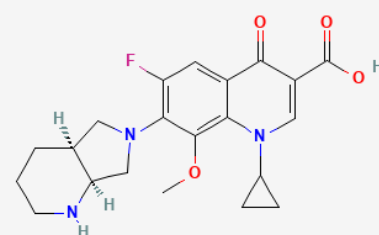
### Ciprofloxacin

add 6-F (tissue penetration) + 7-piperazine (gram-negatives, Pseudomonas)



### Levofloxacin

tricyclic, single active isomer → respiratory cover



### Moxifloxacin

8-OMe + bulky ring → anaerobes; ⚠ no urinary use

## MEDCHEM

**SAR** ★ The pharmacophore is the **1,4-dihydro-4-oxo-3-pyridine carboxylic acid** - the **3-carboxyl** and the **4-keto** together are what grip the DNA-gyrase complex. Reduce the 2,3-double bond or the 4-keto and the molecule is **dead**; substitute at C2 and it can no longer form the enzyme-substrate complex.

**SAR** ● Almost every fluoroquinolone ADR is a named position on the ring. **C5** -NH<sub>2</sub> or -CH<sub>3</sub> → **QT prolongation and phototoxicity**. **C8** as C-F or C-Cl → **phototoxicity** (F worse than Cl); as C-OCH<sub>3</sub> → better anaerobe cover and *less* resistance. **C7 piperazine binds GABA receptors** → dizziness, insomnia, headache - and methylating that piperazine nitrogen reduces it. **N1 (R1)** governs **CYP1A2 inhibition**, which is the theophylline interaction; a bulkier R1 inhibits less.

**SAR** The upgrades: **C6 fluorine** raises both gyrase inhibition and lipophilicity, so the drug penetrates the cell wall - that one atom is the whole 'fluoro-'. **C7 heterocycle** sets the spectrum (azabicyclic > pyrrolidine > piperazine for gram-positives) and stretches the half-life.

**Physicochem** The **same 3-carboxyl and 4-keto that bind gyrase also chelate Zn<sup>2+</sup>, Fe<sup>2+</sup>, Ca<sup>2+</sup>, Mg<sup>2+</sup>, Al<sup>3+</sup>** - the drug's business end and its milk/antacid/iron interaction are one and the same feature. Chelated drug is less soluble and less absorbed.

**SAR** ★ **Levofloxacin is not a new molecule** - it is the **S-isomer** of ofloxacin, and it is **8–125× more active** than the R. Half of racemic ofloxacin was always dead weight.

## DRUGS — BY GENERATION

| DRUG                                                              | WHAT IT TREATS                                                                                       |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Nalidixic acid</b> (1st)                                       | the parent quinolone — all others are derivatives of it                                              |
| <b>Norfloxacin · Ofloxacin · Ciprofloxacin · Rufloxacin</b> (2nd) | gram-negative infections · UTI · gastroenteritis                                                     |
| <b>Levofloxacin</b> (3rd)                                         | respiratory + urinary infections, now with gram-positive cover                                       |
| <b>Moxifloxacin · Sitafloracin · Prulifloxacin</b> (4th)          | respiratory & skin infections · anaerobes ( <i>moxi</i> )                                            |
| <b>Gemifloxacin · Gatifloxacin</b>                                | <i>S. pneumoniae</i> respiratory infections · bacterial conjunctivitis ( <i>gati</i> , topical only) |

## SPECTRUM & USES

**Gram-negative** — excellent for Enterobacteriaceae, *H. influenzae*, *M. catarrhalis*, *Neisseria* (cipro = levo > gati > moxi); *P. aeruginosa* → **cipro/levo best** · **Gram-positive** — only the newer agents (levo, moxi, gemi) reach *S. pneumoniae* incl. PRSP · **Atypicals** — all FQs; ★ **drug of choice for *Legionella pneumophila*** · **Anaerobes** — moxifloxacin only · **Other** — *M. tuberculosis*, MAC, *B. anthracis*

Clinically: sinusitis & ABECB · **CAP** (levo, moxi) · nosocomial pneumonia (cipro, levo) · cystitis/pyelonephritis/prostatitis (cipro, levo) · skin · osteomyelitis · intra-abdominal · gastroenteritis · STDs · TB

## ADVERSE EFFECTS

- **Tendon rupture** ★ and **articular cartilage damage / arthralgia** — *contraindicated in children, pregnancy and breastfeeding*
- **QTc prolongation** ★ — withdrew grepafloxacin and sparfloxacin from the market
- **Aortic aneurysm / dissection** — atherosclerosis, hypertension, Marfan, Ehlers-Danlos, elderly
- CNS: headache, agitation, insomnia, dizziness; rarely **hallucinations and seizures in the elderly**
- Peripheral neuropathy
- Hepatotoxicity — LFT rise (*withdrew trovafloxacin*)
- GI ~5% — nausea, vomiting, diarrhea, dyspepsia
- Phototoxicity — mostly older agents (halogen at position 8)
- Dysglycemia — **gatifloxacin** (*why systemic use was withdrawn*)

## INTERACTIONS

- **Di- and trivalent cations** — **ALL FQs** ★: Zn, Fe, Ca, Al, Mg (antacids, sucralfate, ddi, enteral feeds) chelate the drug and wreck oral absorption → **clinical failure**. *Separate by 2–4 h, FQ first.*
- **Ciprofloxacin** inhibits metabolism of **theophylline** and **cyclosporine** → toxicity (*strong CYP1A2 inhibitor*)
- **Warfarin** — idiosyncratic effect, all FQs

## CAUTIONS

- ⚠ Minimal **CSF** penetration — not for meningitis
- ⚠ Not removed by haemodialysis
- ⚠ Resistance emerges readily: gyrase/topo IV mutations (QRDR), plasmid *qnr* proteins, Aac(6')-Ib-cr acetyltransferase, efflux pumps, porin loss

## ADMINISTRATION

- **Good oral bioavailability across the class — except norfloxacin**, which never reaches systemic levels 
- **Cmax at 1–2 h**; taking it **with food delays the peak** — it doesn't block absorption (*the chelators do*)
- **Dosage forms:** Norfloxacin tab · Rufloxacin tab · **Ciprofloxacin** tab, IV, eye drop · **Ofloxacin** tab, eye drop, **ear drop** · **Levofloxacin** tab, IV, eye drop · **Gatifloxacin — eye drop only** (*systemic use withdrawn: severe dysglycaemia*) · **Moxifloxacin** tab, IV · Prulifloxacin & Sitafloracin tab, **BID**
-  The chelation trap is what actually ruins these at a counter: keep antacids, iron, calcium and dairy about **2 hours** clear of the dose, and ask what else they're taking rather than assuming.



# Fluoroquinolones — Agent by Agent

## WHAT SEPARATES THEM

| AGENT                               | THE DISTINGUISHING FACT                                                                                                            |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Norfloxacin</b>                  | does <b>not</b> reach systemic levels — gastroenteritis, gonorrhea only                                                            |
| <b>Rufloxacin</b>                   | UTIs and prostatitis                                                                                                               |
| <b>Ciprofloxacin</b>                | most active vs <b>g(–), incl. <i>P. aeruginosa</i></b> — keep it for that · high arthropathy rate · <b>strong CYP1A2 inhibitor</b> |
| <b>Ofloxacin</b>                    | racemic mixture · UTI, URI (otitis media), LRI, skin, gastroenteritis, MTB                                                         |
| <b>Levofloxacin</b>                 | the <b>S-enantiomer of ofloxacin</b> → better activity · reaches <b>g(+) <i>S. pneumoniae</i></b> · <i>H. pylori</i> eradication   |
| <b>Moxifloxacin</b>                 | g(+) cover + anaerobes · <b>non-renal excretion</b> → <b>NOT for UTIs</b> ★ · URI, LRI, skin, MTB                                  |
| <b>Gatifloxacin</b>                 | <b>topical eye use only</b> — systemic withdrawn for severe dysglycemia                                                            |
| <b>Prulifloxacin · Sitafloracin</b> | twice-daily agents — URI, LRI, UTI, skin                                                                                           |

## HALF-LIFE

| SHORT                                        | LONG               |
|----------------------------------------------|--------------------|
| Nor · Oflo · Cipro · Ruflo · Pruli · Sitaflo | <b>Levo · Moxi</b> |

# Metronidazole

## MECHANISM

A **prodrug** activated only by **reduction**. Anaerobes and microaerophiles carry **ferredoxins**; these donate electrons to the nitro group, forming a highly reactive **nitro anion radical** that shreds bacterial DNA. Aerobes have no ferredoxin → **that's the selectivity**.

→ **Downstream**: DNA damage → cell death · **bactericidal**, concentration-dependent → **Resistance**: better oxygen scavenging (higher local O<sub>2</sub> blocks activation) or reduced ferredoxin transcription

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                                                                                                                                                                      |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Metronidazole</b> | anaerobic infections incl. <b>brain abscess</b> · ★ <b>DOC for <i>C. difficile</i></b><br><b>pseudomembranous colitis</b> · bacterial vaginosis · trichomoniasis · amebiasis · giardiasis · <i>H. pylori</i> · rosacea · gingivitis |

## SPECTRUM & USES

**Anaerobes — below the diaphragm**: *Bacteroides* (all) · *Fusobacterium* · *Prevotella* · *Clostridium* (all) · *H. pylori* · **Anaerobic protozoa**: *Trichomonas vaginalis* · *Entamoeba histolytica* · *Giardia lamblia* · *Gardnerella vaginalis*

★ **It DOES penetrate the CSF** — which is why it, not clindamycin, treats **brain abscess** and CNS anaerobes. Intra-abdominal · pelvic · skin/soft tissue · diabetic foot · decubitus ulcer infections.

## ADVERSE EFFECTS

- **CNS — the most serious**: peripheral neuropathy, seizures, encephalopathy. **Requires stopping the drug**.
- GI — nausea, vomiting, stomatitis, **metallic taste**
- **Disulfiram-like reaction with alcohol** ★ — inhibits aldehyde dehydrogenase (ALDH)
- Mutagenicity / carcinogenicity — avoid in pregnancy and breastfeeding

## INTERACTIONS

- **Warfarin** potentiated — CYP2C9 inhibition
- ⚠ Alcohol → disulfiram-like reaction

## CAUTIONS

- ⚠ Pre-existing CNS disorders
- ⚠ Pregnancy / breastfeeding

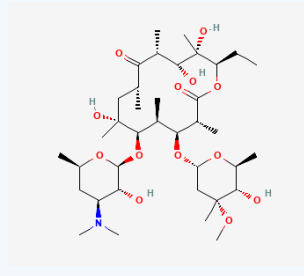
# Macrolides

## MECHANISM

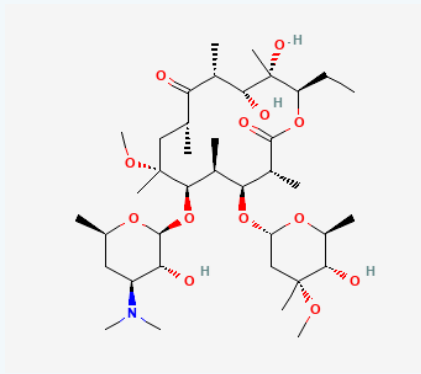
**Reversibly bind the 50S ribosomal subunit** — inhibit elongation and **block translocation**, suppressing RNA-dependent protein synthesis.

→ **Downstream:** bacteriostatic (bactericidal at high concentration against very susceptible organisms) · **time-dependent** activity with a **PAE** → **Resistance:** *mef* efflux pump = **low-level**, 80% of US isolates · *erm* target methylation = **high-level**, cross-resistant across **all macrolides + clindamycin + streptogramin B (MLS<sub>B</sub>)**

## THE SHARED CORE · WHAT CHANGES

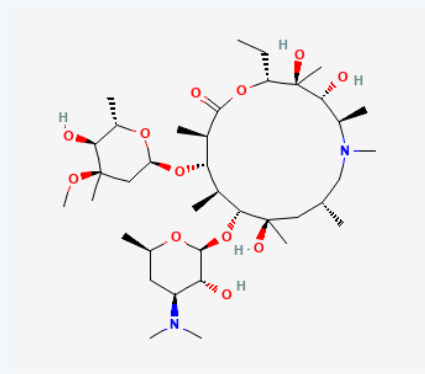


**Erythromycin — the 14-membered macrolactone.** a big lactone ring with two sugars; ⚠ the ring is acid-labile — it cyclises in the stomach, which is the GI upset



### Clarithromycin

6-OMe blocks that acid cyclisation → acid-stable, better absorbed



### Azithromycin

● a nitrogen inserted INTO the ring (15-membered azalide) → huge tissue  $t_{1/2}$ , once daily, and it is the one that doesn't inhibit CYP3A4

## MEDCHEM

**SAR** The three parts do three jobs: the **macrolactone ring** binds, **desosamine** is the weakly basic amino sugar ( $pK_a \approx 8$ ) - the free base is not water-soluble, so every injectable form is a salt - and **L-cladinose** sits at C3.

**Stability** ● **This is the single best thing in the deck.** In stomach acid, erythromycin's own C6-OH and C9-ketone **cyclise into an internal ketal** - the drug destroys itself. Worse, that degradation product **binds the motilin receptor in the gut**, which is exactly where erythromycin's notorious GI cramping comes from (and why it doubles as a prokinetic). **Clarithromycin is erythromycin with that C6-OH methylated** - the ketal can't form, so it survives acid and absorbs better. One methyl group, whole problem solved.

**SAR Azithromycin is a ring-expansion product** (Beckmann rearrangement inserts a nitrogen → a 15-membered azalide). That expansion is what buys the broader spectrum, the long half-life, the tissue penetration and the **post-antibiotic effect** - the page's PAE has a structural cause.

**Prodrug Dirithromycin** is a prodrug (a hemiaminal that reverts to erythromycylamine).

**Resistance** The deck puts a number on *erm*: mono/dimethylation of **A2058** in 23S rRNA, and an A→G mutation at the same base drops erythromycin/clarithromycin binding **10,000-fold**. A2058 is also the lincosamide and streptogramin-B site - that shared base *is* the MLS<sub>B</sub> phenotype. **Ketolides** answer it by replacing the C3 cladinose with a keto group plus a cyclic carbamate and an alkyl-aryl anchor into domain II.

**Interaction** Macrolides inhibit **CYP3A4 and P-glycoprotein** - that pairing, not CYP alone, is why the interaction list is so long.

## DRUGS

| DRUG                                | WHAT IT TREATS                                                                                             |
|-------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Erythromycin</b>                 | the natural parent — corynebacterial infections · URI · intestinal paralysis                               |
| <b>Clarithromycin</b>               | <i>H. pylori</i> triple therapy · URI/LRI · skin · <b>MAC treatment</b>                                    |
| <b>Azithromycin</b>                 | CAP & AECB · STDs ( <i>single 2 g dose for gonococcal urethritis/cervicitis</i> ) · <b>MAC prophylaxis</b> |
| <b>Roxithromycin</b>                | URI · skin infections ( <i>more active vs Legionella</i> )                                                 |
| <b>Spiramycin ·<br/>Midecamycin</b> | further macrolide members named in the lecture                                                             |

## SPECTRUM & USES

**Gram-positive aerobes** — clarithro > erythro > azithro; MSSA, *S. pneumoniae* (resistance emerging), streptococci · **Gram-negative aerobes** — azithro > clarithro > erythro; *H. influenzae*, *M. catarrhalis*, *Neisseria* — ⚠ **no Enterobacteriaceae at all** · **Atypicals** — excellent; ★ **DOC for Legionella** · **Other** — MAC (azithro & clarithro only), *Treponema*, *Campylobacter*, *Borrelia*, *Bordetella*, *Brucella*, *Pasteurella*  
Clinically: pharyngitis/tonsillitis in **penicillin-allergic** patients · sinusitis, ABECEB, otitis media · **atypical CAP** · uncomplicated skin & soft tissue infections

### ADVERSE EFFECTS

- **GI up to 33%** ★ — nausea, vomiting, diarrhea, dyspepsia. Worst with **erythromycin**, much less with the newer agents (*erythromycin is a motilin receptor agonist — that's why*)
- **Cholestatic hepatitis** — rare, after >1–2 weeks of **erythromycin estolate**
- Thrombophlebitis with IV erythromycin and azithromycin — dilute and give slowly
- **QTc prolongation**
- Ototoxicity — high-dose erythromycin in renal impairment
- Allergy

### INTERACTIONS

- ● **Erythromycin and clarithromycin are CYP inhibitors** ★ — raise levels of theophylline, digoxin, disopyramide, carbamazepine, valproate, cyclosporine, tacrolimus, phenytoin, warfarin, cisapride, terfenadine, astemizole, ergot alkaloids
- ✓ **NOT azithromycin** — no CYP inhibition. This is the exam contrast.

# Macrolides — Agent by Agent

## WHAT SEPARATES THEM

| AGENT                 | DISTINGUISHING FACTS                                                                                                                                                                                                           |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Erythromycin</b>   | acid-labile base · F = 15–45%, food ↓ absorption · <b>t<sub>1/2</sub> 1.5 h</b> · <b>potent CYP3A4 inhibitor</b> · 95% biliary/faecal · estolate salt = liver toxicity                                                         |
| <b>Roxithromycin</b>  | t <sub>1/2</sub> 10.5 h · 65% excreted unchanged in bile · <b>low CYP interaction, low GI ADR</b> · more active vs <i>Legionella</i>                                                                                           |
| <b>Clarithromycin</b> | acid-stable, F = 55% · <b>active metabolite</b> (14-OH-clarithromycin) · <b>potent CYP3A4 inhibitor</b> · <b>the only macrolide partly renally cleared</b> → adjust if CrCl < 30 · taste/smell alteration                      |
| <b>Azithromycin</b>   | F = 38% · <b>tissue:serum = 50:1</b> , t <sub>1/2</sub> ≈ <b>3 days</b> ★ · <b>no CYP inhibition</b> · biliary excretion unmetabolised · ~25% of <i>S. pneumoniae</i> resistant → combine with ceftriaxone in hospitalised CAP |

## SHARED PK CAUTIONS

- ⚠ **Minimal CSF penetration** — none of them treat meningitis
- ⚠ **None** of the macrolides are removed by haemodialysis
- All are hepatically eliminated

## ADMINISTRATION — FOOD IS THE WHOLE STORY HERE ★

| AGENT                 | FOOD                                                                                                                                           | ORAL BIOAVAILABILITY     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Erythromycin</b>   | <b>food may ↓ absorption</b> — base is destroyed by gastric acid ( <i>hence enteric coating</i> ); esters and ester salts are more acid-stable | F = 15–45%, variable     |
| <b>Roxithromycin</b>  | ● <b>food ↓ absorption 21%</b>                                                                                                                 | —                        |
| <b>Clarithromycin</b> | <b>absorbed regardless of food</b> — acid-stable                                                                                               | F = 55%                  |
| <b>Azithromycin</b>   | <b>food ↓ absorption of the CAPSULE</b> ( <i>formulation-specific, not the drug</i> )                                                          | F = 38%                  |
| <b>Telithromycin</b>  | —                                                                                                                                              | <b>PO only</b> , F = 57% |

- Dosage forms: Erythromycin** — base (*acid-labile*), stearate (tab), estolate (cap, syrup), ethylsuccinate (syrup) · **Roxithromycin** — tablet · **Clarithromycin** — tablet, **MR**, syrup, IV · **Azithromycin** — capsule, tablet, powder sachet, powder for reconstitution, prolonged-release granules, IV infusion
- 💡 Only two of them need a timing instruction: **roxithromycin and azithromycin capsules go before food**. Clarithromycin you can hand over with no meal rule at all — say so, or the patient invents one.

# Ketolides (Telithromycin)

## MECHANISM

A macrolide derivative that binds **two domains** on the 50S subunit — ~10× stronger binding to **domain II**, which is what widens its spectrum.

→ **Downstream:** protein synthesis inhibited · bacteriostatic, time-dependent → **Why it matters:** *erm*-methylated (high-level macrolide-resistant) organisms still respond — ★ **telithromycin covers macrolide-resistant *S. pneumoniae***

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                                                            |
|----------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Telithromycin</b> | respiratory infections due to <b>macrolide-resistant <i>S. pneumoniae</i></b> (FDA has since pulled 2 of its indications) |

## SPECTRUM

**Best gram-positive activity of the whole MLSK group** (telithro > clarithro > erythro > azithro) · ⚠️ **poor gram-negative activity** — worse than any macrolide, no *H. influenzae* · atypicals covered · PO only, F = 57%, t<sub>1/2</sub> 10 h

## ADVERSE EFFECTS

- **Severe hepatotoxicity** ★ — the reason indications were withdrawn
- **Ocular — blurred vision, decreased accommodation** (*distinctive; macrolides don't do this*)
- **QTc prolongation**
- GI — diarrhea 10%, nausea 7%, vomiting 3%
- CNS — dizziness 3%, headache 2%

## INTERACTIONS

- CYP inhibitor, same list as erythromycin/clarithromycin

# Lincosamides (Lincomycin & Clindamycin)

## MECHANISM

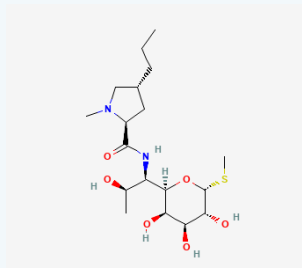
Bind **exclusively to the 50S subunit** at the **P-site**, blocking elongation — a site so close to the macrolide site that the two **competitively inhibit** each other.

→ **Downstream:** bacteriostatic (cidal at high concentration) · gram-negatives are **intrinsically resistant** →

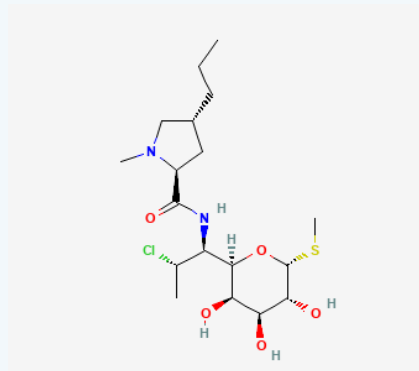
**Resistance:** *erm* methylation kills it (MLS<sub>B</sub>, high-level) · but the *mef* efflux pump pumps out macrolides and

**NOT clindamycin** — so *mef* strains stay clindamycin-susceptible ★

## THE SHARED CORE · WHAT CHANGES



**Lincomycin.** an amino acid joined to a sugar — not a macrolide, but it binds the same 50S site, which is why resistance is shared



### Clindamycin

one OH swapped for Cl → better absorption and far better anaerobic activity

## MEDCHEM

**SAR** ★ **Clindamycin is lincomycin with one atom swapped:** the C7 **–OH becomes –Cl**, and the stereochemistry inverts **7R → 7S**. That is the entire difference between the parent and the drug anyone actually prescribes. The sugar half is a thiomethyl aminooctoside - an 8-carbon sugar, unusual enough to be its own identifier.

**Prodrug Clindamycin phosphate** is an ester prodrug - phosphorylated at the same C7-adjacent hydroxyl for solubility, hydrolysed back in vivo.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                                                                                                 |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Clindamycin</b> | anaerobic infections <b>outside the CNS</b> · skin & soft tissue, esp. in <b>penicillin allergy</b> · <i>C. perfringens</i> , PCP, toxoplasmosis, malaria, bacterial vaginosis |
| <b>Lincomycin</b>  | gram-positive and anaerobic infections · tonsillitis ( <i>with dexamethasone</i> — off-label)                                                                                  |



## SPECTRUM & USES

**Gram-positive aerobes** — MSSA only · *S. pneumoniae* (PSSP only, resistance rising) · streptococci · ★

**Anaerobes ABOVE the diaphragm** — *Peptostreptococcus*, *Actinomyces*, *Propionibacterium*, *Prevotella*, *Fusobacterium*, some *Bacteroides*, *Clostridium* (**not *C. difficile***) · **Other** — *Pneumocystis*, *Toxoplasma*, malaria

★ **Antitoxin effect** — added in **toxic shock syndrome** and **necrotizing fasciitis** from Group A Strep.  
*Mirror this against metronidazole: clindamycin = above the diaphragm and no CSF; metronidazole = below the diaphragm and does reach CSF.*

## ADVERSE EFFECTS

- ***C. difficile* colitis** ★ — clindamycin is **one of the worst offenders**; mild to severe diarrhea, treated with metronidazole
- **Neuromuscular blocking effect** — lincomycin
- GI 3–4% — nausea, vomiting, diarrhea, dyspepsia
- Hepatotoxicity (rare) — raised transaminases
- Allergy (rare)

## CAUTIONS

- ⚠ **Minimal CSF penetration** — never for CNS anaerobes; use metronidazole
- ⚠ Good bone penetration; not removed by haemodialysis

## ADMINISTRATION

- Available **IV and PO** — rapidly and **completely absorbed, F = 90%**, so oral is a real option, not a fallback
- **Food has minimal effect on absorption** — no timing rule
- Hepatically metabolised, **t<sub>1/2</sub> 2.5–3 h**; **not removed by haemodialysis**
- 💡 The counselling point isn't the food, it's the diarrhoea: tell them to finish the course but to come back immediately if diarrhoea turns severe or bloody. This is *the* classic *C. difficile* drug.

# Streptogramins (Quinupristin / Dalfopristin)

## MECHANISM

Two chemically unrelated halves given together, both acting on the **50S subunit**:

- **Group A — dalfopristin** (nonpeptide): distorts the ribosome, **prevents tRNA binding** (*early stage*)
  - **Group B — quinupristin** (cyclic peptide): **blocks translocation** (*late stage*)
- **Downstream**: early **and** late protein synthesis blocked → **synergistic**; bacteriostatic, cidal against some organisms → **Resistance**: altered ribosomal binding sites · enzymatic inactivation

## DRUGS

| DRUG                               | WHAT IT TREATS                                                                                                              |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Quinupristin + Dalfopristin</b> | ★ <b>VRE (<i>E. faecium</i> only) bacteremia</b> · complicated skin & soft tissue infection from MSSA or <i>S. pyogenes</i> |

## SPECTRUM

**Gram-positive** — MSSA and **MRSA**, coagulase-negative staph, *S. pneumoniae* incl. PRSP, streptococci, ⚠ ***Enterococcus faecium* ONLY** — **not \*E. faecalis** · Corynebacterium, Bacillus, Listeria, Actinomyces, Clostridium (*not C. difficile*) · *Gram-negative* — *limited* (Neisseria, Moraxella) · *Atypicals* — Mycoplasma, Legionella\*

## ADVERSE EFFECTS

- **Venous irritation** ★ — bad enough in a peripheral vein that a **central line is generally required**
- **Myalgias and arthralgias** — 2%
- Rash
- ↑ total and unconjugated bilirubin
- GI — nausea, vomiting, diarrhea

# Chloramphenicol

## MECHANISM

A small molecule that **enters both gram-positive and gram-negative bacteria easily**, attaches to the **50S ribosome** and interferes with **peptide bond formation** (peptidyl transferase).

→ **Downstream:** protein synthesis inhibited · very **broad spectrum** — g(+), g(-), atypicals, anaerobes, *Rickettsia* → **PK:** penetrates almost every tissue **including the CNS** — glucuronidation + aryl amine reduction, then urine (major) and bile (minor)

## DRUGS

| DRUG                   | WHAT IT TREATS                                                                                                                               |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chloramphenicol</b> | empiric <b>meningitis</b> treatment · bacterial conjunctivitis ( <i>topical eye</i> ) · otitis media ( <i>topical ear</i> ) · veterinary use |

## ADVERSE EFFECTS

- **Bone marrow toxicity** ★ — two distinct reactions, and the distinction is the exam point:
  - **Bone marrow suppression** — dose-related, predictable (**Type A**), reversible
  - **Aplastic anemia** — idiosyncratic, not dose-related (**Type B**), often fatal
- **Grey baby syndrome** ★ — neonates cannot glucuronidate the drug, so it accumulates
- **Hemolytic anemia in G6PD deficiency**
- Nausea, vomiting, diarrhea

## INTERACTIONS

- **Potent CYP3A4 and CYP2C19 inhibitor** — raises the levels of anything cleared by them

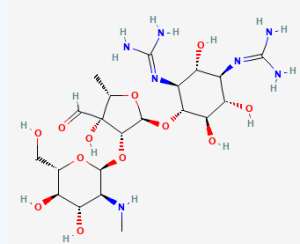
# Aminoglycosides

## MECHANISM

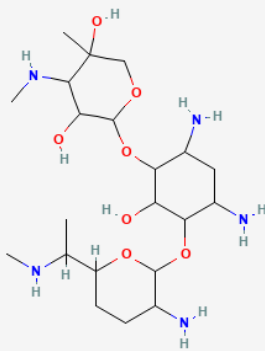
Three steps, and the second explains most of the clinical behaviour:

1. **Cross the cell wall** by passive diffusion — *penicillins/vancomycin damage the wall and speed this up: the basis of synergy*
  2. **Cross the cytoplasmic membrane** by an **oxygen-dependent active transport** (proton-motive force) — EDP-I is rate-limiting. ● **No oxygen → no uptake → anaerobes are intrinsically resistant.**
  3. **Bind irreversibly to 16S rRNA of the 30S subunit** → mistranslation, protein synthesis halted
- **Downstream: bactericidal**, concentration-dependent, with a long **PAE** · narrow spectrum — mainly **aerobic gram-negative bacilli** → **Resistance:** drug-inactivating enzymes (acetyl-, phospho-, nucleotidyl-transferases → high-level) · failure to penetrate · efflux (MexXY) · 30S/16S target mutation

## THE SHARED CORE · WHAT CHANGES

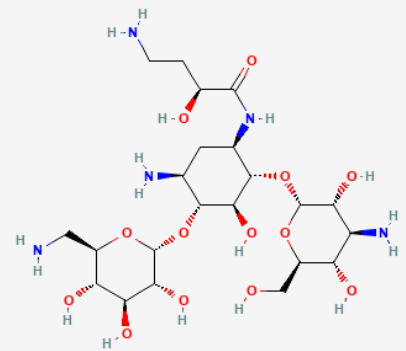


**Streptomycin — the original.** amino sugars on an aminocyclitol. Heavily basic and polycationic → ● no oral absorption, no BBB, renal clearance, and that same charge is the kidney/ear toxicity



**Gentamicin**

2-deoxystreptamine family — the standard



**Amikacin**

an acyl side chain shields it from inactivating enzymes → works on gentamicin-resistant strains

## MEDCHEM

**SAR** ★ **Why amikacin still works when gentamicin doesn't.** Amikacin carries an **HABA** side chain ( $\alpha$ -hydroxy- $\gamma$ -aminobutyryl amide) that **physically blocks the R-factor modifying enzymes from binding**. The bacterium still makes the enzyme; the enzyme just can't reach. The page gives the fact, this is the mechanism.

**SAR** The scaffold is an **aminocyclitol** joined to amino sugars by glycosidic bonds - **2-deoxystreptamine** in the kanamycin/gentamicin/neomycin families, **streptidine** in streptomycin, **spectinamine** in spectinomycin. Gentamicin is not one compound: **C1, C1a and C2** differ only in their purpurosamine sugar.

**Resistance** The three inactivating enzymes, named: **AAC** (N-acetylation), **APH** (O-phosphorylation), **ANT** (O-adenylation). All three work by decorating a functional group so the drug can no longer bind the ribosome.

**Physicochem** ● **Never mix an aminoglycoside and a  $\beta$ -lactam in the same syringe or bag.** The  $\beta$ -lactam **N-acylates C1 of the aminoglycoside** and **both drugs are destroyed**. They synergise in the patient and cancel in the bottle - give them into different compartments. Also: **streptomycin cannot be autoclaved** (its  $\alpha$ -hydroxyaldehyde is unstable) - those solutions are sterilised by ultrafiltration.

## DRUGS

| DRUG                          | WHAT IT TREATS                                                                                                            |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Gentamicin</b>             | UTI · peritonitis · <b>endocarditis</b> (+ cell-wall agent) · plague · tularemia — the default (cheap, long track record) |
| <b>Tobramycin</b>             | UTI · lung infection incl. <b>cystic fibrosis</b> (inhaled) · nosocomial sepsis — best anti- <i>Pseudomonas</i>           |
| <b>Amikacin</b>               | as tobramycin, <b>plus gram-negatives resistant to the others</b> · mycobacterial infections                              |
| <b>Streptomycin</b>           | <b>tuberculosis</b> · endocarditis (+ cell-wall agent) · plague · tularemia                                               |
| <b>Netilmicin · Kanamycin</b> | interchangeable with gentamicin/tobramycin for most gram-negative infections                                              |
| <b>Neomycin</b>               | topical only — minor skin infection · <b>bowel prep</b> before abdominal surgery · bladder irrigation                     |
| <b>Paromomycin</b>            | <b>intestinal amebiasis</b> · cryptosporidiosis · leishmaniasis                                                           |

## SPECTRUM & USES

**Aerobic gram-negative bacilli** — *E. coli*, *Klebsiella*, *Proteus*, *Acinetobacter*, *Citrobacter*, *Enterobacter*, *Morganella*, *Providencia*, *Serratia*, *Salmonella*, *Shigella*, *P. aeruginosa* (amik > tobra > gent) · **Gram-positives only in combination** — staph, viridans strep, *Enterococcus* (gent or strep) · **Mycobacteria** — TB = streptomycin; atypical = streptomycin or amikacin · ● **Anaerobes: never**

**Why they're combined with a cell-wall agent:** ① broaden empiric cover ② synergistic killing ③ prevent resistance emerging. UTI · pneumonia (+  $\beta$ -lactam) · peritonitis · endocarditis (low-dose "synergy" gentamicin) · gram-negative meningitis resistant to  $\beta$ -lactams · tularemia (DOC) · cystic fibrosis · sepsis from MDR organisms.

## ADVERSE EFFECTS

- **NEPHROTOXICITY** ★ — non-oliguric azotemia from **proximal tubule** damage, ↑BUN/Cr. **Reversible if caught early**. Risk: prolonged high troughs, therapy > 2 weeks, existing renal disease, elderly, other nephrotoxins.
- **OTOTOXICITY** ★ — **8th cranial nerve, IRREVERSIBLE**. Same risk factors.
  - **Vestibular** (dizziness, vertigo, ataxia) — streptomycin, gentamicin, tobramycin
  - **Auditory** (tinnitus, hearing loss) — amikacin, netilmicin, gentamicin
- **Neuromuscular junction blockade** — rare but serious: ↓ACh release + Ca<sup>2+</sup> channel blockade → **curare-like respiratory paralysis**. Worst with **neomycin and streptomycin**. *Treat with IV calcium or neostigmine.*
- Hypersensitivity (rare) — rash, fever, eosinophilia, anaphylaxis

## INTERACTIONS

- **Other nephrotoxic / ototoxic drugs** → additive toxicity: **loop diuretics**, vancomycin, cisplatin, cyclosporine, ACEIs, NSAIDs, minocycline
- **Neuromuscular blockers** → respiratory paralysis
- ⚠ **Never mix with a penicillin in the same syringe or line** — chemical antagonism, they crystallise

## CAUTIONS & CONTRAINDICATIONS

- **Pregnancy — fetal ototoxicity (category D)**
- ⚠ Renal disease —  $t_{1/2}$  goes from 2–3 h to > **100 h**; dose must be adjusted
- ⚠ Elderly · concurrent skeletal muscle relaxants
- ⚠ Not absorbed orally (highly polar) — parenteral or topical only; poor CSF penetration

# Aminoglycosides — Dosing & Monitoring

## WHY ONCE-DAILY DOSING WINS

Four properties stack in its favour:

1. Significant **post-antibiotic effect** — killing continues after levels fall below MIC
2. Killing is **concentration-dependent** — one big peak kills more than three small ones (★ **peak:MIC**  $\approx$  8–10:1 is the target)
3. Uptake into **renal tubule cells and the inner ear saturates at low serum levels** — so a high peak adds efficacy without adding toxicity
4. Less adaptive post-exposure resistance

→ **Downstream: high peak** → ↑ **efficacy** · **low trough** → ↓ **nephrotoxicity**. Once-daily is equally effective, cheaper, and *less* nephrotoxic than divided dosing — with a once-daily regimen plasma levels sit below the toxic threshold for >13 h, versus three short windows on an 8-hourly regimen.

## THE TWO REGIMENS

|                   | HIGH-DOSE, EXTENDED-INTERVAL ("ONCE-DAILY")                                 | MULTIPLE-DAILY ("TRADITIONAL")                                   |
|-------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>when</b>       | default                                                                     | when once-daily isn't appropriate                                |
| <b>dosing</b>     | one large dose per interval                                                 | loading dose → maintenance by CrCl                               |
| <b>monitoring</b> | trough-focused                                                              | <b>both peak and trough</b> — peak = efficacy, trough = toxicity |
| <b>avoid in</b>   | CrCl < 25 mL/min · pregnancy, neonates, paediatrics ( <i>data limited</i> ) | —                                                                |

## MONITORING RULES

- ⚠ Most accurate dose adjustment: **two plasma samples several hours apart** (2 h and 12 h post-dose)
- ⚠ Excretion is renal and unchanged — clearance tracks **creatinine clearance** directly
- ⚠ Monitor duration and levels deliberately; these are reserve drugs, used sparingly

## SPECTINOMYCIN — THE AMINOGLYCOSIDE-LOOKALIKE

An **aminocyclitol**: same 16S rRNA / 30S target but a **different binding site**, and it lacks the glycosidic bond and amino sugar. Parenteral. Used for **one thing only** — **gonorrhea** in patients allergic to penicillin or resistant to other drugs.

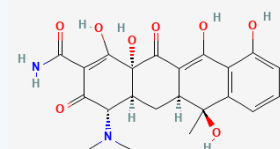
# Tetracyclines

## MECHANISM

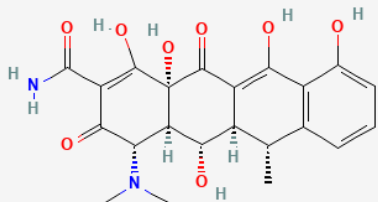
**Reversibly bind the 30S subunit** and block **aminoacyl-tRNA from docking at the A site** of the mRNA-ribosome complex, so the codon-anticodon interaction can't proceed.

→ **Downstream:** protein synthesis stalled → **bacteriostatic** · very broad spectrum → **Resistance:** ↓ accumulation (permeability or **efflux**) · **ribosomal protective proteins** · enzymatic inactivation. Cross-resistance is the rule across the class — ★ **except minocycline**.

## THE SHARED CORE · WHAT CHANGES

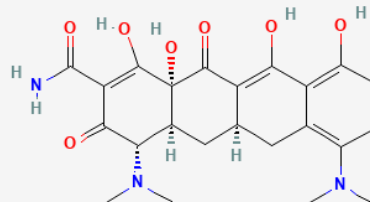


**Tetracycline — the naphthacene core.** four fused rings; the oxygen-rich lower edge is the chelation face that milk, antacids and iron grab



### Doxycycline

more lipophilic → better absorbed, and ● biliary  
not renal → safe in renal failure



### Minocycline

dimethylamino at C7 → best CNS entry, and the vestibular ADR

## MEDCHEM

**SAR ★ The reason doxycycline and minocycline displaced the first generation is one missing hydroxyl.** Tetracycline has a **C6-OH** that **dehydrates in acid** to anhydrotetracycline - degraded, and the degradation product is what caused the classic Fanconi-syndrome poisonings from expired stock. Doxycycline and minocycline have **no C6-OH**, so they cannot take that route: more stable, better absorbed.

**Physicochem** Amphoteric with **three pKa values** (acidic, neutral, basic), and the C11/C12 keto-enol pair **tautomerises**. The molecule is also **non-planar** - rings A and B sit at an angle. That oxygen-rich, charged BCD edge is exactly the metal-binding face.

**Stability** The other degradation route is **C4 epimerisation** - the dimethylamino group flips to 4-epitetracycline, which is inactive. Neither route needs a bacterium; the drug degrades on its own.

**SAR Demeclocycline's phototoxicity comes from its C7 chlorine**, not from the tetracycline core - which is why it is the one singled out for it.



## DRUGS

| DRUG                                                 | WHAT IT TREATS                                                                                    |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Doxycycline</b> (2nd gen)                         | <b>CAP</b> · rickettsial infections · chlamydia · acne · Lyme disease · malaria-adjacent zoonoses |
| <b>Minocycline</b> (2nd gen)                         | ★ preferred for <b>methicillin-resistant staph</b> · occasionally <b>leprosy</b>                  |
| <b>Tetracycline</b> (1st gen)                        | <i>H. pylori</i> (when clarithro/amox/metronidazole can't be used) · general broad-spectrum use   |
| <b>Demeclocycline</b>                                | ★ <b>SIADH</b> — exploits its own side effect (nephrogenic diabetes insipidus)                    |
| <b>Oxytetracycline · Chlortetracycline</b> (1st gen) | eye and skin infections                                                                           |

## SPECTRUM & USES

**Gram-positive** — MSSA, PSSP *S. pneumoniae* (doxy 77–88% susceptible), some streptococci, *Bacillus*, *Listeria*, *Nocardia* · **Gram-negative** — *H. influenzae* (90%), *H. ducreyi* (chancroid), *Campylobacter jejuni*, *H. pylori* · **Anaerobes** — *Actinomyces*, *Propionibacterium* · ★ **The miscellaneous list is where they shine:** *Rickettsia*, *Coxiella*, *Chlamydia*, *Mycoplasma*, *Ureaplasma*, *Legionella*, *Borrelia*, *Treponema*, *Leptospira*, *Bartonella*, *Bordetella*, *Brucella*, *Pasteurella*, *M. fortuitum*

Clinically: CAP (doxy) · **Rocky Mountain spotted fever, Q fever** · chlamydia · **acne** · brucellosis, plague, tularemia, chancroid, pertussis, anthrax, Lyme disease

## ADVERSE EFFECTS

- **Teeth staining and impaired bone development** ★ — chelates calcium in growing tissue. *Contraindicated in children and pregnancy.*
- **Photosensitivity** ★ — exaggerated sunburn, worst with **demeclocycline**
- **Reversible, dose-related nephrogenic diabetes insipidus** — demeclocycline (*the effect that became an indication*)
- **Fanconi-like syndrome** — from **outdated/expired** tetracycline
- Hepatotoxicity — raised transaminases
- GI — nausea, vomiting, diarrhea, pseudomembranous colitis
- Hypersensitivity — rash, pruritus, urticaria, angioedema, anaphylaxis

## INTERACTIONS

- **Di- and trivalent cations** ★ — same trap as the fluoroquinolones: antacids, iron, calcium, dairy chelate the drug → **therapeutic failure**

## PK WORTH REMEMBERING

- ⚠ **Demeclocycline and tetracycline** are excreted **unchanged in urine** → **adjust in renal impairment**
- ✅ **Doxycycline and minocycline** are cleared **non-renally** → **no adjustment needed** — this is why doxycycline is the workhorse
- ⚠ Minimal CSF penetration; minimally removed by haemodialysis

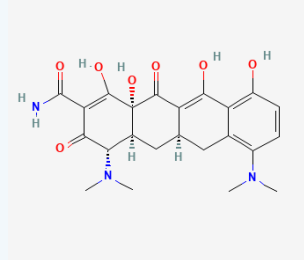
# Glycylcyclines (Tigecycline)

## MECHANISM

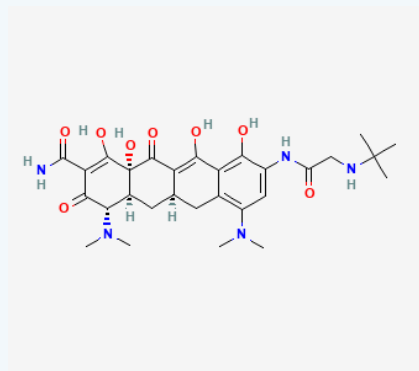
Same 30S A-site block as the tetracyclines, but structurally modified so that **bacterial efflux pumps don't recognise it** and **30S ribosomal modifications don't disable it**.

→ **Downstream:** the two main tetracycline resistance mechanisms are bypassed → activity against MRSA, VSE and resistant gram-negatives · still **bacteriostatic**

## THE SHARED CORE · WHAT CHANGES



**Minocycline — the starting point.** tigecycline is minocycline with one large addition



## Tigecycline

a bulky glycyldamido at C9 → efflux pumps and ribosomal protection proteins can no longer recognise it

## MEDCHEM

**SAR** 'Structurally modified' means specifically: **built from minocycline** - the one tetracycline the class's cross-resistance already spared - with a **glycyldamido** group added at C9. Second generation of the tetracyclines, not a new scaffold.

**Physicochem** Formulated as a **double HCl salt** and given **parenterally only**.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                                                                                                         |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tigecycline</b> | complicated <b>intra-abdominal infection</b> (ESBL) · <i>Acinetobacter baumannii</i> and carbapenemase producers · skin/soft tissue infection ( <b>MRSA</b> ) · occasionally pneumonia |

## SPECTRUM

**Gram-positive** — *S. aureus* (**MSSA and MRSA**), streptococci, *E. faecalis* (VSE), *Listeria* · **Gram-negative** — *Acinetobacter baumannii*, *Aeromonas*, *Citrobacter*, *E. coli*, *Klebsiella*, *Serratia*, *Stenotrophomonas* · **NOT Proteus, NOT Pseudomonas aeruginosa** ★ · **Anaerobes** — *Clostridium perfringens*, *Bacteroides*

## ADVERSE EFFECTS

- GI — nausea, vomiting, diarrhea (*the main complaint*)
- Elevated LFTs

### CAUTIONS

-  **Poor choice for bacteremia**  — it distributes so widely into tissue that **serum concentrations stay low**, and the mechanism is only static
-  **High failure rate in HAP/VAP** — use with caution

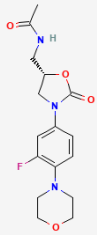
# Oxazolidinones (Linezolid)

## MECHANISM

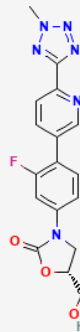
Binds the **50S subunit near its interface with the 30S**, preventing formation of the **70S initiation complex** — it stops protein synthesis before it starts, rather than interrupting elongation.

→ **Downstream:** bacteriostatic (cidal against some), time-dependent · resistance is **rare** (ribosomal methylation) and ★ **cross-resistance with other protein-synthesis inhibitors is unlikely** — a different enough binding site

## THE SHARED CORE · WHAT CHANGES



**Linezolid — the oxazolidinone ring.** a fully synthetic scaffold with no natural relative, which is why there was no pre-existing resistance



**Tedizolid**

added ring system → more potent, once daily

## DRUGS

| DRUG             | WHAT IT TREATS                                                                                                                                                                              |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Linezolid</b> | ★ <b>VRE bacteremia</b> · complicated skin & soft tissue infection (MSSA, <i>S. pyogenes</i> , and may beat vancomycin for MRSA) · CAP from PSSP/MSSA · nosocomial pneumonia from MSSA/MRSA |

## SPECTRUM & PK

**Gram-positive only** — MSSA, **MRSA and VRSA**, coagulase-negative staph, *S. pneumoniae* incl. PRSP, streptococci, ★ *E. faecium* **AND** *E. faecalis*, including **VRE**, *Bacillus*, *Listeria*, *Clostridium* (not *C. difficile*), *P. acnes* · ⚠ **Relatively inactive against gram-negatives**

**100% oral bioavailability** · CSF penetration 30% ·  $t_{1/2}$  4.4–5.4 h · no renal adjustment · not removed by haemodialysis. Reserved for serious infection caused by resistant gram-positives.

## ADVERSE EFFECTS

- **Mitochondrial toxicity on long-term use** ★ — lactic acidosis, peripheral neuropathy, **optic neuritis and blindness**. *Well tolerated short-term; not long-term.*
- **Serotonin syndrome** ★ — linezolid **inhibits monoamine oxidase A**
- **Bone marrow suppression — thrombocytopenia 2–4%**, mostly after >2 weeks. Stop the drug and platelets recover.
- GI 6–8% — nausea, vomiting, diarrhea · headache 6.5%

### INTERACTIONS

-  **Serotonergic drugs (SSRIs etc.)** → serotonin syndrome, via MAO-A inhibition

### CAUTIONS

-  The mechanism is **static** — prefer a cidal agent for severe bacteremia and endocarditis

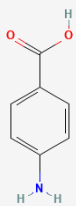
# Folate Synthesis Inhibitors (TMP-SMX)

## MECHANISM

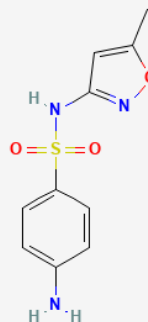
**Sequential blockade** of the folate pathway that bacteria must run themselves — humans just eat folate, which is where the selectivity comes from:

- **Sulfamethoxazole** inhibits **dihydropteroate synthase** — blocks PABA being built into dihydrofolate
  - **Trimethoprim** inhibits **dihydrofolate reductase** — blocks dihydrofolate → tetrahydrofolate
- **Downstream:** no folinic acid → no purines → no DNA. ★ **Each agent alone is bacteriostatic; the combination is bactericidal.** → **Resistance:** develops far more slowly against the pair than against either alone — point mutations in DHPS and/or altered DHFR (*E. coli*, *Klebsiella*, *Proteus*, *H. influenzae*)

## THE SHARED CORE · WHAT CHANGES

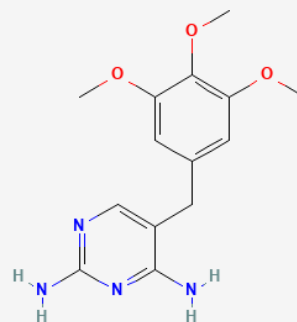


**PABA — the real substrate.** ● the drug below is a fake of this molecule. Put them side by side and the mechanism needs no sentence.



**Sulfamethoxazole**

same shape, same size, wrong molecule → dihydropteroate synthase is fooled



**Trimethoprim**

a different shape entirely — it blocks the NEXT enzyme (DHFR), which is why the two are combined

## MEDCHEM

**SAR** Two rules govern every sulfonamide. **N4 must stay an unsubstituted aromatic amine** - that is the part that impersonates PABA. And the **N1 substituent must be an electron-withdrawing aromatic ring**: it raises the acidity of the sulfonamide N-H (**pKa 4.9**, from 10.4 unsubstituted), which raises both potency and water solubility.

**Physicochem** ● **Crystalluria was a solubility problem, not a toxicity problem.** The early sulfonamides stayed **un-ionised at urine pH**, crystallised in the tubule and cut the kidney. The fix follows straight from the pKa: **alkalinise the urine and push fluids** - ionised drug stays dissolved.

**Metabolism** Cleared by **N4-acetylation** - and acetylating N4 is also what abolishes the activity, since that is the PABA-mimicking nitrogen.

## DRUGS

| DRUG                                                    | WHAT IT TREATS                                                                                                                                                                                                                  |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trimethoprim + Sulfamethoxazole (co-trimoxazole)</b> | ★ <b>DOC for <i>Pneumocystis pneumonia</i> — treatment and prophylaxis</b> · UTI · bacterial prostatitis · ABECB · <i>Salmonella/Shigella</i> /traveller's diarrhea · <i>Nocardia</i> · <i>Stenotrophomonas</i> · toxoplasmosis |

## SPECTRUM & THE RATIO

**Gram-positive** — some *S. pneumoniae*, *S. aureus*, *S. pyogenes*, *Nocardia* · **Gram-negative** — *Acinetobacter*, *Enterobacter*, *E. coli*, *K. pneumoniae*, *Proteus*, *Salmonella*, *Shigella*, *Haemophilus*, *N. gonorrhoeae*, ★  
**Stenotrophomonas maltophilia** · **Other** — *Pneumocystis* · ● **Anaerobes: no activity**

★ **The 1:5 trick** — the optimal synergistic ratio *in serum and tissue* is **1:20 (TMP:SMX)**, achieved by dosing the fixed formulation at **1:5**, because the two drugs distribute differently. Distributes into urine, prostate, **CSF/CNS**; adjust when CrCl < 30.

## ADVERSE EFFECTS

- **Hypersensitivity** ★ — rash, urticaria, and the severe end: **Stevens–Johnson syndrome and toxic epidermal necrolysis**. This is the classic sulfonamide risk.
- **Haematologic** — leukopenia, thrombocytopenia, eosinophilia, **aplastic anemia**, and **hemolytic anemia in G6PD deficiency**
- **Renal** — ↑ creatinine from **two different causes**: a *pseudo-rise* (~20%) because TMP blocks tubular creatinine secretion, and genuine AKI from interstitial nephritis/ATN. **Hyperkalemia**, especially at high dose or in CKD.
- CNS — headache, aseptic meningitis, seizures
- **Kernicterus in the newborn** — sulfonamide displaces bilirubin
- Crystalluria, hematuria
- GI — nausea, vomiting, diarrhea, glossitis

## CAUTIONS

- ⚠ **G6PD deficiency** · **neonates** · sulfonamide allergy
- ⚠ Renal impairment — dose adjust below CrCl 30 mL/min

*Principle behind this: [[pharm-pharmacokinetics]] — tissue accumulation in all three directions — macrolides/FQs in airway tissue (useful), aminoglycosides in kidney and ear (toxic), tetracyclines chelated into bone and teeth (staining).*

*Principle behind this: [[pharm-drug-interactions]] — Tetracycline chelation with dairy and antacids; fluoroquinolone + macrolide → QT prolongation → torsades.*

*Principle behind this: [[pharm-drug-toxicity]] — Sulfonamide granulocytopenia (Type II), sulfamethoxazole and ciprofloxacin (Type IV delayed), and tetracycline phototoxicity at therapeutic dose.\*\**

*Source: 5. Fluoroquinolone and tetracycline and DNA synthesis.pdf — อ.ภาณุรักษ์ ม้ายอุเทศ, PYPH305 Pharmacology 2, 98 slides. Structures omitted.*

# Antimycobacterial Drugs

Mycobacteria are aerobic, non-motile rod-shaped bacilli wrapped in a **thick, hydrophobic, mycolic-acid-rich cell wall**. Three infections matter: **tuberculosis · leprosy · nontuberculous mycobacteria (NTM)**. Everything about the treatment — multiple drugs, for months — falls out of that cell wall.

**Waxy wall (>60% lipid) + efflux pumps (ABC transporters) + intra- and extracellular residence + slow doubling (18–24 h) → poor drug penetration & intrinsic resistance → combination therapy, prolonged course**

## WORDS WORTH SEPARATING

- **Active vs latent TB** — latent is asymptomatic and non-infectious; **5–10% will reactivate**, mostly in the immunocompromised
- **Intensive phase** — kills 99% of the population, drops bacterial load and symptoms
- **Continuation phase** — *sterilizes* the survivors, prevents relapse
- **DOT** (Directly Observed Therapy) — adherence is the whole game; cure rate >85% for drug-susceptible TB when it holds

## FIRST-LINE ANTI-TB DRUGS AT A GLANCE ★

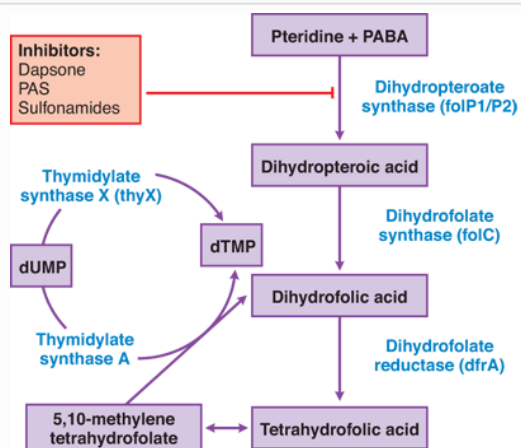
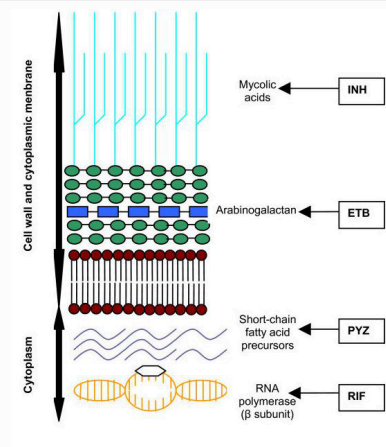
| DRUG                      | MECHANISM                                                   | SIGNATURE TOXICITY                                           |
|---------------------------|-------------------------------------------------------------|--------------------------------------------------------------|
| <b>Isoniazid (H, INH)</b> | inhibits <b>mycolic acid</b> synthesis (InhA / KasA)        | <b>peripheral neuropathy</b> + hepatitis                     |
| <b>Rifampin (R)</b>       | binds β-subunit of <b>DNA-dependent RNA polymerase</b>      | <b>orange-red secretions</b> · flu-like syndrome · hepatitis |
| <b>Pyrazinamide (Z)</b>   | prodrug → pyrazinoic acid; kills <b>only at acidic pH</b>   | <b>hyperuricemia</b> + arthralgia · hepatotoxicity           |
| <b>Ethambutol (E)</b>     | inhibits <b>arabinosyl transferase</b> → no arabinogalactan | <b>optic neuritis</b> — red-green colour blindness           |

## RESISTANCE GENE & CYP FOOTPRINT

| DRUG                | RESISTANCE MECHANISM                                                                 | CYP ACTIVITY                                                               |
|---------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Isoniazid</b>    | ↓ prodrug activation — <b>katG</b> (most common & severe), <i>inhA</i> , <i>kasA</i> | <b>potent inhibitor</b> CYP2C19, 3A4 · CYP2E1 inducer · weak 2D6 inhibitor |
| <b>Rifamycins</b>   | target modification — <b>rpoB</b> (86% at codons 526 & 531)                          | <b>potent inducer</b> CYP1A2, 2B6, 2C8, 2C9, 2C19, 3A4, P-gp               |
| <b>Pyrazinamide</b> | ↓ prodrug activation — <b>pncA</b> · ↑ efflux                                        | not significant                                                            |
| <b>Ethambutol</b>   | target modification — <b>embB</b> · ↑ efflux                                         | not significant                                                            |



The diagram illustrates the structure of a bacterial cell wall and the various targets of antibiotics. The cell wall is shown as a multi-layered structure with the periplasm on the outside and the cytosol on the inside. The outermost layer is the peptidoglycan layer, which is composed of alternating N-acetylmuramic acid (NAM) and N-acetylglucosamine (NAG) units. The inner layer is the cytoplasmic membrane, which is a phospholipid bilayer. Various antibiotics are shown targeting different components of the cell wall and membrane. For example, Cycloserine targets peptidoglycan synthesis, while Vancomycin targets the D-Ala-D-Ala dipeptide. Other antibiotics like Polymyxins and Daptomycin target the cytoplasmic membrane. The diagram also shows the synthesis of ATP, the production of cell wall biosynthesis intermediates like MmpL3, and the synthesis of various antibiotics like Rifampin, Rifampicin, and Linezolid. The diagram is a detailed representation of the bacterial cell wall and its various targets.



# TB Regimens & When First-Line Fails

## REGIMENS

| SETTING                               | REGIMEN                                                                                     | LENGTH             |
|---------------------------------------|---------------------------------------------------------------------------------------------|--------------------|
| <b>Active, drug-susceptible</b>       | <b>2HRZE / 4HR</b> — intensive 2 mo, continuation 4 mo                                      | 6 months           |
| <b>Active, short course</b>           | <b>2HPMZ / 2HPM</b> ( <i>P</i> = rifapentine, <i>M</i> = moxifloxacin)                      | 4 months           |
| <b>Latent</b>                         | <b>3HP</b> (INH + rifapentine, weekly ×12) · <b>4R</b> · <b>3HR</b> · <b>6H</b> · <b>9H</b> | 3–9 months         |
| <b>XDR / treatment-intolerant MDR</b> | <b>BPaLM</b> — bedaquiline + pretomanid + linezolid + moxifloxacin                          | —                  |
| <b>Leprosy MDT</b>                    | rifampin + clofazimine + dapsone                                                            | PB 6 mo · MB 12 mo |
| <b>MAC triple therapy</b>             | rifamycin + ethambutol + macrolide                                                          | —                  |

## WHICH FIRST-LINE DRUG CAUSED THIS ADR? ★

| SIDE EFFECT           | DRUGS             |
|-----------------------|-------------------|
| Rash                  | <b>H, R, Z, E</b> |
| Hepatitis             | <b>H, R, Z</b>    |
| Optic neuritis        | <b>E, H</b>       |
| Peripheral neuropathy | <b>H, R</b>       |
| Thrombocytopenia      | <b>R</b>          |
| Flu-like symptoms     | <b>R</b>          |

## RESISTANCE DEFINITIONS

- **RR-TB** — rifampicin-resistant
- **MDR-TB** — resistant to **isoniazid and rifampin** ★
- **XDR-TB** — MDR **plus** fluoroquinolone **plus** ≥1 second-line injectable

## SECOND-LINE THERAPY IS REQUIRED WHEN

**Drug-resistant TB** (RR / MDR / XDR) · **treatment intolerance** — severe hepatotoxicity or hypersensitivity to first-line · **relapse or treatment failure** despite adherence · **non-compliance** · **drug interactions** with isoniazid / rifampin · **special populations** — e.g. HIV co-infection

# Isoniazid (INH)

## MECHANISM

**Prodrug** activated by mycobacterial catalase-peroxidase (**KatG**) → inhibits the carrier-protein reductase enzymes **InhA** (enoyl-ACP reductase) and **KasA** ( $\beta$ -ketoacyl-ACP synthase) of the fatty-acid elongation cycle → no **mycolic acid** → cell wall disruption → death.

→ **Downstream: bactericidal** against actively dividing bacilli · **bacteriostatic** against dormant ones — which is why it carries the intensive phase

## DRUGS

| DRUG                      | WHAT IT TREATS                  |
|---------------------------|---------------------------------|
| <b>Isoniazid (H, INH)</b> | active TB · latent TB infection |

## ADVERSE EFFECTS

- **Peripheral neuropathy** ★ (*most common*) — INH depletes **pyridoxine (vit B6)** → tingling, numbness, burning in hands & feet. **Prevention: pyridoxine 10–50 mg/day, routine practice.** Risk ↑ in malnutrition, DM, alcoholism, pregnancy, slow acetylators
- ● **Hepatotoxicity** — asymptomatic ↑AST/↑ALT is common; **hepatitis rare but serious.** Risk: age >35 · slow acetylator · co-administered rifampin or alcohol · existing liver disease
- **Seizures** — ↓ pyridoxine → ↓ GABA → loss of CNS inhibition → generalized tonic-clonic (overdose). **Antidote: IV pyridoxine, 1 g per gram of INH ingested** ★
- Psychiatric — irritability, confusion, depression, restlessness (*overdose*)
- GI upset · hypersensitivity — rash, fever, **drug-induced lupus-like syndrome** · rarely agranulocytosis, thrombocytopenia, anemia

## INTERACTIONS

- **Potent CYP2C19 & CYP3A4 inhibitor** · CYP2E1 inducer · weak CYP2D6 inhibitor
- ● **Rifampin or alcohol** → ↑ hepatotoxicity

## CAUTIONS

- ★ **NAT2 polymorphism** — hepatic N-acetylation. *Slow acetylators* (**30% of Thais**, Scandinavians, North African whites) have  $t_{1/2}$  2–5 h vs 1–1.5 h → higher serum INH → **more hepatotoxicity and neuropathy**
- ⚠ **Never as monotherapy** — *katG* resistance emerges rapidly, and usually travels with ethambutol resistance
- Food reduces bioavailability (*otherwise ~90–100%, distributes everywhere including CSF*)

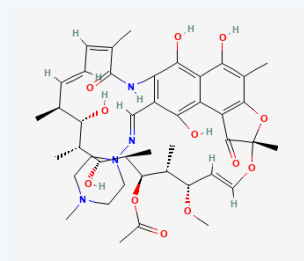
# Rifamycins

## MECHANISM

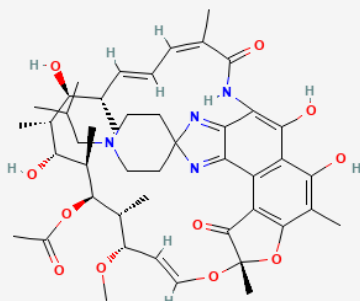
Enters the bacillus concentration-dependently → binds the **β-subunit of DNA-dependent RNA polymerase** → forms a stable drug-enzyme complex → suppresses chain formation in RNA synthesis.

→ **Downstream:** bactericidal, and broad — mycobacteria, *C. difficile*, *S. aureus*, *N. meningitidis*, *N. gonorrhoeae*, *H. influenzae*

## THE SHARED CORE · WHAT CHANGES

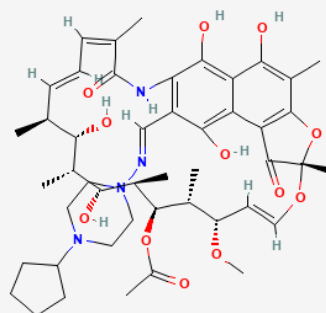


**Rifampicin — the ansamycin core.** a naphthalene bridged by a long aliphatic 'ansa' chain, wrapping RNA polymerase. ● the extended conjugation is also why it is bright red — the orange secretions are the molecule itself



**Rifabutin**

a weaker CYP3A4 inducer → the one used with HIV protease inhibitors



**Rifapentine**

a cyclopentyl group → longer  $t_{1/2}$  → weekly dosing

## MEDCHEM

**SAR** 'Ansamycin' is a shape, not a chemical group: an **aliphatic chain bridging two non-adjacent positions of an aromatic core** - a handle over a ring. Rifampicin and rifapentine are the same 3-formylrifamycin SV converted to a **hydrazone**; the only difference between them is what sits on that hydrazone nitrogen (methyl vs cyclopentane), and that alone sets rifapentine's much longer half-life.

## DRUGS

| DRUG                            | WHAT IT TREATS                                                                                                              |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Rifampin (rifampicin, R)</b> | TB (all phases) · leprosy MDT · MAC · <i>M. kansasii</i> & other NTM                                                        |
| <b>Rifapentine (P)</b>          | long $t_{1/2}$ → <b>weekly latent-TB therapy</b> with INH (3HP) · the 4-month active regimen                                |
| <b>Rifabutin</b>                | equally effective as rifampin — <b>preferred in AIDS patients</b> , fewer CYP interactions (protease inhibitors, efavirenz) |

### ADVERSE EFFECTS

- ★ **Orange-red discoloration** of urine, feces, saliva, sweat, tears — and **teeth (may be permanent)**. Free drug and metabolites are orange. Harmless, but warn the patient
- ● **"Red-man syndrome"** — sign of **overdose**
- **Flu-like syndrome** — high incidence at doses >600 mg once or twice weekly in adults
- **Hepatotoxicity** — ↑ liver enzymes; risk climbs when combined with isoniazid / pyrazinamide
- Hypersensitivity — fever, rash, urticaria, angioedema, hypotension, bronchospasm, conjunctivitis
- Drug-induced **thrombocytopenia** (*rare*) — bruising, bleeding · GI upset

### INTERACTIONS

- ● **The strongest interaction problem in the whole topic.** Strong inducer of **CYP3A4** and **CYP2C19**; moderate **CYP2C9**, **CYP2B6**; also induces **P-glycoprotein / ABCB1**
- → clinically meaningful loss of effect for any 3A4 substrate; use **rifabutin** instead when that matters

### CAUTIONS

- ⚠ **Always in combination** — *rpoB* mutation arises on monotherapy, and rifampin resistance is what defines MDR-TB
- Food ↓ absorption by ~30%; highly lipophilic, crosses the BBB; eliminated mostly in feces (60–65%)

# Pyrazinamide

## MECHANISM

**Prodrug** converted by mycobacterial **pyrazinamidase** (*pncA*) to **pyrazinoic acid**, which inhibits **fatty acid synthase type I**, damages membrane potential and interferes with ATP production.

→ **Only active at acidic pH** — inactive at neutral pH. That's the point: it kills the bacilli sitting in acidic intracellular / inflammatory environments the other drugs reach badly

## DRUGS

| DRUG                    | WHAT IT TREATS                                               |
|-------------------------|--------------------------------------------------------------|
| <b>Pyrazinamide (Z)</b> | active TB — intensive phase of 2HRZE and the 4-month regimen |

## ADVERSE EFFECTS

- **Hyperuricemia** ★ — pyrazinoic acid **competes with uric acid for tubular secretion** → inhibits uric acid excretion
- **Arthralgia / joint pain** (*the clinical face of the hyperuricemia*)
- **Hepatotoxicity**
- Anorexia, nausea, vomiting, dysuria, fever · rash

## CAUTIONS

- ⚠ **Gout** — the hyperuricemia is predictable, not idiosyncratic
- ⚠ Stacks hepatotoxicity with isoniazid and rifampin — the three of them share the liver

## ADMINISTRATION — PYRAZINAMIDE PK

- **Oral bioavailability > 90%** · distributes **widely into tissues and fluids including liver, lung and CSF**.
- **Metabolised hepatically by hydrolysis; eliminated in urine**, half-life **9–10 hours**.
- 💡 CSF penetration is the reason pyrazinamide stays in the regimen for TB meningitis — not every first-line drug gets there.

# Ethambutol

## MECHANISM

Inhibits **arabinosyl transferase** (encoded by the *emb* gene) → arabinogalactan cannot polymerize → impaired mycobacterial cell wall synthesis.

→ **Downstream:** included in the intensive phase largely to **cover undetected INH resistance** until susceptibility is known

## DRUGS

| DRUG                  | WHAT IT TREATS                                                                                 |
|-----------------------|------------------------------------------------------------------------------------------------|
| <b>Ethambutol (E)</b> | active TB · <i>M. kansasii</i> , <i>M. marinum</i> , <b>MAC</b> (with a rifamycin + macrolide) |

## ADVERSE EFFECTS

- **OPTIC NEURITIS** ★ — ↓ visual acuity, **red-green colour blindness**, possible retinal damage. **Dose- and duration-dependent** (prolonged high-dose use). **Usually recovers on withdrawal.** *This is the one fact ethambutol exists for on an exam.*
- Hepatotoxicity
- Headache, confusion, joint pain, hyperuricemia, peripheral neuritis

## CAUTIONS

- ⚠ **Test visual acuity before starting**, and monitor — the patient may not notice early colour loss

# Aminoglycosides (*second-line*)

## MECHANISM

Bind the **30S ribosomal subunit** → inhibit bacterial protein synthesis.

→ **Downstream:** injectable only (poor oral absorption) — **IM: streptomycin, capreomycin** · **IV: amikacin, kanamycin**. No hepatic metabolism; eliminated unchanged by glomerular filtration

## DRUGS

| DRUG                | WHAT IT TREATS                                                |
|---------------------|---------------------------------------------------------------|
| <b>Streptomycin</b> | MDR-TB                                                        |
| <b>Amikacin</b>     | MDR-TB · <i>M. fortuitum</i> complex (with doxycycline) · NTM |
| <b>Kanamycin</b>    | MDR-TB                                                        |
| <b>Capreomycin</b>  | MDR-TB                                                        |

## ADVERSE EFFECTS

-  **Nephrotoxicity**
-  **Ototoxicity**  — resistance to a second-line injectable is part of the **XDR-TB** definition, so losing these drugs matters

## CAUTIONS

-  Renal elimination unchanged — **dose by renal function**; risk compounds with other nephro/ototoxic drugs
- Resistance: *rrs* (16S rRNA) mutation · efflux pump expression



# Fluoroquinolones (*second-line*)

## MECHANISM

Disrupt the **DNA–DNA gyrase complex** → double-strand DNA cleavage → block DNA synthesis.

→ **Downstream**: oral bioavailability 80–90%. **Ciprofloxacin, levofloxacin, ofloxacin** are mainly renally excreted; **moxifloxacin** is hepatically metabolized → **no renal dose adjustment**

## DRUGS

| DRUG                | WHAT IT TREATS                                         |
|---------------------|--------------------------------------------------------|
| <b>Moxifloxacin</b> | MDR-TB · the 4-month regimen · <b>BPaLM</b> for XDR-TB |
| <b>Levofloxacin</b> | MDR-TB                                                 |
| <b>Gatifloxacin</b> | MDR-TB                                                 |
| <b>Ofloxacin</b>    | MDR-TB                                                 |

## ADVERSE EFFECTS

- **Tendonitis** ★
- **QT prolongation**
- **Peripheral neuropathy** · CNS effects — dizziness, confusion, **seizures**
- Dysglycemia · photosensitivity · GI upset

## INTERACTIONS

- ★ **Chelation** — absorption falls sharply with multivalent cations: antacids (Al, Mg), calcium, iron, zinc supplements. **Separate the doses.**

## CAUTIONS

- ⚠ Resistance = target modification in ***gyrA* / \**gyrB*** — *and FQ resistance is a defining component of XDR-TB\**

# Oral Bacteriostatic Second-Line Agents

## MECHANISM

Three separate targets, all **bacteriostatic**, all reserved for MDR-TB regimens:

1. **Ethionamide** — inhibits **mycolic acid** biosynthesis (*same territory as INH*)
  2. **Cycloserine** — inhibits **D-alanine racemase (Alr)** and **D-alanyl-D-alanine synthetase (Ddl)** → blocks peptidoglycan synthesis
  3. **PAS** — inhibits **folate synthesis** plus synthesis of a cell wall component
- **Downstream:** used as regimen components, never alone

## DRUGS

| DRUG                                  | WHAT IT TREATS                                   |
|---------------------------------------|--------------------------------------------------|
| <b>Ethionamide</b>                    | MDR-TB · alternative agent in <i>M. kansasii</i> |
| <b>Cycloserine</b>                    | MDR-TB · alternative agent in <i>M. kansasii</i> |
| <b>Para-aminosalicylic acid (PAS)</b> | MDR-TB                                           |

## ADVERSE EFFECTS

- **Ethionamide** — GI upset (anorexia, N/V, gastric irritation) · **postural hypotension** · mental distress · **hepatitis (5%)**
- **Cycloserine** — ● **neurotoxicity is common** ★: anxiety, depression, confusion, irritability; severe cases **psychosis, seizures, suicidal ideation**. Also headache, dizziness, insomnia; rarely peripheral neuropathy, anemia/leukopenia
- **PAS** — GI upset (common) · hypersensitivity, skin eruption, fever, eosinophilia, haematological abnormality · **N-acetyl-PAS is a hepatotoxin**

## CAUTIONS

- ⚠ Screen for psychiatric history before **cycloserine** — the CNS toxicity is the usual reason it gets stopped
- PAS resistance: mutation in thymidylate synthase (**thyA**)

# Bedaquiline

## MECHANISM

Inhibits **mycobacterial ATP synthase** — the energy plant. **Unique among TB drugs.**

→ **Downstream:** because it attacks energy production rather than growth machinery, it works against **both active and dormant** bacilli ★

## DRUGS

| DRUG               | WHAT IT TREATS                                                          |
|--------------------|-------------------------------------------------------------------------|
| <b>Bedaquiline</b> | <b>XDR-TB</b> and treatment-intolerant MDR-TB — the "B" of <b>BPaLM</b> |

## ADVERSE EFFECTS

- **QT prolongation** ★ — risk of cardiac arrhythmia
- Nausea, headache · hepatotoxicity (*rare*)

## INTERACTIONS

- Metabolized by **CYP3A4** → **avoid CYP3A4 inducers and inhibitors** (*rifampicin is an inducer — the obvious trap*)

## CAUTIONS

- ⚠ Long  $t_{1/2}$  **24–30 h**, protein binding >99% — a QT problem does not clear quickly
- Bioavailability is **increased with food**; eliminated mainly in feces

# Nitroimidazoles — Pretomanid & Delamanid

## MECHANISM

Prodrugs activated by **deazaflavin-dependent nitroreductase (DDN)** → **inhibit mycolic acid synthesis** and release **reactive nitrogen species (NO•)**.

→ **Downstream:** oxidative and nitrosative stress damages DNA, proteins and lipids, and disrupts ATP synthesis / respiration → **bactericidal even against non-replicating bacilli**

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                     |
|-------------------|----------------------------------------------------------------------------------------------------|
| <b>Pretomanid</b> | XDR-TB and treatment-intolerant MDR-TB — the "Pa" of <b>BPaLM</b>                                  |
| <b>Delamanid</b>  | MDR-TB combination regimens, <b>especially where bedaquiline is unavailable or contraindicated</b> |

## ADVERSE EFFECTS

- **Pretomanid** — peripheral **neuropathy**, **myelosuppression**, GI upset
- **Delamanid** —  **QT prolongation**, GI disturbance, hepatotoxicity

## INTERACTIONS

- **Pretomanid** — CYP3A4 substrate; **avoid strong modulators** (inducer: rifampicin · inhibitor: ketoconazole)
- **Delamanid** — metabolized primarily by **plasma albumin**, non-CYP; only minor CYP3A4 → far fewer interactions

## CAUTIONS

-  Delamanid  $t_{1/2}$  ~30–38 h with >99% protein binding — **stacking it with bedaquiline stacks QT risk**
- Pretomanid absorption ↑ with a high-fat meal; excreted in feces (53%) and urine (36%) as inactive metabolites

# Linezolid

## MECHANISM

Binds the **23S portion of the 50S ribosomal subunit** → prevents formation of the **initiation complex** → no bacterial protein synthesis.

→ **Downstream:** oral bioavailability ~100% with good penetration into skin, bone, fat, lung and **CSF** — so IV and oral are effectively interchangeable

## DRUGS

| DRUG             | WHAT IT TREATS                                                 |
|------------------|----------------------------------------------------------------|
| <b>Linezolid</b> | XDR-TB / treatment-intolerant MDR-TB — the "L" of <b>BPaLM</b> |

## ADVERSE EFFECTS

- **Myelosuppression** ★ — especially **thrombocytopenia** with prolonged use (*and TB courses are prolonged*)
- **Peripheral and optic neuropathy** — long-term use
- **Lactic acidosis** (*rare but serious*)
- GI upset — nausea, vomiting, diarrhea

## INTERACTIONS

- **Serotonin syndrome** with **SSRIs or MAO inhibitors** ★

## CAUTIONS

- ⚠ Monitor CBC — the haematologic toxicity is duration-driven, and it stacks with the optic neuropathy risk of **ethambutol**

# Leprosy (Hansen's Disease)

Chronic infection by *Mycobacterium leprae*, predominantly affecting **skin and peripheral nerves**. Treated with **multidrug therapy (MDT)**: rifampin + clofazimine + dapsone.

## CLASSIFICATION DRIVES THE DURATION

| FORM                       | DEFINITION                                                                                                            | MDT LENGTH       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Paucibacillary (PB)</b> | 1–5 skin lesions, <b>no bacilli</b> on skin smear                                                                     | <b>6 months</b>  |
| <b>Multibacillary (MB)</b> | >5 lesions, <b>or</b> nerve involvement (pure neuritis / lesions + neuritis), <b>or</b> bacilli demonstrated on smear | <b>12 months</b> |

# Dapsone

## MECHANISM

Inhibits **dihydropteroate synthase** — the bacterial folic acid synthesis enzyme → disrupts DNA synthesis.  
→ **Downstream: bacteriostatic** against *M. leprae*; broad spectrum beyond it — antibacterial, antiprotozoal (*P. falciparum*, *T. gondii*) and antifungal (*Pneumocystis jirovecii*)

## DRUGS

| DRUG           | WHAT IT TREATS                                                        |
|----------------|-----------------------------------------------------------------------|
| <b>Dapsone</b> | leprosy MDT (PB & MB) · <i>Pneumocystis</i> · toxoplasmosis · malaria |

## ADVERSE EFFECTS

- **Hemolysis** — especially in **G6PD deficiency** ★ (the fact to check before prescribing)
- **Methemoglobinemia**
- Rash, **exfoliative dermatitis**, rarely **dapsone hypersensitivity syndrome**
- Peripheral neuropathy at high doses · hepatotoxicity (rare, monitor)

## CAUTIONS

- ⚠ **Screen for G6PD deficiency**
- ⚠ Crosses the **placenta** and enters **breast milk**
- Metabolized by hepatic **N-acetylation (NAT2)** — same enzyme as isoniazid, so slow acetylators run higher levels.  $t_{1/2}$  20–30 h, 85% renally excreted

## ADMINISTRATION — DAPSONE PK

- **Well absorbed orally** — **70–80% bioavailability**. Widely distributed; ⚠ **crosses the placenta and enters breast milk**.
- **Half-life ~20–30 hours** · metabolised in the liver by **N-acetylation (NAT2)** · **85% excreted in urine**.
- ★ **NAT2 acetylation is the same pathway as isoniazid** — so the fast/slow acetylator split that matters for INH applies here too.
- 💡 The long half-life is what allows once-daily dosing across a multi-month leprosy regimen.

# Clofazimine

## MECHANISM

Two proposed routes: **binds bacterial DNA**, disrupting replication and transcription; and **promotes ROS generation** inside the bacterial cell → cell death.

→ **Downstream**: has genuine **anti-inflammatory** activity alongside the antibacterial effect — useful in leprosy, where the nerve damage is inflammatory

## DRUGS

| DRUG               | WHAT IT TREATS                                        |
|--------------------|-------------------------------------------------------|
| <b>Clofazimine</b> | leprosy (first licensed use) · <b>MDR-TB / XDR-TB</b> |

## ADVERSE EFFECTS

- ★ **Skin discoloration** — **the most common ADR**. Red to brown-black pigmentation with long-term use; **takes months to years to resolve after stopping**. The usual reason patients quit
- Photosensitivity** — increased sunburn risk
- GI — nausea, vomiting, diarrhea, abdominal pain



# Nontuberculous Mycobacteria (NTM)

## MYCOBACTERIUM AVIUM COMPLEX (MAC)

At least two species: *M. avium* and *M. intracellulare* (*pulmonary disease*). Environmental — soil, water, dust — and often an **opportunistic infection**. **Triple therapy**: a **rifamycin** 600 mg + **ethambutol** 25 mg/kg + a **macrolide** (clarithromycin 1000 mg or azithromycin 500 mg).

## OTHER NTM SPECIES

| SPECIES                     | FIRST-LINE THERAPY                     | ALTERNATIVES                                                     |
|-----------------------------|----------------------------------------|------------------------------------------------------------------|
| <i>M. kansasii</i>          | INH + rifampin + ethambutol            | co-trimoxazole, ethionamide, cycloserine, clarithromycin         |
| <i>M. fortuitum</i> complex | amikacin + doxycycline                 | cefoxitin, rifampin, sulfonamide, clarithromycin, co-trimoxazole |
| <i>M. marinum</i>           | rifampin + ethambutol                  | co-trimoxazole, clarithromycin, minocycline, doxycycline         |
| <i>M. malmoense</i>         | rifampin + ethambutol ± clarithromycin | fluoroquinolone                                                  |
| <i>M. haemophilum</i>       | clarithromycin + rifampin + quinolone  | —                                                                |

Principle behind this: *[[pharm-pharmacokinetics]]* — rifampicin is the archetypal enzyme inducer, and its red secretions are excretion by the minor routes (sweat, tears, urine).

Principle behind this: *[[pharm-drug-interactions]]* — Rifampicin as the archetypal inducer of both CYP3A4 and P-glycoprotein; isoniazid on the inhibitor list.

Principle behind this: *[[pharm-drug-toxicity]]* — Isoniazid acetylation by NAT2 — fast and slow acetylators, and the clearance difference that follows.

Principle behind this: *[[pharm-pharmacogenomics]]* — Isoniazid and NAT2 acetylator status — fast vs slow acetylators changing clearance.

Source: 2026 Antimycobacterial drugs.pdf — อ.ภรรัตน์ (Kodcharat Cheevapruk, PhD), PYCH305 Pharmacology II, 79 slides. Structures omitted.

# Antifungal Drugs

Fungi are **eukaryotes with a cell wall** — so every antifungal target is one of the few things fungi have that human cells don't: **ergosterol** (instead of cholesterol) in the membrane, the **glucan cell wall**, and a fungal-specific **CYP (14-demethylase)**. Where selectivity is poor, that's exactly where the toxicity comes from.

**Squalene** → (*squalene epoxidase* — *allylamines block here*) → **squalene epoxide** → **Lanosterol**  
→ (*14-demethylase* — *azoles block here*) → **Ergosterol**

## WHO CAUSES WHAT

| GROUP            | GENERA                                                | HOW IT GROWS                              |
|------------------|-------------------------------------------------------|-------------------------------------------|
| <b>Yeasts</b>    | Candida · Cryptococcus · Trichosporon · Malassezia    | ovoid single cells, bud & divide          |
| <b>Molds</b>     | Aspergillus · Fusarium · Zygomycetes · Dermatophytes  | filamentous — branching hyphae → mycelium |
| <b>Dimorphic</b> | Blastomyces · Coccidioides · Histoplasma · Sporothrix | either form, depending on conditions      |

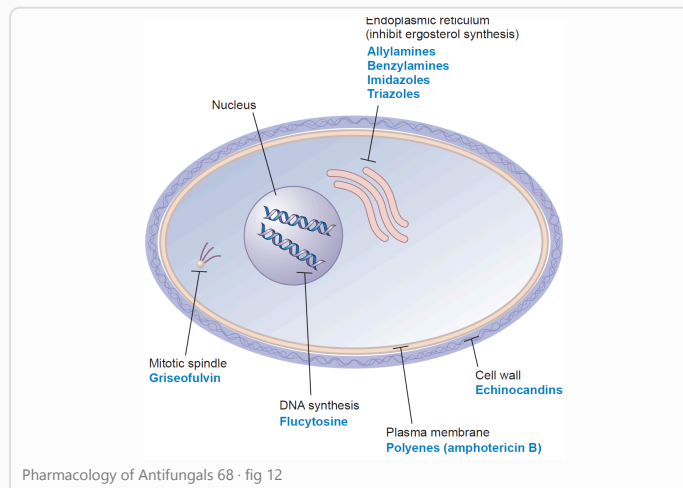
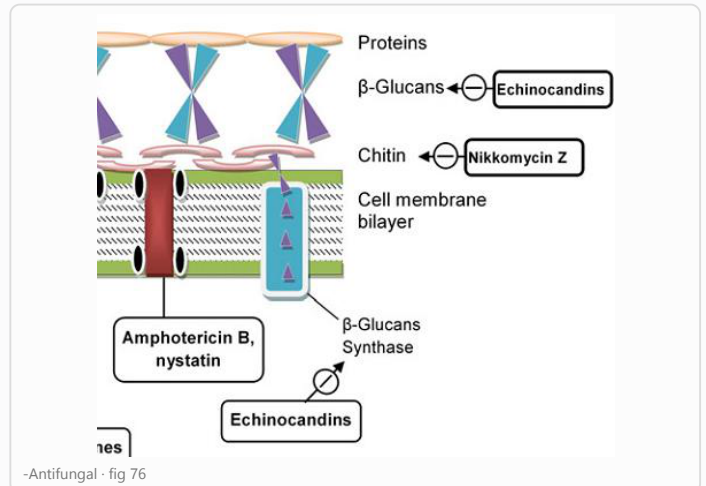
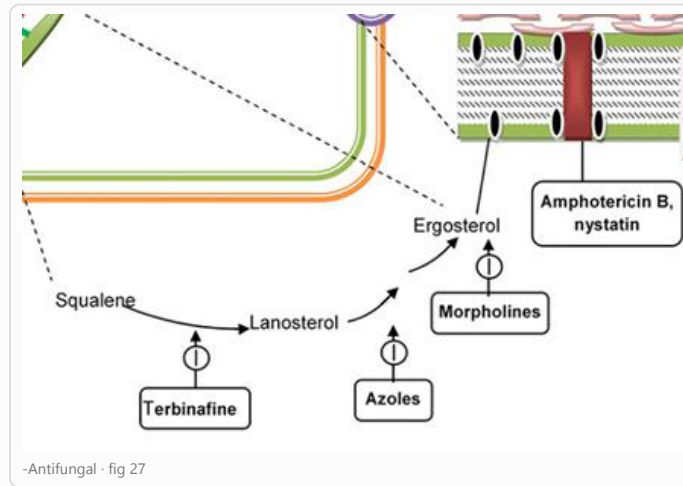
## CLASSIFICATION BY TARGET

| TARGET                                   | CLASSES                                       |
|------------------------------------------|-----------------------------------------------|
| <b>Plasma membrane</b> — bind ergosterol | Polyenes                                      |
| <b>ER</b> — inhibit ergosterol synthesis | Azoles · Allylamines                          |
| <b>Cell wall</b>                         | Echinocandins · Triterpenoids (ibrexafungerp) |
| <b>DNA synthesis</b>                     | Flucytosine                                   |
| <b>Mitotic spindle</b>                   | Griseofulvin                                  |

## TOPICAL OR SYSTEMIC? — DECIDED BY SITE OF INFECTION

| MYCOSIS                                                               | ROUTE                                                                                      |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Superficial</b> (skin, hair) · <b>Cutaneous</b> (skin, hair, nail) | <b>topical</b> — ⚠ <i>except</i> severe cases, <b>tinea unguium</b> , <b>tinea capitis</b> |
| <b>Subcutaneous</b> · <b>Systemic / invasive / deep</b>               | <b>systemic</b>                                                                            |

## OVERVIEW MAPS

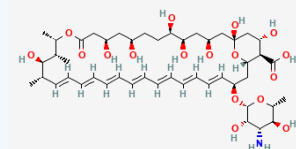


# Polyenes

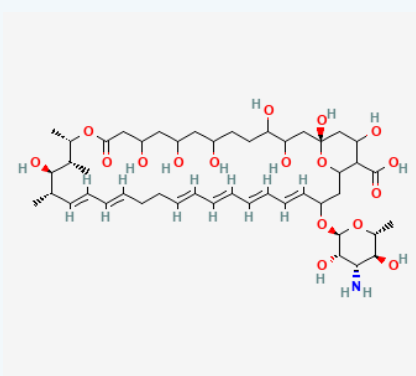
## MECHANISM

Bind **ergosterol** in the fungal cell membrane → **pore formation** → small molecules leak out → cell death.  
→ **Downstream: fungicidal**, concentration-dependent killing with prolonged PAE (*plateaus above 4–10 × MIC*) → **The toxicity is the same mechanism**: amphotericin B also binds **cholesterol** in human membranes

## THE SHARED CORE · WHAT CHANGES



**Amphotericin B.** a macrolactone with one hydrophobic polyene face and one hydroxylated face — it inserts into the membrane and binds ergosterol, forming a pore. ● the same amphipathic face binds human cholesterol → the toxicity



### Nystatin

the same molecule class, too toxic to inject → topical/oral only

## MEDCHEM

**SAR** ★ **Amphotericin B is a molecule with two faces**: the polyene chain is a flat hydrophobic wall, the polyol chain opposite it is hydrophilic. Several molecules stack their hydrophobic faces against membrane lipids and turn their hydrophilic faces inward - **the pore is lined by the drug itself**. That is why the effect is physical and concentration-dependent rather than enzymatic.

**Physicochem** It is excreted **renally and structurally unchanged**, so the kidney sees the intact pore-former for a long time - roughly **30% of patients get nephrotoxicity**. The **liposomal formulation** is a pharmacokinetic fix, not a chemical one: the carrier releases slowly, so less free drug circulates at any moment (↓ nephrotoxicity) and it penetrates the BBB better. It also costs ~40× more (\$196 vs \$5 per 50 mg).

## DRUGS

| DRUG                  | WHAT IT TREATS                                                                                                                         |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Amphotericin B</b> | systemic / invasive mycoses — Candida · Aspergillus · Cryptococcus · dimorphics · Zygomycetes. <b>IV infusion</b> (poor GI absorption) |
| <b>Nystatin</b>       | non-invasive candidiasis — oropharynx, esophagus, vagina. <b>Topical only</b> (toxic parenterally)                                     |
| <b>Natamycin</b>      | fungal eye infection — blepharitis, conjunctivitis, keratitis. Reaches corneal stroma, not intraocular fluid                           |

## USES

**Amphotericin B** — broad-spectrum systemic mycoses incl. CNS (*brain parenchyma levels are believed sufficient, though CSF levels are low*); long  $t_{1/2}$  > 50 h, wide distribution · **Nystatin** — mucosal candidiasis · **Natamycin** — ocular mycosis

## ADVERSE EFFECTS — AMPHOTERICIN B

- **Infusion-related reactions (70–90%)** — fever, chills, rigors, vomiting, hypo/hypertension. Proinflammatory cytokine release; subsides spontaneously in 30–45 min
- **Nephrotoxicity (24–80%)** ★ — azotemia, polyuria, acidosis. Dose-dependent, usually reversible. *Mechanism: afferent arteriolar vasoconstriction → renal hypoperfusion, plus direct tubular toxicity*
- **Hypokalemia (25–75%)** — weakness, arrhythmia · **Hypomagnesemia (30–75%)**. Both track the nephrotoxicity
- **Anemia (75%)** — normocytic, from ↓ EPO

## INTERACTIONS — AMPHOTERICIN B

- **Other nephrotoxic agents** — cyclosporine, tacrolimus · aminoglycosides, vancomycin, teicoplanin · cisplatin, carboplatin, oxaliplatin · diuretics, iodinated contrast
- **Neuromuscular blockers** (atracurium, vecuronium) — the hypokalemia raises **respiratory depression** risk
- **Digoxin** — narrow therapeutic window; hypokalemia → digoxin toxicity

## CAUTIONS & PREVENTION

- **Premedicate 30 min before:** acetaminophen + antihistamine, or IV hydrocortisone. Infuse over **>6 h**. Meperidine (pethidine) shortens rigors
- **Hydrate** — 1 L NSS (e.g. 500 mL before and after). **Monitor serum creatinine,  $K^+$ ,  $Mg^{2+}$** ;  $K^+$  supplement may be needed on prolonged use
- ★ **Liposomal amphotericin B** — kidney tissue concentration **0.2×** conventional → less nephrotoxicity; serum 5–20× higher

## ADMINISTRATION

- **Nystatin** — poorly absorbed through GI tract, skin and mucosa, and **toxic if given parenterally**, so it is a **topical drug only**. Forms: **oral suspension (swish and swallow)** · **vaginal suppository**
- **Natamycin** — poorly absorbed through the GI tract. **Ophthalmic suspension (eye drop)**; reaches effective concentration at the **corneal stroma but not intraocular fluid** ★ — which is exactly why it treats blepharitis/conjunctivitis/keratitis and not endophthalmitis
- 💡 "Swish and swallow" is the instruction patients most often get wrong — hold it in the mouth first, then swallow. Demonstrate it rather than saying it, or it becomes an ordinary drink of medicine.

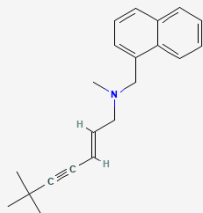
# Allylamines

## MECHANISM

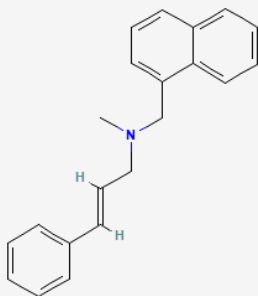
Inhibit **squalene epoxidase** — one step *upstream* of the azole block. Two hits at once: **no ergosterol** and **toxic squalene accumulation** → fungal cell death. (*Drugs: terbinafine, naftifine.*)

→ **Downstream: fungicidal** for dermatophytes · **fungistatic** for *C. albicans*

## THE SHARED CORE · WHAT CHANGES



**Terbinafine.** a lipophilic allylamine that blocks squalene epoxidase — one step EARLIER than the azoles, and outside the CYP family



**Naftifine**

same scaffold, topical

## MEDCHEM

**SAR** Two parts of terbinafine **cannot be changed at all** - the naphthalene ring and the position of the nitrogen. Everything the medicinal chemists were allowed to touch is the tail.

**SAR** ★ **The tert-butyl group is doing two jobs and that is why terbinafine is the oral one.** It raises lipophilicity and absorption - and, because it is bulky and **non-planar**, it obstructs CYP450, which prefers flat substrates. Better absorbed *and* more slowly metabolised, from one group.

**SAR** The inhibition is **reversible and non-competitive (allosteric)**, not competitive. Selectivity is enormous and measured:  $K_i$  **0.03  $\mu\text{M}$**  against *Candida* squalene monooxygenase vs **77  $\mu\text{M}$**  against the rat enzyme - about 2,500-fold.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                |
|--------------------|-----------------------------------------------------------------------------------------------|
| <b>Terbinafine</b> | ★ <b>drug of choice</b> for dermatophytoses, esp. <b>onychomycosis</b> · subcutaneous mycoses |

## USES

Accumulates in — and is released slowly from — **skin, nail and fat**;  $t_{1/2}$  12 h initially but **200–400 h at steady state** → once-daily dosing · ● **Blood concentration stays low** → **no role in systemic mycoses** · Oral bioavailability 40% (first-pass)

### ADVERSE EFFECTS

- **GI (4.9%)** — diarrhea, dyspepsia, nausea
- **Skin (2.3%)** — rash
- **CNS (1.2%)** — headache
- Respiratory (1.0%) · Musculoskeletal (0.8%) · **Taste disturbance (0.7%)**

### INTERACTIONS

Terbinafine is metabolized by **at least 7 CYPs**: - **Rifampicin** (enzyme inducer) → ↓ terbinafine level - **Cimetidine** (enzyme inhibitor) → ↑ terbinafine level

### CAUTIONS

- ⚠ Not recommended in **severe hepatic or renal insufficiency**

# Azoles — Imidazoles

## MECHANISM

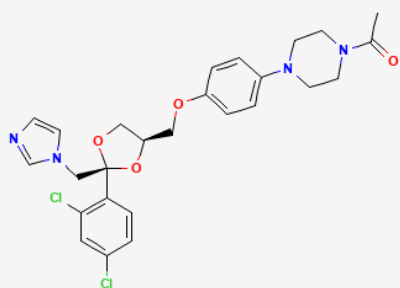
Inhibit **14-demethylase**, a **fungal CYP** — so lanosterol never becomes ergosterol. Two hits: **no ergosterol** and **toxic lanosterol accumulation** → fungal cell death.

→ **Downstream**: imidazoles are the **less selective** half — they hit human sterol synthesis and human CYPs too, so they are **topical drugs**. Triazoles are the selective, systemic half.

## THE SHARED CORE · WHAT CHANGES

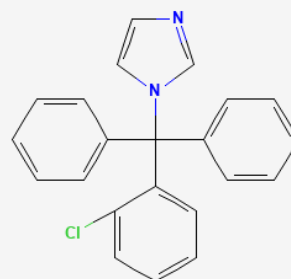


**Imidazole — two nitrogens.** one ring N coordinates the heme iron of fungal CYP51 → ergosterol synthesis stops. ⚠ it also fits human CYPs, which is the interaction problem.



**Ketoconazole**

● the archetypal potent CYP3A4 inhibitor; systemic use largely abandoned



**Clotrimazole**

topical only — too much systemic toxicity

## MEDCHEM

**SAR** ★ The azole nitrogen coordinates directly to the haem iron of 14 $\alpha$ -demethylase, competing with the oxygen the enzyme needs to run the demethylation. Not a substrate mimic - a metal ligand. The rest of the SAR: **halogens at ortho/para** raise activity (F and Cl best), the **C2 oxygen** forms a hydrogen bond and can be swapped for sulfur, and the hydrophobic side chain is the **most modifiable region** - which is where solubility and CYP selectivity get tuned.

**SAR** ● The hepatotoxicity is not a coincidence, it is the target. Lanosterol 14 $\alpha$ -demethylase is a **CYP450**, and all CYP450s share the same haem. An azole that binds one haem binds others - including the human CYPs that do detoxification, so toxins accumulate in the liver. Long-term use needs LFT monitoring.

**Interaction** Ketoconazole is a strong inhibitor of **CYP3A4** (→ triazolam sedation, simvastatin myopathy, amiodarone tachycardia) and **CYP2C9** (→ warfarin bleeding). The **hormone** ADRs come from the same non-selectivity: the CYPs that build steroids are inhibited too → ↓ **aldosterone and cortisol (adrenal insufficiency)**, gynaecomastia, alopecia, ↓ libido.



## DRUGS

| DRUG                                                                                                    | WHAT IT TREATS                                                                      |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Clotrimazole · Miconazole · Econazole · Isoconazole · Sertaconazole · Tioconazole · Luliconazole</b> | topical mycoses — skin, mucosal, vaginal                                            |
| <b>Ketoconazole</b>                                                                                     | topical mycoses · <b>high-dose oral:</b> Cushing disease, sometimes prostate cancer |

### USES — ORAL KETOCONAZOLE

A **lipophilic weak base**: dissolution and absorption need **acidic pH**. Take **with food** (*stimulates gastric acid, slows emptying*) · avoid **antacids, H<sub>2</sub>-blockers, PPIs** · cola (pH 2.5) improves absorption in achlorhydria. High doses block **cortisol and testosterone** synthesis — which is why it treats Cushing disease and prostate cancer.

### ADVERSE EFFECTS — ORAL KETOCONAZOLE

- **Potentially fatal liver injury** ★
- **Adrenal insufficiency** — it suppresses corticosteroid and testosterone synthesis

### INTERACTIONS

- **Strong inhibitor of CYP3A4, CYP2C9, CYP1A2** — harmful interactions
- Any drug that raises gastric pH → ↓ ketoconazole absorption

### CAUTIONS

- ⚠ **Oral ketoconazole is no longer a first-line drug** — hepatotoxicity + adrenal suppression + interactions

### ADMINISTRATION — ACID AND FOOD ARE THE WHOLE PROBLEM ★

- **Orally active**, tablet available. A **lipophilic weak base**, so **dissolution and absorption require an acidic pH**
- **Take with food** — food stimulates gastric acid secretion and slows gastric emptying, both of which help it dissolve
- ● **Avoid anything that raises gastric pH: antacids · H<sub>2</sub> blockers · proton-pump inhibitors**
- **Cola drinks (pH 2.5) improve absorption** in patients with achlorhydria
- 💡 This is the commonest quiet failure at a counter: someone already on a PPI gets an azole, takes it perfectly, and absorbs almost none of it. Ask about acid-suppressing drugs **before** you dispense.

# Azoles — Triazoles

## MECHANISM

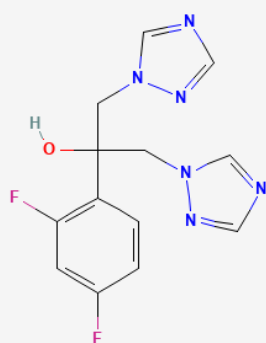
Same **14-demethylase** block as the imidazoles, but triazoles have **much less effect on human sterol synthesis and human CYPs** — "less likely to be toxic," so these are the **systemic** azoles.

→ **Downstream:** generally **fungistatic · fungicidal against *Aspergillus*** — ★ *except fluconazole, which has no Aspergillus activity at all* · time-dependent killing with prolonged PAE

## THE SHARED CORE · WHAT CHANGES

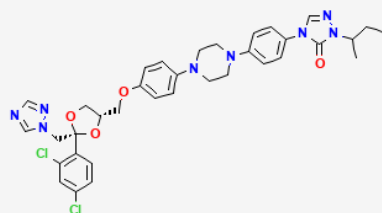


**Triazole — three nitrogens.** one extra N shifts selectivity toward FUNGAL CYP51 and away from human CYPs → safer systemically



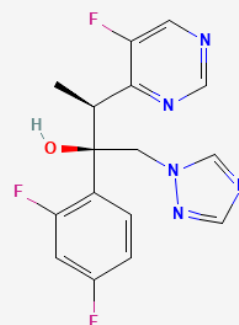
**Fluconazole**

small, water-soluble → excellent CSF entry, renal clearance



**Itraconazole**

big and lipophilic → tissue levels, but no CSF and heavy CYP3A4 interaction



**Voriconazole**

fluoropyrimidine added → *Aspergillus*; ⚠ saturable (zero-order) kinetics

## MEDCHEM

**SAR** The third ring nitrogen is what buys the selectivity the page describes - it also makes the class **more water-soluble**, which is why triazoles come in oral, IV *and* topical forms while the imidazoles are essentially topical.

**Prodrug Isavuconazonium is a prodrug** - it is cleaved to release active isavuconazole plus formaldehyde. The prodrug exists purely to make an insoluble triazole injectable.

## DRUGS

| DRUG                             | WHAT IT TREATS                                                                                                                       |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fluconazole</b> (1st gen)     | Candida · Cryptococcus · CNS mycoses (CSF 80% of serum) · eye · <b>UTI</b> (70–80% excreted unchanged renally)                       |
| <b>Itraconazole</b> (1st gen)    | Candida · Aspergillus · dimorphics · <b>cutaneous &amp; subcutaneous</b> infections (distributes to adipose, bone — <b>not CSF</b> ) |
| <b>Voriconazole</b> (2nd gen)    | invasive aspergillosis · Fusarium · Scedosporium · <b>CNS</b> (CSF 60%, higher in brain tissue) · eye                                |
| <b>Posaconazole</b> (2nd gen)    | broadest spectrum — the only triazole pair with <b>Zygomycetes/Mucorales</b> cover · CNS, bone, eye                                  |
| <b>Isavuconazole</b> (2nd gen)   | <b>invasive aspergillosis · invasive mucormycosis</b> (approved 2015). Prodrug isavuconazonium                                       |
| <b>Oteseconazole</b> (tetrazole) | <b>recurrent vulvovaginal candidiasis</b> (approved 2021). No CYP interaction; $t_{1/2}$ 138 days                                    |

## SPECTRUM — THE PATTERN WORTH KNOWING

| ORGANISM                                           | FLU | ITR | VOR | POS | ISA |
|----------------------------------------------------|-----|-----|-----|-----|-----|
| <i>C. albicans</i>                                 | +   | +   | +   | +   | +   |
| <i>C. glabrata</i> (efflux pump)                   | ±   | ±   | ±   | ±   | ±   |
| <i>C. krusei</i> (impaired 14-demethylase binding) | –   | ±   | +   | +   | +   |
| <i>Cryptococcus neoformans</i>                     | +   | +   | +   | +   | +   |
| <i>Aspergillus</i> spp.                            | –   | +   | +   | +   | +   |
| <i>Fusarium</i> spp.                               | –   | –   | +   | +   | +   |
| Zygomycetes / Mucorales                            | –   | –   | –   | +   | +   |

± = variable activity. Two resistance mechanisms drive the *Candida* gaps: efflux pump (*glabrata*) and impaired drug binding to 14-demethylase (*krusei*).

## ORAL ABSORPTION — THE THING THAT CHANGES HOW YOU DOSE

| TRIAZOLE                                          | ABSORPTION                                                                                                                                                           |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fluconazole · Voriconazole · Isavuconazole</b> | >90%, <b>unaffected by gastric pH and food</b> → free IV ↔ oral switching                                                                                            |
| <b>Itraconazole</b>                               | capsule needs <b>acid and/or food</b> ; oral solution (cyclodextrin) on an <b>empty stomach</b> . BA 22% capsule vs 55% solution                                     |
| <b>Posaconazole</b>                               | suspension BA only 8–47% and <b>saturable</b> — needs acid + fatty food, BID–QID with meals. Gastro-resistant tablet releases in the small intestine and avoids this |

## ADVERSE EFFECTS

- ↑ **Liver enzymes** — class effect of the triazoles (~10%)
- **CNS** — headache, confusion, hallucination
- ★ **Voriconazole** — **visual disturbances (30%)**: altered colour discrimination, blurred vision, bright spots and wavy lines, photophobia. Appear in the **first week and disappear** on continued therapy
- **Voriconazole** — mild rash (17%), including photosensitivity
- ● **Itraconazole** — **negative inotropic effect**: transient ↓ left ventricular ejection fraction; can precipitate or worsen **CHF**

## INTERACTIONS

- Triazoles inhibit human **CYPs** — the whole class carries PK interactions
- **Itraconazole and posaconazole also inhibit P-gp**
- **Isavuconazole is a CYP3A4 substrate**
- ★ **Voriconazole has non-linear PK in adults** (CYP2C19 metabolism saturates). **CYP2C19 polymorphism** — **13% of Thais are poor metabolizers**. In children PK is linear with faster metabolism → **higher doses needed**

## CAUTIONS

- ⚠ **Itraconazole** — avoid in a history of ventricular dysfunction / CHF (both IV and oral)

## ADMINISTRATION — THE FORMULATION CHANGES THE INSTRUCTION, NOT THE DRUG

| DRUG                 | FORM                                                    | HOW TO TAKE IT                                                                                                    | ORAL BIOAVAILABILITY |
|----------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Itraconazole</b>  | <b>capsule</b>                                          | needs <b>acidic pH and/or food</b>                                                                                | <b>22%</b>           |
| <b>Itraconazole</b>  | <b>oral solution</b><br>(cyclodextrin solubiliser)      | <b>empty stomach</b>                                                                                              | <b>55%</b>           |
| <b>Posaconazole</b>  | <b>gastro-resistant tablet</b>                          | pH-dependent polymer holds release until the small intestine — avoids the low-bioavailability problem             | —                    |
| <b>Posaconazole</b>  | <b>oral suspension</b>                                  | needs <b>acidic pH and/or fatty food</b> ; absorption <b>saturates</b> , so it is given <b>BID–QID with meals</b> | <b>8–47%</b>         |
| <b>Isavuconazole</b> | IV and oral ( <i>prodrug: isavuconazonium sulfate</i> ) | <b>no pH or food effect at all</b>                                                                                | <b>&gt; 95%</b>      |

- **Itraconazole** has an active metabolite, **hydroxy-itraconazole**, of equivalent potency
- **Posaconazole** — 20–30% metabolised by **UGT1A4**, 77% excreted in faeces (66% unchanged)
- **Isavuconazole** — **t½ 56–104 h** → **once daily**; urinary excretion very low; widely distributed including vitreous humor, low CSF level but **high brain level**
- 💡 Itraconazole capsule *with* food and itraconazole solution on an *empty* stomach are opposite instructions for the same drug name. **Read the box, not the molecule.**

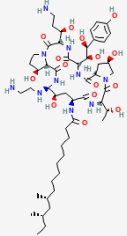
# Echinocandins

## MECHANISM

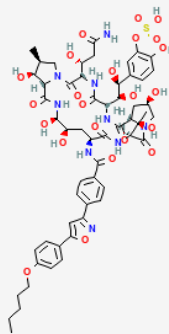
Attack the **fungal cell wall** — disrupt its rigidity. Human cells have no cell wall, so selectivity is near-total: **this is the least toxic class.**

→ **Downstream: fungicidal for *Candida* · fungistatic for *Aspergillus*** · concentration-dependent killing with prolonged PAE

## THE SHARED CORE · WHAT CHANGES



**Caspofungin.** a cyclic hexapeptide with a fatty tail — far too big to absorb orally, so ● IV only, always



**Micafungin**

same peptide + different lipid tail

## MEDCHEM

**SAR** Every substituent on the cyclic hexapeptide has a stated job. **R1's hydrophobic tail** is the membrane anchor - it interacts with fatty acids - and making that tail **linear, aromatic and bulky reduces haemolysis**. A **basic amine at R2 or R4** gives a positive charge that pairs with membrane phosphates and improves solubility, and **R5 as a sulfate** is added purely for plasma solubility.

**Physicochem** The class is **IV only because oral bioavailability is low** - a consequence of being a large peptide, not of instability.

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                     |
|----------------------|------------------------------------------------------------------------------------|
| <b>Caspofungin</b>   | invasive candidiasis · aspergillosis                                               |
| <b>Micafungin</b>    | invasive candidiasis · aspergillosis                                               |
| <b>Anidulafungin</b> | invasive candidiasis · aspergillosis — not metabolized, degrades chemically        |
| <b>Rezafungin</b>    | invasive candidiasis · aspergillosis — $t_{1/2}$ 150 h → <b>once-weekly dosing</b> |

## USES

● **IV only** — oral bioavailability <10% for all four · heavily protein-bound (96–99%+) · ★ **CSF penetration is low across the class** → not for CNS mycoses

## SPECTRUM — A NARROW, SHARPLY DEFINED WINDOW

| ORGANISM                                            | ECHINOCANDINS                                                     |
|-----------------------------------------------------|-------------------------------------------------------------------|
| <i>Candida</i> spp.                                 | + ( <i>C. parapsilosis</i> and <i>C. guilliermondii</i> variable) |
| <i>Aspergillus</i> spp.                             | +                                                                 |
| ★ <i>Cryptococcus neoformans</i>                    | –                                                                 |
| <i>Fusarium</i> · <i>Scedosporium</i> · Zygomycetes | –                                                                 |
| Coccidioides · Blastomyces · Histoplasma            | ± variable                                                        |

## ADVERSE EFFECTS

- Generally **low toxicity** — selective for the fungal cell wall
- **Infusion-related:** phlebitis
- **Histamine-like effects on rapid infusion** — flushing, urticaria, bronchospasm, facial swelling, pruritus

## INTERACTIONS

- **Caspofungin** — alters tacrolimus level (~20%)
- **Micafungin** — mild CYP3A4 inhibitor: ↑ AUC of **nifedipine** (18%) and **sirolimus** (21%)

# Triterpenoids

## MECHANISM

**Inhibit glucan synthase** — the same cell-wall target as the echinocandins, but reached by an **orally active** drug.

→ **Downstream:** cell wall synthesis fails → fungal cell death

## DRUGS

| DRUG                 | WHAT IT TREATS                                                         |
|----------------------|------------------------------------------------------------------------|
| <b>Ibrexafungerp</b> | <b>vulvovaginal candidiasis (VVC)</b> and prevention of its recurrence |

## USES

★ **Orally active** — the point of the class, since every echinocandin is IV-only · metabolized by **CYP3A4** ·  $t_{1/2}$  ~20–30 h

## ADVERSE EFFECTS

- **GI** — abdominal discomfort, diarrhea, nausea, vomiting
- Headache

# Flucytosine

## MECHANISM

Flucytosine (5-FC) is a **pyrimidine analogue**. **Cytosine permease** carries it into the fungal cell, where **cytosine deaminase** converts it to **5-FU** — then two arms:

1. FUMP → FUDP → **FUTP incorporated into RNA** → inhibits protein synthesis
2. **FdUMP inhibits thymidylate synthetase** → inhibits DNA synthesis

→ **Downstream: fungistatic**, time-dependent killing **without PAE** (T>MIC 20–40%) · active **only against *Candida* and *Cryptococcus***

## DRUGS

| DRUG                      | WHAT IT TREATS                                                                     |
|---------------------------|------------------------------------------------------------------------------------|
| <b>Flucytosine (5-FC)</b> | <i>Candida</i> and <i>Cryptococcus</i> infections — ★ <b>always in combination</b> |

## USES

Well absorbed from the GI tract and **widely distributed** — CSF 65–90%, vitreous humor 49% · **80% excreted unchanged renally** · ● **High rate of resistance when used alone** → **combination therapy**

## ADVERSE EFFECTS

*Both trace back to the same thing: gut flora convert 5-FC to 5-FU, the anticancer drug.* - **GI** — nausea, vomiting, **enterocolitis** - ● **Bone marrow suppression** ★ — leukopenia, thrombocytopenia



# Griseofulvin

## MECHANISM

Inhibits **microtubule function** → disrupts assembly of the **mitotic spindle** → fungal mitosis stops. The only antifungal with this target.

→ **Downstream: fungistatic against dermatophytes** only

## DRUGS

| DRUG                | WHAT IT TREATS                                                                                     |
|---------------------|----------------------------------------------------------------------------------------------------|
| <b>Griseofulvin</b> | dermatophytoses — accumulates in <b>skin, hair and nail</b> . ● <b>No role in systemic mycoses</b> |

## USES

Orally active · **high-fat meals increase absorption** · concentrates in keratinized tissue, which is why it works on dermatophytes and nowhere else

## INTERACTIONS

- **Induces CYP3A4** → ↓ **warfarin** level ★
- ● **Disulfiram-like reaction with alcohol**

# Topical Antifungals

## AGENTS USED TOPICALLY, BY CLASS

| CLASS                                 | AGENTS                                                                                                                         |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Azoles</b>                         | Clotrimazole · Econazole · Isoconazole · Ketoconazole · Miconazole · Sertaconazole · Tioconazole                               |
| <b>Allylamines</b>                    | Terbinafine                                                                                                                    |
| <b>Polyenes</b>                       | Nystatin                                                                                                                       |
| <b>Others</b>                         | Tolnaftate · Amorolfine · Ciclopirox · Tavaborole · Undecylenic acid · Gentian violet · Sodium thiosulphate · Selenium sulfide |
| <b>Keratolytic / fungistatic pair</b> | <b>Salicylic acid</b> (keratolytic) + <b>benzoic acid</b> (fungistatic) — Whitfield's ointment, ซีฟิงเบอร์ 28                  |

## ADMINISTRATION — WHICH PREPARATION EXISTS FOR WHICH DRUG

(Trade names omitted per spec rule 4; the preparation is what decides the counselling.)

| CLASS              | DRUG                 | PREPARATIONS                                      |
|--------------------|----------------------|---------------------------------------------------|
| <b>Azoles</b>      | <b>Clotrimazole</b>  | cream · <b>vaginal tablet</b> · <b>troche</b>     |
|                    | <b>Econazole</b>     | cream                                             |
|                    | <b>Isoconazole</b>   | cream                                             |
|                    | <b>Ketoconazole</b>  | cream · <b>shampoo</b> · lotion                   |
|                    | <b>Miconazole</b>    | cream · <b>oral gel</b>                           |
|                    | <b>Sertaconazole</b> | cream · vaginal tablet                            |
|                    | <b>Tioconazole</b>   | cream                                             |
| <b>Allylamines</b> | <b>Terbinafine</b>   | cream                                             |
| <b>Polyenes</b>    | <b>Nystatin</b>      | vaginal tablet · oral suspension · combined cream |

- 💡 The form follows the site, and that's the fastest way to pick one at a counter: **troche or oral gel** for oropharyngeal, **vaginal tablet** for vaginal, **shampoo** for scalp and seborrhoeic dermatitis, **cream** for everything on skin.

Principle behind this: [[pharm-pharmacokinetics]] — voriconazole is zero-order for the same reason, and ketoconazole is the archetypal potent CYP3A4 inhibitor.

Principle behind this: [[pharm-drug-interactions]] — Ketoconazole is both victim and perpetrator — its own absorption dies on a PPI, and it inhibits CYP3A4 and P-gp against everything else.

Source: [Pharmacology of Antifungals\\_68.pdf](#) — Surasak Wichaiyo (PhD), PYCH305 Pharmacology II, 58 slides. Structures omitted.

# Antiparasitic Drugs

Three branches: **antimalarials** · **drugs for other protozoa** · **anthelmintics**. The whole topic hangs on one idea — the parasite has a metabolic step humans don't, and the drug hits that step. Malaria drugs are then sorted by **which metabolic step** and **which life-cycle stage**.

## STAGE VOCABULARY — EVERYTHING ELSE DEPENDS ON THIS

- **Sporozoite** — injected by the mosquito. ★ **No antimalarial kills it** → true prevention of *infection* is impossible; drugs only prevent symptomatic disease.
- **Exoerythrocytic / hepatic stage** — in the liver.
- **Hypnozoite** — the dormant liver form of **P. vivax & P. ovale** → causes **relapse**. Killing it = **radical cure**.
- **Erythrocytic (asexual) stage** — the blood schizont. **This is what causes symptoms** (RBC lysis + merozoite release).
- **Gametocyte** — the sexual form. Killing it blocks **transmission**, not the patient's illness.

## THE FIVE PLASMODIUM SPECIES

| SPECIES              | INCUBATION | NOTE                                         |
|----------------------|------------|----------------------------------------------|
| <b>P. falciparum</b> | 8–12 d     | ● <b>most severe</b> — organ failure & death |
| <b>P. vivax</b>      | 10–15 d    | hypnozoites → relapse                        |
| <b>P. ovale</b>      | 10–15 d    | hypnozoites → relapse                        |
| <b>P. malariae</b>   | 30–40 d    | —                                            |
| <b>P. knowlesi</b>   | 9–12 d     | —                                            |

*P. falciparum + P. vivax = most infections worldwide, including Thailand.*

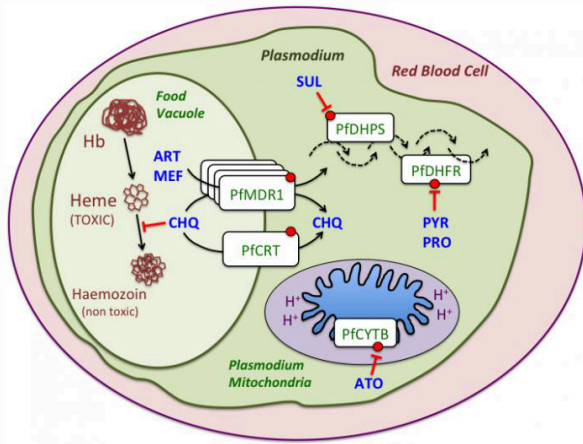
## ANTIMALARIALS BY MECHANISM

| MECHANISM CLASS                       | DRUGS                                                                                                                       | STAGE HIT                   |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Heme metabolism inhibitors</b>     | chloroquine, amodiaquine, piperaquine, quinine/quinidine, mefloquine, artemisinin, lumefantrine, halofantrine, pyronaridine | erythrocytic                |
| <b>Electron transport inhibitors</b>  | primaquine, tafenoquine, atovaquone                                                                                         | liver (± gametocyte)        |
| <b>Protein translation inhibitors</b> | doxycycline, tetracycline, clindamycin                                                                                      | erythrocytic (partner only) |
| <b>Folate metabolism inhibitors</b>   | sulfadoxine–pyrimethamine, proguanil                                                                                        | erythrocytic / hepatic      |

## THE RULE THAT GOVERNS ALL MALARIA THERAPY

- ★ **No single antimalarial covers both liver and erythrocytic stages**, and both can coexist in one patient → **complete elimination needs more than one drug**.
- ● **Never use artemisinin as monotherapy** (WHO).

## OVERVIEW MAPS



Antiparasitic Drugs · fig 05

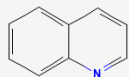
# Aminoquinolines — Chloroquine, Amodiaquine, Piperaquine

## MECHANISM

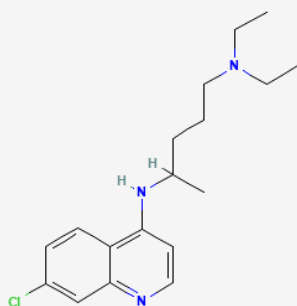
The parasite eats haemoglobin in its **food vacuole**, releasing toxic **ferriprotoporphylin IX (heme)**, which it detoxifies by polymerising into inert **hemozoin**. Chloroquine is a **weak base** → diffuses freely into the vacuole, gets **protonated in the acidic interior and trapped**, then binds ferriprotoporphylin IX and **blocks polymerisation**.

→ **Downstream**: unpolymerised heme accumulates → **ROS, membrane damage, protein dysfunction** → parasite death → **Resistance**: **PfCRT** (food-vacuole membrane, *pfcr* gene) pumps the drug out · **PfMDR1** (P-gp) and **PfMRP** at the plasma membrane — also drives quinine, mefloquine, artemisinin resistance

## THE SHARED CORE · WHAT CHANGES

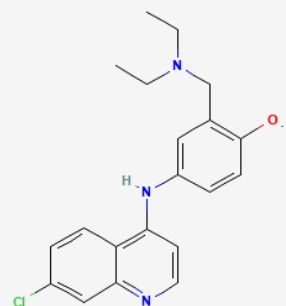


**Quinoline.** the flat aromatic that stacks onto haem inside the parasite's vacuole and blocks haemozoin crystallisation



**Chloroquine**

4-amino + Cl + a basic side chain → trapped in the acidic vacuole and concentrated there



**Amodiaquine**

a phenol on the side chain — active against some chloroquine-resistant strains

## MEDCHEM

**SAR** The side chain is tightly constrained: **2–5 carbons is optimal**, the **tertiary amine is essential** (it is what makes the drug a trappable weak base), and adding a double bond changes nothing while replacing it with a plain alkyl chain **kills activity**. The **C7 chlorine** is the optimum at that position.

**SAR** ★ **The two safer analogues are single edits to chloroquine.** Swap in a **phenol** → amodiaquine, less toxic. Add a **hydroxyl** to the side chain → hydroxychloroquine, less toxic *and* higher plasma concentration - which is why it is the one used long-term in rheumatology.

**SAR** **Piperaquine is two quinolines joined by an aliphatic linker.** Doubling the molecule doubles the lipophilicity, and that alone explains its **20–30 day half-life** - the reason it works as the long partner in an ACT.

**Metabolism** Chloroquine has one stereocentre and is used as the **racemate on purpose** - the deck's note is that the mixture is more beneficial than either single isomer. Cleared by **N-dealkylation via CYP2D6 and CYP3A4**. Beyond hemozoin, it also **intercalates DNA and inhibits topoisomerase II**.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                                                                                                                                        |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chloroquine</b> | ★ <b>DOC for P. ovale &amp; P. malariae</b> (prophylaxis + treatment) · P. vivax (still DOC despite Asian/Pacific resistance) · chloroquine-sensitive P. falciparum · <b>hepatic amoebiasis, RA, SLE, sarcoidosis</b> |
| <b>Amodiaquine</b> | P. falciparum — as <b>artesunate + amodiaquine</b> (ACT partner)                                                                                                                                                      |
| <b>Piperaquine</b> | chloroquine-resistant P. falciparum — as <b>dihydroartemisinin–piperaquine</b> , 1st line for uncomplicated falciparum in Thailand                                                                                    |

## USES

Erythrocytic forms of all species (*falciparum* only if sensitive — otherwise replaced by ACTs) · Amodiaquine & piperaquine survive mainly as **ACT partner drugs** — piperaquine's **very long half-life (~28 d, vs amodiaquine 14 d, mefloquine 14 d, lumefantrine 4 d)** makes it the choice in high-transmission areas.

## ADVERSE EFFECTS

- **Narrow safety margin** — a single **30 mg/kg** dose can be fatal. Acute toxicity comes from **rapid parenteral high doses**.
- **CVS**: hypotension, vasodilation, myocardial suppression, arrhythmia → cardiac arrest
- **CNS**: confusion, convulsions, coma
- ★ **High chronic doses (>250 mg/day): irreversible retinopathy + ototoxicity** — drug accumulates in retinal pigment epithelium and binds melanin → photoreceptor failure; and in the cochlea → hair-cell damage
- Oral: GI upset, headache, visual disturbance, urticaria
- **Pruritus** — most common in dark-skinned patients
- **Amodiaquine**: agranulocytosis, aplastic anaemia, hepatotoxicity → **not for long-term use**

# Quinine & Quinidine

## MECHANISM

Cinchona-bark alkaloids. Two hits: **bind heme and prevent detoxification** (same logic as chloroquine), **and intercalate into DNA** → block strand separation, transcription, translation.

→ **Downstream:** ↓ growth & replication of the **asexual erythrocytic** form. **No effect on hepatic forms.** →

**Resistance:** *pfcr* point mutations

## DRUGS

| DRUG             | WHAT IT TREATS                                                                          |
|------------------|-----------------------------------------------------------------------------------------|
| <b>Quinine</b>   | acute <b>blood-stage</b> malaria · nocturnal leg cramps ( <i>non-antimalarial use</i> ) |
| <b>Quinidine</b> | acute blood-stage malaria                                                               |

## USES

**Acute blood-stage malaria**, usually with **doxycycline or clindamycin** in chloroquine-resistant falciparum ·

★ **Quinine + clindamycin = the 1st-trimester pregnancy regimen** · **Never for prophylaxis** — too toxic, half-life too short (~11 h), more toxic and less effective than chloroquine.

## ADVERSE EFFECTS

- ★ **The dose-related triad: cinchonism · hypoglycaemia · hypotension**
- Cinchonism** = tinnitus, high-tone deafness, headache, nausea, vomiting, visual disturbance, postural hypotension
- GI: nausea, vomiting, abdominal pain, diarrhoea
- QT prolongation** (*rare*)
- The major metabolite **3-hydroxyquinine** is itself possibly toxic

# Mefloquine

## MECHANISM

Exact mechanism **unknown** — probably disrupts **hemozoin polymerisation**. A highly effective **blood schizonticide**.

→ **Downstream**: long half-life (~20 d) + high protein binding + wide tissue distribution → **single-dose regimen**

## DRUGS

| DRUG              | WHAT IT TREATS                                                                    |
|-------------------|-----------------------------------------------------------------------------------|
| <b>Mefloquine</b> | prevention & treatment of <b>drug-resistant P. falciparum</b> and <b>P. vivax</b> |

## USES

Reserved for **drug-resistant** falciparum/vivax · More effective **combined with an artemisinin** (artesunate + mefloquine).

## ADVERSE EFFECTS

- **FDA black-box warning (2013): neurologic & psychiatric toxicity** ★ — vivid dreams/nightmares (common), insomnia, anxiety, depression, hallucinations, seizures, psychosis
- **Cardiac conduction**: bradycardia, QT prolongation, arrhythmia
- Nausea, vomiting, dizziness

## INTERACTIONS

- **Do not co-administer with quinine, quinidine, or halofantrine** — additive cardiac conduction toxicity



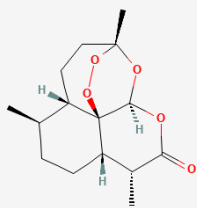
# Artemisinin & Derivatives (ACTs)

## MECHANISM

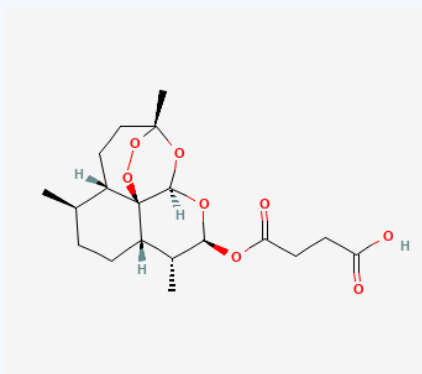
From *Artemisia annua* (ชิงเฮอร์). The **endoperoxide bridge** is cleaved by **iron** in the food vacuole → **free radicals** toxic to the parasite, plus **inhibition of PfATP6** (the plasmodial  $\text{Ca}^{2+}$ -ATPase), which is essential for survival.

→ **Downstream: very potent and fast-acting** — rapid parasite clearance and fever resolution · also kills **blood-stage gametocytes** → ↓ **transmission** → **Resistance**: true resistance not reported; what's seen is **delayed parasite clearance time**, associated with PfATP6 mutations *in vitro*

## THE SHARED CORE · WHAT CHANGES

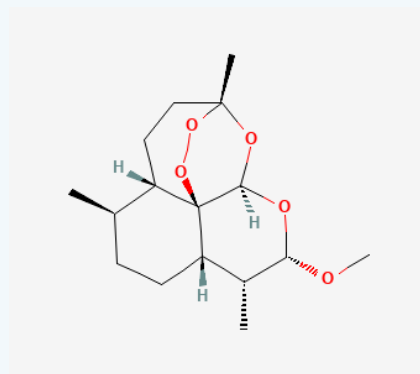


**Artemisinin.** ● the endoperoxide bridge (O–O) is the entire drug — iron cleaves it and the radicals kill the parasite. Remove that bond and nothing works.



**Artesunate**

a succinate ester → water-soluble → IV for severe malaria



**Artemether**

a methyl ether → lipid-soluble → IM/oral

## MEDCHEM

**Physicochem** The two limitations that force the combination therapy are physical: **very low water solubility** (artesunate is the exception - that is what makes it the injectable for severe malaria) and a **plasma half-life of only 1–2 hours**. A drug that clears that fast cannot finish an infection alone, which is the entire design logic of an ACT - fast killer plus slow partner.

## DRUGS

| DRUG                            | WHAT IT TREATS                                                                                                                                           |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Artemisinin</b>              | uncomplicated falciparum malaria, <b>as an ACT only</b> — parent compound; poorly soluble, oral only, low bioavailability                                |
| <b>Artesunate</b>               | ★ <b>severe malaria — the workhorse</b> · uncomplicated falciparum as an ACT — water-soluble, oral/ <b>IV/IM</b> /rectal                                 |
| <b>Artemether</b>               | uncomplicated falciparum malaria as an ACT (artemether–lumefantrine) · severe malaria where IV artesunate is unavailable — lipid-soluble, oral/IM/rectal |
| <b>Dihydroartemisinin (DHA)</b> | uncomplicated falciparum malaria as an ACT (DHA–piperaquine) — water-soluble, oral; also the <b>active metabolite</b> of all the others                  |

## USES

★ **Always as an ACT** — a short-half-life artemisinin (rapid kill) paired with a **long-half-life partner** that mops up survivors → ↑ efficacy, ↓ resistance. **Artemether–lumefantrine (Coartem)** · **artesunate–amodiaquine** · **artesunate–pyronaridine (Pyramax)** · **dihydroartemisinin–piperaquine** · **Not** a prophylactic agent.

## ADVERSE EFFECTS

- Generally **better tolerated than any other antimalarial class**
- ● **Toxic to infants and in the 1st trimester** → use **quinine + clindamycin** instead in T1
- Rare: allergic reaction, neutropenia, anaemia, haemolysis, ↑ liver enzymes
- Rare: auditory impairment / neurotoxicity; **brainstem neuropathy with IM oil-based formulations** (*animal data*)

## CAUTIONS

- ⚠ **Never as monotherapy** (WHO)
- ⚠ **No ACTs in children ≤5 kg or in the 1st trimester**
- ⚠ Artemisinin & artesunate are **CYP2B6 and CYP3A4 inducers**

# Other Heme-Polymerisation Inhibitors

## MECHANISM

Same target as chloroquine — **inhibit polymerisation of heme** — kept in use mainly as **ACT partner drugs** with long half-lives.

→ **Downstream:** effective against **erythrocytic** forms only

## DRUGS

| DRUG                | WHAT IT TREATS                                                                                                                                                                        |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lumefantrine</b> | uncomplicated <i>P. falciparum</i> — <b>only</b> as fixed-dose <b>artemether–lumefantrine (Coartem)</b> , 1st line; $T_{1/2}$ 3–5 d; <b>fatty food</b> ↑ <b>absorption</b> ; few ADRs |
| <b>Pyronaridine</b> | <i>P. falciparum</i> + <b>chloroquine-resistant <i>P. vivax</i></b> — fixed-dose 3:1 with <b>artesunate (Pyramax)</b> ; $T_{1/2}$ 9–13 d; less toxic than chloroquine                 |
| <b>Halofantrine</b> | <i>falciparum</i> — rapidly effective but <b>largely abandoned</b>                                                                                                                    |

## ADVERSE EFFECTS

- **Halofantrine: cardiac toxicity** → limits its use; **embryotoxic in animals** → **contraindicated in pregnancy**
- Lumefantrine and pyronaridine are comparatively well tolerated

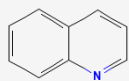
# 8-Aminoquinolines — Primaquine & Tafenoquine

## MECHANISM

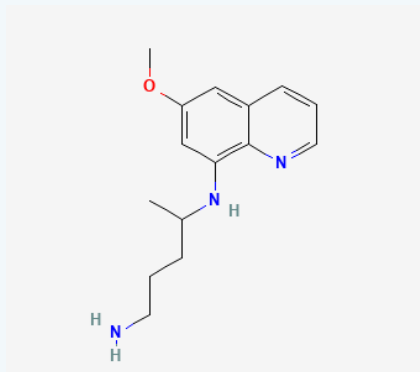
Interfere with **mitochondrial electron transport** in the parasite (disrupting **ubiquinone** function) and are converted to **electrophilic intermediates** → **ROS**. Tafenoquine adds **inhibition of haematin polymerisation** (chloroquine-like).

→ **Downstream:** ★ **the only drugs that kill hypnozoites** — the dormant liver forms of *P. vivax* & *P. ovale*  
 → **radical cure / relapse prevention** → **Also: gametocytocidal** against *P. falciparum* and others · **inactive against asexual blood stages** (primaquine)

## THE SHARED CORE · WHAT CHANGES

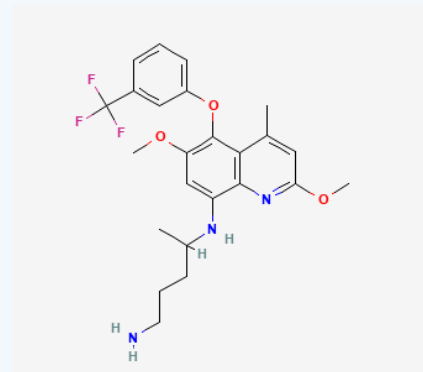


**Quinoline** — same ring, different position. ● the amino group moves from C4 to C8 and the drug changes job entirely: liver hypnozoites, not blood forms



**Primaquine**

⚠ oxidative metabolites → haemolysis in G6PD deficiency



**Tafenoquine**

bulkier, much longer  $t_{1/2}$  → single dose, same G6PD problem

## MEDCHEM

**SAR** ● This is where the **G6PD haemolysis comes from, structurally**. The **8-amino group autoxidises** - the drug generates reactive oxygen species by simply sitting in an oxygenated environment. That ROS is the antiparasitic mechanism *and* the red-cell toxicity; a G6PD-deficient cell has no glutathione reserve to absorb it. The toxicity is not a side effect of the drug, it is the drug.

**Metabolism** Oxidised by **monoamine oxidase** to carboxyprimaquine (the major metabolite), plus N-acetylation.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                                                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primaquine</b>  | ★ <b>DOC for eradicating dormant liver forms of <i>P. vivax</i> &amp; <i>P. ovale</i></b> · gametocytocidal · prophylaxis (CDC daily regimen)                                                                 |
| <b>Tafenoquine</b> | long-acting analogue ( $T_{1/2}$ 15 d, USFDA 2018) — <b>radical cure of <i>P. vivax</i> (<math>\geq 16</math> y)</b> · <b>prophylaxis (<math>\geq 18</math> y)</b> · liver + erythrocytic + gametocyte stages |

## USES

**Radical cure** of acute *P. vivax* / *P. ovale* — given **with a blood schizonticide (usually chloroquine)** to clear the erythrocytic stages and reduce resistance · **Chemoprophylaxis** · **Gametocidal / transmission-blocking**.

## ADVERSE EFFECTS

- ● **Haemolysis in G6PD deficiency** ★ — the defining toxicity of both drugs
- **Methaemoglobinaemia**, mild anaemia, leukocytosis (*high dose*)
- Abdominal distress — **prevented by taking with food**
- **Tafenoquine**: headache, dizziness, back pain, diarrhoea, nausea, motion sickness, depression, anxiety, abnormal dreams, ↑ ALT

## INTERACTIONS

- **Primaquine is a CYP1A2 inducer** — ⚠ caution with **warfarin**

## CONTRAINDICATIONS

- ● **G6PD deficiency** — **screen before use**. Unknown G6PD status also excludes tafenoquine.
- ● **Pregnancy** — the fetus is relatively G6PD-deficient
- ⚠ History of methaemoglobinaemia
- ⚠ Tafenoquine: hypersensitivity to primaquine (chemically similar)

# Atovaquone

## MECHANISM

Plasmodia get ATP from **glycolysis**, not oxidative phosphorylation — but they still need the electron transport chain to **re-oxidise dihydroorotate dehydrogenase (DHOD)**, the enzyme that makes pyrimidines for DNA. Atovaquone selectively blocks the interaction between **reduced ubiquinone and the cytochrome bc1 complex**.

→ **Downstream:** mitochondrial electron transport collapses → **DHOD stalls** → **no pyrimidine synthesis** → **no DNA replication** → **Resistance:** mutation in **cytochrome bc1** → **never use as a single agent**

## DRUGS

| DRUG                                     | WHAT IT TREATS                                                                                                                   |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Atovaquone + proguanil (Malarone)</b> | <b>chemoprophylaxis</b> (adults & children $\geq 11$ kg) · treatment of <b>uncomplicated <i>P. falciparum</i></b> ( $\geq 5$ kg) |

## USES

Effective against **liver stages of *P. falciparum*** — but **not** *P. vivax* hypnozoites · Always partnered: **proguanil** (DHFR inhibitor, synergistic) or **doxycycline**.

## ADVERSE EFFECTS

- Abdominal pain · nausea, vomiting, diarrhoea
- Headache · rash
- Transient  $\uparrow$  serum transaminase or amylase (*rare*)

## CAUTIONS & CONTRAINDICATIONS

- ⚠ Children **<11 kg** · pregnancy · lactation
- ⚠ Concomitant **rifampin** or **tetracycline**

## ADMINISTRATION — ATOVAQUONE

- Used as the **fixed-dose combination atovaquone–proguanil, 250 mg : 100 mg**.
- **Chemoprophylaxis** in adults and children  $\geq 11$  kg; **treatment** of uncomplicated *P. falciparum* in adults and children  $\geq 5$  kg.
- **Weight is the gate, not age** — **below 11 kg it is not used for prophylaxis at all**, and it is out in pregnancy and lactation.

# Antimalarial Translation Inhibitors

## MECHANISM

Antibiotics repurposed against the plasmodial **apicoplast ribosome** — tetracyclines bind the **30S** subunit, clindamycin the **50S**.

→ **Downstream:** parasite protein synthesis fails. **Slow-acting** → **partner drugs only, never alone.**

## DRUGS

| DRUG                              | WHAT IT TREATS                                                                                                                                                   |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Doxycycline / tetracycline</b> | chloroquine-resistant <i>P. falciparum</i> with <b>artesunate or quinine</b> · ★ <b>standard chemoprophylaxis in SE Asia</b> where mefloquine resistance is high |
| <b>Clindamycin</b>                | the same role <b>when tetracyclines are barred</b> — ★ <b>pregnancy and children &lt;8 y</b>                                                                     |

## ADVERSE EFFECTS

- **Doxycycline:** photosensitivity, tooth discolouration, vaginal candidiasis, GI upset
- ● **Clindamycin:** antibiotic-associated diarrhoea and **C. difficile colitis**

# Antimalarial Folate Inhibitors

## MECHANISM

Sequential blockade of the parasite's **own folate pathway** (it cannot scavenge host folate): **sulfadoxine / sulfalene inhibit dihydropteroate synthase**, **pyrimethamine and proguanil inhibit plasmodial dihydrofolate reductase**.

→ **Downstream:** no tetrahydrofolate → **no purine or pyrimidine synthesis** → growth arrest

## DRUGS

| DRUG                                        | WHAT IT TREATS                                                                                                                                           |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sulfadoxine–pyrimethamine (Fansidar)</b> | blood schizont stages of <i>P. falciparum</i> — <b>not</b> gametocytes.<br><b>Slow-acting + rising resistance</b> → <b>no longer recommended</b>         |
| <b>Proguanil</b>                            | <b>hepatic pre-erythrocytic</b> forms of <i>P. falciparum</i> & <i>P. vivax</i> — used <b>synergistically with atovaquone</b> for drug-resistant strains |

## ADVERSE EFFECTS

- **Proguanil:** oral ulceration, **pancytopenia, thrombocytopenia, granulocytopenia**
- Sulfadoxine–pyrimethamine: highly protein-bound → long half-life, prolonged exposure

## CAUTIONS

-  **Proguanil is safe in pregnancy** — including with chloroquine, atovaquone, tetracyclines and other antifolates



# Malaria in Practice

## PREGNANCY & LACTATION

General rule: avoid antimalarials in pregnancy unless needed — safety evidence is thin.

| SITUATION                                 | REGIMEN                                                                                                           |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>P. falciparum, 1st trimester</b>       | ★ Quinine + clindamycin                                                                                           |
| <b>P. falciparum, 2nd–3rd trimester</b>   | Dihydroartemisinin + piperaquine                                                                                  |
| Uncomplicated, chloroquine-sensitive area | <b>Chloroquine</b>                                                                                                |
| Uncomplicated, chloroquine-resistant area | <b>Quinine</b> (vivax) or <b>quinine + clindamycin</b> (falciparum)                                               |
| <b>P. vivax / P. ovale</b>                | chloroquine prophylaxis <b>until delivery</b> , then <b>primaquine</b> — only after confirming <b>G6PD normal</b> |
| <b>Lactation</b>                          | most agents acceptable; prefer chloroquine / hydroxychloroquine                                                   |

## THAILAND — PROPHYLAXIS IS NOT RECOMMENDED

"ไม่แนะนำให้รับประทานยาป้องกันมาลาเรียในประเทศไทย" — five reasons: no drug is 100% preventive · long courses mean real ADR risk · prophylaxis **blurs the clinical picture** if malaria does occur · prevalence is tiny (~1 in 16,000) · **mechanical protection works** — pyrethrin-treated nets, covering clothing, **DEET 35%** (4–6 h).

**Standby emergency treatment** (*carried, not taken routinely*) — **Malarone** or **artesunate + mefloquine**. Take only if: in a risk area > **1 week** · fever with chills, nausea, vomiting, headache, body ache · **no doctor reachable within 12–24 h**. ★ **Still get a blood test afterwards, even if you improve.**

## VACCINES

| VACCINE                                              | TARGET                                                          | EFFICACY                                                                                                             |
|------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>RTS,S/AS01 (Mosquirix)</b> , 2021 — world's first | <b>circumsporozoite protein (CSP)</b> of <i>P. falciparum</i>   | 30–40% vs clinical malaria; ~30% ↓ severe malaria. 3 doses at 5/6/7 months + booster at 18 mo. Short-lived immunity. |
| <b>R21/Matrix-M</b> , 2023                           | same CSP target, <b>higher antigen dose + Matrix-M adjuvant</b> | <b>75%</b> — 3 doses + booster at 12 mo                                                                              |

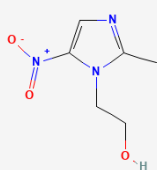
# Nitroimidazoles — Metronidazole & Tinidazole

## MECHANISM

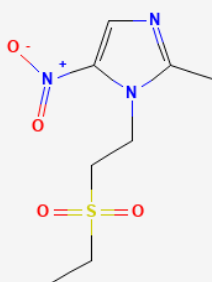
A **prodrug activated only by anaerobes**. Inside the organism, **reduced ferredoxin** (made by **pyruvate:ferredoxin oxidoreductase, PFOR** — an enzyme humans and aerobic bacteria don't have) donates electrons to metronidazole's **nitro group**.

→ **Downstream:** reduced cytotoxic radicals bind **DNA and protein** → **cell death**. The PFOR requirement is exactly why it is **selectively toxic to amoebae and anaerobes** and spares human cells. → **Resistance:** better oxygen scavenging ( $O_2$  competes for the electrons) · ↓ PFOR and ferredoxin levels

## THE SHARED CORE · WHAT CHANGES



**Metronidazole.** ● a prodrug activated by REDUCTION of the nitro group — which only happens in anaerobes. Aerobic cells can't reduce it, so they're untouched.



**Tinidazole**

same nitro group, longer  $t_{1/2}$  → shorter courses

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                                                                                                                                                                                                 |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Metronidazole</b> | <b>amoebiasis</b> ( <i>E. histolytica</i> — colitis + liver abscess) · <b>giardiasis</b> · ★<br><b>trichomoniasis — drug of choice</b> · anaerobic bacterial infections · <i>H. pylori</i> ,<br><i>Campylobacter</i> · pseudomembranous colitis, brain abscess |
| <b>Tinidazole</b>    | same spectrum, 2nd-generation — <b>higher cure rate, better tolerated, shorter course</b> · ★ <b>better than metronidazole for giardiasis (single dose)</b> · works in <b>metronidazole-refractory trichomoniasis</b>                                          |

## USES

It is a **mixed amoebicide** — active both **luminally** (asymptomatic carriers, mild diarrhoea) and **extra-luminally** (severe colitis, liver abscess, systemic spread). ★ **In amoebiasis, always pair it with a luminal agent** — otherwise the gut reservoir reseeds the infection.  $T_{1/2}$ : metronidazole 8 h · tinidazole 10–15 h.

## ADVERSE EFFECTS

- Common: headache, nausea, vomiting, dry mouth, ★ **metallic taste**
- CNS: dizziness, headache, **sensory neuropathy**
- **Stevens–Johnson syndrome** (*rare*)

## INTERACTIONS

- ★ **Alcohol** → **disulfiram-like reaction** ★ — the exam fact. No alcohol during or just after the course.
- **Warfarin** — prolongs prothrombin time
- **Lithium** — precipitates CNS signs of lithium toxicity

## CAUTIONS

- ⚠ **Not in the 1st trimester of pregnancy**
- *Note: it distributes everywhere including CSF — except the placenta.*

## ADMINISTRATION — METRONIDAZOLE & TINIDAZOLE, INDICATION BY INDICATION

| INDICATION                                         | METRONIDAZOLE                                                                                                                                                                   | TINIDAZOLE                                                    |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Amebiasis</b> ( <i>colitis, liver abscess</i> ) | 750 mg orally 3×/day × <b>5–10 days</b>                                                                                                                                         | 2 g once daily × 3 days                                       |
| <b>Trichomoniasis</b>                              | ★ <b>2 g as a single dose</b> — <i>or</i> 250 mg 3×/day × 7 days<br>— <i>or</i> 500 mg twice daily × 7 days ( <i>HIV-infected women</i> )                                       | 2 g once                                                      |
| <b>Giardiasis</b>                                  | 5-day course                                                                                                                                                                    | ★ <b>single dose</b> — more effective than metronidazole here |
| <b>Bacterial vaginosis</b>                         | 500 mg twice daily × 7 days — <i>or</i> one applicatorful (5 g of 0.75% gel) intravaginally once daily × 5 days                                                                 | —                                                             |
| <b>PID</b>                                         | 500 mg orally twice daily × 14 days                                                                                                                                             | —                                                             |
| <b>Anaerobic bacterial infection</b>               | 15 mg/kg IV over 1 hour, then <b>7.5 mg/kg every 6 h</b> × 7–10 days ( <i>IV or oral</i> )                                                                                      | —                                                             |
| <b>Colorectal surgery prophylaxis</b>              | 500 mg or 15 mg/kg IV finished <b>within 1 hour before surgery</b> , then 7.5 mg/kg IV every 6 h × 2 doses — <i>or</i> 1 g orally 3× over 10 hours the afternoon/evening before | —                                                             |

- **Formulations:** metronidazole is **oral, topical (cream/gel), intravaginal and IV**; tinidazole is **oral only**.
- **Food:** ★ **food does not affect oral bioavailability of either** — oral metronidazole is > 80% bioavailable. Both can be taken with food purely to blunt the GI upset.
- ★ **Always pair with a luminal agent for amoebic colitis or liver abscess** — these clear the tissue form, not the cysts left in the gut lumen.
- 💡 So the counselling line is "with or without food, whichever your stomach prefers" — and the one thing that *does* have to be said is **no alcohol**, because of the disulfiram-like reaction.
- 💡 If trichomoniasis comes back after the 2 g single dose, the slide's move is a **second dose in 4–6 weeks**, or tinidazole for a resistant infection — not simply repeating the same course immediately.

# Pentamidine

## MECHANISM

Broad-spectrum against protozoa **and fungi** — inhibits **DNA, RNA, phospholipid and protein synthesis**.  
→ **Downstream**: highly charged → **poorly absorbed orally** → given **IV, IM, or inhaled**

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                                                                                     |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pentamidine</b> | <b>Pneumocystis pneumonia</b> ( <i>P. jirovecii</i> ) · <b>cutaneous leishmaniasis</b> · <b>early-stage African trypanosomiasis</b> ( <i>T. brucei gambiense</i> ) |

## USES

⚠ **Reserved for severe infection** — the toxicity profile rules out casual use.

## ADVERSE EFFECTS

- **Quite toxic.** Serious: **nephrotoxicity**, hepatic impairment, **blood dyscrasias**
- **Metabolic chaos:** **hypoglycaemia** · **hypocalcaemia** · **hypomagnesaemia** · **hyperkalaemia**
- Hypotension, tachycardia, headache — **prevented by slowing the infusion rate**

# Anti-Toxoplasma Agents

## MECHANISM

Sequential folate blockade again, one step apart: **pyrimethamine inhibits dihydrofolate reductase, sulfadiazine inhibits dihydropteroate synthetase.**

→ **Downstream:** folate synthesis collapses → **no DNA or protein synthesis** → **The catch:** the block isn't perfectly selective → **host bone marrow suffers too** → give **leucovorin (folinic acid)** to rescue human cells

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                                    |
|----------------------|---------------------------------------------------------------------------------------------------|
| <b>Pyrimethamine</b> | toxoplasmosis — <b>combine with sulfadiazine or clindamycin</b> , ★ <b>always with leucovorin</b> |
| <b>Sulfadiazine</b>  | toxoplasmosis — combine with pyrimethamine + folic acid                                           |
| <b>Clindamycin</b>   | <b>acute toxoplasmosis</b> — the substitute when <b>sulfonamides aren't tolerated</b>             |

## ADVERSE EFFECTS

- **Bone marrow suppression** — both pyrimethamine and sulfadiazine

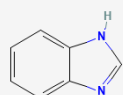
# Benzimidazoles — Albendazole & Mebendazole

## MECHANISM

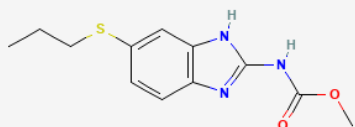
Bind the **colchicine-sensitive site of  $\beta$ -tubulin** in the worm → **microtubule polymerisation fails** → **microtubule-dependent glucose uptake stops** → glycogen stores are not replenished.

→ **Downstream:** the worm runs out of energy → **immobilisation and death** → **Selectivity:** ★ **250–400× more selective for helminth than mammalian tubulin.** Cure rates 60–100%.

## THE SHARED CORE · WHAT CHANGES

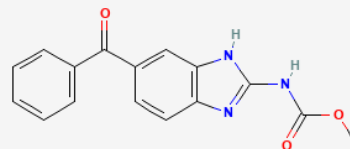


**Benzimidazole.** binds parasite  $\beta$ -tubulin far more tightly than the human version — the whole selectivity sits in that difference



**Albendazole**

a propylthio group → better absorbed → tissue worms too



**Mebendazole**

poorly absorbed on purpose → stays in the gut lumen

## MEDCHEM

**SAR** ★ The class is inactivated by its own metabolism. Substitutions at **C2** and **C5** carry the activity, and the rule is blunt: **R2 = OH means inactive** - so any metabolism that converts R2 to a hydroxyl switches the drug off.

**Physicochem** Both are **practically water-insoluble** and amphoteric (mebendazole logP 2.83, pKa 8.44/3.93; albendazole logP 2.54, pKa 9.51/4.27). Mebendazole is only **~10% absorbed** - which is a feature, not a flaw, since it works locally in the gut. **Taking either with a fatty meal raises absorption**, and chewing helps.

**Metabolism** ● The reason albendazole treats neurocysticercosis and mebendazole cannot. Albendazole's metabolite has a **longer half-life and 70% plasma protein binding**, so it actually reaches tissue; mebendazole's major metabolite is more water-soluble and stays peripheral. Same class, opposite pharmacokinetic fate.

## DRUGS

| DRUG                   | WHAT IT TREATS                                                                                                                |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Albendazole</b>     | GI nematodes · ★ <b>preferred for systemic/tissue infection</b> — <b>neurocysticercosis, hydatid disease, gnathostomiasis</b> |
| <b>Mebendazole</b>     | <b>GI nematodes only</b> — pinworm, roundworm, whipworm, hookworm                                                             |
| <b>Triclabendazole</b> | reserved for <b>liver fluke</b> (พยาธิใบไม้ตับ) and <b>paragonimiasis</b> (พยาธิใบไม้ปอด)                                     |

## USES

★ **The PK explains the split.** Albendazole is **poorly absorbed** — good for gut worms — but a **fatty meal raises absorption 5×** and it is converted by **CYP3A4 to albendazole sulfoxide (more active than the parent)**, so it reaches tissue and treats systemic disease. Mebendazole has **only 22% bioavailability** and heavy CYP3A4 first-pass metabolism **to an inactive compound** → it never leaves the gut usefully.

## ADVERSE EFFECTS

- **Very safe in short courses.** Headache, anorexia, nausea, vomiting, dizziness, diarrhoea
- ⚠ **Long-term use (albendazole): liver dysfunction (↑ liver enzymes) and myelosuppression** → **monitor LFTs and CBC**

## ADMINISTRATION — ALBENDAZOLE & MEBENDAZOLE DOSES

**Albendazole** — *adults and children > 10 kg: 400 mg (2 × 200 mg tablets) · children > 6 months and < 10 kg: 200 mg*

| INFECTION                            | ALBENDAZOLE                                         |
|--------------------------------------|-----------------------------------------------------|
| <b>Pinworm, roundworm, hookworm</b>  | <b>400 mg once</b>                                  |
| <b>Whipworm</b>                      | 400 mg daily × 3 days                               |
| <b>Gnathostomiasis</b>               | 400 mg twice daily × 21 days                        |
| <b>Neurocysticercosis</b>            | 5 mg/kg 3×/day × 1 month                            |
| <b>Larval <i>Taenia solium</i></b>   | 400 mg twice daily × 8–30 days                      |
| INFECTION                            | MEBENDAZOLE                                         |
| <b>Pinworm</b>                       | <b>100 mg once, ★ repeated after 2 weeks</b>        |
| <b>Roundworm, whipworm, hookworm</b> | 100 mg twice daily × 3 days — <i>or</i> 500 mg once |

- ★ **The single-dose regimens are for the gut lumen; the long courses are for tissue.** Neurocysticercosis and hydatid disease get albendazole for weeks because the drug has to reach the larva inside tissue — which is also why albendazole, not mebendazole, is the systemic one.
- 💡 The pinworm repeat at 2 weeks isn't optional detail — it targets the eggs that hatched after the first dose. A household that treats once and stops usually re-presents.

# Ivermectin

## MECHANISM

Activates **glutamate-gated chloride channels** — which exist in invertebrate nerve and muscle, not in humans → **Cl<sup>-</sup> influx** → **hyperpolarisation** → **flaccid paralysis** → **death**. Also binds nematode/insect **GABA receptors**.

→ **Safety basis**: ★ it **does not cross the human blood–brain barrier**, so the mammalian GABA system is untouched

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                                                                                                                                                          |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ivermectin</b> | <b>filarial infections</b> (single annual dose; + <b>albendazole</b> is more effective) · <b>onchocerciasis</b> — ● <i>river blindness</i> · <b>strongyloidiasis</b> · <b>cutaneous larva migrans</b> · roundworm, whipworm, threadworm |

## USES

Mass drug administration for filariasis and onchocerciasis · ⚠ **Not effective against hookworm**. Fatty meals ↑ oral bioavailability; CYP3A4-metabolised; excreted mainly in faeces; T<sub>1/2</sub> 18–36 h.

## ADVERSE EFFECTS

- Common: nausea, dizziness, diarrhoea, rash, mild itching
- ● **Mazzotti reaction** ★ — fever, headache, rash, lymphadenopathy, **especially in onchocerciasis**. It is the **host reaction to rapid parasite death**, not drug toxicity.
- Neurological toxicity — rare in humans; ⚠ caution with **pre-existing brain disorders** (common in dogs)

## ADMINISTRATION — IVERMECTIN DOSES

| INDICATION                                         | DOSE                                                                         |
|----------------------------------------------------|------------------------------------------------------------------------------|
| <b>Filarial infection</b>                          | <b>400 µg/kg single annual dose</b>                                          |
| ★ <i>more effective combination</i>                | ivermectin <b>200 µg/kg</b> + albendazole <b>400 mg</b> , single annual dose |
| <b>Onchocerciasis</b> (" <i>river blindness</i> ") | 150–200 µg/kg single dose, <b>every 6–12 months</b>                          |
| <b>Strongyloidiasis</b>                            | 150–200 µg/kg single dose                                                    |
| <b>Cutaneous larva migrans</b>                     | 200 µg/kg                                                                    |

- ⚠ Ivermectin covers roundworm, whipworm and threadworm — **but not hookworm**.
- 💡 Almost everything here is a *single* dose repeated on a long interval, not a daily course. Patients expecting a week of tablets need telling that one dose is the whole treatment.



# Neuromuscular Paralysing Anthelmintics

## MECHANISM

Three drugs, three ways to jam the worm's neuromuscular junction: 1. **Pyrantel pamoate** — persistent activation of **nicotinic ACh receptors** → **spastic** paralysis 2. **Levamisole** — potent agonist at the **L-type nematode nAChR** → paralysis; also **inhibits fumarate reductase** → ↓ succinate → **ATP depletion** 3.

**Piperazine** — **inhibits ACh at the NMJ** and acts as a **GABA agonist** → Cl<sup>-</sup> influx → hyperpolarisation → **flaccid** paralysis

→ **Downstream:** paralysed but **still alive** — the worms are expelled by normal **peristalsis**

## DRUGS

| DRUG                    | WHAT IT TREATS                                                    |
|-------------------------|-------------------------------------------------------------------|
| <b>Pyrantel pamoate</b> | pinworm · roundworm · hookworm                                    |
| <b>Levamisole</b>       | common <b>roundworm</b> — less effective for threadworm           |
| <b>Piperazine</b>       | <b>alternative</b> for roundworm <b>only</b> — no other helminths |

## ADVERSE EFFECTS

- **Pyrantel:** usually well tolerated
- **Levamisole:** nausea, vomiting, abdominal pain, dizziness, headache, metallic taste — ● **rare agranulocytosis, which is what limits its clinical use**
- **Piperazine:** GI upset, dizziness, headache — ⚠ **rare neurotoxicity**

## CAUTIONS

- ⚠ **Piperazine** — **avoid in:** pregnancy · renal or hepatic impairment · **epilepsy or chronic neurologic disease**

## ADMINISTRATION — LEVAMISOLE DOSE

- **Adults and children ≥ 6 years: 2.5 mg/kg as a single dose**, repeated after **7 days** if needed.
- 💡 Same shape as the other luminal anthelmintics — one dose, and a repeat a week later if the worm is still there rather than a longer first course.

# Praziquantel

## MECHANISM

Binds the **protein-kinase-C-binding site on the  $\beta$ -subunit of the schistosome voltage-gated  $\text{Ca}^{2+}$  channel** →  **$\text{Ca}^{2+}$  homeostasis collapses** → **paralysis and death**. Two supporting hits: it **disrupts the tegument**, unmasking parasite antigens to the host immune system, and **inhibits adenosine uptake** → energy depletion.

→ **Downstream:** broad activity across **trematodes and cestodes** — flukes and tapeworms

## DRUGS

| DRUG | WHAT IT TREATS |
|------|----------------|
|------|----------------|

|                     |                                                                                       |
|---------------------|---------------------------------------------------------------------------------------|
| <b>Praziquantel</b> | ★ <b>schistosomiasis — the main treatment</b> (blood fluke) · liver fluke · tapeworms |
|---------------------|---------------------------------------------------------------------------------------|

## USES

Broad-spectrum trematode + cestode drug · **distributes into CSF** · long  $T_{1/2}$  60–90 h · ✅ **Safe in pregnancy and lactation.**

## ADVERSE EFFECTS

- Abdominal discomfort
- Dizziness
- Headache

## INTERACTIONS

- ⚠️ **CYP enzyme inducers** ↓ **efficacy** — true for praziquantel and albendazole both

# Niclosamide

## MECHANISM

Blocks **oxidative phosphorylation** and **ATP production in the worm's mitochondria**, and **disrupts glucose uptake** in tapeworms.

→ **Downstream:** the worm **starves** — ★ **no paralysis involved**, which is what separates it from the neuromuscular agents

## DRUGS

| DRUG               | WHAT IT TREATS                                                            |
|--------------------|---------------------------------------------------------------------------|
| <b>Niclosamide</b> | tapeworm (cestode) infections — ⚠ <b>not</b> nematodes, <b>not</b> flukes |

## ADVERSE EFFECTS

- Nausea, vomiting
- Pruritus
- Light-headedness

# Diethylcarbamazine (DEC)

## MECHANISM

Three actions: **inhibits parasite cholinergic muscle receptors** · **disrupts microtubule formation** · **alters the helminth surface membrane**.

→ **Downstream**: the altered surface makes the worm **visible to the host immune system** → **enhanced immune killing**. That immune mechanism is also the source of its main ADR.

## DRUGS

| DRUG                      | WHAT IT TREATS                                                                                                                                                   |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diethylcarbamazine</b> | <b>lymphatic filariasis</b> (โรคเท้าช้าง) · <b>loiasis</b> (African eye worm) — often <b>combined with albendazole or ivermectin</b> in mass drug administration |

## ADVERSE EFFECTS

- GI disturbance, joint pain, headache, weakness
- ★ **Allergic reactions driven by the host response to dying filariae — pre-treat with glucocorticoids + antihistamine**
- ✓ **Safe in pregnancy**

# Anthelmintics at a Glance

## WORM TYPE → DRUG

| HELMINTH                      | THAI                                                                                 | DRUGS                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Nematodes (roundworms)</b> | พยาธิตัวกลม — ไล่เดือนกลม, เส้นด้าย, ปากขอ, ตัวจิ๊ด, แส้ม้า                          | <b>benzimidazoles</b> (mebendazole, albendazole) · <b>ivermectin</b> · <b>pyrantel pamoate</b> |
| <b>Cestodes (tapeworms)</b>   | พยาธิตัวแบน — ตืดหมู, ตืดวัว                                                         | <b>praziquantel</b> · <b>niclosamide</b>                                                       |
| <b>Trematodes (flukes)</b>    | พยาธิใบไม้ — ในเลือด ( <i>Schistosoma</i> ), ในตับ ( <i>Opisthorchis viverrini</i> ) | <b>praziquantel</b> · <b>triclabendazole</b>                                                   |

## ANTHELMINTICS BY TARGET

| TARGET                         | DRUGS                                                              |
|--------------------------------|--------------------------------------------------------------------|
| <b>Microtubule function</b>    | albendazole, mebendazole, ( <i>DEC, partly</i> )                   |
| <b>Neuromuscular paralysis</b> | ivermectin, pyrantel pamoate, levamisole, piperazine, praziquantel |
| <b>Energy metabolism</b>       | praziquantel, niclosamide, ( <i>levamisole, partly</i> )           |
| <b>Other</b>                   | diethylcarbamazine                                                 |

Principle behind this: [\[\[pharm-drug-toxicity\]\]](#) — Primaquine + G6PD deficiency → haemolysis in ~10% of black males. The classic pharmacogenetic idiosyncrasy.

Principle behind this: [\[\[pharm-pharmacogenomics\]\]](#) — Primaquine and G6PD deficiency — the oldest pharmacogenetic story in the syllabus (detail on [\[\[pharm-drug-toxicity\]\]](#)).

Source: 2026 Antiparasitic Drugs.pdf — อ.ภชรรัตน์ (Kodcharat Cheevapruk, PhD), PYCH305 Pharmacology II, 92 slides. Structures omitted.

# Anti-Hepatitis Agents

**HBV and HCV** are the main causes of chronic hepatitis, cirrhosis and **hepatocellular carcinoma (HCC)**. The whole topic splits on one fact: **HBV is suppressed, HCV is cured**.

**HCV: oral DAA combination → HCV RNA undetectable 12 weeks after therapy ends → SVR12 → definitive cure in >99%**

## HBV VS HCV — WHY THEY'RE TREATED DIFFERENTLY

|                      | HBV                                                        | HCV                                                  |
|----------------------|------------------------------------------------------------|------------------------------------------------------|
| genome               | <b>DNA</b> virus                                           | <b>RNA</b> virus (Flaviviridae, +ssRNA, 6 genotypes) |
| vaccine              | <b>yes</b> — vaccine-preventable                           | <b>none</b>                                          |
| outcome of treatment | ★ <b>not curative</b> — suppression only                   | ★ <b>curable</b>                                     |
| risk of HCC          | 15–40%                                                     | 6–10%                                                |
| why no cure          | ● <b>no drug destroys cccDNA</b> in the hepatocyte nucleus | —                                                    |

## WHAT EACH HEPATITIS VIRUS GETS

| VIRUS    | DRUGS                                                                 |
|----------|-----------------------------------------------------------------------|
| <b>A</b> | none                                                                  |
| <b>B</b> | PegIFN · nucleoside/nucleotide analogues (NAs)                        |
| <b>C</b> | PegIFN ( <i>historical</i> ) · <b>Direct-Acting Antivirals (DAAs)</b> |
| <b>D</b> | <b>PegIFN-α only</b>                                                  |
| <b>E</b> | none                                                                  |

## ENDPOINTS WORTH SEPARATING

- **HBV goals** — suppress HBV DNA to undetectable · HBeAg (rarely HBsAg) seroconversion positive → negative · normalise ALT
- **HCV goal** — **SVR** (sustained viral response): HCV RNA <15 IU/mL at **12 weeks (SVR12)** or 24 weeks (SVR24) after treatment ends. It is the **surrogate marker of cure**.

# Interferons (IFN- $\alpha$ / PegIFN)

## MECHANISM

Cytokines — a **non-specific antiviral immunomodulator**, not a direct antiviral. They interfere with viral replication inside the host cell (inhibiting viral DNA synthesis) and **enhance the cellular immune response against infected hepatocytes**. Also antiproliferative.

**Pegylation** = polyethylene glycol attached → longer half-life, better efficacy, **SC once weekly** instead of 3×/week.

→ **Downstream**: immune-mediated clearance of infected cells → **finite course (48–52 weeks)**, but the response is **not predictable** and ADRs are heavy.

## DRUGS

| DRUG                                         | WHAT IT TREATS                                                                     |
|----------------------------------------------|------------------------------------------------------------------------------------|
| <b>Interferon alfa-2b</b>                    | chronic HBV — the only option <b>approved in children <math>\geq 1</math> year</b> |
| <b>Pegylated interferon alfa-2a (PegIFN)</b> | chronic HBV — <b>FDA first-line</b> · HCV ( <i>historical</i> ) · HDV              |
| <b>Pegylated interferon alfa-2b</b>          | chronic HBV / HCV                                                                  |

## USES

Chronic **HBV** — finite 48–52-week course · **HDV** — **the only available agent** · **HCV** — superseded by DAAs, kept only as background

## ADVERSE EFFECTS

- ★ **Acute influenza-like syndrome** — starts hours after injection: fever, chills, headache, myalgia, arthralgia, nausea, vomiting, diarrhea
- **Depression**
- **Myelosuppression** — granulocytopenia, thrombocytopenia
- Neurotoxicity — somnolence, confusion, behavioural disturbance, rarely seizures
- Autoimmune disorders — thyroiditis, hypothyroidism
- Uncommon: hypotension, tachycardia
- Others: ↑ hepatic enzymes & triglycerides, alopecia, proteinuria/azotemia, interstitial nephritis, autoantibody formation, pneumonia, hepatotoxicity

*Pegylated IFNs are generally better tolerated than standard IFNs.*

## INTERACTIONS

- ↑ **theophylline** and **methadone** levels
- ● + **didanosine** → hepatic failure
- ● + **zidovudine** → cytopenia

## CAUTIONS

- ● **Not in pregnancy** — abortifacient in primates
- ⚠ Monitor **thyroid function** and **hepatic enzymes** throughout therapy
- ⚠ This toxicity is the whole reason IFN is no longer first-line: oral antivirals are more effective **and** better tolerated ★

# HBV Nucleoside / Nucleotide Analogues (NAs)

## MECHANISM

Directly inhibit **HBV DNA polymerase** → **reverse transcriptase**, and after incorporation into the growing viral DNA cause **chain termination**.

*Entecavir specifically blocks 3 polymerase functions: base priming · reverse transcription of the negative strand · synthesis of the positive strand.*

→ **Downstream**: HBV DNA synthesis stops → viral load falls to undetectable — but **cccDNA survives**, so treatment length is individual and **may be lifelong**. Oral · low ADR · high cost.

## BARRIER TO RESISTANCE — THE SORTING PRINCIPLE ★

|                                    | DRUGS                                      | RESISTANCE                                                                       |
|------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------|
| <b>High barrier</b><br>(preferred) | <b>Entecavir · TDF · TAF</b>               | rare                                                                             |
| <b>Low barrier</b>                 | <b>Lamivudine · Adefovir · Telbivudine</b> | LAM 15–30% at 1 yr, <b>65% at 5 yr</b> · ADV up to 29% at 5 yr · TBV 22% at 1 yr |

## DRUGS

| DRUG                                       | WHAT IT TREATS                                                                                             |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Entecavir (ETV)</b>                     | chronic HBV — <b>first-line</b> ; less effective if lamivudine-refractory (50% resistance in 5 yr)         |
| <b>Tenofovir disoproxil fumarate (TDF)</b> | chronic HBV — <b>first-line</b> ; the choice in <b>pregnancy</b>                                           |
| <b>Tenofovir alafenamide (TAF)</b>         | chronic HBV — prodrug of tenofovir, <b>less renal &amp; bone toxicity</b> ; switch here in renal/bone risk |
| <b>Lamivudine (LAM)</b>                    | chronic HBV ( <i>non-preferred</i> ); last 4 weeks of gestation to prevent vertical transmission           |
| <b>Adefovir dipivoxil (ADV)</b>            | chronic HBV ( <i>non-preferred</i> )                                                                       |
| <b>Telbivudine (TBV)</b>                   | chronic HBV ( <i>non-preferred</i> ); <b>not effective in LAM-resistant HBV</b>                            |

## USES

**Chronic HBV** — acute HBV gets supportive care only · Prevent **mother-to-child transmission** · Prevent **HBV reactivation** · Treat HBV-associated extrahepatic manifestations

## ADVERSE EFFECTS

- **Entecavir** — well tolerated: headache, fatigue, dizziness, nausea, upper abdominal pain
- **Tenofovir** — nausea, abdominal pain, diarrhea, dizziness, fatigue; ● **chronic renal insufficiency** · ↓ **bone mineral density** · **Fanconi's syndrome** (*mostly TDF — TAF is the fix*)
- **Adefovir** — ↑ serum creatinine
- **Lamivudine** — excellent safety profile
- **Telbivudine** — mild peripheral neuropathy, myopathy



## CAUTIONS & MONITORING

- ⚠️ **Monitor ALT + serum HBV DNA in every patient**
- ⚠️ Entecavir, adefovir and tenofovir are **renally excreted** — dose-adjust in renal impairment; care with other nephrotoxic drugs
- ⚠️ On TDF or at renal risk: monitor **eGFR + serum phosphate every 3 months**; if renal or bone disease develops, **switch to ETV or TAF**
- ⚠️ Adefovir is embryotoxic in rats and genotoxic preclinically; telbivudine is acceptable in 2nd/3rd trimester in HBeAg+ mothers
- 🚫 **De novo combinations are NOT recommended** — neither two high-barrier NAs, nor NA + PegIFN. The theory is attractive; the evidence of superiority is missing.

## ADMINISTRATION — HBV NUCLEOS(T)IDE ANALOGUES

| DRUG                                | ADMINISTRATION & PK                                                                                                                  | RENAL                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Entecavir                           | 🚫 <b>oral bioavailability 100% — but decreased by food</b> , so it is taken away from meals. $t_{1/2}$ 128–149 h → <b>once daily</b> | excreted by the kidney → <b>dose adjusted in renal insufficiency</b> ; careful with co-administered drugs that reduce renal function |
| Tenofovir alafenamide (TAF)         | <b>25 mg, one tablet once daily</b>                                                                                                  | ⚠️ <b>switch away if eGFR &lt; 15 mL/min/1.73 m<sup>2</sup></b> ( <i>chronic renal failure</i> )                                     |
| Tenofovir disoproxil fumarate (TDF) | first-line <b>in pregnancy</b>                                                                                                       | considered when <b>eGFR &lt; 15</b> and the patient is <b>not yet on renal replacement therapy</b> and cannot take entecavir         |
| Adefovir dipivoxil                  | oral bioavailability <b>59%</b> , protein binding < <b>5%</b> , $t_{1/2}$ 5–18 h → once daily                                        | renally excreted → 🚫 <b>requires dose adjustment in renal dysfunction</b>                                                            |
| Lamivudine                          |                                                                                                                                      | safely given in <b>decompensated liver disease</b>                                                                                   |
| Telbivudine                         | ★ <b>bioavailability unaffected by food</b> , protein binding 3%, $t_{1/2}$ 15 h                                                     | renally excreted                                                                                                                     |

- 🚫 **Entecavir is the one with a food rule and telbivudine is explicitly the one without** — the same class, opposite instructions, so this is not a class habit you can generalise.
- ⚠️ **Monitoring:** ALT and serum HBV DNA in everyone. For anyone at renal risk on any NA, and for **all patients on TDF** — **eGFR and serum phosphate every 3 months**. A patient on TDF developing renal or bone disease should be **switched to entecavir or TAF**.
- 💡 TAF over TDF is a toxicity decision, not a potency one — **less renal and bone toxicity**, same job.
- 💡 Telbivudine given in the **last 4 weeks of gestation** prevents vertical transmission — but it is **teratogenic**, which is the tension worth knowing before anyone quotes that use.

# Ribavirin

## MECHANISM

**Not fully elucidated** — the lecture lists six candidate mechanisms: inhibiting HCV **RdRp** · depleting **GTP** via inosine-5'-monophosphate dehydrogenase inhibition · enhancing viral mutagenesis · shifting T-helper phenotype **Th2** → **Th1** · inducing IFN-stimulated genes · modulating NK-cell response.

→ **Downstream:** broad, non-specific suppression of DNA and RNA viruses. Alone it is weak — **added to PegIFN it raised SVR from 10% to 50%**, which is its whole historical role.

## DRUGS

| DRUG                   | WHAT IT TREATS                                                                                                                                                          |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ribavirin (RBV)</b> | HCV — add-on to PegIFN ( <i>historical</i> ) or to <b>SOF/VEL in cirrhotic patients</b> ; also active against influenza A/B, parainfluenza, RSV, paramyxoviruses, HIV-1 |

## ADVERSE EFFECTS

- **Hemolytic anemia** ★ — dose-dependent, the defining toxicity
- Depression, fatigue, irritability, insomnia
- Rash, pruritus, cough, nausea

## CAUTIONS & CONTRAINDICATIONS

- **Teratogenic and embryotoxic** — pregnancy, or unwillingness to use contraception, is a contraindication
- ⚠ Shared IFN/RBV contraindications: **decompensated cirrhosis (CTP >6)** · prior hypersensitivity · severe uncontrolled depression · organ transplant (*except liver*) · poorly controlled comorbidity — hypertension, diabetes, cardiovascular disease, COPD, thyroid disease

# HCV NS5B Polymerase Inhibitors

## MECHANISM

**Sofosbuvir** is a prodrug of a **uridine analogue**. It is activated inside the host hepatocyte (GS-331007) and inhibits **NS5B, the HCV RNA-dependent RNA polymerase** — the enzyme that copies the viral genome.

→ **Downstream:** chain termination of HCV RNA synthesis · ★ **highest genetic barrier of all DAA classes** — three or more resistance-associated variants (RAVs) are needed for full resistance. Reported escape: **S282T**.

## DRUGS

| DRUG                    | WHAT IT TREATS                                                                                                                                          |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sofosbuvir (SOF)</b> | HCV — <b>all genotypes</b> , always in combination with another DAA; the backbone of Thai therapy ( <b>SOF/VEL 12 weeks</b> , + ribavirin if cirrhotic) |

## USES

**Pan-genotypic HCV**, in combination — never monotherapy

## ADVERSE EFFECTS

- Fatigue, headache, insomnia, pruritus
- ● **Bradycardia when combined with amiodarone** ★

## INTERACTIONS

- ★ **Not a CYP substrate, inhibitor or inducer** → lowest interaction potential of the DAAs
- Substrate of **P-gp and BCRP** → avoid potent inducers of those transporters

## CAUTIONS

- ⚠ Both sofosbuvir and its metabolite are renally handled — **avoid if eGFR <30 mL/min/1.73 m<sup>2</sup>**
- ⚠ A high-fat meal raises AUC by 67–91%

# HCV NS5A Inhibitors

## MECHANISM

Bind **NS5A** — a non-enzymatic protein needed for the replication complex. Daclatasvir binds its **N-terminus** and blocks **both viral RNA replication and virion assembly**.

→ **Downstream:** replication and packaging both stall ·  **low genetic barrier** — **one or two amino-acid substitutions abolish efficacy** (e.g. **Y93H**), so these are only ever used inside a combination.

## DRUGS

| DRUG                                          | WHAT IT TREATS                                                                                      |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Velpatasvir (VEL)</b>                      | HCV <b>genotypes 1–6</b> , fixed-dose with sofosbuvir ( <b>SOF/VEL</b> ) — the Thai standard        |
| <b>Daclatasvir (DCV)</b>                      | HCV <b>genotype 3</b> with SOF · HCV + <b>HIV coinfection</b> any genotype · advanced liver disease |
| <b>Elbasvir (ELB)</b>                         | HCV <b>genotypes 1a/1b and 4</b> , fixed-dose with grazoprevir; approved with HIV coinfection       |
| <b>Ledipasvir · Pibrentasvir · Ombitasvir</b> | HCV — components of other fixed-dose DAA regimens                                                   |

## ADVERSE EFFECTS

- **Headache** (~22%) and **fatigue** (~15%) — the class pattern, and generally mild
- Nausea, asthenia, insomnia
- Diarrhea with SOF/VEL (5–7%, higher in triple regimens)

## INTERACTIONS

-  **Daclatasvir is a CYP3A substrate and P-gp substrate** — dose must be changed around CYP3A modulators: **strong inhibitors** → **reduce dose** · **moderate inducers** → **increase dose** · **strong inducers (rifamycins, St John's Wort, antiepileptics)** → **contraindicated**
- **Velpatasvir** — P-gp substrate and weak inhibitor, weak BCRP inhibitor, OATP1B1/1B3 substrate, CYP3A4/2C8/2B6 substrate
-  Velpatasvir absorption is **pH-dependent** — acid-suppressing drugs matter

## CAUTIONS

-  **Avoid if eGFR <30 mL/min/1.73 m<sup>2</sup>** (velpatasvir)
-  Daclatasvir absorption drops ~23% with a high-fat, high-calorie meal

# HCV NS3/4A Protease Inhibitors

## MECHANISM

Inhibit the **NS3/4A serine protease**, which processes the HCV polyprotein into its functional non-structural proteins — so the replication machinery is never assembled.

→ **Downstream:** viral RNA formation blocked · ● **low genetic barrier** (1–2 substitutions suffice, e.g. **R155K / D168E**) → combination only. *Grazoprevir is the exception in degree — broader genotype coverage and a higher barrier than the others.*

## DRUGS

| DRUG                                                                          | WHAT IT TREATS                                                                                                 |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Simeprevir (SMV)</b>                                                       | HCV <b>genotype 1 or 4</b> , with sofosbuvir — SVR 97% treatment-naïve non-cirrhotic, <b>83% in cirrhotics</b> |
| <b>Grazoprevir (GRA)</b>                                                      | HCV <b>genotypes 1a/1b and 4</b> , fixed-dose with elbasvir; approved with HIV coinfection                     |
| <b>Voxilaprevir · Glecaprevir · Paritaprevir</b> ( <i>ritonavir-boosted</i> ) | HCV — components of fixed-dose salvage / pan-genotypic regimens                                                |

## ADVERSE EFFECTS

- Headache (~17%), fatigue (~16%), nausea (9–15%)
- ★ **Simeprevir: photosensitivity, rash, itching** — it carries a **sulfonamide moiety**
- **Grazoprevir/elbasvir: ALT elevations up to >5× ULN**
- Ritonavir-boosted paritaprevir/ombitasvir + dasabuvir: pruritus, asthenia, insomnia, **transient ↑ indirect bilirubin**

## INTERACTIONS

- ⚠ **Simeprevir** — CYP3A4-metabolised; substrate of P-gp, MRP2, BCRP, OATP1B1/3, OATP2B1; mild inhibitor of CYP1A2 and intestinal CYP3A
- ● DAA interactions are **numerous and complex**, especially with **HIV drugs and cardiovascular drugs** — check every regimen in an HIV/HCV coinfecting patient

## CAUTIONS

- ⚠ **Avoid if eGFR <30 mL/min/1.73 m<sup>2</sup>** (simeprevir)
- ⚠ **Adherence is the single biggest determinant of cure** — educate the patient before starting

## ADMINISTRATION — HCV REGIMENS AND DOSES (THAI GUIDELINE)

**12 weeks of treatment, with or without HIV:** - **No cirrhosis:** sofosbuvir 400 mg + velpatasvir 100 mg, once daily - **With cirrhosis:** sofosbuvir 400 mg + velpatasvir 100 mg once daily + ribavirin twice daily

**Ribavirin dose is by body weight:**

| BODY WEIGHT | DAILY DOSE   | FREQUENCY   |
|-------------|--------------|-------------|
| < 75 kg     | 1,000 mg/day | twice daily |
| > 75 kg     | 1,200 mg/day | twice daily |

**Simeprevir: 150 mg tablet, once daily — with food.** Bioavailability 62%, 99.9% protein bound,  $t_{1/2}$  10–13 h; CYP3A4-metabolised and a P-gp/OATP substrate. ● **Should not be used if eGFR < 30 mL/min/1.73 m<sup>2</sup>.**

- ★ **Cirrhosis doesn't change the backbone, it adds ribavirin** — and ribavirin is the only weight-banded part of the regimen.
- 💡 Simeprevir carries a **sulfonamide moiety**, which is why photosensitivity, rash and itching show up — sun protection belongs in the counselling, not just the adverse-effect list.

# Putting It Together

## GENETIC BARRIER TO RESISTANCE — WHY HCV IS ALWAYS COMBINATION THERAPY ★

| BARRIER | CLASS                                                           | WHAT IT TAKES TO BREAK            |
|---------|-----------------------------------------------------------------|-----------------------------------|
| High    | Nucleotide NS5B inhibitors ( <i>sofosbuvir</i> )                | ≥3 resistance-associated variants |
| Low     | Non-nucleoside NS5B · NS3 protease inhibitors · NS5A inhibitors | 1–2 amino-acid substitutions      |

## STANDARD THERAPY VS DAAS

|              | STANDARD (IFN / PEGIFN / RBV) | DAAS                             |
|--------------|-------------------------------|----------------------------------|
| specificity  | non-specific                  | <b>specific</b> for HCV proteins |
| duration     | 24–48 weeks                   | <b>12–24 weeks</b>               |
| route        | injection                     | <b>oral</b>                      |
| tolerability | severe side effects           | <b>much better</b>               |

## THE SIX LINES TO WALK OUT WITH

**Interferon is largely replaced — toxicity · HBV = long-term suppression, not cure · HCV = DAAs cure most patients · Tenofovir & entecavir are HBV first-line · During therapy assess renal function · liver function · drug–drug interactions · Adherence matters more than the regimen choice**

Source: Pharmacology Lecture 2026\_Anti-Hepatitis Agents\_Vilasinee (1).pdf — รศ.ดร. วิลาสินี หิรัญพานิชชา โตะ, เกสัชวิทยา 2 (Pharmacology II), 61 slides. Structures omitted.

# Antivirals & Anti-HIV Drugs

Viruses are **obligate intracellular parasites** with no metabolic machinery of their own — they run on the host cell. So every antiviral is a hunt for a **viral protein different enough from a human one** to hit selectively. All current antivirals are **virustatic**: they suppress replication reversibly and **do not touch latent virus**.

**attachment & entry → uncoating → genome replication → integration (*retrovirus*) → protein maturation → assembly & release**

## WHY ANTIVIRAL THERAPY IS HARD

- **Rapid replication** and **rapid mutation** → resistance appears fast
- The virus uses host machinery → hitting it often injures the host cell = **toxicity**
- **Selectivity of antiviral action** = the whole game: exploit the differences between viral and human proteins

## WHERE EACH CLASS HITS THE LIFE CYCLE

| STEP                          | DRUG CLASSES                                                                          |
|-------------------------------|---------------------------------------------------------------------------------------|
| <b>Attachment &amp; entry</b> | HIV entry inhibitors ( <i>maraviroc, enfuvirtide</i> ) · docosanol                    |
| <b>Uncoating</b>              | M2 proton channel inhibitors ( <i>amantadine, rimantadine</i> )                       |
| <b>Genome replication</b>     | nucleoside analogues · non-nucleoside polymerase inhibitors · NRTIs · NNRTIs · INSTIs |
| <b>Maturation</b>             | HIV protease inhibitors                                                               |
| <b>Release</b>                | neuraminidase inhibitors                                                              |
| <b>Immune modulation</b>      | imiquimod · palivizumab                                                               |

## NUCLEOSIDE ANALOGUES — THE SHARED LOGIC

Most drugs that block viral genome replication inhibit a **polymerase**. Every nucleoside analogue is a **prodrug**: it must be phosphorylated (usually to the **triphosphate**) to mimic a natural dNTP.

Two antiviral actions once activated: 1. **Enzyme inhibition** — competes with the natural triphosphate substrate for the polymerase 2. **Chain termination** — gets incorporated into the growing DNA strand and stops elongation

→ **Downstream**: selectivity is decided at **two** points — the **kinase** (does a *viral* enzyme phosphorylate it far better than a host one?) and the **polymerase** (is *viral* DNA synthesis inhibited far harder than host?). ★

**Acyclovir is the most selective** — selective at both steps. **AZT is the least** — non-selective at activation.



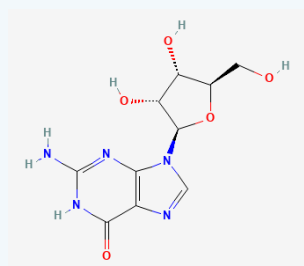
# Antiherpes Agents

## MECHANISM

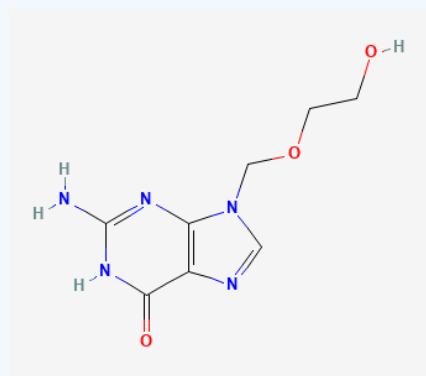
Nucleoside analogues phosphorylated by **viral thymidine kinase** (*acyclovir*, *penciclovir*) — so they are only activated **inside infected cells** — then by host kinases to the triphosphate, which inhibits **viral DNA polymerase**. Trifluridine differs: it is phosphorylated by **host** kinases, which is why it is more toxic and stays topical.

→ **Downstream:** viral DNA polymerase blocked + chain termination → no new viral genome. Latent virus is untouched, so **recurrence still happens**.

## THE SHARED CORE • WHAT CHANGES

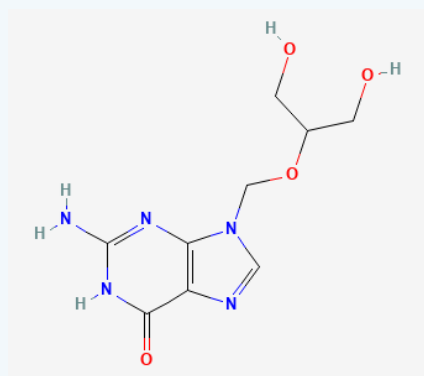


**Guanosine — the real nucleoside.** note the complete sugar ring. ● the drugs below keep the base and throw most of the sugar away.



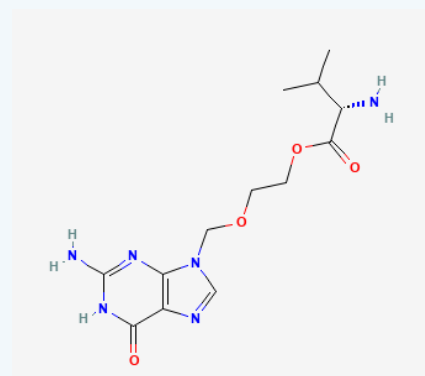
### Aciclovir

an open, incomplete sugar with no 3'-OH → chain terminator.  
Selectivity comes from viral thymidine kinase doing the first phosphorylation — uninfected cells can't



### Ganciclovir

one extra hydroxymethyl → CMV activity, and much more marrow toxicity



### Valaciclovir

aciclovir + a valine ester → carried by peptide transporters → 3–5× the oral bioavailability

## DRUGS

| DRUG                | WHAT IT TREATS                                                                            |
|---------------------|-------------------------------------------------------------------------------------------|
| <b>Acyclovir</b>    | HSV-1 · HSV-2 · VZV                                                                       |
| <b>Valacyclovir</b> | same — the L-valyl <b>prodrug of acyclovir</b> , better oral bioavailability, fewer doses |
| <b>Penciclovir</b>  | HSV-1 · HSV-2 · VZV — topical                                                             |
| <b>Famciclovir</b>  | same — <b>prodrug of penciclovir</b> , oral                                               |
| <b>Trifluridine</b> | HSV <b>keratitis</b> — ophthalmic, incl. <b>acyclovir-resistant</b> HSV                   |
| <b>Docosanol</b>    | HSV — topical; mechanism unknown ( <i>proposed: blocks HSV envelope–membrane fusion</i> ) |

## USES

**HSV** — genital herpes, mucocutaneous herpes, herpes encephalitis · **VZV** — chickenpox, shingles (*oral if immunocompetent · IV if immunocompromised*) · **Prophylaxis** — before immunosuppressants or radiotherapy, high reactivation risk, frequent genital recurrences

Goals: relieve symptoms · shorten the course · prevent complications & recurrence · reduce transmission.

## ADVERSE EFFECTS

- **Acyclovir / valacyclovir:** renal failure (*IV*), thrombocytopenia · **encephalopathic changes**, dizziness, agitation · GI upset
- **Penciclovir / famciclovir:** erythema multiforme · GI upset, headache
- **Trifluridine:** eye irritation, lacrimation, light intolerance
- **Docosanol:** headache, application-site reaction

## CAUTIONS

- ⚠️ Acyclovir  $t_{1/2}$  2–3 h — **adjust the dose in renal impairment**
- ⚠️ Trifluridine is an early, **more toxic** anti-HSV drug — reserve it for the eye

## ADMINISTRATION — ANTIHERPES AGENTS

| DRUG         | ROUTES & SCHEDULE                                                    |
|--------------|----------------------------------------------------------------------|
| Acyclovir    | oral (F 15–20%) — 3–5 times/day · IV · topical 5% cream, 5 times/day |
| Valacyclovir | oral (F 60–70%) — 2–3 times/day                                      |
| Docosanol    | topical 10% cream, every 2 hours                                     |

- ★ **Valacyclovir exists purely to fix acyclovir's bioavailability.** It is the L-valyl ester prodrug; triple the absorption means **twice or three times a day instead of five.**
- ⚠️ Acyclovir's half-life is **2–3 hours** and ● **the dose must be adjusted in renal impairment** — renal failure is its IV-route adverse effect.
- 💡 Five doses a day is the reason acyclovir gets missed; if adherence is the problem, the prodrug is the answer rather than a reminder chart.

# Anti-CMV Agents

## MECHANISM

CMV disease is nearly always **reactivation under deep immunosuppression** (HIV with **CD4 < 50**), and the classic lesion is **CMV retinitis**. Three routes to the same target — **viral DNA polymerase**:

1. **Ganciclovir** — phosphorylated by **viral kinase**, then to triphosphate
2. **Cidofovir** — phosphorylated by **cellular** enzymes (*so it works even when the viral kinase is mutated*)
3. **Foscarnet** — **not** a nucleoside; mimics the **pyrophosphate** product and blocks the pyrophosphate-binding site directly, **no phosphorylation needed**

→ **Downstream**: CMV DNA synthesis halted. ★ Because foscarnet and cidofovir skip the viral kinase, they still work against **nucleoside-analogue-resistant** CMV.

## DRUGS

| DRUG                  | WHAT IT TREATS                                                                                          |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| <b>Ganciclovir</b>    | CMV retinitis — IV                                                                                      |
| <b>Valganciclovir</b> | same — the L-valyl <b>prodrug of ganciclovir</b> , oral, once daily                                     |
| <b>Foscarnet</b>      | CMV retinitis — incl. <b>nucleoside-analogue-resistant</b> CMV                                          |
| <b>Cidofovir</b>      | CMV retinitis — long $t_{1/2}$ , <b>once weekly</b> IV                                                  |
| <b>Fomivirsen</b>     | CMV retinitis, <b>2nd line</b> — intravitreal; designed as an antisense nucleotide, mechanism uncertain |

## ADVERSE EFFECTS

- **Ganciclovir**: ● **bone-marrow suppression** — neutropenia, thrombocytopenia, anemia · fever, phlebitis
- **Foscarnet**: ● **nephrotoxicity (dose-dependent)** · **electrolyte imbalance** —  $\text{Ca}^{2+}$ ,  $\text{PO}_4^{3-}$ ,  $\text{K}^+$ ,  $\text{Mg}^{2+}$  · CNS — headache, hallucination, **seizure** · anemia, fever, GI upset
- **Cidofovir**: ● **nephrotoxicity (dose-dependent)** · neutropenia, metabolic acidosis, ↓ **intraocular pressure** · GI upset, headache, rash
- **Fomivirsen**: ocular inflammation · transiently ↑ **intraocular pressure**

## INTERACTIONS

- ● **Foscarnet + nephrotoxic drugs** (*amphotericin B, aminoglycosides*) — contraindicated
- ● **Foscarnet + imipenem** — seizure risk
- **Cidofovir + any nephrotoxic agent** — contraindicated
- **Fomivirsen** — do not give within **2–4 weeks** of IV or intravitreal cidofovir (*ocular inflammation*)

## CAUTIONS & CONTRAINDICATIONS

- ● **Ganciclovir contraindicated** in severe neutropenia or severe thrombocytopenia
- ⚠ **Foscarnet** — **renal impairment is the dose-limiting toxicity**
- ● **Cidofovir contraindicated** in renal insufficiency

## ADMINISTRATION — ANTI-CMV AGENTS

| DRUG           | ROUTE                                                                       |
|----------------|-----------------------------------------------------------------------------|
| Ganciclovir    | IV                                                                          |
| Valganciclovir | oral (F 60%) — <b>once daily</b> , the l-valyl ester prodrug of ganciclovir |
| Foscarnet      | IV                                                                          |

- ★ Same trick as valacyclovir: **the valyl prodrug is what turns an IV drug into a once-daily tablet.**
- ● **Foscarnet's dose-limiting toxicity is renal impairment**, and it is contraindicated alongside other nephrotoxic drugs (amphotericin B, aminoglycosides) or imipenem (seizure risk). It is the agent kept for CMV that is **resistant to the nucleoside analogues**.
- 💡 Ganciclovir is contraindicated in severe neutropenia or thrombocytopenia — so the blood count, not the viral load, is often what decides which of these two a patient can actually have.

# Influenza — M2 Proton Channel Inhibitors

## MECHANISM

Block **M2**, the viral proton channel that acidifies the interior of the virion. That acidification is required to dissociate the **matrix protein from the viral ribonucleoprotein**.

→ **Downstream: uncoating fails** → the genome never reaches the nucleus. Influenza **A only** — B has no M2.

## DRUGS

| DRUG               | WHAT IT TREATS                                                  |
|--------------------|-----------------------------------------------------------------|
| <b>Amantadine</b>  | influenza A                                                     |
| <b>Rimantadine</b> | influenza A — <b>fewer neurological effects</b> than amantadine |

## ADVERSE EFFECTS

- **Neuroleptic malignant syndrome (NMS)**
- Exacerbation of mental disorders, confusion, insomnia, hallucination
- Orthostatic hypotension, peripheral edema, GI upset

## CAUTIONS

- ⚠ **Largely obsolete** — supplanted by the neuraminidase inhibitors (*widespread resistance + the CNS profile*)

# Influenza — Neuraminidase Inhibitors

## MECHANISM

Inhibit viral **neuraminidase**, the enzyme that cleaves sialic acid so newly made virions can detach from the host cell surface.

→ **Downstream:** new virions stay **stuck to the infected cell** → no spread to neighbouring cells. Influenza **A and B**. ★ Most effective if started **within 48 h of symptom onset** — true for every influenza antiviral.

## DRUGS

| DRUG               | WHAT IT TREATS                                            |
|--------------------|-----------------------------------------------------------|
| <b>Oseltamivir</b> | influenza A & B — oral; <b>treatment + prophylaxis</b>    |
| <b>Zanamivir</b>   | influenza A & B — inhaled; <b>treatment + prophylaxis</b> |
| <b>Peramivir</b>   | influenza A & B — IV; <b>treatment only</b>               |

## ADVERSE EFFECTS

- **Bronchospasm, respiratory depression** — the reason the inhaled route is a problem in asthma/COPD
- GI upset, headache, nasal symptoms

## ADMINISTRATION — INFLUENZA ANTIVIRALS, TREATMENT VS PROPHYLAXIS

| DRUG               | ADULT TREATMENT                                                     | ADULT PROPHYLAXIS                     |
|--------------------|---------------------------------------------------------------------|---------------------------------------|
| <b>Baloxavir</b>   | 40–80 kg: 40 mg once · > 80 kg: 80 mg once                          | ● <b>none</b>                         |
| <b>Oseltamivir</b> | 75 mg capsule twice daily × 5 days                                  | 75 mg capsule daily × 10 days         |
| <b>Zanamivir</b>   | 10 mg (2 inhalations) twice daily × 5 days                          | 10 mg (2 inhalations) daily × 10 days |
| <b>Rimantadine</b> | 200 mg/day in 1–2 doses × 7 days                                    | 200 mg/day in 1–2 doses               |
| <b>Amantadine</b>  | 200 mg/day in 1–2 doses <b>until 24–48 h after symptoms resolve</b> | same as treatment dose                |

- ★ **The pattern is identical across the class: prophylaxis is the same daily amount given once instead of twice, for twice as long.** Treatment 5 days twice daily → prophylaxis 10 days once daily.
- ★ **Baloxavir is a single dose, weight-banded, and has no prophylaxis role at all.**
- 💡 Zanamivir being inhaled is the practical catch — it is the wrong drug for someone who can't use an inhaler, and it is not the one to reach for in bronchospastic disease.

# Influenza — Polymerase Complex Inhibitors

## MECHANISM

The influenza polymerase is a **heterotrimer: PB1 + PB2 + PA**. Two drugs, two different subunit attacks:

1. **Baloxavir** — inhibits the **cap-dependent endonuclease** (PA subunit) used for "**cap snatching**," which influenza needs to initiate its own mRNA synthesis
2. **Favipiravir** — activated to **favipiravir-RTP**, which **mimics both A and G** and inhibits the **RNA-dependent RNA polymerase**

→ **Downstream:** viral mRNA/genome synthesis blocked. Both cover influenza A & B; favipiravir is **broad-spectrum against RNA viruses** (West Nile, yellow fever, flaviviruses, arenaviruses, bunyaviruses, alphaviruses, foot-and-mouth).

## DRUGS

| DRUG                      | WHAT IT TREATS                                                                                       |
|---------------------------|------------------------------------------------------------------------------------------------------|
| <b>Baloxavir marboxil</b> | influenza A & B — prodrug → <b>baloxavir acid</b> ; $t_{1/2} \sim 96$ h → ★ <b>single dose, once</b> |
| <b>Favipiravir</b>        | influenza A & B + broad RNA-virus coverage — prodrug → favipiravir-RTP                               |

## ADVERSE EFFECTS

- **Baloxavir:** diarrhea, bronchitis, headache, nausea
- **Favipiravir:** diarrhea, nausea, vomiting, ↑ **uric acid**
- ● **Favipiravir is teratogenic**

# Other Antivirals

## MECHANISM

Three unrelated agents grouped by "not herpes, not flu, not HIV":

- **Ribavirin** — nucleoside analogue, two proposed actions: inhibits **inosine monophosphate dehydrogenase** (↓ cellular GTP) **and** inhibits **viral RNA polymerase** by mimicking G
  - **Palivizumab** — humanised **monoclonal antibody** against the **A antigenic site on the RSV F (fusion) protein**
  - **Imiquimod** — **TLR7 agonist**: boosts innate immunity, including interferon secretion
- **Downstream**: ribavirin starves and jams RNA synthesis · palivizumab neutralises RSV before fusion · imiquimod makes the **host** clear the lesion rather than attacking the virus directly.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                         |
|--------------------|--------------------------------------------------------------------------------------------------------|
| <b>Ribavirin</b>   | RSV · HCV ( <i>with interferon</i> )                                                                   |
| <b>Palivizumab</b> | RSV prophylaxis in <b>high-risk infants</b> (premature) — IM                                           |
| <b>Imiquimod</b>   | HPV · <b>genital warts</b> (condyloma acuminatum) · basal cell carcinoma · actinic keratosis — topical |

## ADVERSE EFFECTS

- **Ribavirin**: ● **hemolytic anemia**, thrombocytopenia · bradyarrhythmia, hypotension · pancreatitis, hepatotoxicity · bacterial infection, **suicidality** · rash, GI upset, headache, conjunctivitis, fatigue
- **Palivizumab**: URTI, fever, rhinitis, cough, otitis media · rash, diarrhea, vomiting, ↑ aminotransferases
- **Imiquimod**: skin irritation — erythema, superficial erosion/crusting, burning

## CAUTIONS & CONTRAINDICATIONS

- ● **Ribavirin contraindicated**: cardiac disease · **CrCl < 50 mL/min** · severe hepatic impairment



# HIV — Antiretroviral Therapy (ART)

HIV is an **RNA retrovirus**: reverse transcriptase makes a DNA copy of the viral RNA, integrase splices it into the **host genome**, and that provirus directs new virus for life. That's why treatment is lifelong and why **latency is untouchable** — you can only suppress replication.

## THE 5 ANTIRETROVIRAL CLASSES

| CLASS                            | TARGET                                           | WHAT IT STOPS                 |
|----------------------------------|--------------------------------------------------|-------------------------------|
| <b>NRTIs</b>                     | reverse transcriptase ( <i>substrate site</i> )  | genome replication            |
| <b>NNRTIs</b>                    | reverse transcriptase ( <i>allosteric site</i> ) | genome replication            |
| <b>PIs</b> ( <i>-navir</i> )     | HIV-1 protease                                   | viral <b>maturation</b>       |
| <b>Entry inhibitors</b>          | CCR5 · gp41                                      | <b>attachment &amp; entry</b> |
| <b>INSTIs</b> ( <i>-gravir</i> ) | integrase                                        | genome <b>integration</b>     |

## GOALS & MONITORING

Decrease morbidity & mortality · improve quality of life · **restore and preserve immune function** · prevent transmission by driving **HIV RNA to undetectable** · Monitor **plasma viral load + CD4<sup>+</sup> count** · start **before immunodeficiency is evident** · change the regimen if viral load rises

## TRANSMISSION & PRESENTATION

- **3 routes**: sexual · parenteral · perinatal. Condoms cut transmission **~20-fold**
- **Acute**: fever, sore throat, fatigue, weight loss, myalgia · **maculopapular rash on the trunk in 40–80%** · diarrhea, N/V · lymphadenopathy, night sweats · aseptic meningitis in  $\sim\frac{1}{4}$
- ★ **Opportunistic infections track the CD4 count** — e.g. **CMV retinitis at CD4 < 50**

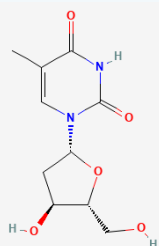
# NRTIs — Nucleoside/Nucleotide RT Inhibitors

## MECHANISM

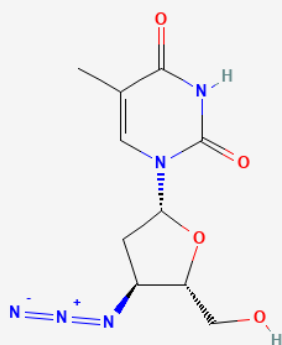
Prodrugs. Phosphorylated intracellularly to the **triphosphate**, which then does both jobs at **reverse transcriptase**: competes with the natural dNTP, and gets **incorporated into the growing DNA chain**. They lack a 3'-OH, so the chain cannot extend.

→ **Downstream: chain termination** → no proviral DNA made → nothing to integrate.

## THE SHARED CORE · WHAT CHANGES

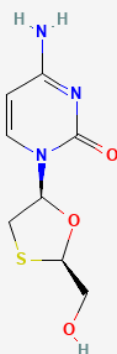


**Thymidine — the real nucleoside.** ● every NRTI is one of these with the 3'-OH removed or replaced. No 3'-OH means the next nucleotide cannot attach → the chain terminates.



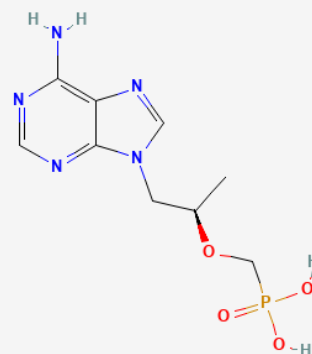
**Zidovudine (AZT)**

3'-OH replaced by an azide → chain terminator



**Lamivudine**

the 3' carbon itself swapped for sulfur



**Tenofovir**

a nucleotide — it arrives already carrying a phosphate, skipping the first activation step

## DRUGS

| DRUG                          | WHAT IT TREATS                                                  |
|-------------------------------|-----------------------------------------------------------------|
| <b>Abacavir (ABC)</b>         | HIV — ● fatal hypersensitivity if <b>HLA-B*5701</b> positive    |
| <b>Zidovudine (AZT / ZDV)</b> | HIV — the <b>first</b> HIV drug; macrocytic anemia, neutropenia |
| <b>Lamivudine (3TC)</b>       | HIV — the best-tolerated backbone partner                       |
| <b>Emtricitabine (FTC)</b>    | HIV — skin hyperpigmentation of palms & soles                   |
| <b>Tenofovir (TDF / TAF)</b>  | HIV — renal toxicity, osteomalacia                              |
| <b>Didanosine (ddI)</b>       | HIV — peripheral neuropathy, pancreatitis                       |
| <b>Stavudine (d4T)</b>        | HIV — peripheral neuropathy, <b>lipodystrophy</b>               |
| <b>Zalcitabine (ddC)</b>      | HIV — peripheral neuropathy, oral ulceration                    |

## ADVERSE EFFECTS

- **Abacavir hypersensitivity reaction — can be fatal.** Test **HLA-B\*5701** before starting (*incidence is low in Asians*). Watch the **first 6 weeks**; **≥2 of** fever · rash · GI symptoms · fatigue/myalgia · respiratory symptoms → **stop immediately and NEVER rechallenge**
- **Peripheral neuropathy** — ddl, d4T, ddC (*the "d-drugs"*)
- **Pancreatitis** — ddl, d4T, ddC, AZT · 3TC in children
- **Bone marrow** — AZT: macrocytic anemia, neutropenia
- **Metabolic** — lactic acidosis, hyperlipidemia, lipodystrophy, DM (*AZT, d4T*) · hyperuricemia (*ddl*)
- **Renal/bone** — TDF: renal toxicity, osteomalacia, hepatotoxicity
- GI upset, headache, rash across the class · possible **MI** with ABC and ddl

## INTERACTIONS

- **Do not stack neuropathic drugs** — ddl + d4T + ddC + **isoniazid**
- **AZT + myelosuppressive drugs** (*ganciclovir, ribavirin*) — additive marrow suppression
- **ddl** — avoid ribavirin and alcohol · **ABC** — avoid alcohol (*metabolised by alcohol dehydrogenase*)

## CAUTIONS

- **Adjust the dose in renal impairment** — ABC aside, most of the class
- **ABC contraindicated in cirrhosis, Child-Pugh 7–12**; adjust in liver impairment

# NNRTIs — Non-Nucleoside RT Inhibitors

## MECHANISM

Bind an **allosteric pocket near — not at — the catalytic site** of reverse transcriptase, locking the enzyme so it cannot join deoxyribonucleotides to the primer-template strand.

→ **Downstream:** reverse transcription blocked. ★ **No phosphorylation needed** — they are not nucleoside analogues, and their structures look nothing like one. HIV-1 only, and a **single point mutation kills the whole class** — never use them alone.

## DRUGS

| DRUG                     | WHAT IT TREATS                                             |
|--------------------------|------------------------------------------------------------|
| <b>Efavirenz (EFV)</b>   | HIV — ★ <b>CNS effects</b> ; bedtime dosing, empty stomach |
| <b>Nevirapine (NVP)</b>  | HIV — ● serious <b>hepatotoxicity</b>                      |
| <b>Rilpivirine (RPV)</b> | HIV — insomnia, depression                                 |
| <b>Etravirine (ETR)</b>  | HIV — <b>after a meal</b> ; for NNRTI-resistant virus      |
| <b>Delavirdine (DLV)</b> | HIV — rash, ↑ liver enzymes                                |

## ADVERSE EFFECTS

- **Rash — the class effect**, all five
- ↑ **liver enzymes** · ● **nevirapine: serious hepatotoxicity / hepatitis**
- ● **Efavirenz CNS toxicity** — psychiatric symptoms, depression, **suicidal ideation** (*dose at bedtime to blunt it*)
- **Rilpivirine:** insomnia, depression
- Headache, N/V, diarrhea across the class

## INTERACTIONS

- ● **Concurrent use of CYP3A4-metabolised drugs is contraindicated for ALL NNRTIs** ★
- **Rilpivirine + proton pump inhibitors** — avoid (*absorption needs acid*)
- **Etravirine** — do not combine with another NNRTI
- Metabolism: EFV & NVP **2B6 + 3A4** · RPV & DLV **3A4** · ETR **3A4, 2C9, 2C19**

## CAUTIONS

- ⚠ **Nevirapine** — adjust in hepatic insufficiency

# Protease Inhibitors (PIs, -navir)

## MECHANISM

Peptidomimetics that inhibit **HIV-1 protease**, the enzyme that cleaves the two precursor polyproteins into the smaller functional fragments a virion needs.

→ **Downstream**: virions are still produced but stay **immature and non-infectious** — this class acts **late**, at maturation, unlike everything else. ★ Almost all are **CYP3A4 substrates**, which is exactly why they get **boosted** with ritonavir or cobicistat.

## DRUGS

| DRUG                               | WHAT IT TREATS                                                    |
|------------------------------------|-------------------------------------------------------------------|
| <b>Atazanavir (ATV)</b>            | HIV — indirect hyperbilirubinemia; <b>PR / QTc prolongation</b>   |
| <b>Darunavir (DRV)</b>             | HIV — a preferred boosted PI; avoid in sulfa allergy              |
| <b>Lopinavir/ritonavir (LPV/r)</b> | HIV — co-formulated, boosted                                      |
| <b>Ritonavir (RTV)</b>             | HIV — mostly used <b>as the booster</b> , not as the active agent |
| <b>Indinavir (IDV)</b>             | HIV — ● <b>nephrolithiasis</b>                                    |
| <b>Saquinavir (SQV)</b>            | HIV — QT prolongation, photosensitivity                           |
| <b>Fosamprenavir (FPV)</b>         | HIV — prodrug of amprenavir                                       |
| <b>Nelfinavir (NFV)</b>            | HIV — GI-heavy, unboosted                                         |
| <b>Tipranavir (TPV)</b>            | HIV — for resistant virus; <b>bleeding risk</b>                   |

## ADVERSE EFFECTS

- **GI is the class signature** — nausea, vomiting, diarrhea, abdominal pain
- **Metabolic** — hypertriglyceridemia, hypercholesterolemia, hyperlipidemia
- ↑ **liver enzymes**, hepatitis · rash, headache
- ● **Indinavir: nephrolithiasis** (*drink ≥ 48 oz fluid daily*) + indirect hyperbilirubinemia
- ● **Atazanavir: PR/QTc prolongation** · **Saquinavir: QT prolongation**
- **Tipranavir**: bleeding risk · **Atazanavir**: peripheral neuropathy

## INTERACTIONS

- ★ **CYP3A4 everywhere** — substrate *and* (for ritonavir/cobicistat) potent inhibitor. Expect interactions with everything CYP3A4-handled
- **Atazanavir** — separate from **cimetidine and other acid-reducing agents by 12 h**
- Food matters: take **with food** (*ATV, DRV, LPV/r, NFV, RTV, TPV*) · **empty stomach** for **indinavir** · avoid high-fat meals with **fosamprenavir**

## CAUTIONS

- ⚠ **Sulfa allergy** — avoid **darunavir** and **tipranavir**
- ⚠ **Hepatic insufficiency** — dose-adjust ATV, IDV, LPV/r, FPV, SQV; **avoid tipranavir entirely**
- ⚠ **Tipranavir** — not for patients at risk of bleeding

## ADMINISTRATION — PROTEASE INHIBITORS, AND THE FOOD RULE THAT DIFFERS FOR EACH

| DRUG                       | ADULT DOSAGE                                                    | FOOD                                     | OTHER                                                                                                               |
|----------------------------|-----------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Atazanavir</b>          | 400 mg once daily<br><b>with ritonavir 100 mg once daily</b>    | <b>with food</b>                         | ● separate from cimetidine and other acid-reducing agents <b>by 12 h</b> · adjust in hepatic insufficiency · CYP3A4 |
| <b>Darunavir</b>           | 600 mg twice daily<br><b>with ritonavir 100 mg twice daily</b>  | <b>with food</b>                         | avoid in <b>sulfa allergy</b> · CYP3A4                                                                              |
| <b>Nelfinavir</b>          | 750 mg three times daily                                        | <b>with food</b>                         |                                                                                                                     |
| <b>Indinavir</b>           | 800 mg twice daily<br><b>with ritonavir 100 mg twice daily</b>  | ● <b>best on an EMPTY stomach</b>        | <b>drink ≥ 48 oz of fluid daily</b> ( <i>nephrolithiasis</i> ) · adjust in hepatic insufficiency                    |
| <b>Lopinavir/ritonavir</b> | 400/100 mg twice daily                                          | <b>with food</b>                         | adjust in hepatic insufficiency · CYP3A4                                                                            |
| <b>Fosamprenavir</b>       | 1400 mg twice daily<br><b>with ritonavir 100 mg twice daily</b> | ● <b>avoid concurrent high-fat meals</b> | prodrug of amprenavir · adjust in hepatic insufficiency                                                             |
| <b>Ritonavir</b>           | 600 mg twice daily<br>( <i>full dose</i> )                      | <b>with food</b>                         |                                                                                                                     |
| <b>Saquinavir</b>          | 1000 mg twice daily<br><b>with ritonavir 100 mg twice daily</b> | <b>within 2 h of a full meal</b>         | <b>use sunscreen</b> — photosensitivity · adjust in hepatic insufficiency                                           |
| <b>Tipranavir</b>          | 500 mg twice daily<br><b>with ritonavir 200 mg twice daily</b>  | <b>with food</b>                         | ● <b>avoid in hepatic insufficiency</b> · avoid in sulfa allergy · <b>not for patients at bleeding risk</b>         |

- **"Take with food" is the class rule and indinavir is the exception** — empty stomach, plus at least 48 oz of fluid a day to keep it from forming stones. Fosamprenavir sits in between: food is fine, a high-fat meal is not.
- ★ **Almost every one of these is boosted with ritonavir**, at 100 mg twice daily in most cases but **200 mg twice daily with tipranavir** and once daily with atazanavir. The booster dose is not a constant.
- 💡 The atazanavir–antacid rule is the one that bites in a pharmacy: **12 hours of separation**, so a patient self-treating reflux with an over-the-counter acid reducer can silently lose the drug.
- 💡 Hepatic insufficiency changes the dose for most of this class and rules tipranavir out entirely — worth checking before anything else here.

# Entry Inhibitors

## MECHANISM

HIV entry is a two-handshake process: **gp120 binds CD4 + a chemokine co-receptor (CCR5 or CXCR4)**, then **gp41** drives fusion of the viral envelope with the host membrane. One drug per handshake:

1. **Maraviroc** — blocks the **CCR5** chemokine receptor
2. **Enfuvirtide** — a peptide that blocks **gp41-mediated fusion**

→ **Downstream:** the virus never gets inside — the only class that acts **before** the cell is infected. ⚠️

Maraviroc **only works on CCR5-tropic strains**; it does nothing against CXCR4-tropic virus, so tropism must be tested first.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                |
|--------------------|-----------------------------------------------------------------------------------------------|
| <b>Maraviroc</b>   | HIV — <b>CCR5-tropic only</b> ; for detectable viral load or <b>multidrug-resistant</b> virus |
| <b>Enfuvirtide</b> | HIV — a peptide, so <b>parenteral twice daily</b> ; for virus uncontrolled on other regimens  |

## ADVERSE EFFECTS

- **Maraviroc:** hepatitis · **myocardial infarction**, orthostatic hypotension · rash, dizziness, ↑ **infection risk (URI)**, fever, cough
- **Enfuvirtide:** renal insufficiency · thrombocytopenia, neutropenia, eosinophilia · peripheral neuropathy, conjunctivitis

## CAUTIONS

- ⚠️ Both are **add-on** agents — always combined with other antiretrovirals, never alone
- Maraviroc is a **CYP3A4** substrate

# INSTIs — Integrase Strand Transfer Inhibitors (-gravir)

## MECHANISM

Block **integrase** at the **strand-transfer** step — the point where the viral DNA ends (the **LTRs**) are spliced into host chromosomal DNA.

→ **Downstream:** no provirus, so the infected cell never becomes a permanent virus factory. Fast viral-load drop, few interactions (*most are cleared by UGT1A1 glucuronidation, not CYP*) — which is why **DTG is Thailand's first-line anchor drug**.

## DRUGS

| DRUG                      | WHAT IT TREATS                                               |
|---------------------------|--------------------------------------------------------------|
| <b>Dolutegravir (DTG)</b> | HIV — ★ <b>first-line in Thailand</b> , once daily           |
| <b>Bictegravir (BIC)</b>  | HIV — co-formulated with <b>FTC + tenofovir alafenamide</b>  |
| <b>Raltegravir (RAL)</b>  | HIV — twice daily; the widest ADR list of the class          |
| <b>Elvitegravir (EVG)</b> | HIV — co-formulated with <b>cobicistat + tenofovir + FTC</b> |

## ADVERSE EFFECTS

- **Class:** insomnia, headache · **hypersensitivity reaction** · ↑ liver enzymes · GI upset
- ● **Raltegravir:** suicidal risk · muscle aches, ↑ **creatinine kinase, rhabdomyolysis** · renal failure
- **Elvitegravir:** renal impairment · **decreased bone mineral density**
- **Bictegravir / dolutegravir:** the mildest — diarrhea, nausea, headache

## INTERACTIONS

- Metabolism: **UGT1A1 glucuronidation** — RAL, DTG, BIC (*low interaction burden*)
- **Elvitegravir** is **CYP3A4 + UGT1A1** and comes **cobicistat-boosted** → inherits the CYP3A interaction profile



# HAART & Pharmacokinetic Boosters

## WHY COMBINATION THERAPY

HIV mutates fast enough that any single drug selects resistance. **HAART = at least 3 drugs**, hitting more than one step of the life cycle at once.

→ **Downstream:** replication suppressed → **plasma HIV RNA undetectable** → immune recovery and prolonged survival. In Thailand, **ART is offered to every infected person at any CD4 count**, provided they are ready for lifelong adherence.

## THAI FIRST-LINE & ALTERNATIVES

| SITUATION           | REGIMEN                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------|
| <b>First-line</b> ★ | <b>2 NRTIs + DTG</b> — single combined tablet, once daily                                    |
| Alternative         | ( <b>TDF</b> or <b>TAF</b> ) + ( <b>3TC</b> or <b>FTC</b> ) + <b>EFV</b> or <b>RPV</b>       |
| TDF/TAF not usable  | NRTI backbone → <b>ABC/3TC</b> or <b>AZT + 3TC</b>                                           |
| DTG not usable      | another INSTI ( <i>EVG, RAL, BIC</i> ) <b>or</b> a boosted PI ( <i>ATV/r, DRV/r, LPV/r</i> ) |

## PK ENHANCERS (BOOSTERS)

**CYP3A4 metabolises most PIs.** Give a **CYP3A inhibitor** alongside and the PI's exposure climbs.

→ **Downstream:** ↑ PI levels → fewer pills, simpler dosing, **better adherence**. **Ritonavir** (/r) and **cobicistat** (/c) are the two — e.g. **DRV/r, ATV/r, DRV/c, ATV/c**. Cobicistat has no antiviral activity of its own; ritonavir does, but is used for the boost.

## COST OF THE REGIMEN

- **Ritonavir** as a booster still brings **GI problems and lipid disturbances**
- Complex regimen, many unwanted effects · **lifelong**, adherence is hard
- Drug interactions + wide inter-individual absorption
- Metabolic and cardiovascular complications
- Some drugs **penetrate the brain poorly** → local viral proliferation in the CNS
- **Cross-resistance** across a class · children and pregnant/breastfeeding women need specific guidelines

*Principle behind this: [[pharm-drug-toxicity]] — Nevirapine on the SJS/TEN list.*

*Principle behind this: [[pharm-pharmacogenomics]] — Abacavir and HLA-B\*57:01 — the screening that dropped hypersensitivity from 7–12% to 0–2%.*

*Source: Antivirals & Anti-HIV drugs \_Warisara.pdf — อ.วริศรา (Warisara Parichatikanond, PhD), PYPH305 Pharmacology II, 108 slides. Structures omitted.*

# Antidiabetic Drugs

Diabetes = glucose is **used less** and **made more** (gluconeogenesis + glycogenolysis) → hyperglycemia. Type 1 = autoimmune  $\beta$ -cell destruction · Type 2 = progressive ↓ insulin secretion on a background of **insulin resistance**. Drugs are chosen not only on glucose-lowering but on **weight, heart, and kidney** effects.

## DIAGNOSIS (ANY ONE, CONFIRMED TWICE)

- **HbA1c  $\geq 6.5\%$**  — glycation of Hb; reflects the **prior ~3 months** of average glucose
- **Fasting plasma glucose  $\geq 126$  mg/dL** ( $\geq 8$  h fast)
- **OGTT 2-h glucose  $\geq 200$  mg/dL** (75 g glucose load)

glucose → GLUT2 into  $\beta$ -cell → ↑ATP →  $K_{\text{catp}}$  channel closes → depolarization →  $\text{Ca}^{2+}$  influx → insulin release

## THE 6 GROUPS BY STRATEGY

| GROUP                        | CLASSES                                                                        |
|------------------------------|--------------------------------------------------------------------------------|
| <b>Insulin analogues</b>     | short / intermediate / long-acting                                             |
| <b>Insulin secretagogues</b> | GLP-1 RA · dual GIP/GLP-1 RA · DPP-4 inhibitors · sulfonylureas · meglitinides |
| <b>Insulin sensitizers</b>   | biguanide (metformin) · thiazolidinediones                                     |
| <b>↓ glucose absorption</b>  | $\alpha$ -glucosidase inhibitors                                               |
| <b>↑ glucose excretion</b>   | SGLT2 inhibitors                                                               |
| <b>Others</b>                | amylin analogues · colessevelam · bromocriptine                                |

*The last three groups are insulin-independent — they work whether or not  $\beta$ -cells still function.*

## POTENCY, WEIGHT, AND THE ADR THAT DEFINES EACH CLASS

| CLASS                                       | ↓HBA1C (%) | BODY WEIGHT | MAJOR ADR                                |
|---------------------------------------------|------------|-------------|------------------------------------------|
| <b>Insulin</b>                              | 1.5–3.5 ★  | ↑           | hypoglycemia · lipodystrophy             |
| <b>GIP/GLP-1 RA</b>                         | 2.0–2.3 ★  | ↓↓          | GI upset                                 |
| <b>Biguanide</b>                            | 1.0–2.0    | neutral / ↓ | GI upset · lactic acidosis               |
| <b>Sulfonylureas</b>                        | 1.0–2.0    | ↑           | <b>hypoglycemia</b>                      |
| <b>GLP-1 RA</b>                             | 0.6–1.7    | ↓           | GI upset                                 |
| <b>DPP-4 inhibitors</b>                     | 0.6–1.5    | neutral     | headache, URI                            |
| <b>Meglitinides</b>                         | 0.5–1.5    | ↑           | hypoglycemia                             |
| <b>TZDs</b>                                 | 0.5–1.5    | ↑           | fluid retention · ↓Hb                    |
| <b><math>\alpha</math>-glucosidase inh.</b> | 0.5–1.0    | neutral     | GI upset                                 |
| <b>SGLT2 inhibitors</b>                     | 0.5–1.0    | ↓           | genitourinary infection · euglycemic DKA |

# Insulin

## MECHANISM

Binds the insulin receptor — a **receptor tyrosine kinase** — which phosphorylates **IRS**, driving **GLUT4 to the cell surface** so glucose is taken up.

→ **Downstream:** muscle ↑ glucose uptake & glycogen storage · fat ↑ lipogenesis, ↓ lipolysis · liver ↓ gluconeogenesis, ↑ glycogen → **Why it must be engineered:** native insulin  $t_{1/2}$  is only **5–6 min** (heavy hepatic clearance), so therapeutic insulins are recombinant analogues restructured to last longer.

## TIME-ACTION PROFILES ★

| INSULIN                            | ONSET        | PEAK           | DURATION |
|------------------------------------|--------------|----------------|----------|
| <b>Regular insulin</b>             | 30 min – 1 h | 2–3 h          | 4–6 h    |
| <b>Aspart · Lispro · Glulisine</b> | < 15 min     | 30 min – 1.5 h | 3–4 h    |
| <b>NPH (isophane)</b>              | 1–2 h        | 6–10 h         | 10–16 h  |
| <b>Detemir</b>                     | 1–4 h        | none           | 12–20 h  |
| <b>Glargine</b>                    | 1–4 h        | none           | 12–24 h  |
| <b>Degludec</b>                    | 1–4 h        | none           | 24–42 h  |

★ **No peak = less hypoglycemia.** The long-acting analogues hold a flat level, which is exactly why they are the safer basal choice; NPH's 6–10 h peak is where its hypoglycemia lives.

## DRUGS

| DRUG                                         | WHAT IT TREATS                                                           |
|----------------------------------------------|--------------------------------------------------------------------------|
| <b>Regular insulin</b>                       | postprandial (bolus) cover · <b>the one given IV</b> in critical illness |
| <b>Insulin aspart / lispro / glulisine</b>   | postprandial (bolus) cover — inject 15–30 min pre-meal                   |
| <b>NPH insulin</b>                           | basal cover, usually twice daily                                         |
| <b>Insulin detemir / glargine / degludec</b> | basal cover, once daily                                                  |

## USES

**Type 1 diabetes** (obligatory) · **Type 2** when other agents fail · the largest HbA1c reduction of any class (1.5–3.5%) · **Standard regimen:** long-acting as basal = **40–50%** of the total daily dose, the rest as short-acting before meals, sized to the anticipated carbohydrate load.

## ADVERSE EFFECTS

- **Hypoglycemia** ★ — the major risk, especially if carbohydrate intake drops
- Modest weight gain
- **Lipodystrophy** at repeated injection sites — lipohypertrophy (raised) or lipoatrophy (sunken). *Rotate injection sites.*
- Allergic reaction — rare; usually to aggregated/denatured insulin or additives (protamine,  $Zn^{2+}$ )

## INTERACTIONS

- ⚠️ **β-blockers mask hypoglycemia** — they blunt the palpitations, tremor, and sweating that warn the patient

## WHAT SPEEDS UP ABSORPTION (AND SO BRINGS HYPOGLYCEMIA FORWARD)

- **Abdomen** absorbs 20–30% faster than the arm
- ↑ subcutaneous blood flow — massage, hot bath, exercise
- **IM instead of SC** (too-deep injection) → faster onset
- Smoking — mainly relevant to inhaled insulin (↑ pulmonary permeability)

# GLP-1 Receptor Agonists

## MECHANISM

**Incretins** (GLP-1 from intestinal L cells, GIP from K cells) are released after a meal and bind **Gs-coupled receptors** on  $\beta$ -cells  $\rightarrow$   $\uparrow$ cAMP  $\rightarrow$  insulin release. Native GLP-1 is destroyed by **DPP-4** within 5 min, so these drugs are restructured to resist it.

$\rightarrow$  **Downstream:** ★ the insulinotropic effect is **glucose-dependent** — at low glucose there is little cAMP amplification, so **hypoglycemia is rare** ·  $\downarrow$  glucagon ·  $\downarrow$  gastric emptying · direct anti-inflammatory, antioxidant, anti-cardiomyocyte-death effects  $\rightarrow$  **Appetite:** stimulates **POMC/CART** neurons and GABA release onto **NPY/AgRP** neurons in the hypothalamic arcuate nucleus  $\rightarrow$  anorexigenic  $\rightarrow$  weight loss

## DRUGS

| DRUG                                                                           | WHAT IT TREATS                                                                              |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Exenatide · Lixisenatide</b> ( <i>exendin-4 derivatives, short-acting</i> ) | T2DM — strong <b>postprandial</b> control; take pre-meal                                    |
| <b>Liraglutide</b>                                                             | T2DM · <b>obesity</b> at higher dose · proven ASCVD + renal benefit                         |
| <b>Dulaglutide</b>                                                             | T2DM · proven ASCVD + renal benefit                                                         |
| <b>Semaglutide</b> ( <i>SC or oral</i> )                                       | T2DM · <b>obesity</b> at higher dose · ASCVD + renal benefit · injectable only: <b>MASH</b> |
| <b>Exenatide-LAR · Albiglutide</b>                                             | T2DM, weekly                                                                                |

## SHORT-ACTING VS LONG-ACTING — THE TRADE-OFF

|                             | SHORT-ACTING ( <i>EXENATIDE, LIXISENATIDE</i> ) | LONG-ACTING ( <i>LIRAGLUTIDE <math>\rightarrow</math> SEMAGLUTIDE</i> ) |
|-----------------------------|-------------------------------------------------|-------------------------------------------------------------------------|
| <b>half-life</b>            | 2–5 h                                           | 13 h (liraglutide) to 5–7 days                                          |
| <b>gastric emptying</b>     | slowed                                          | little effect                                                           |
| <b>postprandial glucose</b> | $\downarrow\downarrow$ strong                   | $\downarrow$ modest                                                     |
| <b>fasting glucose</b>      | $\downarrow$ modest                             | $\downarrow\downarrow$ strong                                           |
| <b>dosing</b>               | before meals                                    | anytime; weight loss 1–4 kg either way                                  |

## USES

**Type 2 diabetes · Obesity / overweight** — liraglutide, semaglutide at higher dose · **ASCVD risk reduction** (MI, ischemic stroke) — liraglutide, dulaglutide, injectable semaglutide · **Slowing renal decline** — same three · **MASH** — injectable semaglutide only

## ADVERSE EFFECTS

- **GI: nausea, vomiting, constipation** ★ — a direct consequence of slowed gastric motility; also diarrhea, bloating, reflux
- Injection-site reactions
- Hypoglycemia — uncommon alone; **risk rises when combined with a sulfonylurea**
- Pancreatitis · nasopharyngitis (*uncommon*)

### CAUTIONS & CONTRAINDICATIONS

-  **Contraindicated** in personal/family history of **medullary thyroid carcinoma (MTC)** or **MEN 2** — sustained GLP-1R activation caused C-cell hyperplasia → MTC in rats; not shown in humans, but the FDA warning stands
-  **Exenatide** (and probably lixisenatide) — avoid in moderate-to-severe renal failure

# Dual GIP/GLP-1 Receptor Agonist

## MECHANISM

**Tirzepatide** — one synthetic peptide agonizing **both GIP and GLP-1 receptors**. Amino iso-butyric acid (not a body amino acid) blocks peptidase attack, and a C-20 fatty di-acid tethers it to albumin →  $t_{1/2} \approx 5$  days, weekly SC injection.

→ **Downstream:** insulin release that is **both glucose-dependent and dose-dependent** · the largest HbA1c drop of any oral/injectable non-insulin agent (**2.0–2.3%**) · weight loss **10–20%**

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                                                         |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tirzepatide</b> | T2DM · <b>obesity</b> (BMI $\geq 30$ , or $\geq 27$ with a risk factor) · moderate-to-severe <b>obstructive sleep apnea</b> in obesity |

## USES

The choice when glucose **and** weight both have to reach target · non-inferior to dulaglutide for ASCVD (SURPASS-CVOT) · reduced MASH in SYNERGY-NASH — a likely future indication

## ADVERSE EFFECTS

- **GI: nausea, vomiting, constipation, abdominal pain, anorexia, dyspepsia, diarrhea** — same profile as GLP-1 RAs
- Injection-site reactions
- Hypoglycemia — very rare

## CAUTIONS

-  **Contraindicated** in personal/family history of **MTC** or **MEN 2** — same C-cell/calcitonin mechanism as the GLP-1 RAs

# DPP-4 Inhibitors

## MECHANISM

Inhibit **DPP-4**, the ectoenzyme (endothelium, T cells, circulating) that degrades endogenous GLP-1 and GIP. Inhibition exceeds **95% for 12 h**.

→ **Downstream**: active GLP-1 rises from 10–20% of total circulating GLP-1 to nearly **100%** — roughly a **2-fold** rise in plasma active GLP-1 → glucose-dependent insulin release → **Ceiling**: you can only amplify the body's own incretin, so the effect is modest (↓HbA1c 0.5–1.5%), **weight-neutral**, with no ASCVD or renal benefit — an **add-on** agent

## DRUGS

| DRUG                                                         | WHAT IT TREATS                                                                             |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Sitagliptin</b>                                           | T2DM — once daily; 74% excreted unchanged renally → <b>reduce dose in renal impairment</b> |
| <b>Vildagliptin</b>                                          | T2DM — the exception: <b>twice daily</b> ( $t_{1/2}$ 1.5–4.5 h)                            |
| <b>Saxagliptin</b>                                           | T2DM — short $t_{1/2}$ but <b>prolonged DPP-4 binding</b> + active metabolite → once daily |
| <b>Linagliptin</b>                                           | T2DM — only 3.9% renal → <b>no dose adjustment in renal impairment</b>                     |
| <b>Alogliptin · Gemigliptin · Teneligliptin · Evogliptin</b> | T2DM — once daily                                                                          |
| <b>Trelagliptin</b>                                          | T2DM — <b>once weekly</b>                                                                  |

## ADVERSE EFFECTS

- Nausea, vomiting, constipation, abdominal pain, diarrhea
- Headache · upper respiratory infection · nasopharyngitis · skin rash
- Uncommon: hypoglycemia · pancreatitis

## INTERACTIONS

- **Saxagliptin** is metabolised by **CYP3A4** — care with strong inhibitors (ketoconazole, itraconazole, clarithromycin, ritonavir, etc.) or inducers. The other gliptins barely touch CYP.

## CAUTIONS

- ⚠ **Renal impairment** — reduce dose for sitagliptin, vildagliptin, saxagliptin, alogliptin, trelagliptin. No adjustment needed for **linagliptin, gemigliptin, teneligliptin, evogliptin**.
- ⚠ **Saxagliptin** — a potential heart-failure risk signal



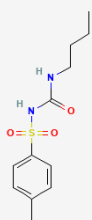
# Sulfonylureas

## MECHANISM

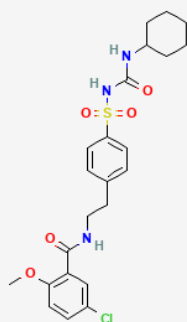
Bind **SUR1** on the  $\beta$ -cell  $K_{\text{catp}}$  channel  $\rightarrow$  channel closes  $\rightarrow$  depolarization  $\rightarrow$   $\text{Ca}^{2+}$  influx  $\rightarrow$  insulin release. **This copies what glucose does — but without needing glucose.**

$\rightarrow$  **Downstream:** insulin secretion that is **not glucose-dependent**  $\rightarrow$  this is exactly why hypoglycemia is frequent and can be severe  $\cdot$   $\downarrow$ HbA1c 1–2%  $\rightarrow$  **Off-target:** glibenclamide and glimepiride also hit **SUR2** in heart and vessels, tending to block **ischemic preconditioning** (the tissue's own adaptation to ischemia)

## THE SHARED CORE • WHAT CHANGES

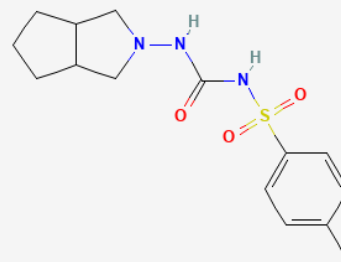


**Tolbutamide — 1st generation.** an arylsulfonylurea. ● the sulfonyl group again — these were discovered from sulfonamide ANTIBIOTICS causing hypoglycaemia



**Glibenclamide**

a large lipophilic tail  $\rightarrow$   $\sim 100\times$  more potent, long acting  $\rightarrow$  ● the highest hypoglycaemia risk



**Gliclazide**

an azabicyclic tail  $\rightarrow$  more SUR1-selective, lower cardiac risk

## MEDCHEM

**SAR** The pharmacophore is literally the **sulfonylurea** unit; everything around it is tuning. **R1 (benzene para position)** sets hypoglycaemic potency - methyl, amino, acetyl, halogen, methylthio,  $\text{CF}_3$  all work, and the bulky **p-( $\beta$ -arylcarboxyamidoethyl)** group is what defines the 2nd generation. **R2 (terminal nitrogen)** is a size window: methyl is **inactive**, ethyl weak, **propyl to hexyl most active**, and 12+ carbons inactive again.

**Physicochem** Weak organic acids ( $\text{pK}_a \approx 5$ ) - which means ⚠ **alkalinising the urine ionises them and speeds their elimination**, a real consideration in sulfonylurea overdose.

**Metabolism** ★ **The 2nd generation's odd profile explained:** that bulky R1 gives much stronger receptor binding, so you get **rapid onset and long duration from a short plasma half-life** - the drug is extensively protein-bound and hepatically recycled, so the plasma number and the clinical duration disagree on purpose. Glimepiride's M-1 metabolite is **still active**; M-2 is not.

## DRUGS

| DRUG                                             | WHAT IT TREATS                              |
|--------------------------------------------------|---------------------------------------------|
| <b>Glibenclamide (glyburide)</b>                 | T2DM — 2nd gen; also binds SUR2             |
| <b>Gliclazide</b>                                | T2DM — 2nd gen, SUR1-selective              |
| <b>Glipizide</b>                                 | T2DM — 2nd gen                              |
| <b>Glimepiride</b>                               | T2DM — 2nd gen, once daily; also binds SUR2 |
| <b>Gliquidone</b>                                | T2DM — 2nd gen                              |
| <b>Tolbutamide · Chlorpropamide · Tolazamide</b> | 1st gen — largely obsolete                  |

## USES

**Type 2 diabetes**, good glucose-lowering (↓HbA1c 1–2%) · **take ~30 min before food** so absorption matches the postprandial rise · no clear cardiovascular or renal benefit — a glucose-only agent

## ADVERSE EFFECTS

- **Hypoglycemia** ★ — the defining ADR; can progress to **coma**
- Weight gain 1–3 kg
- Less common: nausea, vomiting, cholestatic jaundice, agranulocytosis, aplastic & hemolytic anemia, hypersensitivity
- Rare: alcohol-induced flush / **disulfiram-like reaction** — especially chlorpropamide

## INTERACTIONS

Most are cleared by **CYP2C9** ★:

- **CYP2C9 inhibitors** → hypoglycemia — fluconazole, ketoconazole, voriconazole, amiodarone, co-trimoxazole, fluoxetine, fluvoxamine, metronidazole
- **CYP2C9 inducers** → loss of efficacy — carbamazepine, phenobarbital, rifampin, St John's wort, dexamethasone
- **Ethanol** enhances the hypoglycemic effect
- **Sulfonamides, clofibrate, salicylates** displace SUs from plasma protein → transient ↑ free drug

# Meglitinides (Non-sulfonylureas)

## MECHANISM

Bind a site **adjacent to SUR1** on the same  $\beta$ -cell  $K_{catp}$  channel (exact site not fully defined) → insulin release. Same idea as a sulfonylurea, but **fast on, fast off** — onset 15–30 min,  $t_{1/2} \approx 1$  h.

→ **Downstream:** targets the **postprandial** glucose spike specifically; taken 15–30 min before each meal, 2–3 times daily ·  $\downarrow$ HbA1c 0.5–1.5% · still causes hypoglycemia and weight gain, just over a shorter window

## DRUGS

| DRUG               | WHAT IT TREATS                                                                                       |
|--------------------|------------------------------------------------------------------------------------------------------|
| <b>Repaglinide</b> | postprandial hyperglycemia in T2DM — CYP3A4 to inactive metabolites                                  |
| <b>Nateglinide</b> | postprandial hyperglycemia in T2DM — CYP2C9 (70%) + CYP3A4 (30%); <b>fewer hypoglycemic episodes</b> |

## ADVERSE EFFECTS

- **Hypoglycemia** — as with sulfonylureas
- Weight gain

## INTERACTIONS

- **CYP3A4** inhibitors/inducers → repaglinide
- **CYP2C9** and CYP3A4 inhibitors/inducers → nateglinide

## ADMINISTRATION

- ★ **Take 15–30 minutes before each meal, 2–3× a day** — the schedule *is* the mechanism: onset in **15–30 min**,  $t_{1/2} \approx 1$  h, so the drug is built to cover one meal and then get out
- Lowers **HbA1c** by **~0.5–1.5%**
- 💡 The dosing rule and the safety rule are the same sentence: **no meal, no tablet**. A skipped meal with the dose already taken is how these cause hypoglycaemia — say it out loud when dispensing.

# Biguanide — Metformin

## MECHANISM

Inhibits **mitochondrial function** → the **AMP:ATP ratio rises** → activates **AMPK** → the insulin receptor works better. **The only biguanide still in use, and the first-line drug for T2DM.**

→ **Downstream:** ↑ GLUT4-mediated glucose uptake into muscle and fat · ↓ hepatic **gluconeogenesis** · ↓ fatty acid synthesis in liver and fat → **Why first line:** ↓HbA1c 1–2%, **weight-neutral or slightly ↓**, **rarely causes hypoglycemia** (it doesn't force insulin secretion), and it's cheap

## DRUGS

| DRUG             | WHAT IT TREATS                                                                                                       |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Metformin</b> | <b>first-line type 2 diabetes</b> · <b>polycystic ovary syndrome</b> (off-label — PCOS is partly insulin resistance) |

## USES

First-line T2DM · PCOS · possible cardiovascular benefit — anti-inflammatory, ↓ vascular smooth-muscle proliferation; a retrospective cohort showed fewer HF hospitalisations/CV deaths than sulfonylureas, but **no RCT has confirmed it**

## ADVERSE EFFECTS

- **GI, 10–25%** ★ — nausea, indigestion, abdominal cramps, bloating, diarrhea. The most common reason patients stop.
- ↓ **vitamin B12** by 20–30% — impaired absorption of glucose and bile salts
- ● **Lactic acidosis** — rare but severe. Mitochondrial inhibition shifts glucose metabolism toward lactate. Reported almost only alongside **poor tissue perfusion**: renal failure, sepsis, MI, congestive heart failure.

## INTERACTIONS

- **Cationic drugs** — **cimetidine, furosemide, nifedipine** — compete for renal tubular transport → ↓ metformin excretion → ↑ levels

## CAUTIONS

- ⚠ **Renal impairment** — ~60% is excreted unchanged in urine, so reduce the dose. This is the main guard against lactic acidosis.

## ADMINISTRATION

- Absorbed primarily from the **small intestine** · **bioavailability 70–80%** · **Tmax ~2 h** · **plasma t½ 4–5 h** · **does not bind plasma proteins**
- **Excreted unchanged in the urine** → **dose adjustment in renal impairment**, which is what keeps the lactic-acidosis risk down ★
- 💡 The GI upset is dose-related and usually settles: take it **with food**, start low and step up, rather than abandoning the drug in week one — which is what patients do if nobody warns them.

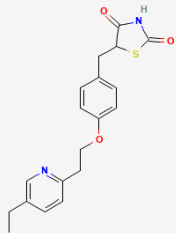
# Thiazolidinediones (TZDs)

## MECHANISM

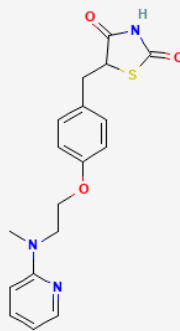
Agonists at **PPAR- $\gamma$** , a nuclear transcription factor that pairs with **RXR** and switches on genes that improve insulin-receptor signalling and **GLUT4**-mediated glucose uptake — the same endpoint as metformin, reached through gene transcription.

→ **Downstream:** insulin-mediated glucose uptake  $\uparrow$  **30–50%** ·  $\downarrow$  HbA1c 0.5–1.5% ·  $\downarrow$  triglycerides but  $\uparrow$  **LDL-C by 18–19%** → **Slow:** because the effect is transcriptional, maximal glucose effect takes **1–3 months**

## THE SHARED CORE · WHAT CHANGES



**Pioglitazone.** the thiazolidine-2,4-dione head is the PPAR- $\gamma$  binding group — the entire class is named for this one ring



### Rosiglitazone

same head, different tail → same mechanism, worse cardiovascular record

## MEDCHEM

**SAR** The glitazone pharmacophore is the **thiazolidine-2,4-dione head** plus a para-substituted phenyl connected through a **methylene bridge**. PPAR $\gamma$ 's natural ligand is a fatty acid (linoleic acid), which is why the tail looks like one.

**SAR** Troglitazone, the first of the class, carried an  **$\alpha$ -tocopherol (vitamin E) fragment** as a built-in antioxidant to retard LDL oxidation - and was withdrawn for **severe hepatotoxicity**. Rosiglitazone followed it off the market in 2010 for cardiovascular harm. Pioglitazone is the survivor, and the one with a lipid-lowering effect.

**Metabolism** Pioglitazone is a **CYP2C9** substrate, cleared as glucuronide and sulfate conjugates.

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                                                                                          |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pioglitazone</b>  | T2DM add-on · <b>MASH</b> — improves liver histology ( $\uparrow$ adiponectin, $\uparrow$ insulin sensitivity, $\downarrow$ fatty-acid influx to liver) |
| <b>Rosiglitazone</b> | T2DM add-on — available in some countries only                                                                                                          |

### ADVERSE EFFECTS

- **Weight gain 2–4 kg**
- **Edema / fluid retention** ★ — TZDs increase **ENaC** in the collecting duct → more Na<sup>+</sup> and water reabsorbed
- ● **~2-fold heart failure risk** with years of exposure
- **Anemia (↓Hb)** — partly dilutional, from the fluid retention
- ↑ **bone fracture risk in women**
- ↑ LDL-C · rarely causes hypoglycemia

### INTERACTIONS

- No known clinically important drug–drug interactions. (*Pioglitazone: CYP2C8/3A4 · rosiglitazone: CYP2C8/9.*)

### CAUTIONS & CONTRAINDICATIONS

- ● **Do not use in moderate-to-severe heart failure**
- ⚠ **Liver impairment** — withhold
- No dose adjustment needed for renal impairment or the elderly

# α-Glucosidase Inhibitors

## MECHANISM

Dietary carbohydrate is broken to **disaccharides**, then cut to absorbable monosaccharides by **α-glucosidase** on small-intestinal microvilli. Blocking that enzyme blocks the last step before absorption.

→ **Downstream:** ↓ postprandial glucose absorption, **completely insulin-independent** · also ↑ GLP-1 release, which may add to the effect · ↓HbA1c 0.5–1.0% — an **add-on** only, no proven CV benefit → **The undigested carbohydrate goes on to the colon** — which is the whole ADR profile

## DRUGS

| DRUG             | WHAT IT TREATS                                                 |
|------------------|----------------------------------------------------------------|
| <b>Acarbose</b>  | postprandial hyperglycemia — barely absorbed                   |
| <b>Voglibose</b> | postprandial hyperglycemia — barely absorbed                   |
| <b>Miglitol</b>  | postprandial hyperglycemia — 50–100% absorbed, renally cleared |

## USES

Add-on for **postprandial** hyperglycemia · take **~30 min before meals**, 2–3 times daily — the enzyme must already be blocked when the food arrives

## ADVERSE EFFECTS

- **Malabsorption, flatulence, diarrhea, abdominal bloating** ★ — undigested carbohydrate fermented in the colon
- Hypoglycemia is very rare alone, but the risk rises when added to **insulin or an insulin secretagogue**

## INTERACTIONS

- **Acarbose** ↓ absorption of **digoxin**
- **Miglitol** ↓ absorption of **propranolol** and **ranitidine**

## CAUTIONS

- ⚠ **Severe renal impairment** — reduce the dose, especially **miglitol** (renally cleared)

## ADMINISTRATION

- ★ **Taken before meals** — they work inside the gut lumen on the carbohydrate that's arriving
- **Acarbose** — minimally absorbed, cleared by the kidney · **Miglitol** — absorption is **saturable**, 50–100% of a dose reaches the circulation, cleared by the kidney
- **Dose reduction in severe renal impairment**
- 💡 Two things worth saying: take it **with the first bite**, and — if they're also on insulin or a secretagogue — hypoglycaemia must be treated with **glucose, not sucrose or sweets**, because this drug is precisely what stops sucrose being broken down.

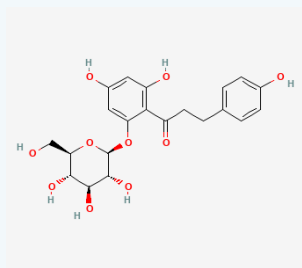
# SGLT2 Inhibitors

## MECHANISM

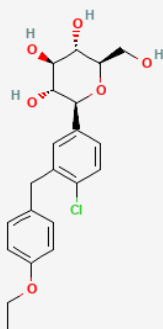
Filtered glucose is normally reabsorbed with  $\text{Na}^+$  in the **proximal convoluted tubule** — **SGLT2 does 90%**, SGLT1 the other 10%. In diabetes SGLT2 is **upregulated**, reabsorbing even more. Blocking it dumps glucose into the urine, **independent of insulin**.

→ **Downstream:** urinary glucose 50–90 g/day (30–50% of filtered load) ·  $\downarrow \text{HbA1c}$  0.5–1.0% · osmotic diuresis  
→ **BP**  $\downarrow$  **2–5/1–2 mmHg** · weight  $\downarrow$  1–3 kg (calorie loss + fluid) → **Cardiac:**  $\downarrow$  preload and afterload; also  $\uparrow$  ketone metabolism for cardiac ATP and inhibition of  **$\text{Na}^+/\text{H}^+$  exchanger 1** →  $\downarrow$  intracellular  $\text{Na}^+/\text{Ca}^{2+}$  → **Renal (tubuloglomerular feedback):** ★ less proximal  $\text{Na}^+$  reabsorption → **more  $\text{Na}^+$  reaches the macula densa** → adenosine → afferent arteriole constricts and efferent relaxes →  $\downarrow$  **glomerular pressure** →  $\downarrow$  **albuminuria**

## THE SHARED CORE · WHAT CHANGES

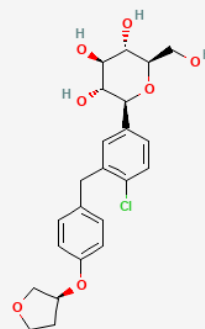


**Phlorizin — the natural lead.** an apple-bark glucoside. ● an O-glycoside, so gut enzymes cleave it — the fix was to replace that O with a C



**Dapagliflozin**

C-aryl glucoside — uncleavable, orally stable, SGLT2-selective



**Empagliflozin**

same C-glucoside trick, highest SGLT2 selectivity

## MEDCHEM

**SAR** ★ The whole class is one fix applied to a natural product. Phlorizin (from apple bark) is an **O-glycoside** - it inhibits SGLT2 but is non-selective (SGLT2  $\text{IC}_{50}$  35.6 nM vs SGLT1 330 nM) and, worse, **intestinal  $\beta$ -glucosidase hydrolyses the acetal** straight off, releasing phloretin which then blocks GLUTs. Replacing the O-glycoside bond with a **C-glycoside** makes it uncleavable: metabolically stable, oral, and far more SGLT2-selective. Every -gliflozin is that same substitution.



## DRUGS

| DRUG                 | WHAT IT TREATS                                                                                              |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Empagliflozin</b> | T2DM · <b>MACE reduction</b> · <b>HFpEF &amp; HFrEF</b> · <b>chronic kidney disease</b>                     |
| <b>Dapagliflozin</b> | T2DM · <b>HFpEF &amp; HFrEF</b> · <b>CKD</b> · MACE only in prior MI                                        |
| <b>Canagliflozin</b> | T2DM · MACE reduction · HF hospitalisation · renal protection — less SGLT2-selective, hence the higher dose |
| <b>Ertugliflozin</b> | T2DM · HF hospitalisation — <b>no renal benefit shown</b>                                                   |
| <b>Sotagliflozin</b> | T2DM · HF · CKD — a <b>dual SGLT1/2</b> inhibitor, so it also blocks gut glucose absorption                 |
| <b>Bexagliflozin</b> | T2DM — no outcome studies                                                                                   |

## USES

**Type 2 diabetes** · **Heart failure** — HFpEF *and* HFrEF (empagliflozin, dapagliflozin, sotagliflozin) · **Chronic kidney disease** — empagliflozin, dapagliflozin · **ASCVD** — empagliflozin, canagliflozin · take in the **morning** to avoid nocturia

## ADVERSE EFFECTS

- **Genital and urinary tract infection, bacterial and mycotic** ★ — the commonest ADR, more in women; glucosuria feeds the organisms. *Counsel on post-void hygiene.*
- **Volume depletion** → **hypotension** in some patients
- ● **Euglycemic diabetic ketoacidosis** — DKA with a normal blood glucose, so it's easy to miss. The drug lowers the insulin:glucagon ratio → the liver makes more ketones. Risk ↑ with **low-carbohydrate/ketogenic diet, dehydration, or an insulin dose reduction.**
- Hypoglycemia is very rare

## INTERACTIONS

- Metabolised by **UGT1A9**. Rifampicin (inducer) ↓ and mefenamic acid (inhibitor) ↑ drug levels, but **no clearly clinically significant interaction** has been shown.

## CAUTIONS

- ⚠ **eGFR < 30 mL/min/1.73 m<sup>2</sup>** — reduce or stop. Not a pharmacokinetic issue: there is simply **less SGLT2 target left to block**, so the glucose-lowering fails. (*HF or CKD indications may continue at usual dose.*)

## ADMINISTRATION — DOSES, AND WHERE THE KIDNEY CUTS THEM OFF

| DRUG                 | USUAL DOSE    | RENAL LIMIT ( <i>GFR ML/MIN/1.73 M<sup>2</sup></i> )                                                       |
|----------------------|---------------|------------------------------------------------------------------------------------------------------------|
| <b>Canagliflozin</b> | 100–300 mg OD | 30–60 → <b>max 100 mg/day</b> · < 30 → <b>avoid</b>                                                        |
| <b>Dapagliflozin</b> | 5–10 mg OD    | < 45 → not recommended in T2DM · < 25 → generally not recommended ( <i>HF/CKD may continue 10 mg/day</i> ) |
| <b>Empagliflozin</b> | 10–25 mg OD   | < 30 → generally not recommended ( <i>HF/CKD: no adjustment</i> )                                          |
| <b>Ertugliflozin</b> | 5–15 mg OD    | < 45 → <b>avoid</b>                                                                                        |
| <b>Bexagliflozin</b> | 20 mg OD      | < 30 → <b>avoid</b>                                                                                        |
| <b>Sotagliflozin</b> | 200–400 mg OD | < 30 → <b>avoid</b>                                                                                        |

- 💡 Two counselling points that stop the common problems: **genital hygiene and fluid intake** (mycotic infection is the usual reason people quit), and a **sick-day rule** — stop it when they're not eating or are dehydrated, because that's the setup for euglycaemic DKA.

# Choosing a Drug — ADA 2026

## MATCH THE DRUG TO THE COMORBIDITY, NOT JUST THE HBA1C

| PATIENT                                                                                                            | FIRST CHOICE                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>ASCVD or at risk</b>                                                                                            | GLP-1 RA <b>or</b> SGLT2 inhibitor with proven outcome data                                                                |
| <b>Heart failure or at risk</b>                                                                                    | <b>SGLT2 inhibitor</b>                                                                                                     |
| <b>Chronic kidney disease or at risk</b>                                                                           | <b>SGLT2 inhibitor</b> or a proven GLP-1 RA                                                                                |
| <b>MASH</b>                                                                                                        | injectable <b>semaglutide, tirzepatide, or pioglitazone</b>                                                                |
| <b>Weight must come down</b>                                                                                       | tirzepatide or GLP-1 RA (semaglutide, dulaglutide, liraglutide); SGLT2i is moderate; metformin & DPP-4i are weight-neutral |
| <b>Glucose-lowering is the priority</b>                                                                            | tirzepatide, GLP-1 RA, or insulin; then SGLT2i, TZD, metformin, sulfonylurea                                               |
| ⚠ <b>Never combine a DPP-4 inhibitor with a GLP-1 RA or tirzepatide</b> — same incretin pathway, no added benefit. |                                                                                                                            |

## CARDIOVASCULAR & RENAL OUTCOMES

| CLASS                   | ASCVD                                                  | HEART FAILURE                             | RENAL                                                    |
|-------------------------|--------------------------------------------------------|-------------------------------------------|----------------------------------------------------------|
| <b>Biguanide</b>        | potential benefit                                      | neutral                                   | potential benefit                                        |
| <b>Sulfonylureas</b>    | neutral                                                | neutral                                   | neutral                                                  |
| <b>TZDs</b>             | potential benefit (pioglitazone)                       | ● ↑ <b>risk</b>                           | neutral                                                  |
| <b>GLP-1 RA</b>         | <b>benefit</b> — liraglutide, dulaglutide, semaglutide | neutral                                   | <b>benefit</b> — same three                              |
| <b>GIP/GLP-1 RA</b>     | benefit                                                | neutral                                   | neutral                                                  |
| <b>DPP-4 inhibitors</b> | neutral                                                | potential risk — saxagliptin              | neutral                                                  |
| <b>SGLT2 inhibitors</b> | <b>benefit</b> — empagliflozin, canagliflozin          | <b>benefit</b> — all except bexagliflozin | <b>benefit</b> — all except ertugliflozin, bexagliflozin |
| <b>Insulin</b>          | neutral                                                | neutral                                   | neutral                                                  |

Principle behind this: *[[pharm-pharmacokinetics]]* — metformin is charged and requires the organic cation transporter to get into cells.

Principle behind this: *[[pharm-drug-action]]* — Insulin is the "replacement" purpose, and its receptor is the enzyme-linked archetype — a membrane tyrosine kinase that dimerises on activation.

Principle behind this: *[[pharm-drug-toxicity]]* — Sulfonylureas are on the delayed-onset overdose list — a hypoglycaemia that arrives hours later.

Source: [Anti-diabetic drugs\\_SW\\_2025.pdf](#) — ผศ.ดร.ภก.สุรศักดิ์ วิชัยโย, ภกปว 305 Pharmacology II, 72 slides; gaps filled from the companion handout [เภสัชวิทยาของยารักษาโรคเบาหวาน.pdf](#). Structures omitted.

# Estrogens, Progestins & Contraceptives

The two female sex steroids, the drugs that block them, and the contraceptives assembled from them. All of them are **nuclear-receptor drugs** — they don't open channels, they change which genes get transcribed, which is why every effect is slow-onset and tissue-dependent.

**hormone → binds nuclear receptor → receptor dimerizes → binds response element on DNA (ERE / PRE) → transcription → physiologic effect**

## THE GROUPS AT A GLANCE

| GROUP                       | WHAT IT DOES                                        | EXAMPLES                               |
|-----------------------------|-----------------------------------------------------|----------------------------------------|
| <b>Estrogens</b>            | replacement, menopause, the estrogen half of a COC  | estradiol, ethinyl estradiol, estetrol |
| <b>SERMs</b>                | agonist in some tissues, antagonist in others       | tamoxifen, raloxifene, toremifene      |
| <b>ER antagonists</b>       | pure block, no agonism                              | fulvestrant, clomiphene                |
| <b>Aromatase inhibitors</b> | cut estrogen <i>production</i> , not the receptor   | anastrozole, letrozole, exemestane     |
| <b>Progestins</b>           | contraception, endometrial protection in HRT        | MPA, levonorgestrel, drospirenone      |
| <b>Antiprogestins</b>       | block PR — abortion, emergency contraception        | mifepristone, ulipristal               |
| <b>Contraceptives</b>       | COC (estrogen + progestin) vs POC (progestin alone) | —                                      |

# Estrogens

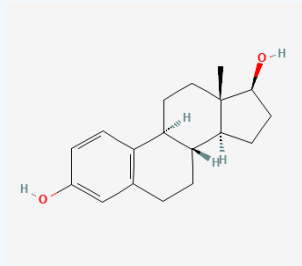
## MECHANISM

Circulates bound to **SHBG**; only the **unbound fraction** enters the cell and binds an intracellular (nuclear) **estrogen receptor** → dimerization → binds **estrogen response elements (EREs)** at DNA promoters → transcription.

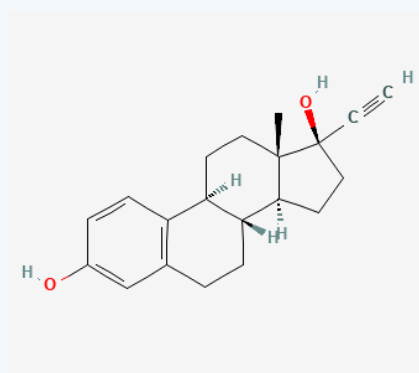
- **ER $\alpha$**  — uterus, vagina, ovary, mammary gland, hypothalamus, endothelium, vascular smooth muscle
- **ER $\beta$**  — prostate, ovary, lung, brain, bone, vasculature

→ **Downstream:** puberty & secondary sexual characteristics · endometrial proliferation · neuroendocrine control of the menstrual cycle · **bone:** ↓ resorption via osteoclast apoptosis, opposes PTH, stimulates osteoblasts, **closes the epiphyses** → **Liver:** ↑ CBG, TBG, **SHBG** · ↑ clotting factors, ↓ PAI-1 · ↑ cholesterol into bile → gallstones → **Metabolic:** ↑ HDL & TG, ↓ LDL & total cholesterol · vascular ↑ NO and ↑ prostacyclin

## THE SHARED CORE · WHAT CHANGES



**Estradiol — the natural estrogen.** an aromatic A-ring with a phenol — that aromatic ring is what distinguishes estrogens from every other steroid



### Ethinylestradiol

- a 17 $\alpha$ -ethynyl group blocks oxidation at C17 → it survives first-pass metabolism → the reason oral contraception works at all

## MEDCHEM

**SAR** ★ **Why oral estradiol barely works and ethinylestradiol does.** Estradiol is destroyed by **oxidation of its 17-OH** on first pass. Putting a **17 $\alpha$ -substituent** there (the ethynyl group) physically blocks that oxidation → oral bioavailability and a much longer half-life. Esterifying the 17-OH instead gives duration; etherifying the **3-OH phenol** gives duration too (mestranol).

**SAR** The receptor's geometry is a measurement: the **3-OH to 17-OH distance must be 10.3–12.1 Å**. That is why diethylstilbestrol works at all despite having no steroid rings - **trans-DES** puts two phenols exactly that far apart, at equal potency to estradiol and more metabolically stable. Cis-DES, with the wrong geometry, does not.

## DRUGS

| DRUG                                              | WHAT IT TREATS                                                                                              |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Estradiol (E2), estrone (E1), estriol (E3)</b> | natural estrogens — replacement therapy, menopausal symptoms                                                |
| <b>Conjugated estrogens</b>                       | menopausal hormone therapy                                                                                  |
| <b>Ethinyl estradiol (EE)</b>                     | the contraceptive estrogen — highest oral bioavailability ★, binds albumin <b>not</b> SHBG → cleared slowly |
| <b>Estetrol (E4)</b>                              | newer COC estrogen — much weaker, $t_{1/2}$ 20–28 h, barely raises SHBG, less thrombosis                    |
| <b>Mestranol, quinestrol</b>                      | synthetic steroidal estrogens                                                                               |
| <b>Diethylstilbestrol, chlorotrianisene</b>       | non-steroidal synthetic estrogens                                                                           |

## USES

**Replacement therapy** — primary hypogonadism (Turner's), oophorectomy · **Menopausal hormone therapy** — hot flush, mood swing, osteoporosis, urogenital atrophy (topical) · **Oral contraception** — EE 20–35 µg/day plus a progestin · Dysmenorrhea / PMS / endometriosis · Suppression of ovarian function · Acne

## ADVERSE EFFECTS

- **Carcinogenic** ★ — breast, endometrial and vaginal cancer. Two routes: **ER-mediated cell proliferation** + **oxidative metabolism to reactive catechol quinones**
- **Thromboembolic disease (DVT)** — ↑ thrombin generation and fibrin clot formation
- Nausea, vomiting — *give with food or before sleep*
- Edema, breast tenderness — *lower the dose*
- Migraine
- Reactivates or worsens endometriosis
- Gallbladder disease

## INTERACTIONS

- **CYP3A4 inducers** — phenytoin, barbiturates, carbamazepine, rifampin → estrogen levels fall
- **Antibiotics** ★ — suppress gut flora → ↓ **enterohepatic recirculation** → ↓ estrogen levels
- Highly protein-bound drugs · drugs that alter GI absorption
- Estrogen agonists / antagonists

## CONTRAINDICATIONS

- **Estrogen-dependent neoplasm** — endometrial or breast carcinoma
- Undiagnosed genital bleeding · liver disease · history of thromboembolic disorder
- **Heavy smokers** — ↑ cardiovascular risk; PAHs in tobacco smoke are potent **CYP1A2 inducers**

## ADMINISTRATION — ESTROGEN ROUTES

- **Oral:** estradiol, conjugated estrogens, esters of estrone and other estrogens, **ethinyl estradiol**.
- **Transdermal patch:** slow, sustained release of the hormone.
- **Intramuscular injection:** estradiol — slow absorption, prolonged duration.
- **Creams/gel:** estradiol and conjugated estrogens · **Vaginal ring:** estradiol.
- ★ **Ethinyl estradiol does not bind SHBG** — it binds albumin instead, so it is **cleared from the body slowly**. That, plus resistance to first-pass metabolism, is why the synthetic is the one used orally.
- 💡 The route decides the first-pass load, and the first-pass load decides the clotting-factor effect. Non-oral routes exist for exactly that reason, not for convenience.

# Selective Estrogen Receptor Modulators (SERMs)

## MECHANISM

Bind the ER, but the drug–receptor complex behaves **differently per tissue** — estrogenic where that's useful (bone, brain, liver), **antagonist where estrogen is dangerous** (breast, endometrium).

→ **Downstream:** breast — blocks the proliferative/carcinogenic drive · bone — antiresorptive · lipids — ↓ total cholesterol and LDL **without** raising HDL

## TISSUE-SPECIFIC ACTIVITY ★

|                   | BREAST | ENDOMETRIUM | BONE |
|-------------------|--------|-------------|------|
| <b>Estrogen</b>   | +++    | +++         | +++  |
| <b>Tamoxifen</b>  | –      | +           | +    |
| <b>Raloxifene</b> | –      | –           | ++   |

*The endometrium column is the whole clinical story — it's why tamoxifen causes endometrial cancer and raloxifene doesn't.*

## DRUGS

| DRUG              | WHAT IT TREATS                                                                   |
|-------------------|----------------------------------------------------------------------------------|
| <b>Tamoxifen</b>  | breast cancer — treatment <b>and</b> prevention                                  |
| <b>Toremifene</b> | breast cancer                                                                    |
| <b>Raloxifene</b> | prevention & treatment of <b>osteoporosis</b> · ↓ risk of invasive breast cancer |

## ADVERSE EFFECTS

- **Tamoxifen** → **endometrial carcinoma** ★ (partial agonist there) — typically used < 5 years
- Both: **DVT and pulmonary embolism** — raloxifene ~**3-fold** increase
- Tamoxifen: hot flashes, cataracts, nausea
- Raloxifene: hot flashes, leg cramps
- ⚠ Raloxifene does **not** relieve menopausal vasomotor symptoms



# Estrogen Receptor Antagonists

## MECHANISM

**Pure** antagonists — no tissue where they act as agonists, which is the line between them and the SERMs.

**Clomiphene** competes with estradiol for receptors in the **hypothalamus / pituitary** → the brain reads estrogen as low → negative feedback is lifted.

→ **Downstream:** ↑ **FSH** secretion → follicular recruitment → **ovulation**

## DRUGS

| DRUG               | WHAT IT TREATS                                                          |
|--------------------|-------------------------------------------------------------------------|
| <b>Clomiphene</b>  | the "fertility pill" — ovulation induction; <b>first-line in PCOS</b> ★ |
| <b>Fulvestrant</b> | pure estrogen-receptor antagonist                                       |

## ADVERSE EFFECTS — CLOMIPHENE

- **Ovarian hyperstimulation syndrome**
- Increased incidence of **multiple births**
- Ovarian cysts
- Hot flash
- Blurred vision

# Aromatase Inhibitors

## MECHANISM

**Aromatase** is the enzyme that converts androgens into estrogens. Blocking it selectively shuts off estrogen **production** rather than blocking the receptor — so estrogen falls everywhere.

→ **Downstream:** estrogen-dependent breast tumours are starved of ligand ★ · **no uterine stimulation** and **no ↑ venous thromboembolism** — the two things that limit tamoxifen

## DRUGS

| DRUG               | WHAT IT TREATS |
|--------------------|----------------|
| <b>Anastrozole</b> | breast cancer  |
| <b>Letrozole</b>   | breast cancer  |
| <b>Exemestane</b>  | breast cancer  |

## USES

**Breast cancer** — first-line, or second-line after tamoxifen · potential use for **ovulation induction**

## ADVERSE EFFECTS

- Hot flashes — from systemic estrogen deprivation
- ● **Bone fracture risk** ★ — they lack estrogen's bone-protective effect, so they're usually given **with a bisphosphonate**
- Effect on plasma lipids unclear

# Progestins

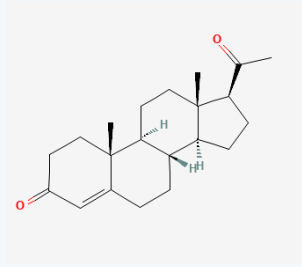
## MECHANISM

Unliganded **PR** (PR-A / PR-B) sits inactive bound to heat-shock proteins (HSP-90, HSP-70, p59).

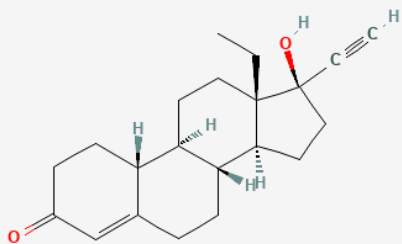
Progesterone binds → HSPs dissociate → receptor is phosphorylated and **dimerizes** → binds **progesterone response elements (PREs)** → recruits coactivators (SRC-1, NcoA-1/-2) → transcription. **Receptor density is itself set by estrogen.**

→ **Downstream:** ↓ **GnRH pulse frequency** (this is the contraceptive action) · counteracts estrogen-driven endometrial proliferation and downregulates breast ER · prepares endometrium for implantation and maintains pregnancy · **thick viscid cervical mucus** → **sperm can't get through** ★ → **Also:** ↑ basal body temperature ~0.6 °C at ovulation · sedation & low mood · ↑ ventilatory response to CO<sub>2</sub> · ↑ basal and post-meal insulin · lipoprotein lipase → fat deposition

## THE SHARED CORE · WHAT CHANGES

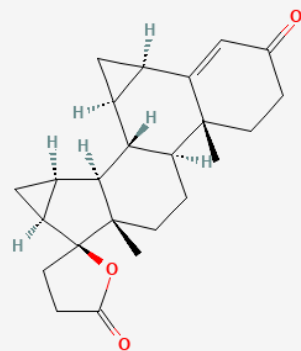


**Progesterone — the natural hormone.** orally almost useless: hepatic metabolism destroys it. Every synthetic below exists to fix that.



### Levonorgestrel

a 19-nortestosterone derivative with a 17 $\alpha$ -ethynyl → orally active, with some androgenic activity



### Drospirenone

derived from spironolactone instead → ● antimineralocorticoid and antiandrogenic; the K<sup>+</sup> warning is structural

## MEDCHEM

**SAR** ● The single biggest edit is removing the C19 methyl - it raises progestin activity 5- to 10-fold, which is what makes the 19-nortestosterone progestins (levonorgestrel and relatives) potent enough for contraception. Adding a **C18 ethyl** pushes it further.

**SAR** As with the estrogens, a **17 $\alpha$ -ethynyl, methyl or ethyl** group blocks metabolism and buys oral bioavailability - and the ethynyl specifically **raises progestin activity while lowering anabolic activity**, which is the whole point of a contraceptive progestin. Acetylating the 17 $\beta$ -OH extends duration. ★

**Removing the 3-keto group keeps progestin activity while dropping the androgenic effects** - that is the design logic behind the less androgenic newer progestins.

## DRUGS

| DRUG                                        | WHAT IT TREATS                                                                                |
|---------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Progesterone</b>                         | the natural hormone — HRT alongside estrogen                                                  |
| <b>Medroxyprogesterone acetate (MPA)</b>    | contraception (depot) · HRT — progesterone-selective, minimal androgenic activity             |
| <b>Norethindrone (norethisterone)</b>       | contraception · postponing menstruation                                                       |
| <b>Norgestrel / levonorgestrel</b>          | contraception · emergency contraception                                                       |
| <b>Norgestimate, desogestrel, gestodene</b> | contraception — potent progestin, <b>low</b> androgenicity                                    |
| <b>Cyproterone acetate</b>                  | <b>anti-androgenic</b> — acne, hirsutism, androgenic baldness                                 |
| <b>Drospirenone</b>                         | <b>anti-androgenic + anti-mineralocorticoid</b> — less fluid retention, favourable CV profile |

## USES

**Contraception** — alone or with estrogen · **HRT with estrogen** — the progestin is there to protect the endometrium from estrogen-driven hyperplasia and carcinoma ★ · **Suppress ovarian function** — dysmenorrhea, endometriosis, diagnosing secondary amenorrhea · **Treat estrogen-driven endometrial hyperplasia / carcinoma** · **Postponing menstruation** — norethisterone, started ≥ 3 days before the expected period, not longer than 10 days · **Emergency contraception** — levonorgestrel

## ADVERSE EFFECTS

- **Drospirenone** → **hyperkalemia** ★ — from the anti-mineralocorticoid activity. Caution with renal disease, ACEIs, ARBs
- Breakthrough bleeding
- May raise blood pressure
- ↓ **HDL** — specifically the progestins with androgenic activity; 19-nor compounds affect lipids most
- Glucose intolerance with prolonged use
- Fluid retention / edema
- With estrogen: more breast epithelial proliferation than estrogen alone

# Antiprogestins

## MECHANISM

Block the **progesterone receptor**, so the endometrium never becomes implantation-ready. **Mifepristone** also antagonizes the **glucocorticoid** receptor; **ulipristal** is a *selective PR modulator* — a **partial agonist** with only weak glucocorticoid antagonism.

→ **Downstream:** endometrium unsuitable for implantation · myometrial contraction and blastocyst detachment (*mifepristone, with a prostaglandin*) · ↓ **LH release** → **ovulation inhibited** (*ulipristal*)

## DRUGS

| DRUG                | WHAT IT TREATS                                                                              |
|---------------------|---------------------------------------------------------------------------------------------|
| <b>Mifepristone</b> | medical abortion in early pregnancy — given <b>with misoprostol within 48 h</b>             |
| <b>Onapristone</b>  | medical abortion — a more selective, <b>pure</b> PR antagonist                              |
| <b>Ulipristal</b>   | emergency contraception — single <b>30 mg</b> dose, effective up to <b>120 h (5 days)</b> ★ |

## USES

Emergency prevention of conception · Induction of labour · Uterine leiomyoma (เนื้องอกกล้ามเนื้อมดลูก) · Endometriosis · Breast cancer

## ADVERSE EFFECTS

- **Mifepristone — vaginal bleeding lasting 8–17 days.** The severe untoward effect
- Mifepristone: abdominal pain, uterine cramps, nausea, vomiting, diarrhea — *driven by the prostaglandin rise*
- Ulipristal: nausea, headache, abdominal pain, fatigue, dizziness, irregular bleeding

## ADMINISTRATION — MIFEPRISTONE

- **400–600 mg/day for 4 days, or 800 mg/day for 2 days.**
- ● **Given with a prostaglandin (misoprostol) within 48 hours** — the prostaglandin drives the myometrial contraction and expulsion; mifepristone alone is only half the regimen.
- ⚠ **Vaginal bleeding lasts 8–17 days**, with abdominal pain, uterine cramps, nausea, vomiting and diarrhoea from the raised prostaglandins.
- 💡 The bleeding duration is the counselling point people are least prepared for — two weeks of bleeding is expected here, not a complication.

# Combined Oral Contraceptives (COC)

## MECHANISM

Estrogen and progestin suppress the gonadotropin axis from both ends:

- **Estrogen** → suppresses **FSH** release → no dominant follicle
  - **Progestin** → ↓ GnRH pulse frequency and **blocks the estrogen-induced LH surge** ★, plus thickens cervical mucus
- **Downstream: no ovulation.** Effectiveness **99.9%** — 0.1% pregnancy rate with correct use

## WHAT'S IN THE PILL

| COMPONENT                     | OPTIONS                                                                 |
|-------------------------------|-------------------------------------------------------------------------|
| <b>Estrogen</b>               | ethinyl estradiol (EE) · estradiol · estetrol (E4)                      |
| <b>1st-gen progestin</b>      | norethindrone acetate, ethynodiol diacetate, lynestrenol, norethynodrel |
| <b>2nd-gen</b>                | levonorgestrel, dl-norgestrel                                           |
| <b>3rd-gen</b>                | norgestimate, gestodene, desogestrel                                    |
| <b>4th-gen / unclassified</b> | drospirenone, cyproterone acetate, dienogest                            |

**Low dose** = EE < 50 µg (20–35 µg) — the standard. High dose (EE ≥ 50 µg) isn't banned but is **clinically obsolete** for contraception.

## REGIMEN TYPES

| TYPE              | HORMONE LEVELS                                                                         |
|-------------------|----------------------------------------------------------------------------------------|
| <b>Monophasic</b> | every active tablet identical — <b>preferred in practice</b> (simpler, equal efficacy) |
| <b>Biphasic</b>   | two hormone levels across the pack                                                     |
| <b>Triphasic</b>  | three hormone levels across the pack                                                   |

*Non-oral combined routes exist too: transdermal patch (1/week), vaginal ring (3 weeks on, 1 off), monthly IM injection.*

## BENEFITS BEYOND CONTRACEPTION

Regularises the cycle · ↓ dysmenorrhea, PMS, endometriosis · ↓ **ectopic pregnancy** · ↓ colon cancer · replaces missing hormone and ↓ osteoporosis around menopause · **Acne / hirsutism** with an anti-androgen progestin (cyproterone, drospirenone, gestodene) · **PMDD** — drospirenone-containing pills

## ADVERSE EFFECTS

- ● **Cardiovascular** ★ — hypertension, ischemic or hemorrhagic stroke, **venous thromboembolism**, worst in **smokers**. Thrombotic risk tracks the **estrogen dose**; E4 appears to lower it
- **Cancer** — ↑ risk of cervical, liver and breast cancer
- Metabolic — impaired glucose tolerance and gallbladder disease (high-dose); estrogen ↑ binding proteins for thyroid hormone, glucocorticoids and sex steroids
- Nausea, edema, headache, **breakthrough bleeding**, acne, hirsutism, weight gain, melanin pigmentation

## INTERACTIONS

- Almost all clinically relevant interactions run through **hepatic CYP450** — enzyme **inducers** drop hormone levels and can cause contraceptive failure

## CONTRAINDICATIONS

- ● Present or past **thromboembolic disease**
- ● Cerebrovascular disease · myocardial infarction · coronary artery disease
- ● Severe liver disorder · past or present liver tumour · impaired liver function
- ● Carcinoma of breast or female reproductive tract
- ● Undiagnosed vaginal bleeding
- ● Pregnancy

# COC Use & Missed Pills

## THE RULE BEHIND EVERY INSTRUCTION

Start on **day 1–5 of the menstrual cycle**, hormone tablets first, follow the arrows on the pack. Take it at the **same time every day** (bedtime is easy to remember) so hormone levels don't fluctuate. More than 24 h late and contraceptive efficacy drops.

→ **The whole goal:** never allow a **hormone-free interval long enough to let ovulation happen**. ★ ≥ 7 **consecutive hormone tablets are needed to suppress ovulation**.

## WHAT TO DO WHEN SHE FORGETS

| SITUATION                                                                             | ACTION                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1 tablet</b> , remembered <b>24–48 h</b> after the due time                        | take the missed tablet immediately · take the next one at its normal time ( <b>2 in one day</b> ) · <b>no backup method needed</b>                                                                                 |
| <b>2 consecutive tablets</b> , > 48 h late                                            | take the <b>most recent</b> missed tablet, <b>discard the earlier ones</b> · continue on schedule (2 in one day) · <b>use backup</b> (condom / abstain) until <b>7 consecutive hormone tablets</b> have been taken |
| Missed in the <b>last hormone week</b> (tablets 15–21)                                | start a <b>new pack immediately</b> after tablet 21 — skip the hormone-free interval                                                                                                                               |
| Missed in <b>week 1</b> (tablets 1–7) + <b>unprotected sex in the previous 5 days</b> | consider <b>emergency contraception</b>                                                                                                                                                                            |

## COUNSELLING POINTS

- ⚠ **Never stop the pack on her own**
- ⚠ Spotting / breakthrough bleeding is expected, especially early — **reassure and follow up**, don't jump to a higher dose



# Progestin-Only Contraceptives (POC)

## MECHANISM

Progestin without estrogen. It only inhibits ovulation in **60–80% of cycles**, so unlike the COC the contraceptive work is mostly **local**.

→ **Downstream: thickened cervical mucus** → ↓ sperm penetration · **altered endometrium** → implantation impaired → **Effectiveness 97–98%** — genuinely lower and less predictable than COC ★

## DRUGS

| DRUG                                                   | WHAT IT TREATS                                           |
|--------------------------------------------------------|----------------------------------------------------------|
| <b>Lynestrenol</b> ( <i>minipill</i> )                 | daily contraception — late-dose window only <b>3 h</b> ★ |
| <b>Desogestrel</b> ( <i>minipill</i> )                 | daily contraception — window <b>12 h</b>                 |
| <b>Drospirenone</b> ( <i>minipill</i> )                | daily contraception — window <b>24 h</b>                 |
| <b>Medroxyprogesterone acetate</b> ( <i>depot IM</i> ) | contraception for <b>3 months</b>                        |
| <b>Levonorgestrel</b> ( <i>implant</i> )               | long-acting contraception, up to <b>5 years</b>          |
| <b>Etonogestrel</b> ( <i>implant</i> )                 | contraception for <b>3 years</b>                         |
| <b>Progestin IUD</b>                                   | local release — contraception for years                  |

## USES

**When estrogen can't be used** · **Breastfeeding women** ★ — no suppression of milk production · Long-acting reversible contraception where daily dosing is unrealistic (implant, depot, IUD)

## ADVERSE EFFECTS

- ★ **Irregular, unpredictable spotting · breakthrough bleeding · amenorrhea** — by far the most common complaint and the usual reason for stopping
- Minipill: acne, vaginal bleeding
- **Depot MPA:** headache, mood changes, weight gain, altered HDL/LDL, **loss of bone density**
- Implants: insertion-site infection, irritation, pain · acne · mood changes
- Thromboembolic events — **less than COC**

## CONTRAINDICATIONS

- ⚠ Breastfeeding **within 6 weeks postpartum**
- Acute DVT or pulmonary embolism
- History of MI or stroke
- Breast cancer
- Severe cirrhosis / hepatoma
- ⚠ Migraine · unexplained vaginal bleeding

# Emergency Contraception

## MECHANISM

A single high progestin (or antiprogestin) hit **before** implantation.

→ **Downstream:** prevention of **ovulation**, of **fertilization**, and of **implantation** — which is why the earlier it's taken the better it works

## OPTIONS

| DRUG                                           | HOW IT'S USED                                                                    |
|------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Levonorgestrel</b>                          | <b>1.5 mg once</b> , or 0.75 mg × 2 twelve hours apart — within <b>72 h</b><br>★ |
| <b>Ulipristal</b>                              | <b>30 mg once</b> — effective up to <b>120 h (5 days)</b> , but sooner is better |
| <b>EE + norgestrel</b> ( <i>Yuzpe method</i> ) | 2 tablets within 72 h, then 2 more after 12 h                                    |

## CAUTIONS

- ⚠ **Not more than 2 packs / 4 tablets per month** — it is not a routine method
- ⚠ Efficacy falls with every hour of delay

## WORTH KNOWING

Oral **piroxicam 40 mg** co-administered with levonorgestrel **improved EC efficacy** in study data — worth considering clinically where levonorgestrel EC is the chosen option.

*Principle behind this: [[pharm-pharmacokinetics]] — enterohepatic recycling (and the antibiotic myth attached to it — real mechanism, no clear clinical evidence of contraceptive failure); estrogen also slows gastric emptying.*

*Principle behind this: [[pharm-drug-interactions]] — Oral contraceptive + penicillin via the enterohepatic shunt — ⚠ and the two lecturers disagree about it, see below.\*\**

*Source: 2569 Estrogen and Progestins.pdf — รศ.ดร.วิลาสินี ทรัพย์พานิชชาโต๊ะ, Pharmacology II, 75 slides. Structures omitted.*

# Thyroid Hormones & Antithyroid Drugs

The thyroid follicle makes **T4 (thyroxine)** and **T3 (triiodothyronine)** from iodide + tyrosine residues on **thyroglobulin (TG)**. Parafollicular C-cells separately make **calcitonin**. Almost every antithyroid drug works by **blocking one specific step of that synthesis line** — so learn the steps first and the drugs fall out of them.

**Hypothalamus TRH → pituitary TSH → thyroid → T3 / T4 → ⊣ negative feedback on hypothalamus & pituitary**

## THE 7 SYNTHESIS STEPS — AND WHAT BLOCKS EACH

| # | STEP                              | WHAT HAPPENS                                                                                                                                     | BLOCKED BY                                                                                           |
|---|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1 | <b>Iodide trapping</b>            | I <sup>-</sup> pulled into the follicle cell against gradient by <b>Na<sup>+</sup>/I<sup>-</sup> symporter (NIS)</b>                             | anion inhibitors (ClO <sub>4</sub> <sup>-</sup> , SCN <sup>-</sup> , TcO <sub>4</sub> <sup>-</sup> ) |
| 2 | <b>Oxidation &amp; iodination</b> | <b>thyroperoxidase (TPO)</b> + H <sub>2</sub> O <sub>2</sub> makes I <sup>·</sup> and Tyr <sup>·</sup> → <b>MIT</b> (one I) / <b>DIT</b> (two I) | <b>thionamides</b> · high-dose iodide                                                                |
| 3 | <b>Coupling</b>                   | MIT + DIT → <b>T3</b> · DIT + DIT → <b>T4</b> (also TPO-dependent)                                                                               | <b>thionamides</b> · high-dose iodide                                                                |
| 4 | <b>Resorption</b>                 | TG colloid taken back into the cell                                                                                                              | —                                                                                                    |
| 5 | <b>Proteolysis</b>                | lysosomal enzymes cut T3/T4 off TG → blood                                                                                                       | <b>high-dose iodide</b>                                                                              |
| 6 | <b>Recycling</b>                  | leftover MIT/DIT de-iodinated, I <sup>-</sup> reused                                                                                             | —                                                                                                    |
| 7 | <b>Peripheral conversion</b>      | T4 → <b>T3</b> (active) by 5'-deiodinase                                                                                                         | <b>PTU</b> · β-blockers · corticosteroids · ipodate                                                  |

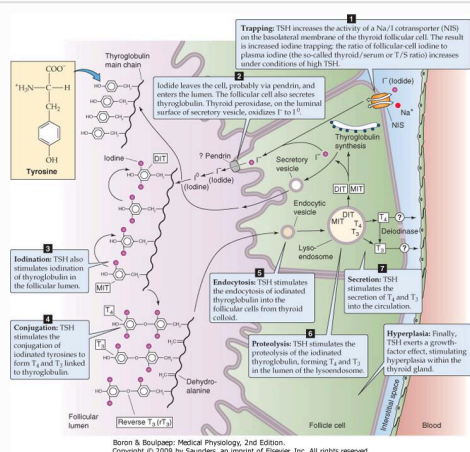
## NUMBERS THE LECTURE STRESSED

- Gland output is mostly **T4** (TG ratio T4:T3 ≈ 5:1) — **>80% of circulating T3 comes from peripheral deiodination of T4** ★
- T3 is 3–4× more potent** than T4 and binds the receptor ~10× more tightly; T4 is effectively the **pro-drug/reservoir**
- Transport: bound to **thyroxine-binding globulin (TBG)** — only **0.04% of T4** and **0.4% of T3** are free
- Receptors are **nuclear (TRs)** → gene transcription. **TR-α**: brain, kidney, gonads, muscle, heart · **TR-β**: pituitary, liver
- rT3** (reverse T3) = the **inactive** deiodination product

## HYPER VS HYPO — THE LAB PATTERN IS THE EXAM ANSWER ★

|                     | HYPERTHYROIDISM                                                                                                                                                  | HYPOTHYROIDISM                                                                                           |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>labs</b>         | <b>T3/T4 ↑ · TSH ↓</b>                                                                                                                                           | <b>T3/T4 ↓ · TSH ↑</b>                                                                                   |
| <b>top cause</b>    | <b>Graves' disease</b> — autoimmune TSH-receptor stimulating antibody                                                                                            | <b>Hashimoto's thyroiditis</b> — autoimmune destruction                                                  |
| <b>other causes</b> | thyroiditis, toxic nodular goiter                                                                                                                                | thyroidectomy, radioiodine damage                                                                        |
| <b>signs</b>        | palpitations, tachycardia, tremor, weight loss despite ↑ appetite, heat intolerance, sweating, irritability, insomnia, goiter, eye signs (lid lag, exophthalmos) | slow speech & movement, cold intolerance, constipation, weight gain, edema, menorrhagia, dry coarse skin |
| <b>in infants</b>   | —                                                                                                                                                                | <b>cretinism</b> — dwarfism + irreversible mental retardation                                            |

## OVERVIEW MAPS



Thyroid-PTH 2024 student · fig 07

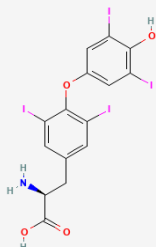
# Thyroid Hormone Replacement

## MECHANISM

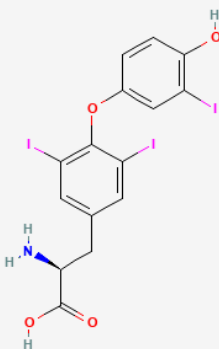
Straight replacement. Given orally, absorbed in **duodenum & ileum**, enters cells, T4 is converted to **T3** by 5'-deiodinase, T3 enters the nucleus and binds **TR** → gene transcription.

→ **Downstream:** ↑ basal metabolic rate, O<sub>2</sub> consumption, thermogenesis · normal growth, CNS and musculoskeletal development · ↓ TSH by negative feedback (*which is what you titrate against*)

## THE SHARED CORE · WHAT CHANGES



**T4 (levothyroxine).** two tyrosines joined, carrying **FOUR** iodines. Mostly a prodrug — the body removes one iodine to make the active hormone.



### T3 (liothyronine)

- one iodine fewer. That single atom is the difference between a 7-day  $t_{1/2}$  storage form and a ~1-day active hormone — and why T4 is what you prescribe

## MEDCHEM

**SAR** The molecule is two rings held **perpendicular** to each other - forced into that twist because the bulky **3,5-iodines** collide with the 2',6'-hydrogens. That geometry is the recognition element. The requirements: **iodine at C3 and C5 of the inner ring** (bromine substitutes, alkyl groups do not), **a free 4'-OH** on the outer ring, **a bridging atom** (O, S or CH<sub>2</sub> - none at all is inactive), and **L-alanine at C1** (the D-isomer and any longer or shorter acid are less active).

**SAR** Outer-ring halogen potency runs **F < Cl < Br < I** - iodine is not incidental to thyroid hormone, it is the top of the series.

**Physicochem** T4 is **tightly bound to plasma carrier protein**, which is what gives it a duration of **several weeks** against T3's several days, and its slower onset. T4 is also **hygroscopic** - the practical reason levothyroxine tablets are storage-sensitive.

## DRUGS

| DRUG                                        | WHAT IT TREATS                                                                                                             |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Levothyroxine (L-T4)</b>                 | <b>hypothyroidism — drug of choice</b> ★                                                                                   |
| <b>Liothyronine (T3)</b>                    | short-term replacement only — TSH suppression while awaiting total body scan in thyroid cancer · myxedema coma (injection) |
| <b>Liotrix (T4:T3 = 4:1)</b>                | hypothyroidism — not for routine replacement                                                                               |
| <b>Desiccated thyroid</b><br>(animal gland) | hypothyroidism — largely obsolete                                                                                          |

### WHY LEVOTHYROXINE WINS

**Long half-life ~7 days** → **once daily and stable levels** · predictable potency · cheap · no animal-protein allergy · easy to assay · and it still supplies T3 by peripheral conversion.

**Why not T3:** half-life ~1 day → multiple doses, transient supra-normal T3 peaks and troughs, more expensive, and **cardiotoxicity** — avoid in cardiac patients. **Why not desiccated thyroid:** unstable, variable T3 content, hard to monitor, allergenic.

### ADVERSE EFFECTS

*Essentially all of these are over-replacement — i.e. iatrogenic hyperthyroidism.* - ● **Cardiovascular** — angina, **arrhythmia / atrial fibrillation**, ↑ BP, tachycardia, heart failure, MI, cardiac arrest - **Osteoporosis** / ↓ **bone mineral density** - Anxiety, nervousness, emotional lability, insomnia, fatigue - Weight loss, ↑ appetite, abdominal cramps, heat intolerance, muscle weakness, alopecia

### INTERACTIONS

- ↓ **absorption:** cholestyramine, colestipol, **iron, calcium, aluminium-containing antacids**, sucralfate
- **Enzyme inducers** (*rifampin, phenobarbital, carbamazepine, phenytoin*) → ↑ metabolism of T3/T4
- Levothyroxine ↓ **sulfonylurea effect** and ↑ **warfarin effect**

### CAUTIONS

- ⚠ **Elderly and cardiac patients are very sensitive** — start low (12.5–25 µg/day), raise slowly. **Angina or arrhythmia appearing = reduce or stop.**
- ⚠ Caution in adrenal disease and diabetes
- ✅ **Safe in pregnancy — never stop it.** Requirement usually *rises*; monitor TSH and re-titrate.

# Thionamides

## MECHANISM

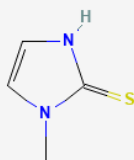
Inhibit **thyroperoxidase (TPO)** → block **iodination (step 2)** and **coupling (step 3)**. They do **not** block iodide uptake. **PTU additionally blocks peripheral 5'-deiodination of T4 → T3**.

→ **Downstream:** no *new* hormone is made — but the gland holds a large pre-formed store, so the clinical effect takes **3–4 weeks** ★. Full course runs 1–2 years, with **60–70% relapse** after stopping.

## THE SHARED CORE · WHAT CHANGES

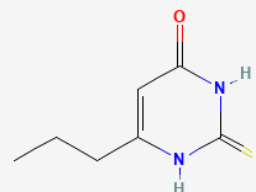


**Thiourea — the thioamide group.** ● the S=C–N group is the drug. Thyroid peroxidase oxidises the sulfur instead of iodide, so no iodine is organified.



**Methimazole**

10× more potent, long acting → once daily, first line



**Propylthiouracil**

⚠ heavily protein-bound → less placental transfer → the first-trimester choice; also blocks peripheral T4→T3

## MEDCHEM

**SAR** Both drugs are ring-stabilised **thioureas** - plain thiourea has the antithyroid activity but is too toxic to use (agranulocytosis), so the class is 2-thiouracil (PTU) and 2-thioimidazole (methimazole) versions of it. ⚠ **The agranulocytosis followed the thiourea into the clinic**; it is a property of the chemistry, not of one brand. Essential for activity: the **C2 thioketo/thioenol group** and an **unsubstituted N1**.

**Prodrug Carbimazole is a prodrug of methimazole** - the carbamate exists to improve the taste and slow the release, nothing more.

**Physicochem** PTU is only slightly water-soluble (readily soluble in alkali), **80–90% albumin-bound**, half-life **1.5 h** → dosed 1–4 times daily. Methimazole is freely water-soluble, essentially **unbound**, half-life **5–6 h** → once or twice daily, and more potent. ⚠ The deck also notes methimazole **does not inhibit peripheral T4→T3 deiodination**, which PTU does - the one pharmacological job PTU still has to itself.

## DRUGS

| DRUG                                  | WHAT IT TREATS                                                                                               |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Methimazole (MMI / thiamazole)</b> | hyperthyroidism, Graves' — <b>more potent (<math>\approx 10\times</math>) and longer acting</b> , once daily |
| <b>Propylthiouracil (PTU)</b>         | hyperthyroidism — preferred in <b>1st-trimester pregnancy</b> , breastfeeding, and thyroid storm             |
| <b>Carbimazole</b>                    | pro-drug converted to methimazole — <b>not available in Thailand</b>                                         |

## USES

**Non-operative therapy of hyperthyroidism / Graves'** — best in young patients with mild disease · **Pre-operative preparation** — combined with iodide · **Thyrotoxic crisis** — combined with propranolol and a large dose of iodide

## ADVERSE EFFECTS

- **AGRANULOCYTOSIS** ★ — the most serious reaction, **0.3–0.6%**, reversible. Higher risk in the **elderly** and on **MMI >40 mg/day**. **Cross-sensitivity between PTU and MMI**, so you cannot simply swap.
  - **MMI: dose-related** · **PTU: idiosyncratic** (*unpredictable, dose-independent*)
- **Hepatitis** → **PTU** · **Cholestatic dysfunction** → **methimazole** ★
- Common: **maculopapular pruritic rash** ± fever, GI upset, arthralgia (*manage with antihistamines*)
- **Hypothyroidism** from overtreatment
- Rare: urticaria, **ANCA-associated vasculitis**, lupus-like reaction, lymphadenopathy, hypoprothrombinemia, polyserositis

## PREGNANCY — THE ONE THEY ASK ABOUT ★

- Both cross the placenta and can accumulate in the fetal thyroid.
- **PTU in the 1st trimester** — more plasma-protein bound → less placental transfer, low in breast milk, less teratogenic. **FDA: PTU is restricted to the 1st trimester** (*hepatotoxicity*).
- ● **Then switch to methimazole** — lower hepatotoxic risk for the rest of the pregnancy.
- ● **MMI embryopathy** if used in the 1st trimester: aplasia cutis, **choanal atresia**, **esophageal atresia**, facial dysmorphism, growth retardation, developmental delay.

## ADMINISTRATION — PTU VS METHIMAZOLE

|                                    | PTU                      | METHIMAZOLE                |
|------------------------------------|--------------------------|----------------------------|
| <b>Plasma protein binding</b>      | 75%                      | none                       |
| <b>Plasma <math>t_{1/2}</math></b> | <b>75 minutes</b>        | <b>4–6 hours</b>           |
| <b>Vd</b>                          | 20 L                     | 40 L                       |
| <b>Concentrated in thyroid</b>     | yes                      | yes                        |
| <b>Dose frequency</b>              | ● <b>1–4 times daily</b> | <b>once or twice daily</b> |
| <b>Transplacental passage</b>      | low                      | low                        |
| <b>Level in breast milk</b>        | low                      | low                        |

- ★ **Both concentrate in the thyroid, which is why the short plasma half-life doesn't dictate the dosing interval** — the drug is where it needs to be long after it has left the blood.
- ● **PTU can be used only in the first trimester of pregnancy** (FDA).
- 💡 Methimazole once or twice daily against PTU up to four times a day is the practical reason methimazole is the default — outside that first trimester.



# Anion Inhibitors

## MECHANISM

Monovalent anions that **competitively inhibit the  $\text{Na}^+/\text{I}^-$  symporter (NIS)** — they block **iodide trapping (step 1)**, the very first step.

→ **Downstream:** no iodide enters the follicle → no substrate for synthesis. ⚠ **Not used in practice** — toxicity outweighs the benefit.

## DRUGS

| DRUG                                                                                      | WHAT IT TREATS                                                 |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Potassium perchlorate (<math>\text{ClO}_4^-</math>)</b>                                | <b>iodide-induced hyperthyroidism</b> — the one residual niche |
| <b>Thiocyanate (<math>\text{SCN}^-</math>)</b>                                            | <i>(mechanism example only)</i>                                |
| <b>Pertechnetate (<math>\text{TcO}_4^-</math>) · Nitrate (<math>\text{NO}_3^-</math>)</b> | <i>(mechanism example only)</i>                                |

## ADVERSE EFFECTS

- **Aplastic anemia** — the reason this class was abandoned

# Iodides

## MECHANISM

At **high dose** (>6 mg/day) iodide does the opposite of what you'd expect: it **inhibits thyroglobulin proteolysis (step 5)** → hormone **release** stops. It also inhibits TPO and **blocks peripheral T4 → T3 conversion**, and it **shrinks the gland and cuts its vascularity**.

→ **Downstream**: the **fastest-acting** antithyroid measure — improvement within **24 h to 2–7 days** ★, which is why it's the pre-operative and thyroid-storm drug.

## DRUGS

| DRUG                                                             | WHAT IT TREATS                                                                                                                 |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Lugol's solution</b> (5% iodine + 10% KI)                     | pre-operative preparation · thyroid storm                                                                                      |
| <b>Saturated solution of potassium iodide (SSKI)</b>             | pre-operative preparation · thyroid storm                                                                                      |
| <b>Potassium iodide (KI)</b>                                     | endemic goiter prophylaxis · <b>thyroid blockade in radiation emergencies</b>                                                  |
| <b>Sodium ipodate / iopanoic acid</b> (iodinated contrast media) | thyroid storm adjunct · alternative when iodide or thionamides are contraindicated — blocks T4 → T3 <b>and</b> hormone release |

## USES

**Pre-operative therapy** — the ↓ vascularity makes the surgeon's job possible · **Thyrotoxic crisis** — with PTU · **Endemic goiter prophylaxis** · **Radiation emergency** — 30–100 mg KI saturates the gland so it can't take up radioactive iodine

## ADVERSE EFFECTS

- **Acute hypersensitivity to iodine** — swelling of lips and eyelids, ● **laryngeal angioedema**, fever, joint pain, petechial hemorrhage
- **Iodism** (*chronic, dose-dependent*) — metallic/unpleasant taste, burning mouth, sore teeth and gums, salivation, acneiform rash, swollen salivary glands, mucous-membrane ulceration, conjunctivitis, rhinorrhea, drug fever
- **Long-term high dose** → **hypothyroidism and goiter**

## CAUTIONS ★

- ● **Escape phenomenon** — the effect simply disappears after ~10–15 days (up to 2–8 weeks). **Short-course only, never as monotherapy**, and abrupt withdrawal causes **severe rebound thyrotoxicosis**.
- ⚠ **Timing vs thionamides**: iodide raises the intraglandular iodine store → **delays thionamide onset**. Give iodide *after* the thionamide has taken effect.
- ● **Blocks radioactive iodine uptake** — avoid entirely if  $^{131}\text{I}$  is planned.
- ● **Not in pregnancy** — crosses the placenta → **fetal goiter**.

# Radioactive Iodine ( $^{131}\text{I}$ )

## MECHANISM

Swallowed as sodium  $^{131}\text{I}$ , rapidly absorbed, **trapped and concentrated in thyroid follicles** like ordinary iodide — then destroys them from the inside. **90% of the therapeutic effect is  $\beta$  radiation**, penetrating only ~0.5 mm, so damage is confined to the gland. (*The  $\gamma$  component is what makes it imageable.*)

→ **Downstream:** thyroid parenchyma destroyed over **2–3 weeks**, maximal effect at **~3 months** (4–18 weeks). Radioactive half-life **8 days**; typical dose **4–15 mCi**; repeat at 6 months if it fails.

## ISOTOPES

| ISOTOPE                            | WHAT IT'S FOR                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------|
| <b><math>^{131}\text{I}</math></b> | <b>treatment</b> — the only isotope used therapeutically for thyrotoxicosis and thyroid cancer |
| <b><math>^{123}\text{I}</math></b> | <b>diagnosis only</b> — 24-h uptake and thyroid imaging, $\gamma$ emitter, $t_{1/2}$ 13 h      |

## USES & ADVANTAGES

**Graves' disease · toxic nodular goiter · thyroid cancer** · the **elderly** · **allergy to PTU** · **recurrent hyperthyroidism** · patients whose systemic disease rules out surgery · patients needing thyrotoxicosis fixed fast (severe heart failure)

**Advantages:** simple · inexpensive · painless · no surgical risk, no scar, no injury to the parathyroids or recurrent laryngeal nerve · **control is permanent**

## ADVERSE EFFECTS

- **Permanent hypothyroidism** ★ — not a complication so much as the price. **12% in year 1 rising to 76% by year 11**;  $\geq 50\%$  within a year on high dose. More than surgery or drugs.
- **Post-radiation thyroiditis** — painful swollen gland. Risk highest in elderly, severely toxic patients not pre-treated with a thionamide.
- **Worsening Graves' ophthalmopathy** — thought to be from thyroid antigen release
- Focal neck soreness · damage to a fetal thyroid · reported thyroid carcinoma / leukemia risk

## CONTRAINDICATIONS & PREPARATION

- **Pregnancy** — and contraception for  **$\geq 6$  months** afterwards · ● **Breastfeeding** (*excreted in milk*)
- **Patients under 20** — the pediatric thyroid is radiation-sensitive → radiation-associated thyroid cancer
- ⚠ **Active, progressive Graves' ophthalmopathy** — treat with a thionamide until the eyes are stable, or give steroid cover first

**Preparation:** restrict dietary iodine + antithyroid drug to euthyroid → **stop the antithyroid drug 3–7 days before** (*propranolol may continue*) → dose by gland size and severity → radiation precautions around other people for ~1 week → **restart the antithyroid drug within 72 h.**

# Adjunctive & Thyroid Storm Therapy

## MECHANISM

None of these touch hormone synthesis — they **blunt the tissue response** while the real drug works. Thyrotoxicosis looks like sympathetic overdrive, so sympatholytics buy symptomatic control immediately instead of in 3–4 weeks.

→ **Downstream:** rapid relief of tremor, palpitations, tachycardia, AF and hypertension · several of them **also block peripheral T4 → T3 conversion**

## DRUGS

| DRUG                                                                  | WHAT IT TREATS                                                                          |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Propranolol</b> ( <i>β-blocker without ISA — agent of choice</i> ) | tachycardia, tremor, palpitations, AF, hypertension · <b>high dose also ↓ T4 → T3</b> ★ |
| <b>Atenolol</b>                                                       | same symptomatic control                                                                |
| <b>Diltiazem</b> ( <i>CCB</i> )                                       | rate control when β-blockers are contraindicated                                        |
| <b>Dexamethasone / hydrocortisone</b>                                 | <b>inhibits T4 → T3</b> · shock prevention in storm                                     |
| <b>Lithium carbonate</b>                                              | inhibits hormone <b>release</b> — used when the patient is <b>iodine-allergic</b>       |

## THYROID STORM — THE EMERGENCY STACK ★

Acute, life-threatening thyrotoxicosis: fever, flushing, sweating, tachycardia, AF, wide pulse pressure, heart failure, CNS signs (*agitation → delirium → coma*), nausea, vomiting, diarrhea, jaundice. Death comes from **heart failure and shock**. Treat immediately, all at once:

- **Propranolol** 1–2 mg IV, or 40–80 mg every 6 h — CVS control
- **Potassium iodide** 10 drops/day *or* **sodium ipodate** 1 g/day — stop hormone release
- **PTU or methimazole** — stop hormone synthesis
- **Hydrocortisone** 50 mg IV every 6 h — prevents shock **and** blocks T4 → T3

## ADVERSE EFFECTS — B-BLOCKERS

- Fatigue, sexual dysfunction
- **Sinus bradycardia, heart block**

## CAUTIONS — B-BLOCKERS

- ⚠ **Asthma** · diabetes · congestive heart failure · depression

Principle behind this: *[[pharm-drug-toxicity]]* — Thyroid hormones, same list, same trap.

Source: 2569\_thyroid\_hormone\_Pharmacology\_2.pdf (30 slides) + 2569\_เอกสารคำสอนเรื่องยาด้าน

ไทรอยด์\_Vilasinee.pdf (18 pp.) — รศ.ดร.วิลาสินี ธีรภูมิพานิช ชำโตะ, PSYCH 305 Pharmacology II. Structures omitted.

# Drugs Regulating Bone-Mineral Homeostasis

**99%** of body calcium sits in bone and teeth as **hydroxyapatite**  $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ ; the **1%** in serum runs muscle contraction, the heartbeat, clotting, nerve conduction and enzyme function. Three hormones — **PTH** · **vitamin D** · **calcitonin** — keep that 1% steady by trading with the bone bank. Every drug here either pushes on that trade or on the bone itself.

↓ serum  $\text{Ca}^{2+}$  → ↑ **PTH** → ① **osteoclast bone resorption** (*the biggest lever*) · ② **renal  $\text{Ca}^{2+}$  reabsorption + phosphate excretion** · ③ **renal activation of vitamin D** → **calcitriol** → **intestinal  $\text{Ca}^{2+}$  absorption** → ↑ serum  $\text{Ca}^{2+}$  → ↓ **PTH** + ↑ **calcitonin** → ↓ serum  $\text{Ca}^{2+}$

## THE THREE CELLS

- **Osteoblast** — builds bone. Also makes **RANKL**, the signal that switches osteoclasts on.
- **Osteoclast** — resorbs bone. **No PTH receptor** — PTH reaches it indirectly, via the osteoblast.
- **Osteocyte** — a buried osteoblast. Secretes **sclerostin**, which brakes bone formation.

## ANTI-RESORPTIVE VS ANABOLIC

| DIRECTION                                         | CLASSES                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Anti-resorptive</b> ( <i>block breakdown</i> ) | Calcium · Vitamin D derivatives · <b>Bisphosphonates</b> · <b>RANKL inhibitors</b> |
| <b>Mixed</b> ( <i>both</i> )                      | Estrogens / STEARs / SERMs · Calcitonin · Vitamin K analogues                      |
| <b>Anabolic</b> ( <i>build new bone</i> )         | <b>PTH analogues</b> · <b>Sclerostin inhibitors</b>                                |
| <b>Not acting on bone directly</b>                | Calcimimetics · Phosphate binders                                                  |

## WHICH ABNORMALITY GETS WHICH DRUG

| PROBLEM                  | DRUGS                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Hypocalcemia</b>      | calcium salts ( <i>IV for crisis</i> ) · vitamin D derivatives · PTH1-84                                              |
| <b>Hypercalcemia</b>     | IV bisphosphonates ( <i>ibandronate, zoledronate, pamidronate</i> ) · salmon calcitonin · calcimimetics               |
| <b>Hyperphosphatemia</b> | phosphate binders                                                                                                     |
| <b>Osteoporosis</b>      | bisphosphonates · denosumab · estrogens/tibolone/raloxifene · teriparatide · romosozumab · calcitonin · menatetrenone |

## WHERE THE ABNORMALITY COMES FROM

- **Hypercalcemia** — hyperparathyroidism · hypervitaminosis D · **cancers** (*squamous cell of head/neck/lung, breast, ovarian, renal*) · milk-alkali syndrome
- **Hypocalcemia** — hypoparathyroidism · vitamin D deficiency · **CKD (2° hyperparathyroidism)** · malabsorption
- **Bone disease** — osteoporosis · Paget's disease (*osteitis deformans*)

# Calcium

## MECHANISM

Pure **substrate** — raises calcium availability. Not optional: every patient on any anti-osteoporotic drug still needs adequate calcium underneath it.

→ **Downstream:** ↑ serum  $\text{Ca}^{2+}$  ·  **$\text{CaCO}_3$  also binds dietary phosphate** in the gut → **Target:** elemental  $\text{Ca}^{2+}$  **800–1,500 mg/day**, higher in pregnancy, renal impairment, or on osteoporosis therapy

## ELEMENTAL CALCIUM CONTENT — THE NUMBER THAT MATTERS

| SALT                      | ELEMENTAL $\text{Ca}^{2+}$                                                    |
|---------------------------|-------------------------------------------------------------------------------|
| <b>Calcium carbonate</b>  | <b>40.0%</b> ★                                                                |
| Calcium acetate           | 25.3%                                                                         |
| Calcium citrate           | 21.0%                                                                         |
| Calcium lactate           | 13.0%                                                                         |
| IV: calcium gluconate 10% | 0.465 mEq $\text{Ca}^{2+}$ /mL                                                |
| IV: calcium chloride 10%  | <b>1.36 mEq <math>\text{Ca}^{2+}</math>/mL</b> ( $\approx 3\times$ gluconate) |

## USES

**Oral** — hypocalcemia (non-crisis) · osteoporosis prevention & treatment · **IV** — **hypocalcemic crisis**, by *slow* infusion

## ADVERSE EFFECTS

- Hypercalcemic disorders · **tissue calcification**
- Flatulence, constipation
- **Metabolic alkalosis**, gastric acid hypersecretion (rebound)
- ● **Rapid IV** → **arrhythmia**. Infuse slowly.
- Vein irritation —  **$\text{CaCl}_2$  > calcium gluconate**

## CAUTIONS

- ★  **$\text{CaCO}_3$  needs gastric acid to dissolve** → **take with food**. Achlorhydria or a PPI kills its absorption; use citrate instead.
- ⚠ Chelates other drugs in the gut — separates from bisphosphonates, tetracyclines, fluoroquinolones

## ADMINISTRATION — CALCIUM SALTS

| ORAL SALT         | ELEMENTAL CALCIUM |
|-------------------|-------------------|
| Calcium carbonate | ★ 40.0%           |
| Calcium acetate   | 25.3%             |
| Calcium citrate   | 21.0%             |
| Calcium lactate   | 13.0%             |

**Recommendation: 800–1,500 mg/day of ELEMENTAL calcium.** Common strengths are **CaCO<sub>3</sub> 625 mg, 1 g**, and **1.5 g (effervescent tablet)**.

- **Calcium carbonate needs gastric acid to be absorbed — take it with food.** ★ That single fact is why it fails in a patient on a PPI, and why calcium citrate is the alternative there.
- **IV, for hypocalcaemic crisis only:** calcium gluconate 10% ( $0.465 \text{ mEq Ca}^{2+}/\text{mL}$ ) · calcium chloride 10% ( $1.36 \text{ mEq Ca}^{2+}/\text{mL}$ ). ● **Slow infusion — rapid IV can cause arrhythmia.** Calcium chloride irritates the vein more than the gluconate.
- 💡 The percentage column is the part patients never see: **1 g of calcium carbonate is 400 mg of calcium**, not 1 g. A dose that looks generous on the label can be half of what was intended.

# Vitamin D Derivatives

## MECHANISM

Activate the **vitamin D receptor**.

→ **Downstream:** ↑ intestinal  $\text{Ca}^{2+}$  absorption (*the main action*) · ↓ **PTH** → **Target:** pro-vitamin D **600–800 IU/day**

## THE ACTIVATION LADDER — THIS IS THE WHOLE CLASS

**D2 / D3 (pro-vitamin)** → liver 25-hydroxylase → **calcidiol, 25(OH)D** (*semi-active*) → kidney  $1\alpha$ -hydroxylase → **calcitriol, 1,25(OH)<sub>2</sub>D** (*active*)

★ **In CKD the kidney step fails** — so cholecalciferol and ergocalciferol barely work. Give a derivative that skips it: **alfacalcidol** ( $1(\text{OH})\text{D}$  — *needs only the liver*) or **calcitriol** (*already active*).

## DRUGS

| DRUG                                              | WHAT IT TREATS                                                             |
|---------------------------------------------------|----------------------------------------------------------------------------|
| <b>Cholecalciferol (D3)</b>                       | supplementation — more effective at raising serum $\text{Ca}^{2+}$ than D2 |
| <b>Ergocalciferol (D2)</b>                        | supplementation                                                            |
| <b>Calcidiol / calcifediol</b><br>(25(OH)D)       | supplementation when liver hydroxylation is impaired                       |
| <b>Alfacalcidol</b> ( $1(\text{OH})\text{D}$ )    | <b>CKD</b> — bypasses the failed renal step                                |
| <b>Calcitriol</b> ( $1,25(\text{OH})_2\text{D}$ ) | CKD · <b>hypocalcemic crisis (IV)</b>                                      |

## USES

Osteoporosis prevention & treatment · Hypocalcemia · **2° hyperparathyroidism**

## ADVERSE EFFECTS

- **Hypercalcemia**
- **Tissue calcification**



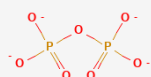
# Bisphosphonates

## MECHANISM

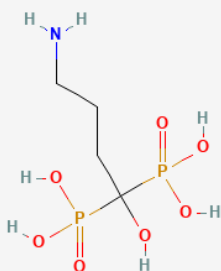
The backbone mimics **pyrophosphate**, a native bone component — so the drug binds hydroxyapatite and **parks in bone for years**, long after it has vanished from plasma. There it **inhibits osteoclast function** through several routes.

→ **Downstream:** ↓ bone resorption · ↓ serum  $\text{Ca}^{2+}$  · effect persists well past the last dose *Nitrogen-containing agents are the modern, more potent ones; the non-nitrogen originals are no longer used in Thailand.*

## THE SHARED CORE · WHAT CHANGES

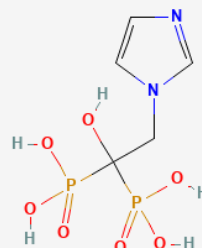


**Pyrophosphate — P–O–P.** bone's own regulator, but enzymes hydrolyse it instantly. ● swap the central O for a CARBON and it becomes indestructible — that is the whole drug class.



### Alendronate

nitrogen-containing → blocks farnesyl pyrophosphate synthase, ~1000× more potent than the non-N ones



### Zoledronic acid

an imidazole ring → the most potent; once-yearly IV

## DRUGS

| DRUG                                 | WHAT IT TREATS                                                                         |
|--------------------------------------|----------------------------------------------------------------------------------------|
| <b>Alendronate</b> (PO, ± vitamin D) | osteoporosis — prevention & treatment                                                  |
| <b>Risedronate</b> (PO)              | osteoporosis                                                                           |
| <b>Ibandronate</b> (PO, IV)          | osteoporosis · <b>hypercalcemia of malignancy</b> · skeletal events in bone metastases |
| <b>Zoledronate</b> (IV)              | osteoporosis · <b>hypercalcemia of malignancy</b>                                      |
| <b>Pamidronate</b> (IV)              | <b>hypercalcemia of malignancy</b>                                                     |
| <i>Non-nitrogen (obsolete)</i>       | clodronate · etidronate · tiludronate                                                  |

## USES

**Osteoporosis** in women *and* men — postmenopausal · glucocorticoid-induced · hormone-deprivation therapy · **Paget's disease** · **Osteolytic bone metastases** · Ectopic ossification (treatment + prophylaxis) · **Hypercalcemia of malignancy** — IV only, high dose, short course

## ADVERSE EFFECTS

- **PO: esophageal & gastric irritation** ★ — this is what drives the whole dosing ritual
- **IV: flu-like symptoms** — fever, myalgia, arthralgia, headache (*mitigate with slow infusion*)
- **Hypocalcemia** — mostly IV
- ● **Osteonecrosis of the jaw (ONJ)** — rare but severe, **IV > PO**
- ● **Atypical femoral fracture** — rare, from over-suppressed remodelling

## INTERACTIONS

- ★ **Multivalent cations block absorption** — calcium, antacids, milk/soy milk, iron. Oral bioavailability is already <1%, and food alone kills it.
- **No CYP interactions** — no systemic metabolism at all

## CAUTIONS

- ★ **Oral dosing rule:** plain **water only**, empty stomach, **stay upright ≥30 min**, do not chew or crush
- ⚠ **Renal excretion** — avoid if renal function is severely impaired (*this is denosumab's opening*)
- ● **Crosses the placenta** — **not for pregnancy**

## ADMINISTRATION — BISPHOSPHONATES ARE THE STRICTEST ORAL DRUG ON THIS PAGE

- **Oral bioavailability is under 1%**, and ● **food inhibits absorption** by forming complexes with multivalent cations.
- ● **Take with water, on an empty stomach** — **no food, no other drugs, no other drinks.**
- **No systemic metabolism, no CYP interactions; excreted mostly in urine** → ⚠ **not appropriate if renal function is severely abnormal.**
- ⚠ **Crosses the placenta** — **not appropriate in pregnancy.**
- 💡 "With water" is literal. A patient taking it with coffee, milk or their morning vitamins is absorbing essentially none of it — and with a baseline under 1%, there is no margin to lose.

# RANKL Inhibitors

## MECHANISM

The osteoblast secretes **RANKL** (*receptor activator of NF- $\kappa$ B ligand*) to switch osteoclasts on. **Denosumab** is a human **IgG2 monoclonal antibody** that binds RANKL with high affinity and mops it up.

→ **Downstream:** osteoclast never activates → ↓ bone resorption → **PK that defines it: non-renal excretion** (*no dose adjustment at low CrCl*) and **no accumulation in bone** — so unlike a bisphosphonate, ★ **the effect stops when the drug stops**, with rebound resorption.

## DRUGS

| DRUG                                                | WHAT IT TREATS                        |
|-----------------------------------------------------|---------------------------------------|
| <b>Denosumab</b> ( <i>60 mg SC every 6 months</i> ) | osteoporosis treatment in women & men |

## USES

**Osteoporosis treatment** — postmenopausal · glucocorticoid-induced · hormone-deprivation therapy · ★ **the choice when renal function is poor**

## ADVERSE EFFECTS

- **Hypocalcemia**
- Musculoskeletal pain (*flu-like*)
- **Hypersensitivity** — it is an antibody

## CAUTIONS

- ⚠ **Never just stop it.** No bone reservoir → resorption rebounds; follow with another agent.

## ADMINISTRATION — DENOSUMAB

- **60 mg subcutaneously every 6 months.**
- ★ **Non-renal excretion** — **no dose adjustment even if CrCl is low**, and no accumulation in bone. That is the practical advantage over the bisphosphonates, which are the opposite on both counts.
- 💡 Twice a year is easy to take and easy to forget. The interval is the adherence risk here, not the dose.

# Estrogens, STEARs & SERMs

## MECHANISM

**Estrogen suppresses osteoclast activity.** Lose it at menopause and resorption accelerates — that *is* postmenopausal osteoporosis. Replacing it protects bone, but the estrogen receptor is everywhere else too:

→ **Bone:** ↑ strength · → **CNS:** ↓ vasomotor symptoms, maintains sexual desire → **Breast / vagina / cervix:** ↑ cell growth → **cancer risk** · → **Cardiovascular:** ↑ clotting factors → **thromboembolism**

**STEARs** and **SERMs** were built to keep the bone effect and drop the rest.

## DRUGS

| DRUG                                           | WHAT IT TREATS                                                  |
|------------------------------------------------|-----------------------------------------------------------------|
| <b>Conjugated equine estrogens</b>             | prevention of postmenopausal osteoporosis                       |
| <b>17β-estradiol · estradiol hemisuccinate</b> | prevention of postmenopausal osteoporosis                       |
| <b>Estradiol valerate + norgestrel</b>         | as above — the progestogen protects the endometrium             |
| <b>Tibolone</b> (STEAR)                        | prevention — <b>with</b> menopausal-symptom relief              |
| <b>Raloxifene</b> (SERM)                       | <b>treatment and prevention</b> — <i>without</i> symptom relief |

## THE COMPARISON ★

|                                | ESTROGENS       | TIBOLONE (STEAR) | RALOXIFENE (SERM)       |
|--------------------------------|-----------------|------------------|-------------------------|
| <b>Bone strength</b>           | ↑               | ↑                | ↑                       |
| <b>Vasomotor symptoms</b>      | improve         | improve          | <b>neutral or worse</b> |
| <b>Androgen-like effects</b>   | neutral         | <b>increase</b>  | neutral                 |
| <b>Breast cancer risk</b>      | <b>increase</b> | decrease         | decrease                |
| <b>Endometrial cancer risk</b> | <b>increase</b> | neutral          | neutral                 |
| <b>Venous thromboembolism</b>  | increase        | increase         | increase                |

## ADVERSE EFFECTS

- **Estrogen-responsive cancers** — breast, endometrial. This is why long-term HRT fell out of favour.
- **Thromboembolism** — the one risk *none* of the three escapes
- **Catch-up phenomenon** — bone benefit is lost rapidly once the drug stops
- Tibolone: androgenic effects

## CAUTIONS

- ⚠ All three sit behind bisphosphonates and denosumab in practice — the risk-benefit rarely wins
- ⚠ Combined oral contraceptives are **not** an osteoporosis treatment

## ADMINISTRATION — ESTROGEN PREPARATIONS

- Estradiol hemisuccinate — patch ·  $17\beta$ -estradiol — gel · conjugated equine estrogens — tablet · estradiol valerate + norgestrel — tablet.
- ⚠️ **Catch-up phenomenon: the benefit is lost rapidly after stopping** — so the route chosen has to be one the patient will keep using, not the one with the best pharmacology on paper.
- 💡 Transdermal versus oral is the decision that gets made here, and "which will you actually stick with for years" is a fair way to make it, because stopping resets the gain.

# Calcitonin

## MECHANISM

A **thyroid peptide hormone** that acts on bone, kidney and gut in the **direction opposite to PTH**.

→ **Downstream:** ↑ bone formation · ↓ bone resorption · ↓ **serum  $\text{Ca}^{2+}$**  *Human calcitonin isn't used clinically — salmon calcitonin is more potent and longer-acting at the human receptor.*

## DRUGS

| DRUG                                                 | WHAT IT TREATS                                                                                   |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Synthetic salmon calcitonin</b> (IV, nasal spray) | hypercalcemia · postmenopausal osteoporosis · Paget's disease · <b>bone pain from osteolysis</b> |

## USES

**Hypercalcemia** · Postmenopausal osteoporosis · **Paget's disease** · ★ **Bone pain associated with osteolysis** — the analgesic niche no other class fills

## ADVERSE EFFECTS

- **Hypocalcemia**
- Nausea, vomiting, **facial flushing**
- **IV:** inflammatory reaction at the injection site
- **Nasal spray:** local nasal symptoms

## ADMINISTRATION — CALCITONIN

- **Dosage forms: IV and nasal spray.**
- ⚠ Each route has its own adverse effect: **IV — inflammatory reactions at the injection site · nasal spray — nasal symptoms.**
- 💡 The nasal spray is what makes calcitonin usable outside hospital, which matters most for the bone-pain indication rather than for osteoporosis.

# Parathyroid Hormone Analogues

## MECHANISM

★ **The pattern of exposure decides the direction.** Endogenous PTH in hyperparathyroidism is **continuous / non-pulsatile** → osteoclast activation → bone loss. Given **pulsatile** — one SC dose a day — the same hormone becomes **anabolic** and builds bone.

→ **Downstream:** ↑ osteoblast bone formation · ↑ serum  $\text{Ca}^{2+}$  *PTH is 84 amino acids, but only the first 34 are needed to activate the receptor.*

## DRUGS

| DRUG                                                   | WHAT IT TREATS                                         |
|--------------------------------------------------------|--------------------------------------------------------|
| <b>Teriparatide</b> ( <i>recombinant PTH1-34, SC</i> ) | osteoporosis treatment                                 |
| <b>PTH1-84</b> ( <i>recombinant human PTH, SC</i> )    | <b>hypoparathyroidism</b> · osteoporosis ( <i>EU</i> ) |

## USES

**Osteoporosis treatment** — SC daily for **18–24 months**, where the effect maxes out · **Hypoparathyroidism** — PTH1-84 as hormone replacement

## ADVERSE EFFECTS

- **Hypercalcemia**
- Arthralgia, pain, nausea
- ● **Osteogenic sarcoma in animals treated >2 years** ★ — this is why the course is capped at 2 years

## CAUTIONS

- ⚠ **Peptide** — **SC only**. Oral dosing would digest it.
- ⚠ **Do not exceed 2 years** of PTH1-34

# Sclerostin Inhibitors

## MECHANISM

**Osteocytes secrete sclerostin**, which blocks the **Wnt/ $\beta$ -catenin pathway** inside osteoblasts — the signalling that drives bone formation. **Romosozumab** (*human IgG2 mAb*) blocks sclerostin, so Wnt/ $\beta$ -catenin runs unopposed.

→ **Downstream:**  $\uparrow\uparrow$  bone formation · partly  $\downarrow$  bone resorption too — the only genuinely dual agent → **PK:** non-renal excretion (*no adjustment at low CrCl*) · no bone accumulation

## DRUGS

| DRUG                                                     | WHAT IT TREATS                                                    |
|----------------------------------------------------------|-------------------------------------------------------------------|
| <b>Romosozumab</b> (210 mg SC monthly $\times$ 12 doses) | osteoporosis in postmenopausal women at <b>high fracture risk</b> |

## USES

**Treatment of osteoporosis in postmenopausal women at high risk of fracture.** ★ The anabolic effect fades after the 12 monthly doses — follow with an anti-resorptive to hold the gain.

## ADVERSE EFFECTS

- **Myocardial infarction · stroke · cardiovascular death** ★ — the class-defining risk
- **Hypocalcemia**
- Musculoskeletal pain, hypersensitivity

## CAUTIONS

- **Avoid with a history of coronary or cerebrovascular disease**
- ⚠ Wnt/ $\beta$ -catenin is not bone-specific — be careful in patients with other pathology



# Vitamin K Analogues & Fluoride

## MECHANISM

**Menatetrenone** = synthetic **vitamin K2**. Vitamin K is the cofactor for making **osteocalcin**, a core bone protein.

→ **Downstream**: ↑ osteocalcin · accelerates osteogenesis · inhibits bone resorption

**Fluoride** works only on **teeth** — inhibits demineralization, promotes remineralization, suppresses plaque bacteria.

## DRUGS

| DRUG                                          | WHAT IT TREATS                                          |
|-----------------------------------------------|---------------------------------------------------------|
| <b>Menatetrenone</b> (PO, with food)          | improves bone mass · relieves back pain in osteoporosis |
| <b>Sodium fluoride</b> (PO, sucked or chewed) | <b>prophylaxis of dental caries — that is all</b>       |

## ADVERSE EFFECTS

- **Menatetrenone**: stomach discomfort, abdominal pain, diarrhea, nausea
- **Fluoride**: **dental fluorosis** — mottling of enamel

## INTERACTIONS

-  **Menatetrenone antagonises warfarin**  — it *is* vitamin K. INR falls; clot risk rises.
- Fluoride: **avoid dairy products** — calcium binds it

## CAUTIONS

-  **Fluoride raises bone mineral density but is NOT indicated for osteoporosis** — the new bone is denser on the scan and not stronger in reality. Classic exam trap.
-  Menatetrenone is fat-soluble → **take with or after food**

## ADMINISTRATION — MENATETRENONE (VITAMIN K<sub>2</sub>)

- **Oral — should be taken with food.**
-  **Interacts with warfarin.** This is vitamin K; it is the antidote pathway, taken as a supplement.
-  Anyone on an anticoagulant needs this flagged before the first dose, not at the first abnormal INR.

# Calcimimetics

## MECHANISM

**Allosteric activators of the calcium-sensing receptor (CaSR)** on the parathyroid gland — they make the gland *think* serum calcium is higher than it is.

→ **Downstream:** ↓ PTH secretion → ↓ bone resorption → ↓ **serum Ca<sup>2+</sup>** and ↓ phosphate *Not a bone drug* — it works on the gland upstream.

## DRUGS

| DRUG                                                                         | WHAT IT TREATS                                                                                                                        |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cinacalcet</b> (PO, small molecule, shorter CaSR activation)              | 2° hyperparathyroidism on dialysis · <b>hypercalcemia of parathyroid carcinoma</b> or hyperparathyroidism when surgery isn't possible |
| <b>Etelcalcetide</b> (IV, octapeptide, longer activation via disulfide bond) | 2° hyperparathyroidism in dialysis patients — given <b>IV at the end of dialysis</b>                                                  |

## USES

**Secondary hyperparathyroidism in CKD patients on dialysis** (*the main indication*) · **Primary hyperparathyroidism / parathyroid carcinoma** where parathyroidectomy isn't an option — cinacalcet

## ADVERSE EFFECTS

- **Hypocalcemia** — the predictable consequence
- Nausea, vomiting, diarrhea

## INTERACTIONS

- ★ **Cinacalcet** — partially metabolised by **CYP3A4**, and a **strong CYP2D6 inhibitor** → raises levels of CYP2D6 substrates such as **metoprolol** (*bradycardia, hypotension*)
- **Etelcalcetide** — no metabolism, **no significant interactions**

## CAUTIONS

- ⚠ Cinacalcet: take **with or right after food**, swallow the tablet **whole** — do not chew or divide
- ⚠ Not an osteoporosis drug. Lowering PTH here is about serum calcium, not bone strength.

## ADMINISTRATION — CALCIMIMETICS

|            | CINACALCET                                                                | ETELCALCETIDE                                     |
|------------|---------------------------------------------------------------------------|---------------------------------------------------|
| Route      | oral                                                                      | IV                                                |
| When       | ● <b>with food or shortly after a meal, swallowed whole — not divided</b> | ★ <b>at the end of dialysis</b>                   |
| Metabolism | partially <b>CYP3A4</b> ; ⚠ <b>strong CYP2D6 inhibitor</b>                | <b>no metabolism, no significant interactions</b> |

- ★ **Etelcalcetide's timing is its whole convenience argument** — given at the end of a dialysis session, it needs no separate visit and no adherence at home.
- 💡 "Whole tablet, not divided" plus "with food" are two instructions that get dropped when a nurse or carer is splitting a dosette box. Worth writing on the label rather than saying once.

# Phosphate Binders

## MECHANISM

Oral, **non-systemic** — they bind phosphate inside the GI tract, both dietary phosphate and the phosphate secreted in digestive juices, so less is absorbed.

→ **Downstream:** ↓ serum phosphate *Why it matters: in 2° hyperparathyroidism phosphate is high and calcium low from failing kidneys; in hypoparathyroidism phosphate is high because there is no PTH to dump it in the urine. Different PTH levels, same fix.*

## DRUGS

| DRUG                                         | WHAT IT TREATS / ITS CATCH                                                                                                  |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Calcium carbonate / citrate / acetate</b> | cheap, and supplies calcium — but ↑ <b>calcium</b> , so wrong in hypercalcemia                                              |
| <b>Aluminum carbonate</b>                    | ● <b>aluminum toxicity</b> · <b>osteomalacia</b>                                                                            |
| <b>Lanthanum carbonate</b>                   | <b>partially absorbed</b> · nausea, vomiting, diarrhea                                                                      |
| <b>Sevelamer</b>                             | <b>non-absorbable</b> — the choice in hypercalcemia. Binds <b>fat-soluble vitamins, bile acid, uric acid</b> too · GI upset |

## USES

Lowering serum phosphate in **2° hyperparathyroidism (CKD)** and **hypoparathyroidism**

## CAUTIONS

- ★ **Take with food** — the phosphate to bind arrives with the meal. Away from food they do nothing.
- ⚠ **Hypercalcemia** → **use a non-calcium binder** (*sevelamer, lanthanum*)
- ⚠ Sevelamer strips fat-soluble vitamins — supplement if used long-term

## ADMINISTRATION — PHOSPHATE BINDERS

- **Oral, with food — always.** They work by binding phosphate **in the GI tract**, so a binder taken away from a meal has nothing to bind.
- 💡 That is the single most common reason a phosphate binder "isn't working": correct drug, correct dose, taken between meals. The timing *is* the mechanism.

Source: *Drugs used for regulating bone-mineral homeostasis 250326 TL.pdf* — อ.ธีรรัตน์ (รศ.ดร.ภก.ธีรรัตน์ เหลืองมั่นคง), *PYCH305 Pharmacology II*, 39 pages / 29 slides. Structures omitted.

# Hypothalamic-Pituitary Hormones

The hypothalamus and **anterior pituitary** are endocrine glands *inside* the CNS. They don't produce the end effect themselves — they release hormones that order a **peripheral gland** to make the hormone that actually does the work. The **posterior pituitary** is the exception: oxytocin and vasopressin act on their target tissue **directly**, and no hypothalamic hormone controls their release.

**hypothalamus → anterior pituitary → peripheral gland → active hormone → physiological effect** ☹ **negative feedback shuts the top two down**

## THE AXES — WHO TELLS WHOM

| HYPOTHALAMUS                     | → ANTERIOR PITUITARY               | → TARGET                         | → END EFFECT                           |
|----------------------------------|------------------------------------|----------------------------------|----------------------------------------|
| <b>CRH</b>                       | ACTH                               | adrenal gland                    | cortisol → ↑ blood glucose             |
| <b>TRH</b>                       | TSH                                | thyroid                          | ↑ metabolism                           |
| <b>GnRH</b> ( <i>pulsatile</i> ) | FSH · LH                           | testis / ovary                   | testosterone · estrogen · progesterone |
| <b>GHRH</b>                      | GH                                 | liver, muscle, cartilage, kidney | IGF-1 → growth                         |
| <b>Dopamine</b> ☹                | <b>inhibits</b> prolactin          | mammary gland                    | ↓ lactation                            |
| <b>Somatostatin</b> ☹            | <b>inhibits</b> GH (+ many others) | —                                | ↓ growth, ↓ GI secretion               |

★ CRH/ACTH only *control* cortisol release — they do **not** raise blood sugar themselves. *Some pituitary hormones act directly too: FSH grows the ovum and drives spermatogenesis with no third gland involved.*

## POSTERIOR PITUITARY — NO HYPOTHALAMIC CONTROLLER

| HORMONE                  | RECEPTOR          | DIRECT ACTION ON TARGET                     |
|--------------------------|-------------------|---------------------------------------------|
| <b>Oxytocin</b>          | oxytocin receptor | uterine contraction · milk letdown          |
| <b>Vasopressin (ADH)</b> | V1 · V2           | vasoconstriction · renal water reabsorption |

## WHY ALMOST NONE OF THESE ARE TABLETS

Hypothalamic/pituitary hormones are **peptides**, so their derivatives are peptides too — made by synthesis, extraction, or recombinant technology, and **destroyed in the gut** → **injection only**. Amino-acid substitutions are engineered in to make the drug bind longer or be degraded slower — and sometimes flip it into a **receptor antagonist**.

★ **The two oral exceptions** — because they are *not* peptides: **dopamine agonists** and **vasopressin antagonists**.

# GnRH Agonists

## MECHANISM

Physiological GnRH is released in **pulses**; that pulsatility is what keeps the pituitary GnRH receptor responsive. GnRH agonists bind long and are degraded slowly → **continuous, non-pulsatile** stimulation → **receptor desensitization / down-regulation**.

→ **Downstream**: gonadotropins ↑ for the **first 7–10 days** (*the flare*), then the pituitary stops releasing FSH/LH → **sex-steroid deprivation**. ★ The drug is an agonist that ends up acting as a suppressor.

## DRUGS

| DRUG               | WHAT IT TREATS                                                                           |
|--------------------|------------------------------------------------------------------------------------------|
| <b>Goserelin</b>   | prostate cancer · ovulation control                                                      |
| <b>Leuprorelin</b> | prostate cancer · endometriosis · uterine fibroids · prep for intrauterine surgery · CPP |
| <b>Triptorelin</b> | prostate cancer · endometriosis · uterine fibroids · CPP                                 |
| <b>Buserelin</b>   | prostate cancer · endometriosis · ovulation control                                      |

## USES

**Suppression of gonadotropin production** — prostate cancer · endometriosis & uterine fibroids · **central precocious puberty (CPP)** · preventing the LH surge during controlled ovarian stimulation · *GnRH replacement in infertility needs a portable pulsatile pump — impractical, so it isn't really done.*

## ADVERSE EFFECTS

- **Flare effect, first 7–10 days** ★ — the initial gonadotropin surge. In prostate cancer it means a transient tumour flare with worsening pain and hot flushes.
- **Sex-hormone deficiency — androgen deprivation**: hot flushes, sweats, gynecomastia, ↓ libido, ↓ hematocrit, ↓ bone density, asthenia
- **Sex-hormone deficiency — menopausal symptoms**: hot flushes, sweats, headache, depression, ↓ libido, vaginal dryness
- Injection-site reactions
- Nausea, vomiting

## CAUTIONS

- ⚠ **Cover the flare** — an **anti-androgen** is usually added at the start of prostate cancer treatment.
- ⚠ Peptide → **injection only** (SC/IM/implant/nasal), never oral.

# GnRH Antagonists

## MECHANISM

Bind the pituitary GnRH receptor and **block it the moment they attach** — no agonist phase, no desensitization step needed.

→ **Downstream:** gonadotropins fall **immediately**. ★ **No flare effect** — this is the whole reason the antagonists exist alongside the agonists.

## DRUGS

| DRUG              | WHAT IT TREATS                                                                              |
|-------------------|---------------------------------------------------------------------------------------------|
| <b>Degarelix</b>  | prostate cancer                                                                             |
| <b>Ganirelix</b>  | preventing the LH surge during controlled ovarian stimulation ( <i>with gonadotropins</i> ) |
| <b>Cetrorelix</b> | preventing the LH surge during controlled ovarian stimulation ( <i>with gonadotropins</i> ) |

## USES

**Suppression of gonadotropin production** — prostate cancer (*degarelix*) · **Preventing the LH surge** in controlled ovarian stimulation for infertility (*ganirelix, cetrorelix, given with gonadotropins*)

## ADVERSE EFFECTS

- Sex-hormone deficiency — androgen deprivation / menopausal symptoms, same as the agonists
- Injection-site reactions · nausea, vomiting
- ★ **No flare effect**

## CAUTIONS

- ⚠ **Adherence is critical** — ganirelix and cetrorelix are daily; a missed dose lets the LH surge through.

# Gonadotropins

## MECHANISM

Replace or supplement the pituitary's own **FSH, LH and CG** at the gonad.

→ **Female:** FSH → follicle development + estrogen · LH → ovulation + progesterone → **Male:** FSH → spermatogenesis · LH → testosterone *CG is a placental glycoprotein nearly identical to LH but with a longer  $t_{1/2}$  — it takes over LH's job in pregnancy.*

## DRUGS

| DRUG                                                               | WHAT IT TREATS                                                                   |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Menotropin (hMG)</b>                                            | FSH + LH mixture 1:1, from urine of postmenopausal women                         |
| <b>Urofollitropin (uFSH)</b>                                       | purified human FSH, from the same urine source                                   |
| <b>Follitropin <math>\alpha</math> / <math>\beta</math> (rFSH)</b> | recombinant FSH — follicle stimulation                                           |
| <b>Corifollitropin <math>\alpha</math></b>                         | recombinant FSH, <b>long-acting</b>                                              |
| <b>Lutropin <math>\alpha</math> (rLH)</b>                          | recombinant LH — <b>only in combination with follitropin <math>\alpha</math></b> |
| <b>Choriogonadotropin <math>\alpha</math> (rhCG)</b>               | recombinant hCG — triggers ovulation                                             |

*Recombinant beats urinary: less protein contamination, less batch-to-batch variability, less local tissue reaction.*

## USES

**Ovulation induction** in female infertility · **Male infertility / hypogonadism** · Cryptorchidism (*outdated*)

## ADVERSE EFFECTS

- **Ovarian hyperstimulation syndrome (OHSS)** ★
  - *Mild:* uncomplicated ovarian enlargement, resolves on its own in a few days
  - *Severe:* ovarian enlargement, **intravascular depletion**, ascites, pulmonary edema, electrolyte imbalance, thromboembolic events
- **Multiple pregnancy** — **5–10% risk** (*background rate is 1%*)

# Dopamine Agonists

## MECHANISM

Dopamine is the hypothalamic hormone that **tonically suppresses prolactin** (and partly GH) release via the pituitary **D2 receptor**. A D2 agonist reproduces that brake.

→ **Downstream:** ↓ prolactin → ↓ lactation · ↓ GH release (*hence its use in acromegaly*) → **Why hyperprolactinemia matters:** excess prolactin suppresses **GnRH** → ↓ FSH/LH → **hypogonadism**

## HYPERPROLACTINEMIA PRESENTS DIFFERENTLY BY SEX

- **Female:** infertility, oligomenorrhea, amenorrhea, **galactorrhea**
- **Male:** loss of libido, erectile dysfunction, infertility

## DRUGS

| DRUG                                             | WHAT IT TREATS                                                                                                      |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Bromocriptine</b> ( <i>ergot derivative</i> ) | hyperprolactinemia · suppression of physiologic lactation · acromegaly · Parkinson's disease · adjunct in type 2 DM |
| <b>Pramipexole</b> ( <i>non-ergot</i> )          | Parkinson's disease — <b>not</b> for hyperprolactinemia                                                             |
| <b>Rotigotine</b> ( <i>non-ergot</i> )           | Parkinson's disease — <b>not</b> for hyperprolactinemia                                                             |

## USES

**Hyperprolactinemia** · Suppression of physiologic lactation · **Acromegaly** · **Parkinson's disease** · Adjunct in type 2 diabetes

## ADVERSE EFFECTS

- **From D2 agonism:** nausea · psychiatric manifestations · headache, light-headedness, **orthostatic hypotension**
- **From 5-HT<sub>2B</sub> agonism: valvular heart disease** ★ — the ergot-derivative problem, *very rare with bromocriptine*

## CAUTIONS

- ★ Not a peptide → **can be taken orally**, unlike almost everything else in this lecture.



# Oxytocin & Oxytocin Antagonists

## MECHANISM

Posterior pituitary releases oxytocin heavily near term. Via the **oxytocin receptor**:

→ **Uterus**: ↑ smooth-muscle contraction → **parturition** — and this runs on ★ **positive feedback**, not negative: contraction drives more oxytocin release → **Breast**: ↑ myoepithelial contraction → **milk letdown**  
*Antagonists block the same receptor to relax the uterus and delay delivery.*

## DRUGS

| DRUG                                                            | WHAT IT TREATS                                                                           |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Oxytocin</b> ( <i>synthetic</i> )                            | initiation & augmentation of labor · stopping vaginal bleeding from <b>uterine atony</b> |
| <b>Atosiban</b> ( <i>peptide oxytocin-receptor antagonist</i> ) | <b>tocolysis</b> in preterm labor                                                        |

## USES

**Induce / augment labor** — oxytocin IV · **Postpartum hemorrhage from uterine atony** — oxytocin IM · **Delay preterm labor (tocolysis)** — atosiban IV

## ADVERSE EFFECTS

- **Excessive uterine contraction** ★ — **fetal distress, uterine rupture**. *This is why fetal heart rate is monitored throughout an oxytocin infusion.*
- **Inadvertent activation of vasopressin receptors** at high dose — fluid retention → **water intoxication** → **hyponatremia**, heart failure, seizures, death

# Growth Hormone

## MECHANISM

GHRH → pituitary GH → GH stimulates **liver, muscle, cartilage and kidney** to produce **IGF-1**, which does most of the actual work.

→ **Bone**: ↑ longitudinal growth — **only until epiphyseal plate fusion** → **Muscle**: ↑ anabolic effects · → **Adipose**: ↑ catabolic effects → **Blood sugar**: ↓ insulin sensitivity (*GH itself is diabetogenic, even though IGF-1 has insulin-like effects*)

## DRUGS

| DRUG                                          | WHAT IT TREATS                                                                    |
|-----------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Somatotropin</b> ( <i>recombinant GH</i> ) | growth hormone deficiency · small-for-gestational-age children · catabolic states |

## USES

**GH deficiency** — idiopathic, **Turner syndrome**, intracranial tumours · **Born small for gestational age** · Severe catabolic state in **AIDS** · **Short bowel syndrome** in patients dependent on TPN after intestinal resection or bypass · Anti-aging remedies (*controversial*)

## ADVERSE EFFECTS

- Peripheral edema
- **Myalgia and arthralgia** — typically hands and wrists
- ★ **ADRs are worse in adults than in children**

## CAUTIONS

- ● **Contraindicated in active malignancy** — it's a growth signal.

# Somatostatin Analogs

## MECHANISM

Somatostatin is the hypothalamic **off-switch** for GH, and it comes from GI cells as well. It inhibits a long list beyond GH: **cortisol, TSH, glucagon, insulin, gastrin, secretin, histamine, gastric acid**, and more. Native somatostatin has a **very short  $t_{1/2}$**  → useless as a drug, so the analogs are engineered for **higher potency and a much longer  $t_{1/2}$** .

→ **Downstream:** ↓ GH & IGF-1 (*acromegaly*) · ↓ splanchnic blood flow & GI secretions (*variceal bleeding, secretory tumours*)

## DRUGS

| DRUG               | WHAT IT TREATS                                         |
|--------------------|--------------------------------------------------------|
| <b>Octreotide</b>  | acromegaly · variceal hemorrhage · secretory neoplasms |
| <b>Lanreotide</b>  | acromegaly · secretory neoplasms                       |
| <b>Pasireotide</b> | acromegaly · secretory neoplasms                       |

## USES

**Acromegaly · Secretory neoplasms** · Prophylaxis of complications after **pancreatic surgery** · **HIV-associated diarrhea** · **Variceal hemorrhage** in cirrhosis with portal hypertension

## ADVERSE EFFECTS

- Abdominal cramps, flatulence
- **Steatorrhea** with bulky bowel movements
- ● **Biliary sludge and gallstones** ★ — from suppressed gallbladder contraction and CCK

# Vasopressin & Analogs

## MECHANISM

Released from the posterior pituitary by **two different triggers**, acting on two receptors:

- **V1** on vascular smooth muscle — released when **BP falls** → vasoconstriction → ↑ BP
- **V2** on renal tubular cells — released when **plasma electrolytes/osmolality rise** → ↑ water reabsorption → ↓ urine volume. ★ This is why it's also called **anti-diuretic hormone (ADH)**.
- **Extrarenal V2-like** — releases **factor VIII and von Willebrand factor** → platelet aggregation → hemostasis

## DRUGS

| DRUG                                                                                              | WHAT IT TREATS                                                                                 |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Vasopressin</b> ( <i>synthetic arginine vasopressin, <math>t_{1/2} \approx 15</math> min</i> ) | diabetes insipidus · <b>vasopressor in resuscitation</b> at high dose, in place of epinephrine |
| <b>Desmopressin (DDAVP)</b> ( $t_{1/2} \approx 2$ h)                                              | diabetes insipidus · <b>hemophilia A</b> · <b>von Willebrand disease</b>                       |

★ **Desmopressin's antidiuretic-to-pressor ratio is ~4,000× vasopressin's** — nearly pure V2. That's what makes it the one you use for a water problem, and vasopressin the one you use for a pressure problem.

*DDAVP = 1-desamino-8-D-arginine vasopressin.*

## USES

**Diabetes insipidus · Resuscitation** — high-dose vasopressin replacing epinephrine · **Hemophilia A and von Willebrand disease** — desmopressin, via factor VIII / vWF release

## ADVERSE EFFECTS

- ● **Hyponatremia** ★ — retaining free water dilutes the sodium
- **Seizures** — the consequence of that hyponatremia

# Vasopressin Antagonists

## MECHANISM

Blocks the **V2** receptor on renal tubular cells — **30-fold higher affinity for V2 than V1** — so free water is excreted without dumping sodium with it.

→ **Downstream:** ↑ free-water excretion → ↑ **serum Na<sup>+</sup>** without a natriuresis. The mirror image of ADH: an **aquaretic**, not a diuretic.

## DRUGS

| DRUG                                          | WHAT IT TREATS                                                      |
|-----------------------------------------------|---------------------------------------------------------------------|
| <b>Tolvaptan</b> ( <i>non-peptide, oral</i> ) | SIADH · hypervolemic & euvoletic hyponatremia · acute heart failure |

## USES

**SIADH** — syndrome of inappropriate ADH · **Hypervolemic & euvoletic hyponatremia** · **Acute heart failure** with elevated vasopressin

## INTERACTIONS

- **CYP3A4 substrate** — inhibitors raise tolvaptan exposure, inducers cut it

## CAUTIONS

-  **Limit treatment to 30 days**  — **hepatotoxicity** risk beyond that.
-  Non-peptide → **orally active**, the second of the lecture's two oral exceptions.

*Principle behind this: [[pharm-cns-basics]] — Tuberoinfundibular DA inhibits prolactin, and the ARH sits outside the BBB. Block D2 and prolactin rises.*

*Source: Hypothalamic-pituitary hormones and their antagonists 010426 TL.pdf — รศ.ดร.ภก.ธีรศักดิ์ เหลืองมันคง, ภกปว 305 Pharmacology II, 23 slides. Structures omitted.*

# Anabolic-Androgenic Steroids & Anti-Androgens

**Androgens** are steroid hormones that produce **male characteristics**. Give them (AAS) for androgen deficiency and wasting states; **block** them (anti-androgens) when there's too much — **prostate cancer · BPH · androgenic alopecia · hirsutism**.

**LH → Leydig cell (testis) → testosterone → (5 $\alpha$ -reductase, in tissue) → DHT → androgen receptor (nuclear receptor) → male characteristics, muscle & bone growth**

## THREE EFFECT-WORDS WORTH SEPARATING

- **Androgenic effects** — making male characteristics: penile & scrotal growth · pubic, axillary, beard hair · spermatogenesis · prostate growth · sexual function · sebaceous glands ( $\rightarrow$  *oily skin, acne*) · larynx & vocal cords thicken (*lower voice, Adam's apple*)
- **Anabolic effects** — body & skeletal growth (*but closes the epiphysis*) ·  $\uparrow$  protein synthesis,  $\downarrow$  protein breakdown,  $\downarrow$  urine nitrogen ·  $\uparrow$  **erythropoietin** ( $\rightarrow$  *this is why they treat anemia*) · cognitive function & sense of well-being
- **Estrogenic effects** — testosterone  $\rightarrow$  estrogen via **aromatase**, acting on the estrogen receptor. Off-target, not through the androgen receptor.

★ **DHT is a more potent androgen-receptor agonist than testosterone.** Every 5 $\alpha$ -reductase story hangs off this.

## THE WHOLE TOPIC AT A GLANCE

| GROUP       | CLASS                               | TARGET                                                                | REPRESENTATIVE DRUGS                                       |
|-------------|-------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------|
| <b>AAS</b>  | Testosterone & esters               | androgen receptor (agonist)                                           | testosterone, undecanoate, enanthate                       |
| <b>AAS</b>  | 17 $\alpha$ -alkylated testosterone | androgen receptor (agonist), orally active                            | methyltestosterone, fluoxymesterone                        |
| <b>AAS</b>  | Anabolic steroids                   | androgen receptor, high anabolic : androgenic                         | mesterolone, oxymetholone, stanozolol, danazol, nandrolone |
| <b>Anti</b> | GnRH analogs                        | pituitary $\rightarrow$ $\downarrow$ LH ( <i>covered separately</i> ) | e.g. triptorelin, ganirelix                                |
| <b>Anti</b> | Steroid synthesis inhibitors        | 17 $\alpha$ -hydroxylase, 17,20-lyase                                 | abiraterone, spironolactone, ketoconazole                  |
| <b>Anti</b> | 5 $\alpha$ -reductase inhibitors    | 5 $\alpha$ -reductase                                                 | finasteride, dutasteride                                   |
| <b>Anti</b> | Steroid AR antagonists              | androgen receptor (block)                                             | cyproterone                                                |
| <b>Anti</b> | Non-steroidal AR antagonists        | androgen receptor (block)                                             | flutamide, bicalutamide, enzalutamide                      |

ANABOLIC ACTIVITY RELATIVE TO TESTOSTERONE

| DRUG                      | ANABOLIC ACTIVITY |
|---------------------------|-------------------|
| Testosterone              | 1                 |
| <b>Methyltestosterone</b> | <b>1</b>          |
| Fluoxymesterone           | 2                 |
| Oxymetholone              | 3                 |
| Nandrolone decanoate      | 2.5–4             |

★ Methyltestosterone sits at **1** — same anabolic activity as testosterone — **which is why it is not indicated for anemia**. The others buy you anabolic effect without buying the same androgenic burden.  
*Ratios derived from animal studies — interpret with caution.*

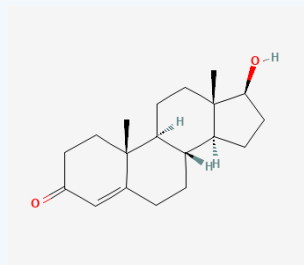
# Testosterone & Testosterone Esters

## MECHANISM

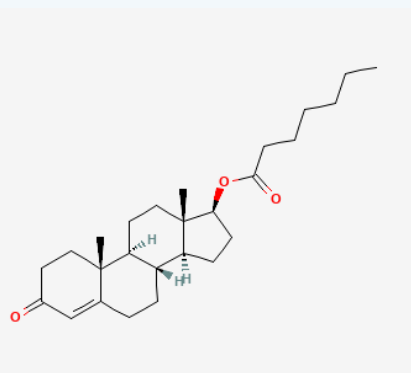
Direct **androgen receptor agonist** — replacement of the missing hormone.

→ **The problem:** testosterone undergoes very heavy **first-pass metabolism** in the liver, so oral testosterone gives inadequate replacement. → **The fix:** bypass the gut (skin gel, or long-acting IM esters), or alkylate the structure so the liver can't clear it (next class).

## THE SHARED CORE • WHAT CHANGES

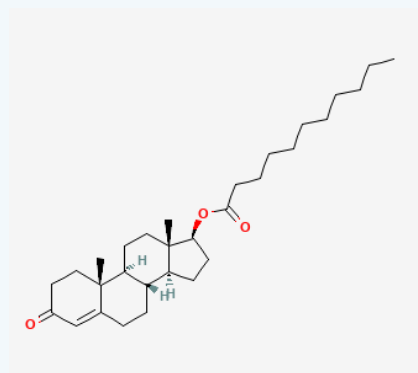


**Testosterone.** ● the free 17 $\beta$ -OH is oxidised in the liver within minutes — plain testosterone is not an oral drug



**Testosterone enanthate**

a 7-carbon ester on that OH → depot IM, released over ~2 weeks



**Testosterone undecanoate**

an 11-carbon ester → absorbed via lymph, bypassing first pass → up to 3 months

## MEDCHEM

**SAR Esterifying the 17 $\beta$ -hydroxyl slows metabolism and stretches duration** - and the longer the ester chain, the longer the depot. That is the entire difference between testosterone propionate, enanthate and undecanoate: same drug, different ester.

**SAR** The deck is blunt about the bodybuilding question: **a complete separation of anabolic from androgenic activity is not possible.** What shifts the ratio is **removing the C19 methyl** (nandrolone) or **inserting oxygen / a hydroxyl** into the ring system (oxandrolone) - both favour anabolic over androgenic, neither abolishes it. An  $\alpha$ -halogen goes the other way and raises androgenic activity.

**Metabolism** Testosterone is readily reduced by **5 $\alpha$ -reductase to dihydrotestosterone** - the reason finasteride and dutasteride exist, and the reason a 5 $\alpha$ -reduction-resistant analogue behaves differently in prostate and scalp than in muscle.



## DRUGS

| DRUG                                                   | WHAT IT TREATS                          |
|--------------------------------------------------------|-----------------------------------------|
| <b>Testosterone</b> ( <i>gel — short duration</i> )    | androgen replacement therapy            |
| <b>Testosterone undecanoate</b> ( <i>IM, also PO</i> ) | androgen replacement — longest interval |
| <b>Testosterone enanthate</b> ( <i>IM</i> )            | androgen replacement                    |
| <b>Testosterone propionate</b> ( <i>IM</i> )           | androgen replacement                    |
| <b>Testosterone cypionate</b> ( <i>IM</i> )            | androgen replacement                    |

## USES

**Androgen replacement therapy (hypogonadism) · Erectile dysfunction · Infertility**

## ADVERSE EFFECTS

- **Excessive androgenic effects** — the whole ADR profile is "too much of the intended action"
- (*Full shared AAS toxicity list is on its own page*)

## ADMINISTRATION — TESTOSTERONE PREPARATIONS

| FORM                                                                  | DURATION                        | TYPICAL SCHEDULE                         |
|-----------------------------------------------------------------------|---------------------------------|------------------------------------------|
| <b>Gel</b>                                                            | <b>short</b> duration of action | <b>apply daily</b>                       |
| <b>IM — testosterone undecanoate</b> ( <i>also available orally</i> ) | <b>longest</b>                  | <b>IM every 10–14 weeks</b> · oral daily |
| <b>IM — testosterone enanthate</b>                                    | long                            | <b>IM every 2–4 weeks</b>                |
| <b>IM — testosterone propionate / cypionate</b>                       | long                            | —                                        |

- ★ **The ester length is the dosing interval.** Propionate, enanthate, undecanoate — the longer the ester chain, the slower the release and the further apart the injections.
- 💡 Gel daily versus an injection every three months is the real choice patients are making, and it is about how steady they want the levels to be, not about potency.

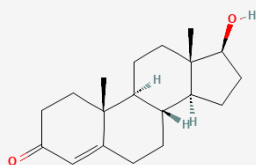
# 17 $\alpha$ -Alkylated Testosterone

## MECHANISM

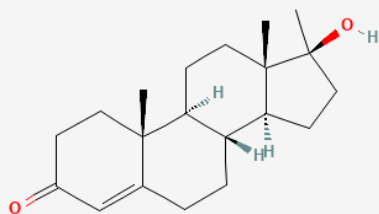
A **17 $\alpha$ -methyl** group blocks hepatic metabolism at the position the liver attacks — so the drug **survives first pass and works orally**.

→ **Downstream:** oral androgen replacement becomes possible → **The price:** the liver now has to handle a molecule built to resist it — **hepatic dysfunction and cholestatic jaundice** ★

## THE SHARED CORE · WHAT CHANGES

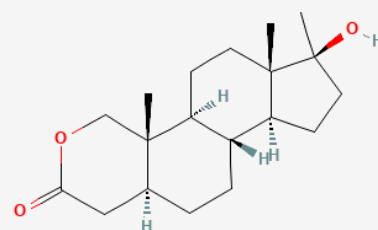


**Testosterone — the starting point.** add a methyl at 17 $\alpha$  and the liver can no longer oxidise C17



### Methyltestosterone

orally active — and **the same alkyl group** that saves it from the liver is what makes the class hepatotoxic



### Oxandrolone

17 $\alpha$ -methyl plus an O in the A-ring → more anabolic, less androgenic

## MEDCHEM

**SAR** ★ **The 17 $\alpha$ -alkyl group is a trade, and the deck states both halves.** A small alkyl at 17 $\alpha$  ( $-\text{CH}_3$ ,  $-\text{CH}_2\text{CH}_3$ ) **blocks first-pass metabolism** → **oral bioavailability** - but it also **lowers intrinsic activity**. You get a drug you can swallow that is weaker than the one you inject, and whose slowed hepatic clearance is precisely why this subgroup carries the hepatotoxicity reputation.

## DRUGS

| DRUG                      | WHAT IT TREATS                                            |
|---------------------------|-----------------------------------------------------------|
| <b>Methyltestosterone</b> | androgen replacement · erectile dysfunction · infertility |
| <b>Fluoxymesterone</b>    | androgen replacement · erectile dysfunction · infertility |
| <b>Oxymetholone</b>       | (also 17-alkylated — see Anabolic Steroids)               |
| <b>Stanozolol</b>         | (also 17-alkylated — see Anabolic Steroids)               |
| <b>Danazol</b>            | (also 17-alkylated — see Anabolic Steroids)               |

## USES

Androgen replacement therapy · Erectile dysfunction · Infertility

### ADVERSE EFFECTS

- Excessive androgenic effects
- ● **Hepatic dysfunction and cholestatic jaundice** ★ — *the defining ADR of this subgroup; risk is far higher than with the injectable esters*

# Anabolic Steroids

## MECHANISM

Androgen-receptor agonists selected for a **high anabolic : androgenic ratio** — you want the muscle, bone and erythropoietin effect without the virilization.

→ **Why the ratio is favourable:** ★ **no potentiation by 5 $\alpha$ -reductase** — they are not converted to the more potent DHT in prostate and skin, so androgenic effects stay lower.

## DRUGS

| DRUG                             | WHAT IT TREATS                                                                                                                                        |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mesterolone</b> (PO)          | androgen deficiency ( <i>potency disturbances, hypogonadism</i> ) · infertility                                                                       |
| <b>Oxymetholone</b> (PO)         | <b>aplastic anemia</b> · cachexia · debilitating states ( <i>recovery from surgery, infection, burns, fractures, severe trauma</i> )                  |
| <b>Stanozolol</b> (IM)           | anemia · catabolic disorders · osteoporosis · breast cancer in postmenopausal women                                                                   |
| <b>Stanozolol</b> (PO)           | <b>hereditary angioedema</b> · vascular manifestations of Behçet's syndrome                                                                           |
| <b>Danazol</b> (PO)              | <b>hereditary angioedema</b> · <b>endometriosis</b> · benign breast disorders                                                                         |
| <b>Nandrolone decanoate</b> (IM) | anemia of chemotherapy & chronic renal failure · anabolic support after debilitating illness · postmenopausal osteoporosis · metastatic breast cancer |

## TWO MECHANISMS THAT AREN'T ANDROGEN-RECEPTOR EFFECTS

★ **Complement system** — stanozolol/danazol **enhance synthesis of C1-inhibitor**, which is the deficient protein in **hereditary angioedema** · ★ **Anti-gonadotropic (progestogenic) activity** — **danazol inhibits GnRH release**, dropping gonadotropins and shutting down the ovarian drive that feeds **endometriosis**

## ADVERSE EFFECTS

- Androgenic effects — **less** than with testosterone itself, but not absent
- Hepatic dysfunction with the 17-alkylated members (*oxymetholone, stanozolol, danazol*)
- ★ **Far more apparent in women** — virilization is the limiting toxicity, not the hepatic effect

# AAS — Shared Effects, Toxicity & Doping

## EFFECTS THAT ARE NOT ANDROGEN-RECEPTOR EFFECTS

**Estrogenic effects** — *only in agents that are aromatase substrates\*\**: gynecomastia, breast hypertrophy, and bone & muscle strength in women.

**Anti-catabolic effects** — *in all agents*: they alter the **lipid profile** (triglyceride, cholesterol, HDL) and **clotting factors**.

→ **Downstream**: dyslipidemia → **atherosclerotic cardiovascular disease**

## ADVERSE EFFECTS OF ANABOLIC-ANDROGENIC STEROIDS

**Males** - Prostatic hyperplasia - **Oligospermia** — high blood testosterone feeds back and suppresses spermatogenesis - Gynecomastia (*aromatase substrates*)

**Females** — ★ *effects are far more apparent here* - **Virilization / masculinization**, hirsutism - Breast hypertrophy

**Both** - Dyslipidemia and atherosclerotic disease - ● **Hepatic dysfunction** — especially 17-alkylated testosterone - **Aggression and hypomania**

## DOPING — BANNED BY WADA

- ⚠ **All androgen derivatives** — epiandrosterone, testosterone, androstenediol, androstenedione, DHEA
- ⚠ **Named AAS** — danazol, fluoxymesterone, gestrinone, mesterolone, methyltestosterone, **nandrolone**, oxymetholone, stanozolol — *plus anything of similar structure or biological effect*
- ⚠ **Other anabolic agents** — **SARMs** (*andarine, LGD-4033/ligandrol, enobosarm/ostarine, RAD140*) · tibolone · clenbuterol · zeranol · zilpaterol

# Steroid Synthesis Inhibitors

## MECHANISM

Inhibit the **CYP enzymes that build testosterone** — **17 $\alpha$ -hydroxylase** and **17,20-lyase**.

→ **Downstream:** ↓ testosterone at its source (*testis, adrenal gland, and the prostate cancer cells themselves*)

→ **● The catch:** those same enzymes sit on the pathway to the **other** steroid hormones — so cortisol and estrogen fall too, and mineralocorticoid precursors back up. **That's where the ADRs come from.**

## DRUGS

| DRUG                       | WHAT IT TREATS                                                             |
|----------------------------|----------------------------------------------------------------------------|
| <b>Abiraterone</b> (PO)    | <b>advanced prostate cancer</b> — always <b>combined with prednisolone</b> |
| <b>Spironolactone</b> (PO) | K <sup>+</sup> -sparing diuretic · <b>off-label: hirsutism</b>             |
| <b>Ketoconazole</b> (PO)   | azole antifungal · <b>off-label: hirsutism</b>                             |

## HOW EACH ONE ACTUALLY BLOCKS ANDROGEN

**Abiraterone** — inhibits **17 $\alpha$ -hydroxylase + 17,20-lyase** → ↓ testosterone & estrogen, ↓ **cortisol**, ↑

**aldosterone** · **Spironolactone** — inhibits **17 $\alpha$ -hydroxylase** *and* is an **androgen receptor antagonist** ·

**Ketoconazole** — inhibits **multiple CYPs** across gonadal and adrenal steroid synthesis (*the least selective of the three*)

## ADVERSE EFFECTS

- **Anti-anabolic-androgenic effects** — the class-wide theme (*loss of libido, erectile dysfunction, gynecomastia*)
- **Anti-estrogenic effects** (*abiraterone*)
- **● Mineralocorticoid effects** (*abiraterone*) — **★ monitor serum K<sup>+</sup> and blood pressure:** hypokalemia, hypertension, fluid retention

## INTERACTIONS & ADMINISTRATION

- **● Abiraterone must be taken on an empty stomach** **★** — a meal raises absorption **more than 10-fold**
- **Prednisolone is co-given with abiraterone** — it replaces the cortisol abiraterone blocks and damps the compensatory drive behind the mineralocorticoid excess

# 5 $\alpha$ -Reductase Inhibitors

## MECHANISM

5 $\alpha$ -reductase converts **testosterone** → **DHT** inside target tissue, and **DHT is the stronger androgen receptor agonist**. Block the enzyme and you cut androgenic effect **only where the enzyme is** — a tissue-selective anti-androgen.

**Where the enzyme lives:** **Type I** — liver, skin, hair follicle · **Type II** — prostate, epididymis, skin, hair follicle  
→ **Downstream:** ↓ DHT in **prostate** (*DHT enlarges the gland* → BPH) and in the **hair follicle** (*DHT terminates hair growth* → androgenic alopecia)

## DRUGS

| DRUG                                            | WHAT IT TREATS                                                                                                                      |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Finasteride</b> ( <i>selective type II</i> ) | <b>BPH</b> (5 mg) · <b>androgenic alopecia</b> (1 mg)                                                                               |
| <b>Dutasteride</b> ( <i>dual type I + II</i> )  | <b>BPH</b> (0.5 mg) — also blocks the type I enzyme in skin & hair follicle                                                         |
| <b>Dutasteride + Tamsulosin</b>                 | <b>BPH</b> — shrink the gland ( <i>5<math>\alpha</math>-RI</i> ) + relax the smooth muscle ( <i><math>\alpha_1</math>-blocker</i> ) |

## USES

**Benign prostatic hyperplasia (BPH) · Androgenic alopecia** — *same drug, same mechanism, different dose*

## ADVERSE EFFECTS

- **Anti-androgenic-anabolic effects** — ↓ libido, erectile dysfunction, gynecomastia
- ★ **Interferes with PSA measurement** — lowers PSA, so a prostate cancer can hide behind a falsely reassuring number

## CAUTIONS

- ● **Contraindicated in pregnancy or possible pregnancy** ★ — blocking DHT disrupts male fetal genital development

# Steroidal Androgen Receptor Antagonists

## MECHANISM

Two actions in one steroid molecule:

1. **Androgen receptor antagonist** — blocks the receptor directly
2. **Progestational effects** — ↓ LH → ↓ endogenous **androgen** production

→ ● **Weakness of the steroidal group:** it is **less selective for the androgen receptor** than the non-steroidal drugs, so it disturbs the other hormone systems more.

## DRUGS

| DRUG                                  | WHAT IT TREATS                                                                                                                                              |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cyproterone</b>                    | reduction of drive in <b>sexual deviations</b> · <b>inoperable prostate carcinoma</b> ( <i>with GnRH analogs</i> ) · <b>severe androgenization in women</b> |
| <b>Cyproterone + Ethinylestradiol</b> | androgen-dependent disease in women ( <i>acne, hirsutism</i> )                                                                                              |

## ADVERSE EFFECTS

- **Anti-anabolic-androgenic effects**
- **Progestational effects** — the extra hormonal burden the non-steroidal drugs don't carry



# Non-Steroidal Androgen Receptor Antagonists

## MECHANISM

Pure androgen receptor blockade — **no other hormone activity**, which is exactly why their ADRs are purely anti-androgenic.

- **1st generation** — inhibits **transcription** of the androgen receptor
  - **2nd generation** — inhibits **translocation** of the androgen receptor and induces receptor apoptosis
- ★ **Registered indication is one thing only: advanced prostate cancer, combined with a GnRH analog** (*block the receptor + shut off the supply*)

## DRUGS

| DRUG                                                      | WHAT IT TREATS                                                                                                           |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Flutamide</b> (1st gen, PO q8h, $t_{1/2}$ 5–6 h)       | advanced prostate cancer                                                                                                 |
| <b>Bicalutamide</b> (1st gen, PO daily, $t_{1/2}$ 7 days) | advanced prostate cancer · <b>off-label:</b> transgender women, hirsutism in women, priapism, precocious puberty in boys |
| <b>Enzalutamide</b> (2nd gen, PO daily, $t_{1/2}$ 5 days) | advanced prostate cancer                                                                                                 |

## ADVERSE EFFECTS

- **Anti-anabolic-androgenic effects** — ★ especially **gynecomastia and hot flushes**
- ● **Flutamide** — **hepatotoxicity** (*and the q8h dosing from its short half-life makes it the least convenient of the three*)
- **Enzalutamide** — **off-target GABA<sub>A</sub> receptor inhibition** → **anxiety, seizure**

## INTERACTIONS

- **Enzalutamide** is a **CYP3A4 and CYP2C8 substrate** — a CYP2C8 inhibitor (*e.g. gemfibrozil*) raises its level, so the dose is **halved**
- **Enzalutamide** is also a **strong CYP3A4 inducer** — it will drop the levels of CYP3A4 substrates given with it

Source: *Anabolic-androgenic steroids and anti-androgens 010426 TL.pdf* — รศ.ดร.ภก.ธีรรัตน์ เหลืองมั่นคง, ภาว 305 Pharmacology II, 31 pages (23 slides). Structures omitted.

# Adrenal Steroids & Their Antagonists

The adrenal cortex makes three steroid families: **mineralocorticoids** (aldosterone — salt & water), **glucocorticoids** (cortisol — metabolism & immunity) and **sex steroids** (DHEA). Both aldosterone and cortisol are **essential for life** — too little kills, so you replace it; too much causes disease, so you antagonize it. Everything on the following pages is one of those two moves.

**stress → hypothalamus CRH → anterior pituitary ACTH → adrenal cortex cortisol → + negative feedback back onto hypothalamus & pituitary = the HPA axis**

## TWO AXES, TWO CONTROLLERS

- **Aldosterone** is driven by **RAAS** (angiotensin II, K<sup>+</sup>) — *not* mainly by ACTH
- **Cortisol** and **adrenal sex steroids** are driven by the **HPA axis**, with a **circadian rhythm** — plasma cortisol ~16 µg/100 mL at 08:00 vs ~4 µg/100 mL at 16:00 (secretion ~10 mg/day)
- **Replacement therapy** = physiological dose · **non-adrenal disorder** = supra-physiological (stress) dose

## ★ RELATIVE ACTIVITY OF THE GLUCOCORTICOIDS — THE TABLE THE EXAM LIVES ON

| DRUG                               | EQUIPOTENT DOSE (MG) | ANTI-INFLAMMATORY | NA <sup>+</sup> RETENTION (MC) | BIOLOGICAL T <sub>1/2</sub> (H) | DURATION     |
|------------------------------------|----------------------|-------------------|--------------------------------|---------------------------------|--------------|
| <b>Hydrocortisone</b> (= cortisol) | 20                   | 1                 | 1                              | 8–12                            | short        |
| <b>Prednisolone</b>                | 5                    | 4                 | 0.8                            | 12–36                           | short/medium |
| <b>Methylprednisolone</b>          | 4                    | 5                 | 0.5                            | 12–36                           | intermediate |
| <b>Triamcinolone</b>               | 4                    | 5                 | 0                              | 12–36                           | intermediate |
| <b>Dexamethasone</b>               | 0.75                 | 30                | 0                              | 36–72                           | long         |
| <b>Betamethasone</b>               | 0.6                  | 25–40             | 0                              | 36–72                           | long         |
| <b>Fludrocortisone</b>             | —                    | 10                | <b>125</b>                     | 12–36                           | —            |

★ Read it as a trade: **the more anti-inflammatory, the less mineralocorticoid**. That is why hydrocortisone (1 : 1) replaces adrenal hormone and dexamethasone (30 : 0) treats inflammation. ● **Glucocorticoids always carry some mineralocorticoid effect** — only "0" rows are truly free of it.

## WHAT HAPPENS IF THE HORMONE IS MISSING

| MISSING               | CONSEQUENCE                |
|-----------------------|----------------------------|
| <b>No aldosterone</b> | volume depletion → death   |
| <b>No cortisol</b>    | stress intolerance → death |

# Mineralocorticoids

## MECHANISM

Agonist at the **nuclear mineralocorticoid receptor** in renal tubular epithelium (also colon and salivary glands) — the same receptor aldosterone uses.

→ **Downstream:** ↑ **Na<sup>+</sup> and water reabsorption** · ↑ **K<sup>+</sup> excretion** into urine → plasma volume ↑ → BP ↑ →

**Note:** synthetic mineralocorticoids also activate the **glucocorticoid receptor**, so they come with cortisol-like effects on top.

## DRUGS

| DRUG                           | WHAT IT TREATS                                                                                                             |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Fludrocortisone</b><br>(PO) | aldosterone <b>replacement</b> — primary adrenal insufficiency (Addison's disease) · salt-losing adrenogenital hyperplasia |

## USES

**Replacement of aldosterone** — primary adrenal insufficiency / **Addison's disease** (*given together with a glucocorticoid*) · **Salt-losing adrenogenital hyperplasia**

## ADVERSE EFFECTS

- **Hypernatremia, hypokalemia, hypertension** — the pharmacology taken too far
- Glucocorticoid-like effects (*it hits the GR as well*)

## ★ WHY ONCE DAILY ON A 3.5-HOUR DRUG

Plasma  $t_{1/2}$  = **3.5 h** but **biological  $t_{1/2}$  = 18–36 h**. The steroid acts through gene transcription, so the effect long outlives the plasma level. **Never read a steroid's dosing interval off its plasma half-life.**

# Mineralocorticoid Receptor Antagonists

## MECHANISM

Block the **mineralocorticoid receptor** — the exact mirror of the page before.

→ **Downstream:** ↓ Na<sup>+</sup> & water reabsorption · ↓ K<sup>+</sup> excretion → K<sup>+</sup> retained (hence "K<sup>+</sup>-sparing diuretic") → ↓ volume, ↓ BP → **Why it matters:** aldosterone drives the pathology of hypertension, heart failure and diabetic CKD, so blocking it is disease-modifying, not just diuretic.

## DRUGS

| DRUG                       | WHAT IT TREATS                                                                                             |
|----------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Spironolactone</b> (PO) | hypertension · heart failure — <i>K<sup>+</sup>-sparing diuretic</i>                                       |
| <b>Finerenone</b> (PO)     | <b>chronic kidney disease in type-2 diabetes</b> — <i>non-steroidal, newer</i>                             |
| <b>Drospirenone</b> (PO)   | a <b>progestin</b> with MR-antagonist activity — oral contraception · moderate acne vulgaris · <b>PMDD</b> |

## USES

**Hypertension · heart failure · chronic kidney disease** — spironolactone, finerenone · **Oral contraception · acne vulgaris · premenstrual dysphoric disorder (PMDD)** — drospirenone

## ADVERSE EFFECTS

- **Hyperkalemia** ★ — the defining one, exact opposite of the mineralocorticoid page
- **Hyponatremia, hypotension**
- Glucocorticoid-like effects
- ★ **Anti-androgen effects** — spironolactone is a steroid and hits androgen receptors too: **gynecomastia** in men, **hirsutism** changes and menstrual irregularity in women

# Glucocorticoids — Actions & Adverse Effects

## MECHANISM

Agonist at the **intracellular / nuclear glucocorticoid receptor**. Hormone binds → hsp complex dissociates → receptor dimer enters the nucleus → binds the **glucocorticoid-response element (GRE)** → changes gene transcription.

→ **Metabolic:** ↑ serum glucose — ↓ glucose uptake & glycogenesis in **muscle**, ↑ **gluconeogenesis** in **liver** · ↑ protein catabolism in muscle & skin to feed that gluconeogenesis · ↑ **lipolysis** in adipose for free fatty acids  
→ **Purpose of all that:** keep **brain and heart** fed with glucose under stress. **The side effects are the same actions, unopposed.** → **Regulatory (the therapeutic one):** ↓ number, distribution and function of white cells · ↓ inflammatory mediators · ↓ **phospholipase A<sub>2</sub>, COX-2, iNOS** · ↓ vascular permeability · ↓ release of endogenous glucocorticoids

## EVERY WANTED EFFECT HAS A MATCHING UNWANTED ONE

| ACTION                            | BENEFIT                               | COST                                                   |
|-----------------------------------|---------------------------------------|--------------------------------------------------------|
| Carbohydrate metabolism           | energy for brain & heart in stress    | <b>hyperglycemia</b>                                   |
| Protein & lipid catabolism        | substrate for gluconeogenesis         | <b>muscle wasting · thin skin · fat redistribution</b> |
| Immune suppression                | ↓ inflammation, allergy, autoimmunity | <b>↑ infection risk</b>                                |
| Suppression of endogenous release | —                                     | <b>adrenal insufficiency</b>                           |

## ADVERSE EFFECTS

- **Short-term use: no severe adverse effects** (drug allergy aside). Risk scales with **dose × duration × potency × route**.
- **Cushing's syndrome** (long-term) — moon face, buffalo hump, central obesity, thin skin
- **Adrenal insufficiency after abrupt withdrawal** ★ — the HPA axis has been switched off and cannot restart in time
- **Bone:** ↓ bone strength — ↓ formation (↓ osteoblast, ↑ osteoclast) · ↓ intestinal Ca<sup>2+</sup> absorption · ↑ renal Ca<sup>2+</sup> excretion
- **Muscle:** ↓ muscle mass · **Growth:** inhibition of child growth
- **Eyes:** ↑ intraocular pressure — glaucoma, cataract · **Brain:** ↑ intracranial pressure
- **CNS:** emotional disturbance, euphoria, sleep disturbance, depression
- **GI:** ↓ mucosal protection → ulcer risk

## CAUTIONS

- ⚠ **No absolute contraindication** — it is a risk-vs-benefit drug, not a forbidden one
- **Never stop abruptly** — **taper** the dose
- ⚠ Use appropriately: right drug for the indication · least systemic exposure · **lowest effective dose · shortest possible time**
- ★ **Imitate the circadian rhythm** — **biggest dose in the morning, none at night**. Evening dosing suppresses the natural morning ACTH surge hardest.
- ⚠ Caution in **severe infection** and **active peptic ulcer**; antagonizes insulin → watch the diabetic

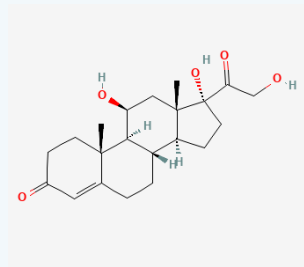
# Glucocorticoids — Systemic

## WHY ROUTE MATTERS

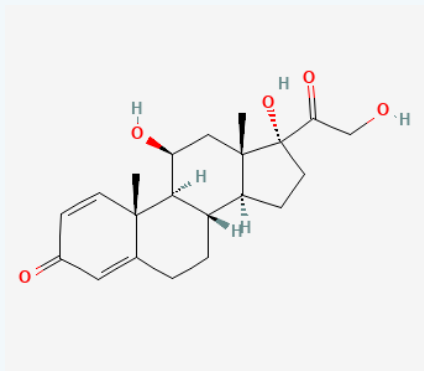
Oral and IV/IM act on **every tissue** (systemic effects), so ADR risk is high. **Intra-articular** and **intralesional** injection delivers a high local dose while limiting how much reaches the blood.

→ **Pharmacokinetics**: rapid, good GI absorption · **topical routes can still be absorbed systemically** · crosses **placenta and breast milk** · mainly hepatic metabolism · excreted in urine as sulfate/glucuronide conjugates → ● **Plasma  $t_{1/2}$  does not predict biological effect** — same rule as fludrocortisone.

## THE SHARED CORE · WHAT CHANGES

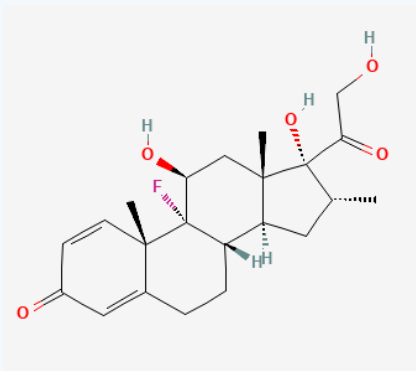


**Cortisol (hydrocortisone).** the natural hormone — the pregnane steroid baseline. Everything below is this with atoms added to boost glucocorticoid action and cut mineralocorticoid action.



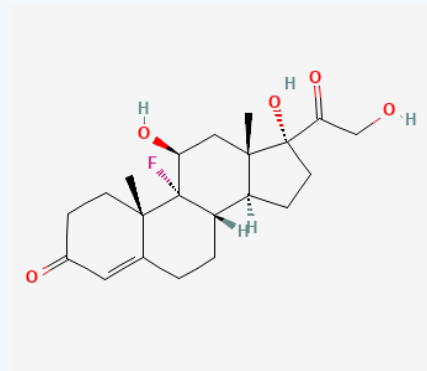
### Prednisolone

one extra 1,2-double bond → 4× the glucocorticoid potency, less salt retention



### Dexamethasone

9α-fluoro + 16α-methyl → ~25× potency and ● essentially zero mineralocorticoid effect



### Fludrocortisone

9α-F without the 16-methyl → the opposite outcome: a mineralocorticoid. One methyl group decides which hormone it is.

## MEDCHEM

**SAR** ★ Every synthetic corticosteroid is the same molecule with a checklist of edits, and each edit does a stated thing. **1,2-double bond ( $\Delta 1$ )** → ↑ glucocorticoid activity (cortisone → prednisone). **11-OH beats 11-keto** - which is exactly why prednisone and cortisone are inactive until the liver reduces them. **9α-fluorine** → raises glucocorticoid *and* mineralocorticoid activity together (fludrocortisone). **16α-methyl** → ↑ glucocorticoid, ↓ **mineralocorticoid** (dexamethasone: the potency without the sodium retention). **6α-methyl or 6α-F** → ↑ glucocorticoid, and 6α-CH<sub>3</sub> also drops the mineralocorticoid.

**SAR** For the topical and inhaled agents the trick is different: a **16,17-ketal (acetonide)** raises glucocorticoid activity, lowers mineralocorticoid activity *and* raises lipophilicity - it keeps the drug in the skin or airway. Replacing an -OH with -Cl raises receptor affinity outright.

## ORAL

| DRUG                          | WHAT IT TREATS                                                                                                                |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Hydrocortisone</b> (10 mg) | ★ <b>drug of choice for cortisol replacement</b> — short-acting, 1:1 GC:MC, closest to the real hormone                       |
| <b>Prednisolone</b> (5 mg)    | ★ <b>drug of choice for non-adrenal disorders</b> — 5–60 mg/day in allergic & inflammatory disease                            |
| <b>Triamcinolone</b> (4 mg)   | inflammatory / immune disease — intermediate                                                                                  |
| <b>Dexamethasone</b> (0.5 mg) | inflammatory disease needing a <b>long, potent</b> action; <b>not</b> used for adrenal replacement (no MC activity, too long) |

## PARENTERAL

| DRUG                                                     | ROUTE & WHAT IT TREATS                                                                  |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Hydrocortisone Na<sup>+</sup> succinate</b>           | IV, IM — ● ★ <b>drug of choice for adrenal crisis</b>                                   |
| <b>Dexamethasone sodium</b>                              | IV, IM — potent systemic anti-inflammatory                                              |
| <b>Methylprednisolone Na<sup>+</sup> succinate</b>       | IV, IM — anti-inflammatory / immunosuppressive; allograft rejection; status asthmaticus |
| <b>Methylprednisolone acetate</b>                        | IM, IA, ID — depot; joints and lesions                                                  |
| <b>Triamcinolone acetonide</b>                           | <b>IA only</b> — joint inflammation                                                     |
| <b>Betamethasone</b> (dipropionate + disodium phosphate) | IM, IA                                                                                  |

## USES

**Replacement therapy** — primary adrenal insufficiency (Addison's) · secondary adrenal insufficiency after glucocorticoid withdrawal · **Inflammatory** — asthma, COPD · **Immune** — RA, OA, IBD, SLE, psoriasis · **Allergic** — dermatitis, rhinitis, conjunctivitis, otitis · **Organ transplantation** · **Adjunct in life-threatening states** — sepsis, cancer · **Other** — vomiting, hemorrhoid, keloid, alopecia areata

★ Glucocorticoids **control, they do not cure** — they suppress the symptom, not the cause.

## PULSE THERAPY

Supra-pharmacological IV doses for **a few days only** — buys faster, stronger control while avoiding the long-term exposure that causes Cushing's syndrome.

## ADMINISTRATION — SYSTEMIC AND PARENTERAL GLUCOCORTICOIDS

Routes available: oral · parenteral — IV, IM, intradermal (ID), intra-articular (IA), intralesional.

| DRUG                                                       | ROUTES     | NOTE                                |
|------------------------------------------------------------|------------|-------------------------------------|
| Hydrocortisone sodium succinate                            | IV, IM     | ★ drug of choice for adrenal crisis |
| Dexamethasone sodium                                       | IV, IM     |                                     |
| Methylprednisolone sodium succinate                        | IV, IM     |                                     |
| Methylprednisolone acetate                                 | IM, IA, ID |                                     |
| Triamcinolone acetonide                                    | IA         |                                     |
| Betamethasone ( <i>dipropionate + disodium phosphate</i> ) | IM, IA     |                                     |

**Local injection uses:** intralesional for alopecia areata and keloid · intra-articular for joint inflammation.

**Pharmacokinetics that shape the route choice:** - Rapidly and well absorbed from the GI tract — which is why oral works at all. - ● **Topical administration can still be absorbed systemically.** "Local" is not the same as "no systemic effect". - **Crosses the placenta and into breast milk.** - Metabolised mainly in the **liver**, excreted in **urine** as sulfate and glucuronide conjugates. - ★ **Plasma half-life cannot be read as duration of biological effect** — the receptor action outlasts the drug in the blood, which is why dosing intervals don't follow the  $t_{1/2}$ .

- 💡 The last point is the one that catches people out: asking "how long does it last" and answering with the half-life gives the wrong number for every steroid on this page.



# Glucocorticoids — Topical & Local

## THE DESIGN PRINCIPLE

Deliver drug where the inflammation is and keep it out of the blood.

→ **Nasal & inhaled** are deliberately built with **high first-pass hepatic metabolism** — the swallowed fraction is destroyed, so **bioavailability is minimal** and systemic ADRs are minimized. → ● **Topical is not "safe"** — enough drug, enough area, enough time and it becomes systemic.

## DERMAL POTENCY CLASSES (WHO + THAI DERMATOLOGY) — PICK BY DISEASE AND BY SKIN SITE

| CLASS         | POTENCY    | EXAMPLE                                                                                                                           | TYPICAL TARGET                                              |
|---------------|------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>I</b>      | ultra high | <b>Clobetasol propionate</b> cream 0.05%                                                                                          | resistant atopic dermatitis · discoid lupus                 |
| <b>II–III</b> | high       | <b>Betamethasone dipropionate</b> oint 0.05% · <b>Betamethasone valerate</b> oint 0.1% · <b>Triamcinolone acetonide</b> oint 0.1% | psoriasis                                                   |
| <b>IV–V</b>   | moderate   | <b>Triamcinolone acetonide</b> cream 0.1% · <b>Fluocinolone acetonide</b> cream 0.025%                                            | atopic dermatitis · nummular eczema · seborrheic dermatitis |
| <b>VI–VII</b> | low        | <b>Triamcinolone acetonide</b> cream 0.02% · <b>Hydrocortisone acetate</b> cream 1% · <b>Prednisolone</b> cream 0.5%              | dermatitis on <b>face, eyelids, diaper area</b>             |

⚠ Same drug, different vehicle and strength = different class. **Ointment > cream > lotion** in potency.

## LOCAL ROUTES AND WHAT THEY ARE FOR

| ROUTE               | DRUGS                                                                                                                                                                                    | WHAT IT TREATS                                                                                                                                                                     |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Oral base</b>    | <b>Triamcinolone acetonide</b> 0.1%                                                                                                                                                      | oral mucosal lesions — recurrent ulcerative stomatitis, erosive lichen planus                                                                                                      |
| <b>Rectal</b>       | <b>Hydrocortisone acetate</b> + cinchocaine · <b>Fluocortolone</b> + lidocaine                                                                                                           | hemorrhoids and complications — fissure, proctitis, perianal eczema ( <i>local anaesthetic added for the pain</i> )                                                                |
| <b>Optic / otic</b> | <b>Dexamethasone</b> (drops, gel, intravitreal) · <b>Prednisolone acetate</b> 1% · <b>Fluorometholone</b> 0.1% · <b>Loteprednol etabonate</b> 0.2 / 0.5%                                 | eye and ear inflammation — uveitis, iritis, conjunctivitis, macular edema. <b>Loteprednol</b> is metabolized to inactive substances in the circulation → <b>no systemic effect</b> |
| <b>Intranasal</b>   | <b>Budesonide</b> · <b>Mometasone furoate</b> · <b>Fluticasone propionate / furoate</b> · <b>Triamcinolone acetonide</b> · <b>Beclometasone</b> · <b>Ciclesonide</b> ( <i>pro-drug</i> ) | allergic rhinitis · nasal polyp · sinusitis (adjunct)                                                                                                                              |
| <b>Inhaled</b>      | <b>Budesonide</b> · <b>Fluticasone propionate / furoate</b> · <b>Beclometasone</b> — usually combined with a <b>β<sub>2</sub>-agonist</b>                                                | <b>maintenance</b> of asthma & COPD · croup · sparing oral steroid                                                                                                                 |

## CAUTIONS

- ★ **All the intranasal steroids have similar efficacy and similar ADRs** — choose on **taste, odour, volume per puff**, i.e. on what the patient will actually keep using
- **Inhaled steroids are maintenance only** — **never for an acute asthma attack**. They are not bronchodilators.
- ⚠ Rinse the mouth after inhaling — reduces the swallowed fraction and local candidiasis
- ⚠ Long-term ocular steroid → ↑ **intraocular pressure, glaucoma, cataract**
- ⚠ Match potency to the site: thin skin (face, eyelids, diaper) takes low potency only

## ADMINISTRATION — TOPICAL POTENCY CLASSES, AND INHALED PREPARATIONS

### Dermal glucocorticoids (*WHO + Thai dermatology classification*)

| CLASS | POTENCY           | PREPARATION                                                                                                                                |
|-------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| I     | <b>ultra high</b> | clobetasol propionate <b>cream 0.05%</b>                                                                                                   |
| II    | high              | betamethasone dipropionate <b>ointment 0.05%</b> · desoximetasone <b>cream 0.025%</b>                                                      |
| III   | high              | betamethasone valerate <b>ointment 0.1%</b> · betamethasone dipropionate <b>cream 0.05%</b> · triamcinolone acetonide <b>ointment 0.1%</b> |
| IV    | moderate          | triamcinolone acetonide <b>cream 0.1%</b> · fluocinolone acetonide <b>cream 0.025%</b>                                                     |
| V     | moderate          | betamethasone dipropionate <b>lotion 0.02%</b>                                                                                             |
| VI    | low               | triamcinolone acetonide <b>cream 0.02%</b> · fluocinolone acetonide <b>solution 0.01%</b>                                                  |
| VII   | low               | hydrocortisone acetate <b>cream 1%</b> · prednisolone <b>cream 0.5%</b>                                                                    |

- ★ **The vehicle changes the class, not just the strength**. Betamethasone dipropionate is **class II as an ointment, class III as a cream and class V as a lotion** — **same molecule**. Ointment > cream > lotion, because occlusion drives penetration.
- **Higher classes** for resistant atopic dermatitis, discoid lupus, psoriasis · **lower classes** for dermatitis of the **diaper area, eyelids and face**, atopic dermatitis, nummular eczema, seborrhoeic dermatitis.
- 💡 That indication split is a thin-skin rule. Face, eyelids and nappy area absorb far more, so they get the weak preparations regardless of how stubborn the rash looks.

### Inhaled glucocorticoids

**Budesonide · fluticasone propionate · fluticasone furoate · beclometasone dipropionate**, for **maintenance of asthma and COPD, croup**, and **replacing or reducing oral steroid therapy**.

- **Budesonide 1 mg/2 mL per tube** — solution for inhalation via nebuliser
- **Fluticasone propionate 50 µg + salmeterol 25 µg per puff** — inhalation aerosol
- **Fluticasone propionate 500 µg + salmeterol 50 µg per puff** — inhalation powder
- **Fluticasone furoate 100 µg + vilanterol 25 µg per puff** — inhalation powder
- ★ **Inhaled steroids are usually combined with a  $\beta_2$ -agonist**, and their tissue distribution behaves like the intranasal ones.
- 💡 The nebuliser solution is the paediatric and acute route — it needs no coordination, which is exactly what a small child or a breathless patient doesn't have.

# Glucocorticoid Antagonists

## MECHANISM

★ The available drugs are cortisol *synthesis* inhibitors, not receptor blockers. They inhibit the adrenal enzymes needed to make cortisol; nothing in current use stops cortisol binding the glucocorticoid receptor.  
→ **Downstream:** ↓ adrenal cortisol synthesis → the excess driving Cushing's syndrome comes down

## DRUGS

| DRUG                | WHAT IT TREATS                                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Ketoconazole</b> | Cushing's syndrome — <b>inhibitor of multiple CYPs</b> , including the steroidogenic ones                                |
| <b>Abiraterone</b>  | Cushing's syndrome — <b>17-hydroxylase</b> and <b>17,20-lyase</b> inhibitor ( <i>so it also cuts adrenal androgens</i> ) |
| <b>Pasireotide</b>  | Cushing's disease — <b>somatostatin analogue</b> , works upstream by suppressing ACTH                                    |

## USES

**Cushing's syndrome / Cushing's disease** — the only indication in this lecture

Source: Adrenal steroids and their antagonists XX0326 TL.pdf — รศ.ดร.ภก.ธีรรัตน์ เหลืองมั่นคง, PYCH305 (ภาว 305) Pharmacology II, 31 slides + handout. Structures omitted.

# Anticancer Agents I

Conventional **cytotoxic chemotherapy** — drugs that kill cells which are dividing fast. They are not selective for cancer; they are selective for *proliferation*. Every shared adverse effect in this lecture falls out of that one fact.

**damage DNA or the mitotic machinery → cell can't complete replication/division → apoptosis**  
→ **but every fast-dividing normal tissue takes the same hit (marrow · GI mucosa · hair follicles)**

## WORDS WORTH SEPARATING

- **Neoplasm / tumour** — abnormal mass from abnormal growth. **Not all tumours are cancerous.**
- **Benign** — well-differentiated, grows by *expansion*, **no metastasis** (suffix *-oma*)
- **Malignant** — anaplastic, atypical, grows by *invasion*, **metastasises**
- **Metastasis** — break away from the primary, travel via blood/lymph, seed a distant site
- **Angiogenesis** — formation of new blood vessels to feed the tumour

## CELL-CYCLE SPECIFICITY — THE ORGANISING PRINCIPLE ★

|                | CCS (CELL CYCLE-SPECIFIC)                               | CCNS (NON-SPECIFIC)                     |
|----------------|---------------------------------------------------------|-----------------------------------------|
| acts           | only in particular phases                               | in <b>all</b> phases                    |
| dose-response  | <b>plateaus</b> — more drug ≠ more kill                 | <b>linear</b> — more drug = more kill   |
| how it's given | low concentration, <b>continuous / prolonged</b> dosing | often a <b>single bolus</b>             |
| classes        | antimetabolites (S) · antimitotics (M)                  | alkylating agents · DNA strand breakers |

## EACH CLASS VS ITS PHASE

| CLASS                                                      | CCS/CCNS             | PHASE HIT                                          |
|------------------------------------------------------------|----------------------|----------------------------------------------------|
| <b>Alkylating agents</b>                                   | CCNS                 | all phases                                         |
| <b>Platinum analogs</b>                                    | CCNS                 | all phases                                         |
| <b>Antifolates</b> (MTX)                                   | CCS                  | <b>S</b>                                           |
| <b>Ribonucleotide reductase inhibitor</b><br>(hydroxyurea) | CCS                  | arrest at/near the <b>G1–S</b> interphase          |
| <b>Pyrimidine analogs</b> (5-FU, Ara-C...)                 | CCS                  | <b>S</b>                                           |
| <b>Purine analogs</b> (6-MP, 6-TG...)                      | CCS                  | <b>S</b>                                           |
| <b>Vinca alkaloids</b>                                     | CCS                  | <b>M</b>                                           |
| <b>Taxanes</b>                                             | CCS                  | <b>M</b>                                           |
| <b>Topoisomerase I inhibitors</b>                          | —                    | <i>(not specified in lecture)</i>                  |
| <b>Topoisomerase II inhibitors</b>                         | CCS                  | cells in <b>S</b> and <b>G2</b> are most sensitive |
| <b>Anthracyclines</b>                                      | —                    | <i>(not specified in lecture)</i>                  |
| <b>Bleomycin</b>                                           | CCS                  | accumulates cells in <b>G2</b>                     |
| <b>Mitomycin C</b>                                         | acts as an alkylator | —                                                  |

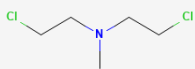
# Alkylating Agents

## MECHANISM

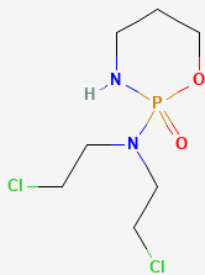
Attach an **alkyl group** to DNA bases — preferentially **N7 of guanine** (also N1/N3 adenine, N3 cytosine, O6 guanine to a lesser degree).

→ **Downstream:** three separate lesions from one chemistry — 1. **Mispairing** — alkylated guanine pairs with thymine during synthesis → mutation 2. **Strand breakage** — N7 alkylation destabilises the imidazole ring 3. **Cross-links** — intrastrand · interstrand · DNA–protein → major disruption of nucleic acid function  
→ **CCNS:** cytotoxic in **every** phase of the cell cycle.

## THE SHARED CORE · WHAT CHANGES

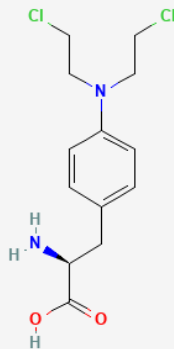


**Mechlorethamine — the nitrogen mustard.** ● the two chloroethyl arms are the drug. Each one closes into a strained ring that attacks DNA guanine — two arms, so it cross-LINKS the strands. Everything below is this warhead with a delivery system bolted on.



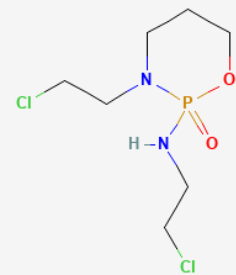
### Cyclophosphamide

an inactive prodrug until CYP2B6 opens it; ● the acrolein by-product is what burns the bladder — hence mesna



### Melphalan

the same arms on phenylalanine → carried in by amino-acid transporters



### Ifosfamide

one arm moved → more acrolein and more neurotoxicity than cyclophosphamide

## DRUGS

| DRUG                                            | WHAT IT TREATS                                                              |
|-------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Mechlorethamine</b> ( <i>N-mustard</i> )     | Hodgkin's & non-Hodgkin's lymphoma                                          |
| <b>Cyclophosphamide</b> ( <i>N-mustard</i> )    | breast · ovarian · NHL · CLL · soft tissue sarcoma · Wilms' · neuroblastoma |
| <b>Ifosfamide</b> ( <i>N-mustard</i> )          | —                                                                           |
| <b>Chlorambucil</b> ( <i>N-mustard</i> )        | CLL · non-Hodgkin's lymphoma                                                |
| <b>Melphalan</b> ( <i>N-mustard</i> )           | <b>multiple myeloma</b> · breast · ovarian                                  |
| <b>Thiotepa</b> ( <i>aziridine</i> )            | breast · ovarian · superficial bladder                                      |
| <b>Busulfan</b> ( <i>alkyl sulfonate</i> )      | <b>CML</b>                                                                  |
| <b>Carmustine (BCNU)</b> ( <i>nitrosourea</i> ) | brain cancer · Hodgkin's & NHL                                              |
| <b>Lomustine (CCNU)</b> ( <i>nitrosourea</i> )  | <b>brain cancer</b>                                                         |
| <b>Dacarbazine</b> ( <i>triazine</i> )          | Hodgkin's · melanoma · soft tissue sarcoma                                  |
| <b>Temozolomide</b> ( <i>triazine</i> )         | <b>brain cancer</b> · melanoma                                              |
| <b>Procarbazine</b> ( <i>methylhydrazine</i> )  | Hodgkin's & NHL · brain tumours                                             |

## ADVERSE EFFECTS

- **Rapidly proliferating tissue takes it** ★ — skin/hair (**alopecia**) · GI (nausea, vomiting, diarrhea) · **bone marrow** (anemia → fatigue, leukopenia → infection, thrombocytopenia → bleeding)
- ● **Hemorrhagic cystitis** — **cyclophosphamide** ★. Caused by **acrolein**, a metabolite directly toxic to bladder mucosa. **Mesna detoxifies acrolein** and prevents it.
- **Busulfan specifically:** skin pigmentation · **pulmonary fibrosis** · adrenal insufficiency
- **Nitrosoureas:** interstitial lung disease · interstitial nephritis (*both rare*)
- Temozolomide — photosensitivity · Procarbazine — hypersensitivity · Dacarbazine — CNS toxicity

## WORTH KNOWING

- ★ **Cyclophosphamide** — high oral bioavailability, **oral = IV in efficacy**. The liver is spared because it converts the drug to *inactive* metabolites there.
- ★ **Nitrosoureas are highly lipid-soluble** → **cross the BBB** — that's why they're the brain-tumour alkylators. O6-guanine alkylation is the one critical for their cytotoxicity, and they are **not cross-resistant** with other alkylating agents.

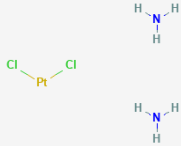
# Platinum Analogs

## MECHANISM

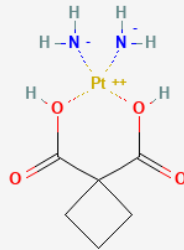
Form **intrastrand and interstrand DNA cross-links**, and bind nuclear + cytoplasmic proteins — alkylating-agent behaviour without being a classical alkylator.

→ **Downstream:** inhibition of DNA synthesis and function · **CCNS**

## THE SHARED CORE · WHAT CHANGES

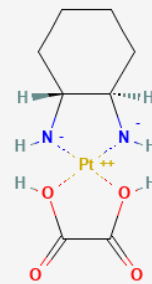


**Cisplatin.** ● not an organic drug at all — a platinum atom with two amines and two leaving groups. Water displaces the chlorides, and the platinum then cross-links guanines in DNA. The LEAVING GROUPS decide toxicity; the platinum core never changes.



### Carboplatin

the chlorides replaced by a stable ring → leaves far more slowly → much less nephro/ototoxicity, more myelosuppression



### Oxaliplatin

a bulky DACH ring stays attached → adducts mismatch repair can't fix → works where cisplatin has failed

## DRUGS

| DRUG                         | WHAT IT TREATS                                                                                                                 |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Cisplatin</b>             | broad range of <b>solid tumours</b> — NSCLC, SCLC, breast, bladder, cholangiocarcinoma, gastroesophageal, head & neck, ovarian |
| <b>Carboplatin</b> (2nd gen) | NSCLC, SCLC, breast, bladder, head & neck, ovarian                                                                             |
| <b>Oxaliplatin</b> (3rd gen) | colorectal cancer (the "OX" of FOLFOX)                                                                                         |

## ADVERSE EFFECTS — THE GENERATIONAL TRADE ★

- **Cisplatin** — ● **nephrotoxicity** (*extensively cleared by the kidney; requires vigorous IV hydration*) · ototoxicity · peripheral sensory neuropathy · N/V · myelosuppression
- **Carboplatin** — **less renal and GI toxicity**; dose-limiting toxicity is **myelosuppression**
- **Oxaliplatin** — dose-limiting toxicity is **peripheral sensory neuropathy**

## CROSS-RESISTANCE

- ★ **Oxaliplatin is NOT cross-resistant** with cisplatin/carboplatin — the basis is a **mismatch-repair defect**. This is why it works where the older platinum drugs have failed.



# Antifolates

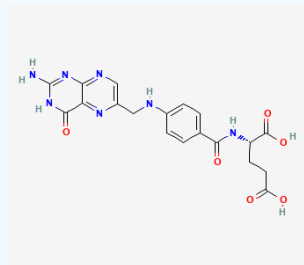
## MECHANISM

Inhibit **dihydrofolate reductase (DHFR)** → no tetrahydrofolate (THF). THF donates the methyl groups for **de novo synthesis of dTMP, purines and methionine**.

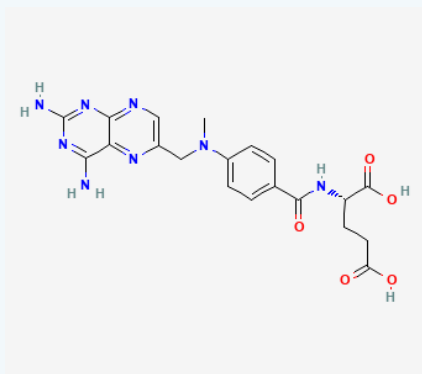
→ **Downstream:** no dTMP, no purines → **DNA, RNA and key protein synthesis stall** · **S-phase specific**, most effective when cells proliferate fast

→ **Polyglutamation:** intracellular **MTX-polyglutamate** is critical to the therapeutic effect — it is **selectively retained inside cancer cells** and inhibits the de novo enzymes even harder.

## THE SHARED CORE · WHAT CHANGES

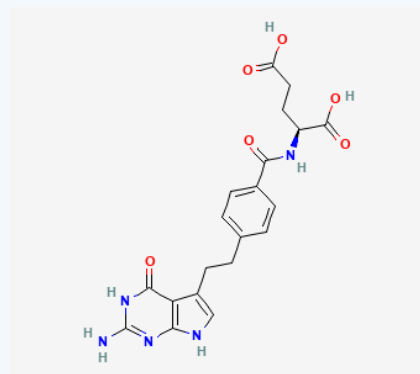


**Folic acid — the real substrate.** pteridine + PABA + glutamate. ● the drug below is this molecule with two atoms changed, and that is enough for DHFR to grab it and stop.



### Methotrexate

an  $\text{-NH}_2$  replaces the 4-OH and an N-methyl is added → binds DHFR  $\sim 1000\times$  tighter than folate, and irreversibly in practice



### Pemetrexed

same idea, a different ring → hits three folate enzymes at once

## DRUGS

| DRUG                      | WHAT IT TREATS                                                                                            |
|---------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Methotrexate (MTX)</b> | breast · head & neck · osteogenic sarcoma · <b>primary CNS lymphoma</b> · NHL · bladder · choriocarcinoma |
| <b>Raltitrexed</b>        | polyglutamated → <b>thymidylate synthase</b> inhibition                                                   |

## ADVERSE EFFECTS

- **Mucositis**, diarrhea
- **Myelosuppression**
- Nausea and vomiting
- Hand-foot syndrome

## INTERACTIONS

- 🚫 MTX is eliminated **90% unchanged by renal excretion** → anything that blocks renal excretion raises MTX toxicity: **aspirin** · NSAIDs · **penicillins** · **cephalosporins** ★
- **Reduce dose in proportion to CrCl.**
- Oral bioavailability is **saturable** above  $\sim 25 \text{ mg/m}^2$  — so high doses go IV (*also intrathecal*).

## LEUCOVORIN RESCUE ★

- **Leucovorin** (folinic acid / 5-formyl-THF) is already-reduced folate — it **bypasses the blocked DHFR** step, so normal cells recover.
- Given alongside **high-dose MTX** to rescue normal tissue from undue toxicity, without rescuing the tumour to the same degree.

## RESISTANCE TO MTX

- ↓ drug transport (↓ *reduced folate carrier* / *folate receptor*) · ↓ formation of toxic polyglutamates · ↑ **DHFR levels** · altered DHFR with lower MTX affinity

## ADMINISTRATION — METHOTREXATE

- **Routes: intravenous, intrathecal, or oral.**
- 🚫 **Oral bioavailability is SATURABLE above  $25 \text{ mg/m}^2$**  — past that point a bigger oral dose does not give a bigger exposure, which is why high-dose regimens go parenteral.
- **Renal excretion (90% unchanged) is the main elimination route**, so 🚫 **the dose must be reduced in proportion to CrCl.**
- ⚠️ **Caution with anything that inhibits renal excretion of MTX — aspirin, NSAIDs, penicillins, cephalosporins.** These are common drugs, which is what makes it dangerous.
- 💡 An NSAID for the joint pain plus methotrexate is the classic accidental overdose: nothing about the MTX dose changed, only the kidney's ability to clear it.

# Ribonucleotide Reductase Inhibitor

## MECHANISM

Inhibits **ribonucleoside diphosphate reductase**, the enzyme that reduces **ribonucleotides** → **deoxyribonucleotides**. Cut off the dNTP supply and DNA synthesis has nothing to build from.

→ **Downstream:** cell-cycle arrest **at or near the G1–S interphase**

## DRUGS

| DRUG                    | WHAT IT TREATS                     |
|-------------------------|------------------------------------|
| <b>Hydroxyurea (HU)</b> | chronic myelogenous leukemia (CML) |

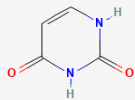
# Pyrimidine Analogs

## MECHANISM

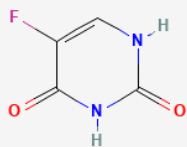
Fake bases and fake nucleosides. They get phosphorylated by the cell's own kinases, then act as **fraudulent substrates** — either poisoning an enzyme or getting built into DNA/RNA.

→ **5-FU**: converted to **FdUMP**, which inhibits **thymidylate synthase (TS)** → no TTP · 5-FdUTP into **DNA** → strand breaks · 5-FUTP into **RNA** → disrupted RNA processing → **Ara-C**: **ara-CTP** competitively inhibits **DNA polymerase  $\alpha$  and  $\beta$**  → blocks DNA synthesis **and repair**. Cellular *retention* of ara-CTP correlates with the kill. **S-phase specific**.

## THE SHARED CORE · WHAT CHANGES

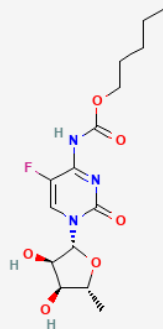


**Uracil — the real base.** ● one hydrogen swapped for fluorine and the whole mechanism follows: thymidylate synthase attacks it, gets stuck, and DNA synthesis stops



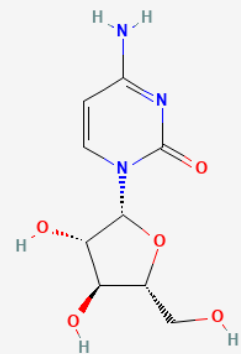
### 5-Fluorouracil

the F at C5 cannot be removed → the enzyme is locked in place



### Capecitabine

an oral prodrug — three enzymatic steps release 5-FU, the last one enriched inside tumour tissue



### Cytarabine

cytosine with the sugar's OH flipped (arabinose, not ribose) → the chain cannot extend

## DRUGS

| DRUG                         | WHAT IT TREATS                                                                        |
|------------------------------|---------------------------------------------------------------------------------------|
| <b>5-Fluorouracil (5-FU)</b> | <b>colorectal</b> ★ · anal · breast · gastroesophageal · head & neck · hepatocellular |
| <b>Capecitabine</b>          | breast · colorectal · gastroesophageal · hepatocellular · pancreatic                  |
| <b>Cytarabine (Ara-C)</b>    | <b>AML</b> · ALL · CML in blast crisis — <b>haematologic only</b>                     |
| <b>Gemcitabine (dFdC)</b>    | pancreatic · bladder · breast · NSCLC · ovarian · NHL · soft tissue sarcoma           |
| <b>Azacitidine</b>           | pyrimidine analog ( <i>deoxynucleoside type</i> )                                     |

## ADVERSE EFFECTS

- **5-FU** — myelosuppression · **mucositis** · diarrhea · N/V · **hand-foot syndrome** · neurotoxicity
- **Capecitabine** — **diarrhea and hand-foot syndrome are the main ones**; myelosuppression, N/V and mucositis are *less* than with 5-FU
- **Cytarabine** — ● **potent myelosuppression** · mucositis · N/V · **cerebellar ataxia / neurotoxicity at high dose** ★
- **Gemcitabine** — N/V, diarrhea, myelosuppression

## THE THREE THINGS TO ACTUALLY KNOW ★

- **5-FU is given IV** — GI absorption is unpredictable and incomplete; half-life is only 10–15 min. 80–85% is destroyed by **dihydropyrimidine dehydrogenase (DPD)**, and ● **~5% of cancer patients are DPD-deficient** → catastrophic toxicity.
- **Leucovorin potentiates 5-FU** (*opposite of the MTX story*) — it raises intracellular CH<sub>2</sub>THF, which is required to lock the **TS-FdUMP-folate ternary complex** together. That's why **FOLFOX = 5-FU + leucovorin + oxaliplatin**.
- **Capecitabine is an oral prodrug** (bioavailability 70–80%) activated by **thymidine phosphorylase**, which tumours express far more of than normal tissue — especially breast and colorectal. Built-in selectivity.
- ⚠ **Ara-C must not be given orally** — cytidine deaminase in GI mucosa and liver destroys it (bioavailability ~20%). IV infusion or intrathecal, and activity is **highly schedule-dependent**. **No activity in solid tumours at all.**

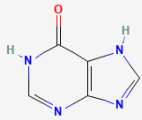
# Purine Analogs

## MECHANISM

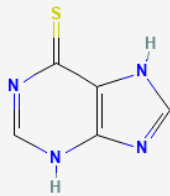
Thiopurines are converted by **HGPRT** into fraudulent nucleotides (TIMP → TGMP).

→ **Downstream:** inhibition of **de novo purine nucleotide synthesis** · incorporation of the triphosphate into RNA and DNA · **S-phase specific**

## THE SHARED CORE · WHAT CHANGES

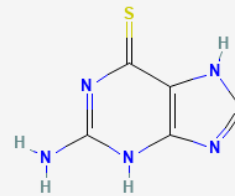


**Hypoxanthine — the real purine.** the base the cell actually salvages. The drugs below are this molecule with one oxygen swapped for sulfur.



**6-Mercaptopurine**

● O→S. Activated by HGPRT into a fake nucleotide; inactivated by TPMT and xanthine oxidase — both of which are the toxicity story



**6-Thioguanine**

the same swap on guanine

## DRUGS

| DRUG                                                                     | WHAT IT TREATS  |
|--------------------------------------------------------------------------|-----------------|
| <b>6-Mercaptopurine (6-MP)</b>                                           | AML             |
| <b>6-Thioguanine (6-TG)</b>                                              | ALL · AML       |
| <b>Azathioprine (AZA)</b>                                                | prodrug of 6-MP |
| <b>Fludarabine · Cladribine · Clofarabine · Nelarabine · Pentostatin</b> | purine analogs  |

## ADVERSE EFFECTS (6-MP)

- **Myelosuppression** — develops **more gradually** than with folate analogs
- **Immunosuppression**
- GI — common: nausea, vomiting, anorexia; rare: stomatitis, diarrhea
- ● **Hepatotoxicity** — jaundice + ↑ hepatic enzymes in **one-third of adult patients**

### INTERACTIONS ★

- **Allopurinol** — 6-MP undergoes first-pass metabolism by **xanthine oxidase** to inactive 6-thiouric acid. Block XO and 6-MP levels spike → **reduce the oral 6-MP dose by 50–75%**. (*Not required for IV dosing — first-pass is bypassed.*)
- **TPMT polymorphism** — 6-MP is also metabolised by thiopurine methyltransferase; ~15% of Caucasians have decreased activity. Testing is recommended in autoimmune-disease patients.

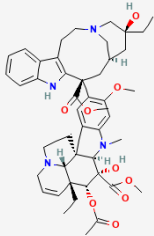
# Vinca Alkaloids

## MECHANISM

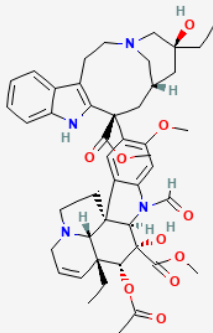
Alkaloids from the periwinkle plant (*Catharanthus rosea*). Bind specifically to  **$\beta$ -tubulin** and **block microtubule polymerization** — the spindle can't be built. **Microtubule-destabilizing**.

→ **Downstream:** no functional mitotic spindle → **cells arrest in mitosis (M phase)** — cell-cycle specific

## THE SHARED CORE · WHAT CHANGES

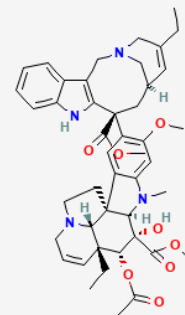


**Vinblastine.** two large alkaloid halves joined together; binds  $\beta$ -tubulin and blocks assembly. Tiny changes on one half redirect the toxicity entirely.



**Vincristine**

● one N-methyl becomes a formyl group — and the dose-limiting toxicity moves from bone marrow to peripheral nerves



**Vinorelbine**

a contracted ring → less neurotoxic, marrow-limited again

## DRUGS

| DRUG               | WHAT IT TREATS                                                                  |
|--------------------|---------------------------------------------------------------------------------|
| <b>Vinblastine</b> | Hodgkin's & NHL · germ cell · breast · Kaposi's sarcoma                         |
| <b>Vincristine</b> | <b>ALL</b> · Hodgkin's & NHL · rhabdomyosarcoma · neuroblastoma · Wilms' tumour |
| <b>Vinorelbine</b> | NSCLC · breast · ovarian                                                        |

## ADVERSE EFFECTS — THE SPLIT MATTERS ★

- **Myelosuppression** — but **very limited with vincristine**
- **Neurotoxicity** — **peripheral neuropathy, paralytic ileus** with vincristine; **rare and mild with vinblastine**
- *Read it as a trade: vincristine spares the marrow and hits the nerves; vinblastine does the reverse.*
- GI — nausea, vomiting, anorexia, diarrhea · constipation (vinorelbine)
- **Alopecia** · mucositis (vinblastine) · **SIADH**

## CAUTIONS

- ● **Intrathecal administration of a vinca alkaloid is FATAL.** They are IV-only. ★
- ⚠ Metabolised by hepatic **CYP3A**; <15% appears unchanged in urine → **cut the dose 50–75% in hepatic dysfunction** (bilirubin > 3 mg/dL).



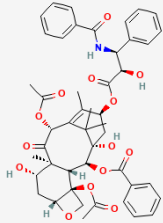
# Taxanes

## MECHANISM

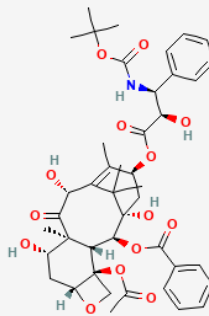
From the Pacific yew. Also bind  $\beta$ -tubulin — but they **block microtubule depolymerization**. **Microtubule-stabilizing**: the spindle is frozen and can't disassemble to pull chromosomes apart.

→ **Downstream: cells blocked in mitosis (M phase)** — cell-cycle specific. *Same target, opposite direction, same outcome as the vincas.* ★

## THE SHARED CORE • WHAT CHANGES



**Paclitaxel**. a rigid diterpene cage that does the OPPOSITE of vincas — it locks microtubules assembled. ● almost insoluble, so it must be given in Cremophor, and that vehicle causes the hypersensitivity



**Docetaxel**

two side-chain changes → more soluble (polysorbate, not Cremophor) and more potent

## DRUGS

| DRUG               | WHAT IT TREATS                                                                        |
|--------------------|---------------------------------------------------------------------------------------|
| <b>Paclitaxel</b>  | breast · NSCLC · SCLC · ovarian · gastroesophageal · prostate · bladder · head & neck |
| <b>Docetaxel</b>   | breast · NSCLC · prostate · gastric · head & neck · ovarian · bladder                 |
| <b>Epothilones</b> | same stabilizing mechanism                                                            |

## ADVERSE EFFECTS

- **Myelosuppression** — **docetaxel** causes more neutropenia than paclitaxel
- **Peripheral neuropathy** — the **dose-limiting toxicity**; **paclitaxel > docetaxel** ★
- ● **Hypersensitivity reactions (~5%)** — premedicate with **dexamethasone + diphenhydramine + an H2 blocker**
- Nausea and vomiting

## INTERACTIONS

- Metabolised by **CYP2C8 + CYP3A4** (paclitaxel) and **CYP3A4 + CYP3A5** (docetaxel); <10% excreted intact in urine → **cut the dose 50–75% in hepatic dysfunction**
- **Cyclosporine** (a P-glycoprotein inhibitor) markedly **delays clearance**

### CAUTIONS

- ★ **Albumin-bound paclitaxel** is *not* associated with hypersensitivity reactions — the reaction is to the vehicle, not the drug.

# Topoisomerase I Inhibitors

## MECHANISM

**Topoisomerase I** relaxes supercoiled DNA by nicking and re-closing **one** strand. These drugs **bind and stabilize the DNA-topoisomerase I complex** — trapping it mid-nick.

→ **Downstream:** the transient single-strand nick becomes **permanent DNA damage**

## DRUGS

| DRUG                       | WHAT IT TREATS                                                                                                   |
|----------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Topotecan</b>           | SCLC · ovarian cancer                                                                                            |
| <b>Irinotecan (CPT-11)</b> | <b>colorectal</b> · NSCLC · SCLC · gastroesophageal                                                              |
| <i>Camptothecin</i>        | not used clinically — the natural parent compound, too toxic and insoluble; the source of topotecan & irinotecan |

## ADVERSE EFFECTS

- **Topotecan** — **myelosuppression (neutropenia = dose-limiting)** · mucositis, diarrhea · less commonly N/V, fever, rash, ↑ transaminases
- **Irinotecan** — ● **two distinct diarrheas** ★
  - **Acute (<24 h): cholinergic syndrome** — diarrhea, diaphoresis, hypersalivation
  - **Delayed diarrhea — the dose-limiting toxicity. Treat with loperamide.**
- Irinotecan also: myelosuppression, N/V

## IRINOTECAN IS A PRODRUG ★

- **Carboxylesterases (CES1/CES2)** convert irinotecan → **SN-38**, which is ~**1000× more potent**.
- **UGT1A1/1A9** glucuronidate SN-38 → inactive **SN-38G**, excreted in bile. Poor glucuronidation = more active drug in the gut = the delayed diarrhea.
- **CYP3A** hepatic metabolism (→ APC, NPC) is the major *inactivating* elimination route.
- Topotecan by contrast is renal — 30–40% of the dose in urine within 24 h.

# Topoisomerase II Inhibitors

## MECHANISM

**Topoisomerase II** changes DNA topology by breaking and rejoining **double**-stranded DNA. Semisynthetic **podophyllotoxin** derivatives (mayapple root) form a **ternary complex with DNA + topoisomerase II** and freeze it there.

→ **Downstream:** double-strand DNA damage · **cells in S and G2 phase are the most sensitive**

## DRUGS

| DRUG                         | WHAT IT TREATS                                                                                            |
|------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Etoposide (VP-16-213)</b> | pediatric leukemia · <b>SCLC</b> · testicular tumours · Hodgkin's & large-cell lymphoma · NSCLC · gastric |
| <b>Teniposide (VM-26)</b>    | same class, same activity profile                                                                         |

## ADVERSE EFFECTS

- **Leukopenia — the dose-limiting toxicity**
- Myelosuppression generally
- Nausea and vomiting
- **Alopecia**

## CAUTIONS

- ⚠️ ~40% is excreted in the urine → **reduce the dose in renal dysfunction.**

# Anthracyclines

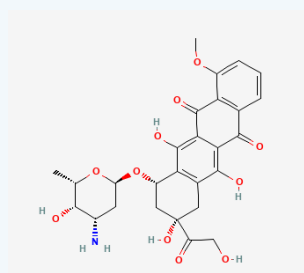
## MECHANISM

From *Streptomyces peucetius* var. *caesius*. **Four mechanisms at once** — this is why they're among the most widely used cytotoxics:

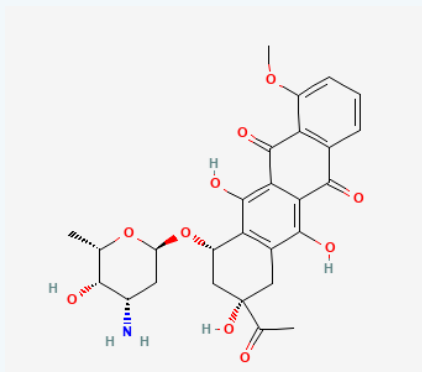
1. **Inhibit topoisomerase II**
2. **Intercalate into DNA** with high affinity → blocks DNA & RNA synthesis, causes **strand scission**
3. **Generate semiquinone and oxygen free radicals** ★
4. Bind cell membranes → alter fluidity and ion transport

→ **Downstream:** mechanism 3 is the one that explains the cardiotoxicity.

## THE SHARED CORE · WHAT CHANGES

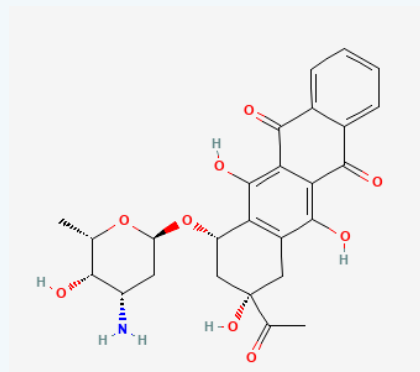


**Doxorubicin.** a flat four-ring system that slides between DNA base pairs, plus a sugar. ● the quinone ring is what redox-cycles into free radicals — that is the cardiotoxicity, and it is separate from the anticancer effect



**Daunorubicin**

one -OH fewer → AML rather than solid tumours



**Idarubicin**

a methoxy removed → more lipophilic → orally usable, better cell entry

## DRUGS

| DRUG                                       | WHAT IT TREATS                                                                                             |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Doxorubicin</b>                         | breast · Hodgkin's & NHL · soft tissue sarcoma · ovarian · NSCLC · SCLC · thyroid · Wilms' · neuroblastoma |
| <b>Daunorubicin</b>                        | AML · ALL                                                                                                  |
| <b>Idarubicin</b> (analog of daunorubicin) | AML · ALL · CML in blast crisis                                                                            |
| <b>Epirubicin</b> (analog of doxorubicin)  | —                                                                                                          |
| <b>Mitoxantrone</b>                        | —                                                                                                          |

## ADVERSE EFFECTS

- **Myelosuppression** — the dose-limiting toxicity of *all* anthracyclines
- **CARDIOTOXICITY, in two forms** ★ — the class-defining fact
  - **Acute** — arrhythmias, conduction abnormalities
  - **Chronic** — **cumulative, dose-dependent dilated cardiomyopathy**, usually at a total adult dose  $\geq 550 \text{ mg/m}^2$ . The heart dilates, weakens, and stops pumping efficiently.
- **Red urine** — **this is the drug colour, NOT hematuria.** ★ Reassure, don't investigate.
- Nausea · **alopecia** · mucositis

# Bleomycin & Mitomycin C

## MECHANISM

**Bleomycin** — a small peptide with a **DNA-binding region** and an **iron-binding domain**. Binds DNA, forms **radicals**, produces **single- and double-strand breaks**. → **Cell-cycle specific: accumulates cells in G2**.

**Mitomycin C** (from *Streptomyces caespitosus*) — **acts as an alkylating agent**, forming DNA cross-links, plus oxygen free radicals. Needs **reductive activation**. → ★ **Hypoxic tumour stem cells** in solid tumours sit in a reducing environment, so they are **more** sensitive to mitomycin than normal or well-oxygenated tumour cells — the rare drug that likes hypoxia.

## DRUGS

| DRUG                    | WHAT IT TREATS                                                                            |
|-------------------------|-------------------------------------------------------------------------------------------|
| <b>Bleomycin</b>        | Hodgkin's & NHL · germ cell cancer · head & neck cancer                                   |
| <b>Mitomycin C</b>      | superficial bladder · gastric · breast · NSCLC · head & neck ( <i>with radiotherapy</i> ) |
| <b>Neocarzinostatin</b> | antitumour antibiotic                                                                     |

## ADVERSE EFFECTS

- **Bleomycin: PULMONARY TOXICITY** → **pulmonary fibrosis**. Dose-limiting, and the incidence rises in patients >70 years. ★ (Note what it does not do: marrow suppression is not its headline.)
- Bleomycin also: allergic reactions · fever · hypotension · skin toxicity · mucositis · alopecia
- Mitomycin**: N/V · myelosuppression · mucositis · anorexia and fatigue · ● **hemolytic uremic syndrome**

# L-Asparaginase

## MECHANISM

An enzyme, not a DNA poison. It **depletes plasma L-asparagine**.

→ **Why it's selective:** most normal tissues make enough of their own L-asparagine for protein synthesis.

**Lymphocytic leukemia cells lack adequate asparagine synthetase** — so they depend on the plasma pool, and starving that pool kills them specifically. ★ *A genuine metabolic selectivity, unlike everything above.*

## DRUGS

| DRUG                  | WHAT IT TREATS                                             |
|-----------------------|------------------------------------------------------------|
| <b>L-Asparaginase</b> | standard agent for <b>acute lymphocytic leukemia (ALL)</b> |

## ADVERSE EFFECTS

- Neurotoxicity
- Hepatotoxicity



# Toxicity, Resistance & the Big Picture

## COMMON & ACUTE TOXICITIES — ORGANS WITH ACTIVE CELL DIVISION

- **Bone marrow suppression** — the reason **most myelosuppressive regimens use 3–4 week dosing intervals** ★
- **GI mucosa** — nausea & vomiting · stomatitis and xerostomia · esophagitis · malabsorption, diarrhea or constipation
- **Dermatologic** — alopecia · dry skin · hand-foot syndrome

★ *The toxicities are the single most important factor limiting the use of potentially curative doses.*

## SPECIFIC ORGAN TOXICITIES — MEMORISE THIS GRID ★

| TOXICITY                  | DRUGS                                                 |
|---------------------------|-------------------------------------------------------|
| <b>Neurotoxicity</b>      | cytarabine · L-asparaginase                           |
| <b>Cardiac toxicity</b>   | doxorubicin · daunorubicin · idarubicin               |
| <b>Nephrotoxicity</b>     | <b>cisplatin</b>                                      |
| <b>Pulmonary toxicity</b> | <b>bleomycin</b>                                      |
| <b>Hepatotoxicity</b>     | L-asparaginase · imatinib · carmustine · methotrexate |

## LONG-TERM COMPLICATIONS

- ⚠ **Secondary malignancies** — unrelated to the original cancer, arising from the very tissues most sensitive to chemo (**marrow, hair follicles, GI epithelium**). May appear **months to years** later; **secondary leukemia and lymphoma** are the greatest risk to adult and childhood survivors.
- ⚠ **Fertility & teratogenicity** — chemotherapy is potentially **gonadotoxic**; risk of an abnormal offspring is highest during ongoing germ-cell exposure.

## MECHANISMS OF CHEMOTHERAPY RESISTANCE

| ROUTE             | WHAT THE CELL DOES                                          |
|-------------------|-------------------------------------------------------------|
| <b>Access</b>     | ↓ drug uptake · <b>efflux pumps</b>                         |
| <b>Metabolism</b> | ↑ drug inactivation                                         |
| <b>Target</b>     | alter the drug target · overexpress it                      |
| <b>Response</b>   | impair the apoptotic pathway · alter cell-cycle checkpoints |

*Principle behind this: [[pharm-drug-action]] — Cytotoxic action as a purpose, plus tubulin as a non-receptor macromolecular target.*

*Principle behind this: [[pharm-drug-interactions]] — Methotrexate — secretion blocked by NSAIDs and probenecid, absorption impaired by neomycin, and cleared faster by urinary alkalinisation in overdose.\*\**

*Principle behind this: [[pharm-pharmacogenomics]] — 6-mercaptopurine and 6-thioguanine, same TPMT pathway, same marrow.*

*Source: Anticancer agents I.pdf — Krisada Sakchaisri, Ph.D., PYCH Pharmacology II, 81 slides. Structures omitted.*

# Anticancer Agents II — Targeted Therapy

Drugs that block cancer growth and spread by interfering with **specific molecules** ("molecular targets") driving proliferation, progression and metastasis. Unlike classical chemotherapy they are **deliberately designed against a chosen target** — and are **often cytostatic**, not cytotoxic. ★

**Tumour carries a driver molecule → drug is designed against that molecule → downstream signalling blocked → growth arrest ± immune killing**

## WORDS WORTH SEPARATING

- **Cytostatic** — stops the cell dividing. What most targeted agents do.
- **Cytotoxic** — kills the cell outright. What classical chemo does.
- **ADCC** — antibody-dependent cellular cytotoxicity (NK cell / macrophage does the killing)
- **CDC** — complement-dependent cytotoxicity (complement MAC does the killing)

## THE SIX TARGETED STRATEGIES

| STRATEGY                              | WHAT IT DOES                                 | EXAMPLE                |
|---------------------------------------|----------------------------------------------|------------------------|
| <b>Signal transduction inhibitors</b> | block proliferation/survival kinases         | imatinib               |
| <b>Angiogenesis inhibitors</b>        | starve the tumour's blood supply             | bevacizumab, sunitinib |
| <b>Immunotherapy</b>                  | recruit the immune system to kill the tumour | rituximab, ipilimumab  |
| <b>Apoptosis-inducing agents</b>      | push the cell into programmed death          | bortezomib             |
| <b>Hormonal therapies</b>             | remove the hormone driving the tumour        | tamoxifen              |
| <b>Gene expression modulators</b>     | re-open transcription → differentiation      | vorinostat, tretinoin  |

## SIGNAL TRANSDUCTION TARGETS ON THE EXAM LIST

| PATHWAY                            | INHIBITOR FAMILIES       |
|------------------------------------|--------------------------|
| <b>Receptor tyrosine kinase</b>    | EGFR · HER2 · ALK · NTRK |
| <b>Fusion / cytoplasmic kinase</b> | BCR-ABL · BTK · JAK      |
| <b>MAPK arm</b>                    | BRAF                     |
| <b>PI3K–AKT–mTOR arm</b>           | PI3K · mTOR              |
| <b>DNA repair / cell cycle</b>     | PARP · CDK4/6            |

# EGFR Inhibitors — Small-Molecule TKIs

## MECHANISM

**EGFR** (= ErbB1/HER1) is a transmembrane receptor tyrosine kinase driving epithelial growth, found at **abnormally high levels in lung and colorectal cancer**. The TKI sits in the receptor's **ATP-binding pocket**, so the kinase can't autophosphorylate its tyrosines.

→ **Downstream**: no receptor phosphorylation → **MAPK signalling off** → ↓ proliferation · ↓ cell-cycle progression · ↑ apoptosis · ↓ invasion & metastasis · ↓ angiogenesis

## DRUGS — GENERATIONS TRACK THE RESISTANCE MUTATION ★

| DRUG                                                          | WHAT IT TREATS                                                            |
|---------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Gefitinib, Erlotinib</b> ( <i>1st gen, reversible</i> )    | EGFR-mutant NSCLC — <b>L858R, Del19</b> activating mutations              |
| <b>Afatinib, Dacomitinib</b> ( <i>2nd gen, irreversible</i> ) | as above + <b>T790M, wild-type/mutant HER2, HER4</b>                      |
| <b>Osimertinib</b> ( <i>3rd gen, irreversible</i> )           | <b>T790M resistance</b> — L858R, Del19, T790M; little activity on wt-EGFR |
| <b>Mobocertinib</b>                                           | <b>exon 20 insertion</b> EGFR                                             |

*The lung-cancer mutations that matter: exon 19 deletion · exon 20 insertion · exon 21 L858R. 4th generation = allosteric inhibition, aimed at the C797S mutation that defeats osimertinib.*

## ADVERSE EFFECTS

- **Rash / acneiform dermatitis** ★ and dry skin — the class signature (EGFR is needed by normal skin)
- Stomatitis
- Diarrhea
- Nausea / vomiting
- **QT prolongation** — mobocertinib
- ➡ Most appear **within the first month**, and are tolerable with supportive meds + dose reduction

## INTERACTIONS

- Mostly **CYP3A4** substrates (gefitinib, erlotinib, osimertinib, mobocertinib) — 3A4 inhibitors/inducers shift exposure. **Dacomitinib is CYP2D6**; **afatinib** is not CYP-cleared (forms covalent adducts).
- Food: erlotinib ↑**AUC 34–66%** · osimertinib ↑AUC 19% · afatinib ↓**AUC 39%** with a high-fat meal
- All are heavily protein-bound (90–99%) and cleared mainly in **feces**

# EGFR Inhibitors — Monoclonal Antibodies

## MECHANISM

The antibody binds the **extracellular** EGFR domain (domain III of the ligand pocket) — the opposite end of the receptor from where the TKIs work.

→ **Three actions:** ① prevents ligand binding & receptor dimerization → no EGFR phosphorylation · ② causes **receptor internalization** → receptor down-regulation · ③ mediates **ADCC** against the tumour cell —  
★ **except panitumumab** (IgG2, poor Fc effector function)

## DRUGS

| DRUG                                      | WHAT IT TREATS                                                                      |
|-------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Cetuximab</b> ( <i>chimeric IgG1</i> ) | EGFR-driven solid tumours — binds domain III; <b>ADCC-capable</b>                   |
| <b>Panitumumab</b> ( <i>human IgG2</i> )  | EGFR-driven solid tumours — <b>no ADCC</b>                                          |
| <b>Necitumumab</b> ( <i>human IgG1</i> )  | targets the <b>S468R variant</b> — for cetuximab/panitumumab- <b>resistant</b> EGFR |

*All three are cleared by proteolytic cleavage, not CYP.  $t_{1/2} \approx 5\text{ d} \cdot 7.5\text{ d} \cdot 14\text{ d}$ .*

## ADVERSE EFFECTS

- **Dermatotoxicity** ★ — acneiform rash, dry skin, exfoliation, skin fissure. ⚠ **Sunlight worsens it.**
- **Infusion reaction** — flushing, fever, chills; severe: **anaphylaxis, bronchospasm, hypotension**
- **Hypomagnesemia** ★ — monitor electrolytes periodically
- ● **Cardiopulmonary arrest**

# How Therapeutic mAbs Actually Kill

## THE TWO IMMUNE MECHANISMS

**ADCC** — an effector cell (**NK cell, monocyte, macrophage**) recognises the antibody-coated tumour cell through its **Fc receptor (FcγRI/II/III)** → releases cytokines and cytotoxic granules → apoptosis.

**CDC** — complement **C1q** binds the antibody's **Fc domain** → complement cascade → **membrane attack complex (MAC)** → cell lysis.

→ **Downstream:** the isotype decides which one you get. **IgG1 and IgG3 have the highest CDC activity; IgG4 can't bind C1q at all**, so it does neither. ★

## WHY THE IGG SUBCLASS IS CHOSEN

|                               | IGG1  | IGG2  | IGG3       | IGG4  |
|-------------------------------|-------|-------|------------|-------|
| relative abundance            | 60%   | 32%   | 4%         | 4%    |
| serum t <sub>1/2</sub> (days) | 21–24 | 21–24 | <b>7–8</b> | 21–24 |
| complement activation         | ++    | +     | +++        | –     |
| binds Fc receptors            | +++   | +/-   | ++++       | ++    |

*IgG3's short half-life: IgG1 blocks FcRn-mediated rescue of IgG3, sending it to the lysosome.*

## WHY ANTIBODIES LAST WEEKS — FCRN RECYCLING

IgG survives 2–4 weeks because **FcRn binds it inside the acidic endosome (pH 6.0–6.5)** and **releases it at neutral pH (7.0–7.5)** back at the cell surface.

→ **Downstream:** IgG is pulled out of the degradation path and recycled — the reason mAb dosing is every few weeks, not daily.

## FOUR WAYS AN ANTIBODY KILLS A TUMOUR

| ROUTE                          | HOW                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------|
| <b>Unarmed mAb</b>             | blocks signalling directly                                                                   |
| <b>Radioimmunoconjugate</b>    | delivers a radioisotope — <sup>90</sup> Y-ibritumomab tiuxetan, <sup>131</sup> I-tositumomab |
| <b>Antibody-drug conjugate</b> | delivers a toxin — gemtuzumab ozogamicin                                                     |
| <b>Immune-mediated</b>         | ADCC / CDC, or <b>checkpoint blockade</b>                                                    |

## ADVERSE EFFECTS — CLASS-WIDE

- Allergic reactions — hives, itching
- Flu-like syndrome — chills, fatigue, fever, myalgia
- Nausea · diarrhea · skin rash
- ● **Infusion reactions** — severe allergy-like; ★ **premedicate** to block them
- Individual agents: heart problems, bleeding, cytopenia, skin problems

# HER2 Inhibitors

## MECHANISM

**HER2 amplification/overexpression** drives **breast and gastric cancer** — excess HER2 forces **homodimerization (HER2:HER2)** and **heterodimerization (HER2:HER3)**, firing a strong pro-tumorigenic cascade. Two ways to break it:

1. **Small molecules (TKIs)** — occupy the **ATP-binding pocket**, kill the kinase activity
  2. **mAbs** — block **dimerization** itself, from outside
- **Downstream:** RAS–ERK and PI3K–AKT off → ↓ survival, ↓ proliferation

## DRUGS

| DRUG                                                | WHAT IT TREATS                                                                                                       |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Trastuzumab</b> ( <i>mAb IgG1, IV</i> )          | HER2+ breast/gastric — blocks <b>ligand-independent</b> signalling · <b>ADCC</b> · prevents HER2 <b>ECD shedding</b> |
| <b>Pertuzumab</b> ( <i>mAb IgG1, IV</i> )           | HER2+ breast — blocks <b>ligand-dependent dimerization</b> · <b>ADCC</b>                                             |
| <b>Lapatinib</b> ( <i>TKI, reversible, oral</i> )   | HER1 + HER2                                                                                                          |
| <b>Neratinib</b> ( <i>TKI, irreversible, oral</i> ) | HER1, HER2, HER4                                                                                                     |
| <b>Tucatinib</b> ( <i>TKI, reversible, oral</i> )   | HER2, HER3                                                                                                           |

★ *Trastuzumab and pertuzumab are complementary — one blocks ligand-independent, the other ligand-dependent dimerization.*

## ADVERSE EFFECTS

- ● **Left ventricular dysfunction** ★ — the defining HER2-class toxicity
- ● **Embryo-fetal toxicity**
- Diarrhea · nausea / vomiting · rash
- Hepatotoxicity
- **Infusion reaction** — trastuzumab, pertuzumab (*the mAbs*)
- **QT prolongation** — lapatinib

## INTERACTIONS

- **Lapatinib and neratinib are CYP3A4 substrates; tucatinib is CYP2C8**
- Food: **lapatinib** ↑AUC 3–4× · **neratinib** ↑1–2× · **tucatinib** ↑1.5× — food effects here are large ⚠
- The mAbs are cleared by **proteolytic cleavage** — no CYP interactions (t<sub>1/2</sub> 12 d and 18 d)

## ADMINISTRATION — HER2 INHIBITORS

| DRUG        | ROUTE | METABOLISM           | T <sub>½</sub>                  | PROTEIN BINDING | FOOD EFFECT  |
|-------------|-------|----------------------|---------------------------------|-----------------|--------------|
| Trastuzumab | IV    | proteolytic cleavage | 12 days (500 mg dose)           |                 | —            |
| Pertuzumab  | IV    | proteolytic cleavage | 18 days                         |                 | —            |
| Lapatinib   | oral  | CYP3A4               | 24 h                            | > 99%           | ● ↑ AUC 3–4x |
| Neratinib   | oral  | CYP3A4               | ~14 h (~22 h active metabolite) | > 99%           | ↑ AUC 1–2x   |
| Tucatinib   | oral  | CYP2C8               | ~8.5 h                          | 97%             | ↑ AUC 1.5x   |

- **Lapatinib with food is a three-to-four-fold exposure change** — that is not a nuance, it is a different dose. The oral HER2 agents have to be taken the same way relative to food every single day.
- ★ **The antibodies are measured in days and the TKIs in hours** — which is why one is an infusion every few weeks and the other is a tablet every day.
- 💡 Tucatinib is the odd one out metabolically (*CYP2C8*, not *3A4*), so the interaction check that clears lapatinib and neratinib doesn't automatically clear it.

# BCR-ABL Inhibitors

## MECHANISM

The **Philadelphia chromosome (Ph+)** — a t(9;22) translocation fusing **BCR** to **ABL** — produces a **constitutively activated tyrosine kinase** that drives **CML**. Most common isoform: BCR exon 13/14 fused to ABL1 exon 2.

→ **Two ways to block it: ATP-competitive** inhibitors sit in the ATP pocket (all the classic TKIs) · **non-ATP-competitive** inhibitors bind the **myristoyl pocket** allosterically (asciminib) ★

## DRUGS

| DRUG                       | WHAT IT TREATS                                                                    |
|----------------------------|-----------------------------------------------------------------------------------|
| <b>Imatinib</b> (1st gen)  | CML — also PDGFR, SCF, c-KIT. Lead compound: <b>low potency, poor specificity</b> |
| <b>Nilotinib</b> (2nd gen) | CML — binds the <b>inactive</b> conformation                                      |
| <b>Dasatinib</b> (2nd gen) | CML — binds the <b>active</b> conformation; also SRC family                       |
| <b>Bosutinib</b> (2nd gen) | CML — binds <b>both</b> conformations                                             |
| <b>Ponatinib</b> (3rd gen) | resistant CML — very broad (VEGFR, PDGFR, FGFR, KIT, RET, FLT3)                   |
| <b>Asciminib</b> (4th gen) | resistant CML — <b>myristoyl pocket</b> , non-ATP-competitive                     |

★ *T315I is the gatekeeper mutation — it appears in the resistance list of every ATP-competitive agent. That's the whole reason ponatinib and asciminib exist.*

## ADVERSE EFFECTS

- **Myelosuppression** ★ — anemia, neutropenia, thrombocytopenia
- **Edema and fluid retention**
- ● **Cardiotoxicity** — LV dysfunction, congestive heart failure, cardiogenic shock
- ● **Embryo-fetal toxicity** — teratogenic, early fetal resorption, nonviable pups
- GI distress — nausea/vomiting, abdominal pain, diarrhea · rash · hepatotoxicity
- **QT prolongation** — nilotinib, dasatinib, asciminib
- **Thrombosis** — ponatinib ⚠ · **Hypertension** — ponatinib, asciminib

## INTERACTIONS

- **CYP3A4** is the shared route (imatinib, dasatinib, bosutinib, ponatinib, asciminib)
- Food effects are opposite ways and clinically real: **nilotinib** ↑**AUC 82%** and **bosutinib** ↑**1.7×** with a high-fat meal, but **asciminib** ↓**AUC 62%** ⚠



## ADMINISTRATION — BCR-ABL INHIBITORS AND FOOD

|                        | IMATINIB                               | NILOTINIB               | DASATINIB     | BOSUTINIB     | PONATINIB          | ASCIMINIB               |
|------------------------|----------------------------------------|-------------------------|---------------|---------------|--------------------|-------------------------|
| <b>Bioavailability</b> | ~98%                                   | 17–44%                  | not known     | ~33%          | not known          | ~73%                    |
| <b>Protein binding</b> | ~95%                                   | ~98%                    | ~96%          | ~96%          | ~99%               | 97.3%                   |
| <b>Metabolism</b>      | CYP3A4                                 | oxidation/hydroxylation | <b>CYP3A4</b> | <b>CYP3A4</b> | <b>CYP3A4</b>      | CYP3A4, UGT2B7, UGT2B17 |
| <b>t<sub>1/2</sub></b> | 18 h ( <i>40 h active metabolite</i> ) | ~17 h                   | <b>3–5 h</b>  | ~22.5 h       | ~24 h              | 5.2 h                   |
| <b>Elimination</b>     | faeces ~68%                            | faeces ~93%             | faeces ~85%   | faeces ~91%   | faeces 87%         | faeces (56.7% intact)   |
| <b>High-fat meal</b>   | no relevant effect                     | ● ↑ <b>AUC 82%</b>      | ↑ AUC 14%     | ↑ AUC 1.7×    | no relevant effect | ● ↓ <b>AUC 62.3%</b>    |

- **Nilotinib and asciminib move in opposite directions on the same meal** — nilotinib's exposure nearly doubles, asciminib's drops by nearly two-thirds. There is no class food rule here; each drug has its own, and getting it backwards is either a toxicity or a treatment failure.
- ★ **Everything is eliminated in the faeces** — so renal function is not the lever it is for most oncology drugs.
- 💡 Imatinib's ~98% bioavailability is why it is the forgiving one; bosutinib at ~33% and nilotinib at 17–44% have far less room for anything that interferes with absorption.

# mTOR Inhibitors

## MECHANISM

**mTOR** is a serine/threonine kinase downstream of **PI3K-AKT** (restrained by **PTEN** — whose loss is a common tumour event). The drug binds **FKBP12**, and that complex inhibits **mTORC1**.

→ **Downstream: G1/S block** (cell cycle stalls) · ↓ angiogenesis · ↑ apoptosis → **Resistance:** incompletely understood — may come through **mTORC2**, which these drugs don't hit

## DRUGS

| DRUG                         | WHAT IT TREATS                                                                                                 |
|------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Sirolimus (rapamycin)</b> | renal-transplant immunosuppression · lymphangioleiomyomatosis — the parent compound; IV and oral               |
| <b>Temsirolimus</b>          | ★ <b>advanced renal cell carcinoma</b> — IV                                                                    |
| <b>Everolimus</b>            | advanced renal cell carcinoma · HR+ breast cancer · neuroendocrine tumours · SEGA in tuberous sclerosis — oral |

*Class applications: renal cell and hepatocellular carcinoma, mantle cell lymphoma.*

## ADVERSE EFFECTS

- **Oral ulceration** (stomatitis) ★
- **Hyperlipidemia** and **hyperglycemia** — the metabolic pair
- Edema
- Anemia, thrombocytopenia

## INTERACTIONS

- All three are **CYP3A4** substrates; sirolimus and everolimus are also **P-gp substrates**
- ⚠ Take the oral form **consistently with or without food** — a high-fat meal shifts sirolimus AUC by 23–35%

## WORTH NOT CONFUSING

- ⚠ **Tacrolimus (FK506)** also binds **FKBP12** — but that complex inhibits **calcineurin**, not mTOR. Same partner protein, completely different target.

## ADMINISTRATION — MTOR INHIBITORS

|                        | SIROLIMUS                     | TEMSIROLIMUS | EVEROLIMUS                    |
|------------------------|-------------------------------|--------------|-------------------------------|
| <b>Route</b>           | <b>IV and oral</b>            | <b>IV</b>    | <b>oral</b>                   |
| <b>Metabolism</b>      | CYP3A4, <b>P-gp substrate</b> | CYP3A4       | CYP3A4, <b>P-gp substrate</b> |
| <b>t<sub>1/2</sub></b> | <b>~60 h</b>                  |              | ~17 h                         |

- ● **The oral dosage form should be taken consistently — with food or without, but always the same way.** A high-fat meal raises AUC by **23–35%**, so switching back and forth is what causes the swing, not the food itself.
- **Everolimus:** a high-fat meal **reduces systemic exposure by 22%**, though **food had no apparent effect on the post-absorption concentration–time profile.**
- 💡 "Consistently" is the whole instruction and it is easy to under-sell. These are trough-monitored drugs; a patient who takes it with breakfast one week and fasting the next produces levels nobody can interpret.

# BRAF Inhibitors

## MECHANISM

**BRAF** sits in the **MAP kinase pathway** (RTK → RAS → BRAF → MEK → ERK). The **V600E** mutation makes it constitutively active. The drug blocks mutant BRAF.

→ **Downstream:** MEK–ERK off → ↓ cellular proliferation → **Where it matters:** melanoma · colorectal cancer · NSCLC · multiple myeloma

## DRUGS

| DRUG               | WHAT IT TREATS                               |
|--------------------|----------------------------------------------|
| <b>Vemurafenib</b> | BRAF <b>V600E/V600D/V600R</b> mutant tumours |
| <b>Dabrafenib</b>  | BRAF <b>V600E/D/R/K</b> mutant tumours       |
| <b>Encorafenib</b> | BRAF <b>V600E/D/K</b> mutant tumours         |

*All three also touch wild-type BRAF and CRAF — which is why non-mutant tissue reacts.*

## ADVERSE EFFECTS

- **Dermatotoxicity** ★ — rash, **skin papilloma**, hyperkeratosis, **photosensitivity**
- Alopecia
- Hepatotoxicity
- ● **Embryo-fetal toxicity**
- **QT prolongation** — vemurafenib, encorafenib

## INTERACTIONS

- **CYP3A4** (vemurafenib, encorafenib); **dabrafenib is CYP2C8 + CYP3A4**
- ⚠ Food moves them in opposite directions: high-fat meal ↑ **vemurafenib AUC 5×** but ↓ **dabrafenib AUC 5×**

# PARP Inhibitors

## MECHANISM — SYNTHETIC LETHALITY ★

**PARPs** repair **single-strand breaks** via base-excision repair. Inhibit PARP → SSBs persist → they collapse into **double-strand breaks**, which normally get fixed by **homologous recombination (HR)**.

→ **The trick:** a cell with **BRCA1/2 mutation has no working HR**. PARP block + BRCA loss = neither repair route works → complex rearrangements → **cell death**. A normal (BRCA-intact) cell survives on HR. **Two individually survivable defects that are lethal together — synthetic lethality.**

## DRUGS

| DRUG               | WHAT IT TREATS                             |
|--------------------|--------------------------------------------|
| <b>Olaparib</b>    | HR-deficient / <b>BRCA-mutated</b> tumours |
| <b>Rucaparib</b>   | HR-deficient / BRCA-mutated tumours        |
| <b>Niraparib</b>   | HR-deficient / BRCA-mutated tumours        |
| <b>Talazoparib</b> | HR-deficient / BRCA-mutated tumours        |

## ADVERSE EFFECTS

- Shared by all four: **nausea · fatigue · vomiting · headache**
- ● **Serious hematologic toxicity (Grade ≥3) — anemia + neutropenia in every agent; ★**  
**thrombocytopenia is the niraparib and talazoparib one**
- Diarrhea · **dysgeusia** (olaparib, rucaparib)
- **LFT elevation** — rucaparib, niraparib
- **Hypertension** and insomnia — niraparib · **Alopecia** — talazoparib

## CAUTIONS

- ● **Avoid in pregnancy** — teratogenicity, embryo-fetal toxicity and death in animal reproduction studies

# CDK4/6 Inhibitors

## MECHANISM

The cell cycle (**G1** → **S** → **G2** → **M**) is driven by cyclin–CDK complexes. **Cyclin D–CDK4/6 phosphorylates retinoblastoma protein (Rb)**, which releases the brake on G1→S.

→ **Downstream:** block CDK4/6 → **Rb stays unphosphorylated** → the cell is held in **G1** → no proliferation

## USES

**Trilaciclib** — a deliberately different use of the same mechanism: given **before** a platinum/etoposide or topotecan regimen in **extensive-stage small cell lung cancer**, to **decrease chemotherapy-induced myelosuppression**. ★ Arresting the marrow in G1 shields it from chemo that only kills dividing cells.

## NOTE ON THIS PAGE

- ⚠ The lecture's CDK4/6 drug list, pharmacokinetics and adverse-effect slides are **images only** — no extractable text. Fill this in from the slide deck itself before the exam.

# Angiogenesis (VEGF) Inhibitors

## MECHANISM

**VEGF-A** and its receptors **VEGFR-1 / VEGFR-2** drive both physiological and pathological angiogenesis. Block either half of the pair:

- **Trap the ligand** — antibody or decoy receptor mops up VEGF-A
  - **Block the receptor** — antibody against VEGFR-2, or an ATP-competitive multikinase TKI
- **Downstream:** no new tumour vasculature → tumour is starved of supply → growth inhibited (*VEGFR-1 binds VEGF-A more tightly than VEGFR-2, but its kinase activity is ~10× lower.*)

## DRUGS

| DRUG                                                                | WHAT IT TREATS                                                      |
|---------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Bevacizumab</b> ( <i>humanized IgG1 vs VEGF-A</i> )              | <b>various cancers</b> — neutralises VEGF-A, t <sub>1/2</sub> ~20 d |
| <b>ziv-Aflibercept</b> ( <i>VEGFR1+2 domains fused to IgG1 Fc</i> ) | <b>wet AMD · diabetic macular edema</b>                             |
| <b>Faricimab</b> ( <i>bispecific IgG1: Ang-2 + VEGF-A</i> )         | <b>AMD · DME</b> — restores vascular stability                      |
| <b>Ramucirumab</b> ( <i>human IgG1 vs VEGFR-2</i> )                 | solid tumours — t <sub>1/2</sub> ~14 d                              |
| <b>Sunitinib</b> ( <i>multikinase TKI</i> )                         | VEGFR1-3, PDGFRα/β, KIT, FLT3, RET, CSF-1R                          |
| <b>Axitinib</b> ( <i>TKI</i> )                                      | VEGFR1-3 — the <b>selective</b> one                                 |
| <b>Lenvatinib</b> ( <i>TKI</i> )                                    | VEGFR1-3, FGFR1-4, PDGFRα, KIT, RET                                 |
| <b>Sorafenib</b> ( <i>multikinase TKI</i> )                         | c-CRAF, BRAF, KIT, FLT-3, RET, VEGFR1-3, PDGFR-β                    |

## ADVERSE EFFECTS — ALL FOLLOW FROM WRECKING VASCULATURE ★

- ● **Vessel injury and bleeding** — including **GI and colonic perforation**
- **Hypertension** — via ↓ nitric oxide production
- ● **Arterial thrombosis** — stroke, myocardial infarction
- Diarrhea · nausea/vomiting · abdominal pain · fatigue
- **Hepatotoxicity** — sunitinib
- **QT prolongation** — sunitinib, lenvatinib, sorafenib

## INTERACTIONS

- Every oral TKI here is **CYP3A4/3A**-cleared and >95% protein-bound; the antibodies are cleared by **proteolytic cleavage**
- **Sorafenib** — high-fat meal ↓ bioavailability 29%

## ADMINISTRATION — VEGF/VEGFR INHIBITORS

**Targeting VEGF-A (all injected):** bevacizumab  $t_{1/2}$  ~20 days · ziv-aflibercept ~5–6 days · faricimab ~7.5 days.

|                             | RAMUCIRUMAB          | SUNITINIB     | AXITINIB       | LENVATINIB   | SORAFENIB                      |
|-----------------------------|----------------------|---------------|----------------|--------------|--------------------------------|
| <b>Protein binding</b>      | n/a                  | ~95%          | > 99%          | 97–99%       | 99.5%                          |
| <b>Metabolism</b>           | proteolytic cleavage | <b>CYP3A4</b> | <b>CYP3A4</b>  | <b>CYP3A</b> | <b>CYP3A4</b>                  |
| <b><math>t_{1/2}</math></b> | ~14 days             | 40–60 h       | <b>2.5–6 h</b> | ~28 h        | 40–60 h                        |
| <b>Elimination</b>          | n/a                  | faeces        | faeces         | faeces       | faeces                         |
| <b>High-fat meal</b>        | none                 | none          | none           | none         | ● ↓ <b>bioavailability 29%</b> |

- ★ **Sorafenib is the only one in this row with a food effect** — and it loses nearly a third of its absorption to a fatty meal, so it is the one that needs a fixed relationship to meals.
- 💡 Axitinib's 2.5–6 hour half-life against sorafenib's 40–60 hours is the difference between a drug that needs twice-daily discipline and one that tolerates a late dose.

# Proteasome Inhibitors

## MECHANISM

The **proteasome** degrades the intracellular proteins that control cell cycle, apoptosis, DNA repair and stress response — including **NF-κB**. **Cancer cells run proteasome activity higher than normal differentiated cells**, which is what makes this selective.

**Bortezomib** binds the **β5 subunit of the 20S core** of the 26S proteasome and **reversibly** inhibits its **chymotrypsin-like activity**.

→ **Downstream:** NF-κB inhibited · ↑ pro-apoptotic proteins · impaired DNA-damage response · cell-cycle arrest · ↓ angiogenesis · **ER stress / unfolded protein response** → **apoptosis**

## ADVERSE EFFECTS

- Common (*each in 30–50% of patients*): mild **maculopapular rash, mucositis, anemia, fatigue**
- Less common: hyperglycemia, hypertriglyceridemia

## INTERACTIONS

- Eliminated by **CYP3A4 and CYP2D6**



# Cancer Vaccines

## MECHANISM

**Preventive vaccine** — prophylaxis, given **before** disease; carries specificity for an infectious agent and induces antibody production.

**Therapeutic vaccine** — treatment, given **after** disease appears; active immunotherapy. Two approaches: **autologous (personalized)** or **allogeneic**.

→ **Downstream:** the patient's own immune system is trained on a tumour antigen and does the killing

## DRUGS

| DRUG                                                    | WHAT IT TREATS                                                              |
|---------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Sipuleucel-T</b> ( <i>autologous, personalized</i> ) | <b>prostate cancer</b> — antigen is <b>prostatic acid phosphatase (PAP)</b> |

## ADVERSE EFFECTS

- Chills, fever, fatigue
- Headache, joint ache, back pain
- Nausea

# Immune Checkpoint Inhibitors

## MECHANISM

**Checkpoint proteins keep immune responses in check** — a set of inhibitory and stimulatory pathways on immune cells. Cancer exploits them to build an **immunosuppressive state that favours immune evasion**. Two axes matter:

- **PD-L1** on the tumour cell binding **PD-1** on the T cell
- **B7-1 / B7-2** on antigen-presenting cells binding **CTLA-4** on the T cell

→ **Downstream:** block the interaction → the brake comes off the T cell → it attacks the tumour. ★ **This drug doesn't touch the cancer cell — it releases the immune system.**

## DRUGS

| DRUG                                     | WHAT IT TREATS                                        |
|------------------------------------------|-------------------------------------------------------|
| <b>Ipilimumab</b> ( <i>anti-CTLA-4</i> ) | releases the <b>priming-phase</b> brake on T cells    |
| <b>anti-PD-1 antibodies</b>              | release the <b>effector-phase</b> brake at the tumour |
| <b>anti-PD-L1 antibodies</b>             | block the tumour's side of the same interaction       |

## ADVERSE EFFECTS

- Common: diarrhea · fatigue · cough · nausea · skin rash · poor appetite · constipation · muscle and joint pain
- **Infusion reaction** — fever, chills, facial flushing, rash, itch, dizziness, wheezing, trouble breathing
- ● **Immune-related adverse events (irAEs)** ★ — the class-defining toxicity: the unleashed immune system attacks **lungs, intestines, liver, hormone-making glands, and kidneys**. The price of taking the brake off is that it comes off everywhere.

# CAR T-Cell Therapy

## MECHANISM

A patient's own **T cells are removed and modified in the laboratory**: the gene for a **chimeric antigen receptor (CAR)** — engineered to bind a specific protein on that patient's cancer cells — is inserted, and the cells are given back.

→ **Downstream**: the returned T cells now recognise the tumour antigen directly and attack it. Classed as an **Advanced Therapy Medicinal Product (ATMP)** — a living drug, not a molecule.

# Anti-Estrogens — SERMs & SERDs

## MECHANISM

In **ER-positive breast cancer**, proliferation is **controlled by estrogen**. Two ways to cut it:

- **SERM (selective estrogen-receptor modulator)** — binds ER and acts **estrogenic in some organs, anti-estrogenic in others** ★ (this tissue-selectivity is the whole point, and the source of the ADRs)
  - **SERD (selective estrogen-receptor downregulator)** — a "**pure anti-estrogen**"; it degrades the receptor rather than just occupying it
- **Downstream:** ER signalling off → proliferation of the ER+ tumour is inhibited

## DRUGS

| DRUG                               | WHAT IT TREATS                                                               |
|------------------------------------|------------------------------------------------------------------------------|
| <b>Tamoxifen</b> ( <i>SERM</i> )   | ER+ breast cancer — the reference anti-estrogen                              |
| <b>Raloxifene</b> ( <i>SERM</i> )  | ER-mediated disease                                                          |
| <b>Fulvestrant</b> ( <i>SERD</i> ) | ER+ breast cancer — <b>pure anti-estrogen</b> , no agonist activity anywhere |

# Aromatase Inhibitors

## MECHANISM

**Aromatase (CYP19)** converts androgens into estrogens — in postmenopausal women this is the *only* remaining estrogen source. Block the enzyme and the supply is cut off upstream of the receptor.

→ **Downstream:** ↓ circulating estrogen → ER+ tumour loses its driver → **Why postmenopausal only:** ★  
premenopausal ovaries still make estrogen directly, so blocking aromatase doesn't remove the source.

## DRUGS

| DRUG                                                | WHAT IT TREATS                                                 |
|-----------------------------------------------------|----------------------------------------------------------------|
| <b>Exemestane</b> ( <i>Type I, steroidal</i> )      | ER+ breast cancer — <b>irreversibly inactivates</b> the enzyme |
| <b>Anastrozole</b> ( <i>Type II, nonsteroidal</i> ) | ER+ breast cancer — <b>reversible</b> binding                  |
| <b>Letrozole</b> ( <i>Type II, nonsteroidal</i> )   | ER+ breast cancer — reversible binding                         |

## USES

**Adjuvant treatment of postmenopausal women with hormone-positive breast cancer** — either as **initial therapy** or **sequenced after tamoxifen**

# Anti-Androgens

## MECHANISM

**Prostate cancer growth is androgen-driven.** These agents antagonise the androgen receptor.

→ **Downstream:** AR signalling blocked → tumour loses its growth signal

## DRUGS

| DRUG                | WHAT IT TREATS           |
|---------------------|--------------------------|
| <b>Flutamide</b>    | advanced prostate cancer |
| <b>Bicalutamide</b> | advanced prostate cancer |
| <b>Nilutamide</b>   | advanced prostate cancer |

## CAUTIONS

- ⚠ **Anti-androgen monotherapy is NOT indicated as first-line** treatment in advanced prostate cancer  
★

# Glucocorticoids

## MECHANISM

Not a targeted agent — a hormone used **cytotoxically**. Glucocorticoids are **lympholytic** and **suppress mitosis in lymphocytes**.

→ **Downstream:** lymphoid tumour cells are killed outright, which is why the indications are all lymphoid

## USES

**Acute leukemia in children · Malignant lymphoma** in children and adults

*The lecture groups progestins with these under "hormones and related agents".*

# Gene Expression Modulators

## MECHANISM

Also called **differentiating agents** — they don't kill the cell, they force it to grow up. **Histone deacetylase inhibitors (HDIs)** block HDAC, so histones stay **acetylated** and chromatin stays open.

→ **Downstream:** transcription resumes → **cell-growth arrest, differentiation and apoptosis**

## DRUGS

| DRUG                                       | WHAT IT TREATS                                                                                  |
|--------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Vorinostat</b> ( <i>HDI</i> )           | <b>cutaneous T-cell lymphoma (CTCL)</b> that persisted, progressed or recurred on other therapy |
| <b>Romidepsin</b> ( <i>HDI</i> )           | CTCL, same setting                                                                              |
| <b>Retinoids</b> ( <i>e.g. tretinoin</i> ) | differentiation therapy                                                                         |
| <b>Arsenic trioxide</b>                    | differentiation therapy                                                                         |



# Target Map

## WHICH TARGET, WHICH DRUG, WHICH TUMOUR TYPE

| TARGET              | DRUGS                                                                           | TUMOUR TYPE |
|---------------------|---------------------------------------------------------------------------------|-------------|
| <b>CD20</b>         | rituximab · <sup>90</sup> Y-ibritumomab tiuxetan · <sup>131</sup> I-tositumomab | hematologic |
| <b>CD33</b>         | gemtuzumab ozogamicin                                                           | hematologic |
| <b>CD52</b>         | alemtuzumab                                                                     | hematologic |
| <b>BCR-ABL</b>      | imatinib · dasatinib                                                            | hematologic |
| <b>Proteasome</b>   | bortezomib                                                                      | hematologic |
| <b>HER2</b>         | trastuzumab · lapatinib                                                         | solid       |
| <b>EGFR</b>         | cetuximab · panitumumab · gefitinib · erlotinib                                 | solid       |
| <b>VEGF / VEGFR</b> | bevacizumab · sorafenib · sunitinib                                             | solid       |

## THE SIX MECHANISMS, ONE DRUG EACH

| MECHANISM                                            | EXAMPLE                         |
|------------------------------------------------------|---------------------------------|
| Block signal transduction driving proliferation      | <b>imatinib, trastuzumab</b>    |
| Block growth of blood vessels to the tumour          | <b>bevacizumab, sunitinib</b>   |
| Modify proteins regulating gene expression           | <b>vorinostat, tretinoin</b>    |
| Deliver toxic molecules specifically to cancer cells | <i>antibody-drug conjugates</i> |
| Induce cancer cells to undergo apoptosis             | <b>bortezomib</b>               |
| Help the immune system destroy cancer cells          | <b>rituximab, ipilimumab</b>    |

Source: *Anticancer agents II.pdf* — Krisada Sakchaisri, Ph.D., PYCH Pharmacology II, 94 slides. Structures omitted.

# Immunomodulators

Tools for **resetting an imbalanced immune system** — hitting one specific point in the immune network per disease, and accepting the trade-off between efficacy and the risks of **infection, cancer and site-specific toxicity**. Push the response down (transplant, autoimmunity) or push it up (immunodeficiency, cancer).

**APC antigen presentation → Signal 1 TCR–MHC → Signal 2 CD28–B7 → calcineurin → NFAT → IL-2 → IL-2R → JAK/STAT → mTOR → T-cell clonal proliferation**

## TWO DIRECTIONS

- **Immunosuppressors** — reduce or inhibit the immune response. Organ transplantation, autoimmune disease, anything with an overactive immune system.
- **Immunostimulators** — augment it. **Specific** ones carry antigenic specificity (vaccines, any antigen); **non-specific** ones act regardless of antigen.

## IMMUNOSUPPRESSANTS — WHERE EACH CLASS HITS THE CHAIN

| CLASS                                  | TARGET                                                       |
|----------------------------------------|--------------------------------------------------------------|
| <b>Lymphocyte-depleting antibodies</b> | lyse the lymphocyte outright (CD52, CD3, thymocyte antigens) |
| <b>IL-2R antagonists</b>               | α-chain of the IL-2 receptor                                 |
| <b>Corticosteroids</b>                 | cytokine gene transcription (IL-1, IL-2, IL-6, TNF, IFN-γ)   |
| <b>Calcineurin inhibitors</b>          | calcineurin → NFAT → IL-2 production                         |
| <b>Antimetabolites</b>                 | purine synthesis → lymphocyte proliferation                  |
| <b>mTOR inhibitors</b>                 | signals <b>past</b> the IL-2 receptor                        |
| <b>Costimulation blocker</b>           | Signal 2 (B7–CD28)                                           |
| <b>JAK inhibitors</b>                  | JAK/STAT signal transduction                                 |

## THE TRANSPLANT TIMELINE

**Pre-transplant immunomodulation → induction antibody therapy → maintenance therapy → minimization**

# Induction Antibodies

## MECHANISM

Heavy immunosuppression in the **peri- and immediate post-transplant** window, to cut acute rejection and blunt memory cells primed against donor antigen. Two strategies:

1. **Depletional** — antibodies bind lymphocyte surface antigens (CD2, CD3, CD4, CD8, CD11a, CD18, CD25, HLA-DR; or CD52) → opsonization and cell lysis
2. **Non-depletional** — competitively block the **α-chain of the IL-2 receptor** on *activated* T cells  
→ **Downstream**: fewer circulating lymphocytes / no IL-2-driven clonal expansion → ↓ acute rejection

## DRUGS

| DRUG                              | WHAT IT TREATS                                                                                                   |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Rabbit ATG (thymoglobulin)</b> | induction · rejection — depletes <b>T</b> >> <b>B</b>                                                            |
| <b>Alemtuzumab</b>                | induction — anti- <b>CD52</b> , depletes <b>T = B = APC</b> , plus NK cells, monocytes, macrophages, eosinophils |
| <b>OKT3</b>                       | induction — depletional                                                                                          |
| <b>Basiliximab</b>                | induction — <b>non-depleting</b> chimeric anti-IL-2Rα                                                            |
| <b>Daclizumab</b>                 | induction — non-depleting                                                                                        |

## CHOOSING BETWEEN THEM

**High-risk recipient** → alemtuzumab or rATG · **Low-risk recipient** (older, history of HCV, HBV, HIV) → basiliximab · Basiliximab gives **equal graft and patient survival** to ATG but supports steroid-free / CNI-avoidance protocols less well · Lecture's graft-failure ordering: **alemtuzumab** << **rATG** << **basiliximab**

## ADVERSE EFFECTS

- **Infusion-related reactions** — reduced by pre-medication
- Alemtuzumab: neutropenia, thrombocytopenia, **autoimmunity**
- rATG: dose is adjusted daily on **WBC / platelet count** — marrow suppression is the limiter

## ADMINISTRATION — INDUCTION ANTIBODY DOSING

- **Rabbit ATG: 1.5 mg/kg IV daily**, *rounded to the nearest 25 mg vial* — **induction total 5 mg/kg; rejection 4–7 doses.** ● **The dose is re-decided every day against the WBC and platelet count.**
- **Alemtuzumab: 30 mg prior to surgery.**
- 💡 Rounding to the vial isn't sloppiness — the vials are 25 mg and there is no way to split one, so the arithmetic ends at the vial. Expect the written dose and the given dose to differ slightly.
- ⚠️ Infusion-related reactions are reduced by **pre-medication**, which is part of the order, not an optional extra.

# B-Cell Targeted Therapy (Desensitization)

## MECHANISM

For the **antibody problem** — a highly sensitized or ABO-incompatible recipient. Strip the antibody (plasmapheresis), then silence the cells making it:

- **IVIg** — feedback inhibition of plasma cells; combined with plasmapheresis it prevents plasma-cell overproduction
  - **Rituximab** — chimeric anti-**CD20**, present on **mature B cells** (*plasma cells questionable*)
  - **Bortezomib** — reversibly binds the chymotrypsin-like subunit of the **26S proteasome**; pro-apoptotic factors are no longer degraded, accumulate, and trigger caspase-mediated apoptosis in neoplastic cells that survive by suppressing exactly those pathways
- **Downstream:** ↓ donor-specific antibody → transplant becomes possible across an antibody barrier

## DRUGS

| DRUG                                             | WHAT IT TREATS                                                    |
|--------------------------------------------------|-------------------------------------------------------------------|
| <b>IVIg</b> ( <i>low dose + plasmapheresis</i> ) | desensitization — high-antibody recipient, ABO-incompatible donor |
| <b>Rituximab</b>                                 | B-cell depletion (anti-CD20)                                      |
| <b>Bortezomib</b>                                | plasma-cell directed                                              |

## ADVERSE EFFECTS

- **IVIg** — osmotic nephropathy, headache, thrombotic complications, aseptic meningitis, bronchospasm, hemolytic anemia
- **Rituximab** — infusion-related reactions, thrombocytopenia
- **Bortezomib** — hypotension, **peripheral neuropathy**, N/V/D, thrombocytopenia, neutropenia

## ADMINISTRATION — DESENSITIZATION DOSING

- **IVIg:** ★ **high dose 2 g/kg (max 140 g) · low dose 100 mg/kg (max 7 g).**
- **Rituximab:** 375 mg/m<sup>2</sup> IV infusion (max 750 mg).
- **Bortezomib:** 1.4 mg/kg × 4 doses.
- ★ **IVIg's two doses are two different strategies** — the **low dose is the one paired with plasmapheresis**, high dose is used alone. A twenty-fold difference is not a titration range.
- ⚠ **IVIg's osmotic nephropathy is a volume-and-rate problem**, which is why the maximum is capped in grams and not left as pure mg/kg.

# Corticosteroids

## MECHANISM

Touch nearly every facet of immune function — the **cell-mediated arm more than the humoral arm** — by inhibiting transcription of cytokine genes (**IL-1, IL-2, IL-6, TNF, IFN- $\gamma$** ) and inducing apoptosis in rapidly dividing leukocytes.

→ **Downstream:** both humoral and cellular immunity disrupted · inflammation inside the graft falls

## DRUGS

| DRUG                 | WHAT IT TREATS                                           |
|----------------------|----------------------------------------------------------|
| <b>Dexamethasone</b> | induction phase, IV, first post-op days                  |
| <b>Prednisone</b>    | maintenance — oral, tapered down over the following days |

## USES

**Induction** — reduce inflammation within the graft · **When no other immunosuppressant can be given** ·

★ Withdrawing prednisone **after >3 months** raises rejection: acute rejection at 1 year **9.8% → 30.8%**

## ADVERSE EFFECTS

- **Acute:** GI upset, insomnia, leukocytosis, mood changes
- **Chronic:** ● **new-onset diabetes after transplant (NODAT)**, weight gain, osteoporosis, hypertension, hyperlipidemia, cushingoid features, acne, hirsutism, impaired wound healing, cataracts, peptic ulcer disease, myopathy

# Calcineurin Inhibitors

## MECHANISM

Both drugs bind an **immunophilin** first, and it is the *complex* that inhibits calcineurin:

- **Cyclosporine** → **cyclophilin**
- **Tacrolimus** → **FK-binding protein**

Calcineurin can then no longer dephosphorylate **NFAT**, so NFAT is never transported into the nucleus.

→ **Downstream:** ↓ **IL-2** and other cytokines → the T cell never gets the order to expand ★ Tacrolimus is **20–30× more potent** than cyclosporine.

## DRUGS

| DRUG                | WHAT IT TREATS                                 |
|---------------------|------------------------------------------------|
| <b>Cyclosporine</b> | maintenance immunosuppression after transplant |
| <b>Tacrolimus</b>   | same — the more potent option                  |

## ADVERSE EFFECTS

- ● **Nephrotoxicity** ★ — the dose-limiting toxicity of the class
- **Neurotoxicity** — encephalopathy, tremor
- Hypertension · hyperkalemia · hypomagnesemia · hypophosphatemia
- Hyperlipidemia · NODAT · osteopenia
- **Cyclosporine only:** gingival hyperplasia, hirsutism, gout
- **Tacrolimus only:** alopecia, diarrhea

## INTERACTIONS

- Metabolized by hepatic **CYP450** — ⚠️ avoid **grapefruit and pomegranate juice**
- Cannot be taken with **calcium, magnesium, iron or antacids**

## CAUTIONS

- ⚠️ Ideally taken on an **empty stomach** — weigh that against compliance
- ⚠️ Target trough levels shift with time from transplant; higher doses may be needed in African-American patients

## ADMINISTRATION — TACROLIMUS

- **Adult dose: 0.1 mg/kg/day split twice daily.** Higher doses may be needed in African American patients.
- **Ideally taken on an empty stomach** — ⚠️ *but weigh that against compliance*; consistency matters more than the ideal.
- ● **Cannot be taken with calcium, magnesium, iron or antacids.**
- **Trough levels vary with how far out from the transplant the patient is** — the target moves, the drug doesn't.
- 💡 The rule that actually protects the graft is **same way every day**: same relationship to food, same time, away from the mineral supplements. An erratic empty stomach is worse than a consistent full one.

# Antimetabolites

## MECHANISM

Starve the lymphocyte of purines, so it cannot proliferate.

- **Azathioprine** — antagonizes purine metabolism → inhibits **B and T** lymphocyte proliferation
  - **Mycophenolic acid** — non-competitive, selective, reversible **IMPDH** inhibitor. IMPDH is the rate-limiting step for guanine nucleotides, and ★ **T and B cells depend on the de novo pathway** — that dependence is where the selectivity comes from.
- **Downstream:** no guanine nucleotides → no clonal expansion

## DRUGS

| DRUG                                     | WHAT IT TREATS                                                 |
|------------------------------------------|----------------------------------------------------------------|
| <b>Azathioprine</b>                      | maintenance immunosuppression — <b>no longer commonly used</b> |
| <b>Mycophenolic acid / mycophenolate</b> | maintenance, as the adjunct to a calcineurin inhibitor         |

## ADVERSE EFFECTS

- **Azathioprine** — bone marrow suppression, malignancy, rash, N/V/D, pancreatitis
- **Mycophenolate** — GI (N/V/diarrhea), bone marrow suppression, GI bleed, **CMV**, lymphoma

## INTERACTIONS

- ● **Azathioprine + allopurinol** ★ — the interaction that matters
- Mycophenolate: better tolerated **with food**; not with calcium, magnesium, iron or antacids

## ADMINISTRATION — MYCOPHENOLATE

- **Rapidly and completely absorbed**, with **enterohepatic recycling**.
- ★ **Better tolerated with food** — the opposite instruction to tacrolimus, on the same prescription.
- ● **Cannot be taken with calcium, magnesium, iron or antacids.**
- **Trough levels are rarely used** (*the APOMYGRE study showed benefit, but it isn't routine*).
- 💡 A transplant patient on both drugs is being told "this one with food, that one without" — worth saying out loud, because the two tablets look like one regimen to them.

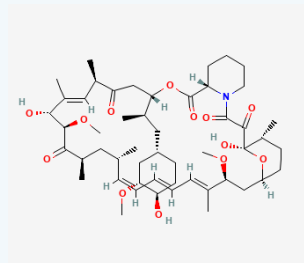
# mTOR Inhibitors

## MECHANISM

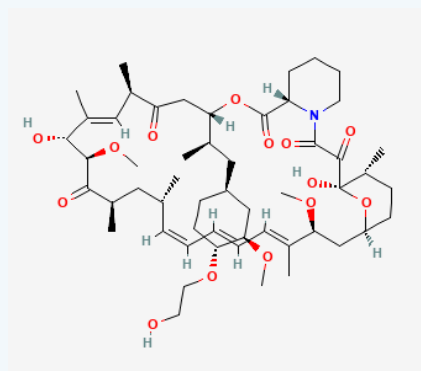
Bind **FK-binding protein** — the same immunophilin tacrolimus uses — but the complex inhibits **mTOR** instead of calcineurin, blocking the intracellular signals **downstream of the IL-2 receptor**.

→ **Downstream:** the T cell sees IL-2 and can do nothing with it → no proliferation → **Different toxicity profile from the CNIs:** marrow and wound problems instead of nephrotoxicity

## THE SHARED CORE · WHAT CHANGES



**Sirolimus (rapamycin).** a large macrolide that must first bind FKBP12 — the DRUG-PROTEIN pair is what inhibits mTOR, not the drug alone



**Everolimus**

one hydroxyethyl group added → better oral bioavailability and a much shorter  $t_{1/2}$

## DRUGS

| DRUG              | WHAT IT TREATS                |
|-------------------|-------------------------------|
| <b>Sirolimus</b>  | maintenance immunosuppression |
| <b>Everolimus</b> | maintenance immunosuppression |

## PLACE IN THERAPY

Reduced use as **initial** maintenance — sirolimus had higher rejection and treatment-failure rates, everolimus higher rejection and more discontinuation for adverse effects · ★ but **lower rates of malignancy**

## ADVERSE EFFECTS

- Myelosuppression — thrombocytopenia, anemia, leukopenia
- ● **Delayed wound healing** (lymphocele) · **delays recovery from acute tubular necrosis**
- Interstitial pneumonitis
- Proteinuria · thrombotic microangiopathy (TMA)
- NODAT · hyperlipidemia · hypokalemia



# Costimulation Blocker

## MECHANISM

A **fusion protein** that antagonizes the **B7 ligands** — **B7-1 (CD80)** and **B7-2 (CD86)** — on APCs, the ligands that normally engage **CD28** on naïve T cells. That is "**Signal 2**".

→ **Downstream:** no costimulation → **no IL-2 release** → the naïve T cell never activates → ⚠ **The gap:** it does **not** eliminate or inhibit **memory T cells**

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                                 |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Belatacept</b> | maintenance immunosuppression in renal transplant — the first <b>intermittent-infusion</b> maintenance therapy |

## USES

May have a role in **non-compliant patients** (*an infusion, not a daily tablet*) · May allow **CNI-free and steroid-free** protocols

## ADVERSE EFFECTS

- **Post-transplant lymphoproliferative disorder (PTLD)** ★ — including CNS involvement
- Infusion reactions
- Aphthous stomatitis
- Proteinuria

# JAK Inhibitors

## MECHANISM

Cytokines don't each get their own pathway — a handful of **Janus kinase** pairs carry all of them, and six **STAT** members then drive transcription. **Tofacitinib** is a small synthetic molecule (~300× smaller than a biologic like etanercept) that sits **reversibly in the ATP site** of the kinase.

→ **Downstream:** one drug damps **many cytokines at once** — T cells, B cells and macrophages all affected. The goal in RA is restoring immune **homeostasis**, not frank immunosuppression.

## WHICH JAK PAIR CARRIES WHAT

| JAK PAIR    | CYTOKINES                                               |
|-------------|---------------------------------------------------------|
| JAK1 + JAK3 | <b>γc family</b> — IL-2, IL-4, IL-7, IL-9, IL-15, IL-21 |
| JAK1 + JAK2 | IL-6, IFN-γ                                             |
| JAK2 + JAK2 | EPO, TPO, GM-CSF                                        |
| JAK2 + TYK2 | IL-12, IL-23                                            |
| JAK1 + TYK2 | type I interferons, IL-10                               |

## DRUGS

| DRUG               | WHAT IT TREATS       |
|--------------------|----------------------|
| <b>Tofacitinib</b> | rheumatoid arthritis |

# Maintenance Regimens

## HOW A REGIMEN IS BUILT

**2 to 3 agents**, combined and then weaned as soon as possible:

| SLOT             | OPTIONS                                             |
|------------------|-----------------------------------------------------|
| <b>Primary</b>   | a calcineurin inhibitor — cyclosporine, tacrolimus  |
| <b>Adjunct</b>   | an antimetabolite — azathioprine, mycophenolic acid |
| <b>Sometimes</b> | a corticosteroid                                    |

★ The reason for combining is **non-overlapping toxicity**, not additive potency.

## DOSE-LIMITING TOXICITY

| DRUG                                 | DOSE-LIMITING TOXICITY                           |
|--------------------------------------|--------------------------------------------------|
| <b>Cyclosporine &amp; tacrolimus</b> | nephrotoxicity, neurotoxicity, dyslipidemia      |
| <b>Sirolimus</b>                     | myelosuppression, hepatic toxicity, dyslipidemia |
| <b>Mycophenolate</b>                 | GI irritation, myelosuppression                  |

## CAUTIONS

- **All of these drugs increase the risk of infection and lymphoma** ★ — the class-wide price
- ⚠ Wean steroids early — but withdrawal after >3 months raises rejection risk

# Immunostimulants

## TWO KINDS

- **Specific** — provide **antigenic specificity** in the immune response: vaccines, or any antigen
- **Non-specific** — act **irrespective of antigenic specificity**, augmenting the response to another antigen or stimulating a component of the immune system

## THE CATEGORIES

| CATEGORY                      | EXAMPLES                                                    |
|-------------------------------|-------------------------------------------------------------|
| <b>Vaccine</b>                | live and non-live                                           |
| <b>Immunoglobulin</b>         | IVIG, specific immunoglobulins                              |
| <b>Myeloid growth factors</b> | G-CSF, GM-CSF, erythropoietins                              |
| <b>Interferons</b>            | IFN- $\alpha$ , IFN- $\beta$ , IFN- $\gamma$                |
| <b>Topical TLR agonist</b>    | imiquimod                                                   |
| <b>Others</b>                 | immunomodulatory imides, checkpoint inhibitors, CAR T-cells |

# Vaccines

## MECHANISM

**Live** vaccines replicate in the host — higher-intensity innate response, more antigen, prolonged antigen persistence. **Non-live** vaccines can't do that, so they need **adjuvants** and **boosters**.

→ **Downstream:** live → higher antibody titre of longer duration, usually immunity from a **single dose**; non-live → multiple doses, and boosters for long-term immunity

## WHAT DETERMINES THE RESPONSE

| FACTOR                        | EFFECT                                                                                                             |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Live vs non-live</b>       | live: higher Ab, longer duration, often one dose · non-live: adjuvant + boosters                                   |
| <b>Polysaccharide antigen</b> | <b>T-cell independent</b> — lower Ab, shorter duration, ★ <b>no booster response</b>                               |
| <b>Protein antigen</b>        | varies with the antigen and its dose (e.g. TT > DT)                                                                |
| <b>Age</b>                    | immature system can't respond to PS antigen · maternal Ab inhibits · conjugates need more doses in infants         |
| <b>Route</b>                  | most non-live must be <b>IM or ID</b> (enough APCs there); mucosal is hard — exception <b>oral cholera vaccine</b> |

## WHAT SETS A SCHEDULE

**Immunological** — minimum age for a response, number of doses, interval between them · **Epidemiological** — susceptibility, disease severity and mortality · **Programmatic** — deliver alongside other interventions, fewer required contacts · **Safety**

## THE SCHEDULING TRADE-OFF

- ★ Aim to protect **before** the most vulnerable age. Starting late gives a **higher response but misses the window**; wider intervals give a **better response but leave a gap**.
- ⚠ Disease epidemiology differs between populations — a schedule that works in one is not the best for another
- ⚠ Special populations: **pregnant women, immunocompromised persons**
- Route, combined vaccines (DTP + Hib + HBV + IPV), timing and spacing, and post-vaccination adverse reactions all have to be accounted for
- Vaccines seldom produce sustained **mucosal** immunity — natural boosting from mucosal infection fills the gap

# Intravenous Immunoglobulin (IVIg)

## MECHANISM — IMMUNOMODULATORY

**Polyclonal human IgG with no specific epitope.** In autoimmune and inflammatory disease it works through Fc, not antigen:

- **FcγR blockade** — saturates activating FcγRs on macrophages → less phagocytosis of opsonized cells (*platelets in ITP*); **upregulates the inhibitory FcγRIIB** on B cells
- **Anti-idiotypic antibodies** neutralize pathogenic autoantibodies (*anti-AChR in MG, anti-platelet in ITP*)
- **Complement inhibition** — binds C3b and C4b → blocks the classical cascade
- **Cytokine shift** — ↓ TNF-α, IL-1β, IL-6, IFN-γ · ↑ IL-10 · ↓ Th1 and Th17, ↑ **Tregs** · inhibits dendritic-cell maturation
- **FcRn saturation** — pathogenic IgG is catabolized instead of recycled

→ **Downstream:** autoantibody scavenged and cleared by the RES, autoreactive B cells suppressed, tissue protected from complement-mediated injury

## USES

**Autoimmune / inflammatory** — ITP · Kawasaki disease · myasthenia gravis · Guillain-Barré ·

**Immunodeficiency** — primary immunodeficiency · pediatric HIV · **Hematology / transplant** — allogeneic bone marrow transplant · CLL · kidney transplant with a high-antibody recipient or an ABO-incompatible donor

## ADVERSE EFFECTS

- **Osmotic nephropathy**
- Thrombotic complications
- Aseptic meningitis
- Headache
- Bronchospasm
- Hemolytic anemia

# Specific Immunoglobulins

## MECHANISM — PASSIVE IMMUNITY

**Preformed IgG**, handed over ready-made. Used in immunodeficiency or to block an infection that has already been contracted.

- **Neutralization** of viruses, bacteria and toxins (*CMV, RSV, measles, tetanus*)
  - **Opsonization** — IgG coats the pathogen → phagocytes recognize and clear it
  - **ADCC** — coated cells are destroyed by NK cells and macrophages via FcγR engagement
  - **Toxin binding** — neutralizes bacterial superantigens and exotoxins (*streptococcal toxic shock*)
- **Downstream:** immediate protection with no immunological memory — it wears off

## DRUGS

| DRUG                                                             | WHAT IT TREATS                                                                                                                                                                                                                          |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rabies immunoglobulin</b> ( <i>ERIG equine / HRIG human</i> ) | post-exposure prophylaxis — transdermal bites or scratches, licks on broken skin. Infiltrated <b>around the bite site</b> , given with the vaccine                                                                                      |
| <b>Rho(D) immunoglobulin</b>                                     | prevention of <b>Rh hemolytic disease of the newborn</b> — Rho(D)-negative mother with an Rho(D)-positive baby, or after miscarriage / ectopic / abortion. Given to the <b>mother</b> at birth. ★ Classed as an <b>immunosuppressor</b> |
| <b>Tetanus immunoglobulin</b>                                    | tetanus                                                                                                                                                                                                                                 |
| <b>CMV immunoglobulin</b>                                        | cytomegalovirus                                                                                                                                                                                                                         |
| <b>Anti-snake venom immunoglobulin</b>                           | envenomation                                                                                                                                                                                                                            |
| <b>Anti-thymocyte immunoglobulin</b>                             | transplant induction and rejection                                                                                                                                                                                                      |

# Myeloid Growth Factors & Erythropoietins

## MECHANISM

Activate the **JAK/STAT** receptor on **myeloid progenitor cells**.

→ **Downstream:** ↑ proliferation and differentiation · activate phagocytic activity of mature neutrophils · prolong neutrophil survival · mobilize hematopoietic stem cells · **GM-CSF also** stimulates erythroid and megakaryocyte progenitors

## DRUGS

| DRUG                        | WHAT IT TREATS                                                                                                                                                                                                               |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Filgrastim (rhG-CSF)</b> | neutropenia from myelosuppressive chemotherapy, bone marrow transplant, severe chronic neutropenia · mobilizing peripheral blood progenitor cells (PBPC) · neutropenia in HIV patients on antiviral / myelosuppressive drugs |
| <b>rhGM-CSF</b>             | neutropenia after BMT and antiviral therapy for AIDS-related CMV · failed BMT or delayed engraftment · PBPC mobilization and post-transplant marrow regeneration                                                             |
| <b>Epoetin alfa</b>         | anemia of renal failure or cancer chemotherapy                                                                                                                                                                               |
| <b>Darbepoetin alfa</b>     | same — long-acting, ~3× <b>epoetin</b>                                                                                                                                                                                       |

## ADVERSE EFFECTS

- **Flu-like symptoms**
- **Bone and muscle pain**
- Erythropoietins: hypertension, headache; hypersensitivity is rare



# Interferons & Imiquimod

## MECHANISM

Interferons act on the **JAK/STAT receptor superfamily** on immune cells, switching on **interferon-stimulated genes (ISGs)** — the antiviral and antiproliferative program.

**Imiquimod** is a topical cream that activates **TLR-7** on **Langerhans cells** → ↑ local **IFN-α** release.

→ **Downstream:** ISG-driven antiviral state · imiquimod turns an interferon response on *only where you put it*

## TWO TYPES, THREE FAMILIES

| TYPE    | MEMBERS      | MADE BY                                                                                         |
|---------|--------------|-------------------------------------------------------------------------------------------------|
| Type I  | IFN-α, IFN-β | induced by viral infection — leukocytes make IFN-α; fibroblasts and epithelial cells make IFN-β |
| Type II | IFN-γ        | T cells                                                                                         |

## DRUGS

| DRUG                                | WHAT IT TREATS   |
|-------------------------------------|------------------|
| <b>Imiquimod</b> ( <i>topical</i> ) | wart · molluscum |

## ADVERSE EFFECTS

- **Interferons** — flu-like symptoms (fatigue, myalgia, headache, fever, chills)
- **Depression**
- **Myelosuppression**
- **Imiquimod** — non-specific inflammation at the application site

## ADMINISTRATION — IMIQUIMOD

- **Cream for local application**, for **warts and molluscum**.
- 💡 It works by provoking a local immune response, so **the inflammation at the site is the drug working**, not a reaction to stop for — patients need telling, or they abandon it in the first week.

# Immunomodulatory Imides

## MECHANISM

**Lenalidomide** binds **cereblon** and drives degradation of the transcription factors **IKZF1/IKZF3**, shutting down **IRF4 / c-Myc** survival signaling in myeloma cells.

→ **Downstream:** direct cell-cycle arrest and apoptosis of myeloma plasma cells · **T- and NK-cell activation** with Th1 cytokines → immune-mediated myeloma killing · inhibits pro-inflammatory cytokines and angiogenesis in the bone-marrow milieu

## DRUGS

| DRUG                | WHAT IT TREATS                                                                |
|---------------------|-------------------------------------------------------------------------------|
| <b>Thalidomide</b>  | ● <b>withdrawn — birth defects</b>                                            |
| <b>Lenalidomide</b> | multiple myeloma · AML — the <b>non-teratogenic</b> derivative of thalidomide |

# Immune Checkpoint Inhibitors

## MECHANISM

The cancer-immunity cycle (*antigen release → presentation → priming & activation → trafficking → tumor infiltration → recognition → killing*) has **two brakes, in two different places**:

- **CTLA-4** — **priming phase**, in the **lymph node**. CTLA-4 competes with CD28 for B7-1/B7-2 on the APC. Blocking it → T-cell activation and proliferation, and a **de novo memory-based** antitumor response.
  - **PD-1 / PD-L1, PD-L2** — **effector phase**, in the **tumor microenvironment**. The tumor displays PD-L1 to switch off the cytotoxic T cell that already found it.
- **Downstream**: release the brake → cytotoxic T cells kill tumor cells — ● **and cause autoimmunity by exactly the same mechanism**

## ADVERSE EFFECTS — IRAES

- ● **Immune-related adverse events** ★ — the "*Achilles heel*" of checkpoint inhibitors: toxicity from dysregulating the host immune system, behaving like autoimmune disease
- **Dose-dependent**, most appear during the initial dosing period — **median onset 5–9 weeks**
- Four major categories: **dermatologic · gastrointestinal · endocrine · hepatic**
- Graded by CTCAE severity, **1 (mild) → 5 (death)**
- ★ irAE development **correlates with treatment response**

## MANAGING IRAES

- Most are **reversible with corticosteroids** — high-dose IV, then taper over **1–3 months**
- Persistent diarrhea / colitis → add **infliximab**
- ⚠ **Exception**: adrenal insufficiency and hypothyroidism need **replacement** hydrocortisone and levothyroxine — *not* immunosuppressive steroid doses
- ★ No evidence that stepping in with steroids curtails the drug's antitumor efficacy

# CAR T-Cell Therapy

## MECHANISM

The patient's own T cells are collected, engineered to express a **chimeric antigen receptor** — an artificial TCR — and infused back. *"A living drug."*

**Construct:** extracellular **scFv** target element + spacer → transmembrane domain → intracellular signaling domain (**co-stimulatory domain** + **CD3ζ activating domain**).

→ **Downstream:** ★ tumor recognition is **HLA-independent** — no HLA typing needed, and it survives the escape routes of MHC and endogenous antigen downregulation

## THE FOUR GENERATIONS

| GENERATION          | WHAT WAS ADDED                                                   | RESULT                                              |
|---------------------|------------------------------------------------------------------|-----------------------------------------------------|
| <b>1st</b>          | —                                                                | poor antitumor effect, short survival               |
| <b>2nd</b>          | one co-stimulatory signal domain                                 | better overall survival, proliferation, persistence |
| <b>3rd</b>          | two co-stimulatory signal domains                                | more cytotoxicity and durability                    |
| <b>4th (TRUCKs)</b> | 2nd gen + gene cassettes for cytokines or co-stimulatory ligands | long-term antitumor effect, high cytokine level     |

## DRUGS

| DRUG                           | WHAT IT TREATS                                                                                                                                                    |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tisagenlecleucel</b>        | CD19-directed, <b>lentiviral</b> , 2nd gen — B-cell precursor <b>ALL</b> , refractory or in second relapse, patients up to 25 years old                           |
| <b>Axicabtagene ciloleucel</b> | CD19-directed, <b>retroviral</b> , 2nd gen — relapsed <b>large B-cell lymphoma</b> (incl. DLBCL) after ≥2 prior treatments. ⚠ <b>Not</b> for primary CNS lymphoma |

## READING THE STEM

**-gen** = gene transfer of genetic material (transduced) · **-lec** = selection and enrichment manipulation · **-leu** = white blood cells · **-cel** = cellular therapy

## USES

Highly effective against **refractory ALL, CLL and DLBCL** — **70–90% complete response rates**, with long-term recognition

## ADVERSE EFFECTS

- **Cytokine release syndrome (CRS)** ★ — hyperimmune activation releasing multiple cytokines: high fever, hypotension, difficulty breathing; can involve **hepatic, renal and cardiac dysfunction and coagulopathy**
- **Neurological toxicity** — headache, encephalopathy, delirium, anxiety, tremor
- **Cross-reaction** — killing of **normal B cells** → hypogammaglobulinemia, infection
- Pyrexia, decreased appetite, hypotension, nausea, diarrhea, vomiting
- Resistance through **CD19 loss**

## CAUTIONS

- **Do not administer to a patient with active infection or an inflammatory disorder**
- ⚠ **Avoid corticosteroids** around the infusion — except in a life-threatening emergency
- Requires **lymphodepleting chemotherapy** (fludarabine + cyclophosphamide) before infusion
- **Not universal** — must be customized for each individual, and the cost is high

## ADMINISTRATION — CAR T-CELL PRODUCTS, DOSE AND THE PROTOCOL AROUND IT

**Tisagenlecleucel** — dose is by weight: -  $\leq 50$  kg:  $0.2\text{--}5.0 \times 10^6$  CAR-positive viable T-cells per kg -  $> 50$  kg:  $0.1\text{--}2.5 \times 10^8$  CAR-positive viable T-cells (*not per kg*) - Infusion bag **10–50 mL**, given **within 30 minutes of thawing** at **10–20 mL/min**. - **Half-life 16.8 days in responders vs 2.52 days in non-responders** — ★ the cells persisting *is* the response.

**Axicabtagene ciloleucel** —  $2 \times 10^6$  CAR-positive viable T-cells per kg, max  $2 \times 10^8$  for patients  $\geq 100$  kg; approximately **68 mL** suspension in one bag, infused **entirely within 30 minutes** by gravity or peristaltic pump. Peak CAR T-cell levels **7–14 days** after infusion.

The protocol around both infusions:

| STEP                         | WHAT                                                                                 | WHEN                                                                                                                                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lymphodepleting chemo</b> | <b>fludarabine 30 mg/m<sup>2</sup> IV + cyclophosphamide 500 mg/m<sup>2</sup> IV</b> | tisagenlecleucel: cyclophosphamide for <b>2 days</b> starting with the first fludarabine dose, infuse cells <b>2–14 days</b> after completion · axicabtagene: <b>5th, 4th and 3rd day before</b> infusion |
| <b>Pre-medication</b>        | <b>acetaminophen 650 mg PO + diphenhydramine 12.5 mg IV or PO</b>                    | <b>30–60 min before</b> (tisagenlecleucel) · <b>1 hour before</b> (axicabtagene)                                                                                                                          |

- **Corticosteroids** **do not use at any time** except a life-threatening emergency; avoid prophylactic use —
- **The steroid ban is the counter-intuitive one.** Everywhere else in this file corticosteroids are the reflex for an infusion reaction — here they would kill the cells that are the treatment.
- 💡 The 30-minute windows (*thaw → infuse, and the whole bag in*) are why this is a scheduled, staffed procedure and not a drug that waits on a pharmacy shelf.

*Principle behind this: [[pharm-pharmacokinetics]] — monoclonal antibodies bypass phase I/II entirely — adalimumab is internalised by receptor-mediated endocytosis and degraded inside the target cell.*

*Principle behind this: [[pharm-pharmacogenomics]] — Azathioprine and TPMT — and the reason a normal immunosuppressant dose behaves like chemotherapy in a deficient patient.*

*Source: Final\_P'Co3\_Immunomodulators\_2026.pdf — อ.อนันต์ชัย (Anunchai Assawamakin, Ph.D.), Pharmacology II, 117 slides. Structures omitted.*

# Pharmacology 3

ภกปว 401 · ภาคต้น · CNS & neuropsychiatry — psychiatry → neurology



# Drugs Acting on GABA-A Receptor

**GABA** is the major **inhibitory** amino-acid neurotransmitter of the CNS — made presynaptically from **glutamate**, released by interneurons distributed throughout the brain. Every drug on this page works by making that inhibition stronger, and almost none of them do it by binding where GABA binds.

**GABA binds GABA-A → Cl<sup>-</sup> channel opens → Cl<sup>-</sup> influx → neuron hyperpolarized → CNS depression**

## TWO GABA RECEPTORS — ONLY ONE IS THE TARGET HERE

- **GABA-A** — a **ligand-gated Cl<sup>-</sup> ion channel**. Heteropentamer (5 subunits). ★ *This page's target.*
- **GABA-B** — a **GPCR (Gi)**. Not addressed by these drugs.

**APL / PAM** — *allosterically potentiating ligand / positive allosteric modulator*: binds a site that is **not** the GABA site and amplifies whatever GABA is already doing. No GABA, no effect.

## WHO BINDS WHERE ON GABA-A

| BINDING SITE                                    | LIGANDS                                                            |
|-------------------------------------------------|--------------------------------------------------------------------|
| <b>Benzodiazepine (BZ) site</b> — α/γ interface | benzodiazepines · zolpidem · <b>flumazenil</b> ( <i>antidote</i> ) |
| <b>Barbiturate site</b>                         | barbiturates                                                       |
| <b>Neurosteroid site</b> — β subunit of BZ2     | allopregnanolone · etifoxine                                       |
| ( <i>not detailed in this lecture</i> )         | ethanol · chloral hydrate · propofol                               |

## A SUBUNIT — ABUNDANCE AND BZP SENSITIVITY

| A SUBUNIT | % OF CNS GABA-A RECEPTORS | BZP SENSITIVE |
|-----------|---------------------------|---------------|
| <b>α1</b> | <b>60</b>                 | Yes           |
| α2        | 15–20                     | Yes           |
| α3        | 10–15                     | Yes           |
| α4        | <5                        | <b>No</b>     |
| α5        | <5                        | Yes           |
| α6        | <5                        | <b>No</b>     |

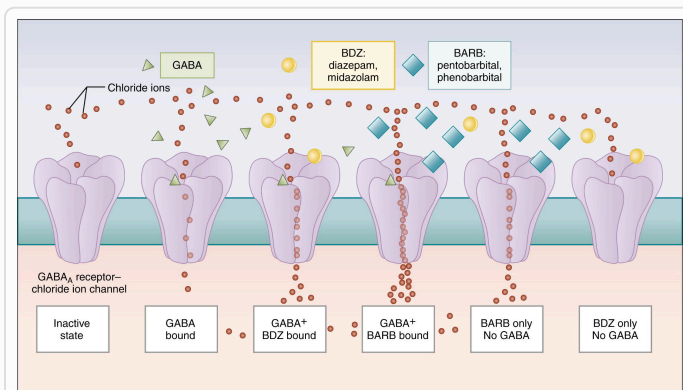


## ★ BZ1 VS BZ2 — THE SUBTYPE SPLIT THAT EXPLAINS EVERYTHING

|            | BZ TYPE I (BZ1 / $\Omega$ 1)                                                   | BZ TYPE II (BZ2 / $\Omega$ 2)                                                                                  |
|------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| definition | <b>contains <math>\alpha</math>1</b>                                           | does <b>not</b> contain $\alpha$ 1                                                                             |
| mediates   | sedation · anticonvulsive · <b>anterograde amnesia</b>                         | anxiolysis ( $\alpha$ 2) · myorelaxation ( $\alpha$ 2, $\alpha$ 3, $\alpha$ 5) · pain inhibition ( $\alpha$ 2) |
| tolerance  | <b>more likely</b>                                                             | less likely                                                                                                    |
| hit by     | BZPs · barbiturates ( <i>both</i> ) · <b>zolpidem</b> ( <i>this one only</i> ) | BZPs · barbiturates                                                                                            |

*This is why zolpidem sedates without much anxiolysis or muscle relaxation — and why tolerance to sedation appears while anxiolysis holds.*

### OVERVIEW MAPS



Week01 Pharmacology of Drugs acting on GABAA receptor · fig 16

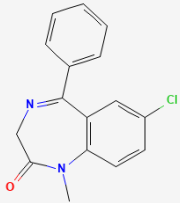
# Benzodiazepines

## MECHANISM

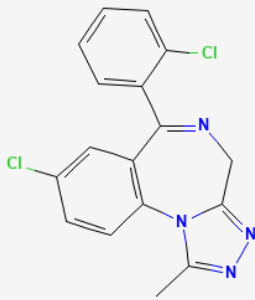
Bind the **benzodiazepine (BZ) site** at the  **$\alpha/\gamma$  interface** — not the GABA site — and act as an **APL / PAM** of GABA-A. They amplify GABA; they cannot open the channel by themselves.

→ **Downstream:**  $\uparrow$  GABA-gated  $\text{Cl}^-$  conductance → CNS depression at **both** BZ1 and BZ2 → sedation/hypnosis · anxiolysis · muscle relaxation · anti-seizure · **anterograde amnesia** · impaired motor performance ·  $\downarrow$  sensitivity to  $\text{CO}_2$  · **REM sleep suppression**

## THE SHARED CORE · WHAT CHANGES

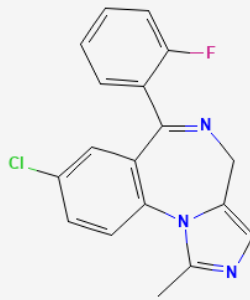


**Diazepam — the 1,4-benzodiazepine prototype.** benzene fused to a seven-membered diazepine. Binds the  $\alpha/\gamma$  interface, not the GABA site — it can only modulate what GABA is already doing.



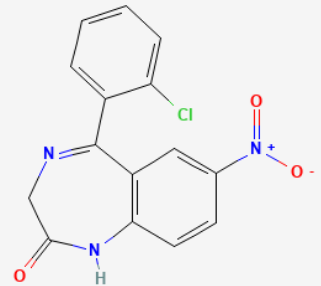
### Triazolam

a triazolo ring fused on → metabolism attacks it fast → ultra-short acting



### Midazolam

triazolo + F, and ● its ring opens and closes with pH — closed and lipophilic at blood pH, which is why it works IV and crosses fast



### Clonazepam

a nitro group → long  $t_{1/2}$ , the antiepileptic end of the family

## MEDCHEM

**SAR** The scaffold is **5-phenyl-1,4-benzodiazepin-2-one** and each ring has a job: **ring A**  $\pi$ -stacks, **ring B** supplies the proton-accepting carbonyl that binds histidine and carries the metabolic sites, **ring C** (the 5-phenyl) does the hydrophobic contact. Requirements: **N at position 1** for the hydrogen bond, a **carbonyl at position 2 coplanar with ring A**, and the C=N at 4,5 - reduce it to C-N and activity falls.

**SAR** Position 7 must carry an **electron-withdrawing group** (halogen or NO<sub>2</sub>) - that is where the potency lives; substituting 6, 8 or 9 instead *lowers* activity. ⚠ Reducing a 7-NO<sub>2</sub> to 7-NH<sub>2</sub> **abolishes activity**, which matters because that reduction is a real metabolic route for the nitro-benzodiazepines. On ring C, a halogen at 2' raises lipophilicity but a halogen at 4' **lowers** activity.

**SAR** ★ **Why alprazolam, triazolam and midazolam behave differently:** fusing an extra electron-rich **triazole or imidazole ring** across positions 1–2 raises receptor affinity outright. And **flumazenil is the same scaffold** with the 5-phenyl removed - the ring that provides the hydrophobic contact is exactly what turns an agonist into a pure antagonist.

**Metabolism** ● **A 3-OH is the single most clinically useful thing on this scaffold.** It barely changes potency but it lets the drug go **straight to glucuronidation**, skipping CYP entirely - that is oxazepam, lorazepam and temazepam, and it is why they are the benzodiazepines used in liver disease and in the elderly. Everything else runs through **CYP3A4/2C19** N-dealkylation and hydroxylation, and most of the class funnels into oxazepam anyway.

**Metabolism** ⚠ **Flurazepam's own half-life is 2 hours; its metabolite's is 47–100 hours.** N-desalkylflurazepam is what actually keeps the patient sedated the next day - the parent drug is nearly irrelevant to the clinical picture.

## DRUGS

| DRUG                    | WHAT IT TREATS                                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Triazolam</b>        | early insomnia · sedation before procedures ( <i>short acting</i> )                                                   |
| <b>Midazolam</b>        | early insomnia · sedation before operation/intubation · status epilepticus ( <i>oral</i> )                            |
| <b>Temazepam</b>        | late insomnia ( <i>intermediate acting</i> )                                                                          |
| <b>Lorazepam</b>        | late insomnia · <b>status epilepticus</b> ( <i>injection</i> ) · febrile seizure                                      |
| <b>Diazepam</b>         | <b>status epilepticus</b> ( <i>injection</i> ) · febrile seizure · muscle relaxation · anxiety ( <i>long acting</i> ) |
| <b>Clonazepam</b>       | <b>panic disorder</b> · febrile seizure · muscle relaxation · <b>restless legs syndrome</b>                           |
| <b>Alprazolam</b>       | <b>panic disorder</b> · anxiety                                                                                       |
| <b>Chlordiazepoxide</b> | anxiety ( <i>long acting</i> )                                                                                        |
| <b>Clorazepate</b>      | anxiety ( <i>long acting</i> )                                                                                        |
| <b>Clobazam</b>         | febrile seizure                                                                                                       |

## EQUIVALENT DOSE & POTENCY

Lower equivalent dose = higher potency.

| LOW POTENCY      | EQ. DOSE (MG) | HIGH POTENCY      | EQ. DOSE (MG) |
|------------------|---------------|-------------------|---------------|
| Bromazepam       | 3             | <b>Alprazolam</b> | 0.5           |
| Chlordiazepoxide | 10            | <b>Clonazepam</b> | 0.25          |
| Clobazam         | 10            | <b>Lorazepam</b>  | 1             |
| Clorazepate      | 7.5           | <b>Midazolam</b>  | 2             |
| Diazepam         | 5             | <b>Triazolam</b>  | 0.25–0.5      |
| Prazepam         | 15            |                   |               |

## METABOLISM ★

| ROUTE                            | DRUGS                                                        | NOTE                                       |
|----------------------------------|--------------------------------------------------------------|--------------------------------------------|
| <b>CYP3A4</b>                    | clonazepam, triazolam, midazolam, alprazolam<br>(+ zolpidem) | main pathway                               |
| <b>CYP2C19 and/or 3A4</b>        | most other BZPs                                              |                                            |
| <b>UGT —<br/>glucuronidation</b> | <b>lorazepam, temazepam</b>                                  | → <b>inactive</b> metabolites, no CYP step |

● **Desmethyldiazepam** (nordazepam) — a major **active** metabolite of several BZPs, **t<sub>1/2</sub> up to 100 h**. This, not the parent drug, is why some BZPs linger.

*Onset bands: rapid 15–30 min · intermediate 30–45 min · slow 45–90 min. Half-life bands: short <10 h · intermediate 10–36 h · long >48 h.*

## ADVERSE EFFECTS

- **Hangover / excessive daytime sedation**
- Impaired motor performance, **ataxia**
- Memory interference — **anterograde amnesia** ★; long-term use in the **elderly** may ↑ **dementia** risk
- Disturbed sleep architecture — **REM suppression** (*barbiturate* >> *BZP* >> *Z-drugs*) and **N3–4 suppression** (*BZP* >> *Z-drugs*)
- **Paradoxical effects** — disinhibited behaviour, complex sleep-related behaviour. *Most likely with high-dose, high-potency BZPs and zolpidem.*

## INTERACTIONS

- ● **Other CNS depressants** — shift the whole dose–response curve left; sedation becomes ataxia becomes **respiratory depression** at doses that would otherwise be safe
- **Ethanol** — CNS depression *plus* **cross-tolerance**
- **CYP3A4 / CYP2C19** substrates — anything altering those enzymes moves BZP levels (*lorazepam and temazepam bypass this — UGT only*)

## CAUTIONS

- ● **Contraindicated in obstructive sleep apnea (OSA)**
- ⚠ Elderly on long-term therapy — dementia risk
- ⚠ Anxiety indication covers most disorders **except PTSD and OCD**

# Benzodiazepine Tolerance & Withdrawal

## WHY TOLERANCE HAPPENS

Long-term BZP exposure → **GABA-A receptor down-regulation** → **tolerance**.

→ **Downstream**: tolerance develops to the **sedative** effect but **not to the anxiolytic** effect ★ · risk of **dependence** · **cross-tolerance with alcohol** · abrupt stop → withdrawal

## WITHDRAWAL SYMPTOMS — CNS EXCITATION (MOTOR · SENSORY · PSYCHIATRIC · AUTONOMIC)

| FREQUENCY       | SYMPTOMS                                                                                                                                  |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frequent</b> | rebound <b>anxiety/insomnia</b> , irritability, poor concentration, muscle ache, abdominal cramp, weakness, tremor, palpitation, sweating |
| <b>Common</b>   | nausea, depression, ataxia, hyperreflexia, blurred vision, fatigue                                                                        |
| <b>Rare</b>     | delirium, confusion, psychosis, panic, <b>seizure</b>                                                                                     |

## HOW TO STOP THEM

- ★ **Taper gradually over 4–8 weeks.** Immediate discontinuation after long-term use → withdrawal.

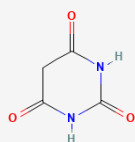
# Barbiturates

## MECHANISM

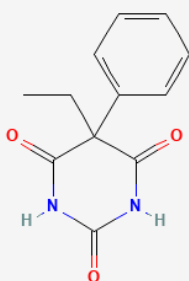
Bind the **barbiturate binding site** and act as an **APL** of GABA-A — *and*, ● **at high dose, act as a direct agonist** of the receptor. That second property is the whole danger: they no longer need GABA.

→ **Downstream:** a **much steeper dose-response curve** than BZPs — anticonvulsant → anxiolytic → sedation → ataxia → **coma/anaesthesia** → **respiratory depression**, with no meaningful ceiling

## THE SHARED CORE · WHAT CHANGES

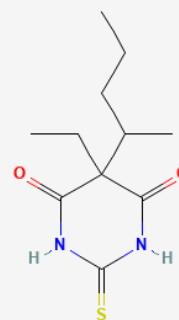


**Barbituric acid.** inactive by itself. Everything below is this ring with two groups added at C5 — and ● unlike BZDs these open the channel with no GABA present, which is the whole safety difference



**Phenobarbital**

phenyl + ethyl → long acting, and the classic enzyme inducer



**Thiopental**

● one O swapped for S → hugely lipid-soluble → in the brain in seconds, then redistributed out

## MEDCHEM

**SAR** ★ **The drug has to be a weak acid, and both C5 positions have to be filled.** With only **one** substituent at C5 the pKa sits near 4 - the molecule is fully anionic at pH 7.4 and **cannot cross the blood-brain barrier at all**. Disubstituted, with **R5 + R5 totalling 7–9 carbons**, is the optimum. Too few carbons and it never gets in; too many and activity falls off.

**SAR N-methylation raises lipophilicity** → faster onset, shorter duration (phenobarbital → mephobarbital). But methylate *both* nitrogens and the acidic proton is gone - **non-acidic means inactive**. The lipophilicity ladder is printed in the deck: amobarbital logP 1.46 · butobarbital 1.60 · phenobarbital 2.07 · pentobarbital 2.10 · secobarbital 2.36.

**Physicochem** The free acid is **lipid-soluble and water-insoluble**; the injectable is the **enolate salt** - water-soluble, but it **precipitates if the line pH drops** and it decomposes by base-catalysed hydrolysis. Both facts are IV-compatibility problems, not shelf-life trivia.

**Metabolism** ● **Thiopental's metabolite is pentobarbital.** **Oxidative desulfuration** puts the oxygen back, converting the ultrashort drug into a short-acting barbiturate - so a long thiopental infusion accumulates an active, longer-lasting drug. Phase 1 is oxidation/hydroxylation, N-demethylation and desulfuration; phase 2 is glucuronide or sulfate.

## DRUGS

| DRUG                 | WHAT IT TREATS                                             |
|----------------------|------------------------------------------------------------|
| <b>Phenobarbital</b> | CNS depressant / anticonvulsant (วัตถุออกฤทธิ์ประเภทที่ 4) |
| <b>Pentobarbital</b> | CNS depressant (วัตถุออกฤทธิ์ประเภทที่ 3)                  |
| <b>Thiopental</b>    | CNS depressant / anaesthesia                               |

## ADVERSE EFFECTS

- **Respiratory depression, coma** — reachable at therapeutic-adjacent doses
- **Strongest REM sleep suppression** of any group here (*barbiturate* >> *BZP* >> *Z-drugs*)
- Shares the BZP profile: daytime sedation, motor impairment, ataxia

## CAUTIONS

- **There is NO specific antidote for barbiturates.** Flumazenil does nothing here.
- ⚠ Combining with any other CNS depressant compounds an already steep curve

# Z-Drugs

## MECHANISM

**Zolpidem** binds the same **BZ site** as the benzodiazepines and acts as an **APL** — but is selective for **BZ1 ( $\alpha 1$ -containing)** receptors.

→ **Downstream:** sedation/hypnosis · anticonvulsive · anterograde amnesia — **without** the BZ2 effects (anxiolysis, myorelaxation, pain inhibition) ★

## DRUGS

| DRUG             | WHAT IT TREATS                                                          |
|------------------|-------------------------------------------------------------------------|
| <b>Zolpidem</b>  | <b>early insomnia</b> ( <i>short acting; วัตถุออกฤทธิ์ประเภทที่ 2</i> ) |
| <b>Zaleplon</b>  | insomnia                                                                |
| <b>Indiplon</b>  | insomnia                                                                |
| <b>Zopiclone</b> | insomnia                                                                |

## ADVERSE EFFECTS

- **Least** sleep-architecture disruption of the three groups — REM and N3–4 suppression are smallest with Z-drugs
- **Paradoxical effects** — complex sleep-related behaviour, disinhibition. ⚠ Zolpidem is named alongside high-potency BZPs as a top offender.
- 🚫 Contraindicated in **obstructive sleep apnea**, like the BZPs

## INTERACTIONS

- **Zolpidem is a CYP3A4 substrate** — same interaction surface as midazolam/triazolam/alprazolam
- Other CNS depressants, including ethanol



# Etifoxine

## MECHANISM

Two actions at once — the only drug here that does both:

1. Binds the **neurosteroid binding site** (*on the  $\beta$  subunit of BZ2*) and acts as an **APL**
2. **Increases synthesis of allopregnanolone**, the endogenous neurosteroid that occupies that same site  
→ **Downstream**: potentiation aimed at **BZ2** receptors — the subtype that carries **anxiolysis, myorelaxation and pain inhibition**, not the  $\alpha 1$  sedation/amnesia package

## WHAT THE LECTURE DOES NOT SAY

The slides give etifoxine's **binding site and mechanism only** — no drug table, no stated indication, no ADR list. Its BZ2 targeting is the reason it appears on an anxiolysis lecture; treat any indication you read elsewhere as coming from elsewhere.

# Flumazenil

## MECHANISM

Binds the **benzodiazepine (BZ) site** on GABA-A and reverses what is sitting there.

→ **Downstream: specific antidote for benzodiazepines and zolpidem** — because both act at that exact site

## DRUGS

| DRUG              | WHAT IT TREATS                                     |
|-------------------|----------------------------------------------------|
| <b>Flumazenil</b> | <b>benzodiazepine overdose · zolpidem overdose</b> |

## THE LIMIT OF THE ANTIDOTE ★

- **NO specific antidote exists for barbiturates or ethanol.** They don't use the BZ site, so flumazenil cannot displace them.

# Clinical Applications

## PSYCHIATRY

| INDICATION                                                              | DRUGS                                                               |
|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Early insomnia</b> ( <i>short acting</i> )                           | triazolam · midazolam · <b>zolpidem</b>                             |
| <b>Late insomnia</b> ( <i>intermediate acting</i> )                     | temazepam · lorazepam                                               |
| <b>Anxiety disorders</b> ★ ( <i>quick onset — except PTSD and OCD</i> ) | long acting: chlordiazepoxide · clorazepate · diazepam · clonazepam |
| <b>Panic disorder</b>                                                   | alprazolam · clonazepam                                             |
| <b>Alcohol withdrawal syndrome</b>                                      | benzodiazepines ( <i>cross-tolerance with alcohol</i> )             |
| Side effects of psychotropic drugs                                      | benzodiazepines                                                     |

## NEUROLOGY & OTHERS

| INDICATION                                    | DRUGS                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------|
| <b>Status epilepticus</b> ●                   | midazolam ( <i>oral</i> ) · diazepam or lorazepam ( <i>injection</i> ) |
| <b>Febrile seizure</b>                        | diazepam · clobazam · clonazepam · lorazepam                           |
| <b>Muscle relaxation</b>                      | diazepam · clonazepam                                                  |
| <b>Restless legs syndrome</b>                 | clonazepam                                                             |
| <b>Sedation before operation / intubation</b> | short acting: triazolam · midazolam ( <i>and chloral hydrate</i> )     |

## ACROSS EVERY INDICATION

- **Contraindicated in obstructive sleep apnea**
- ⚠ Anxiety: works fast, but **not** for PTSD or OCD
- ⚠ Any long course needs a **4–8 week taper**, not a stop

Principle behind this: *[[pharm-drug-action]]* — Benzodiazepines don't open the channel — they modify how it responds to GABA. The cleanest example of modulation rather than agonism, and where inverse agonism at a receptor site is most visible.

Principle behind this: *[[pharm-drug-toxicity]]* — Barbiturates running anxiolysis → sedation → somnolence → coma is the cleanest dose-dependent continuum in the syllabus.

Principle behind this: *[[pharm-cns-basics]]* — Diazepam is the positive allosteric modulator archetype — it does nothing without GABA, which is exactly what "allosteric" buys.

Source: *Week01 Pharmacology of Drugs acting on GABAA receptor.pdf* — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สม), Pharmacology 3 (ภทป 401), 25 slides. Structures omitted.

# Analgesics I — Opioids & Paracetamol

Two completely different ways to kill pain. **Opioids** hijack the body's own descending inhibitory pathway at **Gi-coupled opioid receptors**. **Paracetamol** does not touch them — it is a CNS-acting COX inhibitor with an extra metabolite that works on the same descending pathway from the other side.

**Ascending** (*glutamate · substance P · CGRP*) → **pain perceived** — **Descending inhibitory** (*serotonin · norepinephrine · endogenous opioids*) → **pain suppressed**

## THREE KINDS OF PAIN, BY MECHANISM

- **Nociceptive / inflammatory** — activation or sensitization of the nociceptor
- **Neuropathic** — lesion/injury *to the pain pathway itself*, leading to maladaptation
- **Nociplastic (dysfunctional)** — abnormal pain **processing**, no lesion required

## ANALGESIC CLASSIFICATION

| GROUP                            | MEMBERS                                                                          |
|----------------------------------|----------------------------------------------------------------------------------|
| <b>Opioid analgesics</b>         | this page                                                                        |
| <b>Non-opioid analgesics</b>     | <b>paracetamol</b> · NSAIDs / coxibs                                             |
| <b>Adjuvants (co-analgesics)</b> | antidepressants · antiepileptics · muscle relaxants · steroids · bisphosphonates |

## OPIOID RECEPTORS — ONE SIGNALLING STORY FOR ALL FOUR

Subtypes **μ (MOP) · δ (DOP) · κ (KOP) · NOP (OPRL1)**. All are **GPCRs coupled to Gi**, and all do the same three things:

1. **Inhibit adenylate cyclase**
2. **Inhibit Na<sup>+</sup> and Ca<sup>2+</sup> channels**
3. **Enhance K<sup>+</sup> channels**

→ **Downstream**: less depolarization, less transmitter release → analgesia at **four sites at once**: peripheral nociceptive terminal · central terminal of the nociceptive fibre · dorsal horn cell · higher centres (*thalamus, somatosensory cortex*)

## ★ RECEPTOR DISTRIBUTION → WHICH EFFECT COMES FROM WHICH RECEPTOR

| EFFECT                                | M | Δ  | K | NOP                |
|---------------------------------------|---|----|---|--------------------|
| Analgesia — <b>supraspinal</b>        | ✓ | —? | — | <b>anti-opioid</b> |
| Analgesia — <b>spinal</b>             | ✓ | ✓  | ✓ | ✓                  |
| Analgesia — peripheral                | ✓ | —  | ✓ | —                  |
| Sedation                              | ✓ | —  | ✓ | —                  |
| <b>Respiratory depression</b> ●       | ✓ | —  | — | —                  |
| Euphoria                              | ✓ | —  | — | —                  |
| <b>Dysphoria &amp; hallucinations</b> | — | —  | ✓ | —                  |
| Pupil constriction (miosis)           | ✓ | —  | ✓ | —                  |
| ↓ GI motility & secretion             | ✓ | ✓  | ✓ | —                  |
| <b>Physical dependence</b>            | ✓ | —  | — | —                  |

Read the  $\mu$  column: respiratory depression, euphoria and dependence are  $\mu$ -only. That is why  $\mu$  potency and  $\mu$  danger cannot be separated.

## RECEPTOR SELECTIVITY

Legend: + agonist · ± partial agonist · – antagonist · repetition = strength · · = cell didn't resolve in the source table.

| DRUG                                               | M           | Δ | K   |
|----------------------------------------------------|-------------|---|-----|
| <b>Tramadol · codeine</b> (weak)                   | +           | ? | ?   |
| <b>Morphine</b> · oxymorphone · dextropropoxyphene | +++         | + | +   |
| <b>Methadone</b>                                   | +++         | · | ·   |
| <b>Meperidine (pethidine)</b>                      | ++          | + | +   |
| <b>Fentanyl</b> · sufentanil                       | +++         | · | ·   |
| <b>Nalbuphine</b>                                  | –           | · | ++  |
| <b>Pentazocine</b>                                 | ±           | + | ++  |
| <b>Buprenorphine</b>                               | ± (partial) | – | –   |
| <b>Naloxone · naltrexone</b>                       | ---         | – | --- |

# Weak Opioids

## MECHANISM

**μ agonists with an analgesic ceiling effect** — past a point, more drug buys no more analgesia. Both are prodrug-dependent on **CYP2D6**:

- **Codeine** → CYP2D6 → **morphine** (*the actual analgesic*)
- **Tramadol** → CYP2D6 → **M1**, whose μ affinity is close to morphine's — **plus** direct **NE transporter** and **5-HT transporter** inhibition

→ **Downstream**: μ activation *and*, for tramadol, reinforcement of the **descending inhibitory pathway** (NE from locus coeruleus, 5-HT from raphe nuclei) ★

## DRUGS

| DRUG            | WHAT IT TREATS                                                           |
|-----------------|--------------------------------------------------------------------------|
| <b>Codeine</b>  | mild–moderate pain ( <i>max 240–360 mg/day; initial 15–30 mg q4–6h</i> ) |
| <b>Tramadol</b> | mild–moderate pain ( <i>max 400 mg/day</i> )                             |

## ADVERSE EFFECTS — TRAMADOL SPECIFICALLY

- Nausea, vomiting, dizziness
- Palpitation
- Constipation
- ● **Reduces the seizure threshold**

## INTERACTIONS

- ★ **CYP2D6 inhibitors attenuate codeine's analgesic effect** — no conversion, no morphine, no analgesia
- **CYP2D6 polymorphism** changes both efficacy and safety of codeine

## RENAL DOSING

- **Codeine** — adjust in diminished renal function, same as morphine
- **Tramadol** — 30% excreted unchanged, 60% as metabolites; **mild–moderate renal impairment raises M1 by 20–40%**
  - CrCl **10–30 mL/min** → interval to **12 h**, max **200 mg/day**
  - CrCl **<10 mL/min** → **50 mg q12h**
  - ⚠ **Geriatric: not more than 300 mg/day**

## ADMINISTRATION — CODEINE DOSING

- **Initial 15–30 mg every 4–6 hours · maximum 240–360 mg/day.**
- **Renal dose adjustment is the same as for morphine** — which follows, because ★ **codeine is a prodrug metabolised by CYP2D6 to morphine.**
- ● **CYP2D6 inhibitors blunt codeine's analgesia entirely**, and **CYP2D6 polymorphism decides both efficacy and safety** — a poor metaboliser gets no relief, an ultra-rapid one gets a morphine overdose.
- 💡 This is why "codeine didn't work for me" is a pharmacogenetic statement, not a complaint about pain tolerance. Reaching for a higher dose is the wrong response.

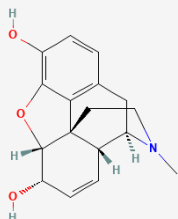
# Strong Opioids

## MECHANISM

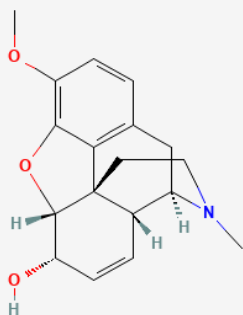
Full  $\mu$  agonists with **NO analgesic ceiling effect** — the dose-response keeps climbing, and so does respiratory depression. ● *This is the whole risk profile in one sentence.*

→ **Downstream:** analgesia · sedation · euphoria · miosis · constipation · **respiratory depression** · **physical dependence**

## THE SHARED CORE · WHAT CHANGES

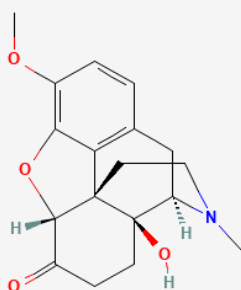


**Morphine — the morphinan core.** a rigid five-ring cage with a phenol at C3 and an alcohol at C6. ● Almost every classical opioid is this molecule with C3 or C6 modified.



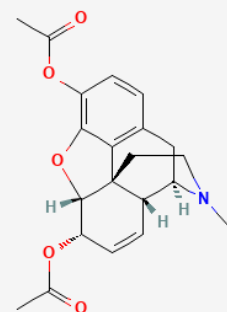
### Codeine

C3-OH methylated → ● inactive until CYP2D6 removes that methyl and turns it back into morphine



### Oxycodone

C3-methyl + C6 ketone + 14-OH  
→ orally reliable, more potent



### Diamorphine (heroin)

both OHs acetylated → far more lipophilic → hits the brain much faster; the classic 'structure sets the abuse potential' case

## MEDCHEM

**SAR** The opiate pharmacophore is three requirements: **a flat area for a flat aromatic ring, a cavity for a second set of rings sitting perpendicular to it, and an anionic site for the protonated amine.**

Everything from morphine to fentanyl is that same T-shape, progressively simplified - morphinans drop ring E, benzomorphans drop C as well, the piperidines (meperidine, fentanyl) drop B too.

**SAR** ● **Agonist vs antagonist is decided by one substituent on the ring-D nitrogen. Methyl → agonist. Allyl or cyclopropylmethyl → antagonist** - that is the entire chemical difference between morphine and naloxone/naltrexone. And a **quaternary** nitrogen has no analgesic activity at all, because it cannot cross into the CNS: that is exactly why **methylnaltrexone** reverses opioid constipation in the gut without touching the analgesia.

**SAR** The ring-C edits are a potency ladder, and the deck prints it: morphine 1× → dihydromorphine 1.2× → **hydromorphone 5–6×** → **oxymorphone 10×**. The recipe is replacing the **6-OH** with a ketone (plus saturating the 7,8-double bond) and adding a **14-OH** for an extra hydrogen bond. Buprenorphine's very high affinity comes from a C7 substituent adding both hydrophobic contact and a hydrogen bond.

**Metabolism** ★ **Codeine is morphine with the phenolic OH methylated** - and that OH is *required* for activity, so codeine is barely an analgesic until **CYP2D6 O-demethylates it back to morphine**. Only about 5–10% is converted. Heroin goes the other way: acetylating both hydroxyls raises lipophilicity so it reaches the brain faster, then esterases strip the acetyls back off.

**SAR** **Tramadol is not really one drug.** Its (–) enantiomer inhibits noradrenaline reuptake, its (+) inhibits serotonin reuptake, and it is sold as the **racemate on purpose** because the mixture is better tolerated than either half.

## DRUGS

| DRUG                                                    | WHAT IT TREATS                                                                        |
|---------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Morphine</b>                                         | moderate–severe pain — the reference standard ( <i>UGT-metabolised</i> )              |
| <b>Oxycodone</b>                                        | moderate–severe pain ( <i>CYP2D6 + CYP3A4, active metabolites</i> )                   |
| <b>Methadone</b>                                        | moderate–severe pain — <b>very long t<sub>1/2</sub> 8–60 h</b> , no active metabolite |
| <b>Meperidine (pethidine)</b>                           | acute pain only — ⚠ <b>not for chronic use</b>                                        |
| <b>Fentanyl</b> ( <i>and sufentanil, remifentanil</i> ) | severe pain — <b>very potent μ agonist</b>                                            |
| <b>Heroin</b>                                           | ( <i>listed by the lecture; no clinical indication given</i> )                        |

## ★ THE THREE THAT HAVE THEIR OWN PERSONALITY

| DRUG                          | WHAT MAKES IT DIFFERENT                                                                                                                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Morphine</b>               | UGT → <b>M3G</b> ( <i>neurotoxic, and a μ antagonist</i> ) + <b>M6G</b> ( <i>40–50× more potent at μ</i> ). Nearly all of it and its metabolites leave via the <b>kidney</b> → ● accumulation in renal impairment. |
| <b>Meperidine (pethidine)</b> | <b>Rapid euphoria</b> from high lipophilicity → highly addictive · <b>anticholinergic</b> · <b>inhibits 5-HTT</b> · metabolite <b>norpethidine is neurotoxic, t<sub>1/2</sub> 15–30 h</b> · renally excreted.      |
| <b>Fentanyl</b>               | IV: onset <b>1 min</b> , duration <b>30–60 min</b> . Transdermal: duration <b>3 days</b> , t <sub>1/2</sub> after removal <b>13–22 h</b> . <b>CYP3A4</b> → <b>inactive</b> metabolite.                             |



### INTERACTIONS

- ● **CYP3A4 inhibitors + fentanyl** → ↑ fentanyl level → **fatal respiratory depression**
- **Pethidine inhibits 5-HTT** — a serotonergic drug wearing an opioid's coat

# Partial Agonists & Mixed Agonist–Antagonists

## MECHANISM

Not full  $\mu$  agonists — each is a **partial agonist or outright antagonist at  $\mu$**  while retaining  **$\kappa$  agonism**.

→ **Downstream**: analgesia with a **built-in ceiling**, but  $\kappa$  activity brings its own package — **dysphoria and hallucinations** are  $\kappa$  effects, not  $\mu$  effects ★

## DRUGS

| DRUG                 | WHAT IT TREATS                                                                           |
|----------------------|------------------------------------------------------------------------------------------|
| <b>Buprenorphine</b> | pain — <b>partial <math>\mu</math> agonist</b> , $\delta$ and $\kappa$ <b>antagonist</b> |
| <b>Nalbuphine</b>    | pain — <b><math>\mu</math> antagonist</b> , strong <b><math>\kappa</math> agonist</b>    |
| <b>Pentazocine</b>   | pain — <b>partial <math>\mu</math> agonist</b> + $\delta/\kappa$ agonist                 |

# Opioid Antagonists

## MECHANISM

Pure **antagonists at  $\mu$  and  $\kappa$**  — they displace whatever is sitting on the receptor.

→ **Downstream:** immediate reversal of opioid effects, **including respiratory depression**

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                               |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Naloxone</b>   | <b>opioid overdose</b> ( <i>parenteral; antagonist activity lasts only 1–2 h, <math>t_{1/2}</math> 1 h</i> ) |
| <b>Naltrexone</b> | <b>alcoholism</b> ( <i>oral, 24 h</i> )                                                                      |

## THE TRAP ★

- **Naloxone's 1–2 h of action is shorter than most opioids it reverses.** The patient can wake up, then go back under when the naloxone wears off before the agonist does.

# Opioid Adverse Effects

## COMMON

- **Nausea / vomiting** — structure-related *and* serotonin-mediated
- **Constipation** — ★ **long-lasting and NOT tolerated**. Chronic opioid use should carry a **laxative**.
- **Sedation**
- ● **Respiratory depression** — rare with appropriate use, but the effect that kills

## LESS COMMON

- **Hypotension** and **itching** — from **opioid-induced histamine release**
- **Urinary retention**
- **Euphoria and addiction** — rare in appropriate use; ⚠ **pethidine is highly addictive**
- **Miosis** — ★ also **not tolerated**, which is why pinpoint pupils stay diagnostic
- Delirium, psychosis, **myoclonic jerks**

## PRECIPITATING FACTORS FOR RESPIRATORY DEPRESSION ●

- **Benzodiazepines** and other **CNS depressants**
- **Asthma · COPD**
- **Brain trauma**

# Opioid Tolerance & Withdrawal

## WHY TOLERANCE HAPPENS

Long-term opioid use →  $\mu$  receptor down-regulation → tolerance.

→ **Downstream:** tolerance develops to **different effects to different degrees** (*constipation and miosis barely at all*) · **cross-tolerance between opioids is incomplete** ★ · abrupt cessation → **withdrawal**, with severity  $\propto$  **potency** × **dose** × **duration**

## WHAT THAT MEANS IN PRACTICE

- ⚠ **Incomplete cross-tolerance** — when rotating opioids, the equianalgesic dose is an *over*-estimate; a fully tolerant patient is not equally tolerant to the new drug.
- ⚠ Never stop abruptly after long-term use.

# Opioid Pharmacokinetics & Equianalgesia

## DURATION, HALF-LIFE, ACTIVE METABOLITES

| DRUG             | DURATION (H) | ELIMINATION T <sub>1/2</sub> (H) | ACTIVE METABOLITES |
|------------------|--------------|----------------------------------|--------------------|
| Codeine          | 4–6          | 3                                | Yes                |
| Tramadol         | 3–5          | 6                                | Yes                |
| Morphine         | 4–5          | 2.3                              | Yes                |
| Meperidine       | 3–4          | 3                                | Yes                |
| Oxycodone        | 3–5          | 3                                | Yes                |
| <b>Methadone</b> | 4–5          | <b>23</b>                        | No                 |
| Fentanyl         | 0.5–1        | 3.7                              | No                 |
| Buprenorphine    | 4–6          | 5                                | No                 |
| Nalbuphine       | 4–5          | 5                                | No                 |
| Pentazocine      | 3–5          | 4                                | No                 |
| Naloxone         | 1–2          | 1                                | No                 |
| Naltrexone       | 24           | 4                                | Yes                |

★ Immediate-release forms of the common opioids all have **short half-lives and short duration** — **methadone is the outlier (t<sub>1/2</sub> 8–60 h)**, and its duration of *analgesia* is far shorter than its half-life, which is exactly how it accumulates. ● **The active forms of most common opioids are excreted by the kidney.**

## METABOLISM & URINARY EXCRETION

| DRUG          | METABOLISM       | ACTIVE METABOLITES | URINE  |
|---------------|------------------|--------------------|--------|
| Codeine       | <b>CYP2D6</b>    | Yes                | 90%    |
| Tramadol      | <b>CYP2D6</b>    | Yes                | 90%    |
| Morphine      | <b>UGT</b>       | Yes                | 90%    |
| Meperidine    | CYP2B6, CYP3A4   | Yes                | 70%    |
| Oxycodone     | CYP2D6 + CYP3A4  | Yes                | 19–64% |
| Methadone     | CYP2D6, 3A4, 1A2 | No                 | <10%   |
| Fentanyl      | <b>CYP3A4</b>    | No                 | 75%    |
| Buprenorphine | CYP3A4, UGT      | No                 | 30%    |
| Nalbuphine    | UGT + CYPs       | No                 | —      |

## EQUIANALGESIC DOSES (MG)

*Lower number = more potent. Note how far oral sits above parenteral for morphine and codeine.*

| DRUG            | PARENTERAL | ORAL      |
|-----------------|------------|-----------|
| <b>Fentanyl</b> | <b>0.1</b> | —         |
| Oxymorphone     | 1          | 10        |
| Hydromorphone   | 1.5        | 7.5       |
| Oxycodone       | —          | 20        |
| <b>Morphine</b> | <b>10</b>  | <b>30</b> |
| Hydrocodone     | —          | 30        |
| Meperidine      | 75–100     | 300       |
| Tramadol        | 80–100     | 120       |
| Codeine         | 100–120    | 200–240   |

# Paracetamol

## MECHANISM

A **reversible competitive COX inhibitor, preferential for COX-2 (COX-2 > COX-1)**, acting **predominantly in the CNS**. It inhibits a **different step** from NSAIDs/coxibs: it works as a **reducing substrate in the peroxidase reaction** that converts **PGG<sub>2</sub> → PGH<sub>2</sub>**, cutting **PGE<sub>2</sub>** synthesis.

★ **Why it has no anti-inflammatory effect:** the inhibition is **attenuated by high arachidonic acid concentration and oxidative stress** — precisely the conditions inside inflamed tissue.

→ **Plus AM404** (*N-arachidonoylphenolamine, formed via FAAH*): ↑ availability of the endocannabinoid **anandamide** · modulates **TRPV1** and **T-type calcium channels** · ↑ **serotonergic and opioidergic tone in the descending inhibitory pain pathway**

→ **Downstream:** analgesia for mild–moderate pain · **antipyresis** by normalizing the set point of the hypothalamic thermoregulation centre · sweating

## PARACETAMOL VS NSAIDS

| EFFECT                   | NSAIDS | PARACETAMOL |
|--------------------------|--------|-------------|
| Antipyretic              | ✓      | ✓           |
| Analgesic                | ✓      | ✓           |
| <b>Anti-inflammatory</b> | ✓      | X ★         |

## PROS & CONS

**Pros** (*at recommended dose*) — relatively safe across most patient conditions · **few drug interactions** ·

**Cons** — **limited analgesic effect** · **no anti-inflammatory effect** · ⚠ long-term use still carries risk

## METABOLISM — WHERE THE DANGER LIVES

- **90–95% → UGT and SULT** → glucuronide and sulfate conjugates (*non-toxic*)
- **5–10% → CYP2E1** → ● **NAPQI** (*N-acetyl-p-benzoquinone imine — the toxic intermediate*)
- NAPQI is normally detoxified fast by **glutathione (GSH)** via **GST** → cysteine and mercapturic acid conjugates, excreted in urine
- **Downstream (overdose): glutathione depletion or CYP2E1 induction** → NAPQI accumulates → **covalent binding to hepatocellular protein** → **hepatotoxicity** ●

## ★ CUT THE DOSE TO 50–75% WHEN CYP2E1 IS OVERACTIVE

- **Chronic alcoholism** / regular intake **>3 drinks/day**
- **Enzyme inducers** — phenytoin · carbamazepine · rifampin
- **Malnutrition** (*less glutathione to spend*)



# Paracetamol Overdose & Antidote

## DOSES THAT MATTER

|                           | AMOUNT                                      |
|---------------------------|---------------------------------------------|
| <b>Toxic</b>              | <b>7.5–10 g</b> intake, or <b>150 mg/kg</b> |
| <b>Fatal</b>              | single ingestion of <b>20–25 g</b>          |
| Hepatic injury from NAPQI | begins <b>within the first 24 h</b>         |

## ● TIMELINE — THE DECEPTIVE MIDDLE

| TIME POST-INGESTION | WHAT HAPPENS                                                                                |
|---------------------|---------------------------------------------------------------------------------------------|
| <b>12–24 h</b>      | nausea / vomiting / diarrhoea, diaphoresis, anorexia                                        |
| <b>24–48 h</b>      | ⚠ <b>all symptoms resolve</b> — while <b>AST, ALT, bilirubin and prothrombin time climb</b> |
| <b>72–96 h</b>      | <b>peak hepatitis</b> symptoms                                                              |
| <b>7–8 days</b>     | recovery                                                                                    |

★ **The 24–48 h window is the trap.** The patient feels better exactly while the liver is failing — treat on the labs and the ingested dose, never on how well they look.

## ANTIDOTE

| DRUG                    | WHAT IT TREATS                                                               |
|-------------------------|------------------------------------------------------------------------------|
| <b>N-acetylcysteine</b> | <b>paracetamol overdose</b> — replenishes the glutathione that mops up NAPQI |

Principle behind this: *[[pharm-drug-action]]* — Meperidine (full) vs pentazocine and buprenorphine (partial) — and buprenorphine being more potent yet less efficacious than morphine, the proof that potency ≠ efficacy.

Principle behind this: *[[pharm-drug-toxicity]]* — Paracetamol → NAPQI → glutathione depletion → hepatic necrosis is the pathological-toxicity archetype, and naloxone appears in the ABCDE "D" step as an empirical trial.

Principle behind this: *[[pharm-pharmacogenomics]]* — Codeine is a CYP2D6 prodrug — PM gets no analgesia, UM gets morphine toxicity. The two case reports are the strongest teaching material in the whole principles set.

Principle behind this: *[[pharm-cns-basics]]* — Buprenorphine as  $\mu$  partial agonist; endorphin, enkephalin and dynorphin in the analgesia row of Concept 5.

Source: 00 Pharmacology of Analgesics I Opioids.pdf (26 slides) + 00 Pharmacology of Analgesics I Paracetamol.pdf (11 slides) — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สม), Pharmacology 3 (ภกปว 401), self-study pack 2. Structures omitted.

# CNS Stimulants and Drugs Affecting Appetite

⚠ This page is a placeholder. There is no content here yet — nothing below is missing by accident.

## WHY THIS PAGE IS EMPTY

The lecture slides for this topic **have not been uploaded to Microsoft Teams** (team code `gwwtmy3`) as of **2 August 2026**. Every other week's slides in this course came from อ.พงศธร, who posted the whole set; this topic belongs to a different lecturer who hasn't posted yet.

|             |                                                               |
|-------------|---------------------------------------------------------------|
| Week        | 3 — Wednesday <b>19 ส.ค. 69</b> , 10:00–12:00, R207           |
| Lecturer    | อ.วิลาสินี (รศ.ดร.วิลาสินี หิรัญพาณิชย์ ชาติตะ)               |
| Exam weight | <b>6.43%</b> of the course · <b>2 exam hours</b> · สอบกลางภาค |

★ **This is examined material, not an optional topic.** It carries the same midterm weight as Drugs acting on GABA-A receptor and as Antipsychotics.

## WHAT TO DO WHEN THE SLIDES LAND

Drop the PDF in `Matcha/matcha-inbox/`. The pipeline is: `convert_corpus.py` → this file gets rewritten as a real drug-class page → rebuild both PDFs.

**Nothing else changes** — the slug `drug-class-cns-stimulants` is already wired into the builder's SYLLABUS list in taught position, so the page order, the สารบัญ and the bookmarks all stay put.

*Principle behind this: `[[pharm-cns-basics]]` — Amphetamine and methamphetamine as releaser + reuptake inhibitor, with the CNS/PNS sum spelled out. (Page still a placeholder pending slides.)\*\**

*Source: none — slides not yet released by the lecturer. Placeholder written 2026-08-02 from `Syllabus Pharmacology 3_2569.pdf`.*

# Drug Abuse and Addiction

⚠️ **This page is a placeholder. There is no content here yet — nothing below is missing by accident.**

## WHY THIS PAGE IS EMPTY

The lecture slides for this topic **have not been uploaded to Microsoft Teams** (team code `gwwtmy3`) as of **2 August 2026**. Every other week's slides in this course came from อ.พงศธร, who posted the whole set; this topic belongs to a different lecturer who hasn't posted yet.

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| <b>Week</b>        | 4 — Wednesday <b>26 ส.ค. 69</b> , 10:00–12:00, R207           |
| <b>Lecturer</b>    | อ.อนันต์ชัย (อ.ดร.อนันต์ชัย อัสวเมธิน)                        |
| <b>Exam weight</b> | <b>6.43%</b> of the course · <b>2 exam hours</b> · สอบกลางภาค |

★ **This is examined material, not an optional topic.** Same midterm weight as Drugs acting on GABA-A receptor and as Antipsychotics.

## WHAT TO DO WHEN THE SLIDES LAND

Drop the PDF in `Matcha/matcha-inbox/`. The pipeline is: `convert_corpus.py` → this file gets rewritten as a real drug-class page → rebuild both PDFs.

**Nothing else changes** — the slug `drug-class-drug-abuse-addiction` is already wired into the builder's SYLLABUS list in taught position, so the page order, the สารบัญ and the bookmarks all stay put.

*Principle behind this: `[[pharm-cns-basics]]` — Mesolimbic DA → euphoria → addiction is the mechanism that page is about. (Also a placeholder.)\*\**

*Source: none — slides not yet released by the lecturer. Placeholder written 2026-08-02 from `Syllabus Pharmacology 3_2569.pdf`.*

# Anxiolytics and Hypnotics

Two problems, one lecture, one overlapping drug list. **Anxiety** = an over-driven **amygdala** producing fear, worry and autonomic symptoms. **Insomnia** = a **sleep-wake switch** that won't flip. The GABA-A drugs sit in both columns — their receptor pharmacology is on [[drug-class-gaba-a]] and is not repeated here.

**Cortical glutamate/NMDA input → amygdala output ↑ → fear + worry + autonomic symptoms — braked by GABA-A interneurons · 5-HT1A from raphe nuclei · N-type  $\text{Ca}^{2+}$  channels**

## DISORDERS OF FEAR AND WORRY — THREE FAMILIES

- **Anxiety disorders** — GAD · social anxiety disorder (social phobia) · specific phobia · panic disorder · separation anxiety disorder · selective mutism
- **Trauma- and stress-related** — **PTSD** · acute stress disorder
- **Obsessive-compulsive and related** — **OCD** · trichotillomania · hoarding disorder

*Fear* → phobia, panic. *Worry* → anxious misery, apprehensive expectation, obsession. Circuitry: **amygdala + cortico-striato-thalamo-cortical (CSTC) pathways**.

## WHAT ANXIETY ACTUALLY LOOKS LIKE

| SYMPTOM GROUP        | FEATURES                                                                                                                                                               |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mood</b>          | anxious mood, fear                                                                                                                                                     |
| <b>Autonomic</b>     | abdominal cramp, stomachache, peptic ulcer, nausea, vomiting, urination, defecation, bradycardia · palpitation, tremor, sweating, <b>BP elevation</b> , pupil dilation |
| <b>Other somatic</b> | medically unexplained chest pain · freezing, facial expression of fear                                                                                                 |

## ★ SIX WAYS TO TREAT ANXIETY — AND WHERE EACH ONE HITS

| GROUP                                                    | TARGET                                                   | ONSET            |
|----------------------------------------------------------|----------------------------------------------------------|------------------|
| <b>Antidepressants</b> — TCA · SSRI · SNRI               | ↑ serotonin ( <i>reuptake inhibition</i> )               | <b>1–2 weeks</b> |
| <b>Benzodiazepines</b>                                   | GABA-A <b>APL</b>                                        | <b>quick</b>     |
| <b>Pregabalin</b>                                        | blocks <b>N-type <math>\text{Ca}^{2+}</math> channel</b> |                  |
| <b>Etifoxine</b>                                         | neurosteroid site on GABA-A → see [[drug-class-gaba-a]]  |                  |
| <b>Buspirone</b>                                         | <b>5-HT1A</b> partial agonist                            | <b>1–2 weeks</b> |
| <b>Others</b> — $\beta$ -blockers ( <i>propranolol</i> ) | autonomic output                                         |                  |

*Part of the target is the psychosomatic manifestation of anxiety — autonomic nervous system imbalance — rather than the mood itself.*

# Antidepressants for Anxiety

## MECHANISM

Block reuptake of **serotonin** (SSRI), **serotonin + noradrenaline** (SNRI, TCA), or block its breakdown (**MAOI**) — raising 5-HT tone onto the amygdala circuit.

→ **Downstream:** anxiolysis with a **1–2 week onset** ★ — nothing happens on day one, which is why a BZP is often run alongside at the start and withdrawn once these take hold

## DRUGS — FDA-APPROVED / OFF-LABEL ANXIETY INDICATIONS

| DRUG                                                                                                | WHAT IT TREATS                                |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Fluoxetine</b> (SSRI)                                                                            | PD · <i>off-label</i> : GAD, SAD              |
| <b>Sertraline</b> (SSRI)                                                                            | PD, SAD · <i>off-label</i> : GAD              |
| <b>Citalopram</b> (SSRI)                                                                            | <i>off-label only</i> : GAD, PD, SAD          |
| <b>Escitalopram</b> (SSRI)                                                                          | GAD · <i>off-label</i> : PD, SAD              |
| <b>Paroxetine</b> (SSRI)                                                                            | <b>PD, SAD, GAD</b> — the broadest label      |
| <b>Fluvoxamine</b> (SSRI)                                                                           | <i>off-label only</i> : GAD, PD, SAD          |
| <b>Duloxetine</b> (SNRI)                                                                            | GAD · <i>off-label</i> : PD, SAD              |
| <b>Venlafaxine XR</b> (SNRI)                                                                        | GAD · <i>off-label</i> : PD, SAD              |
| <b>Desvenlafaxine</b> (SNRI)                                                                        | <i>off-label only</i> : GAD, PD, SAD          |
| <b>Clomipramine · Imipramine · Desipramine · Nortriptyline</b> (TCA)                                | <i>off-label only</i> : GAD, PD, SAD          |
| <b>Phenelzine</b> (MAOI)                                                                            | <i>off-label only</i> : GAD, PD, SAD          |
| <b>Mirtazapine</b> (5-HT <sub>2</sub> , 5-HT <sub>3</sub> , $\alpha_2$ , H <sub>1</sub> antagonist) | <i>off-label only</i> : anxiety, GAD, PD, SAD |

## ADMINISTRATION — THERAPEUTIC DOSE RANGES IN ANXIETY (MG/DAY)

| CLASS                  | DRUG                                                                                           | DOSE RANGE                                                      |
|------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>SSRIs</b>           | fluoxetine · sertraline · citalopram · escitalopram · paroxetine · paroxetine ER · fluvoxamine | 20–60 · 50–200 · 20–40 · <b>10–20</b> · 20–60 · 27–75 · 100–300 |
| <b>SNRIs</b>           | duloxetine · venlafaxine XR · desvenlafaxine                                                   | 30–60 · <b>75–300</b> · 50–100                                  |
| <b>TCA</b> s           | clomipramine · imipramine · desipramine · nortriptyline                                        | 100–250 · 100–300 · 100–200 · 50–150                            |
| <b>MAOIs</b>           | phenelzine                                                                                     | 30–90                                                           |
| <b>Mixed</b>           | mirtazapine                                                                                    | 15–45                                                           |
| <b>GABAergic</b>       | pregabalin · gabapentin                                                                        | 150–600 · <b>600–2,400</b>                                      |
| <b>Benzodiazepines</b> | clonazepam · alprazolam · lorazepam · chlordiazepoxide · oxazepam                              | <b>1–2</b> · 1–4 · 2–6 · 20–100 · 30–60                         |
| <b>Antipsychotics</b>  | trifluoperazine · olanzapine · quetiapine                                                      | 2–6 · 5–15 · 50–300                                             |
| <b>β-blockers</b>      | propranolol                                                                                    | 60–120                                                          |

- ★ **Most of this table is off-label.** FDA approval for an anxiety disorder is the exception — the TCAs, MAOIs, mirtazapine, gabapentinoids and most antipsychotics here carry none, yet all have real dose ranges in use.
- 💡 The spread between classes is enormous — escitalopram at 10–20 mg against gabapentin at up to 2,400 mg. Dose size says nothing about potency; it is a property of the molecule, not the effect.

# Benzodiazepines in Anxiety

## MECHANISM

**GABA-A positive allosteric modulator (APL)** at the BZ site — full receptor detail on [[drug-class-gaba-a]].

→ **Downstream:** anxiolysis with a **quick onset** — the reason they cover the antidepressant lag · **PTSD** and **OCD** are less responsive to BZPs ★

## DRUGS

| DRUG                    | WHAT IT TREATS                                      |
|-------------------------|-----------------------------------------------------|
| <b>Clonazepam</b>       | <b>PD</b> · <i>off-label:</i> anxiety, GAD, PD, SAD |
| <b>Alprazolam</b>       | anxiety, <b>PD</b> · <i>off-label:</i> GAD, PD, SAD |
| <b>Lorazepam</b>        | anxiety · <i>off-label:</i> GAD, PD, SAD            |
| <b>Chlordiazepoxide</b> | anxiety · <i>off-label:</i> GAD, PD, SAD            |
| <b>Oxazepam</b>         | anxiety · <i>off-label:</i> GAD, PD, SAD            |

## CAUTIONS

- **Rebound anxiety on discontinuation** ★ — anxiety overshoots its pre-treatment level when the drug stops. Antidepressants do not show this rebound on the same time course.
- ⚠ Not the answer for **PTSD** or **OCD**

# Pregabalin & Gabapentin

## MECHANISM

Pregabalin is taught as a **blocker of the N-type calcium channel** on the amygdala circuit — one of the three brakes on that pathway. ⚠️ The reference table in the same lecture is more careful: mechanism "**unclear, may modulate  $\text{Ca}^{2+}$  channels.**" Both statements are the lecture's.

→ **Downstream:** ↓ presynaptic  $\text{Ca}^{2+}$  entry → ↓ excitatory transmitter release → anxiolysis

## DRUGS

| DRUG              | WHAT IT TREATS                 |
|-------------------|--------------------------------|
| <b>Pregabalin</b> | <i>off-label:</i> GAD, SAD     |
| <b>Gabapentin</b> | <i>off-label:</i> GAD, SAD, PD |



# Buspirone

## MECHANISM

**5-HT<sub>1A</sub> receptor partial agonist** — acts on the serotonergic input the raphe nuclei send into the amygdala, not on GABA-A.

→ **Downstream:** anxiolysis with a **1–2 week onset** ★ — slow like an antidepressant, so it is not a rescue drug

## DRUGS

| DRUG             | WHAT IT TREATS                         |
|------------------|----------------------------------------|
| <b>Buspirone</b> | <b>anxiety</b> · <i>off-label:</i> GAD |

# Other Agents for Anxiety

## DRUGS

| DRUG                                                                | WHAT IT TREATS                                                               |
|---------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Propranolol</b> ( $\beta 1$ , $\beta 2$ antagonist)              | <i>off-label</i> : anxiety, PD, SAD — aimed at the <b>autonomic</b> symptoms |
| <b>Hydroxyzine</b> ( $H1$ antagonist)                               | <b>anxiety</b> · <i>off-label</i> : GAD, PD, SAD                             |
| <b>Trifluoperazine</b> ( $D2$ antagonist)                           | <b>anxiety</b> · <i>off-label</i> : GAD, PD, SAD                             |
| <b>Olanzapine · Quetiapine</b> ( $D2$ , $5-HT_2$ , $H1$ antagonist) | <i>off-label</i> : anxiety, GAD                                              |
| <b>Etifoxine</b>                                                    | anxiety — neurosteroid-site APL, see [[drug-class-gaba-a]]                   |

# Insomnia & Sleep Architecture

## CLASSIFYING INSOMNIA — THREE AXES

- **By cause** — primary · secondary (e.g. obstructive sleep apnea)
- **By onset** ★ — **early** = sleep onset · **late** = sleep maintenance. *This is the axis that picks the drug: short-acting for early, intermediate for late.*
- **By time** — transient / short-term · chronic

## WHAT THE TWO SLEEP STAGES ARE FOR

- **NREM N3–4** — **restorative** sleep
- **REM** — **dreaming** · **learning and memory** ★

*Which is why a hypnotic that suppresses N3–4 or REM buys sleep time and loses sleep quality.*

## THE HYPNOTIC DRUG GROUPS

| GROUP                                      | DRUGS                                                           |
|--------------------------------------------|-----------------------------------------------------------------|
| <b>Benzodiazepines</b>                     | see [[drug-class-gaba-a]]                                       |
| <b>Z-drugs</b>                             | zolpidem                                                        |
| <b>Melatonin and analogues</b>             | melatonin · ramelteon                                           |
| <b>Orexin receptor antagonists (DORAs)</b> | lemborexant · suvorexant · daridorexant                         |
| <b>Chloral hydrate</b>                     |                                                                 |
| <b>Sedating antihistamines</b>             | hydroxyzine · diphenhydramine                                   |
| <b>Sedating antidepressants</b> (low dose) | amitriptyline · doxepin · trazodone · mirtazapine · agomelatine |
| <b>Sedating antipsychotics</b> (low dose)  | quetiapine · olanzapine                                         |

# The Sleep–Wake Switch

## FLIP-FLOP SWITCH MODEL

Two mutually inhibiting camps. The **sleep-promoting centre** — **VLPO** (ventrolateral preoptic nucleus, **GABA**) — versus the **arousal centres**, which project to thalamus/cortex. Whichever side wins suppresses the other, so the system snaps between states instead of drifting.

**Melatonin** is the **opener of the sleep gate**: it triggers GABA release from VLPO, which inhibits the monoaminergic nuclei. **Orexin** (from LH) does the opposite — it activates most of them and holds wakefulness.

→ **Downstream**: every hypnotic works on one of three handles — **strengthen VLPO GABA** (BZP, Z-drugs, via GABA-A) · **open the gate earlier** (melatonin, ramelteon) · **cut the orexin drive** (DORAs)

## THE AROUSAL CENTRES

| NUCLEUS                                                    | TRANSMITTER      | RECEPTOR(S)                                                                |
|------------------------------------------------------------|------------------|----------------------------------------------------------------------------|
| <b>LH</b> — lateral hypothalamus                           | <b>orexin</b>    | OX1, OX2                                                                   |
| <b>Raphe</b> nuclei                                        | 5-HT             | wake: various · <b>REM-off: 5-HT<sub>2A/2C</sub></b>                       |
| <b>LC</b> — locus coeruleus                                | NE               | wake: various · <b>REM-off: <math>\alpha</math>1, <math>\alpha</math>2</b> |
| <b>TMN</b> — tuberomamillary nucleus                       | <b>histamine</b> | H1                                                                         |
| <b>VTA</b> — ventral tegmental area                        | DA               | D1, D2                                                                     |
| <b>NBM</b> — nucleus basalis of Meynert                    | ACh              | M1                                                                         |
| <b>PPT/LDT</b> — pedunculopontine / laterodorsal tegmental | ACh              | M1                                                                         |

vs. **VLPO** — the sleep-promoting centre, **GABA**, acting through **GABA-A**.

## FIRING ACROSS THE SLEEP CYCLE ★

- **Raphe (5-HT), LC (NE), TMN (His)** — fire in wakefulness, fall to **zero in REM**. These are the **REM-off** neurons.
- **PPT/LDT (ACh)** — the **REM-on** group; its silent window is NREM, not REM.
- **TMN histamine** → **H1** is the receptor every sedating antihistamine is switching off.

# Melatonin & Melatonin Receptor Agonists

## MECHANISM

A **neurohormone from the pineal gland**, essential for **circadian rhythm**. Binds **MT1 and MT2**, triggering **GABA release from the VLPO** → opening of the **sleep gate**.

→ **Downstream**: sleep onset shifted earlier, circadian rhythm re-anchored — **not** sedation. Endogenous secretion peaks **02:00–04:00** with a **1.5–2 h lag** from onset of release, and is **reduced in the elderly** ★ — which is exactly who the drug is for.

## DRUGS

| DRUG                                          | WHAT IT TREATS                         |
|-----------------------------------------------|----------------------------------------|
| <b>Melatonin</b> ( <i>prolonged release</i> ) | <b>primary insomnia in the elderly</b> |
| <b>Ramelteon</b>                              | insomnia — MT1/MT2 agonist             |

## WHY IT BEHAVES DIFFERENTLY FROM A BZP

- **Less sedative effect**
- **Less tolerance and withdrawal**
- PK: short  $t_{1/2}$  1–2 h; ⚠ **bioavailability is poor and variable**

## PATIENT INFORMATION ★

- Take **1–2 h before** the desired sleep time
- Use it at a **regular time** every day
- ⚠ May take **days to weeks** to shift the circadian rhythm — it is not a "tonight" drug

# Orexin Receptor Antagonists (DORAs)

## MECHANISM

Orexin binds **OX1/OX2** and activates most of the monoaminergic arousal nuclei — that is what **maintains wakefulness**. ● *Loss of orexin neurons causes narcolepsy*. DORAs block **both** receptors instead of pushing GABA.

→ **Downstream**: arousal drive removed → sleep — a fundamentally different route from every GABA-A drug

## DRUGS

| DRUG                | WHAT IT TREATS |
|---------------------|----------------|
| <b>Lemborexant</b>  | insomnia       |
| <b>Suvorexant</b>   | insomnia       |
| <b>Daridorexant</b> | insomnia       |

## ★ ADVANTAGES OVER BENZODIAZEPINES / Z-DRUGS

|                                   | DORAS                                                    |
|-----------------------------------|----------------------------------------------------------|
| <b>Sleep architecture</b>         | <b>less N3 and REM suppression</b> — may even enhance it |
| <b>Tolerance &amp; withdrawal</b> | less associated                                          |
| <b>Respiratory depression</b>     | <b>low risk</b>                                          |
| <b>Motor side effects</b>         | none                                                     |

## PATIENT ADVICE

- Take it and **go to bed immediately**
- ⚠ Allow at least **7 hours** of sleep time before you need to be up

# Prescribing & Patient Information

## APPLIES TO EVERY HYPNOTIC ★

- Take it at the **appropriate time before** the desired sleep time — this varies by drug
- ● **Avoid driving or operating heavy machinery**
- Allow a **sufficient night** — **at least 6–8 h**. Too short a window is what produces the hangover.

## DRUGS WITH A US FDA INDICATION FOR INSOMNIA — METABOLISM & HALF-LIFE

| DRUG               | METABOLISM                                       | T <sub>½</sub> (H)                            |
|--------------------|--------------------------------------------------|-----------------------------------------------|
| <b>Estazolam</b>   | CYP3A4                                           | 10–24                                         |
| <b>Flurazepam</b>  | CYP3A4 ( <i>probable</i> )                       | ● <b>47–100</b> ( <i>active metabolite</i> )  |
| <b>Temazepam</b>   | <b>glucuronide conjugation</b> — no CYP step     | 3.5–18.4                                      |
| <b>Triazolam</b>   | CYP3A4                                           | 1.5–5.5                                       |
| <b>Quazepam</b>    | CYP3A4 ( <i>minor: 2C9, 2C19</i> )               | ● <b>39–73</b> ( <i>active metabolite</i> )   |
| <b>Zolpidem</b>    | CYP3A4, CYP2C9                                   | 1.4–4.5                                       |
| <b>Zaleplon</b>    | <b>aldehyde oxidase</b> ( <i>minor: CYP3A4</i> ) | <b>1</b>                                      |
| <b>Eszopiclone</b> | CYP3A4, CYP2E1                                   | 6                                             |
| <b>Ramelteon</b>   | CYP1A2 ( <i>minor: 2C, 3A4</i> )                 | 1–2.6                                         |
| <b>Doxepin</b>     | CYP2C19, CYP2D6 ( <i>minor: 1A2, 2C9</i> )       | 15.3–31 ( <i>parent + active metabolite</i> ) |

★ CYP3A4 dominates this list — so any 3A4 inhibitor turns a short hypnotic into a long one. The exceptions are temazepam (glucuronidation), zaleplon (aldehyde oxidase) and ramelteon (CYP1A2). The long half-lives with active metabolites — flurazepam, quazepam — are the daytime-hangover drugs.

Principle behind this: *[[pharm-cns-basics]]* — Orexin's projections to TMN, RN, VTA, LC and PPT/LDT — the mechanism behind orexin antagonists; melatonin and GABA on the opposing side.

Source: *Week05 Pharmacology of Hypnotics and Anxiolytics.pdf* — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สม), *Pharmacology 3 (ภกปว 401)*, 25 slides. Structures omitted.

# Antidepressants and Lithium

Every antidepressant class does the same thing at the top of the chain — **raise monoamine tone** — and differs almost entirely in **what else it hits on the way**. The receptors a drug blocks *by accident* are its side-effect profile. Learn the receptor map once and every class falls out of it.

↑ **synaptic monoamine** → **postsynaptic receptor activation** → **CREB** → **BDNF / TrkB** → **Bcl-2** → **neurogenesis & synaptogenesis in hippocampus + prefrontal cortex** → **antidepressant effect**  
(2–3 weeks)

## MAJOR DEPRESSIVE DISORDER — THE SYMPTOM SET

**Core:** depressed mood · loss of interest/pleasure **Plus:** fatigue · concentration · sleep · psychomotor change · guilt/worthlessness · appetite/weight · **● suicidality**

## FOUR HYPOTHESES OF WHAT DEPRESSION IS

| HYPOTHESIS                    | THE CLAIM                                          |
|-------------------------------|----------------------------------------------------|
| <b>Monoaminergic</b>          | too little monoamine transmission                  |
| <b>Signalling adaptation</b>  | the post-receptor signalling cascade is maladapted |
| <b>Neurotrophic</b>           | loss of <b>BDNF</b> -driven neuroplasticity        |
| <b>Chronic stress-induced</b> | sustained stress drives the above                  |

*The three monoamine sources: locus coeruleus (NE) · raphe nuclei (5-HT) · ventral tegmental area (DA).*

## ★ THE 8 CLASSES

| # | CLASS                | CORE MECHANISM                                                       |
|---|----------------------|----------------------------------------------------------------------|
| 1 | <b>MAOIs / RIMAs</b> | block monoamine oxidase                                              |
| 2 | <b>TCAs</b>          | block 5-HTT + NET — <i>and everything else</i>                       |
| 3 | <b>SSRIs</b>         | block 5-HTT selectively                                              |
| 4 | <b>SARIs</b>         | block 5-HTT + <b>antagonise 5-HT<sub>2A/2C</sub></b>                 |
| 5 | <b>SNRIs</b>         | block 5-HTT + NET, cleanly                                           |
| 6 | <b>NDRIs</b>         | block NET + DAT ( <i>weak</i> )                                      |
| 7 | <b>NaSSAs</b>        | block <b>α<sub>2</sub></b> autoreceptor + 5-HT <sub>2A/2C</sub> , H1 |
| 8 | <b>Miscellaneous</b> | melatonergic · NRI · SPARI · multimodal · NMDA antagonist            |



# Why Antidepressants Take Weeks

## MECHANISM

Reuptake blockade is **immediate** — the clinical effect is not. Sustained elevation of monoamines activates postsynaptic receptors → **CREB** → transcription of **BDNF** and its receptor **TrkB** → **Bcl-2** → **neurogenesis and synaptogenesis** of hippocampal and cortical neurons.

→ **Downstream:** ★ **onset 2–3 weeks** — the delay is the time it takes to build new synapses, not the time it takes to raise 5-HT. This is why the pharmacology and the therapeutics disagree on day one.

# MAO Inhibitors (MAOIs and RIMAs)

## MECHANISM

Block **monoamine oxidase**, the enzyme that degrades monoamines, so more transmitter survives in the terminal. Two generations:

1. **Irreversible, non-selective** — permanent inactivation of both MAO-A and MAO-B
2. **RIMA — reversible inhibitor of MAO-A** — selective for MAO-A (*at low dose*) and displaceable  
→ **Downstream:** ↑ 5-HT, NE, DA · ● **cheese reaction** with the irreversible drugs — dietary **tyramine** is normally destroyed by gut MAO; without it, tyramine floods in and triggers a hypertensive crisis. Reversibility is what buys the RIMAs safety here.

## DRUGS

| DRUG                                           | WHAT IT TREATS                           |
|------------------------------------------------|------------------------------------------|
| <b>Tranylcypromine</b> ( <i>irreversible</i> ) | depression                               |
| <b>Iproniazid</b> ( <i>irreversible</i> )      | depression                               |
| <b>Moclobemide</b> ( <i>RIMA</i> )             | depression — MAO-A selective at low dose |

## ADVERSE EFFECTS

- **Cheese (tyramine) reaction** ★ — irreversible, non-selective agents
- Lower seizure threshold — **less likely** than TCAs
- **MAOI + linezolid** → **serotonin syndrome**

# Tricyclic Antidepressants (TCAs)

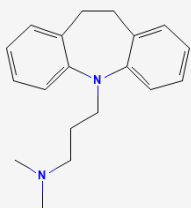
## MECHANISM

The **dirty** class — it blocks the therapeutic targets and four families of bystander targets at once:

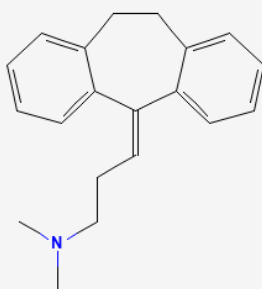
- **5-HTT, NET** — the antidepressant action
- **H1, muscarinic, 5-HT2A,  $\alpha$ 1** — the side-effect action
- **Voltage-gated  $\text{Na}^+$  channels** and **cardiac  $\text{K}^+$  (hERG)** — the lethal action
- **NMDA receptors**

→ **Downstream:** every TCA adverse effect is predictable from which of these it hits hardest ★

## THE SHARED CORE · WHAT CHANGES

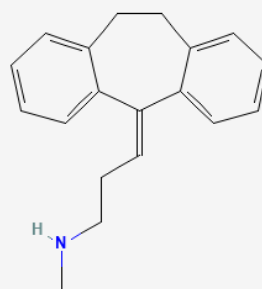


**Imipramine — the dibenzazepine prototype.** three fused rings with a basic side chain. ● the shape resembles the phenothiazine antipsychotics, which is exactly why TCAs also hit H1, M and  $\alpha$ 1 — the side-effect profile is structural



**Amitriptyline**

tertiary amine → more serotonergic, more sedating, more anticholinergic



**Nortriptyline**

one methyl removed (secondary amine) → more noradrenergic, better tolerated

## MEDCHEM

**SAR ★ The tricyclic ring must be bent, not flat.** A coplanar tricyclic has **no action**; the angle between the rings is what creates affinity for the noradrenaline and serotonin transporters. That same bulky three-ring geometry is also what drags in the antimuscarinic, antihistaminic and antipsychotic activity - the side-effect profile is inseparable from the pharmacophore, which is why the SSRIs were built by **breaking the tricycle open**.

**SAR ● The amitriptyline-vs-nortriptyline difference is one methyl group, and it is predictable.** A **tertiary** terminal amine gives 5-HT > NE selectivity *and* the strongest anticholinergic/antihistaminic load; a **secondary** amine gives NE > 5-HT and less of it. The side chain must be **three carbons** - branching it or enlarging the amine kills activity.

**Metabolism** The demethylation that converts a tertiary TCA to a secondary one is a real metabolic step, so a 3° drug partly becomes its own 2° relative in vivo. Cleared by multiple CYPs (N-demethylation, aromatic hydroxylation) with high protein binding - the pharmacokinetic half of the **narrow therapeutic index**.

**SAR** Historical note the deck makes explicit: **imipramine was derived from chlorpromazine**. The antidepressants and the phenothiazine antipsychotics are the same scaffold family, which is why their side effects rhyme.

## ★ RECEPTOR BLOCKED → EFFECT

| BLOCKED                                     | EFFECT                                  |
|---------------------------------------------|-----------------------------------------|
| <b>NET</b>                                  | antidepressant · palpitation            |
| <b>5-HTT</b>                                | antidepressant · nausea/vomiting        |
| <b>Muscarinic (M1, M2, M3)</b>              | anticholinergic effects · sedation      |
| <b>5-HT2A</b>                               | sedation · weight gain                  |
| <b>H1</b>                                   | sedation · weight gain                  |
| <b>α1</b>                                   | orthostatic hypotension                 |
| <b>Voltage-gated Na<sup>+</sup> channel</b> | proarrhythmic · <b>PR</b> prolongation  |
| <b>Cardiac K<sup>+</sup> (hERG)</b>         | proarrhythmic · <b>QTc</b> prolongation |

## DRUGS

| DRUG                                 | WHAT IT TREATS                                             |
|--------------------------------------|------------------------------------------------------------|
| <b>Amitriptyline</b>                 | depression · <b>migraine prevention</b>                    |
| <b>Nortriptyline</b>                 | depression · smoking cessation                             |
| <b>Clomipramine</b>                  | depression · <b>OCD</b>                                    |
| <b>Imipramine</b>                    | depression                                                 |
| <b>Maprotiline</b>                   | depression — ⚠️ notable seizure risk                       |
| <b>Doxepin</b>                       | depression · insomnia — a <b>TCA sleep aid</b> at low dose |
| <b>Melitracen</b> (with flupentixol) | depression                                                 |

## SIDE-EFFECT PROFILE OF THE COMMON TCAS

| AGENT                | SEDATION | ANTICHOLINERGIC | HYPOTENSION |
|----------------------|----------|-----------------|-------------|
| <b>Amitriptyline</b> | ++++     | ++++            | ++++        |
| <b>Clomipramine</b>  | +++      | +++             | +++         |
| <b>Imipramine</b>    | +++      | +++             | +++         |
| <b>Doxepin</b>       | +++      | ++              | ++          |
| <b>Nortriptyline</b> | ++       | ++              | +           |

+ mild · ++ moderate · +++ high affinity. ★ *Nortriptyline is the tolerable one; amitriptyline is maximal on all three.\*\**

## ADVERSE EFFECTS

- **Overdose causes sudden death** ★ — the Na<sup>+</sup>-channel and hERG blockade. This is the single fact that keeps TCAs out of first line.
- **Reduced seizure threshold**
- **QTc prolongation** — TCAs are the most likely class
- **Weight gain** — TCAs are among the most likely
- Anticholinergic load · sedation · orthostatic hypotension

### CAUTIONS — THE ELDERLY ★

More sensitive on every axis: **reduced cardiovascular reflex** (*falls from orthostatic hypotension*) · **sedation** · **cognitive impairment** · **constipation** · **benign prostatic hypertrophy** (*anticholinergic retention*).

- Metabolism: mostly **CYP2D6**, some also **CYP2C19** and **CYP1A2** — see [[drug-class-cyp450-inducers-inhibitors]]

# SSRIs

## MECHANISM

Block **5-HTT** with much **higher affinity and selectivity than TCAs** — and essentially nothing else. ⚠️ The one exception: **high-dose paroxetine has anticholinergic activity**.

→ **Downstream:** the entire SSRI side-effect profile is *serotonin acting where you didn't want it* — not off-target receptor blockade. Same molecule, many brain regions, many 5-HT receptor subtypes.

## DRUGS

| DRUG                                 | WHAT IT TREATS                                      |
|--------------------------------------|-----------------------------------------------------|
| <b>Fluoxetine</b>                    | depression — ★ the long-half-life one               |
| <b>Fluvoxamine</b>                   | depression                                          |
| <b>Paroxetine</b>                    | depression — ⚠️ <b>anticholinergic at high dose</b> |
| <b>Sertraline</b>                    | depression                                          |
| <b>Escitalopram</b> (and citalopram) | depression                                          |

The class shares its indications rather than splitting them: anxiety disorders · PTSD · OCD · eating disorders · PMDD · premature ejaculation · menopausal vasomotor symptoms — see Clinical Applications below.

## PHARMACOKINETICS

| DRUG                | ACTIVE METABOLITE                                     | T <sub>1/2</sub> PARENT | STEADY STATE     | KINETICS  | ↑ IN ELDERLY |
|---------------------|-------------------------------------------------------|-------------------------|------------------|-----------|--------------|
| <b>Fluoxetine</b>   | <b>norfluoxetine</b> ( <i>t</i> <sub>1/2</sub> 7–9 d) | <b>2–3 d</b>            | ● <b>30–60 d</b> | nonlinear | yes          |
| <b>Paroxetine</b>   | —                                                     | 21 h                    | 4–5 d            | nonlinear | yes          |
| <b>Sertraline</b>   | N-desmethylertraline (2–4 d)                          | 26 h                    | 4–5 d            | linear    | yes          |
| <b>Fluvoxamine</b>  | —                                                     | 15 h                    | 3 d              | nonlinear | no           |
| <b>Escitalopram</b> | —                                                     | 35 h                    | 7–8 d            | linear    | no           |

Protein binding >95% for fluoxetine, paroxetine, sertraline; ~77% fluvoxamine, ~80% escitalopram. Hepatic disease raises levels ×3 (fluoxetine), ×2 (sertraline, escitalopram), ×1.8 (paroxetine), ×1.6 (fluvoxamine). Renal disease only matters for sertraline (×1.5).

★ Fluoxetine's month-long steady state is why it is both the hardest to titrate and the least likely to cause discontinuation symptoms — it tapers itself.

## ADMINISTRATION — SSRI DOSING AND PK

|                                               | FLUOXETINE                              | PAROXETINE   | SERTRALINE                        | FLUVOXAMINE   | ESCITALOPRAM  |
|-----------------------------------------------|-----------------------------------------|--------------|-----------------------------------|---------------|---------------|
| <b>Typical dose (mg/day)</b>                  | <b>10–60</b>                            | <b>20–50</b> | <b>50–200</b>                     | <b>50–300</b> | <b>20–60</b>  |
| <b>Elimination half-life</b>                  | ● <b>2–3 days</b>                       | 21 h         | 26 h                              | 15 h          | 35 h          |
| <b>Active metabolite <math>t_{1/2}</math></b> | <b>norfluoxetine</b><br><b>7–9 days</b> | —            | N-desmethylsertraline<br>2–4 days | —             | —             |
| <b>Time to steady state</b>                   | ● <b>30–60 days</b>                     | 4–5 days     | 4–5 days                          | 3 days        | 7–8 days      |
| <b>Linearity</b>                              | nonlinear                               | nonlinear    | <b>linear</b>                     | nonlinear     | <b>linear</b> |
| <b>Protein binding</b>                        | > 95%                                   | > 95%        | > 95%                             | ≈77%          | ≈80%          |
| <b>Raised in the elderly</b>                  | yes                                     | yes          | yes                               | no            | no            |
| <b>Hepatic disease</b>                        | ×3                                      | ×1.8         | ×2                                | ×1.6          | ×2            |
| <b>Renal disease</b>                          | no change                               | no change    | ×1.5                              | no change     | no change     |

- **Fluoxetine's 30–60 days to steady state is the single most consequential number here.** A dose changed today is still changing levels two months later — and after stopping, it is still washing out for weeks, which is why it is the one that doesn't cause discontinuation syndrome and the one with the longest gap before an MAOI.
- ★ **Sertraline and escitalopram are the linear ones** — double the dose, double the level. With the nonlinear three, a dose increase can raise levels far more than expected.
- 💡 Hepatic disease multiplies exposure across the whole class (*fluoxetine* ×3); renal disease barely touches it except sertraline. Liver first, kidney second, when adjusting here.

# The 5-HT Effect Map

## ★ WHERE THE EXTRA SEROTONIN GOES — EVERY SSRI EFFECT IN ONE TABLE

| RECEPTOR  | REGION                          | EFFECT                                                |
|-----------|---------------------------------|-------------------------------------------------------|
| 5-HT3     | CTZ                             | nausea, vomiting                                      |
| 5-HT3     | GI tract                        | nausea, vomiting, <b>loose stool</b>                  |
| 5-HT3     | blood vessel                    | headache                                              |
| 5-HT2A    | nigrostriatal tract             | <b>akathisia</b> , mild EPS                           |
| 5-HT2A    | ventral horn of spinal cord     | ● <b>inhibits sexual reflex</b> — orgasm, ejaculation |
| 5-HT2A/2C | ventromedial hypothalamus (VMH) | promotes satiety                                      |
| 5-HT2A/2C | PVN, supraoptic nucleus         | <b>vasopressin release</b> → hyponatremia             |
| 5-HT2A/2C | reticular activating system     | <b>insomnia</b> , ↓ REM sleep                         |
| 5-HT2A    | tuberoinfundibular tract        | ↑ <b>prolactin</b> release                            |
| 5-HT4     | heart                           | palpitation                                           |
| 5-HT1A    | amygdala                        | ✓ <b>anxiolysis, anti-OCD</b>                         |
| 5-HT1A    | prefrontal cortex               | ✓ <b>antidepressant</b>                               |
| 5-HT1A    | dorsal horn of spinal cord      | ✓ inhibits pain transmission                          |

★ 5-HT1A is where the benefit lives; 5-HT2A and 5-HT3 are where the complaints live. This is the whole design brief for the newer classes — keep 5-HT1A, block 5-HT2A/2C and 5-HT3. Receptor groundwork: *[[drug-class-antihistamine-serotonin]]*.



# Serotonin Syndrome & Discontinuation

## TWO OPPOSITE FAILURE MODES

**Serotonin syndrome** = too much serotonin, acutely. **Discontinuation syndrome** = the brain adapted to too much and lost it too fast.

→ **Downstream:** one is an emergency, the other is unpleasant — do not confuse them ★

## ● SEROTONIN SYNDROME — CAUSES

- **Drug–drug interaction** — multiple serotonergic agents stacked
- **Overdose** of an SSRI or SNRI
- **Genetics**

## SEROTONIN DISCONTINUATION (WITHDRAWAL) SYMPTOMS

Occur on **abrupt stop after chronic use**. ★ **Not life-threatening.**

- Flu-like symptoms
- Insomnia · hyperarousal
- Nausea
- Imbalance
- Sensory disturbance

★ *Lower risk with fluoxetine — its long half-life is a built-in taper.*

# SARIs — Serotonin Antagonist and Reuptake Inhibitors

## MECHANISM

Block **5-HTT** (*the antidepressant part*) while **antagonising 5-HT<sub>2A/2C</sub>** (*removing the insomnia, sexual and EPS complaints of a plain SSRI*) — plus **H<sub>1</sub>** and **α<sub>1</sub>**.

→ **Downstream:** sedation and weight gain from H<sub>1</sub>/5-HT<sub>2A</sub> · orthostatic hypotension from α<sub>1</sub> · ★ **low-dose trazodone works as a sleep aid with no anticholinergic effect** — which is why it is the hypnotic of choice when anticholinergic load is the problem

## DRUGS

| DRUG             | WHAT IT TREATS                                                                              |
|------------------|---------------------------------------------------------------------------------------------|
| <b>Trazodone</b> | depression · <b>insomnia</b> ( <i>low dose</i> ) — see [[drug-class-anxiolytics-hypnotics]] |

## ADVERSE EFFECTS

- Sedation, weight gain
- Orthostatic hypotension and ● **priapism**
- EPS, delayed ejaculation

# SNRIs

## MECHANISM

Block **5-HTT and NET** — the TCA targets **without** the H1 / muscarinic /  $\alpha 1$  / Na<sup>+</sup>-channel baggage.

→ **Downstream:** everything an SSRI does, **plus** the noradrenergic layer

## WHAT THE NORADRENERGIC HALF ADDS

| SITE                                  | EFFECT                                                                |
|---------------------------------------|-----------------------------------------------------------------------|
| <b>CNS</b>                            | zest, energy                                                          |
| <b>RAS</b>                            | insomnia, ↓ REM sleep                                                 |
| <b>Dorsal horn of spinal cord</b>     | <b>inhibits pain transmission</b> ★ — the neuropathic-pain indication |
| <b>Paraventricular nuclei (PVN)</b>   | appetite suppression                                                  |
| <b>Periphery — ↑ sympathetic tone</b> | ↑ <b>BP</b> · constipation                                            |

## DRUGS

| DRUG                   | WHAT IT TREATS                                           |
|------------------------|----------------------------------------------------------|
| <b>Venlafaxine</b>     | depression · GAD                                         |
| <b>Duloxetine</b>      | depression · GAD · <b>neuropathic pain, fibromyalgia</b> |
| <b>Desvenlafaxine</b>  | depression · <b>menopausal vasomotor symptoms</b>        |
| <b>Milnacipran</b>     | depression · fibromyalgia                                |
| <b>Levomilnacipran</b> | depression                                               |

## INTERACTIONS

- **Duloxetine is a moderate CYP2D6 inhibitor**
- Hyponatremia and bleeding risk, as for SSRIs

# NDRI<sub>s</sub>

## MECHANISM

Block **NET** and, **weakly**, **DAT** — producing only a **40–50% rise in DA**. No serotonergic action at all.

→ **Downstream:** ★ **beneficial in patients whose complaint is cognitive or attentional** · sexual side effects **less than SSRIs and SNRIs** (*no 5-HT<sub>2A</sub> at the spinal ventral horn*)

## DRUGS

| DRUG             | WHAT IT TREATS                        |
|------------------|---------------------------------------|
| <b>Bupropion</b> | depression · <b>smoking cessation</b> |

## ADVERSE EFFECTS

- Sympathetic side effects, **insomnia**
- Reduced appetite
- ● **Reduces seizure threshold** ★ — the defining limitation
- Sexual side effects < **SSRIs, SNRIs** (*this is the selling point*)

## INTERACTIONS

- **Moderate-to-potent CYP2D6 inhibitor** — see [[drug-class-cyp450-inducers-inhibitors]]

# NaSSAs

## MECHANISM

The **indirect** route.  **$\alpha_2$  receptors** are the brake — as autoreceptors on locus coeruleus NE neurons and heteroreceptors on raphe 5-HT neurons. Blocking  $\alpha_2$  **releases the brake**, so both transmitters rise in hippocampus and PFC without touching a reuptake transporter.

Also block **H1, 5-HT<sub>2A</sub>, 5-HT<sub>2C</sub>** · **mirtazapine** additionally blocks **5-HT<sub>3</sub>** · **mianserin** additionally blocks  **$\alpha_1$** .

→ **Downstream:**  $\uparrow$  NE + 5-HT → neurogenesis, synaptogenesis → antidepressant effect · ★ blocking 5-HT<sub>2A/2C</sub> and 5-HT<sub>3</sub> pre-empts the insomnia, sexual and nausea complaints — the 5-HT that rises is steered onto **5-HT<sub>1A</sub>**

## DRUGS

| DRUG               | WHAT IT TREATS                                                          |
|--------------------|-------------------------------------------------------------------------|
| <b>Mirtazapine</b> | depression · <b>insomnia</b> ( <i>low dose</i> ) · appetite stimulation |
| <b>Mianserin</b>   | depression · insomnia ( <i>low dose</i> )                               |

## ADVERSE EFFECTS

- **Sedation, weight gain** ★ — mirtazapine is among the worst offenders for weight
- ✅ Low dose works as a sleep aid **with no anticholinergic effect**

# Miscellaneous Antidepressants

## FIVE ONE-DRUG CLASSES

| CLASS                                                 | DRUG                                | MECHANISM                                                                                                                 | WHY IT EXISTS                                                                         |
|-------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Melatonergic agonist + serotonergic antagonist</b> | <b>Agomelatine</b>                  | <b>MT1/MT2 agonist</b> · block 5-HT <sub>2A/2C</sub>                                                                      | ★ <b>low rate of adverse reactions, including weight gain</b>                         |
| <b>NRI</b>                                            | <b>Reboxetine</b>                   | block <b>NET</b>                                                                                                          | ⚠️ sympathetic and activating side effects are the concern                            |
| <b>SPARI</b>                                          | <b>Vilazodone</b>                   | block 5-HTT · <b>5-HT<sub>1A</sub> partial agonist</b>                                                                    | steers 5-HT onto the useful receptor                                                  |
| <b>Multimodal</b>                                     | <b>Vortioxetine</b>                 | block 5-HTT · 5-HT <sub>1A</sub> partial agonist · <b>5-HT<sub>3</sub>, 5-HT<sub>6</sub>, 5-HT<sub>7</sub> antagonist</b> | ★ multiple mechanisms → <b>improved cognitive domains in MDD</b>                      |
| <b>NMDA receptor antagonist</b>                       | <b>Ketamine</b> ( <i>low dose</i> ) | block <b>NMDA</b> → altered glutamate transmission → <b>activates mTOR</b> → rapid synapse formation                      | ● <b>antidepressant effect within 24 h of IV infusion, in treatment-resistant MDD</b> |

★ Ketamine is the exception to the whole 2–3 week rule — it builds the synapses directly via mTOR instead of going the slow monoamine → CREB → BDNF route.

# Cross-Class Adverse Effects

## ★ WHO CAUSES WHAT — THE COMPARISON THE EXAM WANTS

- **Weight gain** — most likely **TCAs**, **mirtazapine** >>> trazodone
- **QTc prolongation** — most likely **TCAs**
- **Lower seizure threshold** — most likely **TCAs**, **maprotiline**, **bupropion**; *less likely* MAOIs, SSRIs
- **Hyponatremia** (*especially the elderly*) — **SSRI / SNRI** ★ (*vasopressin release via 5-HT<sub>2A/2C</sub> at PVN/SO<sub>N</sub>*)
- **Increased bleeding risk** — **SSRI / SNRI**

# Metabolism & Drug Interactions

## PHARMACODYNAMIC — SEROTONIN SYNDROME ●

Stacking serotonergic drugs is the classic cause:

- SSRIs, SNRIs
- Tramadol, pethidine
- Ergotamine, triptans
- MAO inhibitor + linezolid

TCA + **cholinergic drugs** are the other PD pair — TCA + **bethanechol**, TCA + **cholinesterase inhibitors** (*opposing actions at muscarinic receptors*).

## PHARMACOKINETIC — WHAT ANTIDEPRESSANTS DO TO OTHER DRUGS

| DRUG                              | 1A2 | 2C9 | 2C19 | 2D6 | 3A4 |
|-----------------------------------|-----|-----|------|-----|-----|
| <b>Fluoxetine</b>                 | •   | +++ | ++   | +++ | +   |
| <b>Fluvoxamine</b>                | +++ | ++  | +++  | •   | ++  |
| <b>Paroxetine</b>                 | •   | •   | •    | +++ | •   |
| <b>Sertraline</b>                 | •   | •   | •    | +   | •   |
| <b>Escitalopram</b>               | •   | •   | •    | +   | •   |
| <b>Venlafaxine</b>                | •   | •   | •    | +   | +   |
| <b>Duloxetine</b>                 | •   | •   | •    | ++  | •   |
| <b>Desvenlafaxine</b>             | •   | •   | •    | •   | •   |
| <b>Bupropion</b>                  | ?   | ?   | ?    | +++ | ?   |
| <b>Mirtazapine · Vortioxetine</b> | •   | •   | •    | •   | •   |

• no/minimal (<20%) · + mild (20–50%) · ++ moderate (50–150%) · +++ substantial (>150%) increase in the plasma level of a co-administered substrate.

★ **Significant CYP2D6 inhibitors: fluoxetine · paroxetine · bupropion · duloxetine.** Fluoxetine inhibits most CYPs except 1A2. Fluvoxamine inhibits most CYPs except 2D6. ★ **Desvenlafaxine, mirtazapine and vortioxetine are the clean ones** — reach for these when the patient is already on a complicated regimen. Full CYP table: [\[\[drug-class-cyp450-inducers-inhibitors\]\]](#).



## WHAT CLEARS THE ANTIDEPRESSANT ITSELF

| DRUG                                          | MAJOR PATHWAY   | OTHER                         |
|-----------------------------------------------|-----------------|-------------------------------|
| <b>TCA</b> s                                  | CYP2D6, CYP2C19 | 3A4, 1A2                      |
| <b>Citalopram · Escitalopram</b>              | <b>CYP2C19</b>  | 3A4, 2D6                      |
| <b>Fluoxetine</b> (→ <i>norfluoxetine</i> )   | CYP2D6          | 2C9, 2C19, 3A4                |
| <b>Fluvoxamine</b>                            | CYP1A2, CYP2D6  |                               |
| <b>Paroxetine</b>                             | CYP2D6          | 3A4                           |
| <b>Sertraline</b>                             | <b>CYP2B6</b>   | 2C19, 2C9, 3A4, 2D6           |
| <b>Venlafaxine</b>                            | CYP2D6          | 3A4                           |
| <b>Duloxetine</b>                             | CYP2D6, CYP1A2  |                               |
| <b>Desvenlafaxine · Milnacipran</b>           | ★ <b>renal</b>  | UGT, 3A4                      |
| <b>Bupropion</b> (→ <i>hydroxybupropion</i> ) | <b>CYP2B6</b>   |                               |
| <b>Mirtazapine</b>                            | CYP2D6, CYP3A4  | 1A2, UGT                      |
| <b>Agomelatine</b>                            | <b>CYP1A2</b>   | 2C9, 2C19                     |
| <b>Reboxetine · Vilazodone · Trazodone</b>    | CYP3A4          |                               |
| <b>Vortioxetine</b>                           | CYP2D6          | 3A4, 2C19, 2C9, 2A6, 2C8, 2B6 |

★ *Desvenlafaxine and milnacipran are renally cleared — the two that dodge CYP interactions entirely, and the two that need a dose thought in renal impairment.*

# Clinical Applications of Antidepressants

## BEYOND DEPRESSION — INDICATION → DRUG OF CHOICE

| INDICATION                      | DRUGS                                                      |
|---------------------------------|------------------------------------------------------------|
| Major depressive disorder       | all classes                                                |
| Anxiety disorders               | SSRIs, venlafaxine, duloxetine                             |
| PTSD                            | SSRI                                                       |
| OCD                             | ★ clomipramine and SSRI                                    |
| Eating disorders                | binge eating — SSRIs · bulimia nervosa — SSRIs, TCAs       |
| Insomnia ( <i>sleep aid</i> )   | trazodone, mirtazapine, agomelatine, TCA                   |
| Smoking cessation               | ★ bupropion, nortriptyline                                 |
| Premenstrual dysphoric disorder | SSRIs                                                      |
| Neuropathic pain & fibromyalgia | SNRIs, TCAs                                                |
| Menopausal vasomotor symptoms   | SSRIs, desvenlafaxine                                      |
| Premature ejaculation           | SSRIs ( <i>the sexual side effect, used deliberately</i> ) |
| Migraine prevention             | ★ amitriptyline                                            |

Anxiety-specific detail lives on [[drug-class-anxiolytics-hypnotics]].

# Lithium

## MECHANISM

An **alkaline metal monovalent cation similar to sodium** — which is why the kidney handles it like  $\text{Na}^+$  and why every sodium-moving drug changes its level. A **mood stabiliser**: reduces the severity and shortens the duration of mood episodes in bipolar disorder. It works at **two levels at once** — neurotransmitter function and intracellular signalling — and has **both antimanic and antidepressive** effect.

→ **Downstream**: ★ **GSK-3 $\beta$  greatly reduced** · inositol, cAMP, protein kinase C reduced · **BDNF and Bcl-2 increased**, pro-apoptotic p53/BAX reduced — the same neurotrophic endpoint the antidepressants reach by a different road

## WHAT LITHIUM MOVES

| SYSTEM                               | EFFECT OF LITHIUM        |
|--------------------------------------|--------------------------|
| <b>5-HT function</b>                 | <b>greatly increased</b> |
| <b>Acetylcholinesterase function</b> | <b>greatly increased</b> |
| Sodium function · GABA function      | increased                |
| <b>Dopamine function</b>             | <b>reduced</b>           |
| Inositol · cAMP · protein kinase C   | reduced                  |
| <b>GSK-3<math>\beta</math></b>       | ● <b>greatly reduced</b> |
| <b>BDNF · Bcl-2</b>                  | increased                |
| Pro-apoptotic proteins (p53, BAX)    | reduced                  |

## DRUG

| DRUG           | WHAT IT TREATS                                                                                                                                                 |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lithium</b> | <b>acute manic episode</b> · prevention of recurrent manic and depressive episodes · <b>rapid cycling</b> · ★ <b>prevention of suicide in bipolar disorder</b> |

## ● **NARROW THERAPEUTIC INDEX** ★

- **Therapeutic level 0.6–1.2 mEq/L** — the number to remember
- PK: well absorbed into systemic and brain tissue · **no protein binding** · ★ **~95% excreted unchanged in urine** · clearance **depends on renal blood flow**
- **No metabolism at all** means no CYP interactions — every lithium interaction is **renal**

## **ADVERSE EFFECTS — BY SYSTEM**

- **GI** — nausea/vomiting, in the **first 2–3 weeks**
- **CNS** — **fine tremor (15–53%)** ★ · slow thinking, memory problems · somnolence · **weight gain — 20% of patients gain more than 10 kg**
- **Renal** — ● **inability to concentrate urine** → polyuria, polydipsia → **nephrogenic diabetes insipidus** ★
- **Endocrine** — **hypothyroidism**
- **CVS** — cardiac **T-wave inversion**
- **Cutaneous** — pruritic maculopapular rash, typically in the **first month**

# Lithium — Interactions & Pregnancy

## WHY EVERY INTERACTION IS A KIDNEY INTERACTION

Lithium is cleared unchanged by the kidney, so anything that changes **glomerular blood flow** or **tubular Na<sup>+</sup>/Li<sup>+</sup> handling** moves the level.

→ **Downstream:** ↓ GFR → lithium accumulates → toxicity · ↑ GFR → lithium is lost → loss of effect

## ● THE TWO MECHANISMS

### 1. Drugs that change glomerular blood flow

| DRUG                   | GLOMERULAR PRESSURE & GFR | LITHIUM LEVEL |
|------------------------|---------------------------|---------------|
| NSAIDs                 | reduced                   | ● ↑           |
| ACEIs, ARBs            | reduced                   | ● ↑           |
| Theophylline, caffeine | increased                 | ↓             |

### 2. Drugs/states that change Li<sup>+</sup> reabsorption

- **Diuretics — hydrochlorothiazide, loop diuretics** ★: ↑ Na<sup>+</sup> and Li<sup>+</sup> excretion → ↑ Na<sup>+</sup> and Li<sup>+</sup> reabsorption → ● ↑ **blood lithium**
- **Dehydration:** ↑ Na<sup>+</sup> and Li<sup>+</sup> reabsorption → ● ↑ **blood lithium**

Cross-refs: [\[\[drug-class-nsaids-prostanoids\]\]](#) · [\[\[drug-class-ace-arb-dri\]\]](#) · [\[\[drug-class-diuretics\]\]](#).

## ● PREGNANCY & LACTATION

- **Avoid in pregnant and breast-feeding women**
- ★ **Cardiac malformation — Ebstein's anomaly — at 0.05–0.1% with first-trimester exposure**
- Risk of **infant toxicity** through breast milk

Principle behind this: [\[\[pharm-pharmacogenomics\]\]](#) — TCAs, same CYP2D6 recommendation; and fluoxetine + haloperidol across CYP2D6 and CYP3A4 is the lecture's worked DDGGL.

Principle behind this: [\[\[pharm-cns-basics\]\]](#) — Amitriptyline's promiscuity (NET/SERT + α1, M1, H1, 5-HT2A + Na<sup>+</sup>/K<sup>+</sup> channels), fluoxetine at SERT, venlafaxine at both, mirtazapine as an α2/5-HT2A/H1 antagonist, and MAO-A vs MAO-B selectivity.

Source: [Week05-06 Pharmacology of Antidepressants and Lithium.pdf](#) — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สม), Pharmacology 3 (ภาค 401), 50 slides. Structures omitted.

# Antipsychotics

Every antipsychotic does one therapeutic thing — **turn down D2 signalling in the mesolimbic tract** — and the D2 receptor is in **four** tracts, not one. So the therapeutic effect and three of the four signature adverse effects are *the same drug action in a different place*. Everything else the molecule touches (**5-HT<sub>2A</sub> · H<sub>1</sub> · M<sub>1</sub> ·  $\alpha$ <sub>1</sub>**) is its **promiscuity**, and that is the rest of the side-effect profile. Learn the two axes and every agent on the list falls out of them.

**D2 blockade** → **mesolimbic**:  **positive symptoms** ↓ · **nigrostriatal**: **EPS** · **tuberoinfundibular**: **hyperprolactinemia** · **CTZ**: **antiemetic**

## THE DISEASE

**Psychosis / psychotic disorders** (โรคจิต, อาการทางจิต) · **Schizophrenia** (โรคจิตเภท) — defined by being **out of reality** and **lack of insight**.

*Obsolete terms for this drug group: neuroleptic drugs, major tranquillizers.*

## ★ WHY ONE DISEASE NEEDS DOPAMINE BOTH RAISED AND LOWERED

| PATHWAY                                                          | DOPAMINE STATE           | SYMPTOMS                                                                                                                      |
|------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Mesolimbic</b> (→ limbic, nucleus accumbens, D <sub>2</sub> ) | <b>hyperdopaminergia</b> | <b>Positive</b> — hallucination, delusion, paranoid, grandiose, hostility                                                     |
| <b>Mesocortical</b> (→ PFCx, D <sub>1</sub> )                    | <b>hypodopaminergia</b>  | <b>Negative</b> (apathy, social withdrawal) · <b>cognitive</b> (disorganized speech) · <b>affective</b> (anxiety, depression) |

★ *This split is the whole problem of the field: a pure D<sub>2</sub> antagonist fixes the hyperdopaminergic half and can do nothing for — or worsen — the hypodopaminergic half.*

## THE CLASSIFICATION

| GENERATION                             | SUBGROUP                                                                                      |
|----------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Conventional (first generation)</b> | <b>low potency</b> · <b>high potency</b>                                                      |
| <b>Atypical (second generation)</b>    | <b>serotonin–dopamine antagonists (SDAs)</b> · <b>D<sub>2</sub> partial agonists</b> · others |

# The D2 Map — One Receptor, Six Outcomes

## MECHANISM

D2 receptors sit in six places that matter, and a systemic D2 antagonist reaches all of them. There is no way to block one tract only — **selectivity is anatomical, and the drug has none.**

→ **Downstream:** one wanted effect (*mesolimbic*), two classic adverse effects (*nigrostriatal*, *tuberoinfundibular*), and three incidental ones ★

## ★ BRAIN AREA BLOCKED → WHAT YOU GET

| AREA                                    | PHARMACOLOGICAL EFFECT                             |
|-----------------------------------------|----------------------------------------------------|
| <b>Mesolimbic pathway</b>               | ✓ <b>antipsychotic</b> — positive symptoms         |
| <b>Mesocortical pathway</b>             | ? (as lectured — no established effect)            |
| <b>Nigrostriatal pathway</b>            | ● <b>motor</b> — extrapyramidal side effects (EPS) |
| <b>Tuberoinfundibular pathway</b>       | ● <b>hyperprolactinemia</b>                        |
| <b>Chemoreceptor trigger zone (CTZ)</b> | antiemetic                                         |
| <b>Reticular activating system</b>      | somnolence                                         |
| <b>Lateral hypothalamus</b>             | ↑ appetite                                         |

## ★ THE OTHER AXIS — RECEPTOR BLOCKED → EFFECT

| RECEPTOR                  | EFFECT OF BLOCKADE                                                                                                                            |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>H1</b>                 | sedation · weight gain                                                                                                                        |
| <b>M1</b> (central)       | interferes with memory and cognition · sedation · ✓ <b>anti-EPS</b>                                                                           |
| <b>M2–M4</b> (peripheral) | dry mouth, blurry vision, constipation, <b>urinary retention</b> (M3) · <b>tachycardia</b> (M2) · interferes with <b>insulin release</b> (M3) |
| <b>5-HT2A, 5-HT2C</b>     | possible ↑ appetite/weight · <b>improved REM sleep</b>                                                                                        |
| <b>α1</b>                 | postural hypotension, dizziness, syncope, nasal congestion, <b>priapism</b> · improved REM sleep                                              |

★ *Central M1 blockade is anti-EPS — which is why a drug that is dirty at muscarinic receptors gives you fewer movement problems and more anticholinergic ones. The two columns of the profile trade against each other. Receptor groundwork: [[drug-class-cholinergic]] · [[drug-class-adrenergic]] · [[drug-class-antihistamine-serotonin]].*

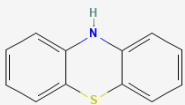
# Conventional (First-Generation) Antipsychotics

## MECHANISM

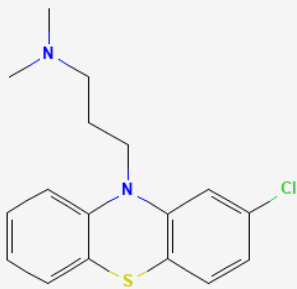
**D2 antagonism**, and that is the entire therapeutic story. Agents differ on two independent axes: **D2 blocking potency** and **degree of promiscuity** (*how much muscarinic,  $\alpha$ 1, H1 and 5-HT2A blockade comes along for the ride*).

→ **Downstream:** ★ the two axes run **opposite** to each other in practice — the high-potency agents give you EPS and prolactin, the low-potency agents give you sedation, anticholinergic load and orthostasis. Pick your poison, not your efficacy.

## THE SHARED CORE · WHAT CHANGES

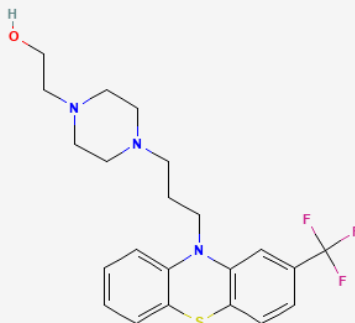


**Phenothiazine**, the parent tricycle. Potency and side effects both track the group at position 2 and the side chain at 10.



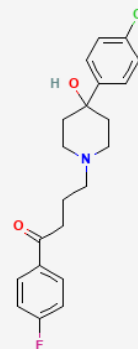
### Chlorpromazine

aliphatic chain → low potency, high sedation and  $\alpha$ -blockade



### Fluphenazine

piperazine +  $\text{CF}_3$  → high potency → more EPS, less sedation



### Haloperidol

● a butyrophenone — a different scaffold entirely, reaching the same D2 pocket

## D2 BLOCKING POTENCY

| POTENCY | DRUGS                                                                  |
|---------|------------------------------------------------------------------------|
| Low     | Chlorpromazine · Thioridazine                                          |
| Medium  | Perphenazine · Zuclopenthixol                                          |
| High    | Haloperidol · Flupenthixol · Fluphenazine · Pimozide · Trifluoperazine |

## DRUGS

| DRUG                   | WHAT IT TREATS                                                               |
|------------------------|------------------------------------------------------------------------------|
| <b>Chlorpromazine</b>  | psychotic disorders · <b>nausea/vomiting</b>                                 |
| <b>Thioridazine</b>    | psychotic disorders — ⚠️ most likely to prolong <b>QTc</b>                   |
| <b>Perphenazine</b>    | psychotic disorders                                                          |
| <b>Zuclopenthixol</b>  | psychotic disorders                                                          |
| <b>Haloperidol</b>     | psychotic disorders · nausea/vomiting — 🚫 <b>laryngospasm with injection</b> |
| <b>Flupenthixol</b>    | psychotic disorders                                                          |
| <b>Fluphenazine</b>    | psychotic disorders                                                          |
| <b>Pimozide</b>        | psychotic disorders                                                          |
| <b>Trifluoperazine</b> | psychotic disorders                                                          |

## ADVERSE EFFECTS — THE SIX THAT DEFINE THE CLASS

- 🚫 **Extrapyramidal side effects (EPS)** ⭐ — nigrostriatal D2
- 🚫 **Hyperprolactinemia** — tuberoinfundibular D2
- **Sedation** — H1, M1, RAS
- **Weight gain and cardiometabolic abnormalities** — H1, 5-HT2A/2C, lateral hypothalamus
- **Anticholinergic effects** — M2–M4
- **Orthostatic hypotension** —  $\alpha_1$



# Extrapyramidal Side Effects

## MECHANISM

Blocking nigrostriatal D2 removes dopaminergic inhibition of the striatum — the same lesion Parkinson's disease produces by killing the neuron instead of blocking the receptor.

→ **Downstream:** four syndromes, separated by **when** they appear — and the last one does not behave like the first three ★

## ★ THE FOUR EPS SYNDROMES

| ONSET       | TIMING         | DISORDER                    | DESCRIPTION                                                                            | TREATMENT                                                                                                                         |
|-------------|----------------|-----------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Acute       | hours–days     | <b>Akathisia</b>            | subjective feeling of motoric restlessness                                             | <b>propranolol</b> · benzodiazepines                                                                                              |
| Acute       | hours–days     | <b>Dystonia</b>             | sustained muscular contraction                                                         | <b>anticholinergics</b> ( <i>parenteral</i> )                                                                                     |
| Acute       | weeks          | <b>Parkinsonism</b>         | bradykinesia, rigidity, <b>symmetrical</b> tremor                                      | anticholinergics · <b>amantadine</b>                                                                                              |
| <b>Late</b> | 3 months–years | ● <b>Tardive dyskinesia</b> | late-onset choreiform, involuntary abnormal movements — commonly <b>mouth and face</b> | switch to another atypical · <b>valbenazine, deutetrabenazine</b> ·<br>⚠ <b>anticholinergics are ineffective or make it worse</b> |

Treatment cross-refs: *[[drug-class-adrenergic]]* (propranolol) · *[[drug-class-anxiolytics-hypnotics]]* · *[[drug-class-cholinergic]]*.

## ● ACUTE DYSTONIA — WHAT IT ACTUALLY LOOKS LIKE

- **Neck** — torticollis
- **Eyes** — oculogyric crisis
- **Tongue**
- **Larynx** — laryngospasm
- **Jaw**

★ Especially after injection of a high-potency D2 antagonist.

## ANTICHOLINERGICS USED TO MANAGE EPS

| DRUG                   | WHAT IT TREATS                     |
|------------------------|------------------------------------|
| <b>Benztropine</b>     | acute EPS — dystonia, parkinsonism |
| <b>Trihexyphenidyl</b> | acute EPS — dystonia, parkinsonism |
| <b>Diphenhydramine</b> | acute EPS — dystonia, parkinsonism |

## ● NEUROLEPTIC MALIGNANT SYNDROME (NMS)

A rare and severe form of EPS. Three features:

- **Altered consciousness**
- **Muscle rigidity**
- **Hyperthermia and rhabdomyolysis**

**Treatment:** discontinue the D2 antagonist · supportive treatment · give **bromocriptine** (*dopamine agonist*) and **dantrolene**.

# Hyperprolactinemia

## MECHANISM

Dopamine from the **tuberoinfundibular** tract is the physiological **brake** on the pituitary lactotroph. Block D2 there and the brake is released.

→ **Downstream:** ↑ prolactin → ↓ **estrogen** — and most of the damage comes from the low estrogen, not the high prolactin ★

## CONSEQUENCES

- **Breast** — gynecomastia, galactorrhea
- **Sexual function** — decreased libido, impaired arousal, delayed orgasm
- **Menstruation** — irregular menorrhea, amenorrhea, **infertility**
- ● **Bone** — decreased bone mineral density → **osteopenia, osteoporosis** (*the estrogen consequence, and the one that outlasts the drug*)

# Atypical (Second-Generation) Antipsychotics

## MECHANISM — THE SDAS

Block **5-HT<sub>2A</sub>** and **D<sub>2</sub>** in the same molecule, with **affinity 5-HT<sub>2A</sub> »»»» D<sub>2</sub>**. The trick: 5-HT<sub>2A</sub> sits on GABAergic interneurons that **inhibit** dopamine neurons, so **blocking 5-HT<sub>2A</sub> enhances dopamine release** — selectively where the dopamine tone is already the problem in reverse.

- **Mesolimbic** — D<sub>2</sub> blockade still reduces positive symptoms (*the 5-HT<sub>2A</sub> rescue does not undo it*)
- **Nigrostriatal** — restored DA release → ★ **less EPS**
- **Mesocortical** — restored DA at D<sub>1</sub> → **preserves or improves negative, cognitive and affective symptoms**

**Other DA-enhancing mechanisms at PFCx: 5-HT<sub>1A</sub> partial agonism** (*some SDAs*) → ↑ 5-HT, NE, DA and ACh in prefrontal cortex, and ↑ 5-HT in amygdala and hippocampus on chronic use · blockade of **5-HT<sub>2C</sub>, 5-HT<sub>6</sub>, 5-HT<sub>7</sub>** · blockade or partial agonism of **D<sub>3</sub>**.

→ **Downstream:** same antipsychotic effect, less motor cost, and a real answer to the negative symptoms

## ADVANTAGES OVER CONVENTIONAL AGENTS

**Similar** for positive symptoms · **less EPS**, and less or no need for anticholinergics · **preserve or improve cognitive symptoms** · **more effective for negative and affective symptoms**

## DRUGS

| DRUG                                                   | WHAT IT TREATS                                                                                    |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Risperidone</b> (SDA)                               | psychotic disorders — ⚠️ high prolactin for an atypical                                           |
| <b>Paliperidone</b> (SDA)                              | psychotic disorders — the <b>active metabolite of risperidone</b>                                 |
| <b>Clozapine</b> (SDA)                                 | psychotic disorders — 🚫 agranulocytosis, myocarditis, seizures                                    |
| <b>Olanzapine</b> (SDA)                                | psychotic disorders · bipolar · <b>sleep aid</b> (low dose)                                       |
| <b>Quetiapine</b> (SDA)                                | psychotic disorders · bipolar mania & depression · MDD · anxiety · <b>sleep aid</b> (low dose)    |
| <b>Ziprasidone</b> (SDA)                               | psychotic disorders — ⚠️ QTc                                                                      |
| <b>Lurasidone</b> (SDA)                                | psychotic disorders · <b>bipolar depression</b>                                                   |
| <b>Aripiprazole</b> (D2 partial agonist)               | psychotic disorders · bipolar · MDD adjunct                                                       |
| <b>Brexpiprazole</b> (D2 partial agonist)              | psychotic disorders · MDD adjunct                                                                 |
| <b>Cariprazine</b> (D2 partial agonist)                | psychotic disorders · bipolar depression                                                          |
| <b>Amisulpride</b> (combined D2/D3 antagonist)         | psychotic disorders — ⚠️ QTc                                                                      |
| <b>Lumateperone</b> (DA, 5-HT and glutamate modulator) | psychotic disorders                                                                               |
| <b>Pimavanserin</b> (5-HT <sub>2A</sub> antagonist)    | ★ <b>psychosis in Parkinson's disease</b> — the one that treats psychosis without a D2 antagonist |

Also marketed outside Thailand: iloperidone · molsaperidone · asenapine (SDAs).

# The Three Atypical Families

## ★ READ THE SUFFIX, PREDICT THE PROFILE

| FAMILY                                                            | CLASS                        | RECEPTOR SIGNATURE                                                                                                                                                  | WHAT THAT MEANS                                                                        |
|-------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>-done</b> ( <i>risperi-, paliperi-, ilo-, luras-, zipra-</i> ) | SDA                          | <b>higher D2</b> affinity than other atypicals<br>· <b>high <math>\alpha 1</math></b> · <b>low muscarinic</b>                                                       | most EPS + most prolactin of the atypicals · orthostasis · little anticholinergic load |
| <b>-pine</b> ( <i>cloza-, olanza-, quetia-, asena-</i> )          | SDA                          | <b>high 5-HT2A, 5-HT2C, H1, <math>\alpha 1</math></b> · <b>lower D2</b> · <b>moderate-high muscarinic</b> · 5-HT1A partial agonism ( <i>clozapine, quetiapine</i> ) | least EPS and least prolactin ·<br>● <b>sedation and weight gain are the price</b>     |
| <b>-praz</b> ( <i>aripip-, brexpip-, cari-</i> )                  | D2/D3 <b>partial agonist</b> | high D2, D3, 5-HT1A, 5-HT2A · <b>low <math>\alpha 1</math> and muscarinic</b>                                                                                       | cleanest profile across the board                                                      |

★ The suffix is not cosmetic — it tracks the chemical family, and the chemical family tracks the receptor fingerprint. This is the single highest-yield thing on the topic.

# D2 Receptor Partial Agonists

## MECHANISM

**Aripiprazole, brexpiprazole, cariprazine. Very high affinity for D2, but intrinsic activity below dopamine's — about 30–60%.** So the drug occupies the receptor and delivers a fixed, moderate signal regardless of what the endogenous dopamine is doing.

→ **Downstream:** ★ **functional antagonist where dopamine is high (mesolimbic → positive symptoms ↓), functional agonist where dopamine is low (mesocortical, nigrostriatal, tuberoinfundibular) —** which is why this family sits at the bottom of nearly every adverse-effect column

# Comparative Adverse Effects

## AXIS 1 — THE D2 CONSEQUENCES

| DRUG                | AKATHISIA | PARKINSONISM | DYSTONIA | TARDIVE DYSKINESIA | HYPERPROLACTINEMIA |
|---------------------|-----------|--------------|----------|--------------------|--------------------|
| Chlorpromazine      | ++        | ++           | ++       | +++                | +                  |
| Thioridazine        | +         | +            | +        | +                  | ++                 |
| Perphenazine        | ++        | ++           | ++       | ++                 | ++                 |
| <b>Fluphenazine</b> | +++       | +++          | +++      | +++                | +++                |
| <b>Haloperidol</b>  | +++       | +++          | +++      | +++                | +++                |
| <b>Pimozide</b>     | +++       | +++          | ++       | +++                | +++                |
| Trifluoperazine     | ++        | ++           | ++       | ++                 | ++                 |
| Lurasidone          | ++        | ++           | ++       | ++                 | +                  |
| Paliperidone        | ++        | ++           | ++       | ++                 | +++                |
| Risperidone         | ++        | ++           | ++       | ++                 | +++                |
| Ziprasidone         | ++        | +            | +        | +                  | ++                 |
| <b>Clozapine</b>    | +         | +            | +        | +                  | +                  |
| Olanzapine          | ++        | ++           | +        | +                  | ++                 |
| <b>Quetiapine</b>   | +         | +            | +        | +                  | +                  |
| <b>Aripiprazole</b> | ++        | +            | +        | +                  | +                  |
| Brexipiprazole      | ++        | +            | +        | +                  | +                  |
| Cariprazine         | ++        | +            | +        | +                  | +                  |

+ seldom · ++ sometimes · +++ often.

## AXIS 2 — THE PROMISCUITY CONSEQUENCES

| DRUG                                                                                               | ANTICHOLINERGIC | ORTHOSTASIS | SEDATION | WEIGHT GAIN |
|----------------------------------------------------------------------------------------------------|-----------------|-------------|----------|-------------|
| <b>Chlorpromazine</b>                                                                              | +++             | +++         | +++      | ++          |
| <b>Thioridazine</b>                                                                                | +++             | +++         | +++      | ++          |
| Perphenazine                                                                                       | ++              | ++          | ++       | ++          |
| Fluphenazine                                                                                       | +               | +           | +        | ++          |
| Haloperidol                                                                                        | +               | +           | +        | ++          |
| Pimozide                                                                                           | +               | +           | +        | +           |
| Trifluoperazine                                                                                    | ++              | +           | +        | ++          |
| Lurasidone                                                                                         | +               | +           | ++       | +           |
| Paliperidone                                                                                       | +               | ++          | +        | ++          |
| Risperidone                                                                                        | +               | ++          | ++       | ++          |
| Ziprasidone                                                                                        | +               | ++          | ++       | +           |
|  <b>Clozapine</b> | +++             | +++         | +++      | +++         |
| <b>Olanzapine</b>                                                                                  | ++              | ++          | +++      | +++         |
| Quetiapine                                                                                         | ++              | ++          | +++      | ++          |
| <b>Aripiprazole</b>                                                                                | +               | +           | +        | +           |
| Brexpiprazole                                                                                      | +               | +           | ++       | +           |
| Cariprazine                                                                                        | ++              | +           | ++       | ++          |

★ Read the two tables against each other and the trade is obvious: haloperidol, fluphenazine, pimozide are maximal on axis 1 and minimal on axis 2; chlorpromazine, thioridazine, clozapine are the mirror image. Aripiprazole is low on both — that is the entire selling point of the partial agonists.



# Serious & Drug-Specific Adverse Effects

## ● THE SERIOUS ONES

- **Laryngospasm** — haloperidol injection
- **QTc prolongation** — most antipsychotics, ★ most likely **thioridazine, ziprasidone, amisulpride**
- ● **Agranulocytosis, myocarditis** — **clozapine**
- ● **Lower seizure threshold** — **clozapine**

## THE DRUG-SPECIFIC NUISANCES

- **Sialorrhea** — clozapine (*the odd one out: a drug with heavy anticholinergic load that still produces drooling*)
- **Dyspepsia** — ziprasidone, aripiprazole
- **Headache** — ziprasidone

# Pharmacokinetics & Interactions

## WHAT CLEARS THEM

- ★ **Risperidone** → **CYP2D6** → **9-hydroxyrisperidone (paliperidone)**, itself an **active** drug marketed separately.  $t_{1/2}$  risperidone **3–20 h**, paliperidone **21–30 h**.  $T_{max}$  within 1 h.
- **Ziprasidone** — only ~1/3 by CYP; ~2/3 by **aldehyde oxidase** ★ (*the one that mostly sidesteps CYP interactions*)
- **Quetiapine** → **norquetiapine (N-desalkylquetiapine)** — an active metabolite with its own binding profile, not just a cleared waste product
- Atypicals in general are handled by **CYP1A2 · 2C9/2C19 · 2D6 · 3A4** — so inducers and inhibitors of those move the levels: [[drug-class-cyp450-inducers-inhibitors]]

## ADMINISTRATION — RISPERIDONE

- **Available as tablets, dispersible tablets and oral solution.  $T_{max}$  within 1 hour.**
- **Mainly metabolised by CYP2D6 to 9-hydroxyrisperidone (paliperidone) — an active metabolite.  $t_{1/2}$ : risperidone 3–20 h · paliperidone 21–30 h.**
- ★ **The active metabolite is itself a marketed drug** — which is why the effect outlasts risperidone's own half-life, and why a CYP2D6 poor metaboliser shifts the parent/metabolite balance rather than losing the effect.
- 💡 The dispersible tablet is the one that matters in practice: it removes the "did they actually swallow it" question in an acutely unwell patient.

# Clinical Applications

## INDICATION → DRUG

| INDICATION                       | DRUGS                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Psychotic disorders</b>       | schizophrenia, schizoaffective disorder · substance-induced psychotic disorder · psychosis in late dementia |
| <b>Delirium</b>                  | antipsychotics                                                                                              |
| <b>Behavioral problems</b>       | autism · <b>agitation in dementia</b>                                                                       |
| <b>Bipolar mania</b>             | antipsychotics                                                                                              |
| <b>Bipolar depression</b>        | ★ <b>quetiapine, aripiprazole, lurasidone, cariprazine</b>                                                  |
| <b>Major depressive disorder</b> | monotherapy — <b>quetiapine</b> · adjunct — <b>aripiprazole, brexpiprazole, quetiapine, olanzapine</b>      |
| <b>Anxiety disorders</b>         | <b>quetiapine</b>                                                                                           |
| <b>Nausea / vomiting</b>         | <b>chlorpromazine, haloperidol</b> (CTZ D2)                                                                 |
| <b>Sleep aid</b>                 | low-dose <b>quetiapine, olanzapine</b> (H1)                                                                 |

Mood-disorder context: *[[drug-class-antidepressants-lithium]]* · sedative context: *[[drug-class-anxiolytics-hypnotics]]*.

Principle behind this: *[[pharm-drug-toxicity]]* — Tardive dyskinesia as duration-dependent rather than dose-dependent toxicity; chlorpromazine phototoxicity.

Principle behind this: *[[pharm-pharmacogenomics]]* — CYP2D6 PM and UM both need dose adjustment — explicit in the lecture, and a class where the therapeutic window is already unforgiving.

Principle behind this: *[[pharm-cns-basics]]* — The four dopamine tracts ARE that page. D2 blockade in mesolimbic is the therapy; in nigrostriatal it's EPS; in tuberoinfundibular it's hyperprolactinaemia. Plus aripiprazole as D2 partial agonist and the receptor-affinity logic in full.

Source: *Week07 Pharmacology of Antipsychotics.pdf* — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สัสม), Pharmacology 3 (ภกปว 401), 35 slides. Structures omitted.

# Local and General Anesthetics

⚠️ This page is a placeholder. There is no content here yet — nothing below is missing by accident.

## WHY THIS PAGE IS EMPTY

The lecture slides for this topic **have not been uploaded to Microsoft Teams** (team code `gwwtmy3`) as of **2 August 2026**. Every other week's slides in this course came from อ.พงศธร, who posted the whole set; this topic belongs to a different lecturer who hasn't posted yet.

|             |                                                |
|-------------|------------------------------------------------|
| Week        | 9 — Wednesday 14 ต.ค. 69, 10:00–12:00, R207    |
| Lecturer    | อ.อานัฐชัย (ผศ.ดร.อานัฐชัย มายอุเทศ)           |
| Exam weight | 7.5% of the course · 2 exam hours · สอบปลายภาค |

⚠️ **The syllabus flags this week specifically:** the cabinet has designated government offices in Bangkok to work from home that day — "ให้รออาจารย์ผู้สอนแจ้งต่อไป". The session may move.

## WHAT TO DO WHEN THE SLIDES LAND

Drop the PDF in `Matcha/matcha-inbox/`. The pipeline is: `convert_corpus.py` → this file gets rewritten as a real drug-class page → rebuild both PDFs.

**Nothing else changes** — the slug `drug-class-anesthetics` is already wired into the builder's `SYLLABUS` list in taught position, so the page order, the สารบัญ and the bookmarks all stay put.

*Principle behind this: `[[pharm-pharmacokinetics]]` — thiopental redistribution — quick on, quick off, because the drug MOVED, not because it was eliminated.*

*Source: none — slides not yet released by the lecturer. Placeholder written 2026-08-02 from `Syllabus Pharmacology 3_2569.pdf`.*

# Neurodegenerative Disease Drugs

Two diseases in one lecture, and they share only their skeleton: a **misfolded protein** accumulates, a **specific neuron population** dies, and the transmitter that population made stops arriving. **AD** loses **acetylcholine** from the basal forebrain; **PD** loses **dopamine** from the substantia nigra. Every drug here is **symptomatic replacement of a lost transmitter** — ★ nothing on this page cures or stops the disease.

**misfolded protein accumulates → ROS · aseptic brain inflammation · excitotoxicity → selective neuronal death → transmitter lost → symptoms appear (*long after the neurons started dying*)**

## ★ THE TWO DISEASES SIDE BY SIDE

|                              | ALZHEIMER'S DISEASE (AD)                                                  | PARKINSON'S DISEASE (PD)                    |
|------------------------------|---------------------------------------------------------------------------|---------------------------------------------|
| <b>Misfolded protein</b>     | insoluble <b>Aβ</b> (+ <i>hyperphosphorylated tau</i> )                   | <b>α-synuclein</b> ( <i>Lewy body</i> )     |
| <b>Neurons lost</b>          | nucleus basalis of Meynert (NBM), cholinergic nuclei, hippocampus, cortex | <b>substantia nigra pars compacta (SNc)</b> |
| <b>Transmitter lost</b>      | <b>acetylcholine</b>                                                      | <b>dopamine</b>                             |
| <b>Core deficit</b>          | memory loss                                                               | motor dysfunction                           |
| <b>Loss at symptom onset</b> | 20% mild–moderate · 50% moderate–severe                                   | <b>50–60%</b> of SNc dopamine neurons       |

## HOW A MISFOLDED PROTEIN BECOMES A LESION

**Native protein** → (*mutation, external factors*) → **misfolded monomer** → **oligomer** → **protofibril** → **insoluble aggregate** → intra/extracellular deposition → **neurotoxicity**.

Opposed by **molecular chaperones** and **clearance** — the disease is that pathway losing. ★ *This is why the newest AD drugs target the protofibril, not the finished plaque.*

# Alzheimer's Disease — The Cholinergic Story

## THE SYNAPSE THE DISEASE DESTROYS

Choline + acetyl-CoA --**ChAT**--> **ACh** → released onto **M1** and **nAChR** (*with M2 as the presynaptic brake*) → degraded by **AChE** and **BuChE** back to choline, which the **choline transporter** recycles.

→ **Downstream:** ACh signalling drives **memory consolidation & recall** plus a **trophic effect** (BCL-2, NMDA). Degeneration of this pathway = loss of memory and intellectual ability ★ — and it is the *only* step in the whole cascade current drugs can reach. Receptor groundwork: [[drug-class-cholinergic]].

## PATHOPHYSIOLOGY & SYMPTOMS

**Pathology:** extracellular **amyloid plaque** (*Aβ aggregate*) · intracellular **neurofibrillary tangle** (*hyperphosphorylated tau*). AD is the **most common form of dementia**.

**Cognitive decline** — aphasia, apraxia, agnosia, disturbed executive function · impaired **activities of daily living (ADL)** **BPSD** (*behavioral & psychological symptoms of dementia*) — **behavioral:** disinhibition, aggression, wandering · **psychological:** depression, anxiety, insomnia, psychosis

# Cholinesterase Inhibitors (ChEIs)

## MECHANISM

Block **AChE** (and for some, **BuChE**) at the synapse — the surviving cholinergic neurons keep making ACh, so stopping its degradation raises what is left in the cleft.

→ **Downstream:** ↑ cholinergic transmission at M1/nAChR → improved memory consolidation and recall.

**Symptomatic only** — it does nothing about the neuron loss, so the benefit fades as the substrate dies.

## DRUGS

| DRUG                                                        | WHAT IT TREATS                                 |
|-------------------------------------------------------------|------------------------------------------------|
| <b>Donepezil</b>                                            | AD — <b>mild, moderate and severe</b> dementia |
| <b>Rivastigmine</b>                                         | AD — <b>mild, moderate and severe</b>          |
| <b>Galantamine</b>                                          | AD — <b>mild to moderate</b>                   |
| <b>Ginkgo biloba</b> standardized extract ( <i>Egb761</i> ) | AD                                             |

*First-generation agents physostigmine and tacrine are historical.*

## ★ PHARMACODYNAMICS — WHAT SEPARATES THE THREE

|                                                | DONEPEZIL | RIVASTIGMINE                | GALANTAMINE |
|------------------------------------------------|-----------|-----------------------------|-------------|
| AChE inhibition                                | ✓         | ✓ <b>pseudoirreversible</b> | ✓           |
| BuChE inhibition                               | —         | ✓                           | —           |
| <b>α7 nAChR</b> allosteric potentiating ligand | —         | —                           | ✓           |
| <b>σ receptor</b> agonist                      | ✓         | —                           | —           |

*They are not interchangeable: rivastigmine is the dual AChE/BuChE agent, galantamine adds a nicotinic action on top of the esterase block.*

## ★ PHARMACOKINETICS — AND WHY IT DECIDES THE INTERACTIONS

| PARAMETER             | DONEPEZIL      | RIVASTIGMINE         | GALANTAMINE    |
|-----------------------|----------------|----------------------|----------------|
| Tmax (hr)             | 3–5            | 0.5–2                | 0.5–1          |
| Protein binding (%)   | <b>96</b>      | 40                   | 18             |
| Plasma half-life (hr) | <b>70–80</b>   | <b>2</b>             | 5–7            |
| Metabolism            | CYP2D6, CYP3A4 | ★ <b>non-hepatic</b> | CYP2D6, CYP3A4 |
| Dosing                | <b>OD</b>      | BID                  | BID            |

★ *Donepezil's 70–80 h half-life is why it is once daily; rivastigmine's non-hepatic clearance is why it is the one that sidesteps every CYP interaction below.\*\**

# ChEI Adverse Effects & Interactions

## ADVERSE EFFECTS — ALL OF THEM ARE JUST "TOO MUCH CHOLINERGIC TRANSMISSION"

**Peripheral — cholinergic side effects** - Muscle cramp - Nausea, vomiting, diarrhea, abdominal cramp -

**Bradycardia, bronchoconstriction** - ⚠️ May increase the risk of **peptic ulcer**

**CNS** - Extrapyrimal symptoms (*mild*) - Sleep disturbance - Weight loss - 🚫 Seizure

## PHARMACODYNAMIC INTERACTIONS

- 🚫 **Anticholinergic drugs — direct antagonism of the therapy.** ⭐ Sedating antihistamines (*chlorpheniramine, brompheniramine, triprolidine*) · hyoscine · **benztropine, trihexyphenidyl** · orphenadrine · **tricyclic antidepressants** · **conventional antipsychotics**
- **Cholinergic drugs** (*bethanechol*) — additive cholinergic excess
- **Antihypertensives —  $\beta$ -blockers, verapamil, diltiazem:** additive **bradycardia**

Cross-refs: [\[\[drug-class-cholinergic\]\]](#) · [\[\[drug-class-antihistamine-serotonin\]\]](#) · [\[\[drug-class-antidepressants-lithium\]\]](#) · [\[\[drug-class-antipsychotics\]\]](#) · [\[\[drug-class-adrenergic\]\]](#) · [\[\[drug-class-ccb\]\]](#).

## PHARMACOKINETIC INTERACTIONS

**Donepezil and galantamine** are metabolized by **CYP2D6 and CYP3A4** — rivastigmine is not. - **Inducers** — CYP3A4: **phenytoin, carbamazepine, rifampin** - **Inhibitors** — CYP2D6: **fluoxetine, paroxetine** · CYP3A4: **ketoconazole, clarithromycin**

Full table: [\[\[drug-class-cyp450-inducers-inhibitors\]\]](#).



# NMDA Receptor Antagonist — Memantine

## MECHANISM

In **moderate to severe AD**, excess synaptic glutamate **tonically over-activates NMDA receptors**, which creates constant **background noise** and drowns out long-term potentiation. Memantine **partially blocks NMDA with only moderate affinity** — enough to silence the tonic noise, but weak enough that a **phasic** burst of glutamate still displaces it.

→ **Downstream:** ★ **improved LTP in the hippocampus** — the signal-to-noise argument, not simple receptor blockade. (*A high-affinity blocker would kill the signal along with the noise.*)

## DRUGS

| DRUG             | WHAT IT TREATS                                                                                    |
|------------------|---------------------------------------------------------------------------------------------------|
| <b>Memantine</b> | <b>moderate to severe AD</b> — ★ excreted <b>unchanged by the kidney</b> , so no CYP interactions |

## ADVERSE EFFECTS

- Most are **not severe** — dizziness, constipation, headache
- Sedation, confusion, hallucination

# Anti-Amyloid Monoclonal Antibodies

## MECHANISM

The first agents aimed at the **pathology** rather than the surviving transmitter. **Lecanemab** binds **A $\beta$  protofibrils** — the most neurotoxic form of the protein.

→ **Downstream:** reduces brain amyloid plaque · **slows clinical and cognitive decline by roughly 30% over 18 months** ★ — slowing, still not stopping.

## DRUGS

| DRUG              | WHAT IT TREATS                                               |
|-------------------|--------------------------------------------------------------|
| <b>Lecanemab</b>  | <b>early AD</b> — mild dementia or mild cognitive impairment |
| <b>Aducanumab</b> | AD — anti-amyloid mAb                                        |
| <b>Donanemab</b>  | AD — anti-amyloid mAb                                        |

*Other novel targets:  $\beta$ -secretase (BACE) inhibitors — verubecestat ·  $\gamma$ -secretase inhibitors — semagacestat.*

## ● ARIA — THE REASON THESE ARE NOT ROUTINE

**Amyloid Related Imaging Abnormalities**, monitored by **MRI**: - **ARIA-E** — edema / swelling - **ARIA-H** — microhemorrhage / bleeding

**Conditions of use for lecanemab:** ⚠ **ApoE  $\epsilon$ 4 non-carriers or heterozygotes only** · **confirmed amyloid pathology** required.

# Parkinson's Disease — The Dopaminergic Story

## THE SYNAPSE THE DISEASE DESTROYS

Tyrosine --TH--> **L-dopa** --DDC--> **dopamine** → released onto **D2** (*pre- and postsynaptic*) → cleared by **DAT** reuptake and destroyed by **COMT** and **MAO-B**. The projection runs **SNC** → **caudate/putamen**, and its job is **movement initiation and control**.

→ **Downstream:** ★ every PD drug class is one arrow on this diagram — **make more** (*levodopa*), **destroy less** (*MAO-B, COMT inhibitors*), **bypass the neuron** (*D2 agonists*), or **rebalance what is left** (*amantadine, anticholinergics*).

## SYMPTOMS

**Motor** — **bradykinesia** · **resting tremor** · **rigidity** · **postural instability** · also micrographia, freezing, masked face, hypophonia, drooling **Non-motor** — cognitive deficit · psychiatric (*depression, anxiety, insomnia, psychosis*) · autonomic dysfunction

⚠ *Blocking nigrostriatal D2 reproduces this pharmacologically — drug-induced parkinsonism, see [[drug-class-antipsychotics]]. Same lesion, receptor instead of neuron.*

# Levodopa + Dopa Decarboxylase Inhibitor

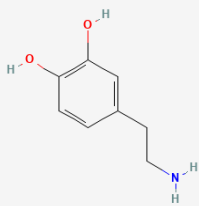
## MECHANISM

Dopamine cannot cross the blood–brain barrier; **levodopa** can. Absorbed in the **small intestine via the large neutral amino acid transporter (LNAAT/LAT)**, it uses the *same* transporter to cross the **BBB**, then **DDC** inside surviving nigrostriatal neurons converts it to dopamine. **The gold standard for PD.**

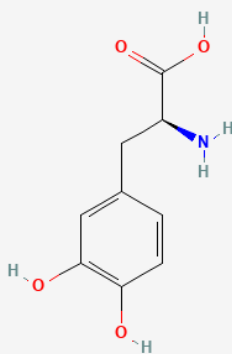
**The problem:** of the dose given, only ~30% reaches the blood and **1–3% reaches the brain** — the rest is decarboxylated to dopamine *in the periphery*, where it causes side effects and no benefit.

→ **Downstream:** adding a **peripheral DDC inhibitor** (*carbidopa, benserazide* — *they do not cross the BBB*) raises the brain share to ★ ~10%: ↓ **peripheral dopamine side effects** · ↑ **brain dopamine availability.**

## THE SHARED CORE · WHAT CHANGES

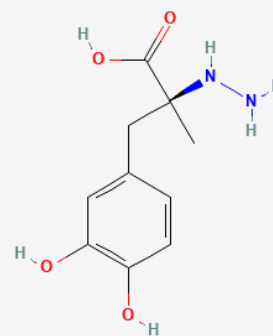


**Dopamine — what you actually want in the brain.** ● and it cannot get there. No transporter carries it and it's too polar to diffuse across the BBB.



**Levodopa**

one carboxyl group added and it becomes an **AMINO ACID** → the LAT1 transporter carries it across, then the brain decarboxylates it back to dopamine. That single –COOH is the whole drug.



**Carbidopa**

levodopa plus a hydrazine → blocks the same decarboxylase but ● cannot cross the BBB, so it only stops the conversion in the periphery

## DRUGS

| DRUG                          | WHAT IT TREATS                |
|-------------------------------|-------------------------------|
| <b>Levodopa + carbidopa</b>   | PD — the standard combination |
| <b>Levodopa + benserazide</b> | PD — the standard combination |

## ADVERSE EFFECTS

**Peripheral** (*levodopa* → *dopamine outside the brain*) - **Cardiovascular** — tachycardia, **orthostatic hypotension** - **GI** — nausea, vomiting, constipation

**Central** (↑ *dopamine at the dopaminergic tracts you were not aiming for*) - **Mesolimbic** · **tuberoinfundibular** · **CTZ** · **lateral hypothalamus** · **RAS**

★ *This is the exact mirror of the D2-blockade map in [[drug-class-antipsychotics]] — same five places, arrow reversed.*

### INTERACTIONS — MOSTLY AT THE GUT, NOT THE LIVER

-  **Protein-rich diet** — competes for the **same LNAAT transporter**; absorption changes by **>80%** 
- **Metal ions** — absorption changes by **>50%**
- **Drugs that delay gastric emptying delay the onset: anticholinergics · opioids · GLP-1 and analogues**

*Cross-refs: [[drug-class-analgesics-opioids]] · [[drug-class-antidiabetic]] · [[drug-class-gi-motility]].*

### PATIENT COUNSELLING

-  **Avoid taking with a high-protein meal**
- Take **at least 30 min before** a meal, or **1–2 h after**
- Adequate fluid — ideally **200 ml**
- **Maintain an upright position**
- Warn about **nausea/vomiting** and **orthostatic hypotension**

# Levodopa Motor Fluctuation

## WHY THE DRUG STOPS WORKING SMOOTHLY

**Early PD:** onset **0.5–1 h**, initial duration of action  $\geq$ **3–6 h** — and efficacy falls over time. Three things drive the decay:

1. **Continuing SNc neuronal loss** → **loss of buffering capacity** — fewer neurons left to store dopamine and release it steadily
  2. → **narrowing of the therapeutic level** of levodopa
  3. **Pulsatile dopamine stimulation** → **abnormal postsynaptic response**
- **Downstream:** ★ the response follows the plasma curve instead of smoothing it out — the patient starts feeling every dose.

## THE FLUCTUATION PATTERNS

**Delayed-on** — the dose takes too long to work **Wearing-off effect** — benefit fades before the next dose  
**On-off phenomenon** — abrupt, unpredictable switching **Nocturnal freezing** **Peak-dose dyskinesia** — involuntary movement at maximum concentration

## TREATING "WEARING OFF" IN ADVANCED PD

↑ **dose frequency** of levodopa · switch standard formulation to **controlled-release** · **add a COMT inhibitor**  
· **add a dopamine agonist** · **add an MAO-B inhibitor** · **add amantadine**

★ *Every option is the same idea — flatten the curve, either by dosing more evenly or by slowing the destruction of what is already there.*

# MAO-B Inhibitors

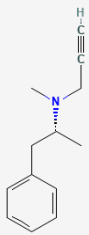
## MECHANISM

**Selectively inhibit MAO-B**, the isoform that degrades **dopamine** (*MAO-A handles 5-HT and NE; tyramine sits between them*). Blocking it protects the dopamine that is already inside the neuron.

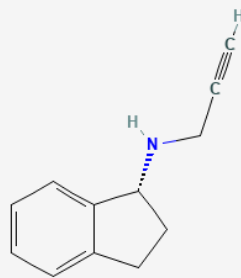
- **Irreversible** — **selegiline, rasagiline**
- **Reversible** — **safinamide**

→ **Downstream:** ↑ **intracellular dopamine** · ↑ **phasic dopamine release**. Usable as a **single agent in early PD but not in advanced PD** (*there has to be a neuron left to protect*) — otherwise add-on to reduce motor fluctuation.

## THE SHARED CORE · WHAT CHANGES



**Selegiline.** ● the propargyl ( $C\equiv C$ ) group is a suicide warhead — MAO-B oxidises it and gets permanently bonded. That's why the effect outlasts the drug by weeks.



**Rasagiline**

the same propargylamine on an indane ring — ⚠️ and no amphetamine metabolites, unlike selegiline

## DRUGS

| DRUG              | WHAT IT TREATS                                           |
|-------------------|----------------------------------------------------------|
| <b>Selegiline</b> | early PD · motor fluctuation add-on — ⚠️ <b>insomnia</b> |
| <b>Rasagiline</b> | early PD · motor fluctuation add-on                      |
| <b>Safinamide</b> | PD — the <b>reversible</b> one                           |

## ★ TWO SIMILAR DRUGS, OPPOSITE METABOLITES

| DRUG              | ENZYME           | METABOLITE    | CONSEQUENCE                                               |
|-------------------|------------------|---------------|-----------------------------------------------------------|
| <b>Selegiline</b> | CYP2B6 / CYP2C19 | l-amphetamine | ⚠️ <b>insomnia</b> — take no later than 14:00             |
| <b>Rasagiline</b> | CYP1A2           | aminoindan    | ↑ <b>Bcl-2</b> → possible <b>disease-modifying effect</b> |

★ *The whole clinical difference between the two irreversible agents is what they break down into.*

# COMT Inhibitors

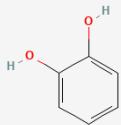
## MECHANISM

Block **COMT**, the other enzyme that destroys levodopa/dopamine — so less of a levodopa dose is lost before it reaches the brain.

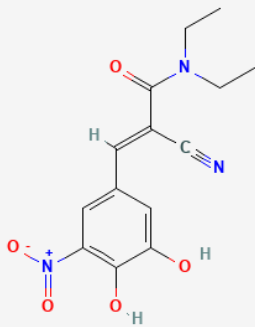
→ **Downstream:** ↓ peripheral conversion of levodopa · ↑ **synaptic dopamine** · ↓ **on-off phenomenon**. ●

**Only ever in combination with levodopa — never monotherapy**, because there is nothing to protect without it.

## THE SHARED CORE · WHAT CHANGES

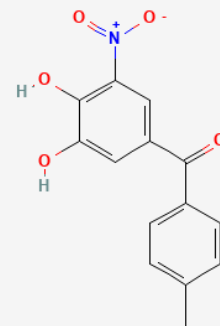


**Catechol — what COMT methylates.** COMT recognises this pair of hydroxyls. Give it a catechol it cannot process and the enzyme is occupied.



**Entacapone**

a nitrocatechol — ● peripheral only, so it just stops levodopa being wasted before it reaches the brain



**Tolcapone**

lipophilic enough to enter the CNS too — and hepatotoxic, which is why it's second line ⚠

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                                                    |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Entacapone</b> | PD — <b>add-on to levodopa</b> for motor fluctuation · also given as a fixed <b>carbidopa + levodopa + entacapone</b> combination |



# Dopamine Receptor Agonists

## MECHANISM

Skip the dying neuron entirely — stimulate **postsynaptic D2** directly, so no synthesis, storage or release is required. Split into **ergots** and **non-ergots**, and the difference is **what else they hit**:

- **Bromocriptine** (*ergot*) — **D2 agonist** · **α1 blocker** · **α2 agonist** · **5-HT2B agonist**
- **Pramipexole, ropinirole, rotigotine** (*non-ergots*) — **D2 agonist** · **D3 agonist** · antioxidant

→ **Downstream**: ★ the ergots' extra receptors are the entire safety difference — **α1 blockade** → **orthostatic hypotension**, **5-HT2B agonism** → **valvular fibrosis**. Receptor groundwork: [[drug-class-adrenergic]].

## DRUGS

| DRUG                                    | WHAT IT TREATS                      |
|-----------------------------------------|-------------------------------------|
| <b>Bromocriptine</b> ( <i>ergot</i> )   | PD · <b>hyperprolactinemia</b>      |
| <b>Cabergoline</b> ( <i>ergot</i> )     | PD                                  |
| <b>Pergolide</b> ( <i>ergot</i> )       | PD                                  |
| <b>Lisuride</b> ( <i>ergot</i> )        | PD                                  |
| <b>Pramipexole</b> ( <i>non-ergot</i> ) | PD — ★ also <b>depression in PD</b> |
| <b>Ropinirole</b> ( <i>non-ergot</i> )  | PD                                  |
| <b>Rotigotine</b> ( <i>non-ergot</i> )  | PD                                  |
| <b>Piribedil</b> ( <i>non-ergot</i> )   | PD                                  |
| <b>Apomorphine</b> ( <i>non-ergot</i> ) | PD                                  |

## USES

**Early PD** — used to **delay levodopa therapy** · **With levodopa** — ↓ **motor fluctuation** · **Depression in PD** — especially **pramipexole** · **Hyperprolactinemia** — **bromocriptine** (see [[drug-class-pituitary-hormones]], and the **D2-blockade** cause in [[drug-class-antipsychotics]])

## ADVERSE EFFECTS

All of them come from **D2 receptor activation** — including in places you did not want it. - ★ **Impulse control disorders** — gambling, punding (*the signature ADR of this class*) - ● **Sudden sleep attack** · somnolence - **Orthostatic hypotension** — ⚠ prominent with **ergots**, **less with non-ergots** - ● **Valvular fibrosis** — **ergots only**, via 5-HT2B - **Peripheral edema** — from peripheral **D2/D3** stimulation

# Amantadine & Anticholinergics

## AMANTADINE — THREE MECHANISMS, THREE JOBS

- **Dopamine transporter inhibitor + dopamine releaser** → treats **PD symptoms**
  - **NMDA receptor blocker** → ★ treats **motor fluctuation** (*this is why it is added for dyskinesia, not for tremor*)
  - **σ receptor agonist** → antidepressant effect, neuroprotection
- **Downstream: primarily excreted unchanged in the urine** — no hepatic metabolism to interact with.

## ANTICHOLINERGICS — REBALANCING, NOT REPLACING

**Trihexyphenidyl, benztropine.** Losing dopamine leaves striatal cholinergic tone relatively unopposed; blocking muscarinic transmission restores the balance.

→ **Downstream:** some benefit for **resting tremor** and **muscle rigidity** — and ⚠ **risk > benefit** in PD, which is the lecturer's verdict. (*Note the collision: these are the same drugs used to treat antipsychotic-induced EPS — [[drug-class-antipsychotics]] — and the same drugs that antagonise ChEIs in AD.*)

## DRUGS

| DRUG                   | WHAT IT TREATS                                      |
|------------------------|-----------------------------------------------------|
| <b>Amantadine</b>      | PD symptoms · <b>motor fluctuation / dyskinesia</b> |
| <b>Trihexyphenidyl</b> | PD — resting tremor, rigidity ⚠ risk > benefit      |
| <b>Benztropine</b>     | PD — resting tremor, rigidity ⚠ risk > benefit      |

## WHY ANTICHOLINERGICS LOST

- **Anticholinergic side effects** (*see [[drug-class-cholinergic]]*)
- ● **Cognition** — the last thing a PD patient can afford to spend
- **Delayed-on** — they slow gastric emptying, so levodopa arrives late

*Principle behind this: [[pharm-drug-interactions]] — Oxybutynin cancelling donepezil — the receptor-mediated PD interaction, in the exact patient group where both get prescribed. Memantine is the uncompetitive antagonist.*

*Principle behind this: [[pharm-cns-basics]] — Parkinson's is the nigrostriatal tract, treated at five different steps of Concept 2. Levodopa crosses via LAT (dopamine can't). Pergolide vs bromocriptine at 5-HT2B. Galantamine's dual mechanism.\*\**

*Source: Week10-11 Pharmacology of Drugs Used in Neurodegenerative diseases.pdf — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สม), Pharmacology 3 (ภาค 401), 47 slides. Structures omitted.*

# Antiepileptics

Also called **antiseizure medications (ASMs)** or **antiseizure drugs (ASDs)**. Epilepsy is a disorder of **neuronal excitability**, and excitability is set by exactly two families of channel: the **voltage-gated** ones ( $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{K}^+$ ) and the **ligand-gated** ones (glutamate  $\rightarrow$  NMDA/AMPA, GABA  $\rightarrow$  GABA-A). ★  
**Every drug on this page is one arrow on that diagram** — damp the excitatory side, or strengthen the inhibitory side.

↓  $\text{Na}^+/\text{Ca}^{2+}$  influx *or* ↓ glutamate release *or* ↑  $\text{Cl}^-$  influx via GABA-A  $\rightarrow$  ↓ neuronal depolarization  $\rightarrow$  fewer action potentials  $\rightarrow$  seizure suppressed

## SEIZURE VS EPILEPSY — AND THE TYPES

**Major types by onset** - **Focal (partial) onset** — retained awareness or impaired awareness - **Generalized onset**

**Symptoms: motor** — tonic, clonic, atonic, myoclonic · **non-motor**

**Named syndromes / states: status epilepticus** · **absence seizure** (*generalized non-motor*) · **Dravet's syndrome** · **Lennox–Gastaut syndrome**

## ★ THE SIX MECHANISM FAMILIES

| FAMILY                                               | TARGET                                | DRUGS                                                                                                                      |
|------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b><math>\text{Na}^+</math> channel blockers</b>     | VGSC                                  | phenytoin, fosphenytoin · carbamazepine, oxcarbazepine, eslicarbazepine · lamotrigine · zonisamide, rufinamide, lacosamide |
| <b><math>\text{Ca}^{2+}</math> channel blockers</b>  | <b>N-type</b><br>( $\alpha 2\delta$ ) | gabapentin, pregabalin, mirogabalin                                                                                        |
|                                                      | <b>T-type</b><br>( <i>pore</i> )      | ethosuximide, valproate                                                                                                    |
| <b>GABA-A allosteric potentiating ligands (APLs)</b> | GABA-A                                | benzodiazepines ( <i>clonazepam, clobazam, diazepam, midazolam</i> ) · phenobarbital                                       |
| <b>GABA turnover</b>                                 | GABA-T · GAT-1                        | vigabatrin ( <i>GABA-T inhibitor</i> ) · tiagabine ( <i>GAT-1 inhibitor</i> )                                              |
| <b>Glutamate receptor blockers</b>                   | AMPA                                  | perampanel                                                                                                                 |
| <b>SV2A modulators</b>                               | SV2A                                  | levetiracetam · brivaracetam                                                                                               |
| <b>Broad spectrum (multimodal)</b>                   | several at once                       | valproate · topiramate, felbamate, cenobamate · zonisamide, rufinamide · cannabidiol                                       |

## THE TWO SYNAPSES BEING TUNED

**Inhibitory (GABAergic):** glutamate --GAD--> **GABA** → released onto **GABA-A** ( $Cl^-$  influx) → cleared by the **GAT-1** reuptake transporter into neuron and astrocyte → destroyed by **GABA-T** to succinic semialdehyde.

**Excitatory (glutamatergic):** depolarization opens **N-type / P/Q-type VGCC** →  $Ca^{2+}$  entry → **SV2A**-dependent vesicle release of **glutamate** → **NMDA** and **AMPA** → further  $Na^+/Ca^{2+}$  entry → more APs. Cleared by astrocytic **EAAT** (*excitatory amino acid transporter*).

→ **Downstream:** ★ a drug can hit **either synapse at any station** — the channel, the transporter, the degrading enzyme or the receptor — which is why the classes below look unrelated but all end in the same place. Receptor groundwork: [[drug-class-gaba-a]].

## ADMINISTRATION — WHAT THESE ACTUALLY COME AS IN THAILAND

| DRUG                    | PREPARATIONS                                                                                                        |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Carbamazepine</b>    | tablet 200 mg · syrup 100 mg/5 mL · <b>CR tablet 200, 400 mg</b>                                                    |
| <b>Phenytoin</b>        | capsule 100 mg · chewable tablet 50 mg · solution for injection 50 mg/mL                                            |
| <b>Fosphenytoin</b>     | solution for injection <b>75 mg/mL = 50 mg phenytoin equivalent</b> ★                                               |
| <b>Rufinamide</b>       | tablet 200 mg                                                                                                       |
| <b>Lacosamide</b>       | tablet 50, 100 mg · solution for infusion 10 mg/mL                                                                  |
| <b>Zonisamide</b>       | tablet 100 mg                                                                                                       |
| <b>Lamotrigine</b>      | tablet 25, 50, 100 mg                                                                                               |
| <b>Oxcarbazepine</b>    | tablet 300, 600 mg                                                                                                  |
| <b>Phenobarbital</b>    | tablet 32.5, 65 mg                                                                                                  |
| <b>Clobazam</b>         | tablet 5 mg                                                                                                         |
| <b>Gabapentin</b>       | capsule 100, 300, 400 mg · tablet 600 mg                                                                            |
| <b>Pregabalin</b>       | capsule 25, 75, 150 mg                                                                                              |
| <b>Mirogabalin</b>      | tablet 2.5, 5, 10, 15 mg                                                                                            |
| <b>Perampanel</b>       | tablet 2, 4, 8 mg                                                                                                   |
| <b>Levetiracetam</b>    | tablet 250, 500 mg · solution for injection 100 mg/mL ( <i>500 mg/5 mL</i> )                                        |
| <b>Topiramate</b>       | tablet 25, 50, 100 mg                                                                                               |
| <b>Sodium valproate</b> | <b>CR tablet 500 mg · enteric-coated tablet 200 mg</b> · oral solution 200 mg/mL · powder for injection 400 mg/4 mL |

- **Fosphenytoin is written in phenytoin equivalents, not in mg of itself** — 75 mg/mL of the vial *is* 50 mg of phenytoin. Reading the vial number as the dose overdoses the patient by half.
- 💡 The CR and enteric-coated forms exist to blunt peak-related side effects, so they are the ones that must be swallowed whole. Crushing a valproate CR tablet or a carbamazepine CR turns a controlled release into a full dose at once.
- 💡 Syrup, chewable tablet and oral solution are the paediatric/swallowing-difficulty routes — worth knowing they exist before telling a family the only option is a tablet.

# Na<sup>+</sup> Channel Blockers

## MECHANISM

**Reversible blockers of voltage-gated Na<sup>+</sup> channels (NaV)**, and **most are non-selective across NaV subtypes**. They do not plug an open channel — they **bind the inactivated state** and hold it there, so the block deepens the faster the neuron is firing (*use-dependent blockade* — *demonstrated for lamotrigine*).

- **High affinity to the FAST inactivation state** — ★ **most VGSC-blocking AEDs**
- **High affinity to the SLOW inactivation state** — ★ **lacosamide**, on its own

→ **Downstream**: a normally firing neuron is barely touched; a **high-frequency, seizure-firing** neuron is silenced. Selectivity comes from the firing rate, not from the drug.

## NAV SUBTYPES — WHY THE ADRS LAND WHERE THEY DO

**By amino-acid homology**: **NaV1.1, 1.2** brain · **NaV1.6** brain, cerebellum · **NaV1.3, 1.7, 1.8, 1.9** peripheral nerve · **NaV1.4** skeletal muscle · **NaV1.5** cardiac tissue.

⚠ *Non-selective blockade therefore reaches NaV1.5 — this is the origin of the class's cardiac rhythm abnormalities.*

## DRUGS ACTING ON THE VGSC CYCLE

| DRUG                           | WHAT IT TREATS                                                                                  |
|--------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Phenytoin, fosphenytoin</b> | epilepsy — focal & generalized tonic-clonic · status epilepticus                                |
| <b>Carbamazepine</b>           | epilepsy — focal & generalized tonic-clonic · <b>neuropathic pain</b> · <b>bipolar disorder</b> |
| <b>Oxcarbazepine</b>           | epilepsy · <b>neuropathic pain</b> — the lower-rash carbamazepine                               |
| <b>Eslicarbazepine acetate</b> | epilepsy                                                                                        |
| <b>Lamotrigine</b>             | epilepsy · <b>bipolar disorder</b> · absence                                                    |
| <b>Lacosamide</b>              | epilepsy — ★ the <b>slow-inactivation</b> agent                                                 |
| <b>Rufinamide</b>              | epilepsy                                                                                        |
| <b>Zonisamide</b>              | epilepsy                                                                                        |
| <b>Topiramate, valproate</b>   | <i>also block VGSC — see the broad-spectrum pages</i>                                           |

## CLASS ADVERSE EFFECTS

- **Cardiac rhythm abnormalities** (*non-selective block reaches NaV1.5*)
- **Hyponatremia** — ★ **carbamazepine and oxcarbazepine** specifically

# Phenytoin

## THE PHARMACOKINETICS ARE THE DRUG

- **Absorption variable and incomplete**
- **90% protein bound** — narrow therapeutic index on top
- **95% hepatic metabolism by CYP2C9 (major) and CYP2C19 (minor)** — and that pathway is ● **SATURABLE**
- ● **Non-linear pharmacokinetics** ★ — once the enzyme saturates, a small dose increase produces a disproportionate jump in plasma level

**Sources of the variability: CYP2C9 polymorphism** · metabolic rate **increased** in younger children, pregnancy, during menses, acute trauma · **decreased** in advanced age.

→ **Downstream:** ★ **strong enzyme inducer**, saturable clearance, high protein binding and a narrow window — the reason phenytoin is the classic therapeutic-drug-monitoring drug.

## ADVERSE EFFECTS

- **Dose-related CNS** — dizziness, sedation, ataxia, **nystagmus, diplopia**
- ● **Arrhythmia if the IV infusion rate is too high**
- ★ **Gingival hyperplasia, hirsutism, oily skin, acne** — the cosmetic signature
- Rash and idiosyncratic reactions
- ● **Purple glove syndrome (PGS)** — with **peripheral IV** administration

# Carbamazepine & Oxcarbazepine

## CARBAMAZEPINE — METABOLISM THAT FIGHTS BACK

**98% metabolized in the liver by CYP3A4**, and it is a ● **potent CYP3A4 inducer** that induces other enzymes as well — including **its own** metabolism (**autoinduction**).

→ **Downstream:** ★ **time-dependent PK** — half-life is **~30 h in the first month**, then falls to **12–24 h** as the induction matures. The dose that worked at week 2 is under-dosing by week 6.

**The toxicity branch:** carbamazepine → **epoxide** → (*epoxide hydrolase*) → hydroxy metabolite. If the reactive intermediate binds protein instead, it forms a **hapten complex** → **immune response** → **hypersensitivity** and **hepatotoxicity**.

## OXCARBAZEPINE — THE SAME DRUG WITH THE EPOXIDE PROBLEM REMOVED

Its metabolism avoids the epoxide route, so it causes **less CYP induction** and **less rash**.

→ **Downstream:** rash incidence **CBZ ~7% → OXC ~4–5%**, and milder when it happens. ⚠️ **But cross-reactivity on switching CBZ → OXC is 25–30%** ★ — "switch to oxcarbazepine" is *not* a safe move after a serious carbamazepine reaction.

## ADVERSE EFFECTS — CARBAMAZEPINE

- **Dose-related CNS effects similar to phenytoin, but more sedating**
- **Abnormal liver enzymes are common** (*usually transient*) — ● but **fatal hepatitis** can develop
- ● **HLA-B\*1502 positive greatly increases the risk of serious cutaneous reactions** ★ — **SJS** and **TEN**. *Relevant in Thai and other Southeast/East Asian populations.*
- **Hyponatremia** — via **augmentation of ADH**

# Lamotrigine

## MECHANISM — FOUR ACTIONS, WHICH IS WHY IT IS ALSO A MOOD DRUG

- **Block VGSC and VGCC** → **antiepileptic** and **anti-manic** effects
- **5-HT1A agonist** → **antidepressant** effect
- **Blocks monoamine reuptake (high dose)** → antidepressant effect

**PK:** metabolized by **UDP-glucuronosyltransferase (UGT1A4, UGT2B7)** to the **2-N-glucuronide** conjugate — a **major inactive metabolite** cleared renally. **No CYP involvement.**

→ **Downstream:** ★ the multi-mechanism profile is exactly why lamotrigine crosses over into **bipolar disorder** — see [[drug-class-antidepressants-lithium]].

## ADVERSE EFFECTS

- **Less sedating** than the older agents
- **● SERIOUS RASH ★** — the defining risk, and it is **dose-dependent**. Higher in:
  - **Pediatric patients**
  - **Co-administration of valproate** (*valproate inhibits UGT → lamotrigine levels rise*)
  - **Exceeding the recommended dose**

## THE RULE THAT FOLLOWS FROM IT

- **⚠ Start low and titrate slowly. ★** Lamotrigine's slow titration is not caution for its own sake — it is the *only* thing that prevents the rash.
- **⚠ On valproate, the titration has to be slower still.**



# Ca<sup>2+</sup> Channel Blockers

## TWO WAYS TO BLOCK A CALCIUM CHANNEL

The VGCC is a **pore-forming  $\alpha 1$  subunit** plus **auxiliary subunits ( $\alpha 2\delta$ ,  $\beta$ , and sometimes  $\gamma$ )**. AEDs split by *which part they grab*:

- ★ **Bind the  $\alpha 2\delta$  auxiliary subunit** → **gabapentinoids** (" $\alpha 2\delta$  ligands") → block the **N-type** channel *indirectly*
- Bind the ion pore itself** → **valproate and ethosuximide** → **T-type** · **topiramate** → **N-type**

**What the Ca<sup>2+</sup> was doing:** neuronal excitability · activation of Ca<sup>2+</sup>-dependent enzymes (*protein kinase, MAPK, phospholipases*) · neurite growth · **neurotransmitter release** · **central sensitization**.

→ **Downstream:** ↓ presynaptic Ca<sup>2+</sup> entry → ↓ **glutamate release** from **hyperexcited** neurons only. ★ **T-type block is the absence-seizure mechanism; N-type/ $\alpha 2\delta$  block is the neuropathic-pain mechanism.**  
Channel groundwork: [[drug-class-ccb]].

## WHERE THE SUBTYPES LIVE

**L-type** — CaV1.1 skeletal muscle · CaV1.2 heart, brain, smooth muscle, adrenal, kidney · CaV1.3 brain, kidney, pancreas · CaV1.4 retina **P/Q-type** — CaV2.1 brain, pituitary, kidney · **N-type** — CaV2.2 brain, nervous system ★ · **R-type** — CaV2.3 brain, heart, pituitary **T-type** — CaV3.1 brain, kidney, nervous system ★ · CaV3.2 brain, heart, kidney, liver · CaV3.3 brain

## DRUGS

| DRUG                | WHAT IT TREATS                                                                                   |
|---------------------|--------------------------------------------------------------------------------------------------|
| <b>Gabapentin</b>   | epilepsy · <b>neuropathic pain</b>                                                               |
| <b>Pregabalin</b>   | epilepsy · <b>neuropathic pain</b> · ★ <b>generalized anxiety disorder</b> · <b>fibromyalgia</b> |
| <b>Mirogabalin</b>  | <b>neuropathic pain only</b> — ⚠ <b>not licensed as an antiepileptic</b>                         |
| <b>Ethosuximide</b> | <b>absence seizure</b> ( <i>T-type</i> )                                                         |
| <b>Valproate</b>    | broad spectrum, incl. <b>absence</b> ( <i>T-type</i> ) — see its own page                        |

## ★ THE THREE GABAPENTINOIDS — THE DIFFERENCES ARE ALL PHARMACOKINETIC

|                              | GABAPENTIN             | PREGABALIN                          | MIROGABALIN                    |
|------------------------------|------------------------|-------------------------------------|--------------------------------|
| <b>Absorption</b>            | ● <b>SATURABLE</b>     | non-saturable across the dose range | non-saturable                  |
| Tmax                         | 2–3 h                  | 0.8–1.4 h                           | 0.5–2 h                        |
| Elimination t <sub>1/2</sub> | 5–7 h                  | 5–6.5 h                             | 2–3 h                          |
| Protein binding              | 3%                     | <b>none</b>                         | 25%                            |
| Metabolism                   | <b>not metabolized</b> | 0.9% N-methylated                   | negligible                     |
| Renal excretion              | 76–81%                 | <b>90–99%</b>                       | 96% ( <i>76.4% unchanged</i> ) |
| Dosing frequency             | q8h                    | q8–12h                              | q12–24h                        |

★ **Gabapentin's saturable absorption is the whole clinical point:** bioavailability falls with dose — **900 mg → 60% · 1200 mg → 47% · 2400 mg → 34% · 3600 mg → 33%**. Doubling the dose does not double the exposure. Pregabalin has no such ceiling.

⚠ **All three are renally cleared and all three need dose reduction in renal insufficiency**, and none of them touch CYP — so no metabolic interactions.

## ADVERSE EFFECTS

- Weight gain
- Peripheral edema

## ADMINISTRATION — THE GABAPENTINOID DOSE RANGES

|                      | GABAPENTIN                                | PREGABALIN                | MIROGABALIN               |
|----------------------|-------------------------------------------|---------------------------|---------------------------|
| <b>Starting dose</b> | 100–900 mg/day                            | <b>75–150 mg/day</b>      | 5 mg                      |
| Dosing frequency     | every 8 h                                 | every 8–12 h              | every 12–24 h             |
| Usual effective dose | 1200–2400 mg                              | 150–600 mg                | <i>(not established)</i>  |
| <b>Max dose</b>      | <b>3600 mg</b> ( $\leq 1200$ mg per dose) | 600 mg                    | 30 mg                     |
| Renal impairment     | ● dose reduction required                 | ● dose reduction required | ● dose reduction required |

- ★ **Gabapentin's per-dose cap of 1200 mg is the saturable absorption showing up as a rule** — beyond that the extra drug largely isn't absorbed, which is also why it has to be given three times a day while mirogabalin can go once or twice.
- 💡 All three are titrated up from a low start rather than begun at the effective dose; dizziness and sedation are worst early, and starting at the target dose is how a patient stops taking it in week one.

# GABA Enhancers

## TWO ROUTES TO MORE INHIBITION

1. **GABA-A allosteric potentiating ligands (APLs)** — do not bind where GABA binds; they make GABA's own effect on the receptor bigger → ↑  $\text{Cl}^-$  influx → **hyperpolarization**. **Benzodiazepines** (*clonazepam, clobazam, diazepam, midazolam*) · **phenobarbital**
  2. **GABA turnover** — leave more GABA sitting in the synapse: **Vigabatrin** — **GABA-T inhibitor** (*blocks GABA breakdown*) · **Tiagabine** — **GAT-1 inhibitor** (*blocks GABA reuptake*)
- **Downstream:** ↑ inhibitory tone across the whole CNS — effective, but **non-selective**, which is why this family is the sedating one. Full receptor pharmacology: [[drug-class-gaba-a]] · [[drug-class-anxiolytics-hypnotics]].

## DRUGS

| DRUG                       | WHAT IT TREATS                                                  |
|----------------------------|-----------------------------------------------------------------|
| <b>Phenobarbital</b>       | epilepsy — focal, generalized tonic-clonic · status epilepticus |
| <b>Clonazepam</b>          | epilepsy — incl. <b>absence</b> , atonic/tonic, myoclonic       |
| <b>Clobazam</b>            | epilepsy — focal onset                                          |
| <b>Diazepam, midazolam</b> | <b>status epilepticus</b> ( <i>injection</i> )                  |
| <b>Nitrazepam</b>          | atonic / tonic seizure                                          |
| <b>Vigabatrin</b>          | epilepsy — GABA-T inhibitor                                     |
| <b>Tiagabine</b>           | epilepsy — GAT-1 inhibitor                                      |

## PHENOBARBITAL — PHARMACOKINETICS

Metabolized by **CYP2C9 / CYP2C19**; ~20–25% **excreted unchanged in urine**;  $t_{1/2}$  **36–118 h** — the longest of the conventional four.

→ **Downstream:** ● **strong enzyme inducer**. Combined with the long half-life, its interactions arrive slowly and leave slowly.

## ADVERSE EFFECTS — PHENOBARBITAL

- **Sedation** · ● **respiratory depression** (*also with injectable benzodiazepines*)
- Rash and other idiosyncratic reactions
- ● **Reduced IQ with long-term use** ★ · **may induce a hyperactive child**

# Broad Spectrum — Valproate

## MECHANISM — FOUR ACTIONS AT ONCE

- Block VGSC
- Block VGCC (T-type) — ★ this is why it works in **absence seizure**
- GABA-A APL
- Inhibit GABA-T — more GABA left in the synapse

**PK: 90–95% protein bound** (*concentration-dependent*) · **95% hepatic metabolism** by **β-oxidation** and **CYP450** (*UGT1A6/1A9/2B7, CYP2C9, CYP2C19*). A **hepatotoxic metabolite** is produced along the way.

→ **Downstream:** ★ the broadest single agent in the lecture — the only one that covers **absence, generalized tonic-clonic, focal, myoclonic and atonic/tonic** — and the reason it also serves as a **mood stabiliser** and a **migraine preventive**.

## DRUGS

| DRUG                                    | WHAT IT TREATS                                                                                                 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Sodium valproate / valproic acid</b> | epilepsy — <b>broad spectrum</b> , incl. <b>absence</b> · <b>bipolar disorder</b> · <b>migraine prevention</b> |

## ADVERSE EFFECTS — THE PRICE OF THE BROAD SPECTRUM

- **Dose-related CNS effects** like the other AEDs · **tremor**
- **GI in ~16%** — anorexia, nausea, vomiting · ● **severe pancreatitis**
- **Weight gain 8–18%** · **alopecia 2–12%**
- **Thrombocytopenia**, bleeding
- ● **Hepatotoxicity** — ⚠️ **especially under 2 years of age**
- ● **Valproate-induced encephalopathy with hyperammonemia**
- ● **TERATOGENICITY** ★ — **neural tube defects**; major malformations **≈10%** with first-trimester exposure. **The highest teratogenic risk of the antiepileptics.**

## INTERACTIONS — VALPROATE IS THE ONE INHIBITOR IN THE CLASS

- ● **Inhibits UGT** → ↑ **lamotrigine** (*the rash risk*), ↑ **lorazepam**
- **Inhibits CYP2C9** → ↑ **phenytoin**, ↑ **phenobarbital**
- ⚠️ **90–95% protein bound** and concentration-dependent — displacement interactions matter

Full table: *[[drug-class-cyp450-inducers-inhibitors]]*.

# Broad Spectrum — Topiramate

## MECHANISM — FOUR ACTIONS, A DIFFERENT FOUR

- Block VGSC
- Block VGCC (*N-type, at the pore*)
- Block AMPA / kainate receptors
- GABA-A APL

**PK: low protein binding 15–41% · ★ 70% eliminated unchanged in the urine** — negligible hepatic metabolism, so almost no CYP interactions in the outward direction.

→ **Downstream:** broad-spectrum cover plus **migraine prevention** — and the ADR profile is the mirror image of valproate's (*weight loss, not gain*).

## DRUGS

| DRUG              | WHAT IT TREATS                                           |
|-------------------|----------------------------------------------------------|
| <b>Topiramate</b> | epilepsy — broad spectrum · ★ <b>migraine prevention</b> |

## ADVERSE EFFECTS

- ★ **Cognitive impairment, slow thinking** — the signature complaint
- Paresthesia
- Weight loss
- Hypohydrosis (*reduced sweating*)
- Kidney stones
- Glaucoma

# AMPA Receptor Blocker — Perampanel

## MECHANISM

**Non-competitive AMPA receptor blocker** — it does not compete with glutamate for the site, so a surge of glutamate cannot out-compete it.

**PK: mainly metabolized by CYP3A4** — ⚠ therefore vulnerable to every CYP3A4 inducer in this class (*carbamazepine, phenytoin, phenobarbital*).

→ **Downstream:** ↓ postsynaptic excitatory transmission at the fast glutamate receptor.

## DRUGS

| DRUG              | WHAT IT TREATS                   |
|-------------------|----------------------------------|
| <b>Perampanel</b> | epilepsy — AMPA receptor blocker |

## ADVERSE EFFECTS

- Aggression, irritability ★
- Psychosis, suicidal ideation

# SV2A Modulators — Levetiracetam

## MECHANISM

Binds **SV2A**, the **synaptic vesicle glycoprotein 2A**, and **inhibits glutamate release** — it works on the vesicle, *before* the transmitter ever reaches a receptor. ★ It **preferentially acts at the excitatory synapse**, which is why it is not a general sedative.

**PK:** ★ **mainly excreted unchanged by the kidney** — no hepatic metabolism, so **no CYP interactions at all**. That, plus the clean PK, is why it is used so widely.

→ **Downstream:** ↓ presynaptic glutamate release without touching  $\text{Na}^+$ ,  $\text{Ca}^{2+}$  or GABA.

## DRUGS

| DRUG                 | WHAT IT TREATS                                           |
|----------------------|----------------------------------------------------------|
| <b>Levetiracetam</b> | epilepsy — focal, generalized tonic-clonic, atonic/tonic |
| <b>Brivaracetam</b>  | epilepsy — SV2A modulator                                |

## ADVERSE EFFECTS

- **Aggression, irritability, psychosis** ★ — the behavioural ADR it shares with perampanel. *Clean pharmacokinetics, dirty behaviour: this is the trade-off.*

# Matching Drug to Seizure Type

## ★ THAI CPG OF EPILEPSY (2559) — WHAT IS USED FOR WHAT

| SEIZURE TYPE                         | DRUGS USED                                                                                                                                                                                                                   |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adults, partial (focal) onset</b> | carbamazepine · phenytoin · sodium valproate · phenobarbital · clonazepam · lamotrigine ( <i>elderly</i> ) · topiramate · levetiracetam · gabapentin ( <i>elderly</i> ) · oxcarbazepine · zonisamide · clobazam · pregabalin |
| <b>Children, partial onset</b>       | carbamazepine · phenytoin · phenobarbital · sodium valproate · clonazepam · topiramate · lamotrigine · oxcarbazepine · zonisamide · clobazam                                                                                 |
| <b>Generalized tonic-clonic</b>      | phenobarbital · sodium valproate · phenytoin · carbamazepine · clonazepam · lamotrigine · topiramate · levetiracetam · gabapentin · oxcarbazepine · clobazam                                                                 |
| <b>Absence epilepsy</b><br>★         | <b>sodium valproate · clonazepam · lamotrigine</b> — <i>the narrowest list on the table</i>                                                                                                                                  |
| <b>Juvenile myoclonic epilepsy</b>   | sodium valproate · topiramate                                                                                                                                                                                                |
| <b>Atonic / tonic seizure</b>        | sodium valproate · clonazepam · topiramate · lamotrigine · nitrazepam · levetiracetam                                                                                                                                        |

★ **Read it by mechanism, not by memory: sodium valproate appears in every row** because it is the only agent that blocks  $\text{Na}^+$ , T-type  $\text{Ca}^{2+}$  and boosts GABA. The **narrow-spectrum  $\text{Na}^+$  blockers** (carbamazepine, phenytoin, oxcarbazepine) appear in the **focal and tonic-clonic** rows and **never in absence** — a pure  $\text{Na}^+$  block does nothing to the thalamic **T-type** current that generates absence.



# Class-Wide Adverse Effects

## THE ADRS ALMOST EVERY AED SHARES

**General CNS** - Dizziness, fatigue, **sedation** - **Diplopia**, blurred vision, nystagmus, ataxia - **Cognitive disturbance** — memory problems, slow thinking, ⚠️ **lower IQ in long-term use** - Weight change

**Immune-mediated** - 🟡 **Rash and serious skin reactions** — maculopapular rash → **Stevens-Johnson syndrome (SJS)** → **toxic epidermal necrolysis (TEN)** - **Agranulocytosis**, aplastic anemia, thrombocytopenia - **Hepatitis**, hepatic failure - 🟡 **Antiepileptic hypersensitivity syndrome (AHS)**

## ★ WHY AHS HAPPENS — AND WHY "AROMATIC" IS THE WORD THAT MATTERS

**Aromatic** anticonvulsants are oxidised to reactive **arene oxide** intermediates. **Epoxide hydrolase** normally detoxifies them; when it cannot keep up, the intermediate **binds protein**, forms a **hapten complex**, and the immune system attacks the adduct.

→ **Downstream:** ⚠️ **hypersensitivity and hepatotoxicity** — and **cross-reactivity between the aromatic AEDs**, because they all converge on the same reactive intermediate. **Non-aromatic AEDs do not share it.**

## EFFECT ON BODY WEIGHT

| WEIGHT GAIN                                                                 | NO EFFECT                                               | WEIGHT LOSS                                               |
|-----------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
| carbamazepine · oxcarbazepine ·<br><b>gabapentinoids</b> · <b>valproate</b> | lamotrigine · lacosamide ·<br>levetiracetam · phenytoin | tiagabine · <b>topiramate</b> ·<br>zonisamide · felbamate |

## ★ THE SPECIFIC ADR OF EACH AGENT — THE EXAM MAP

| DRUG                                              | ITS OWN ADR                                                                                           |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Phenobarbital</b>                              | reduced IQ · hyperactive child · <b>respiratory depression</b>                                        |
| <b>Benzodiazepines</b><br>(injection)             | <b>respiratory depression</b>                                                                         |
| <b>Na<sup>+</sup> channel blockers</b><br>(class) | <b>cardiac rhythm abnormalities</b> · <b>hyponatremia</b> (carbamazepine, oxcarbazepine)              |
| <b>Phenytoin</b>                                  | <b>gingival hyperplasia</b> , hirsutism, oily skin, acne                                              |
| <b>Gabapentinoids</b>                             | weight gain · <b>peripheral edema</b>                                                                 |
| <b>Valproate</b>                                  | weight gain, alopecia · thrombocytopenia & bleeding · hepatitis, pancreatitis · <b>teratogenicity</b> |
| <b>Topiramate</b>                                 | paresthesia, <b>cognitive slowing</b> · kidney stone · glaucoma · weight loss                         |
| <b>Levetiracetam, perampanel</b>                  | <b>psychosis</b> , behavioural problems                                                               |

# Pharmacokinetics of the Conventional Four

## ★ CARBAMAZEPINE · PHENOBARBITAL · PHENYTOIN · VALPROATE

|                                | CARBAMAZEPINE                                      | PHENOBARBITAL                                   | PHENYTOIN                                        | VALPROATE                                                      |
|--------------------------------|----------------------------------------------------|-------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------|
| <b>Therapeutic serum conc.</b> | 4–12 µg/mL (+ epoxide 0.2–2.0)                     | 15–40 µg/mL                                     | <b>10–20 µg/mL total</b> , 1–2 free              | 50–100 µg/mL total, 5–12.5 free                                |
| Oral absorption                | 85%                                                | 70–90%                                          | 90–100%                                          | <b>100%</b>                                                    |
| Vd (L/kg)                      | 0.8–2                                              | 0.5–1.0                                         | 0.5–1.0                                          | <b>0.1–0.2</b>                                                 |
| <b>t<sub>1/2</sub> (h)</b>     | 12–17                                              | <b>36–118</b>                                   | 7–42                                             | 6–17                                                           |
| <b>Protein binding</b>         | 76%                                                | 55%                                             | <b>90%</b>                                       | <b>90–95%</b> (conc-dependent)                                 |
| Elimination                    | hepatic 100% — <b>CYP3A4</b> major, CYP1A2, CYP2C8 | renal 20% unchanged · hepatic CYP2C9, 2C19, 2E1 | hepatic 98% — <b>CYP2C9</b> major, CYP2C19 minor | hepatic 95% — <b>β-oxidation</b> , UGT1A6/1A9/2B7, CYP2C9/2C19 |
| <b>Active metabolite</b>       | ★ <b>CBZ-10,11-epoxide</b>                         | none                                            | none                                             | —                                                              |

## ⚠ WHY THE CONVENTIONAL AGENTS ARE THE DIFFICULT ONES

- **Narrow therapeutic index** — all four
- **High plasma protein binding** — **phenytoin, valproate**
- **Mainly metabolized by CYP450** — and ● **phenytoin's is non-linear**
- **They induce or inhibit CYP450 themselves: CBZ, PHT, PB are inducers** of CYP450 and UGT · **valproate is an inhibitor** of CYP2C9 and UGT

## HOW MUCH LIVER EACH AED ACTUALLY USES

| PREDOMINANTLY HEPATIC                                                                                             | PARTIALLY                            | NEGLIGIBLE                                           |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------|
| carbamazepine (3A4) · oxcarbazepine (3A4) · phenytoin (2C9) · phenobarbital (2C9) · valproate · lamotrigine (UGT) | lacosamide (2C19) · zonisamide (3A4) | ★ <b>topiramate · gabapentinoids · levetiracetam</b> |

★ The right-hand column is the interaction-free column — that is the practical reason the newer agents get chosen for polypharmacy patients.

# Enzyme Induction & Inhibition

## ● AED INDUCERS — THE BIGGEST INTERACTION PROBLEM IN THIS LECTURE

**Broad-spectrum inducers:** carbamazepine · phenytoin · phenobarbital / primidone.

They **induce the synthesis of new enzyme protein**, so the effect is ★ **slow in onset and slow in offset** — days to weeks each way, not one dose. Stopping the inducer is as dangerous as starting it.

**What gets destroyed by induction:** - ● **Oral contraceptive levels** → contraceptive failure (*see [[drug-class-estrogen-progestins]]*) - ● **Immunosuppressant levels** — **ciclosporin, tacrolimus** (*see [[drug-class-immunomodulators]]*) - ● **Warfarin levels** (*see [[drug-class-antithrombotics]]*) - **Folate** - **Vitamin D & calcium** (*see [[drug-class-bone-mineral]]*)

*Full inducer/inhibitor lists: [[drug-class-cyp450-inducers-inhibitors]].*

## AED INHIBITORS — THE OTHER DIRECTION

- **Valproate** → **UGT** inhibition: ↑ **lamotrigine**, ↑ **lorazepam** · **CYP2C9** inhibition: ↑ **phenytoin**, ↑ **phenobarbital**
- **Topiramate and oxcarbazepine** → **CYP2C19**: ↑ **phenytoin**
- **Cannabidiol** → **CYP1A2**, **2B6**, **2C9**, **2D6**, **3A4** and **P-glycoprotein**: ↑ **phenytoin**, ↑ **clobazam**

★ *Note who keeps appearing on the receiving end: phenytoin. Saturable, non-linear, narrow-index — it is the drug that cannot absorb a small nudge.*

# Beyond Epilepsy

## WHERE THESE DRUGS GET USED OUTSIDE SEIZURES

**Migraine prevention** — valproate, topiramate · **Bipolar disorder** — valproate, carbamazepine, lamotrigine (see [[drug-class-antidepressants-lithium]]) · **Neuropathic pain** — carbamazepine, oxcarbazepine ( $\text{Na}^+$  channel route) and gabapentin, pregabalin, mirogabalin ( $\alpha 2\delta$  route) · **Anxiety disorder** — pregabalin (see [[drug-class-anxiolytics-hypnotics]])

*Principle behind this: [[pharm-pharmacokinetics]] — phenytoin saturates CYP2C9, so it follows zero-order kinetics — past that point a small dose rise causes a large jump in level.*

*Principle behind this: [[pharm-drug-toxicity]] — Three separate hits: valproate's saturable elimination in overdose, and lamotrigine + carbamazepine on the SJS/TEN list.*

*Principle behind this: [[pharm-cns-basics]] — Concept 2's excitability half — carbamazepine and phenytoin on  $\text{Na}^+$  channels, gabapentin and pregabalin on  $\text{Ca}^{2+}$  channels. And phenytoin/phenobarbital as P-gp substrates, which is why their brain uptake lags their lipophilicity.*

*Source: Week11-12 Pharmacology of Antiepileptics.pdf — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สม), Pharmacology 3 (ภาว 401), 66 slides. Structures omitted.*

# Migraine Headache

★ The entire topic splits in two: **ABORTIVE** (stop this attack) vs **PREVENTIVE** (reduce how often attacks happen). They are different drugs, different targets and different rules of use — and a few agents (the **gepants**) sit in both columns. Everything below hangs on one anatomical unit: the **trigeminovascular system (TGVS)** and the peptide it releases, **CGRP**.

brain hyperexcitability (*brain stem, hypothalamus, PAG*) → cortical spreading depression (CSD) → aura → TGVS activation/sensitization → neuropeptide release → vasodilation + neurogenic inflammation + peripheral sensitization → headache → central sensitization → associated symptoms

## DIAGNOSIS — ICHD-3 FEATURES

≥5 episodic headaches lasting 4–72 h, with:

**At least two of:** unilateral location · **pulsating (throbbing) quality** · moderate or severe intensity · aggravated by — or causing avoidance of — routine physical activity (*walking, stairs*)

**Any one of:** nausea and/or vomiting · **photophobia and phonophobia**

★ – Aura = common migraine · + Aura = classic migraine

## THE TRIGEMINOVASCULAR SYSTEM — WHERE MIGRAINE PAIN PHYSICALLY IS

Trigeminal nerve → trigeminal ganglion (TG) → trigeminocervical complex (TCC), also called the trigeminal nucleus caudalis (TNC).

The pain-sensitive structures are **cranial blood vessels**: arteries at the base of the brain and their major branches · **dural and meningeal arteries** · venous sinuses.

→ **Downstream:** ★ the brain itself doesn't hurt — the **vessels and their trigeminal innervation** do. That is why every abortive drug on this page acts on a vessel, a trigeminal nerve ending, or the TNC.

## ★ CGRP — THE MOLECULE THE WHOLE MODERN HALF OF THIS LECTURE IS BUILT ON

**Calcitonin gene-related peptide** — a **37-amino-acid** multifunctional peptide, acting as an **autacoid** and a **neurotransmitter**, and a **potent vasodilator**: **10× prostaglandins** and **10–100× ACh, NO and substance P**.

The receptor: a **GPCR coupled to Gs** with a composite structure — **CLR** (*calcitonin receptor-like receptor*) + **RAMP-1** (*receptor activity-modifying protein 1*) + **RCP** → **adenylate cyclase** → ↑ **cAMP**.

Where it acts:

| SITE                    | EFFECT OF CGRP                                                                       |
|-------------------------|--------------------------------------------------------------------------------------|
| Vascular smooth muscle  | <b>vasodilation</b>                                                                  |
| Endothelium             | <b>NO synthesis</b> → more vasodilation                                              |
| Trigeminal nerve ending | ↑ excitability → <b>peripheral sensitization</b>                                     |
| TNC neuron              | ↑ excitability → <b>central sensitization</b>                                        |
| Immune cells (WBC)      | NO, <b>PGE<sub>2</sub></b> , <b>PGI<sub>2</sub></b> → <b>neurogenic inflammation</b> |
| Myenteric neurons       | ↑ GI motility and secretion                                                          |

→ **Downstream:** three roles in headache — ↑ **neuronal excitability** · **vasodilation** · **neurogenic inflammation**. ⚠ CGRP and its receptor are **also widely distributed in the CVS and GI tract**, which is where the off-target effects come from. The prostanoid arm links to [[drug-class-nsaids-prostanoids]].

## ★ THE WHOLE ARSENAL — ABORTIVE VS PREVENTIVE

| ABORTIVE ( <i>STOP THIS ATTACK</i> )                                                                                                                                                                                                                | PREVENTIVE ( <i>FEWER ATTACKS</i> )                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Specific — 5-HT<sub>1B/1D/1F</sub> agonists</b>                                                                                                                                                                                                  | <b>Anticonvulsants</b> — topiramate, valproate, gabapentin    |
| · <b>Ergots:</b> ergotamine, dihydroergotamine                                                                                                                                                                                                      | <b>Antidepressants</b> — TCAs, SNRIs                          |
| · <b>Triptans:</b> sumatriptan, eletriptan, almotriptan, frovatriptan, naratriptan, rizatriptan, zolmitriptan                                                                                                                                       | <b>β-blockers</b> — propranolol                               |
| · <b>Ditan:</b> lasmiditan ( <i>5-HT<sub>1F</sub> selective</i> )                                                                                                                                                                                   | <b>Calcium channel blockers</b> — flunarizine, verapamil      |
| <b>Specific — gepants:</b> rimegepant, ubrogepant, zavegepant                                                                                                                                                                                       | <b>Others</b> — pizotifen, candesartan                        |
| <b>Combination:</b> sumatriptan 50 mg + naproxen 250–500 mg                                                                                                                                                                                         | <b>Gepants</b> — rimegepant, atogepant                        |
| <b>Non-specific:</b> paracetamol, aspirin, ibuprofen, naproxen, diclofenac                                                                                                                                                                          | <b>Monoclonal antibodies</b> — anti-CGRP / anti-CGRP receptor |
| ★ <b>Available in Thailand (migraine-specific abortives):</b> ergotamine tartrate + caffeine · <b>sumatriptan</b> · <b>eletriptan</b> · <b>rimegepant</b> . Non-specific abortives are ordinary analgesics — see [[drug-class-nsaids-prostanoids]]. |                                                               |

# Ergots

## MECHANISM — AN AGONIST THAT HITS FAR TOO MANY RECEPTORS

**Agonist at 5-HT<sub>1B/1D</sub>**, but ★ **ergotamine is non-selective** — it is also an agonist at  $\alpha_1$ , D<sub>2</sub> and 5-HT<sub>2B</sub> receptors.

**5-HT<sub>1B/1D/1F</sub> are Gi-coupled** — they **inhibit adenylate cyclase** → ↓ **cAMP**, the exact mirror of what the Gs-coupled CGRP receptor does.

**Where the subtypes sit:** - **5-HT<sub>1B</sub>** — trigeminal nerve, **large cranial vessels**, and ⚠ **other vascular beds incl. the coronary artery** - **5-HT<sub>1D</sub>** — trigeminal nerve - **5-HT<sub>1F</sub>** — trigeminal nerve and **TNC**

→ **Downstream: 5-HT<sub>1B</sub> → vasoconstriction · 5-HT<sub>1D/1F</sub> → ↓ neuronal excitability and sensitization.**

Receptor groundwork: [[drug-class-antihistamine-serotonin]]. ● The **off-target  $\alpha_1$ , D<sub>2</sub> and 5-HT<sub>2B</sub> agonism is the whole toxicity profile** — systemic vasoconstriction and fibrosis.

## DRUGS

| DRUG                                               | WHAT IT TREATS                   |
|----------------------------------------------------|----------------------------------|
| <b>Ergotamine</b> ( <i>usually with caffeine</i> ) | acute migraine attack — abortive |
| <b>Dihydroergotamine (DHE)</b>                     | acute migraine attack — abortive |

## ADVERSE EFFECTS & INAPPROPRIATE USE

- ● **ERGOTISM** ★ — a **fatal condition from systemic vasoconstriction** after ergot ingestion (*overdose*):
  - abdominal and chest pain
  - **cold, numbness, pain or tingling of the extremities**
  - **cyanosis and possible gangrene**
  - headache, psychosis
  - nausea, cramping, diarrhea
- **Long-term use** → **tolerance** → **rebound headache / medication-overuse headache**
- ● **Valvular, retroperitoneal and pleural fibrosis** — the 5-HT<sub>2B</sub> signature

## ● INTERACTIONS — THE DEADLIEST BLOCK ON THIS PAGE

**Ergotamine (and eletriptan) + CYP3A4 inhibitors** → ★ **must not be used within 3 days of a potent CYP3A4 inhibitor** — this can produce **fatal ergotism**: - **HIV antivirals** — darunavir, indinavir, lopinavir, nelfinavir, ritonavir, saquinavir - **Macrolides** — clarithromycin, erythromycin (*azithromycin excepted*) - **Azole antifungals** — itraconazole, ketoconazole, fluconazole - **Others** — imatinib, fluvoxamine

**Ergotamine / eletriptan / sumatriptan + any serotonin-raising drug** → **serotonin syndrome**. ⚠ **Do not co-administer a triptan and an ergot within 24 h of each other.**

*Full inducer/inhibitor lists: [[drug-class-cyp450-inducers-inhibitors]].*

## CONTRAINDICATIONS & PRECAUTIONS

● **Contraindicated:** - **Pregnancy** - **Vascular disease** · **uncontrollable hypertension** - **Renal or liver dysfunction** - **Familial hemiplegic migraine** · **basilar migraine**

⚠ **Precautions:** concomitant **CYP3A4 inhibitors or inducers** · **diabetes** · **peripheral arterial disease**

# Triptans & Lasmiditan

## MECHANISM — THE SAME FAMILY, CLEANED UP

Triptans are agonists at 5-HT<sub>1B</sub>/1D/1F — more potent and more selective than the ergots, with less affinity for 5-HT<sub>1B</sub> and more potency at 5-HT<sub>1F</sub> than ergotamine.

Lasmiditan is a potent, selective 5-HT<sub>1F</sub> agonist.

→ **Downstream:** ↓ cAMP at the trigeminal nerve and TNC → ↓ **neuronal excitability and sensitization**, plus **cranial vasoconstriction via 5-HT<sub>1B</sub>**. ★ **Read the class as a slide along one axis:** ergots = 5-HT<sub>1B</sub>-heavy + off-target → most vasoconstriction · triptans = selective 1B/1D/1F · **lasmiditan = 1F only** → **the neuronal action without the 5-HT<sub>1B</sub> vascular one.**

## DRUGS

| DRUG                                                                         | WHAT IT TREATS                                                                   |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Sumatriptan</b>                                                           | acute migraine — abortive ( <i>metabolized by MAO-A</i> )                        |
| <b>Eletriptan</b>                                                            | acute migraine — ⚠ <b>CYP3A4</b> , so it inherits the ergot 3-day rule           |
| <b>Almotriptan · frovatriptan · naratriptan · rizatriptan · zolmitriptan</b> | acute migraine — abortive                                                        |
| <b>Sumatriptan 50 mg + naproxen 250–500 mg</b>                               | acute migraine — the combination product                                         |
| <b>Lasmiditan</b>                                                            | acute migraine — selective <b>5-HT<sub>1F</sub></b> agonist ( <i>a "ditan"</i> ) |

## ADVERSE EFFECTS

- ★ **"Triptan sensations"** — the characteristic cluster:
  - paraesthesias
  - flushing
  - tingling
  - neck pain
  - mild transient chest pressure
- **Tolerance · withdrawal / rebound headache · medication-overuse headache** with overuse (*shared with ergots and ditans*)

## CAUTIONS

- 🚫 **Contraindicated in vascular disease and uncontrolled hypertension** — same reason as the ergots: **5-HT<sub>1B</sub> is on the coronary artery too**
- ⚠ **Eletriptan follows the ergot rule** — no potent CYP3A4 inhibitor within 3 days
- ⚠ **No triptan within 24 h of an ergot**, and watch every serotonergic co-medication



# Gepants — CGRP Receptor Antagonists

## MECHANISM

**Block the CGRP receptor** — so the  $G_s \rightarrow$  adenylate cyclase  $\rightarrow$  **cAMP rise never happens**: no vasodilation, no peripheral or central sensitization, no neurogenic inflammation.

**History worth knowing:** ⚠️ **olcegepant and telcagepant** were developed until the early 2000s and **terminated for hepatotoxicity** — the current generation is the fix for that.

→ **Downstream:** ★ **no direct vasoconstricting side effects** (*they antagonise the dilator instead of constricting the vessel*) · generally **minimal adverse effects** · 🚫 **they do not induce tolerance or medication-overuse headache, unlike ergots / triptans / ditans** · and they can be used **both abortively and preventively** — the only class on this page that does both.

## DRUGS

| DRUG              | WHAT IT TREATS                                                                                 |
|-------------------|------------------------------------------------------------------------------------------------|
| <b>Rimegepant</b> | ★ <b>both</b> — acute migraine with or without aura <b>and</b> prevention of episodic migraine |
| <b>Ubrogepant</b> | acute migraine with or without aura                                                            |
| <b>Zavegepant</b> | acute migraine with or without aura ( <i>intranasal</i> )                                      |
| <b>Atogepant</b>  | <b>prevention</b> of episodic and chronic migraine                                             |

## PHARMACOKINETICS

|                        | ATOGEPAANT                             | RIMEGEPANT                         | UBROGEPANT                                        | ZAVEGEPANT                         |
|------------------------|----------------------------------------|------------------------------------|---------------------------------------------------|------------------------------------|
| <b>Tmax (h)</b>        | 1–2                                    | 1–1.5                              | 1.5                                               | <b>0.5</b>                         |
| <b>t½ (h)</b>          | <b>11</b>                              | <b>11</b>                          | 5–7                                               | 6–7                                |
| <b>High-fat meal</b>   | AUC & Cmax change ~20%, Tmax unchanged | Cmax 42–53%, AUC 32–38%            | Cmax ~22%, AUC unchanged, <b>Tmax delayed 2 h</b> | —                                  |
| Vd (L)                 | 292                                    | 120                                | 350                                               | <b>1774</b>                        |
| <b>Protein binding</b> | <b>none</b>                            | 96%                                | 87%                                               | 90%                                |
| <b>Elimination</b>     | <b>CYP3A4</b>                          | <b>CYP3A4</b> ( <i>minor 2C9</i> ) | <b>CYP3A4</b>                                     | <b>CYP3A4</b> ( <i>minor 2D6</i> ) |

⚠️ **All four run through CYP3A4** — the interaction watch-list is the same one that makes ergotamine dangerous, even though the consequence here is far milder.

## ADVERSE EFFECTS

- ★ **No severe hepatotoxicity at therapeutic doses** — the problem that killed the first generation
- Generally **mild** — **nausea, vomiting**
- Long-term use:** constipation · **blood pressure increase** in some cases

## USE LIMITS

- ⚠️ **Rimegepant** — abortive use as needed, max once daily; **no established safety beyond 18 doses / 30 days**
- ⚠️ **Ubrogepant, zavegepant** — **no established safety beyond 8 migraines treated / 30 days**

## ADMINISTRATION — GEPANTS, AND THE FOOD EFFECT THAT SEPARATES THEM

|                        | ATOGEANT                                   | RIMEGEPANT                                                           | UBROGEPANT                | ZAVEGEPANT                  |
|------------------------|--------------------------------------------|----------------------------------------------------------------------|---------------------------|-----------------------------|
| <b>Formulation</b>     | tablet                                     | <b>orally disintegrating tablet (ODT)</b>                            | tablet                    | <b>nasal spray</b>          |
| <b>Tmax (h)</b>        | 1–2                                        | 1–1.5                                                                | 1.5                       | <b>0.5</b>                  |
| <b>t½ (h)</b>          | 11                                         | 11                                                                   | 5–7                       | 6–7                         |
| <b>High-fat meal</b>   | ↓ AUC and Cmax <b>20%</b> , Tmax unchanged | ↓ Cmax <b>42–53%</b> , ↓ AUC <b>32–38%</b> , Tmax <b>delayed 2 h</b> | ↓ Cmax 22%, AUC unchanged | —                           |
| <b>Protein binding</b> | none                                       | 96%                                                                  | 87%                       | 90%                         |
| <b>Elimination</b>     | CYP3A4                                     | CYP3A4 ( <i>minor 2C9</i> )                                          | CYP3A4                    | CYP3A4 ( <i>minor 2D6</i> ) |

### DOSE

|                                           |                                                                                                                                                          |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Atogepant</b><br>( <i>prevention</i> ) | <b>episodic: 10, 30 or 60 mg orally daily · chronic: 60 mg orally daily</b>                                                                              |
| <b>Rimegepant</b>                         | <b>acute: 75 mg orally as needed, max 75 mg/24 h · prevention: 75 mg orally every other day</b> · ⚠️ no established safety above <b>18 doses/30 days</b> |
| <b>Ubrogepant</b><br>( <i>acute</i> )     | <b>50–100 mg orally as needed, a second dose 2 h later if needed, max 200 mg/24 h</b> · ⚠️ no established safety above <b>8 migraines/30 days</b>        |
| <b>Zavegepant</b><br>( <i>acute</i> )     | <b>10 mg intranasal</b> · ⚠️ no established safety above <b>8 migraines/30 days</b>                                                                      |

- **Rimegepant is the one a fatty meal wrecks** — half the peak concentration and the peak arrives two hours late, which for an acute migraine drug is the difference between working and not. Take it away from a heavy meal.
- ★ **Zavegepant's Tmax of 0.5 h is the whole argument for the nasal spray** — fastest in, and it bypasses a stomach that a migraine has already shut down. 💡 That is also why it's the sensible choice for someone whose attacks come with vomiting.
- ★ **Rimegepant is the only one that does both jobs** — as-needed for an attack and every-other-day for prevention, same 75 mg tablet.
- ⚠️ All four run through **CYP3A4**.

# Abortive Therapy — Rules of Use

## ★ ERGOTAMINE VS TRIPTAN VS GEPANT — THE COMPARISON THE OBJECTIVES ASK FOR

|                             | ERGOTAMINE                   | SUMATRIPTAN  | ELETRIPTAN    | RIMEGEPANT                                           |
|-----------------------------|------------------------------|--------------|---------------|------------------------------------------------------|
| <b>Oral bioavailability</b> | ● 1–5%                       | 15%          | 50%           | <b>64%</b>                                           |
| Tmax (oral)                 | 0.5–2 h                      | 2.5 h        | 1–1.5 h       | 1–1.5 h                                              |
| <b>Half-life</b>            | 3 h                          | 2.5 h        | 3.6–6.9 h     | <b>11 h</b>                                          |
| <b>Major metabolism</b>     | <b>CYP3A4</b>                | <b>MAO-A</b> | <b>CYP3A4</b> | <b>CYP3A4</b>                                        |
| Usual initial dose          | 1–2 mg                       | 50–100 mg    | 40 mg         | 75 mg                                                |
| <b>Interval to repeat</b>   | 30 min                       | 2 h          | 2 h           | ★ none — no repeat                                   |
| <b>Maximum</b>              | <b>6 mg/day · 10 mg/week</b> | 200 mg/24 h  | 80 mg/24 h    | 75 mg/24 h (or 75 mg every other day for prevention) |

★ **The weekly cap on ergotamine is the tell** — it is the only agent here whose limit is set by cumulative vascular toxicity rather than by daily dose.

## ★ WHY "TAKE IT AT ONSET" IS PHARMACOLOGY, NOT NAGGING

**Give any abortive as soon as possible — within 1 hour of headache onset.**

**Early in the attack:** throbbing pain only. **Later:** cutaneous allodynia develops and the **receptive field expands** — central sensitization is established, and the picture becomes throbbing pain + **photophobia, phonophobia, nausea/vomiting and aggravation by movement**.

→ **Downstream:** once central sensitization has set in, a drug acting at the periphery is arriving after the fire has spread. **The timing window is the mechanism.**

## ⚠ OVERUSE OF ERGOTS, TRIPTANS AND DITANS

- Tolerance
- Withdrawal / rebound headache
- ● **Medication-overuse headache** — the treatment becomes the disease
- ★ **Gepants are the exception** — they do not do this.

## ADMINISTRATION — ABORTIVE DOSING, AND WHAT'S ACTUALLY AVAILABLE IN THAILAND

Available migraine-specific abortives: ergotamine tartrate 1 mg + caffeine 100 mg tablet · sumatriptan tablet 50 mg · eletriptan film-coated tablet 40 mg · rimegepant ODT 75 mg.

|                             | ERGOTAMINE           | SUMATRIPTAN | ELETRIPTAN | RIMEGEPANT                                           |
|-----------------------------|----------------------|-------------|------------|------------------------------------------------------|
| <b>Oral bioavailability</b> | ● 1–5%               | 15%         | 50%        | 64%                                                  |
| <b>Tmax (oral)</b>          | 0.5–2 h              | 2.5 h       | 1–1.5 h    | 1–1.5 h                                              |
| <b>Half-life</b>            | 3 h                  | 2.5 h       | 3.6–6.9 h  | 11 h                                                 |
| <b>Metabolism</b>           | CYP3A4               | MAO-A       | CYP3A4     | CYP3A4                                               |
| <b>Usual initial dose</b>   | 1–2 mg               | 50–100 mg   | 40 mg      | 75 mg                                                |
| <b>Interval to repeat</b>   | 30 min               | 2 h         | 2 h        | ● no repeat                                          |
| <b>Maximum</b>              | 6 mg/day, 10 mg/week | 200 mg/24 h | 80 mg/24 h | 75 mg/24 h (or 75 mg every other day for prevention) |

- **Ergotamine is the one with a WEEKLY cap as well as a daily one** — 6 mg/day *and* 10 mg/week. That weekly ceiling is the guard against ergotism, and it is the number patients don't know exists.
- ★ **Ergotamine's 1–5% oral bioavailability** is why it is combined with caffeine (which improves absorption) and why the oral dose looks small next to its effect.
- ★ **Sumatriptan is the odd one metabolically** — MAO-A, not CYP3A4, which is where its interaction profile diverges from everything else in the row.
- 💡 The repeat interval matters more than the dose in practice: 30 minutes for ergotamine, 2 hours for the triptans, and **never** for rimegepant. A patient who redoses a gepant at 2 hours out of triptan habit is over the daily maximum in one attack.

# Preventive Therapy — Classical Oral Agents

## WHY PREVENT AT ALL

**Frequent headache episodes themselves induce TGVS sensitization** — the attacks make the system easier to trigger.

**What the preventive drugs do** (*time- and dose-dependent — they are not fast*): - strengthen the **descending inhibitory pain pathway** - reduce **TGVS activation / sensitization** - reduce **brain hyperexcitability** - reduce the **frequency of cortical spreading depression**

→ **Downstream:** ⚠️ prevention **never replaces abortive therapy** — acute medication must still be provided for breakthrough attacks.

## THE CLASSES, AND WHERE EACH ONE ALREADY LIVES IN THIS WIKI

| CLASS                           | DRUGS                                                      | PAGE                                   |
|---------------------------------|------------------------------------------------------------|----------------------------------------|
| <b>Anticonvulsants</b>          | <b>topiramate</b> , valproate, gabapentin                  | [[drug-class-antiepileptics]]          |
| <b>Antidepressants</b>          | <b>TCAs (amitriptyline)</b> , SNRIs ( <i>venlafaxine</i> ) | [[drug-class-antidepressants-lithium]] |
| <b>β-adrenoceptor blockers</b>  | <b>propranolol</b> , metoprolol, nadolol                   | [[drug-class-adrenergic]]              |
| <b>Calcium channel blockers</b> | flunarizine, verapamil                                     | [[drug-class-ccb]]                     |
| <b>Serotonin antagonist</b>     | pizotifen                                                  | [[drug-class-antihistamine-serotonin]] |
| <b>ARB / ACEI</b>               | candesartan, lisinopril                                    | [[drug-class-ace-arb-dri]]             |
| <b>Neurotoxin</b>               | onabotulinumtoxinA ( <i>chronic migraine only</i> )        | —                                      |
| ★ <b>Newer, CGRP-targeted</b>   | <b>rimegepant, atogepant · monoclonal antibodies</b>       | <i>see below</i>                       |

## EVIDENCE AND STRENGTH OF RECOMMENDATION

| DRUG                                       | CLASS                | TARGET DAILY DOSE | EVIDENCE    | RECOMMENDATION |
|--------------------------------------------|----------------------|-------------------|-------------|----------------|
| <b>Topiramate</b>                          | anti-epileptic       | 100 mg            | <b>High</b> | <b>Strong</b>  |
| <b>Propranolol</b>                         | β-blocker            | 80–160 mg         | <b>High</b> | <b>Strong</b>  |
| <b>Metoprolol</b>                          | β-blocker            | 100–200 mg        | <b>High</b> | <b>Strong</b>  |
| <b>Amitriptyline</b>                       | antidepressant       | 10–100 mg         | <b>High</b> | <b>Strong</b>  |
| Nadolol                                    | β-blocker            | 80–160 mg         | Moderate    | Strong         |
| Candesartan                                | ARB                  | 16 mg             | Moderate    | Strong         |
| Gabapentin                                 | anticonvulsant       | 1200–3600 mg      | Moderate    | Strong         |
| Divalproex sodium                          | anti-epileptic       | 500–1500 mg       | High        | Weak           |
| Flunarizine                                | CCB                  | 10 mg             | High        | Weak           |
| Pizotifen                                  | serotonin antagonist | 1.5–4.0 mg        | High        | Weak           |
| Venlafaxine                                | antidepressant       | 150 mg            | Low         | Weak           |
| Lisinopril                                 | ACEI                 | 20 mg             | Low         | Weak           |
| Verapamil                                  | CCB                  | 120–480 mg        | Low         | Weak           |
| OnabotulinumtoxinA ( <i>chronic only</i> ) | neurotoxin           | 155–195 units     | n/a         | n/a            |

★ The four "high evidence + strong recommendation" agents are the ones to know: topiramate · propranolol · metoprolol · amitriptyline.

## ★ ADRS OF THE FOUR CLASSICAL ORAL PREVENTIVES

(A = amitriptyline · P = propranolol · T = topiramate · V = valproate) - **Weight gain** — amitriptyline, valproate - **Weight loss, paresthesia, slow thinking** — topiramate - **Depressed mood, impotence** — propranolol - **Sedation, fatigue** — all four - **Xerostomia, constipation, anticholinergic effects** — amitriptyline - ⚠ **Reduced seizure threshold** — amitriptyline

## INTERACTIONS

- **Elimination is via cytochrome P450** — **CYP2D6** for amitriptyline and propranolol — and **phase II enzymes** for topiramate.
- ⚠ **DDI potential is HIGH** for this whole oral group, and they all **need dose titration**.

See [[drug-class-cyp450-inducers-inhibitors]].

# Anti-CGRP Monoclonal Antibodies

## MECHANISM — TWO TARGETS, ONE PATHWAY

- **Against the CGRP receptor:** erenumab
- **Against the peptide itself (CGRP- $\alpha$  and  $\beta$ ):** fremanezumab · galcanezumab · eptinezumab

→ **Downstream:** the same blockade the gepants achieve, but with **antibody kinetics** — weeks-to-months half-life, **eliminated by proteolysis, not CYP450**, so ★ **essentially no pharmacokinetic drug interactions**.

## DRUGS

| DRUG                                                     | WHAT IT TREATS                                                                      |
|----------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Erenumab</b> ( <i>human IgG2, anti-receptor</i> )     | prevention of episodic / chronic migraine — SC                                      |
| <b>Fremanezumab</b> ( <i>humanised IgG2, anti-CGRP</i> ) | prevention of episodic / chronic migraine — SC                                      |
| <b>Galcanezumab</b> ( <i>humanised IgG4, anti-CGRP</i> ) | prevention of episodic / chronic migraine · ★ <b>episodic cluster headache</b> — SC |
| <b>Eptinezumab</b> ( <i>humanised IgG1, anti-CGRP</i> )  | prevention of episodic / chronic migraine — IV                                      |

## PHARMACOKINETICS & DOSING

|                               | ERENUMAB          | FREMANEZUMAB                                   | GALCANEZUMAB                                                                                                   | EPTINEZUMAB    |
|-------------------------------|-------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------|
| <b>Route</b>                  | SC                | SC                                             | SC                                                                                                             | IV             |
| Tmax (days)                   | 6                 | 5–7                                            | 5                                                                                                              | n/a            |
| Vd (L)                        | 3.86              | 6                                              | 7.3                                                                                                            | n/a            |
| <b>t<sub>1/2</sub> (days)</b> | 21                | <b>40–48</b>                                   | 28                                                                                                             | 31             |
| <b>Dosing</b>                 | 70–140 mg monthly | 225 mg monthly <b>or</b> 675 mg every 3 months | migraine: 240 mg then 120 mg monthly · cluster: 300 mg at onset, then monthly to the end of the cluster period | every 3 months |

## ★ MABS VS THE CLASSICAL ORAL PREVENTIVES

|                                 | MABS                                                                | ORAL PREVENTIVES                                                                      |
|---------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Selectivity</b>              | <b>High</b>                                                         | Low                                                                                   |
| Administration                  | SC / IV                                                             | oral                                                                                  |
| <b>t<sub>1/2</sub></b>          | <b>weeks to months</b>                                              | < 24 h                                                                                |
| Dosing frequency                | every <b>1–3 months</b>                                             | ≥ 1 × daily                                                                           |
| <b>Elimination</b>              | <b>proteolysis</b>                                                  | <b>CYP450</b> (CYP2D6 — amitriptyline, propranolol) · phase II (topiramate)           |
| <b>PK interaction potential</b> | <b>low to none</b>                                                  | <b>high</b>                                                                           |
| Titration                       | <b>not needed</b>                                                   | needed                                                                                |
| <b>Onset of prophylaxis</b>     | ★ <b>rapid — within a week</b>                                      | <b>slow — several weeks to months</b>                                                 |
| Known ADRs                      | <b>injection-site pain</b> ; ⚠ long-term significance still unknown | weight change, sedation, cognitive and mood effects ( <i>see the ADR list above</i> ) |
| <b>Immunogenicity</b>           | <b>Yes</b>                                                          | No                                                                                    |

## ADMINISTRATION — ANTI-CGRP MONOCLONAL ANTIBODIES

|                               | ERENUMAB                 | FREMANEZUMAB                                                | GALCANEZUMAB                                                                                                                | EPTINEZUMAB                   |
|-------------------------------|--------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Route</b>                  | <b>SC</b>                | <b>SC</b>                                                   | <b>SC</b>                                                                                                                   | ★ <b>IV</b>                   |
| <b>T<sub>max</sub> (days)</b> | 6                        | 5–7                                                         | 5                                                                                                                           | —                             |
| <b>t<sub>1/2</sub> (days)</b> | 21                       | <b>40–48</b>                                                | 28                                                                                                                          | 31                            |
| <b>Dose</b>                   | <b>70–140 mg monthly</b> | <b>225 mg monthly or 675 mg (225 mg × 3) every 3 months</b> | <b>migraine: 240 mg loading, then 120 mg monthly · cluster: 300 mg at onset, then monthly until the cluster period ends</b> | <b>1000 mg every 3 months</b> |

- ★ **Fremanezumab's 40–48 day half-life is what makes the quarterly option possible** — same total drug, three injections in one sitting instead of three visits.
- ★ **Galcanezumab is the only one with a loading dose**, and the only one dosed differently for cluster headache, where it is given **at the onset of the cluster period** rather than continuously.
- 💡 Erenumab and the other SC agents can be self-injected at home; eptinezumab is an infusion, so the quarterly schedule comes with a clinic visit attached.
- 💡 ⚠ **The eptinezumab figure of 1000 mg reads oddly against the rest of the table** — it is an order of magnitude above the other antibodies. That is what the slide states; **check it against the paper before you use the number**, because if it is a transcription slip it is the kind that survives into an exam answer.

Principle behind this: *[[pharm-cns-basics]]* — Ergotamine at 5-HT<sub>1B/1D</sub> — and the 5-HT<sub>2B</sub> valvulopathy story sits one receptor away.

Source: *Week13 Pharmacology of Drugs Used in Migraine Headache.pdf* — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สม), Pharmacology 3 (ภกปว 401), 38 slides. Structures omitted.





# Analgesics II — Two Separate Territories

★ This lecture is two topics stapled together, and they do not share a mechanism. **Part 1 — neuropathic pain:** pain generated by a damaged nervous system itself, which is **not responsive to NSAIDs/coxibs and only weakly responsive to morphine**, so it needs a completely different drug list. **Part 2 — centrally acting muscle relaxants:** drugs that lower muscle tone by acting in the spinal cord and brain, not at the muscle. The only thing linking them is that several agents (NaV blockers, NMDA blockers,  $\alpha_2$  ligands) show up in both.

## ★ NOCICEPTIVE VS NEUROPATHIC — WHY THE USUAL ANALGESICS FAIL

|                        | NOCICEPTIVE PAIN                               | NEUROPATHIC PAIN                                                                        |
|------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Origin</b>          | tissue damage detected by an intact nociceptor | <b>damage / disease of the somatosensory system itself</b>                              |
| <b>NSAIDs / coxibs</b> | work                                           | ● <b>not responsive</b>                                                                 |
| <b>Morphine</b>        | works                                          | ⚠ <b>less responsive</b>                                                                |
| <b>Time course</b>     | resolves with the tissue                       | ★ <b>may persist beyond the pathology</b>                                               |
| <b>Quality</b>         | aching, throbbing, well-localised              | burning, tingling, <b>electric-shock-like</b> , with numbness <i>alongside</i> the pain |

Opioid pharmacology itself: [[drug-class-analgesics-opioids]].

# Neuropathic Pain — the Disease

nerve injury / disease → peripheral sensitization (*channel + receptor upregulation, immune mediators*) → ectopic firing → central sensitization (*loss of spinal inhibition + glial activation*) → maladaptive plasticity — pain outlives the injury

## SYMPTOMS — THE VOCABULARY THE EXAM USES

**Negative symptoms** (*loss of function*) — numbness · hypoalgesia

**Positive symptoms** (*gain of function*), in two families: - **Spontaneous** — arises with no stimulus at all -

**Continuous**: burning, tingling - **Paroxysmal**: sharp shooting, lancinating, **electrical-shock-like** - **Abnormal sensation: paresthesia** (*abnormal but not unpleasant*) · **dysesthesia** (*abnormal and unpleasant*) - **Stimulus-dependent** — needs a trigger - **Hyperalgesia** — an exaggerated response to a painful stimulus - **Allodynia** — pain from a stimulus that shouldn't hurt at all

★ **Negative and positive symptoms coexist in the same territory** — a numb foot that also burns is the signature, and it's what makes neuropathic pain diagnosable from the history.

**Common examples: painful diabetic neuropathy · chemotherapy-induced neuropathy.**

## ★ PATHOPHYSIOLOGY — MALADAPTIVE PLASTICITY OF THE SOMATOSENSORY SYSTEM

**Peripheral sensitization** (*at the injured nerve*): - **Sensitization of TRP channels** — TRPV1, TRPM8, TRPA1 (*TRP = transient receptor potential ion channels*) - **Upregulation of voltage-gated sodium channel (NaV) subtypes** → ectopic, spontaneous firing - **Upregulation of the  $\alpha_2\delta$ -1 subunit of the N-type calcium channel** → more transmitter release - **Proinflammatory cytokines** — TNF- $\alpha$ , IL-1 $\beta$ , IL-6 — and NGF released from **immune cells surrounding the nerve**

**Central sensitization** (*at the spinal cord*) — **loss of central inhibition**: - **Dysregulation of the descending inhibitory pathways (NE and 5-HT)** - ↓ **opioid binding sites** - ↓ **GABA inhibitory efficiency** - **Glial cell activation** → inflammatory mediators that **facilitate spinal pain transmission**

→ **Downstream:** ★ **every drug class below is named on this list.** NaV upregulation → sodium channel blockers ·  $\alpha_2\delta$ -1 upregulation → gabapentinoids · lost NE/5-HT descending inhibition → antidepressants · TRPV1 sensitization → capsaicin · NMDA-driven central sensitization → ketamine. ● The lost opioid binding sites are exactly why morphine underperforms here.

## THE WHOLE ARSENAL

| CLASS                                                         | DRUGS                                                      | TARGET                                                                      |
|---------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Antiepileptics — NaV blockers</b>                          | carbamazepine, oxcarbazepine (+ <i>topical lidocaine</i> ) | <b>voltage-gated Na<sup>+</sup> channel</b>                                 |
| <b>Antiepileptics — <math>\alpha_2\delta</math>-1 ligands</b> | gabapentin, pregabalin, mirogabalin                        | <b>N-type Ca<sup>2+</sup> channel <math>\alpha_2\delta</math>-1 subunit</b> |
| <b>Antidepressants — TCAs</b>                                 | amitriptyline, nortriptyline                               | <b>NET + 5-HTT</b> (+ NMDA, NaV)                                            |
| <b>Antidepressants — SNRIs</b>                                | duloxetine, venlafaxine, milnacipran                       | <b>NET + 5-HTT</b>                                                          |
| <b>Opioids</b>                                                | tramadol, methadone                                        | <b><math>\mu</math> receptor + NET/5-HTT</b><br>(methadone also NMDA)       |
| <b>Others</b>                                                 | <b>topical capsaicin · ketamine · cannabinoids</b>         | TRPV1 · NMDA · —                                                            |

Class pages: [[drug-class-antiepileptics]] · [[drug-class-antidepressants-lithium]] · [[drug-class-analgesics-opioids]].

### ★ THE DORSAL-HORN SYNAPSE — ONE PICTURE THAT HOLDS EVERY DRUG

**Ascending arm:** the **peripheral nociceptive fiber** fires via **NaV**, opens the **N-type calcium channel**, and releases **glutamate** onto **NMDA receptors** on the **dorsal horn neuron** → spinal pain transmission → higher centres.

**Descending arm:** the **raphe nuclei** send **5-HT** and the **locus coeruleus** sends **NE** down onto that same dorsal horn neuron, acting at **5-HT<sub>1A</sub>** and  **$\alpha_2$**  receptors to suppress transmission — and both transmitters are cleared by their reuptake transporters, **5-HTT** and **NET**.

→ **Downstream:** two ways to win. **Turn the ascending arm down** — NaV blockers and TCAs on the sodium channel, gabapentinoids on the N-type calcium channel, ketamine/methadone/TCAs on NMDA. Or **turn the descending arm up** — block NET and 5-HTT with **TCAs, SNRIs, tramadol or methadone** so NE and 5-HT stay in the synapse longer.

# Sodium Channel Blockers

## MECHANISM

**Accumulation and sensitization of NaV1.3, NaV1.7, NaV1.8 and NaV1.9** on the injured nerve produces neuropathic symptoms — ★ **especially the paroxysmal, sharp-shooting pain**. These drugs **block those NaV channels at the peripheral nociceptive fiber**, so the ectopic firing never leaves the nerve.

→ **Downstream:** ⚠ **most antiepileptics that block NaV are non-selective** — they hit cardiac and central sodium channels too, which is where the ADRs come from. Full antiepileptic pharmacology: [[drug-class-antiepileptics]].

## DRUGS

| DRUG                                          | WHAT IT TREATS                                                                    |
|-----------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Carbamazepine</b>                          | neuropathic pain — especially <b>paroxysmal sharp shooting / lancinating</b> pain |
| <b>Oxcarbazepine</b>                          | neuropathic pain — same target                                                    |
| <b>Lidocaine</b> ( <i>transdermal patch</i> ) | <b>localised</b> neuropathic pain — applied over the painful area                 |

## ADVERSE EFFECTS

- **Somnolence**
- **Dizziness**
- **Ataxia**
- **Rash**
- ● **Arrhythmia** — the lidocaine patch, from systemic sodium-channel blockade

# Gabapentinoids ( $\alpha_2\delta$ -1 Ligands)

## MECHANISM

Bind the  $\alpha_2\delta$ -1 subunit of the N-type voltage-gated calcium channel — the subunit that gets **upregulated** after nerve injury. Less calcium entry at the presynaptic terminal means **less glutamate released from the C-fiber**, and **reduced neuronal hyperexcitability**.

→ **Downstream:** ★ **they are not calcium-channel blockers in the cardiovascular sense** — they bind an auxiliary subunit whose expression the injury itself created, which is why they are selective for the pathological state rather than for normal transmission. See [[drug-class-antiepileptics]].

## DRUGS

| DRUG               | WHAT IT TREATS   |
|--------------------|------------------|
| <b>Gabapentin</b>  | neuropathic pain |
| <b>Pregabalin</b>  | neuropathic pain |
| <b>Mirogabalin</b> | neuropathic pain |

## ADVERSE EFFECTS

- Edema
- Weight gain
- Somnolence
- Dizziness
- Ataxia, incoordination

# Antidepressants — TCAs & SNRIs

## MECHANISM — REPAIRING THE DESCENDING INHIBITORY PATHWAY

**Block NET and 5-HTT** → more NE and 5-HT in the dorsal horn → ★ **the descending inhibitory pain pathway is enhanced**, which is precisely the system that central sensitization degraded.

★ **TCAs do more than the SNRIs**: on top of the transporter block they also **block NMDA receptors** and **NaV channels** — three mechanisms in one molecule, which is why they remain first-line despite the side effects.

→ **Downstream**: the analgesic effect is **independent of the antidepressant effect** — same drugs, different job. Full class: [[drug-class-antidepressants-lithium]].

## DRUGS

| DRUG                       | WHAT IT TREATS                                          |
|----------------------------|---------------------------------------------------------|
| <b>Amitriptyline</b> (TCA) | neuropathic pain — NET/5-HTT + <b>NMDA</b> + <b>NaV</b> |
| <b>Nortriptyline</b> (TCA) | neuropathic pain — same triple mechanism                |
| <b>Duloxetine</b> (SNRI)   | neuropathic pain                                        |
| <b>Venlafaxine</b> (SNRI)  | neuropathic pain                                        |
| <b>Milnacipran</b> (SNRI)  | neuropathic pain                                        |

## ADVERSE EFFECTS — SPLIT BY SUBCLASS

**TCAs**: - Anticholinergic effects - Sedation - ● **QT interval prolongation**

**SNRIs**: - Nausea, vomiting - Insomnia - Tachycardia - ⚠ **Increased blood pressure**

# Opioids — Tramadol & Methadone

## MECHANISM — WHY THESE TWO AND NOT MORPHINE

All opioids here are  $\mu$  agonists — but neuropathic pain is **less responsive to morphine**, because the injury itself **reduced spinal opioid binding sites**.

★ **Tramadol and methadone are the exceptions because they are not purely opioids:** both **block NET and 5-HTT**, so they recruit the descending inhibitory pathway the same way a TCA or SNRI does. **Methadone additionally blocks NMDA receptors.**

→ **Downstream:** the non-opioid mechanisms are what earn them a place on this list. Core opioid pharmacology, tolerance and dependence: [[drug-class-analgesics-opioids]].

## DRUGS

| DRUG             | WHAT IT TREATS                                                  |
|------------------|-----------------------------------------------------------------|
| <b>Tramadol</b>  | neuropathic pain — $\mu$ agonist + NET/5-HTT block              |
| <b>Methadone</b> | neuropathic pain — $\mu$ agonist + NET/5-HTT block + NMDA block |

## ADVERSE EFFECTS

- **Tramadol** — nausea, vomiting
- ● **Methadone** — QT interval prolongation



# Other Agents — Capsaicin, Ketamine, Cannabinoids

## ★ TOPICAL CAPSAICIN — AN AGONIST USED AS A DESTROYER

**Activates and then desensitizes TRPV1**, and with continued use **temporarily degenerates the intraepidermal C-fibers** themselves — the nerve endings generating the pain are pruned away.

**The clinical consequences follow directly from that:** - ⚠️ **An initial burning sensation on application** — you are *agonising* TRPV1 before you desensitize it - ★ **Analgesic onset takes 1–2 weeks** — you are waiting for fiber degeneration, not for a receptor to occupy - **Effective concentration for neuropathic pain is at least 0.075%**

→ **Downstream:** ★ **the burning is the mechanism working, not a failure** — the single most counselling-relevant fact on this page, because a patient who stops on day 1 never reaches the effect.

## KETAMINE

**Blocks NMDA receptors** — the receptor driving central sensitization at the dorsal horn neuron.

→ **Downstream:** the same target methadone and the TCAs reach partially, hit directly and hard — which is also why the ADR is a **dissociative** one.

## DRUGS

| DRUG                                                           | WHAT IT TREATS                                            |
|----------------------------------------------------------------|-----------------------------------------------------------|
| <b>Capsaicin</b> ( <i>topical, <math>\geq 0.075\%</math></i> ) | localised neuropathic pain — <b>TRPV1 desensitization</b> |
| <b>Ketamine</b>                                                | neuropathic pain — <b>NMDA antagonist</b>                 |
| <b>Cannabinoids</b>                                            | neuropathic pain                                          |

## ADVERSE EFFECTS

- **Capsaicin** — **burning sensation, irritation** at the application site
- ● **Ketamine** — **dissociative psychosis**

# Centrally Acting Muscle Relaxants — Overview

★ **Territory 2 starts here.** "Muscle relaxant" is a classification by *site*, not by effect — and the **centrally acting** group works in the spinal cord and brain, nowhere near the neuromuscular junction.

## CLASSIFICATION — CENTRAL VS PERIPHERAL

| GROUP               | SUBCLASS         | DRUGS                                                                                                                                |
|---------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ★ Centrally acting  | —                | orphenadrine · tolperisone, eperisone · diazepam, clonazepam · baclofen · tizanidine · riluzole ( <i>approved for spasm in ALS</i> ) |
| Peripherally acting | depolarizing     | succinylcholine                                                                                                                      |
|                     | non-depolarizing | curare alkaloid derivatives                                                                                                          |
|                     | direct-acting    | dantrolene                                                                                                                           |

*Everything below this line is the centrally acting group — the peripheral agents belong to neuromuscular-blocker pharmacology, not to this lecture.*

## ★ SPASTICITY VS SPASM — TWO DIFFERENT PROBLEMS

|             | MUSCLE SPASTICITY                                                                                 | MUSCLE SPASM                 |
|-------------|---------------------------------------------------------------------------------------------------|------------------------------|
| Lesion is   | CENTRAL                                                                                           | PERIPHERAL                   |
| Caused by   | cerebral infarction · spinal cord tumor                                                           | musculoskeletal disturbances |
| Presents as | clonus · stiffness · rigidity and tightness · difficulty walking, moving and occasionally talking | pain and/or disability       |
| Ends in     | chronic disability                                                                                | pain and/or disability       |

## ★ THE SPINAL CIRCUIT — WHERE EACH DRUG SITS

Two independent effects come out of the same drug list, from two different sites: - Actions at the CNS, ventral horn, motor nerve and skeletal muscle → the **MUSCLE RELAXANT** effect - Actions at the sensory nerve and dorsal horn → the **ANALGESIC** effect

The tone-setting circuit at the ventral horn: a **glutamate neuron** (*la fibre and corticospinal tract*) drives the **α motor neuron** through **AMPA and NMDA** receptors → **increased muscle tone**. Opposing it, a **glycine interneuron** (*glycine receptor*) and a **GABA interneuron** (*GABA-A and GABA-B*) → **decreased muscle tone**.

→ **Downstream:** ★ ● **muscle relaxants have NO anti-inflammatory effect.** They relax and they relieve pain — they do not treat the inflammation causing the spasm, so pairing one with an anti-inflammatory is a mechanism decision, not a habit.

★ MECHANISM TABLE — THE EXAM'S FAVOURITE

| DRUG                           | MECHANISM                                                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Orphenadrine</b>            | <b>non-selective NaV blocker · NMDA receptor blocker</b> ( <i>moderate affinity</i> ) · <b>muscarinic receptor antagonist</b>                                                         |
| <b>Tolperisone · Eperisone</b> | <b>non-selective NaV blockers</b> + inhibit <b>Ia fiber discharge</b> · <b>CaV blockers</b> → ↓ transmitter release from C-fiber · ↑ <b>blood flow to muscle</b> ( <i>eperisone</i> ) |
| <b>Diazepam · Clonazepam</b>   | <b>GABA-A allosteric potentiating ligand</b>                                                                                                                                          |
| <b>Baclofen</b>                | <b>GABA-B agonist</b>                                                                                                                                                                 |
| <b>Tizanidine</b>              | <b>α<sub>2</sub> agonist</b>                                                                                                                                                          |
| <b>Riluzole</b>                | <b>non-selective NaV blocker · NMDA receptor blocker</b>                                                                                                                              |

# Orphenadrine

## MECHANISM — THREE RECEPTORS, THREE CONSEQUENCES

**Non-selective NaV blockade · NMDA receptor blockade** (*moderate affinity*) · **muscarinic receptor antagonism**.

→ **Downstream:** ★ it produces **both analgesic and anticholinergic effects** — the first two mechanisms give the analgesia, the third gives the entire side-effect profile. **Read its ADR list straight off its receptor list.**

## DRUGS

| DRUG                              | WHAT IT TREATS                                  |
|-----------------------------------|-------------------------------------------------|
| <b>Orphenadrine</b>               | muscle spasm — <b>100 mg twice daily</b> orally |
| <b>Orphenadrine + paracetamol</b> | muscle spasm with pain — the fixed combination  |

## ADVERSE EFFECTS

- **Cardiovascular** — palpitations, tachycardia
- **Anticholinergic, and common:**
  - constipation
  - urinary retention
  - confusion

## CAUTIONS

- ⚠ **Use with caution in heart failure, tachycardia or arrhythmias** — the CV effects
- ● **Chronic use must be tapered on discontinuation** — do not stop it abruptly

## INTERACTIONS

- **Orphenadrine + cholinergic drugs** — direct pharmacodynamic opposition
- **+ CNS depressants** — the class-wide interaction (*see the shared PK/interaction page below*)

## ADMINISTRATION — ORPHENADRINE

- **100 mg orally twice daily.**
- ● **Chronic use requires a taper on discontinuation** — it is not a drug to simply stop.
- ⚠ Use with caution in **heart failure, tachycardia or arrhythmias** (palpitations, tachycardia), and expect the anticholinergic set: constipation, urinary retention, confusion.
- 💡 The taper is the part that gets missed, because a muscle relaxant doesn't feel like a drug anyone needs to come off carefully.

# Tolperisone & Eperisone

## MECHANISM

**Non-selective NaV blockers** that also **inhibit Ia fiber discharge** — cutting the afferent limb of the stretch reflex that maintains tone. Plus **CaV blockade**, which **reduces neurotransmitter release from the C-fiber**.

**Eperisone additionally increases blood flow to the muscle.**

→ **Downstream:** they attack the **muscle spasm reflex** at both ends — the Ia sensory input that triggers it and the C-fiber transmission that makes it hurt.

## DRUGS

| DRUG               | WHAT IT TREATS                                                    |
|--------------------|-------------------------------------------------------------------|
| <b>Tolperisone</b> | muscle spasm — $t_{1/2}$ 1.5 h                                    |
| <b>Eperisone</b>   | muscle spasm — $t_{1/2}$ 1.5–2 h; also ↑ <b>muscle blood flow</b> |

# Benzodiazepines — Diazepam & Clonazepam

## MECHANISM

**GABA-A allosteric potentiating ligands** — they do not open the channel themselves, they make the GABA already present work harder at the **GABA-A** receptor on the inhibitory interneuron circuit of the ventral horn.

→ **Downstream:** ★ **more GABA-A inhibition** → **decreased muscle tone**. Note the contrast that the next page turns on: **benzodiazepines = GABA-A · baclofen = GABA-B** — same transmitter, different receptor, different drug. Full class: [[drug-class-anxiolytics-hypnotics]] · receptor detail: [[drug-class-gaba-a]].

## DRUGS

| DRUG              | WHAT IT TREATS                                   |
|-------------------|--------------------------------------------------|
| <b>Diazepam</b>   | muscle spasm / spasticity — ⚠ $t_{1/2}$ 50–100 h |
| <b>Clonazepam</b> | muscle spasm / spasticity — ⚠ $t_{1/2}$ 19–60 h  |

## CAUTIONS

- ⚠ **By far the longest half-lives in this group** — accumulation and daytime sedation are the practical limit on their use as relaxants
- + **CNS depressants** — the class-wide interaction, and it bites hardest here

# Baclofen

## MECHANISM

**A GABA-B agonist** — ★ its **chemical structure resembles GABA**, which is exactly why it reaches the receptor. It **binds GABA-B receptors to decrease synaptic transmission in the spinal cord**.

→ **Downstream: a potent CNS depressing effect** — the therapeutic action and the ADR list are the same phenomenon at different doses. Contrast [[drug-class-gaba-a]]: the benzodiazepines potentiate an ionotropic **GABA-A** chloride channel, baclofen is a direct agonist at the metabotropic **GABA-B** receptor.

## DRUGS

| DRUG            | WHAT IT TREATS                                                                                                  |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Baclofen</b> | spasticity — <b>start 5 mg three times daily</b> , titrate up to <b>80 mg/day in 3–4 divided doses</b> (adults) |

## PHARMACOKINETICS — THE REASON IT NEEDS RENAL MONITORING

| BACLOFEN                                                                            |                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Half-life</b>                                                                    | <b>short — 2–6 h</b> ( <i>3–4 h in the class PK table</i> ) |
| <b>Elimination</b>                                                                  | ★ <b>excreted mainly by the kidney UNCHANGED (70–85%)</b>   |
| ⚠ <b>Almost no metabolism means renal function <i>is</i> the dose</b> — monitor it. |                                                             |

## ADVERSE EFFECTS

- Weakness
- Dizziness, drowsiness, vertigo
- Hypotension
- Insomnia

## ● WITHDRAWAL — THE DANGEROUS FACT ON THIS PAGE

**Sudden discontinuation can cause: - High fever - Confusion, hallucinations - Seizures - ★**

**Exacerbation of muscle rigidity** — the treated symptom comes back worse than baseline

→ **Taper off baclofen, often over several weeks. Also monitor renal function** throughout.

# Tizanidine

## MECHANISM

**A centrally acting  $\alpha_2$  adrenergic agonist.** It **increases presynaptic inhibition of motor neurons by blocking glutamate release** — turning down the excitatory drive onto the  $\alpha$  motor neuron rather than boosting inhibition.

→ **Downstream:** ⚠️  **$\alpha_2$  agonism is not confined to the spinal cord** — the same receptor action elsewhere is what produces the hypotension and bradycardia below.

## DRUGS

| DRUG              | WHAT IT TREATS                                                                               |
|-------------------|----------------------------------------------------------------------------------------------|
| <b>Tizanidine</b> | spasticity / muscle spasm — <b><math>\alpha_2</math> agonist, <math>t_{1/2}</math> 2–3 h</b> |

## ADVERSE EFFECTS

- Hypotension
- Bradycardia
- Excessive sedation

## ● INTERACTIONS — THE ONE TO ACTUALLY MEMORISE

Metabolized extensively in the liver by CYP1A2 to inactive metabolites.

- ● CYP1A2 inhibitors are **CONTRAINDICATED** with tizanidine — **fluvoxamine · ciprofloxacin**
- CYP1A2 inducers — **smoking · omeprazole · lansoprazole** → loss of effect
- ⚠️ + **drugs that cause vasodilation** —  **$\alpha_1$  blockers · CCBs and other vasodilators** → additive hypotension

See *[[drug-class-cyp450-inducers-inhibitors]]*.

## CAUTIONS

- ⚠️ **Requires gradual dose reduction to stop therapy** — like orphenadrine and baclofen, not a drug you simply stop



# Riluzole

## MECHANISM

**Non-selective NaV blocker** and **NMDA receptor blocker** — mechanically the same pair orphenadrine uses, without the muscarinic arm.

→ **Downstream:** its niche is defined by indication rather than by mechanism.

## DRUGS

| DRUG            | WHAT IT TREATS                                                               |
|-----------------|------------------------------------------------------------------------------|
| <b>Riluzole</b> | ★ approved for spasm in amyotrophic lateral sclerosis (ALS) — $t_{1/2}$ 12 h |

## CAUTIONS

- ⚠ **A high-fat meal reduces absorption** — the only food effect flagged in this class

# Muscle Relaxants — Shared PK & Interactions

## HALF-LIVES — THE PROPERTY THAT ACTUALLY DRIVES DOSING

| DRUG                | T <sub>1/2</sub> (H) | OTHER PK                                           |
|---------------------|----------------------|----------------------------------------------------|
| <b>Tolperisone</b>  | 1.5                  | —                                                  |
| <b>Eperisone</b>    | 1.5–2                | —                                                  |
| <b>Tizanidine</b>   | 2–3                  | ★ <b>CYP1A2</b> → inactive metabolites             |
| <b>Baclofen</b>     | 3–4 (2–6 h reported) | ★ <b>excreted unchanged, mainly renal (70–85%)</b> |
| <b>Riluzole</b>     | 12                   | ⚠ <b>high-fat meal reduces absorption</b>          |
| <b>Orphenadrine</b> | 14–20                | CYP metabolism ( <i>not fully characterised</i> )  |
| <b>Clonazepam</b>   | ⚠ 19–60              | —                                                  |
| <b>Diazepam</b>     | ● 50–100             | —                                                  |

★ **Read the spread as one fact:** tolperisone/eperisone/tizanidine are short enough to need multiple daily doses, while the **benzodiazepines sit 20–50× longer** — which is the real reason they're not the routine choice for ordinary muscle spasm.

## INTERACTIONS — PHARMACODYNAMIC VS PHARMACOKINETIC

**Pharmacodynamic (PDDI):** - ● **ALL muscle relaxants + CNS depressants** — the class-wide, always-applies interaction - **Orphenadrine + cholinergic drugs** — its muscarinic antagonism works against them - **Tizanidine + anything vasodilating** —  $\alpha_1$  blockers · CCBs and other vasodilators → hypotension

**Pharmacokinetic (PKDI):** - ● **CYP1A2 inhibitors + tizanidine** — fluvoxamine, ciprofloxacin — **contraindicated** - **CYP1A2 inducers + tizanidine** — smoking, omeprazole, lansoprazole

★ **Only tizanidine has a named PK interaction in this lecture.** Everything else in the group interacts by *sedation*, not by *enzyme*. Inducer/inhibitor lists: [[drug-class-cyp450-inducers-inhibitors]].

*Principle behind this: [[pharm-cns-basics]] — Dorsal horn 5-HT<sub>1A</sub> and  $\alpha_2$  inhibit pain transmission, which is why an antidepressant is an analgesic there; gabapentin and pregabalin arrive via LAT and act on Ca<sup>2+</sup> channels.*

*Source: Week14 Pharmacology of Analgesics II Drug used in Neuropathic Pain & Centrally Acting Muscle Relaxants.pdf — อ.พงศธร (อ.ดร.พงศธร มีสวัสดิ์สม), Pharmacology 3 (ภาค 401), 25 slides. Structures omitted.*